Accelerating Variance Reduction: Optimal Complexity of Finite-Sum Optimization

Project report, Statistical Learning Theory

Beline Yuan Paul Ruiz Alliata Louis Darrigol

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Abstract

SAGA minimizes averages of n smooth functions with condition number κ in $O((n+\kappa)\log(1/\varepsilon))$ component gradients, whereas span-based methods with i.i.d. sampling need $\Omega((n+\sqrt{n\kappa})\log(1/\varepsilon))$. We ask whether accelerating variance reduction needs more than Nesterov momentum, and where the SAGA proof of Bach's *Learning Theory from First Principles* (Proposition 5.9) is loose. We write the proofs of SAGA and loopless Katyusha as chains of inequalities evaluated exactly along trajectories. On synthetic logistic regression ($n = 1000$, $10 \le \kappa \le 10^5$, one data set per covariance spectrum, five sampling seeds), SAGA's exact contraction exponent, averaged over passes 2–20, is 3.2–220 times the certified one; for $\kappa \ge 10^4$, cocoercivity and strong convexity with global constants carry 77–96% of the logarithmic gap in proof order, whereas the best step and Lyapunov weight gain at most a factor 4/3. No fixed preconditioner improves the global condition number of logistic regression; only local constants at the optimum account, heuristically, for the per-pass gains of dense metrics costing $O(d^2)$ per iteration. Loopless Katyusha's certified one-step contraction is pessimistic in exponent by a factor 2.8–4.7, and its negative momentum acts at a single proof step, an exact cancellation; without it the method diverges on hard instances with $\kappa \ge 10^3$ but is twice as fast on a logistic-regression problem ($\kappa = 10^4$, one seed). On four quadratic instances ($n = 16$), log–log slopes of gradient counts over $\kappa \ge 10n$ are 0.95–1.02 for SAGA (0.99 inside the certified window) and 0.49–0.52 for loopless Katyusha.

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1 Introduction

Finite sums and their complexity. Empirical risk minimization leads to minimizing over $\mathbb{R}^d$ an average $F = \frac{1}{n}\sum_{i=1}^n f_i$ of n functions. As in Chapter 5 of [11], each f_i is convex and R^2-smooth, F is μ-strongly convex, $\kappa = R^2/\mu$ (Section 2), and the cost of a method is the number of component gradients f_i' it evaluates to reach a relative accuracy ε. In the worst case, gradient descent needs of order $n\kappa\log(1/\varepsilon)$ of them and Nesterov's accelerated method $n\sqrt{\kappa}\log(1/\varepsilon)$, whereas variance-reduced methods such as SAG [52, 77], SVRG [43] and SAGA [16] need only $O((n+\kappa)\log(1/\varepsilon))$: the product of n and κ is replaced by a sum [11]. Proposition 5.9 of [11], restated in Section 2, proves this for SAGA with the step $1/(4R^2)$: a Lyapunov function combining the squared distance to the minimizer and the error of the gradient memory contracts in expectation by the factor $1-\min\{1/(3n), 3\mu/(16R^2)\}$ per iteration.

Two gaps. This guarantee leaves two gaps. The first is algorithmic: randomized incremental methods that sample one component i.i.d. per step and whose iterates stay in the span of the observed gradients need of order $(n+\sqrt{n\kappa})\log(1/\varepsilon)$ component gradients in the worst case [50], the $\sqrt{n\kappa}\log(1/\varepsilon)$ term survives without the span assumption when $\kappa \gtrsim n$ [87], and this order is attained by Katyusha [2] and its loopless version L-Katyusha [47], which improve on the guarantee of SAGA by a factor of order $\sqrt{\kappa/n}$ when $\kappa \gg n$. The second gap separates the theorem from the algorithm: the bound of Proposition 5.9 decreases by at most a factor $3/2$ per effective pass (n iterations), whereas on ℓ_2-regularized logistic regression the contraction rate of SAGA with the same step is several times larger, and up to 220 times larger when κ is large (Figure 1 and Section 5.3).

Question A: what it takes to accelerate variance reduction. Nesterov's momentum applied to a variance-reduced estimator, such as the SVRG estimator $f_i'(\theta)-f_i'(w)+F'(w)$ with w a reference point where the full gradient is known, is not known to accelerate with one sampled component per iteration [67, 2]; Katyusha adds a *negative momentum* that pulls the query point back towards w. We ask what exactly the lower bound states and on which instances it is attained (A1), whether and how this *naive acceleration* fails (A2), where the negative momentum enters a Lyapunov proof of L-Katyusha written in the style of the book (A3), and whether the exponents 1 and 1/2 of κ predicted for the gradient counts of SAGA and L-Katyusha are observed on hard instances (A4).

Question B: where the proof of Proposition 5.9 loses. Besides identities, the proof of Proposition 5.9 uses six inequalities (I1–I6, Section 2), two of which replace quantities that depend

on the iterate by the global constants R^2 (cocoercivity, I2) and μ (strong convexity, I4). We ask which inequalities explain the gap between the certified and the observed contraction, for various conditionings and anisotropies of the data (B3); whether trajectory-dependent constants such as the effective curvature $\mu_{\text{eff}}(\theta) = \langle \theta - \eta_*, F'(\theta) \rangle / \|\theta - \eta_*\|^2$ describe the convergence better (B2); and whether SAGA in a fixed quadratic metric (a preconditioner) improves the algorithm or only its analysis (B1).

Hypotheses. The project proposal turned these questions into seven hypotheses, which Section 6 assesses one by one. (A1) Span-based methods with i.i.d. uniform sampling need $\Omega((n + \sqrt{n\kappa}) \log(1/\varepsilon))$ component gradients; for an arbitrary index sequence the $\sqrt{n\kappa}$ term survives and the $n \log(1/\varepsilon)$ term weakens by a factor $\log(n/\kappa)$ when $\kappa \ll n$. (A2) Naive acceleration, Nesterov's momentum applied to the SVRG estimator, is the baseline that (A3) corrects: implicitly, it does not accelerate. (A3) The negative momentum of loopless Katyusha makes the variance absorbable and yields the optimal rate, with a Lyapunov proof in the style of the book. (A4) On the hard instance of (A1), the number of gradients to a fixed accuracy grows like κ for SAGA and like $\kappa^{1/2}$ for L-Katyusha when $\kappa \gg n$. (B1) Running SAGA in a fixed quadratic metric P shows which steps of the proof are too loose. (B2) Trajectory-dependent constants such as μ_{eff} are significantly tighter than μ, so that the strong-convexity step, or another specific inequality, explains most of the gap between the certified and the observed contraction. (B3) An experimental diagnosis, for several curvatures, locates the loose steps of the proof. The proposal also planned three figures (convergence against the bound, tightness of each inequality, bottleneck as a function of the geometry), which are Figures 1–3.

Method. We treat proofs as objects that can be measured. Each proof is rewritten as a chain of labelled elementary inequalities that starts from the conditional expectation of its Lyapunov function after one step. As the sampled index is uniform over n values, every conditional expectation is computed exactly from the current state by enumerating the n indices (and the coin of the loopless methods); along trajectories this gives the exact one-step contraction, the slack of each inequality (a *tightness ratio* in $[0, 1]$) and the share of the gap that each inequality accounts for. For acceleration we use quadratic instances built, like the lower-bound constructions, from chain quadratics with minimizers known in closed form (the lower bounds of Appendix A1 are proved on one of them), and random block quadratics as a control.

Contributions.

- **Anatomy of Proposition 5.9** (Section 4.1, Appendix B3). We write the proof as an exact one-step identity, Proposition B3.5 (exact conditional contraction), followed by I1–I6, attach a tightness ratio to each inequality (Definition B3.8), and prove in Proposition B3.11 (attribution of the gap) that undoing I1–I5 one at a time gives a monotone chain from the certified to the exact contraction, hence an additive attribution of the logarithmic gap. By Proposition B3.16 (optimal constants of the proof structure) and Remark B3.18, optimizing the step and the Lyapunov weight within the proof gains at most a factor 3 (at most 4/3 when $\kappa \geq 9n/16$), the optimal step and rate being those of [40]; and no analysis can remove the exponent 1 of κ for SAGA with a constant step, since a two-dimensional instance requires at least $\frac{\kappa-1}{4} \log \frac{1}{2\varepsilon}$ iterations whatever the step (Proposition B3.17).
- **Where the proof loses** (Sections 5.3 and 5.4). On synthetic logistic regression ($n = 1000$, $d = 50$, $\kappa \in \{10, \ldots, 10^5\}$, covariance spectra $j^{-\alpha}$ with $\alpha \in \{0, 1, 2\}$, 5 seeds, passes 2–20), the exact contraction exponent exceeds the certified one by a factor 3.2–3.7 for $\kappa \leq 10^2$ and 19–220

at $\kappa = 10^5$. In the cumulative attribution of Proposition B3.11 in proof order, for $\kappa \leq 10^2$, I1 (square norm) and I2 (cocoercivity) carry 0.39–0.44 and 0.32–0.37 of the logarithmic gap; for $\kappa \geq 10^4$, I2 (0.46–0.71, growing with the anisotropy) and I4 (strong convexity, 0.25–0.36) carry 0.77–0.96 of it; the largest shares of I3 and I5 are 0.54% and 22%. At $\kappa = 10^3$, the textbook constants are within 17% of the best ones of the proof, those of Proposition B3.16, while SAGA ($\alpha = 1$, 3 seeds) still converges at $\gamma R^2 = 4$, eight times the supremum $\frac{1}{2}$ of the steps γR^2 admitted by Proposition B3.14 (certified rate for general (γ, Δ)).

- **Effective constants** (Sections 4.2 and 5.5, Appendix B2). The constants $\mu_{\text{eff}}(\theta)$ and $L_{\text{eff}}(\theta)$ of Definition B2.2 (effective constants μ_{eff} and L_{eff}) turn I4 and I2 into identities at the current iterate. We prove $\mu \leq \mu_{\text{eff}} \leq L_{\text{eff}} \leq R^2$ (Lemma B2.6), show in Proposition B2.12 that the global strong-convexity constant of ℓ_2-regularized logistic regression is exactly λ, relate μ_{eff} to the Hessian at the optimum by self-concordance in Corollary B2.15 (effective curvature versus curvature at the optimum), and prove trajectory-dependent versions of Proposition 5.9, Proposition B3.19 (a-posteriori rates with effective constants) and Proposition B3.21 (a-priori rate on a region). Along SAGA (one run per configuration), at $\kappa = 10^4 = 10n$ and $\alpha \in \{0, 1\}$, μ_{eff}/μ approaches $\lambda_{\min}(F''(\eta_*))/\mu$ (9.94 against 9.86, 4.97 against 4.97) and the error stays concentrated on the flattest direction of the Hessian, which already carries 89–98% of the initial squared error, contrary to a heuristic prediction of our own; for $\alpha = 2$, whose bottom spectrum is nearly degenerate, neither SAGA nor gradient descent aligns. Remark B2.22 (outcome of prediction (P2)) reports the nine runs.
- **Preconditioning** (Sections 4.3 and 5.6, Appendix B1). By the change of variables of Proposition B1.5 (P-SAGA as SAGA on a transformed problem), Proposition 5.9 holds in a fixed metric P with relative constants (Corollary B1.12), but for ℓ_2-regularized logistic regression no P improves the global condition number, $\kappa_P \geq \kappa$ by Proposition B1.21 (no free lunch for global constants): the gains can only be trajectory-dependent; for $P = \hat{\Sigma} + \lambda I$ the trajectory bound of Corollary B1.23 (leverage-score bound along trajectories) sees the spectrum only through ridge leverage scores but pays the smallest per-sample curvature, and the observed gains are accounted for only by the local constants at the optimum of Lemma B1.27 (limits of the effective constants at η_*), used heuristically along the runs. With $\kappa = 10^3$ (5 seeds), this statistical metric and the Hessian at the optimum (an oracle benchmark), at $O(d^2)$ cost per iteration, raise the contraction exponent of $\|\theta_t - \eta_*\|_P^2$ from 1.00 to 1.11 per pass for $\alpha = 1$ and from 0.61 to 1.24–1.26 for $\alpha = 2$, whereas the transferred bound with global constants ranks them below the identity.
- **Lower bounds and a certified hard instance** (Section 4.4, Appendix A1). We restate the finite-sum lower bounds with their exact oracle assumptions in the notation of the book (Section A1.2). The first clause of hypothesis A1 holds for span-based methods with i.i.d. sampling: it is Theorem A1.8 (Lan–Zhou lower bound; Theorem 3 and Corollary 3 of [50]). Its second clause holds only in this sense: for oblivious methods (whose parameters do not depend on the oracle answers), up to logarithmic factors, and for loopless methods with a full-gradient coin, full-gradient steps can divide the $n \log(1/\varepsilon)$ term by at most $1 + (\log(n/\kappa))_+$, a gain only when $\kappa \ll n$ (part (c) of Theorem A1.11 and Theorem A1.12). By Theorem A1.9 (Woodworth–Srebro lower bounds; Theorems 4 and 8 of [87]), deterministic methods, cyclic ones included, cannot gain such a factor when $\kappa > 73$, and for randomized methods with fully adaptive queries only an additive n is proved; whether randomly reshuffled span-based methods, or deterministic ones when $\kappa \leq 73$, can beat $n \log(1/\varepsilon)$ is not settled by these results. On a block-separable instance with a closed-form minimizer (Proposition A1.14), we prove the lower bounds under an explicit dimension condition (Theorem A1.19, Corollary A1.20 and Proposition A1.18), bound the gradient count of SAGA from both sides in Corollary A1.23 (SAGA sandwich), and

delimit in Proposition A1.26 (testable prediction of hypothesis A4) the window of κ in which the exponents 1 and 1/2 of SAGA and L-Katyusha are certified. Beyond it the slope of SAGA against the nominal κ bends (from 0.99 inside the window to 0.95 over $\kappa \geq 10n$) and returns close to 1 against $\kappa_{\text{eff}} = R^2/\lambda_{\min}(F'')$ (1.03), as part (iii) of this proposition predicts, whereas that of L-Katyusha hardly bends (0.50 to 0.49) and rises slightly above 1/2 against κ_{eff} (0.53) (Section 5.9).

- **L-Katyusha in the style of the book** (Sections 4.5 and 5.7, Appendix A3). In the proof of Theorem A3.4 (contraction and rate of L-Katyusha; Lemma 10 and Theorem 11 of [47]), the negative momentum acts at a single step, Proposition A3.14 (Bregman-gap cancellation by the negative momentum): an exact cancellation between the inner-product part of the variance bound of the estimator and the term created by the pull towards the reference point. Without the pull, Proposition A3.19 (closing by smoothness and strong convexity costs κ) shows that the same argument certifies at best a contraction $1 - \kappa^{-2}$ per iteration, a limit of the argument, not a lower bound (Remark A3.20). Measured exactly, the certified one-step contraction is never violated and its exponent is pessimistic by a factor 2.8–4.7 (three problems, 3 seeds; 3.7–9.3 for the closed-form rate of Theorem A3.4), against 5.2 and 15.7 for Proposition 5.9 on the same logistic problems.
- **Naive acceleration and exponents** (Sections 5.8 and 5.9). On a block-chain instance ($n = 16$ chains of length 64, $\kappa = 10^4$, one seed; the lower bound is not certified on this instance), L-Katyusha without negative momentum diverges, and the best naive scheme of a grid needs $9.6 \cdot 10^4$ gradients to reach the relative accuracy 10^{-10}, against $3.2 \cdot 10^4$ for L-Katyusha with its untuned parameters; on logistic regression ($n = 1000$, $\kappa = 10^4$, $\alpha = 1$) the ranking is reversed, the best naive scheme needing less than half the gradients of L-Katyusha. Over sweeps of κ on four quadratic instances ($n = 16$, 2 or 3 seeds), the log–log slopes of the gradient counts over $\kappa \geq 10n$ are 0.95–1.02 for SAGA (0.99 inside the window of Proposition A1.26 on the shared-regularizer instance) and 0.49–0.52 for L-Katyusha; on the two instances where the naive scheme was run, re-tuned at every κ, its slope is 0.81, and L-Katyusha without negative momentum diverges for every $\kappa \geq 10^3$.

All numbers are produced from fixed seeds by the scripts of the accompanying code (Appendix R).

Outline. Section 2 fixes the notation and restates Proposition 5.9; Section 3 reviews related work; Section 4 states the main theoretical results; Section 5 reports the experiments; Section 6 assesses the hypotheses and discusses limitations and open questions. The five theory notes form Appendices B3, B2, B1, A1 and A3, named after the hypotheses, with complete statements and proofs. Section 4 restates their main results, without the proofs. Sections 4–6 treat the questions of type B before those of type A, because the exact instrumentation built for Proposition 5.9 is then reused for L-Katyusha.

2 Setting, notation and the result under study

Finite sums. We consider

$$F(\theta) = \frac{1}{n} \sum_{i=1}^{n} f_i(\theta), \qquad \theta \in \mathbb{R}^d, \tag{1}$$

and follow the notation of [11]: $f_i' = \nabla f_i$, $F' = \nabla F$, η_* is the minimizer of F, each f_i is convex and R^2-smooth (f_i' is R^2-Lipschitz continuous), F is μ-strongly convex, and $\kappa = R^2/\mu$ is the condition

number. Note that $R^2 \geq L_F$, the smoothness constant of F, so that $\kappa \geq L_F/\mu$; this distinction matters for the comparison between stochastic and full-gradient methods. Smoothness and convexity of each f_i give *cocoercivity* (Proposition 5.4 of the book), $\|f_i'(\theta) - f_i'(\eta)\|^2 \leq R^2 \langle f_i'(\theta) - f_i'(\eta), \theta - \eta\rangle$, which is the only property of the f_i used by the SAGA proof.

The main instance: ℓ_2-regularized logistic regression. For features $x_i \in \mathbb{R}^d$ and labels $y_i \in \{-1, 1\}$,

$$f_i(\theta) = \log\left(1 + e^{-y_i x_i^\top \theta}\right) + \frac{\lambda}{2}\|\theta\|^2, \qquad f_i'(\theta) = -y_i\,\sigma(-y_i x_i^\top \theta)\,x_i + \lambda\theta, \tag{2}$$

with $\sigma(u) = (1 + e^{-u})^{-1}$. Since the second derivative of the logistic loss is $\sigma(u)\sigma(-u) \leq 1/4$, each f_i is L_i-smooth with $L_i = \|x_i\|^2/4 + \lambda$, and we take $R^2 := \max_i L_i$ throughout; as Remark B3.7 (the constant R^2 for logistic regression) details, the book's "features bounded by R" is loose by a factor close to 4 when $\lambda \ll \max_i \|x_i\|^2$. The global strong-convexity constant of F is exactly $\mu = \lambda$ (Proposition B2.12): the data term of the Hessian, $\frac{1}{n}\sum_i \sigma'(y_i x_i^\top \theta)\, x_i x_i^\top$, is positive semidefinite and vanishes along directions where all margins go to infinity, so that no larger constant is valid globally, whereas locally (near η_*) the curvature is $\lambda_{\min}(F''(\eta_*)) \geq \lambda$, as Proposition B3.22 (local strong convexity of logistic regression) quantifies. This gap between global and local curvature is the object of hypothesis B2.

Synthetic problems are generated with $x_i = \Sigma^{1/2} g_i$, $g_i \sim \mathcal{N}(0, I_d)$, where Σ has eigenvalues $j^{-\alpha}$, $j = 1, \ldots, d$ (so $\alpha = 0$ is isotropic and the covariance condition number is d^α), a random rotation, and a global rescaling so that $\max_i \|x_i\| = 1$; labels are $y_i = \operatorname{sign}(x_i^\top w + \varepsilon_i)$ with $\operatorname{std}(Xw) = 2$ and $\varepsilon_i \sim \mathcal{N}(0, 1)$ (on the data sets of Section 5, $n = 1000$, $d = 50$, data seed 0, the labels differ from $\operatorname{sign}(x_i^\top w)$ for 15.0%, 16.6% and 15.7% of the samples when $\alpha = 0$, 1 and 2); the regularization $\lambda = (1/4)/(\kappa - 1)$ fixes $\kappa = R^2/\mu$ exactly. The minimizer η_* is computed by a damped Newton method until $\|F'(\eta_*)\| \leq 10^{-15}$ or stagnation, which bounds $\|\eta_* - \eta_{\text{exact}}\| \leq \|F'(\eta_*)\|/\mu$ by strong convexity, and $f_i'(\eta_*)$ is stored for all i. For the cached results, the worst of the fifteen problems of Section 5.3 stopped at $\|F'(\eta_*)\| = 1.3 \cdot 10^{-12}$, where the Armijo test of the line search had reached the rounding level of F (Appendix R).

SAGA and Proposition 5.9. SAGA [16] keeps a memory $z_i^{(t)} \in \mathbb{R}^d$ of the last gradient seen for each i. At iteration t an index $i(t)$ is drawn uniformly in $\{1, \ldots, n\}$ and

$$\begin{aligned} \theta_t &= \theta_{t-1} - \gamma\,\omega_t, \qquad \omega_t = f_{i(t)}'(\theta_{t-1}) + \frac{1}{n}\sum_{i=1}^n z_i^{(t-1)} - z_{i(t)}^{(t-1)}, \\ z_{i(t)}^{(t)} &= f_{i(t)}'(\theta_{t-1}), \qquad z_j^{(t)} = z_j^{(t-1)} \quad (j \neq i(t)). \end{aligned} \tag{3}$$

The direction is unbiased, $\mathbb{E}_{i(t)}[\omega_t] = F'(\theta_{t-1})$.

Proposition 5.9 (convergence of SAGA, [11]). *If $z_i^{(0)} = f_i'(\theta_0)$ for all i and $\gamma = 1/(4R^2)$, then*

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \leq \left(1 - \min\left\{\frac{1}{3n}, \frac{3\mu}{16R^2}\right\}\right)^t \left(1 + \frac{n}{4}\right)\|\theta_0 - \eta_*\|^2.$$

The proof uses the Lyapunov function

$$V_t = \|\theta_t - \eta_*\|^2 + \Delta \sum_{i=1}^n \left\|z_i^{(t)} - f_i'(\eta_*)\right\|^2, \qquad \Delta = 3\gamma^2, \tag{4}$$

and consists of six inequalities, which we label once and for all:

label	name	statement
I1	square norm	$\mathbb{E}_{i(t)} \|\omega_t\|^2 \le 2\,\mathbb{E}_{i(t)} \|a_t\|^2 + 2\,\mathbb{E}_{i(t)} \|b_t\|^2$ with $a_t = f'_{i(t)}(\theta_{t-1}) - f'_{i(t)}(\eta_*)$, $b_t = \omega_t - a_t$
I2	cocoercivity	$\frac{1}{n}\sum_i \|f'_i(\theta_{t-1}) - f'_i(\eta_*)\|^2 \le R^2\,\langle F'(\theta_{t-1}), \theta_{t-1} - \eta_*\rangle$ (used twice)
I3	variance $\le$ second moment	$\mathbb{E}_{i(t)} \|b_t\|^2 \le \frac{1}{n}\sum_i \left\|z_i^{(t-1)} - f'_i(\eta_*)\right\|^2$
I4	strong convexity	$\langle F'(\theta_{t-1}), \theta_{t-1} - \eta_*\rangle \ge \mu \|\theta_{t-1} - \eta_*\|^2$
I5	min of two rates	$\rho_D D + \rho_M \Delta M \le \max(\rho_D, \rho_M)(D + \Delta M)$, with $D = \|\theta_{t-1} - \eta_*\|^2$, $M = \sum_i \left\|z_i^{(t-1)} - f'_i(\eta_*)\right\|^2$, and $\rho_D = 1 - \frac{3\mu}{16R^2}$, $\rho_M = 1 - \frac{1}{3n}$ the coefficients left by I1–I4
I6	initialization	$\sum_i \|f'_i(\theta_0) - f'_i(\eta_*)\|^2 \le nR^4 \|\theta_0 - \eta_*\|^2$

The cumulative attribution of Section 5 applies them in the order I1, I3, I2, I4, I5 (*proof order*: the proof uses I2 in both recursions, the first time right after I1, and the attribution applies it to both at once, after I3). With $\Delta = 3\gamma^2$, I6 gives $V_0 \le (1 + \frac{3n}{16}) \|\theta_0 - \eta_*\|^2$, which the statement of Proposition 5.9 rounds up to $1 + \frac{n}{4}$. Section 4 rewrites the proof as an exact one-step identity, Proposition B3.5 (exact conditional contraction), followed by this chain, defines a *tightness ratio* in $[0, 1]$ for each inequality (Definition B3.8), and shows how replacing the inequalities one at a time by the measured quantities attributes the gap between the certified contraction factor and the exact one-step contraction to each step (Proposition B3.11).

Other algorithms. We also implement gradient descent (step $1/L_F$), Nesterov's accelerated gradient with constant momentum $(\sqrt{L_F/\mu} - 1)/(\sqrt{L_F/\mu} + 1)$, loopless SVRG [40, 47] (estimator $f'_i(\theta) - f'_i(w) + F'(w)$ with the reference point w refreshed with probability $p = 1/n$), loopless Katyusha [47] with the parameters of its Theorem 11 (Theorem 4.6 of arXiv:1901.08689v2),[1] and two "naive" accelerations of the SVRG estimator: Nesterov momentum applied to the estimator, and L-Katyusha with the negative-momentum weight (θ_2 in the notation of [47]) set to zero and its other parameters unchanged, an instance of Definition A3.15 (naive accelerated SVRG). Gradient evaluations are counted as one per component gradient and n per full gradient. SAGA pays n for its initial sweep and one per iteration. L-SVRG, L-Katyusha and the naive schemes pay n initially, two per iteration ($f'_{i(t)}$ at the current point and at the reference point, the latter charged as if it were recomputed, although the code stores the n gradients $f'_i(w)$) and n per refresh of the reference point, i.e. three per iteration on average when $p = 1/n$ (Section 5.1). Proposition A1.26 (testable prediction of hypothesis A4), like the other bounds of Appendix A1, counts L-Katyusha with stored reference gradients: n for the initial full gradient $F'(w_0)$, then two per iteration on average. Measured counts are about $\frac{3}{2}$ times the counts of that convention, and log–log slopes are essentially unaffected.

Hard instances. For hypotheses A2–A4 we use quadratic instances with $n = 16$ components (Section 5.1). The first, denoted (H), is the block-separable construction behind the lower bound of [50] (Theorem A1.8), as built by our code: n blocks B_i of m coordinates and $f_i(\theta) = n\,h(\theta_{B_i})$, where

[1] In the main text and in the appendices, the assumptions, lemmas and theorems of [47] are numbered as in the proceedings version listed in the bibliography (ALT 2020, PMLR 117). Its Assumptions 1–2, Theorem 5 (L-SVRG), Lemmas 6–10 and Theorem 11 (L-Katyusha) are Assumptions 1.1–1.2, Theorem 3.5, Lemmas 4.1–4.5 and Theorem 4.6 of version 2 of the arXiv preprint (arXiv:1901.08689v2; version 1 numbers them differently); algorithms, equations, sections and appendices are numbered identically in both.

$h(x) = c\,(\frac{1}{2}x^\top T x - e_1^\top x) + \frac{\mu_b}{2}\|x\|^2$ is a scaled Nesterov chain quadratic, T is tridiagonal with 2 on the whole diagonal and -1 off it, and $\mu_b = 1$: the regularizer of f_i acts on its own block only. Each f_i is then $n\lambda_{\max}(cT + \mu_b I)$-smooth and $F = \sum_i h(\theta_{B_i})$ is μ-strongly convex with $\mu = \lambda_{\min}(cT + \mu_b I)$, so that $\kappa = R^2/\mu > n$. The constant $c = \mu_b(\kappa - n)/(n\lambda_{\max}(T) - \kappa\lambda_{\min}(T))$, computed from the extreme eigenvalues of T (which requires $n < \kappa < n\lambda_{\max}(T)/\lambda_{\min}(T)$), makes $\kappa = R^2/\mu$ hold exactly with this μ, which exceeds μ_b (by 30% at $\kappa = 10^5$, $m = 256$). The minimizer, the same on every block, solves the tridiagonal system $(cT + \mu_b I)x = ce_1$ and is computed by a direct solve; its closed form is that of part (b) of Proposition A1.14 (closed-form minimizer), with $Q = 1 + 4c/\mu_b$. The lower bounds that Appendix A1 proves on a hard finite-sum instance, Proposition A1.18 (the $\sqrt{n\kappa}$ term for any index sequence), Theorem A1.19 (Lan–Zhou bound on the instance), Corollary A1.20 (iteration count and dimension requirement) and Corollary A1.21 (initial full gradient), as well as Proposition A1.26, which delimits the window of certified exponents, concern a variant of (H), denoted (S) and defined in Section 4.4: one chain per component as in (H), but with the regularizer $\frac{\mu}{2}\|\theta\|^2$ shared by all components, which allows every $\kappa > 1$, the constant $c = \mu(\kappa - 1)/4$, and the last diagonal entry of the chain modified so that the minimizer is an exact geometric sequence. None of these results is proved for (H) itself. Section 5.9 also uses the aligned variant (A) of (S), all of whose components are equal to one chain, and block quadratics with random spectra (R).

3 Related work

Variance-reduced methods. In SAG [52, 77] the iterate follows a biased average of stored gradients, in SVRG [43] stochastic gradients are corrected by a full gradient computed at a periodically refreshed snapshot, and SAGA [16] stores one gradient per component and uses an unbiased direction; SDCA [78], Finito and MISO [17, 59], SARAH [66] and loopless versions of SVRG [40, 47] complete the family, surveyed in [35]. These methods need $O((n+\kappa)\log(1/\varepsilon))$ component gradients, with κ defined from the largest smoothness constant of the components (Finito and MISO under a big-data condition $n \gtrsim \kappa$). The proof of Proposition 5.9 belongs to a family of analyses whose Lyapunov function adds to $\|\theta - \eta_*\|^2$ a weighted sum of squared errors of the stored gradients, or of bounds on them [75, 40, 41, 29, 47], whereas the original analysis of SAGA [16] uses Bregman divergences of the components; all certify contraction factors of the form $1 - \min\{\Theta(1/n), \Theta(1/\kappa)\}$, with constants that differ by small factors between analyses. The analysis closest to the book is that of [40], which uses the same square-norm inequality ("we sacrifice tightness") and whose optimal step and rate are exactly the best ones that the proof structure of Proposition 5.9 can certify (Proposition B3.16). Two sources of slack have been identified: this square-norm inequality, and cocoercivity with the worst-case constant $\max_i L_i$ (our R^2) instead of the average of the L_i, which non-uniform sampling exploits [88, 75, 63], or of an expected-smoothness constant [33, 34]. Practical steps also exceed the theoretical ones: for SAG, the step $1/L$ "always converged and often performed better than the step size of $1/16L$ suggested by our analysis" [77]. We add a measurement along trajectories of the slack of every inequality of the proof (the tightness ratios of Definition B3.8), which also shows that strong convexity with the global constant μ carries a sizeable share of the logarithmic gap when $\kappa \geq 10n$ (Section 5.3).

Acceleration of variance-reduced methods. Indirect acceleration wraps a linearly convergent method in an inexact accelerated proximal-point loop: Catalyst [54, 55] and APPA [27] turn the complexity of SAGA or SVRG into $(n + \sqrt{n\kappa})\log(1/\varepsilon)$ up to logarithmic factors, and accelerated proximal SDCA [79] has the same inner–outer structure. Direct optimal methods include APCG [56], SPDC [90], the primal–dual and gradient-extrapolation methods of [50, 51], Varag [49], Point-

SAGA [15], which uses proximal steps and no momentum, and Katyusha [2]. Katyusha writes Nesterov's method in the linear-coupling form of [3] and adds to the coupling a pull towards the snapshot, the "Katyusha momentum", which keeps the query point close to the point where the estimator has no variance. MiG [92] keeps only this negative momentum, SSNM [91] transfers it to SAGA, and L-Katyusha [47] replaces the outer loop by a coin flip and admits a one-step Lyapunov proof, extended to arbitrary sampling in [72]; estimate sequences give the same rate [48]. Our note on L-Katyusha (Appendix A3) reproduces the proof of [47] with its constants (Theorem A3.4); it adds the presentation in the style of the book, the identification of the single exact cancellation performed by the negative momentum (Proposition A3.14), and exact measurements of the certified contraction, of the variance bound and of the cancelled term along trajectories (Section 5.7).

Why naive acceleration fails. Nesterov's momentum on the SVRG estimator is the accelerated proximal SVRG of [67], which with one sample per iteration has the complexity of SVRG; growing mini-batches give a complexity of order $(n + \min\{\kappa, n\sqrt{\kappa}\})\log(1/\varepsilon)$ [68]. In [2], this scheme is identified with Katyusha without negative momentum and found, after careful tuning, to be sometimes faster, but on some tasks "without Katyusha momentum the algorithm does not even enjoy an accelerated convergence rate". Acceleration without a reference point is obtained in [21], with mini-batches of size of order $n^{2/3}$ and a dependence on n of order $n^{2/3}$ instead of $\sqrt{n}$. The mechanisms invoked are error accumulation in accelerated methods [76, 18], an inner product in the tight variance bound of the SVRG estimator, which the negative momentum cancels [2, 91], and the weakness of classical momentum with stochastic gradients [45]; the semidefinite certificate of [42] for Katyusha is stated for negative-momentum weights of at least $1/5$. To our knowledge, no lower bound excludes the acceleration of the naive scheme, and we do not claim one: Proposition A3.19 (closing by smoothness and strong convexity costs κ) only shows that, without the cancellation, the Lyapunov argument of L-Katyusha certifies at best a contraction $1 - \kappa^{-2}$ per iteration. Our experiments agree with [2]: divergence, or an exponent of κ of about 0.8 even with re-tuning at every κ, on block quadratics, and competitive speed on logistic regression (Sections 5.8 and 5.9).

Lower bounds for finite sums. Lower bounds descend from the oracle model of [64] and from Nesterov's chain quadratic, on which a method whose iterates stay in the span of the observed gradients discovers one coordinate per gradient [65, Theorem 2.1.13] (a "zero chain" in the terminology of [14]). For finite sums, deterministic incremental methods need $\Omega(n + \sqrt{n\kappa}\log(1/\varepsilon))$ gradients [1]; span-based methods that sample the components i.i.d. from a fixed distribution need $\Omega((n + \sqrt{n\kappa})\log(1/\varepsilon))$, in a dimension that grows with the number of iterations [50]; the $\sqrt{n\kappa}\log(1/\varepsilon)$ term remains without the span assumption for randomized algorithms with proximal oracles, for κ larger than a constant multiple of n (the $n\log(1/\varepsilon)$ term becoming an additive n), and deterministic ones need $\Omega(n\sqrt{\kappa}\log(1/\varepsilon))$ at small enough ε [87]; and the $\sqrt{n\kappa}$ term holds, up to logarithmic factors, in every dimension for oblivious methods, which include SAGA with a constant step [5]. The $n\log(1/\varepsilon)$ term is proved for i.i.d. sampling: full-gradient steps can divide it by $1 + \log(n/\kappa)$ when $\kappa \ll n$, and, for oblivious methods, not by more, up to logarithmic factors [38]; the corresponding bounds hold for loopless protocols such as those of L-SVRG and L-Katyusha in a dimension of order $\sqrt{\kappa/n}\log(1/\varepsilon)$ only [37]. For deterministic methods, cyclic ones included, the deterministic bound of [87] already excludes a gain of this type when $\kappa > 73$; whether randomly reshuffled span-based methods, or deterministic ones when $\kappa \leq 73$, can beat $n\log(1/\varepsilon)$ is not settled by these results. Our note (Appendix A1) restates these results with their exact hypotheses in the notation of the book, corrects hypothesis A1 accordingly, and builds an instance in the spirit of [1, 50] (block-separable chains with a shared regularizer, so that every $\kappa > 1$ is allowed) with

the explicit dimension condition of Corollary A1.20 (iteration count and dimension requirement) and the explicit window of certified exponents of Proposition A1.26 (testable prediction of hypothesis A4). Among the works we reviewed, the only comment on the empirical exponent is in [91]: on two data sets, multiplying κ by 10 slows accelerated methods down by about $3 \approx \sqrt{10}$ and SAGA significantly more; we fit the exponents over several decades of κ on lower-bound instances (Section 5.9).

Tightness of convergence proofs and computer-assisted analyses. Performance estimation computes the worst case of a method by semidefinite programming [22], exactly for smooth strongly convex classes thanks to interpolation conditions [84]; integral quadratic constraints certify linear rates with small linear matrix inequalities [53]; the best rate certifiable by a quadratic Lyapunov function can be computed [83], also for stochastic methods [82] and automatically [85, 86], and PEPit [31] implements these tools. For variance reduction, the jump-system analysis of [41] certifies SAGA-type methods with a small linear matrix inequality whose Lyapunov function has the structure of the one of Proposition 5.9, the dissipativity analysis of [42] treats SVRG and Katyusha, and the SAGA example distributed with PEPit (file `PEPit/examples/stochastic_and_randomized_convex_minimization/saga.py`, documented at `https://pepit.readthedocs.io`) finds the one-step contraction proved in [16] tight for its Bregman Lyapunov function at the parameters it tests; the dual multipliers of such programs indicate which inequalities of a worst-case proof are active [30]. These tools compute the worst case over a class of functions, whereas our tightness ratios measure how far each inequality of a given proof is from equality along one trajectory of one instance. The two views are complementary: a guarantee can have the right order in the worst case (Proposition B3.17 shows that the exponent of κ in Proposition 5.9 cannot be improved for SAGA with a constant step) and still be very pessimistic on the instances met in practice (Section 5.3). Our optimization of (γ, Δ), Proposition B3.16 (optimal constants of the proof structure), is the analytic counterpart of a Lyapunov search within the two-parameter family of the book's proof; to our knowledge, the exact worst-case one-step contraction of the Lyapunov function of Proposition 5.9 has not been computed, and we do not compute it.

Local curvature and effective constants. The effective curvature $\mu_{\text{eff}}(\theta)$ of Definition B2.2 (effective constants μ_{eff} and L_{eff}) is the constant of the restricted secant inequality [89] at the point θ, and linear convergence under such one-point or growth conditions is classical [44, 62]; the SAGA certificate of [41] only needs a sector condition at the minimizer, in line with our trajectory-dependent versions of Proposition 5.9, the a-posteriori rates of Proposition B3.19 and the a-priori rate on a region of Proposition B3.21. Our L_{eff} is a trajectory-wise version of the expected-smoothness constant [33, 34], for which the valid value given in [34] for single-element uniform sampling is again $\max_i L_i$. For logistic regression, self-concordance compares Hessians at nearby points [8, 81], averaged stochastic gradient descent adapts to the smallest eigenvalue of the Hessian at the optimum [10], the curvature at the optimum can be bounded through a measure of non-separability [26], SAGA-type methods reach a local regime after identifying the active manifold of composite problems [70], and average-case analyses replace κ by the spectral density of the Hessian [69]. Faster-than-predicted convergence attributed to local curvature has been reported repeatedly [54, 48, 91]. Two results limit what effective constants can give: one-point conditions alone do not allow acceleration [36], and oblivious methods need μ to accelerate [4]; our trajectory-dependent statements concern SAGA with its fixed step. In the literature we reviewed we found no measurement of μ_{eff}, or of the slack of each inequality of a proof, along SAGA trajectories; Sections 5.3 and 5.5 provide them.

Preconditioning of stochastic methods. For a fixed symmetric positive definite matrix P, the step $\theta \mapsto \theta - \gamma P^{-1} g$ is a mirror step for the reference function $\frac{1}{2}\theta^\top P \theta$, governed by relative smoothness and relative strong convexity [58, 12]; Bregman-SAGA with a quadratic reference function has a rate of the form of Proposition 5.9 with relative constants [19] (compared with ours in Remark B1.13), and a change of variables transfers Euclidean proofs to the fixed metric [46, 28]. For ridge regression, preconditioning by the regularized covariance reduces the relevant condition number to the dimension with sampling proportional to the smoothness constants, or to n times the largest ridge leverage score with uniform sampling [28]. Preconditioned SVRG and Katyusha X [57] and the methods of PROMISE [25] are analysed with relative or quadratic-regularity constants, the latter defined relative to the Hessian at a reference point and close to one near the optimum. Quasi-Newton and diagonal metrics [13, 61, 74] are analysed through eigenvalue bounds on the metric only, so that their guarantees are worse than the Euclidean ones even when they help in practice, and statistical preconditioning by subsampled Hessians has mostly been studied for distributed optimization [80, 39]. In the worst case, second-order information on the components does not remove the $\sqrt{n\kappa}$ term [7], and relative smoothness alone does not allow acceleration [20]. Our note (Appendix B1) redoes the gradient-memory proof of Proposition 5.9 in the P-metric, which gives Corollary B1.12 (Proposition 5.9 transferred to P-SAGA), and shows in Proposition B1.21 (no free lunch for global constants) that, unlike for ridge regression, no fixed P improves the global relative condition number of logistic regression, so that the gains of preconditioning are entirely trajectory-dependent there.

4 Theory

This section states the main results of Appendices B3, B2, B1, A1 and A3, which contain the complete statements and proofs. The statements are reproduced verbatim, with the numbers they have in the appendices (equation numbers included) and without their proofs. Each ends with a bracketed pointer to the appendix: "proof:" and the page of its proof when the appendix proves it, and otherwise the page where the appendix states it (definitions, assumptions, and the results that the appendices quote from the literature without proof). In this section, as in the rest of the report, a result of the appendices is referred to by its number and its name, as in Proposition B3.16 (optimal constants of the proof structure), or by its number alone when its name or its content has just been given. Proposition 5.9, the result of the book under study, is restated in Section 2 under its number in the book.

The notation is that of Section 2, and $\mathbb{E}_{i(t)}$ is the expectation over $i(t)$ given the past. For logistic regression, $R_x = \max_i \|x_i\|$ (so $R^2 = R_x^2/4 + \lambda$, $\mu = \lambda$), $\hat{\Sigma} = \frac{1}{n}\sum_i x_i x_i^\top$, and $\ell(u) = \log(1 + e^{-u})$ is the logistic loss. Each subsection ends with what is known and what is new.

4.1 Proposition 5.9 as a chain of inequalities

Hypothesis B3 asks which of the inequalities I1–I6 of the proof of Proposition 5.9 explain the gap between its certified contraction factor $\rho_{\text{bound}} = 1 - \min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}$ and the contraction observed along SAGA. This subsection states the instruments of the diagnosis (an exact one-step identity, a tightness ratio for each inequality, and an attribution of the gap to the inequalities) and delimits what a better analysis could certify: the best constants of the proof, a lower bound on the exponent of κ, and a trajectory-dependent version of the rate.

Exact identity and tightness ratios. Fix $t \geq 1$ and write $(\theta, z) = (\theta_{t-1}, z^{(t-1)})$ for the state before the t-th draw and $\mathcal{F}_{t-1}$ for the σ-field generated by $i(1), \dots, i(t-1)$, so that $\mathbb{E}_{i(t)} = \mathbb{E}[\cdot \mid \mathcal{F}_{t-1}]$. Let $D_{t-1} = \|\theta - \eta_*\|^2$, $M_{t-1} = \sum_i \|z_i - f_i'(\eta_*)\|^2$, $G_{t-1} = \langle F'(\theta), \theta - \eta_* \rangle$, $S_{t-1} = \sum_i \|f_i'(\theta) - f_i'(\eta_*)\|^2$, $\Omega_{t-1} = \mathbb{E}_{i(t)} \|\omega_t\|^2$ and $B_{t-1} = \mathbb{E}_{i(t)} \|b_t\|^2 = \frac{1}{n} \sum_i \|f_i'(\eta_*) - z_i + \bar{z}\|^2$, where $\bar{z}$ is the mean of the z_i and b_t is as in I1; the index $t-1$ is dropped when no confusion is possible. With a free weight $\Delta > 0$ in place of $3\gamma^2$ in (4), $V_{t-1} = D_{t-1} + \Delta M_{t-1}$. All these quantities are explicit averages over the n values of $i(t)$. Before any inequality is applied, the conditional expectation of the Lyapunov function is an exact function of them, which serves as the reference for every certified factor:

Proposition B3.5 (exact conditional contraction). *For every $t \geq 1$ and every $\gamma, \Delta > 0$, almost surely,*

$$\mathbb{E}_{i(t)}[V_t] = D_{t-1} - 2\gamma G_{t-1} + \gamma^2 \Omega_{t-1} + \Delta \Big[\Big(1 - \frac{1}{n}\Big) M_{t-1} + \frac{1}{n} S_{t-1} \Big], \tag{B3.7}$$

so that, on $\{V_{t-1} > 0\}$, the exact conditional contraction factor is

$$\rho_{\text{exact}}(t) := \frac{\mathbb{E}_{i(t)}[V_t]}{V_{t-1}} = 1 - \frac{2\gamma G_{t-1} - \gamma^2 \Omega_{t-1} + \Delta\,(M_{t-1} - S_{t-1})/n}{D_{t-1} + \Delta M_{t-1}}. \tag{B3.8}$$

[proof: Appendix B3, p. 79]

Thus $\rho_{\text{exact}}(t)$ is computable from the state at the cost of one pass over the n components, and Section 5.3 records it along SAGA trajectories. In this notation I1–I4 read $\Omega \leq \frac{2}{n} S + 2B$, $\frac{1}{n} S \leq R^2 G$, $B \leq \frac{1}{n} M$ and $G \geq \mu D$, and I5 replaces the two coefficients left by I1–I4 by the larger one. Each inequality receives a ratio, its smaller side over its larger side, equal to 1 exactly when the inequality is an equality at the current state:

Definition B3.8 (tightness ratios). *For a state $(\theta, z) = (\theta_{t-1}, z^{(t-1)})$ with $\theta \neq \eta_*$, define, with the quantities of Section B3.2,*

$$r_{\text{sq}} := \frac{\Omega}{\frac{2}{n} S + 2B} \quad \textit{(I1)}, \qquad r_{\text{coco}} := \frac{S/n}{R^2 G} = \frac{L_{\text{eff}}}{R^2}, \quad L_{\text{eff}} := \frac{S}{nG} \quad \textit{(I2)},$$

$$r_{\text{var}} := \frac{B}{M/n} = 1 - \frac{n \|\bar{z}\|^2}{M} \quad \textit{(I3)}, \qquad r_{\text{sc}} := \frac{\mu D}{G} = \frac{\mu}{\mu_{\text{eff}}}, \quad \mu_{\text{eff}} := \frac{G}{D} \quad \textit{(I4)},$$

$$r_{\text{min}} := \frac{\big(1 - \frac{3\mu}{16R^2}\big) D + \big(1 - \frac{1}{3n}\big) \Delta M}{\big(1 - \min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}\big)(D + \Delta M)} \quad \textit{(I5)}, \qquad r_{\text{init}} := \frac{\sum_i \|f_i'(\theta_0) - f_i'(\eta_*)\|^2}{nR^4 \|\theta_0 - \eta_*\|^2} \quad \textit{(I6)},$$

with the convention $r_{\text{var}} := 1$ if $M = 0$ (then (I3) is the equality $0 \leq 0$; no convention is needed for r_{sq}, since $S > 0$ when $\theta \neq \eta_*$, Lemma B3.9(i)). Each ratio is the smaller side divided by the larger side of the corresponding inequality of Section B3.3 (for (I4), $G \geq \mu D$, this is $\mu D/G$); $r_{\text{sq}}, r_{\text{coco}}, r_{\text{var}}, r_{\text{sc}}$ depend on the state only (not on γ, Δ), r_{min} also depends on (Δ, n, μ, R^2).

[Appendix B3, p. 81]

The effective constants $\mu_{\text{eff}} = G/D$ and $L_{\text{eff}} = S/(nG)$ turn I4 and I2 into identities at the current iterate. By Lemma B3.9 (ranges of the tightness ratios), for $\theta \neq \eta_*$ the ratios of I1, I2, I4 and I5 lie in $(0, 1]$, r_{var} lies in $[0, 1]$ and $\mu \leq \mu_{\text{eff}} \leq L_{\text{eff}} \leq R^2$ ($r_{\text{init}} \in (0, 1]$ when $\theta_0 \neq \eta_*$); moreover $r_{\text{sq}} = \frac{1}{2}$ exactly when the gradient changes $a_i = f_i'(\theta) - f_i'(\eta_*)$ and the memory errors $b_i = f_i'(\eta_*) - z_i + \bar{z}$ are uncorrelated under $i(t)$. As Remark B3.13 (the square-norm step discards the control variate) explains, one expects r_{sq} close to $\frac{1}{2}$ when the memory is stale and $r_{\text{sq}} \leq \frac{1}{2}$, possibly $\ll \frac{1}{2}$, when it is nearly fresh.

Representation and attribution. Inserting the ratios into (B3.7) gives the two coefficients of $\mathbb{E}_{i(t)}[V_t]$ exactly, and shows what the proof does: it substitutes worst-case values for four quantities that depend on the state.

Proposition B3.10 (exact contraction in terms of the ratios). *Let $\theta_{t-1} \neq \eta_*$ and $M_{t-1} > 0$, and let $\gamma, \Delta > 0$. Then*

$$\mathbb{E}_{i(t)}[V_t] = c_D\, D_{t-1} + c_M\, \Delta M_{t-1}, \qquad \begin{aligned} c_D &= 1 - 2\gamma\,\mu_{\text{eff}}\Big(1 - \gamma\, r_{\text{sq}} L_{\text{eff}} - \frac{L_{\text{eff}}\Delta}{2\gamma}\Big), \\ c_M &= 1 - \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\, r_{\text{sq}}\, r_{\text{var}}\Big), \end{aligned} \tag{B3.14}$$

where $\mu_{\text{eff}}, L_{\text{eff}}, r_{\text{sq}}, r_{\text{var}}$ are evaluated at the state $(\theta_{t-1}, z^{(t-1)})$. For the textbook constants $\gamma = \frac{1}{4R^2}$, $\Delta = 3\gamma^2$,

$$c_D = 1 - \frac{\mu_{\text{eff}}}{2R^2}\Big(1 - \frac{L_{\text{eff}}}{R^2}\Big(\frac{r_{\text{sq}}}{4} + \frac{3}{8}\Big)\Big), \qquad c_M = 1 - \frac{1}{n}\Big(1 - \frac{2}{3}\, r_{\text{sq}}\, r_{\text{var}}\Big). \tag{B3.15}$$

Substituting the worst-case values $r_{\text{sq}} = r_{\text{var}} = 1$, $L_{\text{eff}} = R^2$, $\mu_{\text{eff}} = \mu$ gives exactly the two coefficients $1 - \frac{3\mu}{16R^2}$ and $1 - \frac{1}{3n}$ of (B3.13). [proof: Appendix B3, p. 82]

With the textbook constants, the parameter rate $1 - c_D$ is $\frac{\mu_{\text{eff}}}{2R^2}$ times a bracket in $[\frac{3}{8}, 1)$, so that r_{coco} and r_{sq} change it by at most a factor $\frac{8}{3}$, and the memory rate $1 - c_M$ lies between $\frac{1}{3n}$ and $\frac{1}{n}$, so that r_{sq} and r_{var} change it by at most a factor 3; only μ_{eff}/μ, which lies in $[1, \kappa]$ by Lemma B3.9 (ranges of the tightness ratios) and is therefore not bounded independently of κ, and the step I5, which replaces the weighted mean $(c_D D + c_M \Delta M)/V_{t-1}$ by $\max\{c_D, c_M\}$, can produce larger factors. Undoing the substitutions one at a time, in either of two orders, gives monotone chains from the certified to the exact factor, hence an attribution of the gap to the individual inequalities:

Proposition B3.11 (attribution of the gap). *Let $\theta_{t-1} \neq \eta_*$, $M_{t-1} > 0$, $\gamma, \Delta > 0$ with $\alpha := 1 - \gamma R^2 - \frac{R^2\Delta}{2\gamma} > 0$, and write D, M, G, S, B, Ω, V for the state quantities. Define the coefficient functions*

$$c_D(r, L, m) := 1 - 2\gamma m\Big(1 - \gamma r L - \frac{L\Delta}{2\gamma}\Big), \qquad c_M(r, v) := 1 - \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta} r v\Big),$$

and

$$\rho_{\text{bound}}(\gamma, \Delta) := \max\{c_D(1, R^2, \mu),\, c_M(1, 1)\} = 1 - \min\Big\{2\gamma\mu\alpha,\ \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\Big)\Big\},$$

which is the worst-case coefficient of Proposition B3.14 (it is < 1 iff $2\gamma^2 < \Delta$) and equals the ρ_{bound} of (I5) for the textbook constants; in the rest of this proposition we write ρ_{bound} for $\rho_{\text{bound}}(\gamma, \Delta)$.

Order A (I4 $\to$ I2 $\to$ I3 $\to$ I1 $\to$ I5; the min is kept until the end).

$$\rho_0^A := \max\{c_D(1, R^2, \mu),\, c_M(1,1)\} = \rho_{\text{bound}},$$

$$\rho_1^A := \max\{c_D(1, R^2, \mu_{\text{eff}}),\, c_M(1,1)\} = \max\Big\{1 - 2\gamma\alpha\frac{G}{D},\; 1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\Big\},$$

$$\rho_2^A := \max\{c_D(1, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(1,1)\} = \max\Big\{1 - 2\gamma\frac{G}{D} + (2\gamma^2+\Delta)\frac{S}{nD},\; 1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\Big\},$$

$$\rho_3^A := \max\{c_D(1, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(1, r_{\text{var}})\} = \max\Big\{1 - 2\gamma\frac{G}{D} + (2\gamma^2+\Delta)\frac{S}{nD},\; 1 - \frac{1}{n} + \frac{2\gamma^2 B}{\Delta M}\Big\},$$

$$\rho_4^A := \max\{c_D(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(r_{\text{sq}}, r_{\text{var}})\}$$
$$= \max\Big\{1 - 2\gamma\frac{G}{D} + \frac{2\gamma^2 r_{\text{sq}} S}{nD} + \frac{\Delta S}{nD},\; 1 - \frac{1}{n} + \frac{2\gamma^2 r_{\text{sq}} B}{\Delta M}\Big\},$$

$$\rho_5^A := \frac{c_D(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}})\, D + c_M(r_{\text{sq}}, r_{\text{var}})\, \Delta M}{D + \Delta M} = \rho_{\text{exact}}(t),$$

with $r_{\text{sq}} = \Omega/(2S/n + 2B)$.

Order B (I5 $\to$ I4 $\to$ I2 $\to$ I3 $\to$ I1; reverse proof order, the min is undone first).

$$\rho_0^B := \rho_{\text{bound}},$$

$$\rho_1^B := \frac{(1 - 2\gamma\mu\alpha)D + \big(1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\big)\Delta M}{V} \quad \big(= r_{\min}\,\rho_{\text{bound}} \text{ for the textbook constants}\big),$$

$$\rho_2^B := \frac{D - 2\gamma\alpha G + \big(1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\big)\Delta M}{V},$$

$$\rho_3^B := \frac{D - 2\gamma G + (2\gamma^2+\Delta)\frac{S}{n} + \frac{2\gamma^2}{n}M + \big(1 - \frac{1}{n}\big)\Delta M}{V},$$

$$\rho_4^B := \frac{D - 2\gamma G + (2\gamma^2+\Delta)\frac{S}{n} + 2\gamma^2 B + \big(1 - \frac{1}{n}\big)\Delta M}{V},$$

$$\rho_5^B := \frac{D - 2\gamma G + \gamma^2\Omega + \Delta\frac{S}{n} + \big(1 - \frac{1}{n}\big)\Delta M}{V} = \rho_{\text{exact}}(t).$$

Then both sequences are nonincreasing:

$$\rho_{\text{bound}} = \rho_0^A \ge \rho_1^A \ge \rho_2^A \ge \rho_3^A \ge \rho_4^A \ge \rho_5^A = \rho_{\text{exact}}(t), \qquad \rho_{\text{bound}} = \rho_0^B \ge \rho_1^B \ge \cdots \ge \rho_5^B = \rho_{\text{exact}}(t),$$

and consequently the nonnegative increments $\delta_k := \rho_{k-1} - \rho_k$ *(additive attribution) and, when* $\rho_{\text{exact}}(t) > 0$, $\nu_k := \log(\rho_{k-1}/\rho_k)$ *(attribution in nats per iteration) satisfy*

$$\sum_{k=1}^{5}\delta_k = \rho_{\text{bound}} - \rho_{\text{exact}}(t), \qquad \sum_{k=1}^{5}\nu_k = \log\frac{1}{\rho_{\text{exact}}(t)} - \log\frac{1}{\rho_{\text{bound}}}, \qquad \prod_{k=1}^{5}\frac{\rho_k}{\rho_{k-1}} = \frac{\rho_{\text{exact}}(t)}{\rho_{\text{bound}}}.$$

In order A the increments are attributed to (I4), (I2), (I3), (I1), (I5) in this order; in order B to (I5), (I4), (I2), (I3), (I1). [proof: Appendix B3, p. 83]

The increments ν_k add up to the gap between the exact and the certified contraction exponents, and the *share* of an inequality is its increment divided by this gap. The shares depend on the order. Order B gives each inequality its full effect on the weighted contraction; in order A, undoing I4 and I2, which only lower c_D, has no effect as long as c_M is the larger coefficient ($\kappa < \frac{9n}{16}$ for the textbook constants), and undoing I3, which only lowers c_M, has none as long as c_D is. By Remark B3.12 (reading the two attribution orders), comparing the two orders shows which component of V_t is the bottleneck of the bound. Section 5.3 measures the ratios and the shares along SAGA trajectories on logistic regression.

Optimal constants and the exponent of κ. The identities and I1–I3 do not involve γ or Δ, so the same chain certifies a rate for every pair (γ, Δ) of an explicit admissible set; optimizing over this set measures how much of the gap is due to the constants $\gamma = \frac{1}{4R^2}$, $\Delta = 3\gamma^2$ of Proposition 5.9.

Proposition B3.14 (certified rate for general (γ, Δ))**.** *Let $\gamma, \Delta > 0$ satisfy*

$$\alpha(\gamma, \Delta) := 1 - \gamma R^2 - \frac{R^2 \Delta}{2\gamma} > 0 \qquad \text{and} \qquad \frac{2\gamma^2}{\Delta} < 1.$$

Then for all $t \geq 1$, almost surely,

$$\mathbb{E}_{i(t)}[V_t] \leq \Big(1 - \min\Big\{2\gamma\mu\,\alpha(\gamma, \Delta),\ \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\Big)\Big\}\Big) V_{t-1}, \tag{B3.16}$$

and both arguments of the min lie in $(0, \frac{1}{2}]$ and $(0, \frac{1}{n})$ respectively. [proof: Appendix B3, p. 85]

Writing $\Delta = c\gamma^2$ separates the two constants: the memory rate depends on c only, and for fixed c the parameter rate is a concave function of the step.

Corollary B3.15 (parametrization $\Delta = c\gamma^2$)**.** *Write $\Delta = c\gamma^2$ with $c > 2$. Then $\alpha = 1 - \gamma R^2(1 + \frac{c}{2})$, the condition $\alpha > 0$ reads $\gamma < \frac{2}{R^2(2+c)}$, and (B3.16) becomes*

$$\begin{aligned} &\mathbb{E}_{i(t)}[V_t] \leq \big(1 - \min\{\tau_D(\gamma, c), \tau_M(c)\}\big) V_{t-1}, \\ &\tau_D(\gamma, c) := 2\gamma\mu\Big(1 - \gamma R^2\Big(1 + \frac{c}{2}\Big)\Big), \qquad \tau_M(c) := \frac{1 - 2/c}{n}. \end{aligned} \tag{B3.17}$$

For fixed c, $\gamma \mapsto \tau_D(\gamma, c)$ is a concave parabola, maximal at

$$\gamma^\star(c) = \frac{1}{R^2(2+c)}, \qquad \tau_D(\gamma^\star(c), c) = \frac{\mu}{R^2(2+c)} = \frac{1}{\kappa(2+c)},$$

while τ_M does not depend on γ. Hence the best rate that (B3.17) certifies for a given c is

$$\tau(c) := \min\Big\{\frac{1}{\kappa(2+c)},\ \frac{c-2}{cn}\Big\}.$$

[proof: Appendix B3, p. 86]

Since $c > 2$, no weight is admissible when $\gamma R^2 \geq \frac{1}{2}$. At the textbook step every $c \in (2, 6)$ is admissible, and the best of these weights certifies a rate asymptotically equivalent to $\frac{2}{3n}$, twice the textbook $\frac{1}{3n}$, as $n/\kappa \to \infty$ (Section B3.5). Maximizing $\tau(c)$ over c gives the best constants of the proof structure in closed form:

Proposition B3.16 (optimal constants of the proof structure)**.** *Let $s := n/\kappa = n\mu/R^2$. The maximum over $c > 2$ and admissible γ of the rate $\min\{\tau_D, \tau_M\}$ in (B3.17) is attained at*

$$\begin{aligned} c^\star &= \frac{s + \sqrt{s^2 + 16}}{2}, \qquad \Delta^\star = c^\star(\gamma^\star)^2, \\ \gamma^\star &= \frac{1}{R^2(2 + c^\star)} = \frac{2}{R^2\big(4 + s + \sqrt{s^2 + 16}\big)} = \frac{2}{4R^2 + n\mu + \sqrt{n^2\mu^2 + 16R^4}}, \end{aligned}$$

and equals

$$\tau^\star = \frac{1}{\kappa(2 + c^\star)} = \frac{2}{4\kappa + n + \sqrt{n^2 + 16\kappa^2}}, \qquad \frac{1}{n + 4\kappa} \ \leq\ \tau^\star \ \leq\ \frac{4 - 2\sqrt{2}}{n + 4\kappa} \approx \frac{1.172}{n + 4\kappa}.$$

In particular $\tau^\star \sim \frac{1}{4\kappa}$, $\gamma^\star \sim \frac{1}{4R^2}$ and $c^\star \to 2$ when $n/\kappa \to 0$, and $\tau^\star \sim \frac{1}{n}$, $\gamma^\star \sim \frac{1}{n\mu}$ and $c^\star \sim n/\kappa$ when $n/\kappa \to \infty$. [proof: Appendix B3, p. 86]

These are the optimal rate and step of [40, Theorem 1 and Corollary 1] for SAGA: their Lyapunov function has the same two-term form, but its memory term averages upper bounds $H_i \geq \|z_i - f_i'(\eta_*)\|^2$ obtained from Bregman divergences, and their proof uses the same I1 and I3 but applies strong convexity and cocoercivity through function values; the two chains differ, but their optima coincide (Section B3.5). The ratio of $\tau^\star$ to the textbook rate $\min\{\frac{1}{3n}, \frac{3}{16\kappa}\}$ lies in $[1.05, 3)$. For $\kappa \geq \frac{9n}{16}$ it is at most $\frac{4}{3}$: there the textbook constants are within $\frac{4}{3}$ of the best of the proof structure (the ratio is $\frac{32}{3(5+\sqrt{17})} \approx 1.17$ for $n = 1000$, $\kappa = 10^3$; Section 5.4). For $n = 1000$ and $\kappa \leq 10^2$ the constants matter more: the optimal constants, at a smaller step, raise the certified rate by a factor 2.42 ($\kappa = 10^2$) or 2.94 ($\kappa = 10$), of the same size as the speed-ups 3.2–3.7 of the exact over the certified exponent measured there at the textbook step (Section 5.3), as noted in Remark B3.18 (optimizing (γ, Δ) gains at most a factor 3). At the textbook step itself, the best weight Δ gains a factor 1.81 or 1.98 (Section 5.4). This optimality is relative to the proof structure (Lyapunov function $D + \Delta M$, inequalities I1–I5): another Lyapunov function or another chain could certify more. Whatever the analysis, however, this Lyapunov function cannot certify more than $\frac{1}{n}$ per iteration: the memory recursion (B3.5) of Lemma B3.4 (exact one-step identities), $\mathbb{E}_{i(t)}[M_t] = (1 - \frac{1}{n})M_{t-1} + \frac{1}{n}S_{t-1}$, gives $\mathbb{E}_{i(t)}[M_t] \geq (1 - \frac{1}{n})M_{t-1}$, hence $\mathbb{E}[V_t] \geq \Delta(1 - \frac{1}{n})^t M_0$, and no bound $\mathbb{E}[V_t] \leq c_0 \rho^t V_0$ valid for all t, with c_0 independent of t and $M_0 > 0$, has $1 - \rho > \frac{1}{n}$, whatever $\Delta > 0$. When $\kappa \leq \frac{9n}{16}$, where the textbook rate is $\frac{1}{3n}$, a factor 3 is thus the most that any analysis of $\mathbb{E}[V_t]$ can gain.

Conversely, the exponent 1 of κ is intrinsic to SAGA with a constant step, not to its analysis. The following instance satisfies Assumption B3.1 (convex smooth components, strongly convex sum), the setting of Section 2, and (B3.1) in its statement is the SAGA iteration (3):

Proposition B3.17 (lower bound for SAGA with any constant step)**.** *Let $n \geq 1$, $0 < \mu < R^2$, $d = 2$, and $f_i(\theta) = \frac{1}{2}(\mu\theta_1^2 + R^2\theta_2^2)$ for all $i \leq n$. Assumption B3.1 holds with these constants, $F = f_1$, $F'(\theta) = Q\theta$ with $Q = \operatorname{diag}(\mu, R^2)$, and $\eta_* = 0$. Run SAGA (B3.1) with any constant step $\gamma > 0$ (which may depend on μ) and any deterministic initial memory $z^{(0)}$. Then for all $t \geq 0$,*

$$\mathbb{E}[\theta_t] = (I - \gamma Q)^t \theta_0 \qquad \text{and} \qquad \mathbb{E}\,\|\theta_t - \eta_*\|^2 \;\geq\; \|\mathbb{E}[\theta_t] - \eta_*\|^2 \;\geq\; \Big(\frac{\kappa - 1}{\kappa + 1}\Big)^{2t} \min\{\theta_{0,1}^2, \theta_{0,2}^2\}. \tag{B3.18}$$

In particular, for $\theta_0 = (r, r)$ with $r \neq 0$ and $\varepsilon < \frac{1}{2}$, reaching $\mathbb{E}\,\|\theta_t - \eta_*\|^2 \leq \varepsilon\,\|\theta_0 - \eta_*\|^2$ requires $t \geq \frac{\kappa - 1}{4} \log \frac{1}{2\varepsilon}$, whatever the step. [proof: Appendix B3, p. 88]

The argument only uses that the direction is unbiased and that F' is affine: on any quadratic F the mean SAGA iterate is the gradient-descent iterate with the same step, whatever the memory, and the variance only adds to the error. For $\kappa \gg n$ the textbook bound is within the factor $\frac{64}{3}$ of this bound on its leading term, and within $\frac{8}{3}$ of the bound $(2\kappa - \frac{1}{2}) \log \frac{1}{2\varepsilon}$ that the same argument gives for the textbook step; caveat (ii) of Remark B3.18 derives both comparisons.

Trajectory-dependent version. Strong convexity I4 and cocoercivity I2 are only used at the current iterate, so the proof also certifies rates in which effective constants replace μ and R^2; this is the hook for hypothesis B2 (Section 4.2). Regard $\mu_{\text{eff}}(\theta) = \langle F'(\theta), \theta - \eta_* \rangle \,/\, \|\theta - \eta_*\|^2$ and $L_{\text{eff}}(\theta) = \frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|^2 \,/\, \langle F'(\theta), \theta - \eta_* \rangle$ as functions of $\theta \neq \eta_*$ (they do not depend on the memory), and write $\mu_{\text{eff}}(t-1) = \mu_{\text{eff}}(\theta_{t-1})$ and $L_{\text{eff}}(t-1) = L_{\text{eff}}(\theta_{t-1})$:

Proposition B3.19 (a-posteriori rates with effective constants)**.** *Let $\gamma = \frac{1}{4R^2}$ and $\Delta = 3\gamma^2$. For*

every $t \geq 1$, almost surely on $\{\theta_{t-1} \neq \eta_*\}$:

$$\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \Big(1 - \min\Big\{\frac{1}{3n},\ \frac{3\,\mu_{\text{eff}}(t-1)}{16R^2}\Big\}\Big)V_{t-1}, \tag{B3.19}$$

$$\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \Big(1 - \min\Big\{\frac{1}{3n},\ \frac{\mu_{\text{eff}}(t-1)}{2R^2}\Big(1 - \frac{5\,L_{\text{eff}}(t-1)}{8R^2}\Big)\Big\}\Big)V_{t-1}, \tag{B3.20}$$

and the coefficient in (B3.20) satisfies $\frac{3\mu_{\text{eff}}(t-1)}{16R^2} \leq \frac{\mu_{\text{eff}}(t-1)}{2R^2}\big(1 - \frac{5L_{\text{eff}}(t-1)}{8R^2}\big) \leq \frac{\mu_{\text{eff}}(t-1)}{2R^2} \leq \frac{1}{2}$, so that both contraction factors lie in $[\frac{1}{2}, 1)$. More generally, for γ, Δ as in Proposition B3.14,

$$\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \Big(1 - \min\Big\{2\gamma\,\mu_{\text{eff}}(t-1)\Big(1 - \gamma L_{\text{eff}}(t-1) - \frac{L_{\text{eff}}(t-1)\Delta}{2\gamma}\Big),\ \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\Big)\Big\}\Big)V_{t-1}. \tag{B3.21}$$

On $\{\theta_{t-1} = \eta_*\}$ one has $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \big(1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\big)V_{t-1}$, so that (B3.19)–(B3.21) hold there with any value assigned to $\mu_{\text{eff}}(t-1)$ and $L_{\text{eff}}(t-1)$. [proof: Appendix B3, p. 90]

In the second bound, which uses both effective constants, μ_{eff} enters linearly, whereas a small L_{eff} helps by at most a factor $\frac{8}{3}$, as explained in Remark B3.20 (the step is fixed in advance). These factors are random ($\mathcal{F}_{t-1}$-measurable) and correlated with V_{t-1}, so the conditional bounds cannot be iterated: the product of the factors along a trajectory, which Figure 1 plots, does not bound $\mathbb{E}[V_T]$ in general. An a-priori rate needs a deterministic bound on the effective constants over a region Θ of the parameter space, and follows from the supermartingale $\rho_\Theta^{-(t\wedge\tau)}V_{t\wedge\tau}$, where τ now denotes the exit time from Θ (not a rate as in $\tau(c)$ and $\tau^\star$):

Proposition B3.21 (a-priori rate on a region). *Let $\Theta \subset \mathbb{R}^d$ and*

$$\mu_\Theta := \inf_{\theta\in\Theta\setminus\{\eta_*\}} \mu_{\text{eff}}(\theta) \ \geq \mu, \qquad L_\Theta := \sup_{\theta\in\Theta\setminus\{\eta_*\}} L_{\text{eff}}(\theta) \ \leq R^2,$$

$$\rho_\Theta := 1 - \min\Big\{\frac{1}{3n},\ \frac{\mu_\Theta}{2R^2}\Big(1 - \frac{5L_\Theta}{8R^2}\Big)\Big\} \ \leq \ 1 - \min\Big\{\frac{1}{3n}, \frac{3\mu_\Theta}{16R^2}\Big\},$$

and let $\tau := \inf\{t \geq 0 : \theta_t \notin \Theta\}$ (with $\inf\emptyset = +\infty$). For $\gamma = \frac{1}{4R^2}$, $\Delta = 3\gamma^2$ and every $T \geq 0$,

$$\mathbb{E}\big[V_T\,\mathbf{1}_{\{\tau>T\}}\big] \leq \rho_\Theta^T\,V_0. \tag{B3.22}$$

In particular, if $\theta_t \in \Theta$ for all t almost surely, then $\mathbb{E}[V_T] \leq \rho_\Theta^T V_0$ and $\mathbb{E}\,\|\theta_T - \eta_*\|^2 \leq \rho_\Theta^T(1 + \frac{n}{4})\,\|\theta_0 - \eta_*\|^2$ for the book's initialization. [proof: Appendix B3, p. 91]

For logistic regression, the self-concordance-type bound $|\ell'''| \leq \ell''$ [8, 10] gives an explicit lower bound on μ_Θ over balls around η_*:

Proposition B3.22 (local strong convexity of logistic regression). *Let $f_i(\theta) = \ell(y_i x_i^\top\theta) + \frac{\lambda}{2}\,\|\theta\|^2$ as in Remark B3.7, $\lambda > 0$, $R_x := \max_i \|x_i\|$, and $\psi(x) := \frac{1-e^{-x}}{x}$ for $x > 0$, $\psi(0) := 1$ (ψ is decreasing on $[0,\infty)$, with values in $(0,1]$). Then for every $\theta \neq \eta_*$,*

$$\mu_{\text{eff}}(\theta) \ \geq \ \lambda + \psi\big(R_x\,\|\theta - \eta_*\|\,\big)\big(\lambda_{\min}(F''(\eta_*)) - \lambda\big) \ \geq \ \max\big\{\lambda,\ \psi\big(R_x\,\|\theta - \eta_*\|\,\big)\lambda_{\min}(F''(\eta_*))\big\},$$

where $F''(\eta_*) = \lambda I + \frac{1}{n}\sum_i \ell''(y_i x_i^\top\eta_*)x_i x_i^\top \succeq \lambda I$. Consequently, on the ball $\Theta_r = \{\theta : \|\theta - \eta_*\| \leq r\}$, the constant μ_{Θ_r} of Proposition B3.21 satisfies

$$\mu_{\Theta_r} \geq \lambda + \psi(R_x r)(\lambda_{\min}(F''(\eta_*)) - \lambda) \geq \max\{\lambda, \psi(R_x r)\lambda_{\min}(F''(\eta_*))\};$$

in particular $\mu_{\Theta_r} \geq \frac{1}{2}(\lambda + \lambda_{\min}(F''(\eta_*))) \geq \max\{\lambda, \frac{1}{2}\lambda_{\min}(F''(\eta_*))\}$ as soon as $R_x r \leq \frac{3}{2}$ (since $\psi(3/2) \approx 0.518$). [proof: Appendix B3, p. 92]

Applied from the first time the iterate enters the ball of radius $\frac{3}{2R_x}$ around η_* (by the strong Markov property of $(\theta_t, z^{(t)})$, with the Lyapunov value at that time in place of V_0), Proposition B3.21 (a-priori rate on a region) gives, until the next exit, a contraction factor at most $1-\min\{\frac{1}{3n}, \frac{3}{32R^2}(\lambda+\lambda_{\min}(F''(\eta_*)))\}$: the parameter term of Proposition 5.9 is multiplied by $\frac{1}{2}(1+\lambda_{\min}(F''(\eta_*))/\lambda)$, which changes the rate only when that term binds ($\kappa > \frac{9n}{16}$). From $\theta_0 = 0$, the starting point lies outside this ball in all our problems with $\kappa \geq 10^2$ ($\|\eta_*\| \geq 5.9$, $R_x = 1$). In the fifteen configurations of Section 5.3, $\lambda_{\min}(F''(\eta_*))/\lambda$ ranges from 1.0003 to 69.9 and is below 2 in nine of them: six have $\kappa \leq 10^2$, where the memory term binds and the rate is unchanged, and the other three gain less than a factor $\frac{3}{2}$.

Known and new. Proposition 5.9 and its proof are those of [11]; Lyapunov functions adding memory errors, or upper bounds on them, to the squared distance are used in [40, 41]; the optimal constants of this proof structure (Proposition B3.16) are those of [40]; for oblivious step schedules depending only on R^2, the exponent 1 of κ follows from [6, Theorem 2]. To our knowledge, the use of the exact identity as an instrument, the reading of the proof as a substitution of worst-case values with monotone attribution chains, the elementary lower bound for every n and every constant step, and the trajectory-dependent bounds are not in the sources we reviewed in this form.

4.2 Effective curvature and smoothness along trajectories

Hypothesis B2 asks whether the constants of the strong-convexity step I4 and of the cocoercivity step I2, evaluated along the trajectory actually followed by the algorithm, describe its convergence better than their worst-case values μ and R^2. Appendix B2 works under the following assumption, in which L is the smoothness constant L_F of Section 2, and defines the two constants that make I4 and I2 identities at the current iterate.

Assumption B2.1 (convex R^2-smooth f_i, μ-strongly convex F). *(i) Each $f_i : \mathbb{R}^d \to \mathbb{R}$ is convex, differentiable and R^2-smooth, i.e. f_i' is R^2-Lipschitz-continuous.*

(ii) $F = \frac{1}{n}\sum_{i=1}^n f_i$ is μ-strongly convex with $\mu > 0$; its (unique) minimizer is η_*, so that $F'(\eta_*) = 0$ (Proposition 5.2 of [11]). We denote by $L \leq R^2$ the smoothness constant of F (F' is L-Lipschitz-continuous; $L \leq R^2$ because an average of R^2-Lipschitz maps is R^2-Lipschitz), and by $\kappa = R^2/\mu$ the condition number used in the book.

(iii) Whenever second derivatives appear (Lemma B2.5, Proposition B2.8, Section B2.3 and after), each f_i is twice continuously differentiable.

[Appendix B2, p. 95]

Definition B2.2 (effective constants μ_{eff} and L_{eff}). *For $\theta \neq \eta_*$, define*

$$\mu_{\text{eff}}(\theta) \;=\; \frac{(\theta-\eta_*)^\top F'(\theta)}{\|\theta-\eta_*\|^2}, \tag{B2.1}$$

$$L_{\text{eff}}(\theta) \;=\; \frac{\frac{1}{n}\sum_{i=1}^n \|f_i'(\theta)-f_i'(\eta_*)\|^2}{(\theta-\eta_*)^\top F'(\theta)}. \tag{B2.2}$$

Along a trajectory $(\theta_t)_{t\geq 0}$ we abbreviate $\mu_{\text{eff}}(t) = \mu_{\text{eff}}(\theta_t)$ and $L_{\text{eff}}(t) = L_{\text{eff}}(\theta_t)$. [Appendix B2, p. 95]

Both quotients are well defined and positive, by Lemma B2.4 (μ_{eff} lies between μ and L) and by Lemma B2.6 below, and μ/μ_{eff} and L_{eff}/R^2 are the tightness ratios r_{sc} and r_{coco} of I4 and I2 in Definition B3.8 (tightness ratios).

Ranges and directional limits. For C^2 functions, μ_{eff} is the curvature, in the direction of the error, of an averaged Hessian, and both constants lie between the worst-case ones.

Lemma B2.5 (integral form of the effective curvature). *Assume F is twice continuously differentiable and let, for $\theta \neq \eta_*$,*

$$\bar{H}(\theta) \;=\; \int_0^1 F''(\eta_* + s\,v)\,ds \qquad (v = \theta - \eta_*,\; u = v/\|v\|).$$

Then $\bar{H}(\theta)$ is symmetric, $\mu I \preccurlyeq \bar{H}(\theta) \preccurlyeq LI$, $F'(\theta) = \bar{H}(\theta)\,v$, and

$$\mu_{\text{eff}}(\theta) \;=\; \frac{v^\top \bar{H}(\theta)\,v}{\|v\|^2} \;=\; u^\top \bar{H}(\theta)\,u, \tag{B2.5}$$

i.e. $\mu_{\text{eff}}(\theta)$ is the Rayleigh quotient, in the direction of the error, of the Hessian averaged along the segment $[\eta_*, \theta]$. [proof: Appendix B2, p. 97]

Lemma B2.6 (L_{eff} lies between μ_{eff} and R^2). *Under Assumption B2.1, for all $\theta \neq \eta_*$, with $v = \theta - \eta_*$,*

$$\|F'(\theta)\|^2 \;\le\; \frac{1}{n}\sum_{i=1}^n \|f_i'(\theta) - f_i'(\eta_*)\|^2 \qquad \textit{(Jensen / gradient diversity)}, \tag{B2.6}$$

$$\frac{1}{n}\sum_{i=1}^n \|f_i'(\theta) - f_i'(\eta_*)\|^2 \;\le\; R^2\, v^\top F'(\theta) \qquad \textit{(averaged co-coercivity, eq. (5.31) of the book)}, \tag{B2.7}$$

$$\mu \;\le\; \mu_{\text{eff}}(\theta) \;\le\; \frac{\|F'(\theta)\|}{\|v\|} \;\le\; \frac{\|F'(\theta)\|^2}{v^\top F'(\theta)}. \tag{B2.8}$$

Consequently

$$\mu \;\le\; \mu_{\text{eff}}(\theta) \;\le\; \frac{\|F'(\theta)\|^2}{v^\top F'(\theta)} \;\le\; L_{\text{eff}}(\theta) \;\le\; R^2, \tag{B2.9}$$

and, more precisely, with $h_i = f_i'(\theta) - f_i'(\eta_*)$ (the vectors a_i of Section B3.2) and $\bar{h} = \frac{1}{n}\sum_i h_i = F'(\theta)$,

$$L_{\text{eff}}(\theta) \;=\; \frac{\|F'(\theta)\|^2 + \frac{1}{n}\sum_{i=1}^n \|h_i - \bar{h}\|^2}{v^\top F'(\theta)}. \tag{B2.10}$$

If moreover each f_i is L_i-smooth, then $L_{\text{eff}}(\theta) \le \max_i L_i$ (so one may take $R^2 = \max_i L_i$). [proof: Appendix B2, p. 97]

The effective condition number $L_{\text{eff}}/\mu_{\text{eff}}$ is thus at least 1, and (B2.10) writes L_{eff}, and hence the tightness ratio $r_{\text{coco}} = L_{\text{eff}}/R^2$ of I2, as the sum of two terms: the diversity of the gradient differences h_i, which vanishes iff all h_i coincide, and the part of R^2 that the full gradient uses in the direction of the error. Near the optimum, the two constants become functions of this direction.

Proposition B2.8 (directional limits at the optimum). *Assume each f_i is twice continuously differentiable, and write $H_* = F''(\eta_*)$ and $M_* = \frac{1}{n}\sum_{i=1}^n f_i''(\eta_*)^2$. For a unit vector u define*

$$\Lambda(u) \;=\; \frac{u^\top M_* u}{u^\top H_* u}.$$

Then, as $\theta \to \eta_*$ (with $u = (\theta - \eta_*)/\|\theta - \eta_*\|$ allowed to vary),

$$\mu_{\text{eff}}(\theta) - u^\top H_* u \;\to\; 0, \qquad L_{\text{eff}}(\theta) - \Lambda(u) \;\to\; 0,$$

uniformly in the direction. In particular, if $\theta \to \eta_*$ along a fixed direction u, then $\mu_{\text{eff}}(\theta) \to u^\top H_* u$ (the Rayleigh quotient of the Hessian at the optimum in the direction of approach) and $L_{\text{eff}}(\theta) \to \Lambda(u)$. Moreover, for every unit u,

$$u^\top H_* u \;\le\; \frac{u^\top H_*^2 u}{u^\top H_* u} \;\le\; \Lambda(u) \;\le\; R^2, \tag{B2.11}$$

consistently with (B2.9). [proof: Appendix B2, p. 98]

As u ranges over the unit sphere, $u^\top H_* u$ spans $[\lambda_{\min}(H_*), \lambda_{\max}(H_*)]$ and equals the local strong-convexity constant $\lambda_{\min}(H_*)$ only on the bottom eigenspace of H_*: near η_*, the slack of I4 is set by the direction in which the algorithm approaches η_*.

Logistic regression. For ℓ_2-regularized logistic regression, the global strong-convexity constant μ_{glob} (the largest μ for which F is μ-strongly convex) is exactly λ, whatever the data, but, when the features span $\mathbb{R}^d$, the value λ is approached only at infinity.

Proposition B2.12 (the global strong convexity constant is λ)**.** *For ℓ_2-regularized logistic regression, let $I_0 = \{i : x_i \neq 0\}$ (if $I_0 = \emptyset$ then $F'' \equiv \lambda I$ and there is nothing to prove).*

(a) There exists a unit vector $\hat{d} \in \mathbb{R}^d$ with $x_i^\top \hat{d} \neq 0$ for every $i \in I_0$; in fact Lebesgue-almost every direction has this property.

(b) For every such $\hat{d}$, with $\delta(\hat{d}) = \min_{i \in I_0} \left|x_i^\top \hat{d}\right| > 0$, and every $t \ge 0$,

$$\lambda \;\le\; \lambda_{\min}\big(F''(t\hat{d})\big) \;\le\; \lambda + R_x^2\, e^{-t\,\delta(\hat{d})}. \tag{B2.20}$$

(c) Consequently $\mu_{\text{glob}} = \inf_\theta \lambda_{\min}(F''(\theta)) = \lambda$: the global strong convexity constant of F is exactly λ, whatever the data.

(d) The infimum is attained at some θ if and only if $\operatorname{span}\{x_1, \dots, x_n\} \neq \mathbb{R}^d$; in that case $\lambda_{\min}(F''(\theta)) = \lambda$ for every θ. If the features span $\mathbb{R}^d$, then $\lambda_{\min}(F''(\theta)) > \lambda$ for every θ and the value λ is only approached as $\|\theta\| \to \infty$; it is in fact approached along every *ray: for every unit vector $\hat{d}$ and every $t \ge 0$,*

$$\lambda_{\min}\big(F''(t\hat{d})\big) \;\le\; \lambda + \frac{1}{n}\sum_{i=1}^{n} \ell''\big(t\, y_i x_i^\top \hat{d}\big)(x_i^\top \hat{d})^2 \;\le\; \lambda + R_x^2\, e^{-t\,\delta'(\hat{d})}, \tag{B2.21}$$

where $\delta'(\hat{d}) = \min\{\left|x_i^\top \hat{d}\right| : x_i^\top \hat{d} \neq 0\}$, with the conventions $\delta'(\hat{d}) = +\infty$ and $e^{-\infty} = 0$ when $x_i^\top \hat{d} = 0$ for all i; hence $\lim_{t\to\infty} \lambda_{\min}(F''(t\hat{d})) = \lambda$ for every direction $\hat{d}$. Directions as in (a) are those along which the whole matrix $F''(t\hat{d}) - \lambda I$ tends to 0; along a general ray,

$$F''(t\hat{d}) - \lambda I \;\xrightarrow[t\to\infty]{}\; \frac{1}{4n} \sum_{i:\, x_i^\top \hat{d} = 0} x_i x_i^\top,$$

a matrix which is singular because $\hat{d}$ lies in its kernel.

[proof: Appendix B2, p. 101]

Corollary B2.13 (strong convexity on a bounded region). *In ℓ_2-regularized logistic regression, for every $D > 0$ and every θ with $\|\theta\| \le D$,*

$$\lambda_{\min}\big(F''(\theta)\big) \;\ge\; \lambda + \ell''(R_x D)\,\lambda_{\min}(\hat{\Sigma}) \;\ge\; \lambda + \tfrac{1}{4}e^{-R_x D}\lambda_{\min}(\hat{\Sigma}). \tag{B2.22}$$

Moreover $\|\eta_*\| \le \sqrt{2\log 2/\lambda}$, so that the ball $\{\|\theta\| \le 2\sqrt{2\log 2/\lambda}\}$ contains the ball of radius $\sqrt{2\log 2/\lambda}$ around η_*. [proof: Appendix B2, p. 102]

An algorithm whose iterates stay in a bounded region thus never experiences the curvature λ unless $\lambda_{\min}(\hat{\Sigma}) = 0$. Gradient descent with $\gamma \le 1/L$ stays in the ball of radius $\|\theta_0 - \eta_*\|$ around η_*; with the step and initialization of Proposition 5.9, the SAGA iterates are almost surely bounded but not, in general, confined to a deterministic ball (Section B2.3). Near η_*, Lemma B2.14 (generalized self-concordance of the logistic loss) yields a comparison of μ_{eff} with the curvature at the optimum in the direction of the error.

Corollary B2.15 (effective curvature versus curvature at the optimum). *For ℓ_2-regularized logistic regression, let $p_i = \sigma'(y_i x_i^\top \eta_*)$, $H_* = F''(\eta_*) = \frac{1}{n}\sum_i p_i x_i x_i^\top + \lambda I$ and $p_{\min} = \min_i p_i$. For every $\theta \neq \eta_*$, with $u = (\theta - \eta_*)/\|\theta - \eta_*\|$,*

$$\lambda + e^{-R_x\|\theta-\eta_*\|}\left(u^\top H_* u - \lambda\right) \;\le\; \mu_{\text{eff}}(\theta), \tag{B2.24}$$

$$\mu_{\text{eff}}(\theta) \;\le\; \lambda + e^{R_x\|\theta-\eta_*\|}\left(u^\top H_* u - \lambda\right), \tag{B2.25}$$

and

$$u^\top H_* u \;\ge\; \lambda + p_{\min}\, u^\top \hat{\Sigma} u \;\ge\; \lambda + p_{\min}\lambda_{\min}(\hat{\Sigma}). \tag{B2.26}$$

In particular, $\liminf_{\theta\to\eta_*} \mu_{\text{eff}}(\theta) \ge \lambda + p_{\min}\lambda_{\min}(\hat{\Sigma})$, and for $\|\theta - \eta_*\| \le 1/R_x$ one has $\mu_{\text{eff}}(\theta) \ge \lambda + e^{-1}p_{\min}\lambda_{\min}(\hat{\Sigma})$. If moreover $\theta \in S = \operatorname{span}\{x_1, \dots, x_n\}$ (which holds along the whole trajectory of gradient descent and of SAGA when $\theta_0 \in S$, Lemma B2.16), then $u \in S$, and $\lambda_{\min}(\hat{\Sigma})$ may be replaced, in (B2.26) and in the two consequences, by $\lambda^+_{\min}(\hat{\Sigma})$, the smallest nonzero eigenvalue of $\hat{\Sigma}$ (which is positive even when $d > n$). [proof: Appendix B2, p. 103]

Once $R_x\|\theta_t - \eta_*\| \le 1$, $\mu_{\text{eff}}(\theta_t)/\lambda$ is therefore within a factor $e^{\pm 1}$ of $u_t^\top H_* u_t/\lambda$, where u_t is the direction of the error. This ratio is close to 1 in every direction under strong regularization, $\lambda \gg \lambda_{\max}(\hat{\Sigma})/4$, which forces $\kappa \ll n$; under intermediate regularization, $\lambda_{\min}(H_* - \lambda I) \ll \lambda \ll \lambda_{\max}(H_* - \lambda I)$, typical of anisotropic features, it is close to 1 along the bottom eigenvectors of H_* and of order $\lambda_{\max}(H_*)/\lambda$ along the top ones; under weak regularization with well-conditioned features, $\lambda \ll p_{\min}\lambda_{\min}(\hat{\Sigma})$, it is large in every direction (Section B2.3).

Which curvature does an algorithm meet? On a quadratic, gradient descent concentrates its error on the bottom eigenspace of the Hessian, unless the initial error is orthogonal to it. In the statement, $\pi_k(t) = (q_k^\top v_t)^2/\|v_t\|^2$ is the share of $\|v_t\|^2$ carried by the eigenvector q_k (the spectral profile (B2.28)).

Proposition B2.17 (alignment of gradient descent with the flattest direction). *Let $F(\theta) = F(\eta_*) + \frac{1}{2}(\theta-\eta_*)^\top H(\theta-\eta_*)$, where H is symmetric positive definite with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_d = \lambda_{\min} > 0$ and orthonormal eigenvectors $q_1, \dots, q_d$; let $E_{\min}$ be the eigenspace of $\lambda_{\min}$, P the orthogonal projector onto $E_{\min}$, and $\lambda' = \min\{\lambda_k : \lambda_k > \lambda_{\min}\}$ the second smallest distinct eigenvalue (if $H = \lambda_{\min} I$, then $P = I$ and the statement is trivial). Let $\theta_t = \theta_{t-1} - \gamma F'(\theta_{t-1})$ be gradient descent,*

$v_t = \theta_t - \eta_*$, *and* $\pi_k(t)$ *the spectral profile* (B2.28) *of the error. Let* $0 < \gamma \le 1/\lambda_1$ *and* $v_0 \neq 0$*; if* $\gamma = 1/\lambda_1$*, assume moreover that* v_0 *does not lie in the eigenspace of* λ_1 *(otherwise* $\theta_t = \eta_*$ *for all* $t \ge 1$*). Then* $v_t \neq 0$ *for every* t*, and:*

(a) $v_t = (I - \gamma H)^t v_0$*, i.e.* $q_k^\top v_t = (1 - \gamma\lambda_k)^t\, q_k^\top v_0$ *for all* k*, and* $\mu_{\mathrm{eff}}(\theta_t) = \sum_k \lambda_k \pi_k(t)$ *with* $\pi_k(t) \propto (1 - \gamma\lambda_k)^{2t}(q_k^\top v_0)^2$*.*

(b) (Monotonicity) $t \mapsto \mu_{\mathrm{eff}}(\theta_t)$ *is non-increasing.*

(c) (Alignment) If $Pv_0 \neq 0$ *(true for Lebesgue-almost every* v_0*), then with* $\rho_{\mathrm{gap}} = (1 - \gamma\lambda')/(1 - \gamma\lambda_{\min}) \in [0, 1)$*,*

$$\frac{\|v_t - Pv_t\|}{\|Pv_t\|} \;\le\; \rho_{\mathrm{gap}}^{t}\, \frac{\|v_0 - Pv_0\|}{\|Pv_0\|},$$

so that $v_t / \|v_t\|$ *converges to* $E_{\min}$ *at the geometric rate* ρ_{gap}*, and*

$$0 \;\le\; \mu_{\mathrm{eff}}(\theta_t) - \lambda_{\min} \;\le\; (\lambda_1 - \lambda_{\min})\, \rho_{\mathrm{gap}}^{2t}\, \frac{\|v_0 - Pv_0\|^2}{\|Pv_0\|^2} \;\xrightarrow[t\to\infty]{}\; 0. \tag{B2.29}$$

(d) (Invariant subspaces) If S *is a linear subspace with* $HS \subseteq S$ *and* $v_0 \in S$*, then* $v_t \in S$ *for all* t*, and (a)–(c) hold with* H *replaced by its restriction* $H|_S$*: the sums run over the eigenvectors* $q_k \in S$ *only,* $\lambda_{\min}$ *is replaced by* $\lambda_{\min}(H|_S) = \min\{\lambda_k : q_k \in S\}$*,* $E_{\min}$ *and* P *by the corresponding eigenspace within* S *and its orthogonal projector,* λ' *by the second smallest distinct eigenvalue of* $H|_S$*, and the hypothesis* $Pv_0 \neq 0$ *by its analogue for the restricted projector.*

[proof: Appendix B2, p. 106]

The alignment time is of order $1/(\gamma(\lambda' - \lambda_{\min}))$ iterations, which can exceed the horizon of an experiment when the spectrum is dense near $\lambda_{\min}$; monotonicity (b) is then the robust prediction, as Remark B2.18 (scope of Proposition B2.17) explains. For SAGA, the mean error follows gradient descent exactly, while the second moment adds the estimator noise accumulated along the trajectory.

Lemma B2.21 (SAGA on a quadratic: exact moment recursions)**.** *Let* $f_i(\theta) = \frac{1}{2}\theta^\top H_i \theta - b_i^\top \theta + c_i$ *with* H_i *symmetric,* $0 \preccurlyeq H_i \preccurlyeq R^2 I$*, and* $H = \frac{1}{n}\sum_i H_i \succcurlyeq \mu I$*. Consider SAGA with step* γ *and memory* $z_i^{(t)}$ *as in the book, write* $v_t = \theta_t - \eta_*$*, and define the memory errors* $e_i^{(t)} = z_i^{(t)} - f_i'(\eta_*)$ *and* $\bar e^{(t)} = \frac{1}{n}\sum_i e_i^{(t)}$*. Let* $i(t)$ *be uniform on* $\{1, \dots, n\}$ *and independent of the past* $\mathcal{F}_{t-1} = \sigma(i(1), \dots, i(t-1))$*.*

(a) (State recursion) $v_t = (I - \gamma H)v_{t-1} + \gamma\xi_t$ *with*

$$\xi_t = \big(e_{i(t)}^{(t-1)} - \bar e^{(t-1)}\big) - \big(H_{i(t)} - H\big)v_{t-1}, \qquad \mathbb{E}_{i(t)}[\xi_t] = 0,$$

and $e_{i(t)}^{(t)} = H_{i(t)} v_{t-1}$*,* $e_j^{(t)} = e_j^{(t-1)}$ *for* $j \neq i(t)$*. Moreover* $\xi_t = F'(\theta_{t-1}) - \omega_t$ *is minus the error of the SAGA gradient estimator* ω_t*.*

(b) (Mean) $\mathbb{E}[v_t] = (I - \gamma H)^t v_0$*: the mean error follows exactly the gradient descent recursion.*

(c) (Second moment) With $C_t = \mathbb{E}_{i(t)}[\xi_t \xi_t^\top] = \frac{1}{n}\sum_{i=1}^n g_i^{(t)} g_i^{(t)\top}$*, where* $g_i^{(t)} = (e_i^{(t-1)} - \bar e^{(t-1)}) - (H_i - H)v_{t-1}$ *(so that* C_t *is the conditional covariance of* ω_t *given* $\mathcal{F}_{t-1}$*),*

$$\mathbb{E}[v_t v_t^\top] = (I - \gamma H)\,\mathbb{E}[v_{t-1} v_{t-1}^\top]\,(I - \gamma H) + \gamma^2 \mathbb{E}[C_t],$$

and therefore, for every eigenpair (λ_k, q_k) *of* H,

$$\mathbb{E}\big[(q_k^\top v_t)^2\big] = (1-\gamma\lambda_k)^{2t}(q_k^\top v_0)^2 + \gamma^2 \sum_{s=1}^{t} (1-\gamma\lambda_k)^{2(t-s)}\, q_k^\top \mathbb{E}[C_s]\, q_k. \tag{B2.30}$$

[proof: Appendix B2, p. 108]

Assuming that the noise covariance decays at the certified rate of Proposition 5.9, which is slower than every mode of the mean dynamics, a quasi-stationary heuristic predicted that the error of SAGA stays spread over the directions fed by the noise, with $\mu_{\text{eff}}(\theta_t)$ fluctuating above $\lambda_{\min}(H_*)$ (prediction (P2) of Section B2.4; (P1) predicts that along gradient descent $\mu_{\text{eff}}(\theta_t)$ decreases to $\lambda_{\min}(H_*)$ when the features span $\mathbb{R}^d$). In nine runs on logistic regression ($n = 1000$, $d = 50$, one seed; SAGA with $\gamma = \frac{1}{4R^2}$, gradient descent with step $1/L$), (P1) is confirmed up to the speed of convergence, whereas (P2) is confirmed only in the four runs with $9n\lambda_{\min}(H_*)/(16R^2) = \frac{9}{4}n\gamma\lambda_{\min}(H_*) > 1$ ($\kappa = 10^2$, all α; $\kappa = 10^3$, $\alpha = 0$). It is refuted for $\kappa = 10^4$, $\alpha \in \{0, 1\}$, where SAGA, started from an error that already carries 89–98% of its squared norm on the flattest direction of H_*, keeps and sharpens this alignment, like gradient descent; the three other runs are intermediate. Thus (P2) holds only for κ at most of the order of n, and not in every such run. Remark B2.22 (outcome of prediction (P2)) reports the nine runs and explains the alignment heuristically; Section 5.5 discusses them.

How the effective constants enter the bound. Keeping I4 and I2 as identities in the proof of Proposition 5.9 gives the a-posteriori one-step factors of Proposition B3.19 (a-posteriori rates with effective constants), rederived as Proposition B2.23 (trajectory-dependent one-step contraction). As noted after Proposition B3.19, $\mu_{\text{eff}}(\theta_{t-1})$ enters them linearly, whereas, at the fixed step $\gamma = \frac{1}{4R^2}$, $L_{\text{eff}} < R^2$ improves the parameter rate by a factor at most $\frac{8}{3}$; moreover, once μ_{eff} exceeds $16R^2/(9n)$, the memory rate $\frac{1}{3n}$ binds and no further gain in curvature shows in the bound, as Remark B2.24 (what each constant can explain) details. In order B of Proposition B3.11 (attribution of the gap), undoing I4 and I2 replaces μ and R^2 by μ_{eff} and L_{eff}; for $\kappa \geq 10^4$ ($n = 1000$, $d = 50$, $\alpha \in \{0, 1, 2\}$, 5 seeds) these two steps carry 77–96% of the logarithmic gap (Section 5.3). In order A, which keeps the minimum until the end and whose first two steps give the two one-step factors of Proposition B3.19, I4 and I2 carry 59–96% of the gap in the same runs (Section 5.5).

Known and new. That algorithms meet the local curvature is classical: the local strong-convexity constant tends to $\lambda_{\min}(H_*)$ [11, Section 5.2.3], averaged SGD adapts to it on logistic regression [10] through self-concordance [8], $\mu_{\text{eff}}(\theta)$ is the constant of the restricted secant inequality at θ [89], and L_{eff} is a trajectory-wise version of the expected-smoothness constant [33, 34]. To our knowledge, the precise form of Proposition B2.12 (the global strong convexity constant is λ) and Corollary B2.15 (effective curvature versus curvature at the optimum), and the comparison of the directions selected by gradient descent and SAGA, whose heuristic prediction the experiments partly refute, are not in the sources we reviewed in this form; the integral form, the ranges and the directional limits (Lemma B2.5, Lemma B2.6 and Proposition B2.8) are elementary consequences of standard facts, collected here for the trajectory constants.

4.3 Preconditioned SAGA

Hypothesis B1 runs SAGA in a fixed quadratic metric to see which steps of the proof of Proposition 5.9 are too loose, and asks whether the metric improves the algorithm or only its analysis. Let P be a fixed symmetric positive definite matrix, $\text{cond}(P) = \lambda_{\max}(P)/\lambda_{\min}(P)$ its condition number,

$\|v\|_P = (v^\top P v)^{1/2}$ the norm of parameters and $\|g\|_{P^{-1}} = (g^\top P^{-1} g)^{1/2}$ the dual norm of gradients and memories. The algorithm of Definition B1.3 (P-SAGA: SAGA in the metric of P) keeps the direction ω_t and the memory of SAGA (3) and, by Lemma B1.4 (closed form, unbiasedness, Bregman form), steps $\theta_t = \theta_{t-1} - \gamma P^{-1}\omega_t$; it is Bregman-SAGA [19] with the reference function $h(\theta) = \frac{1}{2}\theta^\top P \theta$. A linear change of variables reduces it to SAGA:

Proposition B1.5 (P-SAGA as SAGA on a transformed problem). *Let $u := P^{1/2}\theta$, $\tilde f_i(u) := f_i(P^{-1/2}u)$ and $\tilde F := \frac{1}{n}\sum_i \tilde f_i = F \circ P^{-1/2}$. Then:*

(i) $\nabla \tilde f_i(u) = P^{-1/2} f_i'(P^{-1/2}u)$ *and, when f_i is C^2,* $\nabla^2 \tilde f_i(u) = P^{-1/2} f_i''(P^{-1/2}u) P^{-1/2}$*; the same formulas hold for $(\tilde F, F)$.*

(ii) $\tilde F$ *is strongly convex and* $\tilde u_* := P^{1/2}\eta_*$ *is its unique minimizer.*

(iii) Let $(\theta_t, z^{(t)})_{t\ge 0}$ be generated by P-SAGA (Definition B1.3) with step γ and index sequence $(i(t))_{t\ge 1}$, and let $u_t := P^{1/2}\theta_t$, $\tilde z_i^{(t)} := P^{-1/2} z_i^{(t)}$. Then $(u_t, \tilde z^{(t)})_{t\ge 0}$ is generated by Euclidean SAGA ($P = I$) on $(\tilde f_i)_{i\le n}$ with the same *step γ and the* same *index sequence, started from $(u_0, \tilde z^{(0)})$.*

(iv) For all t and i, $\|u_t - \tilde u_*\|^2 = \|\theta_t - \eta_*\|_P^2$ and $\left\|\tilde z_i^{(t)} - \nabla \tilde f_i(\tilde u_*)\right\|^2 = \left\|z_i^{(t)} - f_i'(\eta_*)\right\|_{P^{-1}}^2$; in particular $z_i^{(0)} = f_i'(\theta_0)$ for all i iff $\tilde z_i^{(0)} = \nabla \tilde f_i(u_0)$ for all i.

[proof: Appendix B1, p. 119]

Every step of the proof of Proposition 5.9 thus holds for P-SAGA, in the norms $\|\cdot\|_P$ and $\|\cdot\|_{P^{-1}}$ and with the constants of the transformed functions, written here for C^2 functions (for C^1 functions, see Remark B1.10):

Definition B1.7 (relative constants and relative condition number). *We set*

$$R_P^2 := \max_{1\le i\le n} \sup_{\theta\in\mathbb{R}^d} \lambda_{\max}\big(P^{-1/2} f_i''(\theta) P^{-1/2}\big),$$
$$\mu_P := \inf_{\theta\in\mathbb{R}^d} \lambda_{\min}\big(P^{-1/2} F''(\theta) P^{-1/2}\big), \qquad \kappa_P := \frac{R_P^2}{\mu_P}. \tag{B1.6}$$

By Lemma B1.6, every $\tilde f_i$ of Proposition B1.5 is R_P^2-smooth and $\tilde F$ is μ_P-strongly convex; in the terminology of [39, Def. 1] and [58], f_i is R_P^2-smooth and F is μ_P-strongly convex relative to *the reference function $h = \frac{1}{2}\|\cdot\|_P^2$ (whose Hessian is P). For $P = I$ these are the usual constants, and R^2, μ, κ always denote the Euclidean ($P = I$) constants.* [Appendix B1, p. 121]

Proposition 5.9 then transfers verbatim, with these constants in place of (R^2, μ):

Corollary B1.12 (Proposition 5.9 transferred to P-SAGA). *Let P be SPD and let R_P^2, μ_P be the constants of (B1.6) (or any constants such that each $\tilde f_i$ of Proposition B1.5 is R_P^2-smooth and $\tilde F$ is μ_P-strongly convex). Run P-SAGA (Definition B1.3) with $\gamma = 1/(4R_P^2)$ and define*

$$V_t^P := \|\theta_t - \eta_*\|_P^2 + 3\gamma^2 \sum_{i=1}^n \left\|z_i^{(t)} - f_i'(\eta_*)\right\|_{P^{-1}}^2,$$
$$\rho_P := 1 - \min\left\{\frac{1}{3n}, \frac{3\mu_P}{16 R_P^2}\right\} = 1 - \min\left\{\frac{1}{3n}, \frac{3}{16\kappa_P}\right\}. \tag{B1.11}$$

Then for every $t \geq 1$,

$$\mathbb{E}_{i(t)}\left[V_t^P\right] \leq \rho_P\, V_{t-1}^P, \tag{B1.12}$$

and if $z_i^{(0)} = f_i'(\theta_0)$ for all i,

$$\mathbb{E}\,\|\theta_t - \eta_*\|_P^2 \leq \rho_P^t\Big(1 + \frac{n}{4}\Big)\,\|\theta_0 - \eta_*\|_P^2\,. \tag{B1.13}$$

Consequently

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \leq \mathrm{cond}(P)\,\rho_P^t\Big(1 + \frac{n}{4}\Big)\,\|\theta_0 - \eta_*\|^2$$

(the condition number of P enters the constant, not the rate), and $t \geq \max\{3n, \frac{16}{3}\kappa_P\}\log\frac{2n}{\varepsilon}$ iterations guarantee $\mathbb{E}\,\|\theta_t - \eta_*\|_P^2 \leq \varepsilon\,\|\theta_0 - \eta_*\|_P^2$, i.e. a running time of order $c_P\,(3n + \frac{16}{3}\kappa_P)\log\frac{2n}{\varepsilon}$, where c_P is the cost of one iteration (Section B1.5). [proof: Appendix B1, p. 122]

The metric acts on the rate only through κ_P; $\mathrm{cond}(P)$ enters only the constant of the Euclidean bound. By Lemma B1.8 (comparison with the Euclidean constants), $\kappa/\,\mathrm{cond}(P) \leq \kappa_P \leq \kappa\,\mathrm{cond}(P)$, so that a fixed metric changes the certified rate by at most the factor $\mathrm{cond}(P)$, in either direction. In the proof written in the P-metric (Section B1.3.1), I1 and I3 change only their norm, I2 and I4 their norm and their constant (R_P^2, μ_P), I5 depends on P only through $3\mu_P/(16R_P^2)$, and I6 through both norms and R_P^4. The tightness ratios of I2 and I4 in the P-geometry (Definition B1.14) are $r_{\mathrm{coco}}^P(t) = L_{\mathrm{eff}}^P(\theta_{t-1})/R_P^2$ and $r_{\mathrm{sc}}^P(t) = \mu_P/\mu_{\mathrm{eff}}^P(\theta_{t-1})$, with the effective constants $\mu_{\mathrm{eff}}^P(\theta) = \langle F'(\theta), \theta - \eta_*\rangle\,/\,\|\theta - \eta_*\|_P^2$ and $L_{\mathrm{eff}}^P(\theta) = \frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|_{P^{-1}}^2\,/\,\langle F'(\theta), \theta - \eta_*\rangle$ $(\theta \neq \eta_*)$. By Remark B1.16 (effective one-step contraction), if I2 and I4 are replaced by the identities that define these constants, the parameter part of the one-step bound (B1.28) contracts by a factor between $1 - \mu_{\mathrm{eff}}^P/(2R_P^2)$ and $1 - 3\mu_{\mathrm{eff}}^P/(16R_P^2)$, with μ_{eff}^P evaluated at θ_{t-1}: the rate seen by the proof is governed by $R_P^2/\mu_{\mathrm{eff}}^P(\theta_{t-1})$.

Global constants of logistic regression. For ℓ_2-regularized logistic regression, μ_P is explicit and R_P^2 is bounded on both sides ($\sigma_1 \geq \cdots \geq \sigma_d$ denote the eigenvalues of $\hat{\Sigma}$, not the per-sample curvatures $\sigma_i'(\theta)$ of the lemma):

Lemma B1.11 (relative constants of logistic regression)**.** *Let $f_i(\theta) = \ell(y_i x_i^\top\theta) + \frac{\lambda}{2}\,\|\theta\|^2$ with $\ell(u) = \log(1 + e^{-u})$, $y_i \in \{-1, 1\}$, $x_i \in \mathbb{R}^d$, $\lambda > 0$, and let $\sigma(u) = (1 + e^{-u})^{-1}$. Then $\ell'(u) = -\sigma(-u)$ and $\ell''(u) = \sigma'(u) = \sigma(u)\sigma(-u) = \big(2 + e^u + e^{-u}\big)^{-1} \in (0, \frac{1}{4}]$, which equals $\frac{1}{4}$ at $u = 0$ and is strictly decreasing in $|u|$. Writing $\sigma_i'(\theta) := \sigma'(y_i x_i^\top\theta)$,*

$$f_i'(\theta) = \ell'(y_i x_i^\top\theta)\,y_i x_i + \lambda\theta, \qquad f_i''(\theta) = \sigma_i'(\theta)\,x_i x_i^\top + \lambda I, \qquad F''(\theta) = \frac{1}{n}\sum_{i=1}^n \sigma_i'(\theta)\,x_i x_i^\top + \lambda I. \tag{B1.8}$$

For every SPD P,

$$\max\left\{\frac{1}{4}\max_i \|x_i\|_{P^{-1}}^2\,,\ \frac{\max_i \|x_i\|^2/4 + \lambda}{\lambda_{\max}(P)}\right\} \ \leq\ R_P^2 \ \leq\ \frac{1}{4}\max_i \|x_i\|_{P^{-1}}^2 + \frac{\lambda}{\lambda_{\min}(P)}, \tag{B1.9}$$

and more precisely $R_P^2 \geq \frac{1}{4}\,\|x_i\|_{P^{-1}}^2 + \lambda\,\frac{x_i^\top P^{-2} x_i}{x_i^\top P^{-1} x_i}$ for every i with $x_i \neq 0$;

$$\mu_P = \lambda\,\lambda_{\min}(P^{-1}) = \frac{\lambda}{\lambda_{\max}(P)}. \tag{B1.10}$$

In particular, for $P = I$: $R^2 := \frac{1}{4}\max_i \|x_i\|^2 + \lambda$ is a valid smoothness constant of every f_i, $\mu = \lambda$, and $\kappa = R^2/\mu = \max_i \|x_i\|^2/(4\lambda) + 1$; these are the Euclidean constants used in the whole project. [proof: Appendix B1, p. 122]

Comparing the lower bound on R_P^2 with μ_P gives the central negative result:

Proposition B1.21 (no free lunch for global constants)**.** *For ℓ_2-regularized logistic regression and* every *SPD matrix P,*

$$\kappa_P \;\geq\; \frac{\max_i \|x_i\|^2}{4\lambda} + 1 = \kappa, \tag{B1.33}$$

with κ the Euclidean condition number (B1.30)*; consequently $\rho_P \geq \rho_I$ for every P. In particular, for $P = \hat{\Sigma} + \lambda I$, $\kappa \leq \kappa_P \leq \kappa \, \frac{\sigma_1+\lambda}{\sigma_d+\lambda}$.* [proof: Appendix B1, p. 130]

The supremum in R_P^2 is attained on the decision boundary of a sample, where its data curvature is $\frac{1}{4}x_i x_i^\top$, whereas the infimum in μ_P is approached far from all of them, where $F'' \to \lambda I$ is isotropic; no single metric fits both, and with global constants the transferred Proposition 5.9 never certifies a faster rate than for $P = I$ on this problem. For the square loss, by contrast, $P = \hat{\Sigma} + \lambda I$ brings κ_P into $[d_\infty(\lambda), d_\infty(\lambda) + 1]$ (the ridge setting of [28]; Section B1.4). Here, with X the matrix of rows $x_i^\top$, $\mathrm{lev}_i(\lambda) = x_i^\top (X^\top X + n\lambda I)^{-1} x_i$ are the ridge leverage scores [23, 9], $d_\infty(\lambda) = n \max_i \mathrm{lev}_i(\lambda)$ and $d_{\mathrm{eff}}(\lambda) = \sum_i \mathrm{lev}_i(\lambda) = \mathrm{tr}(\hat{\Sigma}(\hat{\Sigma}+\lambda I)^{-1})$; by Lemma B1.17 (leverage scores), $d_{\mathrm{eff}}(\lambda) \leq \min\{d_\infty(\lambda), d\}$ and $d_\infty(\lambda) < n$.

Trajectory-dependent constants. For logistic regression, any benefit of a metric must therefore come from the trajectory-dependent constants, which are explicit in terms of the per-sample curvatures averaged on the segment $[\eta_*, \theta]$:

Proposition B1.22 (effective strong convexity from segment curvatures)**.** *For the logistic regression of Lemma B1.11, every $\theta \neq \eta_*$, $\delta := \theta - \eta_*$, and $\bar{\sigma}_i'(\theta) := \int_0^1 \sigma'\big(y_i x_i^\top (\eta_* + s\delta)\big)\, ds \in (0, \frac{1}{4}]$,*

$$f_i'(\theta) - f_i'(\eta_*) = \bar{\sigma}_i'(\theta)\,(x_i^\top \delta)\, x_i + \lambda\delta, \qquad \big\langle F'(\theta), \theta - \eta_* \big\rangle = \frac{1}{n}\sum_{i=1}^n \bar{\sigma}_i'(\theta)\,(x_i^\top \delta)^2 + \lambda\,\|\delta\|^2 . \tag{B1.34}$$

Consequently, with $m(\theta) := \min_i \bar{\sigma}_i'(\theta) \;\geq\; \min_i \sigma'\big(\max\{|x_i^\top \eta_*|, |x_i^\top \theta|\}\big) > 0$:

(i) for $P = \hat{\Sigma} + \lambda I$, the effective strong convexity $\mu_{\mathrm{eff}}^P(\theta) := \langle F'(\theta), \delta\rangle \,/\, \|\delta\|_P^2$ is the convex combination

$$\mu_{\mathrm{eff}}^P(\theta) = \frac{\frac{1}{n}\sum_i \bar{\sigma}_i'(\theta)(x_i^\top \delta)^2 + \lambda\,\|\delta\|^2}{\frac{1}{n}\sum_i (x_i^\top \delta)^2 + \lambda\,\|\delta\|^2} \;\in\; [\,m(\theta),\, 1\,] \tag{B1.35}$$

of the numbers $\bar{\sigma}_1'(\theta), \ldots, \bar{\sigma}_n'(\theta), 1$, and more precisely

$$\mu_{\mathrm{eff}}^P(\theta) \;\geq\; \frac{m(\theta)\,\sigma_1 + \lambda}{\sigma_1 + \lambda} \;\geq\; \max\Big\{m(\theta),\; \frac{\lambda}{\sigma_1+\lambda}\Big\} = \max\{m(\theta), \mu_P\}; \tag{B1.36}$$

(ii) for $P = I$, $\mu_{\mathrm{eff}}(\theta) = \frac{1}{n}\sum_i \bar{\sigma}_i'(\theta)\,(x_i^\top \delta/\,\|\delta\|)^2 + \lambda \in [\,m(\theta)\sigma_d + \lambda,\; \sigma_1/4 + \lambda\,]$.
[proof: Appendix B1, p. 130]

A sample with a large margin makes $m(\theta)$ small ($\sigma'(u) \approx e^{-|u|}$) but has a small weight in (B1.35) unless $\theta - \eta_*$ is aligned with it, so that $\mu_{\mathrm{eff}}^P(\theta)$ is typically much larger than $m(\theta)$; the global version of (B1.36) is only $\mu_P = \lambda/(\sigma_1 + \lambda)$, because $\inf_\theta \min_i \sigma'(y_i x_i^\top \theta) = 0$. The smoothness side is controlled by the leverage scores, by a multiple of the effective dimension when they are incoherent:

Assumption B1.18 (coherence of the leverage scores)**.** *There is $C \geq 1$ such that the ridge leverage scores* (B1.29) *satisfy* $\max_i \mathrm{lev}_i(\lambda) \leq C\, d_{\mathrm{eff}}(\lambda)/n$*, i.e. $d_\infty(\lambda) \leq C\, d_{\mathrm{eff}}(\lambda)$: no data point has a leverage score larger than C times the average leverage score.* [Appendix B1, p. 128]

Since $\|x_i\|^2_{(\hat\Sigma+\lambda I)^{-1}} = n\,\mathrm{lev}_i(\lambda)$ (Lemma B1.17(i)), Lemma B1.11 gives, for the statistical metric:

Proposition B1.20 (global constants of the statistical preconditioner). *For $P = \hat\Sigma + \lambda I$,*

$$\frac{d_\infty(\lambda)}{4} \;\le\; R_P^2 \;\le\; \frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda} \;\le\; \frac{d_\infty(\lambda)}{4} + 1 \;\le\; \frac{C\,d_{\mathrm{eff}}(\lambda)}{4} + 1, \tag{B1.31}$$

the last inequality under Assumption B1.18, and

$$\mu_P = \frac{\lambda}{\sigma_1+\lambda}, \qquad \kappa_P = \frac{R_P^2}{\mu_P} \in \Big[\frac{d_\infty(\lambda)}{4}\cdot\frac{\sigma_1+\lambda}{\lambda},\; \Big(\frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda}\Big)\frac{\sigma_1+\lambda}{\lambda}\Big]. \tag{B1.32}$$

[proof: Appendix B1, p. 129]

Dividing (B1.31) by (B1.36) bounds the effective relative condition number:

Corollary B1.23 (leverage-score bound along trajectories). *Let $\kappa^P_{\mathrm{eff}}(\theta) := R_P^2/\mu^P_{\mathrm{eff}}(\theta)$ be the effective relative condition number at θ (the quantity governing the effective contraction (B1.28)). Then, for $P = \hat\Sigma + \lambda I$ and $m(\theta)$ as in Proposition B1.22,*

$$\kappa^P_{\mathrm{eff}}(\theta) \;\le\; \Big(\frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda}\Big)\frac{\sigma_1+\lambda}{m(\theta)\sigma_1+\lambda} \;\le\; \frac{d_\infty(\lambda)/4+1}{m(\theta)} \;\le\; \frac{C\,d_{\mathrm{eff}}(\lambda)/4+1}{m(\theta)}, \tag{B1.37}$$

the last inequality under Assumption B1.18, whereas for $P = I$ the same argument only gives

$$\frac{\max_i \|x_i\|^2/4+\lambda}{\sigma_1/4+\lambda} \;\le\; \kappa_{\mathrm{eff}}(\theta) \;\le\; \frac{\max_i \|x_i\|^2/4+\lambda}{m(\theta)\sigma_d+\lambda}, \qquad \textit{with} \qquad \frac{\max_i \|x_i\|^2/4+\lambda}{\sigma_1/4+\lambda} \;\ge\; \frac{\mathrm{tr}\,\hat\Sigma + 4\lambda}{\sigma_1+4\lambda}. \tag{B1.38}$$

[proof: Appendix B1, p. 131]

In the last two bounds of (B1.37), the eigenvalues of $\hat\Sigma$ appear only through $d_\infty(\lambda) \le C\,d_{\mathrm{eff}}(\lambda)$, and $d_{\mathrm{eff}}(\lambda)$ counts the directions not killed by the regularization, whatever the dynamic range of the spectrum. The price is the smallest segment curvature $m(\theta)$, and on the instances of Section 5.6 it dominates. There, as $\theta \to \eta_*$, $m(\theta) \to \min_i w_i$ with $w_i = \sigma'(y_i x_i^\top \eta_*)$, and $\min_i w_i = 7.4\cdot 10^{-3}$ ($\alpha = 1$) and $6.1\cdot 10^{-3}$ ($\alpha = 2$), so that the first bound in (B1.37) tends to $1.7\cdot 10^3$ and $1.2\cdot 10^3$: above $\kappa = 10^3$, and above the upper bound in (B1.38) for $P = I$ with the same $m(\theta)$ (968 and 999). On these instances this rigorous bound therefore predicts no gain of $\hat\Sigma + \lambda I$.

Jacobi and Newton metrics; local constants. The Jacobi metric $P_{\mathrm{J}} = \mathrm{diag}(\hat\Sigma) + \lambda I$ costs $O(d)$ per iteration, like $P = I$, whereas a dense P costs $O(d^2)$ per iteration and a precomputation of order $nd^2 + d^3$ (Section B1.5). With $\hat\Sigma_{jj} > 0$ the diagonal entries of $\hat\Sigma$, x_{ij} the entries of x_i and $C_{\mathrm{corr}} = \mathrm{diag}(\hat\Sigma)^{-1/2}\hat\Sigma\,\mathrm{diag}(\hat\Sigma)^{-1/2}$ the correlation matrix of the features, the constants of P_{J} compare with those of $\hat\Sigma + \lambda I$ as follows:

Proposition B1.19 (constants of the Jacobi preconditioner). *(i)* $\|x_i\|^2_{P_{\mathrm{J}}^{-1}} = \sum_{j=1}^d \frac{x_{ij}^2}{\hat\Sigma_{jj}+\lambda}$ *and*

$$\frac{1}{n}\sum_i \|x_i\|^2_{P_{\mathrm{J}}^{-1}} = \sum_{j=1}^d \frac{\hat\Sigma_{jj}}{\hat\Sigma_{jj}+\lambda} =: d_{\mathrm{eff}}^{\mathrm{diag}}(\lambda) \;\ge\; d_{\mathrm{eff}}(\lambda).$$

(ii) $R^2_{P_{\mathrm{J}}} \le \frac{1}{4}\max_i \|x_i\|^2_{P_{\mathrm{J}}^{-1}} + \frac{\lambda}{\min_j \hat\Sigma_{jj}+\lambda}$ *and* $\mu_{P_{\mathrm{J}}} = \frac{\lambda}{\max_j \hat\Sigma_{jj}+\lambda}$.

(iii) $\lambda_{\min}(C_{\mathrm{corr}})\,P_{\mathrm{J}} \preceq \hat{\Sigma} + \lambda I \preceq \lambda_{\max}(C_{\mathrm{corr}})\,P_{\mathrm{J}}$; *consequently* $\lambda_{\min}(C_{\mathrm{corr}})\,n\,\mathrm{lev}_i(\lambda) \le \|x_i\|^2_{P_{\mathrm{J}}^{-1}} \le \lambda_{\max}(C_{\mathrm{corr}})\,n\,\mathrm{lev}_i(\lambda)$ *for all* i.

[proof: Appendix B1, p. 129]

By (iii), the constants of P_{J} are those of $\hat{\Sigma}+\lambda I$ up to the factor $\mathrm{cond}(C_{\mathrm{corr}})$, in either direction: P_{J} removes the scale anisotropy of the features but not their correlations, and nothing orders it with respect to $P = I$. The oracle Newton metric $H = F''(\eta_*)$, which requires η_*, uses the curvatures $w_i \in (0, \frac{1}{4}]$ at the optimum: with $W = \mathrm{diag}(w_1, \dots, w_n)$ and $\hat{\Sigma}_w = \frac{1}{n}X^\top W X$ (eigenvalues $\sigma_1^w \ge \cdots \ge \sigma_d^w$), $H = \hat{\Sigma}_w + \lambda I$, and its constants involve the leverage scores $\mathrm{lev}_i^w(\lambda) = w_i x_i^\top (X^\top W X + n\lambda I)^{-1} x_i$ of the weighted design and $d_\infty^w(\lambda) = n \max_i \mathrm{lev}_i^w(\lambda)$:

Proposition B1.26 (constants in the Newton metric). *For* $P = H = F''(\eta_*)$*:*

(i) $\|x_i\|^2_{H^{-1}} = n\,\mathrm{lev}_i^w(\lambda)/w_i$, *hence*

$$\max_i \frac{n\,\mathrm{lev}_i^w(\lambda)}{4w_i} \;\le\; R_H^2 \;\le\; \max_i \frac{n\,\mathrm{lev}_i^w(\lambda)}{4w_i} + \frac{\lambda}{\sigma_d^w + \lambda} \;\le\; \frac{d_\infty^w(\lambda)}{4\min_i w_i} + 1, \qquad \mu_H = \frac{\lambda}{\sigma_1^w + \lambda}. \tag{B1.42}$$

(ii) For $\theta \neq \eta_*$, $\delta := \theta - \eta_*$ *and* $\bar{\sigma}_i'(\theta)$ *as in Proposition B1.22,*

$$\mu_{\mathrm{eff}}^H(\theta) = \frac{\frac{1}{n}\sum_i \bar{\sigma}_i'(\theta)(x_i^\top\delta)^2 + \lambda\|\delta\|^2}{\frac{1}{n}\sum_i w_i(x_i^\top\delta)^2 + \lambda\|\delta\|^2} \;\in\; \Big[\min\Big\{1, \min_i \frac{\bar{\sigma}_i'(\theta)}{w_i}\Big\},\; \max\Big\{1, \max_i \frac{\bar{\sigma}_i'(\theta)}{w_i}\Big\}\Big], \tag{B1.43}$$

a convex combination of $\bar{\sigma}_1'/w_1, \dots, \bar{\sigma}_n'/w_n, 1$; *in particular* $\mu_{\mathrm{eff}}^H(\theta) \to 1$ *as* $\theta \to \eta_*$, *and* $\mu_{\mathrm{eff}}^H(\theta) \ge \mu_H$ *always.*

[proof: Appendix B1, p. 133]

The Newton metric makes strong convexity perfect near the optimum, but R_H^2 still pays the supremum over θ: sample i enters it with its largest curvature $\frac{1}{4}$ in the numerator and its curvature w_i at the optimum in the metric, hence the factor $1/w_i$ in (B1.42). Near the optimum, the realized smoothness constant does not pay this factor, and the realized strong convexity is a local curvature:

Lemma B1.27 (limits of the effective constants at η_*). *Let the* f_i *be* C^2, $H_i := f_i''(\eta_*)$, $H := F''(\eta_*) = \frac{1}{n}\sum_i H_i \succ 0$, *and let* $\theta \to \eta_*$ *with* $\delta := \theta - \eta_*$ *satisfying* $\delta/\|\delta\| \to v$, $\|v\| = 1$. *Then, for every SPD* P,

$$\begin{aligned} L_{\mathrm{eff}}^P(\theta) &\longrightarrow \frac{\frac{1}{n}\sum_i v^\top H_i P^{-1} H_i v}{v^\top H v}, \\ \mu_{\mathrm{eff}}^P(\theta) &\longrightarrow \frac{v^\top H v}{v^\top P v} \in \big[\lambda_{\min}(P^{-1/2}HP^{-1/2}), \lambda_{\max}(P^{-1/2}HP^{-1/2})\big]. \end{aligned} \tag{B1.44}$$

Moreover $\frac{1}{n}\sum_i H_i P^{-1} H_i \succeq HP^{-1}H$, *so the limit of* L_{eff}^P *is at least* $\lambda_{\min}(P^{-1/2}HP^{-1/2})$, *and for* $P = H$

$$\lim L_{\mathrm{eff}}^H(\theta) = 1 + \frac{\frac{1}{n}\sum_i \big\|H^{-1/2}(H_i - H)v\big\|^2}{\big\|H^{1/2}v\big\|^2} \;\ge\; 1, \tag{B1.45}$$

with equality iff $H_i v = Hv$ *for all* i *(homogeneous per-sample curvature in the direction* v*). For the logistic regression of Lemma B1.11, in addition,* $\lim L_{\mathrm{eff}}^H(\theta) \le \max\{d_\infty^w(\lambda), 2\}$ *for*

$P = H$, $\lim L^P_{\text{eff}}(\theta) \le d_\infty(\lambda)/4 + \frac{3}{2}$ *for* $P = \hat\Sigma + \lambda I$, *and, for* $P = I$, *with* $c_i := x_i^\top v$ *and* $B_v := \frac{1}{n}\sum_i w_i^2 c_i^2 \|x_i\|^2$,

$$\lim L_{\text{eff}}(\theta) = \frac{\lambda^2 + 2\lambda\, v^\top \hat\Sigma_w v + B_v}{\lambda + v^\top \hat\Sigma_w v}, \qquad (v^\top \hat\Sigma_w v)^2 \le B_v \le \frac{\max_i \|x_i\|^2}{4}\, v^\top \hat\Sigma_w v = (\kappa - 1)\lambda\, v^\top \hat\Sigma_w v \tag{B1.46}$$

(this is the exact formula $L_{\text{eff}} = (\lambda^2 + 2\lambda A + B)/(\lambda + A)$ *of Lemma B2.10, in the limit, where its* A *and* B *become* $v^\top \hat\Sigma_w v$ *and* B_v*); in particular* $\lambda + v^\top \hat\Sigma_w v \le \lim L_{\text{eff}}(\theta) \le \lambda + (\kappa + 1)\, v^\top \hat\Sigma_w v$.
[proof: Appendix B1, p. 134]

Only these local constants account for the gains observed in Section 5.6. Whatever the direction of the error, $\liminf_{\theta\to\eta_*} \mu^P_{\text{eff}}(\theta) \ge \lambda_{\min}(M_P)$ with $M_P = P^{-1/2} F''(\eta_*) P^{-1/2}$, so that $R_P^2/\lambda_{\min}(M_P)$ bounds $\limsup_{\theta\to\eta_*} \kappa^P_{\text{eff}}(\theta)$. For $P = \hat\Sigma + \lambda I$, $\lambda_{\min}(M_P)$ is the smallest over directions v of a weighted average of $w_1, \dots, w_n, 1$ (weights $(x_i^\top v)^2/n$ and $\lambda\|v\|^2$), not their minimum: on those instances it is 0.117 ($\alpha = 1$) and 0.109 ($\alpha = 2$), 16 and 18 times $\min_i w_i$, and $R_P^2/\lambda_{\min}(M_P) = 166$ and 103, against $R^2/\lambda_{\min}(F''(\eta_*)) = 576$ and 981 for $P = I$ (R_P^2 being the upper bound in (B1.9), as in the experiments); over passes 15–20 the measured $\kappa^P_{\text{eff}}(\theta_{t-1})$ are about 84 and 31, against 460 and 878 for $P = I$ (geometric means over states and seeds). These limits hold only as $\theta \to \eta_*$, and their use as rates along a whole run is heuristic; so used, the values 166 and 103 are below $\frac{9n}{16} = 562.5$ ($n = 1000$), and by Remark B1.24 (memory-bottleneck regime) the memory term $\frac{1}{3n}$ of the rate then binds. Through these limits, Section B1.6 also predicts which step is the loosest late in a run (prediction (P2), heuristic where it assumes the direction of the error): I2 for $P = I$, I4 with the global μ_P for the dense metrics (Table 4 in Appendix B1).

Known and new. Relative smoothness [58, 12], Bregman-SAGA (whose rate has the same structure, with constants $(\frac{1}{2}, \frac{1}{8})$ instead of $(\frac{1}{3}, \frac{3}{16})$ [19, Corollary 1]; Remark B1.13), the change of variables for a fixed metric [46, 28] and statistical preconditioning [39] are known. To our knowledge, Proposition B1.21 (no free lunch for global constants) and Corollary B1.23 (leverage-score bound along trajectories) are not in the sources we reviewed in this form; the first shows that, for logistic regression, the benefit of a fixed metric is entirely trajectory-dependent, and the second does not certify the gains observed in Section 5.6, which only the local constants of Lemma B1.27 (limits of the effective constants at η_*) account for, and only heuristically.

4.4 Lower bounds and the hard instances

Oracle classes, known bounds and hypothesis A1. Costs are counted in component gradients, a full gradient costing n; Table 1 collects the guarantees of the methods compared in this report and the main lower bounds. The lower bounds of the literature differ in how an algorithm may choose the indices of its queries and combine the answers; the following classes make these assumptions exact.

Definition A1.2 (linear-span protocol and classes (S1)–(S3))**.** *An algorithm follows the* linear-span protocol *(Nesterov's "black-box" assumption, [65] Section 2.1, relation (3.41) of [50], relation (1.2) of [38]) if, when it has queried the oracle at the pairs* (θ_{s-1}, i_s), $s = 1, \dots, t$, *its iterate satisfies*

$$\theta_t \in \theta_0 + \operatorname{span}\left\{ f'_{i_1}(\theta_0),\ f'_{i_2}(\theta_1),\ \dots,\ f'_{i_t}(\theta_{t-1}) \right\}. \tag{A1.1}$$

We distinguish three classes according to the index sequence (i_s)*:*

(S1) span-based with i.i.d. sampling*:* (i_s) *is i.i.d. from a fixed distribution* p *on* $\{1, \dots, n\}$ *(uniform in the project), independent of the oracle answers [50, 38]. This contains SGD, SAG,*

Finito/MISO, SDCA-without-duality, SAGA started from the zero memory $z^{(0)}=0$, and, on block-separable problems, randomized block-coordinate methods. With the book's initialization $z_i^{(0)}=f_i'(\theta_0)$, SAGA is an (S1) method preceded by one full gradient at θ_0 *(n deterministic queries); Lemma A1.16(ii) and Corollary A1.21 treat this variant.*

(S2) span-based with an arbitrary index sequence*: (i_s) is any sequence, deterministic (cyclic, or n consecutive queries at the same point, which is a full gradient), random, or adaptive. This contains GD, Nesterov's accelerated gradient (AGD), cyclic incremental methods, and the hybrid methods SVRG, SARAH, L-SVRG, Katyusha and L-Katyusha, which mix i.i.d. single queries with full gradients.*

(S3) no span restriction*: arbitrary deterministic algorithms [1, 87] or arbitrary randomized algorithms [87], and the* oblivious canonical linear iterative *(CLI) class of [5], in which new iterates are linear combinations $\sum_j O_F(\theta^{(j)};\alpha^{(j)})$ of oracle outputs (gradients possibly multiplied by fixed matrices, coordinate steps, full gradients) whose coefficients $\alpha^{(j)}$ are chosen* without looking at the oracle answers *(they may be random). Fixed step sizes, fixed momentum and fixed sampling schemes are allowed, line searches and quasi-Newton updates are not.*

[Appendix A1, p. 143]

Lower bounds for (S1) therefore apply to SAGA, up to its initial sweep, but not to L-SVRG or L-Katyusha, whose full gradients are n queries at one point; those for (S2) apply to GD, AGD, SAGA, L-SVRG and L-Katyusha. On the chain constructions below, a span method reveals at most one new coordinate per query, so that span-based bounds require a dimension of the order of the number of queries they certify.

Hypothesis A1 (Section 1) asserts the bound $\Omega((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon})$ for class (S1) and a weakened form for arbitrary index sequences. Its first part is the theorem of Lan and Zhou, quoted with the displays of arXiv:1507.02000v3, whose sum convention gives the same function and bounds as the journal version [50]. On their instance (A1.7), m components act on disjoint blocks of dimension $\tilde{n}$, each a Nesterov chain f_i, convex with an L_i-Lipschitz gradient, $L_i=\mu(\mathcal{Q}-1)$, plus its own regularizer; $n=m\tilde{n}$ is the total dimension, and each block has condition number at most $\mathcal{Q}$.

Theorem A1.8 (Lan–Zhou lower bound; Theorem 3 and Corollary 3 of [50])**.** *Let $\mathcal{Q}$ be the parameter of the instance (A1.7) and $q=\frac{\sqrt{\mathcal{Q}}-1}{\sqrt{\mathcal{Q}}+1}$. The iterates of any randomized incremental gradient method of class (S1) (Definition A1.2) applied to (A1.7) satisfy*

$$\frac{\mathbb{E}\left\|x^k-x^*\right\|^2}{\left\|x^0-x^*\right\|^2}\;\geq\;\frac{1}{2}\exp\Big(-\frac{4k\sqrt{\mathcal{Q}}}{m(\sqrt{\mathcal{Q}}+1)^2-4\sqrt{\mathcal{Q}}}\Big)\qquad\text{for all}\quad n\geq\underline{n}(m,k),$$

$$\underline{n}(m,k):=\frac{m\log\big[(1-(1-q^2)/m)^k/2\big]}{2\log q},$$

where $n=m\tilde{n}$ is the total dimension. Consequently the number of gradient evaluations needed by such a method to find $\bar{x}$ with $\mathbb{E}\left\|\bar{x}-x^*\right\|^2\leq\varepsilon$ cannot be smaller than

$$\Omega\Big\{\big(m+\sqrt{m\,\mathcal{C}}\big)\log\frac{\left\|x^0-x^*\right\|^2}{\varepsilon}\Big\},\qquad\mathcal{C}=\frac{L}{\mu},\;L=\sum_{i=1}^m L_i,$$

if n is sufficiently large; precisely (their display (3.45) in arXiv:1507.02000v3) $k\geq\big[\frac{m}{2}\big(\frac{\sqrt{\mathcal{Q}}}{2}+1\big)-1\big]\log\frac{\left\|x^0-x^*\right\|^2}{2\varepsilon}$.

[Appendix A1, p. 148]

Written as an average of m components, m times the summands of (A1.7), the instance has $R^2 = m\mu\mathcal{Q}$, so that $n_{\text{book}} = m$, $\kappa = m\mathcal{Q}$ and, as $\mathcal{C} = m(\mathcal{Q}-1)$, $m + \sqrt{m\mathcal{C}} \asymp n_{\text{book}} + \sqrt{n_{\text{book}}\kappa}$: this is the bound $\Omega((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon})$ for class (S1), the first part of A1, on instances with $\kappa \geq n$ and a dimension that grows with the number of queries.

The second part holds only in a restricted sense. Han, Xie and Zhang [37] prove a lower bound for a class $\mathcal{A}$ that, by their account, contains SAGA, L-SVRG and L-Katyusha: at each step, an algorithm of $\mathcal{A}$ samples one index i.i.d. from a fixed distribution, may query the proximal incremental first-order oracle (PIFO, which returns the value, the gradient and a proximal step of a component) of the sampled component at all previous iterates, computes the full gradient at the current point with an independent probability at most c_0/n, for a constant c_0 (the full gradient at the initial point is free), and takes its iterate in the span of the previous iterates and of the answers, projected on the domain. In their theorem, L is the smoothness constant of the components (our R^2), and R is not ours: $2R$ bounds the diameter of the domain.

Theorem A1.12 (Han–Xie–Zhang lower bound; Theorem E.6 of [37]). *Let $n \geq 2$ and consider the class $\mathcal{F}^{\mathrm{C}}(R, L, \mu)$ of finite sums $f = \frac{1}{n}\sum_i f_i$ on a closed convex domain $\mathcal{X}$ of diameter at most $2R$, with L-smooth components and μ-strongly convex f, $\kappa = L/\mu \geq 2$, and let $\varepsilon \leq LR^2/4$ (the hard instance is constrained to the ball $\{\|x\| \leq R\}$, which contains $x_0 = 0$ and x^*; its components are individually μ-strongly convex, their Proposition E.13). The number of PIFO queries that algorithms of class $\mathcal{A}$ (their Definition 3.1, which by their account contains SAGA, L-SVRG and L-Katyusha) need to find $\hat{x} \in \mathcal{X}$ with $\mathbb{E}f(\hat{x}) - \min_{\mathcal{X}} f < \varepsilon$ is*

$$\mathfrak{m}_*^{\mathrm{C}}(\varepsilon, R, L, \mu) = \begin{cases} \Omega\big((n+\sqrt{\kappa n})\log(1/\varepsilon)\big) & \text{for } \kappa = \Omega(n), \\ \Omega\big(n + \frac{n}{1+(\log(n/\kappa))_+}\log(1/\varepsilon)\big) & \text{for } \kappa = O(n), \end{cases}$$

and the construction only requires the dimension to be $O\big(1+\sqrt{\kappa/n}\,\log(1/\varepsilon)\big)$. [Appendix A1, p. 150]

For the methods of class $\mathcal{A}$, and, up to logarithmic factors, for the oblivious methods of part (c) of Theorem A1.11 (Hannah–Liu–O'Connor–Yin bounds; [38]), the lower bound is $\Omega\big(n + (\frac{n}{1+(\ln(n/\kappa))_+} + \sqrt{n\kappa})\ln\frac{1}{\varepsilon}\big)$: the $\sqrt{n\kappa}$ term is untouched, and full gradients can divide $n\log\frac{1}{\varepsilon}$ by at most $1 + (\ln(n/\kappa))_+$, a gain only when $\kappa \ll n$, never below the bare n; Prox-SVRG with the parameters of Theorem A1.11(b) attains this bound when $\kappa = O(n)$. Without these restrictions, only an additive n is proved for randomized algorithms, while deterministic algorithms, cyclic incremental methods included, can need $\Omega(n\sqrt{\kappa}\log\frac{\varepsilon_0}{\kappa\varepsilon})$ when $\kappa > 73$, by Theorem A1.9 (Woodworth–Srebro lower bounds; Theorems 4 and 8 of [87]), which excludes for them any gain over $n\log\frac{1}{\varepsilon}$ at small ε; whether randomly reshuffled span methods, or deterministic ones when $\kappa \leq 73$, can beat $n\log\frac{1}{\varepsilon}$ is not settled by these sources.

A hard instance with a closed-form minimizer. The instance of Theorem A1.8 (Lan–Zhou lower bound; Theorem 3 and Corollary 3 of [50]) gives each component its own regularizer, which forces $\kappa \geq n$. The instance (S) of Appendix A1 keeps one Nesterov chain per component but shares the regularizer among all components, as in [1], so that every $\kappa > 1$ is allowed. With n blocks $\theta^{(i)} \in \mathbb{R}^m$ and $A_m(\zeta)$ the $m \times m$ tridiagonal matrix with -1 off the diagonal and 2 on it, except $(A_m(\zeta))_{mm} = \zeta \in [1,3]$ (see (A1.2)),

$$\begin{aligned} &c = \frac{\mu(\kappa-1)}{4}, \qquad \phi(x) = \frac{1}{2}\langle A_m(\zeta)x, x\rangle - x_1 \quad (x \in \mathbb{R}^m), \\ &f_i(\theta) = c\,\phi(\theta^{(i)}) + \frac{\mu}{2}\|\theta\|^2, \quad i = 1, \dots, n. \end{aligned} \tag{A1.8}$$

The instance (H) of Section 2, used in Figures A3, A2 and A4, has instead a regularizer local to each block and $\kappa > n$. The constants and the minimizer of (S) are explicit.

Proposition A1.13 (constants of the instance). *Let $Q := 1 + \frac{\kappa-1}{n}$. For the instance (A1.8):*

(a) each f_i is μ-strongly convex and R^2-smooth with $R^2 = \mu\kappa$ (its exact Lipschitz constant is $\mu + c\,\lambda_{\max}(A_m(\zeta)) \le \mu\kappa$, with equality only in the limit $m \to \infty$);

(b) $\nabla^2 F = \operatorname{diag}\left(\frac{c}{n}A_m(\zeta) + \mu I, \dots, \frac{c}{n}A_m(\zeta) + \mu I\right)$ is block diagonal; F is μ-strongly convex, its smallest Hessian eigenvalue is $\mu_F := \mu + \frac{c}{n}\lambda_{\min}(A_m(\zeta)) = \mu\big(1 + \frac{Q-1}{4}\lambda_{\min}(A_m(\zeta))\big) \ge \mu$ and its smoothness constant is $L_F = \mu + \frac{c}{n}\lambda_{\max}(A_m(\zeta)) \le \mu Q$;

(c) hence $\kappa = R^2/\mu = 1 + n(Q-1)$ and

$$\frac{L_F}{\mu} \;\le\; Q = 1 + \frac{\kappa - 1}{n} \;\le\; \kappa, \tag{A1.10}$$

so that $(L_F/\mu)/\kappa \le \frac{1}{n} + \frac{1}{\kappa}$: the condition number of F is about n times smaller than κ as soon as $\kappa \gg n$.

[proof: Appendix A1, p. 152]

On (S), full-gradient methods therefore depend on $L_F \le \mu Q$, about R^2/n when $\kappa \gg n$, rather than on R^2: gradient descent and AGD do not pay there the factors n and $\sqrt{n}$ of Table 1, which require aligned components, as in the aligned instance (A) of Figure A4b.

Proposition A1.14 (closed-form minimizer). *For the instance (A1.8), let $Q = 1 + \frac{\kappa-1}{n}$ and $q = \frac{\sqrt{Q}-1}{\sqrt{Q}+1}$.*

(a) (Geometric chain.) If

$$\zeta = \zeta_Q := \frac{\sqrt{Q}+3}{\sqrt{Q}+1} \in (1, 2), \tag{A1.11}$$

then the minimizer is $\eta_* = (x^*, \dots, x^*)$ with $x_j^* = q^j$, $j = 1, \dots, m$, so that

$$\|\eta_*\|^2 = n\,\frac{q^2(1-q^{2m})}{1-q^2}, \qquad F(\eta_*) = -\frac{c\,q}{2} = -\frac{\mu(\kappa-1)\,q}{8}.$$

(b) (Nesterov's chain, $\zeta = 2$.) The minimizer is $\eta_* = (x^*, \dots, x^*)$ with

$$x_j^* = \frac{q^j - q^{2(m+1)-j}}{1 - q^{2(m+1)}} = \frac{\sinh\big((m+1-j)\tau\big)}{\sinh\big((m+1)\tau\big)}, \qquad \tau = \log\frac{1}{q} = 2\,\operatorname{artanh}\frac{1}{\sqrt{Q}}, \quad j = 1, \dots, m.$$

(c) (Any $\zeta \in [1, 3]$.) x^* is the solution of the tridiagonal system $\big(A_m(\zeta) + \frac{4}{Q-1}I\big)x = e_1$, computable in $O(m)$ by the Thomas algorithm (forward elimination, backward substitution), which is numerically stable here because the matrix is symmetric positive definite and diagonally dominant.

In all cases $F(\theta) - F(\eta_*) = \frac{1}{2}\langle \nabla^2 F(\theta - \eta_*), \theta - \eta_* \rangle$ lies between $\frac{\mu_F}{2}\|\theta - \eta_*\|^2$ and $\frac{L_F}{2}\|\theta - \eta_*\|^2$, with μ_F and L_F as in Proposition A1.13(b).

[proof: Appendix A1, p. 152]

Part (a) gives η_* in closed form, so that every error measured on (S) is exact; part (b), with $Q = 1 + 4c/\mu_b$, gives the minimizer of (H) (Section 2).

Started from $\theta_0 = 0$, a span method reveals at most one new coordinate of block j per query to component j by Lemma A1.16 (support under the span protocol), and the coordinates not yet revealed carry an explicit share of $\|\eta_*\|^2$ by Proposition A1.17 (tail bound). Averaging over the blocks by Jensen's inequality gives a bound for every index sequence, the $\sqrt{n\kappa}$ term of A1:

Proposition A1.18 (the $\sqrt{n\kappa}$ term for any index sequence). *On the instance (A1.8) with $\zeta = \zeta_Q$, and Q, q as in Proposition A1.14, consider any method of class (S2) (Definition A1.2) started at*

$\theta_0 = 0$. *After G gradient evaluations in total (full gradients counted as n), of which τ_j are queries to component j, every iterate θ in the span of the answers satisfies*

$$\frac{1}{n}\sum_{j=1}^{n} q^{2\tau_j} \;\ge\; q^{2G/n}, \qquad \textit{hence} \qquad \frac{\|\theta-\eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{q^{2G/n}-q^{2m}}{1-q^{2m}}. \tag{A1.13}$$

Consequently, if $q^{2m} \le \varepsilon < \frac{1}{2}$, reaching $\|\theta-\eta_*\|^2 \le \varepsilon\,\|\eta_*\|^2$ requires

$$G \;\ge\; \frac{n\log\frac{1}{2\varepsilon}}{2\log(1/q)} \;\ge\; \frac{n(\sqrt{Q}-1)}{4}\log\frac{1}{2\varepsilon} \;\ge\; \frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon} \quad (\textit{the last step for } \kappa-1 \ge 2n). \tag{A1.14}$$

[proof: Appendix A1, p. 154]

For the methods with coin flips, such as L-SVRG and L-Katyusha, the bound holds for the expected number of gradient evaluations (Appendix A1, after the proof). AGD on F, which needs $O(n\sqrt{Q}\log\frac{1}{\varepsilon})$ gradients since $L_F \le \mu Q$, attains its order when $\kappa \gtrsim n$. The sequence-free argument cannot produce the $n\log\frac{1}{\varepsilon}$ term, because a full gradient advances all n chains at once. With i.i.d. indices, the number of queries to each block is binomial, and the term appears:

Theorem A1.19 (Lan–Zhou bound on the instance)**.** *On the instance (A1.8) with $\zeta = \zeta_Q$, and Q, q as in Proposition A1.14, let $\theta_0 = 0$ and let the method belong to class (S1) (Definition A1.2) with a fixed sampling distribution p (uniform or not). Then for every $t \ge 0$,*

$$\frac{\mathbb{E}\,\|\theta_t-\eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{\left(1-\frac{1-q^2}{n}\right)^t - q^{2m}}{1-q^{2m}}, \tag{A1.15}$$

and, whenever $q^{2m} \le \frac{1}{2}\left(1-\frac{1-q^2}{n}\right)^t$,

$$\frac{\mathbb{E}\,\|\theta_t-\eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{1}{2}\Big(1-\frac{1-q^2}{n}\Big)^t \;\ge\; \frac{1}{2}\exp\Big(-\frac{4\sqrt{Q}\,t}{n(\sqrt{Q}+1)^2-4\sqrt{Q}}\Big). \tag{A1.16}$$

[proof: Appendix A1, p. 155]

Corollary A1.20 (iteration count and dimension requirement)**.** *In the setting of Theorem A1.19, let $\varepsilon \in (0, \frac{1}{2})$ and assume the block length m satisfies*

$$m \;\ge\; \frac{\log(1/\varepsilon)}{2\log(1/q)}, \qquad \textit{for which it suffices that} \qquad m \;\ge\; \frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon} = \frac{1}{4}\sqrt{1+\frac{\kappa-1}{n}}\,\log\frac{1}{\varepsilon}. \tag{A1.17}$$

Then any method of class (S1) needs

$$t \;\ge\; \Big(\frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}}-1\Big)\log\frac{1}{2\varepsilon} \;\ge\; \Big(\frac{n+\sqrt{n(\kappa-1)}}{8}+\frac{n}{2}-1\Big)\log\frac{1}{2\varepsilon} = \Omega\Big((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon}\Big) \tag{A1.18}$$

gradient evaluations to reach $\mathbb{E}\,\|\theta_t-\eta_*\|^2 \le \varepsilon\,\|\eta_*\|^2$. In particular $t \ge (n-1)\log\frac{1}{2\varepsilon}$ for every $\kappa > 1$, because $(\sqrt{Q}+1)^2 \ge 4\sqrt{Q}$.

[proof: Appendix A1, p. 155]

This proves the first part of A1 on (S), for every $\kappa > 1$, under an explicit dimension condition: by (A1.17) each block must be longer than the number of touches that an accelerated method needs, of order $\frac{1}{4}\sqrt{\kappa/n}\log\frac{1}{\varepsilon}$ for $\kappa \gg n$, so that the total dimension nm is of the order of the number of queries certified. L-Katyusha is not in (S1), because of its full gradients: its lower bound on (S) is Proposition A1.18 (the $\sqrt{n\kappa}$ term for any index sequence), without the $n\log\frac{1}{\varepsilon}$ term.

SAGA and L-Katyusha on the instance (hypothesis A4). Hypothesis A4 predicts that the gradient counts of SAGA and L-Katyusha grow like κ and $\kappa^{1/2}$ when $\kappa \gg n$. The chain bounds above are at most of order $(n+\sqrt{n\kappa})\log\frac{1}{\varepsilon}$; for SAGA, the exponent 1 comes instead from its mean iterate, which moves like gradient descent whatever the memory:

Proposition A1.22 (mean iterate of SAGA on quadratics). *Let the f_i be convex quadratics with Hessians $H_i \succeq 0$, let $\bar{H} = \frac{1}{n}\sum_i H_i = \nabla^2 F$ have eigenpairs $(\nu_l, w_l)_{l=1}^d$ (w_l orthonormal) and $\mu_F = \lambda_{\min}(\bar{H})$, and for $\nu' \geq \mu_F$ let $P_{\nu'} = \sum_{l:\nu_l \leq \nu'} w_l w_l^\top$ be the orthogonal projector onto the sum of the eigenspaces of $\bar{H}$ with eigenvalues at most ν'. For SAGA with step $\gamma > 0$, uniform sampling and any initial memory $z^{(0)}$ (deterministic, or random and independent of the sampling), $e_t = \theta_t - \eta_*$ satisfies $\mathbb{E}[e_t] = (I-\gamma\bar{H})^t e_0$; hence, for every $\nu' \in [\mu_F, 1/\gamma]$,*

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \;\geq\; \big\|(I-\gamma\bar{H})^t(\theta_0-\eta_*)\big\|^2 = \sum_{l=1}^{d}(1-\gamma\nu_l)^{2t}\,\langle\theta_0-\eta_*, w_l\rangle^2 \;\geq\; (1-\gamma\nu')^{2t}\,\|P_{\nu'}(\theta_0-\eta_*)\|^2\,. \tag{A1.20}$$

With $\nu' = \mu_F$ and a simple smallest eigenvalue with unit eigenvector $v_{\min}$, the last term is $(1-\gamma\mu_F)^{2t}\,\langle\theta_0-\eta_*, v_{\min}\rangle^2$. [proof: Appendix A1, p. 158]

For n identical components this is the argument of Proposition B3.17 (lower bound for SAGA with any constant step); here it holds on every quadratic instance. Combined with the proof of Proposition 5.9, run with $\mu_F = \lambda_{\min}(\nabla^2 F)$ in place of μ, it gives a two-sided bound:

Corollary A1.23 (SAGA sandwich). *Consider an instance with convex quadratic components such that $\lambda_{\max}(H_i) \leq R^2$ for all i (each f_i is then convex and R^2-smooth), with μ_F and the projectors $P_{\nu'}$ of Proposition A1.22. Let $\kappa_{\text{eff}} = R^2/\mu_F$, $\gamma = 1/(4R^2)$, and let $T_{\text{SAGA}}(\varepsilon)$ be the number of gradient evaluations (the n of an initial sweep $z_i^{(0)} = f_i'(\theta_0)$ included) that SAGA with step γ and uniform sampling needs to reach $\mathbb{E}\,\|\theta_t-\eta_*\|^2 \leq \varepsilon\,\|\theta_0-\eta_*\|^2$.*

(a) (Lower bound, any initial memory.) If $\|P_{\nu'}(\theta_0-\eta_*)\|^2 \geq \rho\,\|\theta_0-\eta_*\|^2$ for some $\nu' \in [\mu_F, 4R^2]$ and $\rho \in (0,1]$, then for $\varepsilon < \rho$

$$T_{\text{SAGA}}(\varepsilon) \;\geq\; \Big(\frac{2R^2}{\nu'} - \frac{1}{2}\Big)\log\frac{\rho}{\varepsilon},$$

i.e. $(\kappa - \frac{1}{2})\log\frac{\rho}{\varepsilon}$ for $\nu' = 2\mu$ and $(2\kappa_{\text{eff}} - \frac{1}{2})\log\frac{\rho}{\varepsilon}$ for $\nu' = \mu_F$.

(b) (Upper bound.) With the initialization $z_i^{(0)} = f_i'(\theta_0)$ of Proposition 5.9 of [11], the number of iterations needed is at most

$$\max\Big\{3n, \frac{16\kappa_{\text{eff}}}{3}\Big\}\log\frac{1+n/4}{\varepsilon} \qquad (\text{add } n \text{ gradient evaluations for the initial sweep}),$$

and with $z^{(0)} = 0$ the same bound holds with $1+n/4$ replaced by $1 + \frac{3\sum_i \|f_i'(\eta_*)\|^2}{16R^4\|\theta_0-\eta_*\|^2}$.

[proof: Appendix A1, p. 159]

On (S), the eigenvectors of the chain are sampled sine waves, and those of curvature at most 2μ carry a fixed share of the initial error as long as the chain is long compared with the decay length $\approx \sqrt{Q}/2$ of η_*:

Lemma A1.24 (slow-mode mass of the initial error). *Let $m \geq 2$, $\zeta \in [1,2]$ and $s \in (0,1]$. Let $0 < \lambda_1 < \cdots < \lambda_m < 4$ be the eigenvalues of the chain matrix $A_m(\zeta)$ of (A1.2) with orthonormal*

eigenvectors $v_1, \dots, v_m$, and let $\Pi_s = \sum_{k:\lambda_k \le s} v_k v_k^\top$ be the orthogonal projector onto the eigenspaces with eigenvalue at most s. For $x^* = (A_m(\zeta) + sI)^{-1} e_1$ and $K_0 := \lfloor (m+1)\sqrt{s}/\pi \rfloor$,

$$\|\Pi_s x^*\|^2 \;\ge\; \frac{7\pi^2(1-s/4)}{96\, m\, s^2 (2m+1)^2}\, K_0\big(4K_0^2 - 1\big). \tag{A1.23}$$

If moreover $\zeta = \zeta_Q$ and $s = \frac{4}{Q-1}$ (so that $x^* = (q^j)_{j \le m}$ is the block of η_*, Proposition A1.14(a)), the slow-mode fraction $\rho := \|\Pi_s x^*\|^2 / \|x^*\|^2$ satisfies, with $y := (m+1)\sqrt{s}/\pi$,

$$\rho \;\ge\; \frac{7(1-s/4)}{192\pi} \cdot \frac{K_0(4K_0^2-1)}{y^3} \;\ge\; \begin{cases} 1/105 & \text{if } y \ge 2, \\ 1/310 & \text{if } y \ge 1. \end{cases} \tag{A1.24}$$

On the instance (A1.8) with $\theta_0 = 0$, $\|P_{2\mu}(\theta_0 - \eta_*)\|^2 = \rho\, \|\theta_0 - \eta_*\|^2$, where $P_{2\mu}$ is the projector of Proposition A1.22 for $\nu' = 2\mu$. [proof: Appendix A1, p. 161]

These results and the bound of Theorem 11 of [47] for L-Katyusha (Section 4.5 recalls its iteration, its mirror sequence and its reference point, written z_t and w_t there and z^k and w^k below, and its parameters θ_1 and σ), converted into the distance of its mirror sequence in (A1.21)–(A1.22) with an explicit constant $C_0 \ge 1$ that depends only on n, κ and L_F/μ and enters the counts only through $\log \frac{C_0}{\varepsilon}$, give the prediction tested in Section 5.9:

Proposition A1.26 (testable prediction of hypothesis A4). *Fix $n \ge 2$, $m \ge 2$, μ and a relative accuracy $\varepsilon \in (0, \frac{1}{105})$, start at $\theta_0 = 0$, and let κ vary. On the instance (A1.8) with $\zeta = \zeta_Q$ (Q, q as in Proposition A1.14; μ_F, L_F as in Proposition A1.13), let $T_{\mathrm{alg}}(\varepsilon)$ be the number of gradient evaluations (its expectation for L-Katyusha) that an algorithm needs to reach $\mathbb{E}\,\|\theta - \eta_*\|^2 \le \varepsilon\, \|\theta_0 - \eta_*\|^2$: SAGA with $\gamma = \frac{1}{4R^2}$ and $z_i^{(0)} = f_i'(0)$, the n gradients of this sweep being counted; L-Katyusha with the parameters of Theorem 11 of [47], θ being its mirror sequence z^k, charged n gradients for $F'(w^0)$ and, per iteration, one gradient plus n when the reference point is refreshed (probability $p = \frac{1}{n}$; the gradients $f_i'(w^k)$ of each full pass are stored), i.e. 2 per iteration in expectation; GD and AGD on F, n per iteration. Let C_0 be the constant of (A1.21).*

(i) (Plateau, $\kappa \le n$.) For SAGA, under (A1.17) and $\varepsilon < q^2/2$,

$$n + (n-1)\log\frac{q^2}{2\varepsilon} \;\le\; T_{\mathrm{SAGA}}(\varepsilon) \;\le\; n + \max\Big\{3n, \frac{16\kappa}{3}\Big\} \log\frac{1+n/4}{\varepsilon} \;\le\; n + \frac{16n}{3}\log\frac{1+n/4}{\varepsilon},$$

both sides being $\Theta(n\log(1/\varepsilon))$ once $\varepsilon \ll q^2$ and $\varepsilon \ll 1/n$ (the upper bound is $\Theta(n\log(n/\varepsilon))$ in general): the plateau with slope ≈ 0 is a theorem for SAGA. For L-Katyusha only

$$\frac{n\log(1/(2\varepsilon))}{2\log(1/q)} \;\le\; T_{\mathrm{LK}}(\varepsilon) \;\le\; n + 16n\log\frac{C_0}{\varepsilon}, \qquad C_0 \le n + \tfrac{7}{3},$$

is proved (the lower bound under (A1.17)); the two sides are of the same order $n\log(1/\varepsilon)$ only when $\log(1/q) = O(1)$, i.e. when κ is of order n (for $\kappa = n$, $\log(1/q) = 2\,\mathrm{artanh}\,\big((2-\frac{1}{n})^{-1/2}\big)$, which decreases to $\log(3+2\sqrt{2}) \approx 1.763$ as $n \to \infty$: 1.81 for $n = 16$, 1.77 for $n = 100$). For $\kappa \ll n$, $\log(1/q) \approx \log\frac{4n}{\kappa-1}$, and a gain of this logarithmic order over $n\log(1/\varepsilon)$ is not excluded for L-Katyusha; it is realized by GD on F with step $1/L_F$, which contracts the error by $1 - \frac{\mu_F}{L_F} \le 1 - \frac{1}{Q} = \frac{\kappa-1}{n+\kappa-1} \approx \frac{\kappa-1}{n}$ per full gradient, in agreement with Theorem A1.11(c) and with item (ii) of the assessment of hypothesis (A1) in Section A1.2. In the upper bounds, the second term takes over at $\kappa = \frac{9}{16}n$ for SAGA ($\frac{16\kappa}{3} = 3n$) and at $\kappa = \frac{8}{3}n$ for L-Katyusha (its momentum parameter θ_1 reaches $\frac{1}{2}$).

(ii) (Power laws.) Let

$$\kappa_{\max}(m,\varepsilon) := 1 + n\min\Big\{\frac{16m^2}{\log^2(1/\varepsilon)} - 1,\ \frac{(m+1)^2}{\pi^2}\Big\}, \tag{A1.27}$$

where the first term is the sufficient form $m \geq \frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon}$ of the dimension condition (A1.17) solved for κ (so that $q^{2m} \leq \varepsilon$), and the second is the condition $y \geq 2$ of Lemma A1.24 solved for κ. In the window $4n+1 \leq \kappa \leq \kappa_{\max}(m,\varepsilon)$, with $\rho = \rho(m,\kappa)$ the slow-mode fraction of Lemma A1.24,

$$\Big(\kappa - \frac{1}{2}\Big)\log\frac{\rho}{\varepsilon} \ \leq\ T_{\mathrm{SAGA}}(\varepsilon) \ \leq\ n + \frac{16\kappa}{3}\log\frac{1+n/4}{\varepsilon}, \qquad \rho = \rho(m,\kappa) \geq \frac{1}{105}, \tag{A1.28}$$

$$\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon} \ \leq\ T_{\mathrm{LK}}(\varepsilon) \ \leq\ n + 4\sqrt{6}\,\sqrt{n\kappa}\,\log\frac{C_0}{\varepsilon}, \qquad C_0 \leq 1 + \frac{5}{8}Q, \tag{A1.29}$$

so that $T_{\mathrm{SAGA}} = \Theta(\kappa\log(1/\varepsilon))$ and $T_{\mathrm{LK}} = \Theta(\sqrt{n\kappa}\log(1/\varepsilon))$ up to the displayed constants and the $\log(1+\frac{5}{8}Q)$ of (A1.29): κ enters T_{SAGA} and T_{LK} with the exponents 1 and $\frac{1}{2}$ as orders of growth (not as the slope of a line fitted over the window, see Remark A1.27), and $T_{\mathrm{SAGA}}/T_{\mathrm{LK}} \asymp \sqrt{\kappa/n}$. GD on F (step $1/L_F$) and AGD on F have the same two orders of growth: $n(\frac{L_F}{4\mu} - \frac{1}{2})\log\frac{\rho}{\varepsilon} \leq T_{\mathrm{GD}} \leq \frac{nQ}{2}\log\frac{1}{\varepsilon}$ with $L_F \geq \mu(1+\frac{13}{20}(Q-1))$, and $\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon} \leq T_{\mathrm{AGD}} = O(n\sqrt{Q}\log\frac{1}{\varepsilon})$.

(iii) (Saturation.) For $\kappa \gg nm^2$ the chain is exhausted: $\mu_F/\mu = 1 + \frac{\kappa-1}{4n}\lambda_{\min}(A_m(\zeta_Q))$ grows linearly in κ, so

$$\kappa_{\mathrm{eff}} = \frac{R^2}{\mu_F} = \frac{\kappa}{1+\frac{\kappa-1}{4n}\lambda_{\min}(A_m(\zeta_Q))} \ \xrightarrow[\kappa\to\infty]{}\ \frac{4n}{\lambda_{\min}(A_m(1))} = \frac{n}{\sin^2\frac{\pi}{2(2m+1)}} \approx \frac{4n(2m+1)^2}{\pi^2}, \tag{A1.30}$$

because $\zeta_Q \to 1$ and $\lambda_{\min}$ is continuous in the matrix; for every finite κ, Lemma A1.3 gives the two-sided bound $\frac{\kappa}{1+\frac{\kappa-1}{n}\sin^2\frac{\pi}{2(m+1)}} \leq \kappa_{\mathrm{eff}} \leq \frac{\kappa}{1+\frac{\kappa-1}{n}\sin^2\frac{\pi}{2(2m+1)}}$. Then $(2\kappa_{\mathrm{eff}} - \frac{1}{2})\log\frac{\delta}{\varepsilon} \leq T_{\mathrm{SAGA}}(\varepsilon) \leq n + \frac{16\kappa_{\mathrm{eff}}}{3}\log\frac{1+n/4}{\varepsilon}$ for $\varepsilon < \delta$, with the single-mode overlap $\delta = \langle x^*, v_1\rangle^2 / \|x^*\|^2$ (x^*, v_1 as in Lemma A1.24), which tends to $\delta_\infty(m) > 0$ (Remark A1.25): T_{SAGA} saturates at $\Theta(\kappa_{\mathrm{eff}}\log(1/\varepsilon)) = \Theta(nm^2\log(1/\varepsilon))$, while T_{LK} is no longer bounded below by $\sqrt{n\kappa}$ (the hypothesis $q^{2m} \leq \varepsilon$ of Proposition A1.18 fails) and its behaviour depends on the strong-convexity constant used to set the parameter σ of L-Katyusha. Since F is μ_F-strongly convex, Theorem 11 of [47] also applies with $\sigma = \mu_F/R^2$ (the tuning used in the experiments, where $\mu = \lambda_{\min}(\nabla^2 F)$ is passed to the method): (A1.21)–(A1.22) then hold with μ_F and κ_{eff} in place of μ and κ, so that $T_{\mathrm{LK}} = O\big((n+\sqrt{n\kappa_{\mathrm{eff}}})\log\frac{C_0}{\varepsilon}\big)$ with C_0 bounded in κ, and T_{LK} saturates at $O(nm\log\frac{C_0}{\varepsilon})$. With the nominal $\sigma = \mu/R^2$ of (A1.22) no saturation is proved: the upper bound keeps growing like $\sqrt{n\kappa}$, and the momentum $\theta_1 = \sqrt{2n\mu/(3R^2)}$ tends to 0, whereas the value matched to the curvature, $\sqrt{2n\mu_F/(3R^2)}$, does not. Thus T_{SAGA}, and T_{LK} with $\sigma = \mu_F/R^2$, stay bounded as $\kappa \to \infty$, and their log-log slopes against κ tend to 0.

[proof: Appendix A1, p. 163]

Part (ii) certifies the exponents 1 and $\frac{1}{2}$ as orders of growth, not as slopes. For the instance of Figure A4b ($n = 16$, $m = 256$, $\varepsilon = 10^{-8}$) the window is $65 \leq \kappa \leq \kappa_{\max} \approx 4.9\cdot 10^4$, i.e. 2.9 decades; with ρ and C_0 replaced by their bounds $\frac{1}{105}$ and $1+\frac{5}{8}Q$, the two sides of (A1.28) differ there by a factor of about 7.8, and those of (A1.29) by 88 to 115, so that they constrain the end-to-end slope only to $[0.69, 1.31]$ and $[-0.17, 1.22]$, as Remark A1.27 (what to expect in a plot) details; the slopes

measured in Section 5.9 agree with these orders but are not implied by them. Figure A4b counts L-Katyusha as in Table 1 (3 gradients per iteration in expectation) and measures the distance of its gradient sequence y_t: this multiplies the upper bound of (A1.29) by at most $\frac{3}{2}$ and replaces C_0 by a constant $C_y = O(\kappa)$ (Appendix A1, after (A1.22)), without changing the exponents. Part (iii) proves that the counts saturate as $\kappa \to \infty$ only for SAGA and for L-Katyusha tuned with $\lambda_{\min}(\nabla^2 F)$, the tuning of the experiments, and proves nothing of the kind with the nominal μ; the slopes measured beyond $\kappa_{\max}$ are not predicted by part (ii). Logistic regression is not a hard instance: there is no chain, and its effective constants can be much better than R^2 and λ (Section 4.2).

Known and new. The lower bounds and the chain constructions [65, 1, 50] are known, and so is the geometric minimizer given by ζ_Q [50]. To our knowledge, the following are not in the sources we reviewed in this form: the restatement of the bounds with their exact oracle classes and the correction of hypothesis A1; self-contained proofs on a shared-regularizer instance valid for every $\kappa > 1$, including the bound for every index sequence (Proposition A1.18); the explicit dimension condition and window (Corollary A1.20, (A1.27)); and the SAGA sandwich (Corollary A1.23, with Lemma A1.24), which makes the exponent 1 a theorem on the instance (Proposition A1.26).

4.5 L-Katyusha: Lyapunov proof, negative momentum, and naive acceleration

Hypothesis A3 asks where the negative momentum of loopless Katyusha (L-Katyusha) [47] enters a Lyapunov proof written in the style of the book. For hypothesis A2, whether naive acceleration fails (Nesterov's momentum applied to the loopless SVRG estimator, without negative momentum), we ask why the same proof does not close in this case. Appendix A3 rewrites the proof of [47] in the notation of the book, under the following assumption.

Assumption A3.1 (convex R^2-smooth components, μ-strongly convex F). *$F(\theta) = \frac{1}{n}\sum_{i=1}^n f_i(\theta)$ on $\mathbb{R}^d$, where each f_i is convex and R^2-smooth; we write $L := R^2$ so that L is the* per-sample *smoothness constant, as in Assumption 1 of [47]. F is μ-strongly convex, with minimizer η_* and optimal value $F_* := F(\eta_*)$. We write $\kappa := R^2/\mu = L/\mu \geq 1$ and $\sigma := \mu/L = 1/\kappa \in (0, 1]$ (the quantity called σ in [47]).* [Appendix A3, p. 169]

In this subsection $L = R^2$ is the per-sample smoothness constant, not the smoothness constant L_F of F, which Section 4.2 denotes L. L-Katyusha keeps a gradient sequence y_t, a mirror sequence z_t and a reference point w_t at which the full gradient $F'(w_t)$ is known. We keep the notation of [47] for its parameters: θ_1, θ_2 are momentum weights, not iterates, η is a step parameter, not the minimizer η_*, and z_t is the mirror sequence, not the SAGA memory. Let $\theta_1, \theta_2 > 0$ with $\theta_1 + \theta_2 \leq 1$, $p \in (0, 1]$, and

$$\eta := \frac{\theta_2}{(1+\theta_2)\theta_1}. \tag{A3.2}$$

From $y_0 = z_0 = w_0 = \theta_0$, with $F'(w_0)$ computed, an iteration reads

$$x_t = \theta_1 z_t + \theta_2 w_t + (1 - \theta_1 - \theta_2) y_t, \tag{K1}$$

$$g_t = f'_{i(t)}(x_t) - f'_{i(t)}(w_t) + F'(w_t), \tag{K2}$$

$$z_{t+1} = \frac{1}{1+\eta\sigma}\Big(\eta\sigma\, x_t + z_t - \frac{\eta}{L}\, g_t\Big), \tag{K3}$$

$$y_{t+1} = x_t + \theta_1(z_{t+1} - z_t), \tag{K4}$$

$$w_{t+1} = y_t \text{ with probability } p, \qquad w_{t+1} = w_t \text{ otherwise}, \tag{K5}$$

Method or class	Component gradients to reach accuracy ε	Validity and source
GD, step $1/R^2$	$n\kappa \log \frac{1}{\varepsilon}$ $\left(n \frac{L_F}{\mu} \log \frac{1}{\varepsilon}\right.$ with step $\left.1/L_F\right)$	all n, κ; [11, Section 5.4.4]
AGD	$n\sqrt{\kappa} \log \frac{1}{\varepsilon}$ $\left(n\sqrt{L_F/\mu} \log \frac{1}{\varepsilon}\right.$ with $\left.L_F\right)$	all n, κ; [11, Section 5.2.5]
SAGA, $\gamma = \frac{1}{4R^2}$	$n + \max\{3n, \frac{16\kappa}{3}\} \log \frac{1+n/4}{\varepsilon}$	all n, κ; Proposition 5.9; Corollary A1.23
L-SVRG, $\eta = \frac{1}{6R^2}$, $p = \frac{1}{n}$	$n + 3\max\{6\kappa, 2n\} \log \frac{\Phi_0}{\varepsilon}$	all n, κ; [47, Theorem 5]
L-Katyusha	$n + (12n + 15\sqrt{n\kappa}) \log \frac{3R^2(n+3)\|\theta_0 - \eta_*\|^2}{8\varepsilon}$	all n, κ and $\varepsilon \leq \frac{3R^2(n+3)}{16} \|\theta_0 - \eta_*\|^2$; Corollary A3.12
class (S1)	$\Omega\big((n + \sqrt{n\kappa}) \log \frac{1}{\varepsilon}\big)$	dimension $\gtrsim$ number of queries; Theorem A1.8; Corollary A1.20
class (S2), instance (S)	$\frac{\sqrt{n(\kappa-1)}}{8} \log \frac{1}{2\varepsilon}$	$\kappa - 1 \geq 2n$, $q^{2m} \leq \varepsilon < \frac{1}{2}$; Proposition A1.18
any randomized algorithm	$\Omega\big(n + \sqrt{n\kappa} \log(\frac{\varepsilon_0}{\varepsilon}\sqrt{n/\kappa})\big)$	$\kappa > 161n$, no span assumption; Theorem A1.9

Table 1: Guarantees for every instance of the class of Proposition 5.9 (top) and lower bounds (bottom), in component gradients. The accuracy criteria differ only inside the logarithm, by factors polynomial in n and κ (for instance $\frac{3\kappa(n+3)}{4}$ when the L-Katyusha criterion is converted into the relative squared distance by strong convexity): relative squared distance (SAGA, instance (S)), the Lyapunov function Φ of [47, Theorem 5] (L-SVRG), $\mathbb{E}[F(y_t)] - F(\eta_*)$ (L-Katyusha, y_t its gradient sequence, Section 4.5), initial gap ε_0 (last row). Counting convention, that of the experiments (Section 2): the SAGA, L-SVRG and L-Katyusha counts include the initial full gradient, and L-SVRG and L-Katyusha are charged 2 gradients per iteration ($f'_{i(t)}$ at the query point and at the reference point) and n per refresh of the reference point, i.e. 3 per iteration in expectation. With stored reference gradients, the convention of T_{LK} in Appendix A1, an iteration costs 2 in expectation, which gives $n + (8n + 10\sqrt{n\kappa}) \log(\cdot)$ for L-Katyusha and $n + 2\max\{6\kappa, 2n\} \log(\cdot)$ for L-SVRG. The SAGA and L-SVRG counts treat the number of iterations as a real number (rounding it up adds at most one iteration); the L-Katyusha count is that of an integer number of iterations, which its condition on ε ensures. For $\kappa \gg n$ the leading terms are $n\kappa$, $n\sqrt{\kappa}$, κ, κ and $\sqrt{n\kappa}$, the last one matching the lower bounds.

where $i(t)$ is uniform on $\{1, \dots, n\}$ and the coin of (K5) is independent of $i(t)$ and of the past; $\mathbb{E}_{i(t)}$ denotes the expectation over both given (y_t, z_t, w_t), so that $\mathbb{E}_{i(t)}[g_t] = F'(x_t)$. Step (K3) is a mirror step, (K4) a gradient step, and (K1) couples them [3]. The term $\theta_2 w_t$ of (K1), the anchor, is the *negative momentum* of Katyusha [2]: it pulls the query point towards the point where the estimator has no variance. With $\theta_2 = 0$ and a free step η, $x_{t+1} = y_{t+1} + (1 - \theta_1)(y_{t+1} - y_t)$ and the method is a variant of Nesterov's momentum applied to the loopless SVRG estimator, the naive acceleration of hypothesis A2; see Remark A3.16 (relation with Nesterov's method of the book). Theorem 11 of [47] takes

$$\theta_2 = \tfrac{1}{2}, \qquad \theta_1 = \min\left\{\sqrt{\tfrac{2\sigma n}{3}}, \tfrac{1}{2}\right\} = \min\left\{\sqrt{\tfrac{2n}{3\kappa}}, \tfrac{1}{2}\right\}, \qquad p = \tfrac{1}{n}, \qquad \eta = \frac{\theta_2}{(1+\theta_2)\theta_1} = \frac{1}{3\theta_1}, \tag{A3.3}$$

and its proof tracks the Lyapunov function

$$\Psi_t := Z_t + Y_t + W_t,$$
$$Z_t := \frac{L(1+\eta\sigma)}{2\eta}\,\|z_t-\eta_*\|^2, \qquad Y_t := \frac{1}{\theta_1}\big(F(y_t)-F_*\big), \qquad W_t := \frac{\theta_2(1+\theta_1)}{p\theta_1}\big(F(w_t)-F_*\big), \tag{A3.4}$$

the sum of a distance of the mirror sequence, of a function value of the gradient sequence and of a memory term, which measures the quality of the reference point. Finally, $D_F(w,x) := F(w) - F(x) - \langle F'(x), w-x\rangle \geq 0$ denotes the Bregman divergence of F.

The proof certifies a contraction of Ψ in conditional expectation at every step, for all admissible parameters, and a global rate for the parameters (A3.3).

Theorem A3.4 (contraction and rate of L-Katyusha; Lemma 10 and Theorem 11 of [47]). *Under Assumption A3.1, for any $\theta_1, \theta_2 > 0$ with $\theta_1 + \theta_2 \leq 1$, any $p \in (0,1]$, $\sigma = \mu/L$ and η as in (A3.2), the iterates of (K1)–(K5) and the potential (A3.4) satisfy, for all $t \geq 0$,*

$$\mathbb{E}_{i(t)}\big[\Psi_{t+1}\big] \;\leq\; \frac{1}{1+\eta\sigma}\,Z_t + \big(1-\theta_1(1-\theta_2)\big)\,Y_t + \Big(1-\frac{p\theta_1}{1+\theta_1}\Big)\,W_t \qquad \textit{(one-step contraction)}. \tag{A3.5}$$

With the parameters (A3.3), for all $t \geq 0$,

$$\mathbb{E}\big[\Psi_t\big] \leq (1-\rho)^t\,\Psi_0, \qquad \rho := \min\Big\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\Big\} = \begin{cases} \dfrac{1}{\sqrt{24\,n\kappa}} & \text{if } \kappa \geq \frac{8n}{3}, \\ \min\Big\{\dfrac{1}{3\kappa}, \dfrac{1}{4n}\Big\} \;\Big(\geq \dfrac{1}{8n}\Big) & \text{if } \kappa < \frac{8n}{3}, \end{cases} \tag{A3.6}$$

so that in all cases $\rho \geq \min\{1/(4n), 1/\sqrt{24n\kappa}\}$ (global rate). [proof: Appendix A3, p. 175]

We write c_Z, c_Y and c_W for the three coefficients of (A3.5), so that $\mathbb{E}_{i(t)}[\Psi_{t+1}]$ is at most $\max\{c_Z, c_Y, c_W\}\,\Psi_t$, a factor that the proof of (A3.6) bounds by $1-\rho$. The two terms of ρ are lower bounds on $1-c_Z$, the contraction rate of the distance Z_t of the mirror sequence, whose step $\eta/L = 1/(3\theta_1 L)$ grows as θ_1 decreases, and on $1-c_W$, that of the memory term W_t, refreshed with probability $\frac{1}{n}$; for $\kappa \geq \frac{8n}{3}$, the value $\theta_1 = \sqrt{2n/(3\kappa)}$ of (A3.3) balances them up to a factor 2. Along L-Katyusha runs, with $\mathbb{E}_{i(t)}[\Psi_{t+1}]$ computed exactly, (A3.5) holds at every recorded state, and the exact one-step contraction exponent is on average 2.8 to 4.7 times the certified exponent $-\log\max\{c_Z, c_Y, c_W\}$ and 3.7 to 9.3 times $-\log(1-\rho)$ (three problems, 3 seeds; Section 5.7). In gradient evaluations, the rate gives the following count.

Corollary A3.12 (gradient complexity of L-Katyusha). *With the parameters (A3.3), ρ as in (A3.6) and $y_0 = z_0 = w_0 = \theta_0$, for all $t \geq 0$,*

$$\mathbb{E}\big[F(y_t) - F_*\big] \leq \theta_1(1-\rho)^t\Psi_0, \qquad \mathbb{E}\,\|z_t-\eta_*\|^2 \leq \frac{2\eta}{L(1+\eta\sigma)}(1-\rho)^t\Psi_0,$$
$$\theta_1\Psi_0 \leq \frac{3L(n+3)}{8}\,\|\theta_0-\eta_*\|^2\,. \tag{A3.20}$$

Consequently $\mathbb{E}[F(y_t) - F_*] \leq \varepsilon$ as soon as $t \geq \big(4n + \sqrt{24n\kappa}\big)\log\frac{3L(n+3)\|\theta_0-\eta_*\|^2}{8\varepsilon}$. An iteration evaluates $f'_{i(t)}(x_t)$ and $f'_{i(t)}(w_t)$ in (K2) and, with probability $p = 1/n$, the n gradients of a full pass: $2 + pn = 3$ evaluations of some f'_i in expectation. If the n gradients $f'_i(w)$ computed during the pass are stored ($\mathcal{O}(nd)$ memory, as for the table of SAGA), $f'_{i(t)}(w_t)$ is a look-up and the count is

$1+pn=2$ *(this is the "$\mathcal{O}(1+pn)$ times per iteration" of [47]). Adding the n evaluations of $F'(w_0)$ at initialization, for $\varepsilon \le \frac{3L(n+3)}{16}\|\theta_0-\eta_*\|^2$ the expected number of gradient evaluations is at most*

$$\begin{aligned} & n+\big(12\,n+3\sqrt{24}\,\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon} \\ & \qquad \le\ n+\big(12\,n+15\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon} \quad \textit{(without storage)}, \\ & n+\big(8\,n+2\sqrt{24}\,\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon} \\ & \qquad \le\ n+\big(8\,n+10\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon} \quad \textit{(with storage)}, \end{aligned} \tag{A3.21}$$

and in both cases $\mathcal{O}\big((n+\sqrt{n\kappa})\log\frac{nL\|\theta_0-\eta_*\|^2}{\varepsilon}\big)$. [proof: Appendix A3, p. 176]

The count without storage, three gradients per iteration in expectation, is the convention of the experiments and of Table 1; the count with storage is that of the bounds of Appendix A1. The condition on ε lets the constants absorb the rounding of the number of iterations to an integer. For $\kappa \gg n$, the leading term $\sqrt{n\kappa}\log\frac{1}{\varepsilon}$ is of the order of the lower bounds of Section 4.4.

The proof, and where the negative momentum acts. The proof of the one-step contraction (A3.5) chains five lemmas, L1–L5, which reproduce Lemmas 6–10 of [47] and use seven elementary inequalities, listed with their tightness ratios in Table 5 of Appendix A3. L1 (Lemma A3.6) bounds the variance of the estimator; L2 (Lemma A3.7) bounds the mirror step by the three-point identity; L3 (Lemma A3.8) bounds the gradient step by smoothness and Young's inequality; L4 (Lemma A3.10) is the identity (A3.14) for the refresh of w; and L5 (Lemma A3.11) combines them with strong convexity at x_t and convexity towards y_t. The form of L1 matters for hypothesis A3: it bounds the variance by a Bregman divergence, whose inner-product part the negative momentum cancels.

Lemma A3.6 (L1: unbiasedness and variance of the estimator; Lemma 6 of [47]). *For all t, $\mathbb{E}_{i(t)}[g_t]=F'(x_t)$ and*

$$\mathbb{E}_{i(t)}\big\|g_t-F'(x_t)\big\|^2=\frac{1}{n}\sum_{i=1}^n\big\|f_i'(x_t)-f_i'(w_t)\big\|^2-\big\|F'(x_t)-F'(w_t)\big\|^2\ \le\ \frac{1}{n}\sum_{i=1}^n\big\|f_i'(x_t)-f_i'(w_t)\big\|^2 \tag{A3.7}$$

$$\le\ 2L\big(F(w_t)-F(x_t)-\big\langle F'(x_t),w_t-x_t\big\rangle\big)\ =\ 2L\,D_F(w_t,x_t). \tag{A3.8}$$

We call (A3.7) the centering inequality *and (A3.8) the* co-coercivity bound*; their product is the* variance bound

$$\mathbb{E}_{i(t)}\big\|g_t-F'(x_t)\big\|^2\le 2L\,D_F(w_t,x_t). \tag{A3.9}$$

[proof: Appendix A3, p. 172]

It is the one-sided (Bregman) form of co-coercivity in (A3.8), instead of the symmetric form used in the proof of Proposition 5.9, that produces this divergence; Section 5.7 measures the two steps of (A3.9) along trajectories. In the proof of L5 (Lemma A3.11), θ_2 is used for something other than bookkeeping at a single step, Step 6, the identity

$$\frac{\theta_2}{\theta_1}\big\langle F'(x_t),x_t-w_t\big\rangle-\frac{\theta_2}{\theta_1}\Big(F(w_t)-F(x_t)-\big\langle F'(x_t),w_t-x_t\big\rangle\Big)=-\frac{\theta_2}{\theta_1}\big(F(w_t)-F(x_t)\big), \tag{A3.19}$$

in which the first term comes from the anchor $\theta_2 w_t$ of (K1) and the second is the variance term, which enters through L3 and L1:

Proposition A3.14 (Bregman-gap cancellation by the negative momentum)**.** *In the chain of Lemma A3.11, the variance of the estimator enters through L3 and L1 (Lemmas A3.8 and A3.6) as the term* $-\frac{\theta_2}{\theta_1} D_F(w_t, x_t)$ *with* $D_F(w_t, x_t) = \big(F(w_t) - F(x_t)\big) + \langle F'(x_t), x_t - w_t\rangle$*, and the weight* θ_2 *of* w_t *in* (K1) *enters, through the decomposition* (A3.17)*, as the term* $+\frac{\theta_2}{\theta_1}\langle F'(x_t), x_t - w_t\rangle$*. The two inner products cancel identically, and the remainder* $-\frac{\theta_2}{\theta_1}(F(w_t) - F(x_t))$ *is absorbed as follows:* $F(x_t)$ *disappears because the coefficients of* $F(x_t)$ *sum to zero (Step 7 of the proof of Lemma A3.11), and* $\frac{\theta_2}{\theta_1}(F(w_t) - F_*) = \frac{p}{1+\theta_1} W_t$ *is paid by the reference-point potential, which contracts by* (A3.14) *because* w *is refreshed with probability* p*.* [proof: Appendix A3, p. 177]

The inner-product part of $D_F(w_t, x_t)$ must be cancelled rather than absorbed. By smoothness, $D_F(w_t, x_t) \le \frac{L}{2}\|w_t - x_t\|^2$ is a squared distance at the scale L, while the certified decrease per step is of order $\rho\Psi_t$, with $\rho = 1/\sqrt{24n\kappa}$ when $\kappa \ge \frac{8n}{3}$: absorbing this term would require converting a squared distance into function values, at the price of a factor κ, which Proposition A3.19 (closing by smoothness and strong convexity costs κ) makes precise for the naive scheme; and the inner product $\langle F'(x_t), x_t - w_t\rangle$ has no sign and is not a difference of function values that a potential could pay. The negative momentum costs a constant factor: Y_t contracts by $\theta_1(1-\theta_2)$ instead of θ_1, half as fast for $\theta_2 = \frac{1}{2}$. It also ties the step to θ_2, as Remark A3.9 (the step size couples the two momenta) explains: for a general step, Young's inequality in L3 (Lemma A3.8) produces the variance coefficient $\nu/(2L\theta_1)$ with $\nu = \eta\theta_1/(1-\eta\theta_1)$, and the cancellation is exact only for $\nu = \theta_2$, that is, for the step (A3.2): a larger step leaves the residual $\frac{\nu-\theta_2}{\theta_1} D_F(w_t, x_t)$, the extra term of the naive scheme (Proposition A3.18) with $\nu - \theta_2$ in place of ν, and a smaller one wastes part of the pull. Along the L-Katyusha runs of Section 5.7, the cancelled term $\frac{\theta_2}{\theta_1}\langle F'(x_t), x_t - w_t\rangle$ is positive and larger than the certified one-step decrease $(1 - \max\{c_Z, c_Y, c_W\})\Psi_t$ on 9.9%, 5.2% and 19.3% of the recorded states of the three problems; only there could it not have been absorbed into that decrease, since where it is negative, $D_F(w_t, x_t) \le F(w_t) - F(x_t)$ is bounded by a difference of function values.

Naive acceleration (hypothesis A2). Without the anchor, the naive scheme is the same iteration with $\theta_2 = 0$ and a free step.

Definition A3.15 (naive accelerated SVRG)**.** *The* naive scheme $\mathrm{N}(\theta_1, \eta, p)$ *is* (K1)–(K5) *with* $\theta_2 = 0$ *and the step* $\eta > 0$ *taken as a free parameter with* $\eta\theta_1 \le 1$*:*

$$\begin{aligned} &x_t = \theta_1 z_t + (1-\theta_1) y_t, \qquad g_t \text{ as in (K2)}, \qquad z_{t+1} = \frac{\eta\sigma x_t + z_t - \frac{\eta}{L} g_t}{1+\eta\sigma}, \\ &y_{t+1} = x_t + \theta_1(z_{t+1} - z_t), \qquad w_{t+1} = \begin{cases} y_t & w.p.\ p, \\ w_t & w.p.\ 1-p. \end{cases} \end{aligned} \tag{A3.22}$$

We set $\nu := \frac{\eta\theta_1}{1-\eta\theta_1} \in (0, \infty]$*, with* $\nu = \infty$ *in the limiting case* $\eta\theta_1 = 1$*; this case is admissible only with exact gradients* $g_t = F'(x_t)$ *(Proposition A3.17), because the Young step of L3 (Lemma A3.8) requires* $\nu < \infty$ *(Remark A3.9). The potential is the same* Ψ_t *as in* (A3.4) *(with* W_t *defined with an arbitrary constant* $a_W \ge 0$ *in place of* $\frac{\theta_2(1+\theta_1)}{p\theta_1}$*, which is* 0 *when* $\theta_2 = 0$*).* [Appendix A3, p. 178]

With exact gradients, the same lemmas prove acceleration, so that what fails with the estimator is the treatment of its variance.

Proposition A3.17 (acceleration with exact gradients and $\theta_2 = 0$)**.** *Let* $g_t = F'(x_t)$ *in* (A3.22) *(full gradient),* $\theta_2 = 0$*,* $\eta = 1/\theta_1$ *(so* $\eta\theta_1 = 1$ *and* $\nu = \infty$*: the limiting case of Definition A3.15) and*

$\theta_1 = \sqrt{\sigma} = \sqrt{\mu/L}$. *Then for all* t,

$$Z_{t+1} + Y_{t+1} \le \frac{1}{1+\sqrt{\sigma}} Z_t + (1-\sqrt{\sigma})\, Y_t \le \Big(1 - \frac{1}{2}\sqrt{\frac{\mu}{L}}\Big)(Z_t + Y_t), \tag{A3.23}$$

hence $F(y_t) - F_* \le \sqrt{\mu/L}\,(1 - \frac{1}{2}\sqrt{\mu/L})^t (Z_0 + Y_0)$*: the accelerated rate, as for (5.16)–(5.17) in the book.* [proof: Appendix A3, p. 178]

With the estimator, Steps 1–5 and 7 of the proof of L5 (Lemma A3.11) go through, but no step produces the inner product that cancelled the variance term.

Proposition A3.18 (one-step inequality of the naive scheme)**.** *For the scheme* $\mathrm{N}(\theta_1, \eta, p)$ *of Definition A3.15 with* $\eta\theta_1 < 1$ *(so that* $\nu < \infty$*), for all* t,

$$\begin{aligned} \mathbb{E}_{i(t)}\big[Z_{t+1} + Y_{t+1}\big] &\le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1)\, Y_t + \frac{\nu}{2L\theta_1} \mathbb{E}_{i(t)} \left\| g_t - F'(x_t) \right\|^2 \\ &\le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1)\, Y_t + \frac{\nu}{\theta_1} D_F(w_t, x_t), \end{aligned} \tag{A3.24}$$

where the last inequality is the variance bound (A3.9). The extra term $\frac{\nu}{\theta_1} D_F(w_t, x_t)$ *is nonnegative; under Assumption A3.1 it vanishes only when* $w_t = x_t$*, since* $D_F(w_t, x_t) \ge \frac{\mu}{2} \left\| w_t - x_t \right\|^2$ *by the* μ*-strong convexity of* F. [proof: Appendix A3, p. 179]

The decrease of the potential that (A3.24) can certify per step is at most $\frac{\eta\sigma}{1+\eta\sigma} Z_t + \theta_1 Y_t$, plus, with a reference term $a_W(F(w_t) - F_*)$, the expected decrease $p a_W(F(w_t) - F_*) - p a_W(F(y_t) - F_*)$ of that term; none of these involves the query point x_t. Bounding the extra term through smoothness and strong convexity costs a factor κ:

Proposition A3.19 (closing by smoothness and strong convexity costs κ)**.** *Under Assumption A3.1, for the scheme* $\mathrm{N}(\theta_1, \eta, p)$ *of Definition A3.15 with* $\eta\theta_1 < 1$,

$$\frac{\nu}{\theta_1} D_F(w_t, x_t) \;\le\; 2\kappa\nu \Big[\frac{1}{\theta_1}\big(F(w_t) - F_*\big) + \frac{\eta}{1+\eta\sigma} Z_t + (1-\theta_1)\, Y_t \Big], \tag{A3.25}$$

so that, writing $W_t = a_W(F(w_t) - F_*)$ *with* $a_W > 0$ *and using (A3.14) with* $\theta_2(1+\theta_1)/(p\theta_1)$ *replaced by* a_W*,*

$$\mathbb{E}_{i(t)}[\Psi_{t+1}] \le \Big(\frac{1 + 2\kappa\nu\eta}{1+\eta\sigma}\Big) Z_t + \Big(1 - \theta_1 + 2\kappa\nu(1-\theta_1) + p a_W \theta_1\Big) Y_t + \Big(1 - p + \frac{2\kappa\nu}{\theta_1 a_W}\Big) W_t. \tag{A3.26}$$

The coefficient of Z_t *is* < 1 *iff* $2\kappa\nu\eta < \eta\sigma$*, i.e.* $\nu < \frac{1}{2\kappa^2}$*; the coefficient of* Y_t *is* < 1 *only if* $2\kappa\nu(1-\theta_1) < \theta_1$*, i.e.* $\nu < \frac{\theta_1}{2\kappa(1-\theta_1)}$*; the coefficient of* W_t *is* < 1 *iff* $a_W > \frac{2\kappa\nu}{p\theta_1}$ *(substituting this bound in the* Y*-coefficient sharpens the second condition to* $\nu < \frac{\theta_1}{2\kappa(2-\theta_1)}$*, which we do not need). No choice of* (θ_1, η, p, a_W) *makes (A3.26) certify a rate better than* $1 - \kappa^{-2}$ *per iteration: worse than gradient descent with step* $1/L$*, which contracts by* $1 - 1/\kappa$ *per iteration (per gradient evaluation, with* $p = 1/n$*, the comparison is* $1 - \frac{1}{3\kappa^2}$ *against* $1 - \frac{1}{n\kappa}$*, so the naive certificate is worse as soon as* $\kappa > n/3$*), let alone the accelerated* $1 - \Theta(1/\sqrt{n\kappa})$. [proof: Appendix A3, p. 180]

This is a limit of the Lyapunov argument, not a lower bound for the naive scheme, as Remark A3.20 (other routes, and scope of the obstruction) details: bounding the extra term through Z_t instead of μ is no better, paying it by the reference term alone is impossible, and no lower bound specific to the scheme is known to us. It agrees with the literature: with one sample per iteration,

the guarantee of Nesterov's momentum on the SVRG estimator is that of SVRG [67], the anchor-free acceleration of [21] needs mini-batches of size of order $n^{2/3}$, and the semidefinite certificate of [42] is stated for negative-momentum weights of at least $\frac{1}{5}$. In the experiments at $\kappa = 10^4$ (one seed; Section 5.8), L-Katyusha with $\theta_2 = 0$ and otherwise unchanged parameters diverges on the instance (H) ($n = 16$ blocks of length 64) and converges on logistic regression, and Nesterov's momentum applied to the estimator, at its best grid setting, converges on both problems; where they converge, these anchor-free schemes need 8 to 20 times fewer gradient evaluations than L-SVRG. Re-tuned at every κ, Nesterov's momentum applied to the estimator has fitted slopes of about 0.8 in κ on block quadratics (Section 5.9). Prediction (P3) of Section A3.4, that a stable naive scheme keeps the exponent 1 of κ and does no better than L-SVRG, is thus refuted in its second part and not supported in its first: Proposition A3.18 and Proposition A3.19 describe a limitation of the certificate, not the observed behaviour.

Known and new. Theorem A3.4 and its constants are those of [47], the negative momentum is that of [2], and its role of cancelling an inner product of the variance bound is stated in [2, 91]. To our knowledge, the following are not in the sources we reviewed in this form: the rewriting of the proof as a chain of labelled inequalities with tightness ratios, the isolation of the cancellation as a single identity tied to the choice of the step (Proposition A3.14 and Remark A3.9), and Proposition A3.18 and Proposition A3.19, which quantify what the same argument certifies without the anchor.

5 Experiments

5.1 Common protocol

Problems. Logistic regression uses the generator of Section 2 with $n = 1000$, $d = 50$, the covariance spectrum $j^{-\alpha}$, $\alpha \in \{0, 1, 2\}$ (population condition numbers 1, 50 and 2500), and $\lambda = \frac{1}{4}/(\kappa - 1)$, so that $\kappa = R^2/\mu$ exactly, with $R^2 = \max_i L_i = \frac{1}{4} + \lambda$ and $\mu = \lambda$. One data set $(x_i, y_i)_{i \le n}$ is drawn per α (data seed 0) and shared by all values of κ, which only set λ; the seeds reported below are sampling seeds, and all runs start at $\theta_0 = 0$ (SAGA with $z_i^{(0)} = f_i'(\theta_0)$). The nominal κ is a worst case: over the fifteen problems of Section 5.3 (κ from 10 to 10^5), L_F/μ ranges from 1.2 to $8.1 \cdot 10^3$ and $\lambda_{\min}(F''(\eta_*))/\mu$ from 1.0 to 70. The quadratic instances have $n = 16$ components: (H) the block-separable Nesterov chains of Section 2 (normalization of [50] with Nesterov's chain, κ exact, $L_F = R^2/n$, blocks of length $m = 64$ or 256); (R) block quadratics with eigenvalues drawn log-uniformly in $[1, \kappa/n]$, random eigenbases and Gaussian linear terms ($m = 256$, one instance per seed); (S) the instance (A1.8) of Appendix A1 (shared regularizer, $m = 256$), whose nominal κ exceeds the effective $\kappa_{\text{eff}} = R^2/\lambda_{\min}(F'')$ of (A1.30) once the chain is exhausted; (A) the aligned instance of the same appendix, one chain of length $m = 2048$ shared by all components, so that $L_F = R^2$. Minimizers are obtained by direct solves, and every method receives the exact constants of its instance: $R^2 = \max_i \lambda_{\max}(f_i'')$, $\mu = \lambda_{\min}(F'')$ and $L_F = \lambda_{\max}(F'')$ (on (S) the methods are thus tuned with κ_{eff}), and for logistic regression $\mu = \lambda$ and $L_F = \lambda_{\max}(\hat{\Sigma})/4 + \lambda$ with $\hat{\Sigma} = \frac{1}{n}\sum_i x_i x_i^\top$.

Exact one-step quantities and precision floors. No one-step expectation is estimated by sampling: given the state, $i(t)$ takes n equally likely values, so the conditional expectations of Lemma B3.4 (exact one-step identities), hence the exact conditional contraction $\rho_{\text{exact}}(t) = \mathbb{E}[V_t \mid \mathcal{F}_{t-1}]/V_{t-1}$ (Proposition B3.5), the tightness ratios (Definition B3.8), μ_{eff}, L_{eff} and the attribution chains (Proposition B3.11), are computed exactly in $O(nd)$ operations; for L-Katyusha, $\mathbb{E}[\Psi_{t+1} \mid \text{state}]$ enumerates the n indices and both outcomes of the coin that refreshes the reference point

($O(n^2 d)$). They are evaluated every $n/10$ iterations (every iteration when $n = 16$) and checked by the test suite (Appendix R); they are exact given the visited states, but their products along a trajectory are realized quantities, not expectations (Section B3.6). For logistic regression, η_* comes from a damped Newton method run to $\|F'(\eta_*)\| \le 10^{-15}$ or stagnation, so that its squared error is at most $\|F'(\eta_*)\|^2/\mu^2$: at most $2.1 \cdot 10^{-21}$ on the problems of Section 5.3 whose runs reach 10^{-18} (the runs at $\kappa = 10$, $\alpha = 1$ indeed stall at $1.3 \cdot 10^{-21}$), hence a relative error of at most about 10% on $\|\theta_t - \eta_*\|^2 = 10^{-18}$, and at most $7 \cdot 10^{-20}$ at $\kappa = 10^5$, $\alpha = 0$, whose runs stay above about 10^{-13}. Records below the precision of these computations are discarded: SAGA records with $V_t < 10^{-18}$, states with $\|\theta_t - \eta_*\|^2 < 10^{-18}$ in Section 5.5, and L-Katyusha records with $\Psi_t < 10^{-10}$ (the W-term of Ψ amplifies rounding errors by $\theta_2(1+\theta_1)/(p\theta_1) \ge 250$).

Runs, gradient counts and slopes. The steps are $1/(4R^2)$ for SAGA, $1/(6R^2)$ for L-SVRG (Theorem 5 of [47]) and $1/L_F$ for gradient descent (GD) and Nesterov's accelerated gradient (AGD, momentum $(\sqrt{L_F/\mu} - 1)/(\sqrt{L_F/\mu} + 1)$); L-Katyusha uses the parameters of Theorem 11 of [47]: $\theta_2 = \frac{1}{2}$, $\theta_1 = \min\{\sqrt{2n/(3\kappa)}, \frac{1}{2}\}$, $p = \frac{1}{n}$, $\eta = 1/(3\theta_1)$, with $\kappa = R^2/\mu$ computed, like the momentum of AGD, from the exact constants of the instance.

Section	problems	seeds	length of the runs
5.2	book setting, $d = 40$, $n \in \{10^3, 10^4\}$	1	50 passes
5.3	logistic, $\kappa \in \{10, 10^2, \dots, 10^5\}$, $\alpha \in \{0, 1, 2\}$	5	100 passes
5.4	logistic, $\kappa = 10^3$, $\alpha = 1$, $\gamma R^2 \in \{\frac{1}{8}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}, 1, 2, 4\}$	3	40 passes
5.5	logistic, $\kappa \in \{10^2, 10^3, 10^4\}$, $\alpha \in \{0, 1, 2\}$	1	60 passes; GD at most 400 steps
5.6	logistic, $\kappa = 10^3$, $\alpha \in \{1, 2\}$, four metrics P	5	100 passes
5.7	(H) with $m = 64$, $\kappa = 10^4$; logistic, $\kappa \in \{10^3, 10^4\}$, $\alpha = 1$	3	60 passes
5.8	(H) with $m = 64$, $\kappa = 10^4$; logistic, $\kappa = 10^4$, $\alpha = 1$	1	to relative error 10^{-10}
5.9	(H), (R): $\kappa \in \{10^2, 3 \cdot 10^2, \dots, 10^5\}$, 3 seeds; (S): $\kappa \in \{10^2, 3 \cdot 10^2, \dots, 10^6\}$ and (A): $\kappa \in \{10^2, 3 \cdot 10^2, \dots, 10^5\}$, 2 seeds	2–3	to relative error 10^{-8}

The SAGA and L-Katyusha runs of Sections 5.3–5.7 also stop when $\|\theta_t - \eta_*\|^2 \le 10^{-26}$ ($\|\theta_t - \eta_*\|_P^2 \le 10^{-26}$ in Section 5.6: the runs with the statistical and Hessian metrics, and with the Jacobi metric for $\alpha = 1$, stop after 44 to 72 passes, at $\|\theta_t - \eta_*\|^2$ between 10^{-24} and $3 \cdot 10^{-23}$, and the curves of Figure B1 end with the shortest run). The budget of Section 5.8 is 200 passes on logistic regression and $9.2 \cdot 10^4$ passes on (H). In Section 5.9 a run stops at the first recorded iterate with $\|\theta - \eta_*\|^2 \le 10^{-8} \|\theta_0 - \eta_*\|^2$, at an iteration cap (above the theoretical count of the method, or at the level of that of SAGA for the two schemes without a guarantee), at a CPU-time budget, or when it diverges; capped runs would not enter the fits, but in the final sweeps no run was capped. Iterates are recorded every $n/20$ iterations in Sections 5.3, 5.5 and 5.6, every $n/10$ in Sections 5.2, 5.4, 5.7 and 5.8 (every iteration when $n = 16$) and every $n/4$ in Section 5.9; GD and AGD are recorded at every iteration. A component gradient counts 1 and a full gradient n: SAGA pays n initially and 1 per iteration; L-SVRG, L-Katyusha and the naive schemes pay n initially, 2 per iteration ($f_i'(x)$ and $f_i'(w)$, the latter charged as if recomputed although the code stores the n gradients at w) and n per refresh of the reference point (probability $1/n$), i.e. 3 per iteration on average, against 2 with stored reference gradients, the convention of Proposition A1.26 (testable prediction of hypothesis A4) and of the other bounds of Appendix A1: their measured counts are thus about $\frac{3}{2}$ times those of that convention, which changes constants, not slopes; GD and AGD pay n per iteration. Counts to a target are interpolated log-linearly between records, except the counts to 10^{-10} of Section 5.8, which are read at the first record below the target, where

the runs stop. Speeds are contraction exponents per pass (n iterations), $-n\log\rho$ for a factor ρ per iteration (at most about $\frac{1}{3}$ for the bound of Proposition 5.9), and the *speed-up* is the ratio of the exact to the certified exponent; empirical exponents are least-squares slopes of $\log\|\theta_t-\eta_*\|^2$ against t over a window of passes, averaged over seeds. Complexity exponents are least-squares slopes b of $\log\bar{T}$ against $\log\kappa$ (or $\log\kappa_{\text{eff}}$), where $\bar{T}$ is the geometric mean over the seeds of the counts at each κ, fitted over $\kappa\geq 10n$ (K values of κ) unless stated otherwise, and reported as $b\pm h$, with $h=t_{0.975,K-2}\,\text{se}(b)$ the half-width of the 95% Student interval with $K-2$ degrees of freedom. The seeds are independent replicates for the stochastic methods, whereas GD and AGD are deterministic on (H), (S) and (A) (on (R) each seed draws its own instance). Fitting the per-κ means treats every method alike, and the interval then measures the scatter of these means about a single power law over the fitted range, essentially the departure of the curve from a power law: the seed-to-seed spread of the counts is small ($\max/\min-1$ at most 3% for SAGA, L-SVRG, and GD and AGD on (R), 6.4% for L-Katyusha on (H), (R) and (S) and 8.4% on (A), and 26–30% for the naive scheme, whose setting is tuned; error bars of Figure A4). These intervals are not sampling intervals, and no coverage is claimed.

5.2 Validation on the setting of the book

We first reproduce Figure 5.4 of [11]: $d=40$ standard Gaussian features, labels given by the sign of a linear function plus Gaussian noise, $n\in\{1000, 10\,000\}$, $\lambda=\mu=R_x^2/n$, the steps of the book's analysis, $1/R_x^2$ (GD) and $1/(4R_x^2)$ (SAGA), and for stochastic gradient descent (SGD) the decreasing step $1/(R_x^2+\mu t)$, our choice (the book's text gives $1/(\mu t)$ in its Proposition 5.8, and the code distributed with the book for this figure uses $1/(R_x^2\sqrt{t+1}+\mu(t+1))$), where $R_x^2=\max_i\|x_i\|^2$ is here about four times $\max_i L_i$ (72.8 against 18.3 for $n=1000$). SAGA reaches $F(\theta_t)-F(\eta_*)\leq 10^{-8}$ after 13.6 and 11.4 passes and the floor of double precision ($\approx 10^{-16}$) within 30 passes; after 50 passes SGD is at $4.6\cdot10^{-4}$ and $3.5\cdot10^{-4}$, GD at $9.4\cdot10^{-2}$ and 0.14 (Figure 0). In the book's figure, whose axis stops at 10^{-8}, SAGA reaches that level after about 25 and 11 passes, and SGD and GD end between $10^{-2.2}$ and $10^{-1.6}$ (values read from the figure); the text states that SAGA reaches machine precision within 50 passes. The linear convergence of SAGA, its separation from SGD and GD, and its pass count for $n=10\,000$ are reproduced; for $n=1000$ our SAGA is about 1.8 times as fast (13.6 passes instead of about 25); for both values of n our SGD ends lower and our GD higher, and for $n=1000$ SGD and GD do not cross within 50 passes as they do in the book. These differences are compatible with details that the book's text does not state and that its code fixes differently (covariance eigenvalues $1/k$, steps $1/(3R_x^2)$ for SAGA and $1/\lambda_{\max}(\frac{1}{n}\sum_i x_i x_i^\top)$ for GD, besides the SGD step above), which we did not try to match.

5.3 Where the proof of Proposition 5.9 loses (hypothesis B3)

Question and measurements. Which inequalities make $\rho_{\text{bound}}=1-\min\{\frac{1}{3n},\frac{3\mu}{16R^2}\}$ pessimistic, and how does this depend on the conditioning and the anisotropy of the data? We run SAGA with the textbook constants ($\gamma=1/(4R^2)$, $\Delta=3\gamma^2$) on the fifteen problems $\kappa\in\{10,\dots,10^5\}$, $\alpha\in\{0,1,2\}$, and record, with the instruments of Section 4.1, $\rho_{\text{exact}}(t)$, the tightness ratios $r_{\text{sq}}, r_{\text{coco}}, r_{\text{var}}, r_{\text{sc}}, r_{\min}$ of I1–I5 and r_{init} of I6, and the attribution chains of Proposition B3.11 (attribution of the gap), which undo the inequalities one at a time and split the logarithmic gap $\log(1/\rho_{\text{exact}})-\log(1/\rho_{\text{bound}})$ into nonnegative *shares*. Order B (undo I5, I4, I2, I3, I1, i.e. the intermediate rates of the proof applied cumulatively) is shown in the figures; order A (I4, I2, I3, I1, then I5) keeps the minimum until the end. Numbers are averages over passes 2–20 and over the 5 seeds unless stated otherwise.

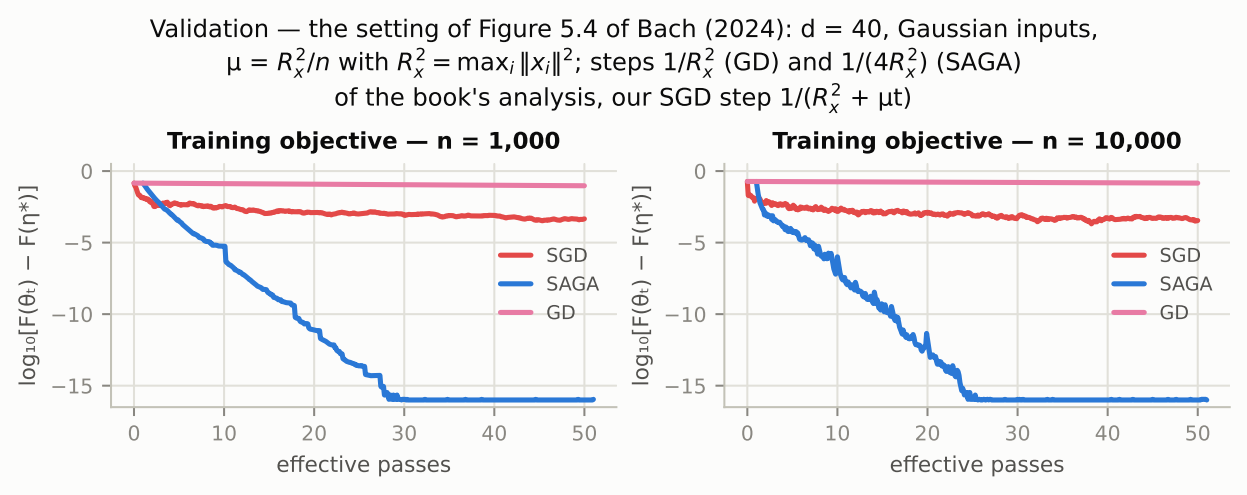


Figure 0: **Validation on the setting of Figure 5.4 of [11].** ℓ_2-regularized logistic regression with $d = 40$ standard Gaussian features, $y = \text{sign}(x^\top w + \text{noise})$, $\lambda = \mu = R_x^2/n$ with the book's constant $R_x^2 = \max_i \|x_i\|^2$, the steps of the book's analysis, GD $1/R_x^2$ and SAGA $1/(4R_x^2)$, and our SGD step $1/(R_x^2 + \mu t)$. Training objective $F(\theta_t) - F(\eta_*)$ against effective passes for $n = 1{,}000$ (left) and $n = 10{,}000$ (right), one run each. SAGA reaches 10^{-8} after 13.6 and 11.4 passes and the floor of double precision ($\approx 10^{-16}$) in fewer than 30 passes.

Size of the gap (Figure 1). The speed-up is 3.2–3.7 for $\kappa \le 10^2$, where the memory term $\frac{1}{3n}$ is the active one, and grows with κ in the parameter-term regime: 5.1, 5.2 and 3.4 at $\kappa = 10^3$, 29, 16 and 6.0 at $\kappa = 10^4$, 220, 100 and 19 at $\kappa = 10^5$, for $\alpha = 0$, 1 and 2 (passes 20–100: 187, 72 and 11 at $\kappa = 10^5$). At $\kappa = 10^4$ and $\alpha = 1$, the mean squared distance decreases from $\|\theta_0 - \eta_*\|^2 = 1954$ to $7.1 \cdot 10^{-9}$ in 100 passes, by 11.4 orders of magnitude, whereas the bound only decreases by the factor $\rho_{\text{bound}}^{10^5} = 0.15$, to $29\,\|\theta_0 - \eta_*\|^2$ with the constant $1 + \frac{3n}{16}$ of the proof ($38\,\|\theta_0 - \eta_*\|^2$ with the $1 + \frac{n}{4}$ of the statement). Along seed 0 ($\alpha = 1$), the product of the exact conditional contractions agrees with the realized V_t within 5% after 100 passes at $\kappa = 10^4$ and 10^5 (11.5 and 6.5 decades of decrease), and within a factor 5.2 where the curves reach the floor at $\kappa = 10^2$ and 10^3 (about 20 decades; within a factor 7.8 all along). At $\kappa = 10$ (not shown in Figure 1) the realized V_t, then carried by its memory term, stalls for several passes before dropping abruptly, and the two differ by a factor 11 after 20 passes and by 24 to $1.2 \cdot 10^3$ beyond 30 passes, i.e. by up to three of 19 decades (a factor of about 2.8 comes from holding the contraction of the fresh memory at $t = 0$ for $n/10$ iterations: the product is already $0.36\,V_t$ at $t = n/10$). The exact one-step identity thus accounts for the observed decay, and the gap lies essentially in the inequalities.

Tightness ratios (Figure 2). I3 and I5 are nearly tight on all fifteen problems: $r_{\text{var}} \ge 0.958$, because the mean stored error $\bar{z}$ is small compared with the individual ones ($r_{\text{var}} = 1 - n\,\|\bar{z}\|^2 / M$), and $r_{\min} \ge 0.9998$. The square-norm step has $r_{\text{sq}} = 0.50$–0.52 for $\kappa \le 10^2$, the factor 2 lost, by item (iii) of Lemma B3.9 (ranges of the tightness ratios), when the gradient changes a_i and the memory errors b_i (the values of a_t and b_t of I1 when $i(t) = i$) are uncorrelated, and 0.075, 0.015 and 0.0035 at $\kappa = 10^5$, where the memory is fresh relative to the slow motion of θ_t, as Remark B3.13 (the square-norm step discards the control variate) explains. Cocoercivity has $r_{\text{coco}} = L_{\text{eff}}/R^2$ in $[0.16, 0.37]$, $[0.11, 0.16]$ and $[0.019, 0.18]$ for $\alpha = 0, 1, 2$: $R^2 = \max_i L_i$ overestimates the constant met along the trajectories by a factor 2.7 to 52. Strong convexity has $r_{\text{sc}} = \mu/\mu_{\text{eff}}$ between 0.74

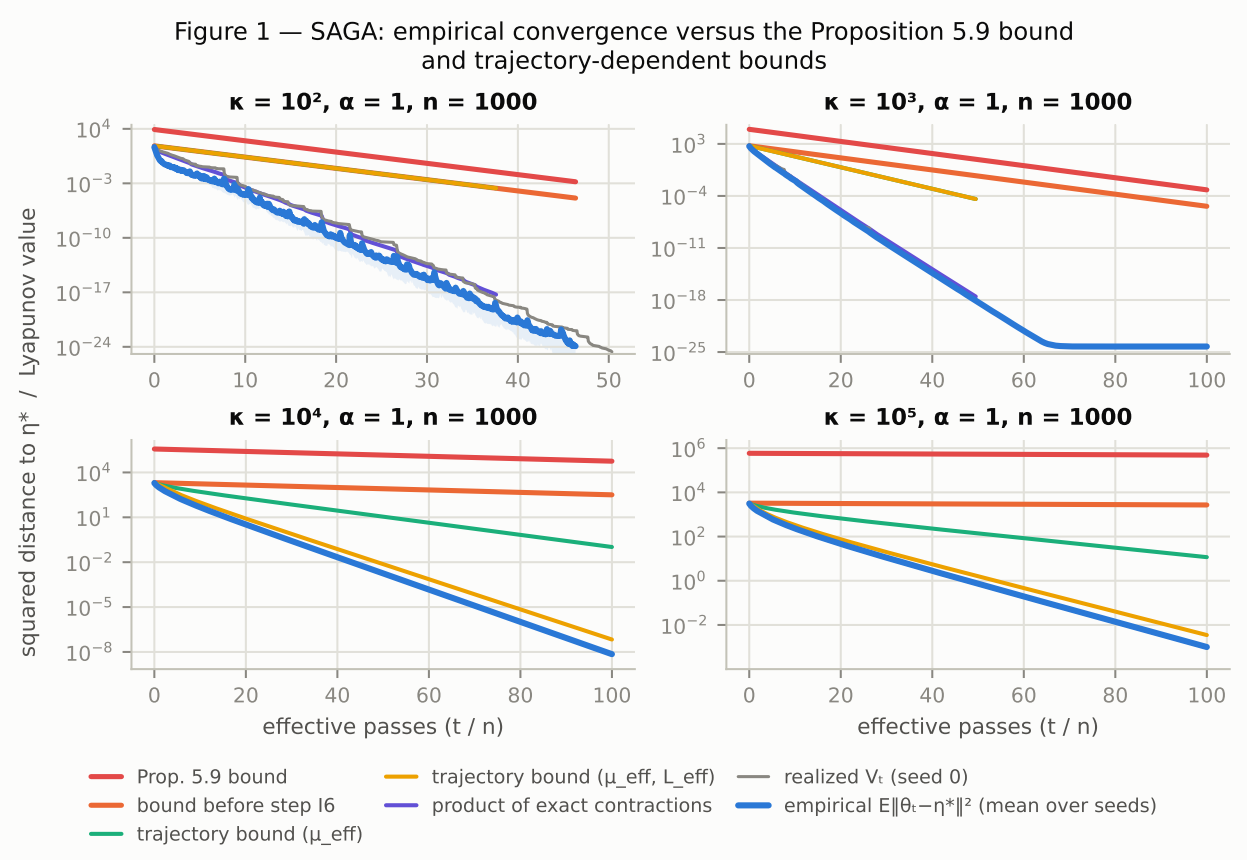


Figure 1: **Figure 1 of the proposal: SAGA against the bound of Proposition 5.9.** Logistic regression, $n = 1{,}000$, $d = 50$, $\alpha = 1$, $\kappa \in \{10^2, 10^3, 10^4, 10^5\}$ (one panel each), step $\gamma = 1/(4R^2)$. Blue: $\mathbb{E}\,\|\theta_t - \eta_*\|^2$ estimated by the mean over 5 seeds (band: min–max); red: the bound of Proposition 5.9 with the constant $1+3n/16$ obtained in its proof (the statement rounds it up to $1+n/4$); orange: the same bound before the initialization step I6, $\rho_{\text{bound}}^t V_0$ (V_t is the Lyapunov function of Section 2); green and yellow: products, along the realized trajectory of seed 0 and from its V_0, of the factors of Proposition B3.19 (a-posteriori rates with effective constants), with μ replaced by $\mu_{\text{eff}}(\theta_{t-1})$ (green) and, in addition, R^2 replaced by $L_{\text{eff}}(\theta_{t-1})$ in the coefficient of the parameter term (yellow); violet: product of the exact conditional contractions $\rho_{\text{exact}}(t)$ along the same trajectory; grey: realized Lyapunov value V_t of seed 0. The factors are computed every $n/10$ iterations and held constant in between. The three trajectory-dependent products (green, yellow, violet) are cut at the first instrumentation time with $V_t < 10^{-18}$, below which the conditional expectations are dominated by rounding; the other curves end with the runs, which stop at $\|\theta_t - \eta_*\|^2 \le 10^{-26}$ (the mean and the two bounds with the shortest of the five). Where the memory term $1/(3n)$ is the active one in the a-posteriori factors ($\kappa \le 10^3$ here, since $\mu_{\text{eff}}/\mu \ge 16\kappa/(9n)$ along these runs; in the certified factor only for $\kappa \le 10^2$, as $\frac{3}{16\kappa} < \frac{1}{3n}$ for $\kappa > \frac{9n}{16}$) the green and yellow curves coincide, and for $\kappa = 10^2$ they also coincide with the orange one; for $\kappa \ge 10^4$ the violet and grey curves are indistinguishable from the blue one and hidden under it. One legend, below the panels, serves all of them.

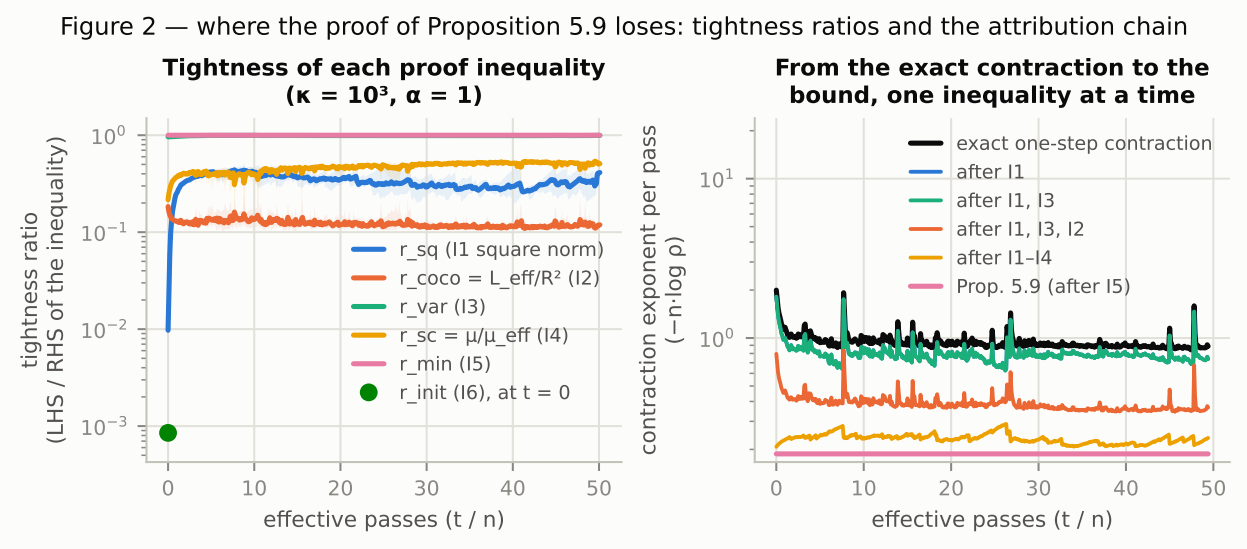


Figure 2: **Figure 2 of the proposal: tightness of each inequality of the proof.** Logistic regression, $n = 1{,}000$, $d = 50$, $\kappa = 10^3$, $\alpha = 1$, textbook constants. Left: tightness ratios of I1–I5 (Definition B3.8: smaller side over larger side, 1 meaning equality), computed from exact conditional expectations every $n/10$ iterations; geometric mean over 5 seeds (band: min–max), records with $V_t < 10^{-18}$ discarded. The ratio of the initialization inequality I6, $r_{\text{init}} = 8.5 \cdot 10^{-4}$ for this problem, is the single marker at $t = 0$, below the other curves. Right: attribution chain along seed 0 (order B of Proposition B3.11, records with $V_t \geq 10^{-18}$): contraction exponent per pass, $-n \log \rho$, of the exact one-step contraction and after applying the inequalities cumulatively in the order of the proof (I1, I3, I2, I4, I5), down to the certified rate of Proposition 5.9. The curve after I1 is hidden under the curve after I1 and I3, because I3 is nearly tight.

and 0.90 at $\kappa = 10$ and 0.012, 0.027 and 0.14 at $\kappa = 10^5$, i.e. $\mu_{\text{eff}}/\mu = 81$, 37 and 6.9 against $\lambda_{\min}(F''(\eta_*))/\mu = 70$, 26 and 2.1 (the iterates are still far from η_* during passes 2–20). Finally r_{init} is $3.1 \cdot 10^{-5}$ to 0.028, but I6 only enters the constant $1 + \frac{n}{4}$.

Attribution (Figure 3). In order B, for $\kappa \leq 10^2$ the gap is carried by I1 (0.39–0.44) and I2 (0.32–0.37), then I5 (0.07–0.22) and I4 (0.03–0.17). For $\kappa \geq 10^4$, cocoercivity dominates (0.46 for $\alpha = 0$, 0.54–0.58 for $\alpha = 1$, 0.62–0.71 for $\alpha = 2$), strong convexity follows (0.25–0.36), and together they carry 0.77–0.96 of the gap, while I1 falls to 0.03–0.16 and I5 to at most 0.073; at $\kappa = 10^3$ and $\alpha = 2$, I2 alone carries 0.83. Over the grid the largest share of I3 is 0.54% and that of I5 is 22% (a ratio as close to 1 as $r_{\min}$ still matters because the exponents are small: at $\kappa = 10$, I5 adds about $2 \cdot 10^{-4}$ per iteration to a certified exponent of $3.3 \cdot 10^{-4}$); the seed-to-seed spread of the shares is below 0.03 for $\kappa \geq 10^4$. Order A also gives I3 a small share (at most 0.006, except 0.063 at $\kappa = 10^3$, $\alpha = 2$) and gives 0.59–0.96 to I4 and I2 together for $\kappa \geq 10^4$, but for $\kappa \leq 10^2$ it gives 0.56–0.61 to the minimum I5, 0.38–0.44 to I1 and nothing to I4 and I2, which only lower the parameter coefficient, not the active one there: as Remark B3.12 (reading the two attribution orders) explains, the attribution is order dependent where the two components of V_t compete.

Interpretation. By Proposition B3.10 (exact contraction in terms of the ratios) the exact parameter coefficient is $1 - \frac{\mu_{\text{eff}}}{2R^2}\left(1 - r_{\text{coco}}\left(\frac{r_{\text{sq}}}{4} + \frac{3}{8}\right)\right)$: relative to the worst case, I4 multiplies the parameter exponent by $\mu_{\text{eff}}/\mu \in [1, \kappa]$ (item (i) of Lemma B3.9), and I1 and I2 together by

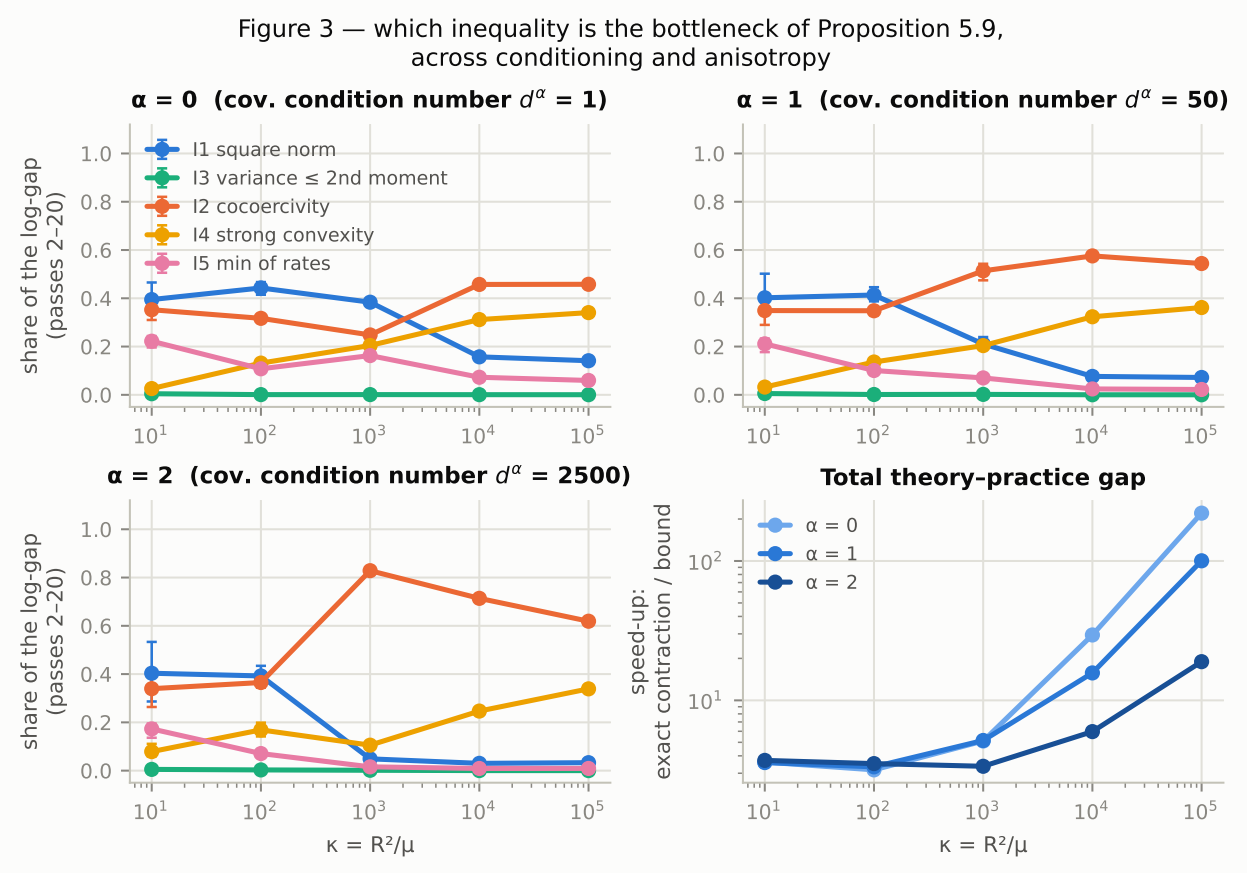


Figure 3: **Figure 3 of the proposal: which inequality is the bottleneck.** Share of the logarithmic gap between the exact one-step contraction and the certified one attributed to each inequality (order B of Proposition B3.11, i.e. cumulative attribution in proof order), averaged over passes 2–20 (records with $V_t \geq 10^{-18}$) and over 5 seeds (error bars: min–max over seeds), as a function of κ, for the covariance spectra $j^{-\alpha}$, $\alpha \in \{0, 1, 2\}$ (top left, top right and bottom left; population condition numbers 1, 50, 2500); logistic regression, $n = 1{,}000$, $d = 50$, textbook constants. Bottom right: total speed-up factor $(-\log \rho_{\text{exact}})/(-\log \rho_{\text{bound}})$ (window average, mean over seeds).

$\left(1 - r_{\text{coco}}(\frac{r_{\text{sq}}}{4} + \frac{3}{8})\right)/\frac{3}{8} \in [1, \frac{8}{3}]$ at the fixed step (Remark B3.20). For $\kappa \geq 10^4$, where the parameter term is active and D_t carries most of V_t ($r_{\min} \approx 1$), inserting the window geometric means of the ratios into this product reproduces 0.91–0.96 of the measured speed-ups: μ_{eff}/μ (between 2.2 and 81) times a factor 2.45–2.62 from I1 and I2. The κ-dependent part of the gap is thus the strong-convexity step, the difference between $\mu = \lambda$ and the curvature met along the trajectory; cocoercivity contributes a bounded factor, whose larger additive share in order B comes from acting on an exponent already multiplied by μ_{eff}/μ. With strongly anisotropic features ($\alpha = 2$) the bottom of the spectrum of $F''(\eta_*)$ stays close to λ, so I4 is closer to tight and I2 dominates the shares. The large factors at $\kappa = 10^5$ are a property of these instances, not of the order of the bound: by Proposition B3.17 (lower bound for SAGA with any constant step) no analysis of SAGA with a constant step can improve the exponent 1 of κ uniformly over the class of Proposition 5.9 (caveat (ii) of Remark B3.18).

5.4 The constants of the proof

Is the gap a matter of the convenient constants $\gamma = 1/(4R^2)$ and $\Delta = 3\gamma^2$? Proposition B3.16 (optimal constants of the proof structure) gives the best pair (γ, Δ) in closed form; by Remark B3.18 (optimizing (γ, Δ) gains at most a factor 3), it improves the certified rate by a factor between 1.05 and 3, and by at most $\frac{4}{3}$ when $\kappa \geq 9n/16$. At $\kappa = 10^3$ the optimum is $\gamma R^2 = 0.219$, $\Delta = 2.56\gamma^2$, with certified rate $2.19 \cdot 10^{-4}$ per iteration against $1.875 \cdot 10^{-4}$, a gain of 17%. On logistic regression ($\kappa = 10^3$, $\alpha = 1$, 3 seeds, slopes over passes 2–40, Figure C) the observed exponent per pass is 0.51, 0.97, 1.14, 1.09, 0.95, 0.66 and 0.027 for $\gamma R^2 = \frac{1}{8}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}$, 1, 2 and 4, whereas the best certificate is 0.18, 0.22 and 0.17 for the first three steps and does not exist for $\gamma R^2 \geq \frac{1}{2}$, where, by Corollary B3.15 (parametrization $\Delta = c\gamma^2$), no Δ is admissible. At the textbook step the observed exponent is 4.5 times the best certificate, and SAGA, fastest around $\gamma R^2 = \frac{1}{3}$, still converges for all three seeds at $\gamma R^2 = 4$, eight times the supremum $\frac{1}{2}$ of the steps γR^2 that the proof admits. The constants of Proposition 5.9 are thus within 17% of the best ones of their proof at this conditioning, and the factor 4.5 left at the textbook step is the slack of the inequalities (Remark B3.18); the admissible range $\gamma R^2 < \frac{1}{2}$ itself comes from absorbing the second moment $\gamma^2 \Omega_{t-1}$ of the step and the memory refresh with the worst-case bounds I1–I3.

These conclusions concern $\kappa = 10^3 \geq \frac{9n}{16}$, where the textbook rate is the parameter term and the constants can gain at most $\frac{4}{3}$. For $\kappa \leq 10^2$, where it is the memory term $\frac{1}{3n}$, the constants account for a larger part of the speed-ups 3.2–3.7 of Section 5.3: at the textbook step, the best weight, $\Delta = c\gamma^2$ with $c \approx 5.04$ ($\kappa = 10^2$) or $c \approx 5.89$ ($\kappa = 10$) instead of $c = 3$, multiplies the certified rate by 1.81 or 1.98, and the optimal constants of Proposition B3.16, at a smaller step, by 2.42 or 2.94, leaving factors 1.75–1.95 and 1.22–1.45, respectively. The exact exponents $-\log \rho_{\text{exact}}$ there (passes 2–20, mean over the seeds) are 1.06–1.24 times $-\log(1 - \frac{1}{n})$, the largest exponent $-\log \rho$ of a bound $\mathbb{E}[V_t] \leq c_0 \rho^t V_0$ with c_0 independent of t, whatever Δ (Remark B3.18): a typical trajectory of V_t decreases faster than its expectation can.

5.5 Effective curvature along the trajectories (hypothesis B2)

Are the constants $\mu_{\text{eff}}(\theta_t)$ and $L_{\text{eff}}(\theta_t)$ of Definition B2.2 (effective constants μ_{eff} and L_{eff}), which make I4 and I2 identities at the current iterate, much better than μ and R^2, and do they explain the gap (Section 4.2)? Along one SAGA run and one GD run per problem ($\kappa \in \{10^2, 10^3, 10^4\}$, $\alpha \in \{0, 1, 2\}$) we record μ_{eff}/μ, L_{eff}/R^2, the share $\pi_{\min}(t)$ of $\|\theta_t - \eta_*\|^2$ carried by the eigenvector $q_{\min}$ of $F''(\eta_*)$ with the smallest eigenvalue, and the effective number of directions $d_{\text{eff}}(t) = 1/\sum_k \pi_k(t)^2$ of the error in the eigenbasis of $F''(\eta_*)$, not to be confused with the ridge effective dimension $d_{\text{eff}}(\lambda)$ of Section 4.3 (Figure B2).

The bound $\mu_{\text{eff}} \geq \mu$ of Lemma B2.4 (μ_{eff} lies between μ and L) holds at every retained state (ten discarded GD states below the floor, at $\alpha = 2$, $\kappa = 10^2$, give μ_{eff}/μ between 0.795 and 0.993, a rounding artefact), and along SAGA μ_{eff}/μ starts between 1.8 and 50. The SAGA runs end after 60 passes (last record at pass 59.9), or after 34 to 49 passes for the five runs that reach the floor ($\kappa \leq 10^3$, except $\alpha = 2$, $\kappa = 10^3$). For $\kappa = 10^4$ and $\alpha \in \{0, 1\}$ the error of SAGA concentrates on $q_{\min}$ ($\pi_{\min} = 0.998$ and 0.9996 at the end, $d_{\text{eff}} = 1.005$ and 1.001; from $\theta_0 = 0$, $\pi_{\min}$ starts at 0.984 and 0.888, and for $\alpha = 0$ it first drops to 0.938 at pass 3.1), and μ_{eff}/μ ends within 1% of $\lambda_{\min}(F''(\eta_*))/\mu$ (9.94 against 9.86, 4.97 against 4.97); GD aligns with $q_{\min}$ as well ($\pi_{\min} = 1.000$ and 0.996 after 400 iterations), its μ_{eff} ending at $\lambda_{\min}(F''(\eta_*))$ and 1.8% above it. For $\kappa = 10^2$ the error of SAGA does not concentrate ($\pi_{\min} \leq 0.05$ over the last five passes, d_{eff} about 18 and 14 on average for $\alpha = 0, 1$) and μ_{eff}/μ fluctuates above $\lambda_{\min}(F''(\eta_*))/\mu$ (for $\alpha = 1$: geometric mean 1.90 over the run, between 1.38 and 3.63 over the last five passes, against 1.10, while GD ends at 1.15);

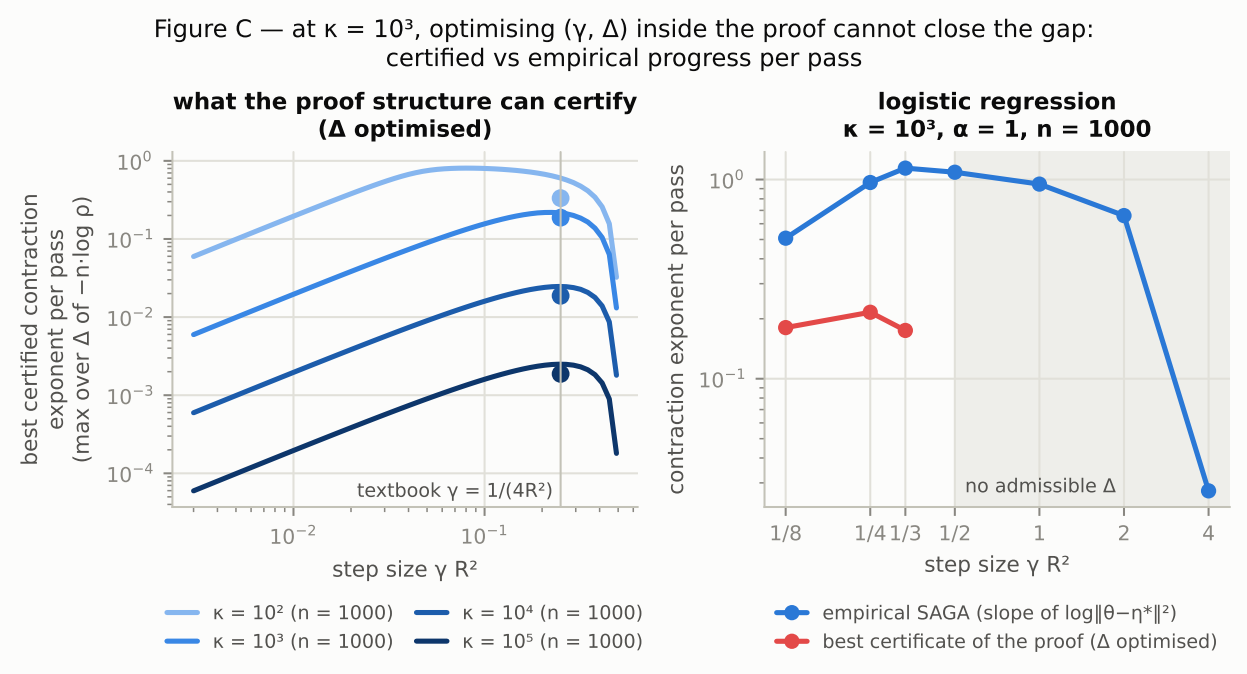


Figure C: **The constants inside the proof.** Left: best contraction exponent per pass that the proof structure of Proposition 5.9 can certify, as a function of the step γR^2, with the Lyapunov weight $\Delta = c\gamma^2$ optimized over c (Corollary B3.15), for $\kappa \in \{10^2, 10^3, 10^4, 10^5\}$ and $n = 1{,}000$; dots: textbook constants $\gamma = 1/(4R^2)$, $\Delta = 3\gamma^2$. Right: logistic regression ($\kappa = 10^3$, $\alpha = 1$, $n = 1{,}000$, $d = 50$), empirical contraction exponent per pass of SAGA (least-squares slope of $\log \|\theta_t - \eta_*\|^2$ over passes 2–40, mean over 3 seeds) and best certificate of the proof at the same step; the proof certifies nothing for $\gamma R^2 \geq \frac{1}{2}$, where no Δ is admissible (shaded region).

at $\kappa = 10^3$ the alignment is partial ($\alpha = 1$: $\pi_{\min} = 0.31$) or absent ($\alpha = 0$: d_{eff} about 17 over the last five passes, while GD is aligned). For $\alpha = 2$ the bottom of the spectrum of $F''(\eta_*)$ is nearly degenerate and neither method aligns with $q_{\min}$ (for $\kappa \geq 10^3$, GD has moreover not converged after 400 iterations: $\|\theta_t - \eta_*\|^2 = 5.7 \cdot 10^{-4}$ and 202). Along GD, μ_{eff} is non-increasing at every step up to rounding, as it is exactly on a quadratic by Proposition B2.17 (alignment of gradient descent with the flattest direction): the exceptions, 13 of 118 and 56 of 400 steps for $\alpha = 0$ and $\kappa = 10^3$, 10^4, are relative increases of at most $7.2 \cdot 10^{-7}$ and $2.7 \cdot 10^{-6}$, at states with $\|\theta_t - \eta_*\|^2 \leq 8.3 \cdot 10^{-14}$ and $8.1 \cdot 10^{-13}$. At the end of every SAGA run the quadratic proxy $\sum_k \lambda_k \pi_k(t)$, with λ_k the eigenvalues of $F''(\eta_*)$, matches μ_{eff} within 1%: near η_*, μ_{eff} is set by the direction of the error, as Proposition B2.8 (directional limits at the optimum) states. The geometric means of L_{eff}/R^2 along SAGA are 0.19–0.38, 0.11–0.16 and 0.012–0.09 for $\alpha = 0, 1, 2$.

The effective curvature thus exceeds μ by at least the factor $\lambda_{\min}(F''(\eta_*))/\mu$ near η_* (up to 9.9 in these runs, and up to 70 over the problems of Section 5.3) and by more far from it, and L_{eff} is 2.6 to 86 times smaller than R^2, as hypothesis B2 states. Prediction (P2) of Section B2.4, that the error of SAGA stays spread over the directions fed by its noise, holds in the four runs with $9n\lambda_{\min}(F''(\eta_*))/(16R^2) > 1$ ($\kappa = 10^2$, and $\kappa = 10^3$ with $\alpha = 0$) and fails at $\kappa = 10^4 = 10n$ for $\alpha \in \{0, 1\}$, where SAGA aligns with $q_{\min}$ like GD; the three other runs are intermediate (partial alignment at $\kappa = 10^3$, $\alpha = 1$; for $\alpha = 2$ and $\kappa \geq 10^3$, a nearly degenerate bottom spectrum, and neither method aligns with $q_{\min}$). In terms of κ alone, (P2) thus holds only for κ at most of the

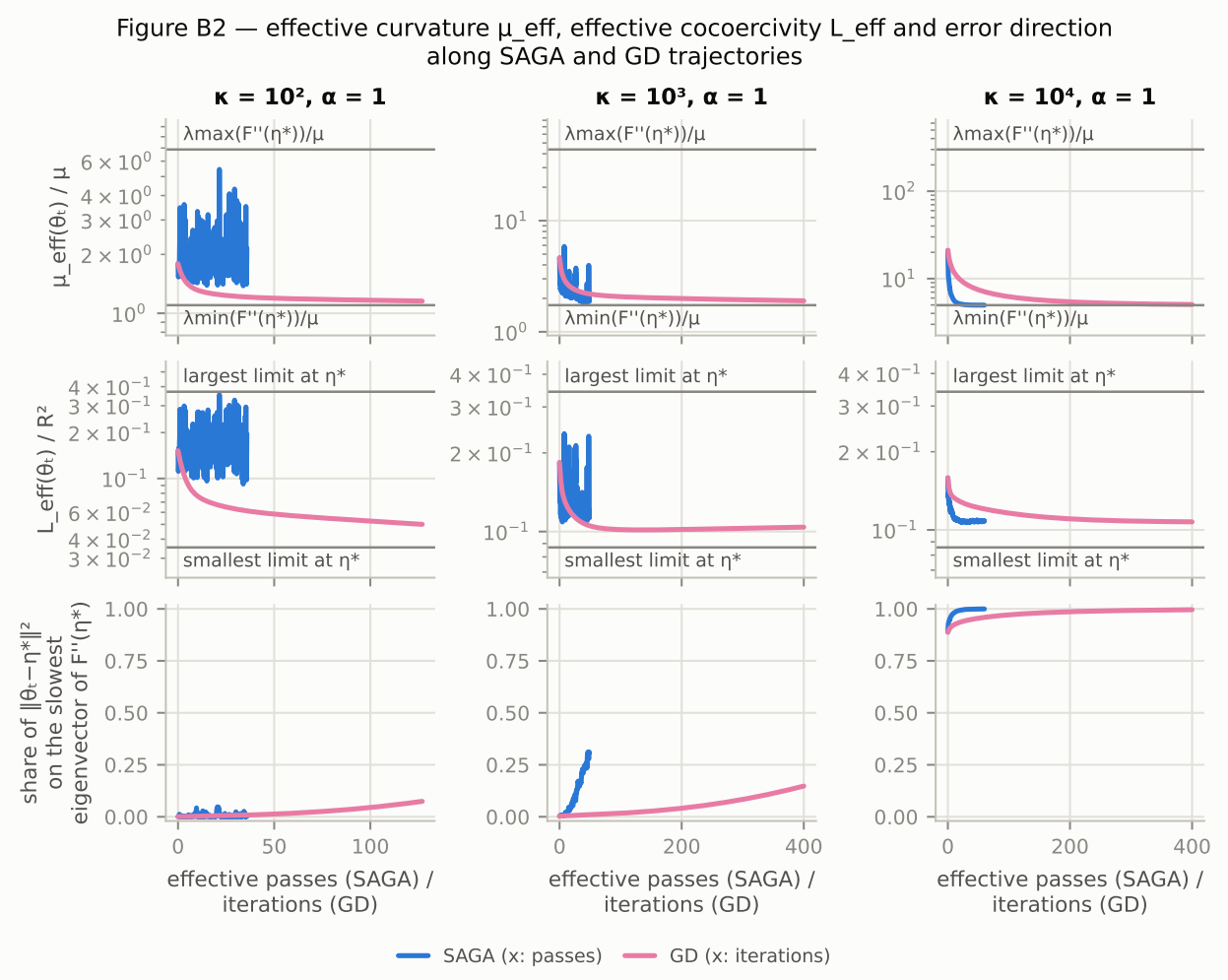


Figure B2: **Hypothesis B2: effective constants along the trajectories.** $\mu_{\text{eff}}(\theta_t)/\mu$ (top), $L_{\text{eff}}(\theta_t)/R^2$ (middle) and share $\pi_{\min}$ of $\|\theta_t - \eta_*\|^2$ carried by the eigenvector of $F''(\eta_*)$ with the smallest eigenvalue (bottom), along SAGA (blue, x-axis in passes, $\gamma = 1/(4R^2)$, at most 60 passes) and gradient descent (magenta, x-axis in iterations, step $1/L_F$, at most 400 iterations), for $\kappa \in \{10^2, 10^3, 10^4\}$ (one column each), $\alpha = 1$, $n = 1{,}000$, $d = 50$, one seed; states with $\|\theta_t - \eta_*\|^2 < 10^{-18}$ are discarded, so that the runs reaching this floor end earlier (SAGA after 36 and 49 passes for $\kappa = 10^2$ and 10^3, GD after 127 iterations for $\kappa = 10^2$). Horizontal lines: $\lambda_{\min}(F''(\eta_*))/\mu$ and $\lambda_{\max}(F''(\eta_*))/\mu$ (top), smallest and largest limits of L_{eff}/R^2 at η_* over directions (middle). The configurations $\alpha \in \{0, 2\}$ are summarized in Section 5.5.

order of n, and not in every such run: $\kappa \approx n$ does not suffice. Remark B2.22 (outcome of prediction (P2)) explains the alignment heuristically: when $n\gamma\lambda_{\min}(F''(\eta_*))$ is small, the iterate moves little during one refresh cycle of the memory, so that the noise decays with the slowest mode of the mean dynamics instead of at the certified rate (the noise spectrum was not recorded, so this mechanism is inferred, not measured). For the bound, the one-step factors of Proposition B3.19 (a-posteriori rates with effective constants) are exactly the first two steps of order A of Proposition B3.11 (attribution of the gap): replacing μ by $\mu_{\text{eff}}(\theta_{t-1})$ recovers 0.25–0.37 of the logarithmic gap for $\kappa \geq 10^4$ (green curves in Figure 1), and replacing in addition R^2 by $L_{\text{eff}}(\theta_{t-1})$ in the parameter coefficient recovers 0.59–0.96 (yellow), whereas for $\kappa \leq 10^2$ they recover nothing: the bound is then saturated by the memory term, as explained in Remark B2.24 (what each constant can explain). These are a-posteriori statements; the a-priori version, Proposition B3.21 (a-priori rate on a region), was not evaluated along the runs (Section 4.1 only computes the rate it gives on a ball around η_*).

5.6 SAGA in a fixed quadratic metric (hypothesis B1)

Does SAGA in the metric of a fixed positive definite P, $\theta_t = \theta_{t-1} - \gamma P^{-1}\omega_t$, converge faster, and which steps of the proof change? By Proposition B1.5 (P-SAGA as SAGA on a transformed problem), the proof transfers with the relative constants (R_P^2, μ_P), which gives Corollary B1.12 (Proposition 5.9 transferred to P-SAGA); for logistic regression, no P improves the global condition number: $\kappa_P \geq \kappa$ by Proposition B1.21 (no free lunch for global constants), restated in Section 4.3. We compare $P = I$, the Jacobi metric $P_{\text{J}} = \operatorname{diag}(\hat{\Sigma}) + \lambda I$, the statistical metric $\hat{\Sigma} + \lambda I$ and the Hessian $F''(\eta_*)$, with $\gamma = 1/(4R_P^2)$, at $\kappa = 10^3$ (Figure B1); here and in the constants below, R_P^2 is the upper bound $\frac{1}{4}\max_i \|x_i\|_{P^{-1}}^2 + \lambda/\lambda_{\min}(P)$ of (B1.9), exact for $P = I$ and at most 5.2% above the supremum of Definition B1.7 (relative constants and relative condition number) otherwise. In this order, the exponents per pass of $\|\theta_t - \eta_*\|_P^2$ (passes 2–20) are 1.00, 0.94, 1.11, 1.11 for $\alpha = 1$ and 0.61, 0.27, 1.26, 1.24 for $\alpha = 2$. The global constants favour $P = I$ instead: $\kappa_P = 1000$, 1899, 5083, 3623 ($\alpha = 1$) and 1000, 2852, 3639, 2089 ($\alpha = 2$) (with the exact supremum, 1000, 1898, 5057, 3611 and 1000, 2841, 3459, 2051), and the certified exponents for $\alpha = 1$ are 0.19, 0.099, 0.037, 0.052. With the local constant $\lambda_{\min}(F''(\eta_*), P)$ (smallest generalized eigenvalue) in place of μ_P, a heuristic substitution that Lemma B1.27 (limits of the effective constants at η_*) supports only as $\theta \to \eta_*$, the relative condition numbers $R_P^2/\lambda_{\min}(F''(\eta_*), P)$ become 576, 641, 166, 83 ($\alpha = 1$) and 981, 2433, 103, 41 ($\alpha = 2$) (the condition numbers $\lambda_{\max}/\lambda_{\min}$ of $F''(\eta_*)$ relative to P are 25, 26, 2.8, 1 and 50, 116, 8.4, 1), and Proposition 5.9 so evaluated gives 0.33, 0.29, 0.33, 0.33 and 0.19, 0.077, 0.33, 0.33 per pass: the observed ordering, up to the near tie of the dense metrics, which saturate the memory term ($\frac{1}{3}$ per pass) while their observed exponents are 3.3–3.8 times larger. In the P-geometry the slack moves from cocoercivity to strong convexity: for $\alpha = 1$, r_{coco} rises from 0.13 to 0.33–0.34 and r_{sc}, computed with the global μ_P, falls from 0.42 to 0.017–0.023, as prediction (P2) of Section B1.6 anticipates.

The gain of a fixed metric is therefore real but entirely trajectory-dependent: invisible in the global constants, which cannot improve for logistic regression, it appears in the local ones. Once the curvature is handled by P, the bound is saturated by its memory term, the regime of Remark B1.24 (memory-bottleneck regime), 3.3 to 3.8 times below the observed exponents. Along the measured trajectories (passes 2–20, in the P-geometry), this factor splits in three when the inequalities are undone in order A of Proposition B3.11 (attribution of the gap), that is, I4, I2, I3, I1, then the min step I5. The memory coefficient $1 - \frac{1}{3n}$ comes from the textbook weight $\Delta = 3\gamma^2$, which absorbs the memory term created by I1 and I3; evaluated exactly by Proposition B3.10 (exact contraction in terms of the ratios), with the measured $r_{\text{sq}} \approx 0.51$–$0.53$ and $r_{\text{var}} \approx 1$, it is $1 - \frac{1}{n}(1 - \frac{2}{3}r_{\text{sq}}r_{\text{var}}) \approx 1 - \frac{0.65}{n}$: a factor 1.95–1.99. The min step I5 charges at the memory rate the parameter part of

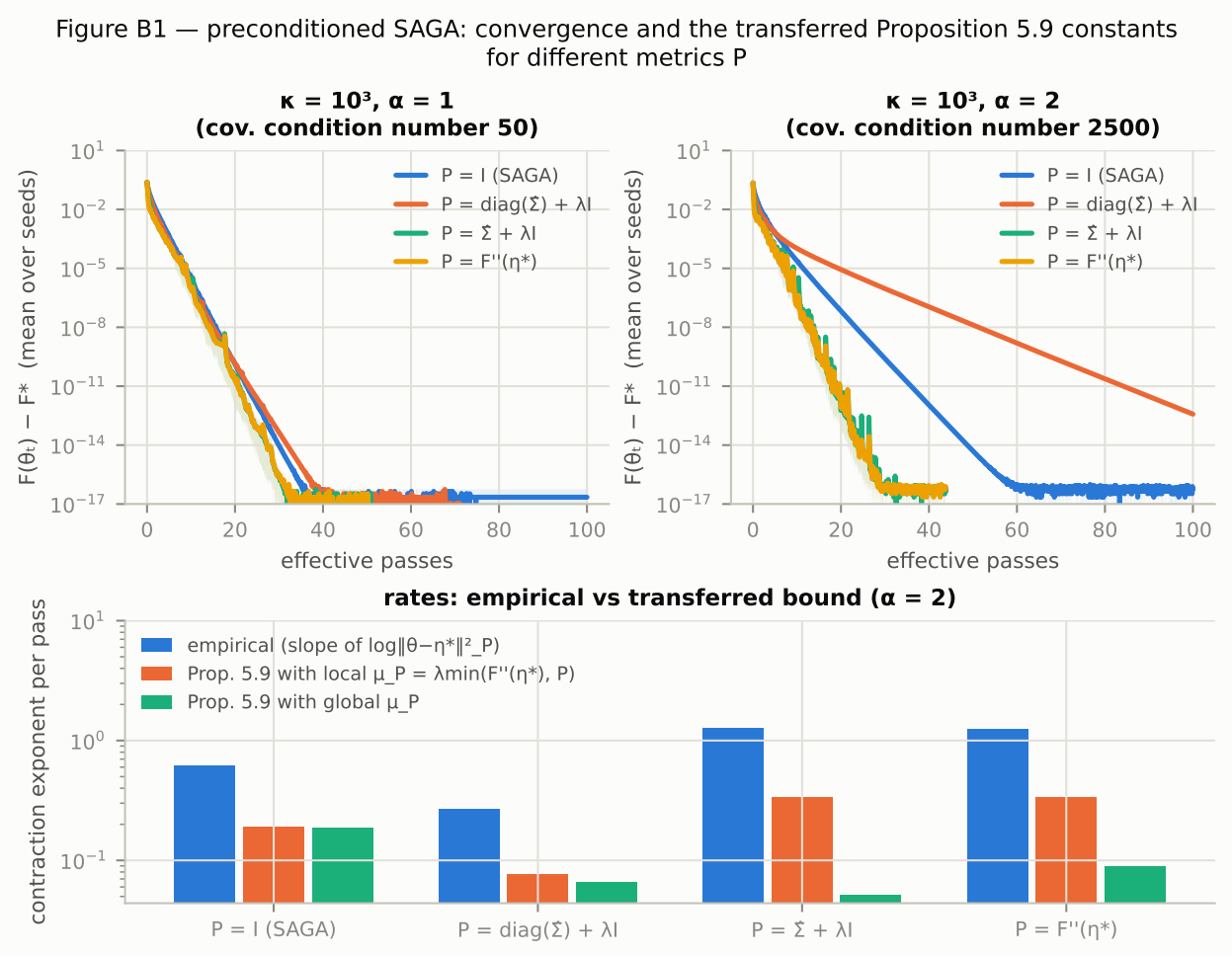


Figure B1: **Hypothesis B1: SAGA in a fixed quadratic metric.** $\theta_t = \theta_{t-1} - \gamma P^{-1}\omega_t$ with $\gamma = 1/(4R_P^2)$, where $R_P^2 = \frac{1}{4}\max_i \|x_i\|_{P^{-1}}^2 + \lambda/\lambda_{\min}(P)$ is the upper bound (B1.9) on the transferred smoothness constant, for $P \in \{I, \mathrm{diag}(\hat{\Sigma})+\lambda I, \hat{\Sigma}+\lambda I, F''(\eta_*)\}$, $\hat{\Sigma} = \frac{1}{n}\sum_i x_i x_i^\top$, on logistic regression ($n = 1{,}000$, $d = 50$, $\kappa = 10^3$). Top: $F(\theta_t) - F(\eta_*)$ for $\alpha = 1$ (left) and $\alpha = 2$ (right), mean over 5 seeds (band: min–max), up to the end of the shortest run (the runs stop at $\|\theta_t - \eta_*\|_P^2 \le 10^{-26}$). Bottom ($\alpha = 2$): contraction exponent per pass of $\|\theta_t - \eta_*\|_P^2$ (least-squares slope over passes 2–20, mean over seeds) against Proposition 5.9 evaluated, heuristically, with the local constant $\lambda_{\min}(F''(\eta_*), P)$ (smallest generalized eigenvalue) and with the global constant μ_P.

V_t^P, which contracts 7 to 21 times faster: a factor 1.55–1.70. The remaining factor, 1.10–1.12, separates the measured slopes from the exact one-step contraction. None of this concerns $\mathbb{E}V_t^P$: by the memory recursion (B3.5), the memory part of $\mathbb{E}V_t^P$ cannot decrease faster than $(1-\frac{1}{n})^t$, so that no bound on $\mathbb{E}V_t^P$ certifies more than about 1 per pass, 3 times the local bound, and a factor 1.10–1.26 of the observed exponents is invisible in the expectation that the analysis controls. The Jacobi metric does not help, and is the slowest for both values of α: the features are randomly rotated, so the diagonal of $\hat{\Sigma}$ carries little of its anisotropy, and Proposition B1.19 (constants of the Jacobi preconditioner) only places P_{J} within a factor $\mathrm{cond}(C_{\mathrm{corr}})$ of $\hat{\Sigma}+\lambda I$, in either direction: nothing in the theory orders P_{J} with respect to $P=I$. The dense metrics cost $O(nd^2)$ per pass, and $P=F''(\eta_*)$ requires η_* (an oracle benchmark).

5.7 The Lyapunov proof of L-Katyusha, measured (hypothesis A3)

Appendix A3 reproduces the proof of [47] in the style of the book, which gives Theorem A3.4 (contraction and rate of L-Katyusha; Lemma 10 and Theorem 11 of [47]), restated in Section 4.5. We measure its one-step contraction (A3.5) (Lemma 10 of [47]), $\mathbb{E}_{i(t)}[\Psi_{t+1}] \le c_Z Z_t + c_Y Y_t + c_W W_t$ with c_Z, c_Y, c_W the three coefficients of Theorem A3.4, exactly along L-Katyusha runs on (H) with $m=64$ and $\kappa=10^4$ ($\theta_1=0.033$) and on logistic regression with $\alpha=1$ and $\kappa=10^3$ ($\theta_1=\frac{1}{2}$) or 10^4 ($\theta_1=0.26$) (Figure A3). All statistics of this section pool the instrumented states of the three seeds from $t=0$ until Ψ_t first falls below 10^{-10} or the run ends, i.e. over the first 16 to 60 passes. The contraction is never violated: the largest value of $(\mathbb{E}_{i(t)}[\Psi_{t+1}] - c_Z Z_t - c_Y Y_t - c_W W_t)/\Psi_t$ over all recorded states is $-6.5\cdot10^{-4}$, $-2.3\cdot10^{-4}$ and $-1.7\cdot10^{-4}$, and that of its consequence $\rho_{\mathrm{exact}}(t) - \max\{c_Z, c_Y, c_W\}$, with $\rho_{\mathrm{exact}}(t) = \mathbb{E}_{i(t)}[\Psi_{t+1}]/\Psi_t$, is $-1.7\cdot10^{-3}$, $-4.0\cdot10^{-4}$ and $-2.8\cdot10^{-4}$. The exact one-step contraction exponent exceeds the certified one, that of $\max\{c_Z, c_Y, c_W\}$, by a factor 4.7, 2.8 and 4.1 (5.9, 2.8 and 4.1 over passes 2–20, the window used for SAGA), against 5.2 and 15.7 for SAGA and Proposition 5.9 on the same logistic problems. The closed-form rate ρ of (A3.6) in Theorem A3.4 gives away a further factor $\frac{4}{3}$ at $\kappa=10^3$ and 2 on the two problems with $\kappa \ge \frac{8n}{3}$ (through $\frac{1}{1+x} \le 1-\frac{x}{2}$): against it the factors are 9.3, 3.7 and 8.2. The variance bound (A3.9) of Lemma A3.6 (L1: unbiasedness and variance of the estimator; Lemma 6 of [47]), $\mathbb{E}_{i(t)}\|g_t - F'(x_t)\|^2 \le 2L\,D_F(w_t, x_t)$, has tightness 0.125–0.15 (geometric means; maxima 0.30–0.59): its centering step (A3.7) is nearly tight (0.94–0.99) and the slack is in its cocoercivity step (A3.8) (0.13–0.15), which uses the worst-case curvature $L=\max_i L_i$ for gradient differences along $w_t - x_t$, where the curvature met is far smaller (on (H) all L_i are equal, so this is not a max-versus-average effect), as for I2. The signed share $c(t) = \langle F'(x_t), x_t - w_t\rangle / D_F(w_t, x_t)$ of the Bregman gap that the anchor removes in the identity (A3.19) of Proposition A3.14 (Bregman-gap cancellation by the negative momentum) is negative on 77%, 77% and 64% of the steps (medians -1.5, -0.61, -0.62) and above 1 on 12%, 8% and 20%: the cancelled inner product is rarely a fraction of the gap, which illustrates why the proof cancels it identically rather than bounding it. Its size relative to the certified one-step decrease, $q(t)$, has medians 1.4, 0.64 and 1.4, and $q(t) \ge 1$ on 63%, 5% and 65% of the steps.

The proof thus holds along the trajectories. Its looseness (2.8 and 4.1 at $\kappa=10^3$ and 10^4 on logistic regression, 4.7 on (H)) is smaller than that of Proposition 5.9 on the same logistic problems (5.2 and 15.7) and grows less between these two values of κ ($\times1.5$ against $\times3.1$); two values of κ, in different regimes of the certificate (c_W binding at $\kappa=10^3$, where $\theta_1=\frac{1}{2}$, and c_Z at 10^4), do not determine how it scales. Both certificates have the optimal order in the worst case, by the lower bounds of Section 4.4 for L-Katyusha and by Proposition B3.17 (lower bound for SAGA with any constant step) for SAGA, so the difference concerns these instances, not the order of the bounds. On the two ill-conditioned problems the cancelled term exceeds the whole certified decrease on about

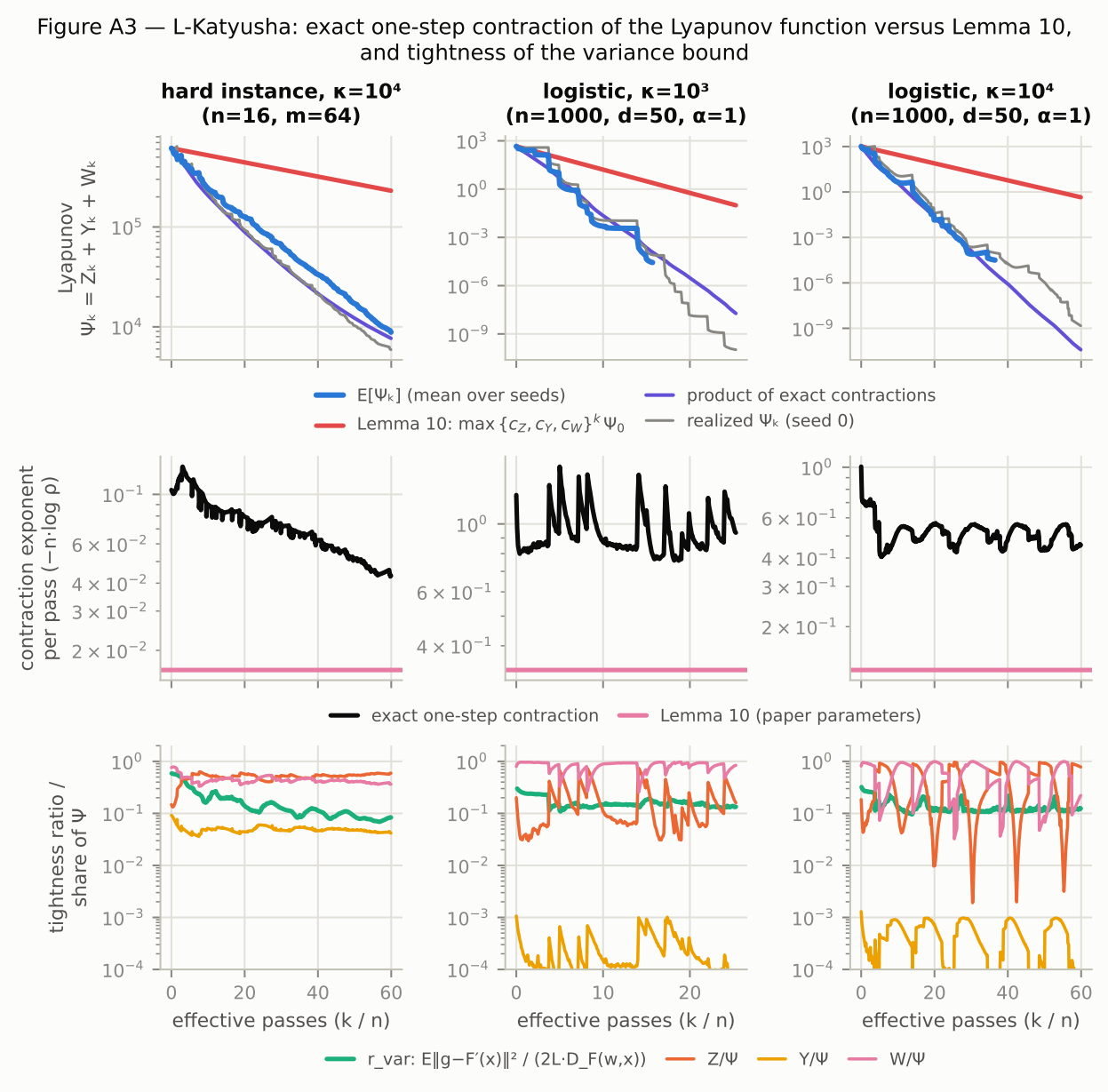


Figure A3: **Hypothesis A3: the Lyapunov proof of L-Katyusha, measured.** L-Katyusha with the parameters of Theorem 11 of [47] ($\theta_2 = \frac{1}{2}$, $\theta_1 = \min\{\sqrt{2n/(3\kappa)}, \frac{1}{2}\}$, $p = \frac{1}{n}$) on (H) ($n = 16$, $m = 64$, $\kappa = 10^4$) and on logistic regression ($n = 1{,}000$, $d = 50$, $\alpha = 1$, $\kappa \in \{10^3, 10^4\}$), 3 seeds, records with $\Psi_t < 10^{-10}$ discarded (the iteration counter is written k in the figure). Top: the Lyapunov function $\Psi_t = Z_t + Y_t + W_t$ (mean over seeds), the contraction $\max\{c_Z, c_Y, c_W\}^t \Psi_0$ certified by Lemma 10 of [47] (inequality (A3.5)), the product of the exact conditional contractions along seed 0 and the realized Ψ_t of seed 0. Middle: exact one-step contraction exponent per pass along seed 0 against the certified one. Bottom (seed 0): tightness of the variance bound $\mathbb{E}\,\|g - F'(x)\|^2 \le 2L\, D_F(w, x)$ and shares of Z, Y and W in Ψ.

two thirds of the steps, as the part of prediction (P4) of Section A3.4 on L-Katyusha anticipates. On most of these steps, however, it has the favourable sign ($c(t) < 0$ on 84% and 70% of them, against none of the 5% of such steps at $\kappa = 10^3$): if $c(t) \le 0$, then $D_F(w_t, x_t) \le F(w_t) - F(x_t)$, a difference of function values, which a reference potential can pay. The sign-aware ratio $q_+(t)$, which counts the cancelled term only where $c(t) > 0$, is at least 1 on 9.9%, 5.2% and 19.3% of the recorded states, that is, on 43%, 23% and 54% of the states with $c(t) > 0$: only there could the cancelled term not have been absorbed into the certified decrease, which is the mechanism behind predictions (P2) and (P3) of Section A3.4 for the naive scheme.

5.8 Naive acceleration (hypothesis A2)

What happens when Nesterov's momentum is combined with the loopless-SVRG estimator $g = f_i'(x) - f_i'(w) + F'(w)$ without the negative momentum of Katyusha? We test L-Katyusha with $\theta_2 = 0$ and the same θ_1, η, p, that is, the scheme of Definition A3.15 (naive accelerated SVRG) with $\eta = 1/(3\theta_1)$, and the two-sequence form $v = \theta + \beta(\theta - \theta_{\text{prev}})$, $\theta^+ = v - \gamma g(v)$, with the reference point reset to v with probability $\frac{1}{n}$, tuned over $\beta \in \{0.5, 0.8, 0.9, 0.95, (\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)\}$ and $\gamma R^2 \in \{\frac{1}{6}, \frac{1}{20}, \frac{1}{60}\}$ (best setting: fewest gradients to the relative error 10^{-6}). The latter is a variant of Definition A3.15 without the drift term of Remark A3.16 (relation with Nesterov's method of the book), with the reference point reset to the query point instead of the gradient-step iterate, and with (β, γ) free; Proposition A3.18 (one-step inequality of the naive scheme) and Proposition A3.19 (closing by smoothness and strong convexity costs κ) are proved for Definition A3.15 only. Both schemes are compared with SAGA, L-SVRG, L-Katyusha with the untuned parameters of its theorem, and AGD, on (H) with $m = 64$, $\kappa = 10^4$, and on logistic regression with $\kappa = 10^4$, $\alpha = 1$ (one seed each, Figure A2). With $m = 64$, the analogue for (H) of the dimension condition (A1.17) of Corollary A1.20 (iteration count and dimension requirement), $q^{2m} \le \varepsilon$ with $q = (\sqrt{Q_b} - 1)/(\sqrt{Q_b} + 1) = 0.938$ and $Q_b = 1 + 4c/\mu_b$ (Section 2), fails at the accuracies 10^{-6} and 10^{-10} used here, which need $m \ge 109$ and $m \ge 181$: the lower bound is not certified on this instance, and the certified window is explored with $m = 256$ in Section 5.9.

On (H), reaching 10^{-10} (counts read at the first record below the target) costs $5.3 \cdot 10^3$ gradients to AGD, $3.2 \cdot 10^4$ to L-Katyusha, $9.6 \cdot 10^4$ to the best naive setting ($\beta = 0.95$, $\gamma R^2 = \frac{1}{6}$), $4.3 \cdot 10^5$ to SAGA and $1.9 \cdot 10^6$ to L-SVRG, while L-Katyusha with $\theta_2 = 0$ diverges (relative error $8.8 \cdot 10^{27}$ after $3.4 \cdot 10^3$ gradients). With the rule $\beta = (\sqrt{\kappa}-1)/(\sqrt{\kappa}+1) = 0.980$ the naive scheme diverges at $\gamma R^2 = \frac{1}{6}$ and needs $4536n$ and $13\,689n$ gradients to reach 10^{-6} at $\gamma R^2 = \frac{1}{20}$ and $\frac{1}{60}$, against $1188n$ for L-Katyusha; the best setting of the grid needs $3445n$, 2.9 times more. AGD is the cheapest here because $L_F = R^2/n$ (Section 5.9). On logistic regression the order of AGD, L-Katyusha and the best naive setting is reversed: reaching 10^{-10} costs $4.7 \cdot 10^4$ gradients to the best naive setting ($\beta = 0.9$, $\gamma R^2 = \frac{1}{6}$), $4.9 \cdot 10^4$ to L-Katyusha with $\theta_2 = 0$, $8.8 \cdot 10^4$ to SAGA, $1.06 \cdot 10^5$ to L-Katyusha ($7.1 \cdot 10^4$, fewer than SAGA, with stored reference gradients) and $4.0 \cdot 10^5$ to L-SVRG, while AGD ($L_F/\mu = 654$) is still at $6.1 \cdot 10^{-8}$ after 200 passes; to 10^{-6}, the best naive setting needs $28n$ against $56n$ for L-Katyusha, and the $\sqrt{\kappa}$ rule fails at $\gamma R^2 = \frac{1}{6}$ (final relative error 1.26 after 200 passes) and needs $39n$ and $42n$ at the smaller steps.

On (H) the negative momentum is what keeps the coupling stable: removing it produces the divergence anticipated by prediction (P2) of Section A3.4 for $\kappa \gg n$, and the momentum scheme must be detuned, which costs it the advantage over L-Katyusha. Proposition A3.18 (one-step inequality of the naive scheme) and Proposition A3.19 (closing by smoothness and strong convexity costs κ) describe the same obstruction in the proof, for the scheme of Definition A3.15 (naive accelerated SVRG): without the anchor the one-step inequality keeps the uncontrolled term $\frac{\nu}{\theta_1} D_F(w_t, x_t)$, and closing it with smoothness and strong convexity costs a factor κ. On a benign logistic prob-

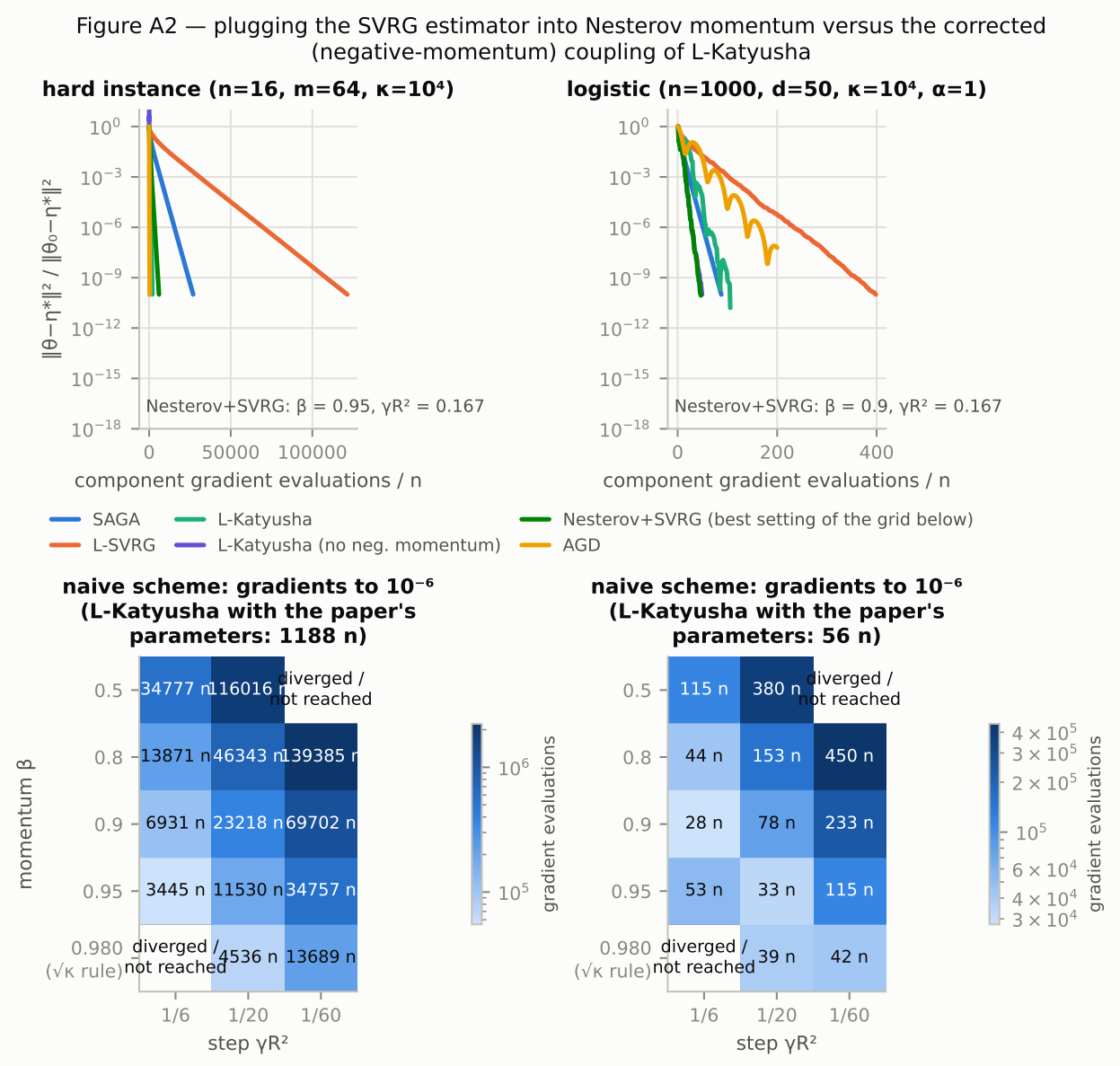


Figure A2: **Hypothesis A2: naive acceleration.** Top: relative error $\|\theta - \eta_*\|^2 / \|\theta_0 - \eta_*\|^2$ (for L-Katyusha, $\theta = y_t$) against component gradient evaluations (counted as in Section 5.1) divided by n, for SAGA, L-SVRG, L-Katyusha (parameters of Theorem 11 of [47]), L-Katyusha without the negative-momentum term ($\theta_2 = 0$), Nesterov momentum applied to the loopless-SVRG estimator (best setting of the grid below, given in each panel) and AGD, on (H) ($n = 16$, $m = 64$, $\kappa = 10^4$) and on logistic regression ($n = 1{,}000$, $d = 50$, $\kappa = 10^4$, $\alpha = 1$); one seed, runs stopped at the first record below 10^{-10}. Bottom: gradient evaluations needed by the naive scheme to reach the relative error 10^{-6} (interpolated log-linearly between records) for each $(\beta, \gamma R^2) \in \{0.5, 0.8, 0.9, 0.95, (\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)\} \times \{\frac{1}{6}, \frac{1}{20}, \frac{1}{60}\}$ (white cells: divergence or budget exhausted); the titles give the count of L-Katyusha with its untuned parameters.

lem, whose effective constants are much better than the worst-case ones ($\mu_{\text{eff}} \approx 5\mu$ near η_* and $L_{\text{eff}} \approx R^2/9$ along SAGA, Section 5.5), the anchor-free schemes converge and are about twice as fast as L-Katyusha with worst-case parameters ($\theta_1 = \sqrt{2n/(3\kappa)} = 0.26$), in line with the report of [2, Section 7.1] that the scheme without negative momentum is sometimes faster after careful tuning but does not accelerate on some tasks. These observations come from two instances, one seed and a grid that favours the naive scheme (fifteen settings against an untuned L-Katyusha); they do not show that it cannot accelerate, a question that no lower bound known to us settles, as noted in Remark A3.20 (other routes, and scope of the obstruction).

5.9 Complexity exponents on hard instances (hypothesis A4)

Question and setting. Hypothesis A4 predicts that, for $\kappa \gg n$, the number T of component gradients needed to reach $\|\theta - \eta_*\|^2 \le \varepsilon \|\theta_0 - \eta_*\|^2$ grows like κ for SAGA and like $\kappa^{1/2}$ for L-Katyusha. On (S), Proposition A1.26 (testable prediction of hypothesis A4) proves $T_{\text{SAGA}} = \Theta(\kappa \log \frac{1}{\varepsilon})$ and $T_{\text{LK}} = \Theta(\sqrt{n\kappa} \log \frac{1}{\varepsilon})$ inside the window $4n + 1 \le \kappa \le \kappa_{\max}(m, \varepsilon)$ of (A1.27), up to the constants of (A1.28)–(A1.29) and the factor $\log(1 + \frac{5}{8}Q)$ of (A1.29); its part (ii) gives GD and AGD the same two exponents, L_F/μ being of the order of $Q = 1 + (\kappa - 1)/n$ by Proposition A1.13 (constants of the instance). Beyond $\kappa_{\max}$ the exhaustion of the chain sets in (part (iii) of Proposition A1.26), and the counts should be read against κ_{eff}, as Remark A1.27 (what to expect in a plot) explains. For $n = 16$, $m = 256$ and $\varepsilon = 10^{-8}$, $\kappa_{\max} \approx 4.9 \cdot 10^4$ on (S) (the dimension term of (A1.27); the slow-mode term is $1.1 \cdot 10^5$); on (A), a single chain of length $m = 2048$, the dimension condition holds in its sufficient form up to $(4m/\log \frac{1}{\varepsilon})^2 \approx 2.0 \cdot 10^5$, beyond the sweep. The bounds of Appendix A1 are proved for (S), not for (H); the analogue of their dimension condition, $q^{2m} \le \varepsilon$ with $q = (\sqrt{Q_b} - 1)/(\sqrt{Q_b} + 1)$ for the per-block parameter $Q_b = 1 + 4c/\mu_b$ of (H) (Section 2), holds up to $\kappa \approx 4.4 \cdot 10^4$ ($q^{2m} = 1.2 \cdot 10^{-10}$ at $\kappa = 3 \cdot 10^4$) and fails at the last point of (H), $\kappa = 10^5$ ($q^{2m} = 1.2 \cdot 10^{-5} > \varepsilon$), which thus lies outside the regime in which a span lower bound would certify the exponent $\frac{1}{2}$. The counts follow the conventions of Section 5.1: for L-Katyusha the error is measured on the gradient sequence y_t and 3 gradients are charged per iteration on average, whereas T_{LK} in Proposition A1.26 refers to the mirror sequence z_t with stored reference gradients, 2 per iteration; the exponents are unaffected, but the measured counts of L-Katyusha are about $\frac{3}{2}$ times those of that convention, and ratios such as $T_{\text{SAGA}}/T_{\text{LK}}$ below are smaller by the same factor.

On (H) and (R) the sweeps also include L-SVRG, L-Katyusha without negative momentum, and the naive scheme of Section 5.8, re-tuned at every pair (instance, κ) over $\beta \in \{0.8, 0.95, (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)\}$ and $\gamma R^2 \in \{\frac{1}{6}, \frac{1}{20}\}$: on (H) on seed 0, the choice being reused for the other seeds, and on (R), where each seed draws a new instance, separately on each seed. Each grid run is capped at 45 s of CPU time; as the grid runs have the same cost per iteration, a run cut by this cap had already used more gradients than the selected one needed in total. Every run reached the target except the 30 runs of L-Katyusha without negative momentum at $\kappa \ge 10^3$, which diverge (Figures A4 and A4b).

Exponents. The slopes $b \pm h$ of Section 5.1, against $x = \kappa$ (nominal) or $x = \kappa_{\text{eff}}$, are, over $x \ge 10n$ unless the window $10n \le \kappa \le \kappa_{\max}$ is indicated (the five values of the nominal κ of (S) from $3 \cdot 10^2$ to $3 \cdot 10^4$; L-SVRG and the naive scheme were run on (H) and (R) only):

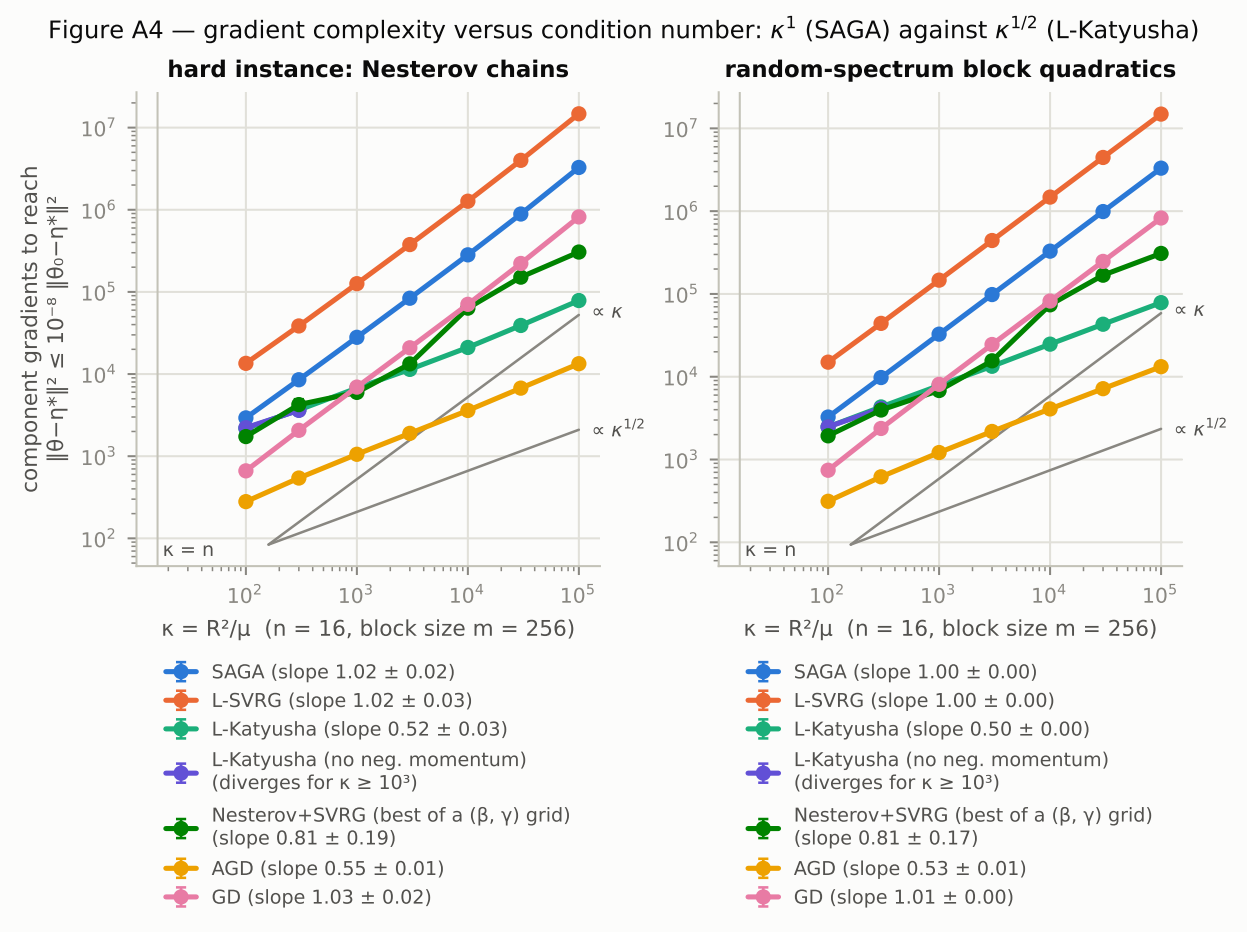


Figure A4: **Hypothesis A4: gradient complexity against κ.** Component gradients (counted as in Section 5.1) needed to reach $\|\theta-\eta_*\|^2 \le 10^{-8}\|\theta_0-\eta_*\|^2$ (for L-Katyusha, $\theta = y_t$) for $\kappa \in \{10^2, 3\cdot 10^2, \dots, 10^5\}$ ($n = 16$) on (H) (block-separable Nesterov chains of length $m = 256$, left) and on block quadratics with random spectra (R) ($m = 256$, right); geometric mean over 3 seeds, error bars: min–max. SAGA uses $\gamma = 1/(4R^2)$, L-SVRG $\gamma = 1/(6R^2)$, L-Katyusha the parameters of Theorem 11 of [47], L-Katyusha without negative momentum $\theta_2 = 0$ with the same θ_1, η, p (it converges only for $\kappa \in \{10^2, 3\cdot 10^2\}$ and diverges for every $\kappa \ge 10^3$, hence no slope), the naive Nesterov+SVRG scheme the best of six settings $(\beta, \gamma R^2) \in \{0.8, 0.95, (\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)\} \times \{\frac{1}{6}, \frac{1}{20}\}$, selected at each κ on seed 0 for (H) and on each seed for (R), whose instance changes with the seed, AGD and GD the step $1/L_F$. Legend: slope $b \pm h$, the least-squares slope of log(gradients) against $\log\kappa$ fitted to the per-κ geometric means over the seeds, for $\kappa \ge 10n$ ($K = 6$), and h the half-width of its 95% Student interval with $K-2$ degrees of freedom, which measures the departure from a single power law, not sampling noise (function `loglog_slope` of `vrlab/analysis.py`). Grey lines: reference slopes 1 and $\frac{1}{2}$; vertical line: $\kappa = n$. The only missing points are those of the divergent runs.

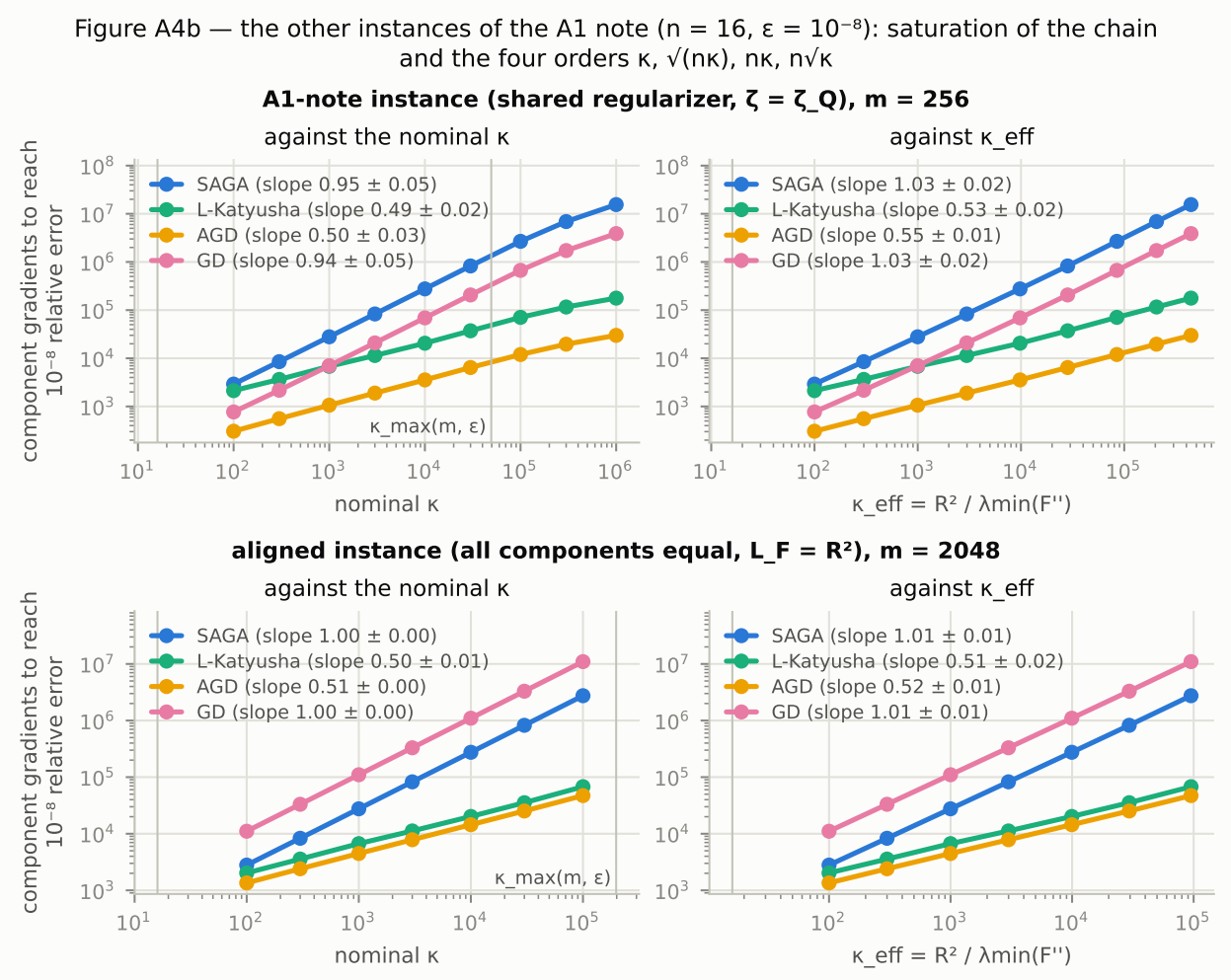


Figure A4b: **Hypothesis A4 on the instances of Appendix A1.** Component gradients needed to reach $\|\theta - \eta_*\|^2 \le 10^{-8}\,\|\theta_0 - \eta_*\|^2$ ($n = 16$; geometric mean over 2 seeds; for L-Katyusha, $\theta = y_t$) for SAGA ($\gamma = 1/(4R^2)$), L-Katyusha (parameters of Theorem 11 of [47]), AGD and GD (step $1/L_F$), all tuned with the exact constants of the instance; L-Katyusha and AGD receive $\lambda_{\min}(F'')$, not μ, as their strong-convexity constant, so that on (S) their parameters depend on κ_{eff}, not on the nominal κ. Top: the instance (S) of (A1.8) (shared regularizer, $\zeta = \zeta_Q$, $m = 256$), $\kappa \in \{10^2, 3\cdot 10^2, \dots, 10^6\}$. Bottom: the aligned instance (A) (all components equal to one chain of length $m = 2048$, $L_F = R^2$), $\kappa \in \{10^2, 3\cdot 10^2, \dots, 10^5\}$, on which GD and AGD pay the factors n and $\sqrt{n}$ that they do not pay on (S). Left: against the nominal κ; right: against $\kappa_{\text{eff}} = R^2/\lambda_{\min}(F'')$. In the left panels the vertical line on the right marks $\kappa_{\max}(m,\varepsilon)$ of (A1.27) ($\approx 4.9\cdot 10^4$, top) and its single-chain analogue $(4m/\log\frac{1}{\varepsilon})^2 \approx 2.0\cdot 10^5$ (bottom, beyond the sweep), beyond which the chain is exhausted and the lower-bound argument for the exponent $\frac{1}{2}$ no longer applies; in every panel the vertical line on the left marks $\kappa = n$. Legend: slope $b \pm h$, the least-squares slope of log(gradients) against the logarithm of the abscissa, fitted to the per-κ geometric means over the seeds for abscissae at least $10n$ ($K = 8$ at the top, 6 at the bottom), and h the half-width of its 95% Student interval with $K - 2$ degrees of freedom (function `loglog_slope` of `vrlab/analysis.py`); the slopes restricted to the window $10n \le \kappa \le \kappa_{\max}(m,\varepsilon)$ are given in Section 5.9. All runs reached the target.

instance, x	K	SAGA	L-SVRG	L-Katyusha	naive scheme	AGD	GD
(H), κ	6	1.022 ± 0.025	1.021 ± 0.026	0.525 ± 0.027	0.806 ± 0.188	0.550 ± 0.010	1.026 ± 0.021
(R), κ	6	1.0025 ± 0.0001	1.002 ± 0.001	0.500 ± 0.005	0.814 ± 0.166	0.526 ± 0.013	1.007 ± 0.004
(S), κ	8	0.945 ± 0.048	–	0.490 ± 0.022	–	0.501 ± 0.027	0.944 ± 0.047
(S), κ, window	5	0.994 ± 0.004	–	0.500 ± 0.020	–	0.530 ± 0.005	0.990 ± 0.008
(S), κ_{eff}	8	1.031 ± 0.017	–	0.533 ± 0.019	–	0.547 ± 0.007	1.029 ± 0.018
(S), κ_{eff}, window	5	1.005 ± 0.015	–	0.506 ± 0.025	–	0.536 ± 0.009	1.002 ± 0.019
(A), κ	6	0.999 ± 0.001	–	0.502 ± 0.015	–	0.511 ± 0.003	0.9991 ± 0.0004
(A), κ_{eff}	6	1.006 ± 0.008	–	0.506 ± 0.018	–	0.515 ± 0.006	1.006 ± 0.007

On (H) and (R) the slopes are close to 1 for SAGA, L-SVRG and GD and to $\frac{1}{2}$ for L-Katyusha and AGD; the largest deviations are those of AGD and of (H), whose local slopes rise over the last two intervals of κ (SAGA 1.04 and 1.08, L-Katyusha 0.56 and 0.58) and whose last point fails the analogue of the dimension condition: without this point, the fits on (H) give 1.008 for SAGA and 0.510 for L-Katyusha. On (S), against the nominal κ, SAGA and GD have the slopes 0.95 and 0.94, with intervals $[0.90, 0.99]$ that exclude 1: the points beyond $\kappa_{\max}$ bend both curves (local slopes 0.98, 0.86 and 0.67 between $3 \cdot 10^4$, 10^5, $3 \cdot 10^5$ and 10^6), whereas inside the window their slopes are $0.994\,[0.990, 0.998]$ and $0.990\,[0.982, 0.998]$. On (A), where every κ lies in the window, they are 0.999, against 0.50 and 0.51 for L-Katyusha and AGD. Several intervals exclude the predicted exponent, by at most 0.01 for SAGA, L-SVRG, L-Katyusha and GD on (H), on (R) and inside the windows, and by up to 0.04 for AGD: they measure the lack of fit of a single power law, and Proposition A1.26 (testable prediction of hypothesis A4) predicts the exponents only as orders of growth, up to constant and logarithmic factors.

The naive scheme has the fitted slopes 0.81 on both (H) and (R), with wide intervals, $[0.62, 0.99]$ and $[0.65, 0.98]$, which nevertheless exclude $\frac{1}{2}$; it does not follow a power law. Its best setting is the same on both instances: $\beta = 0.95$ with $\gamma R^2 = \frac{1}{20}$ at $\kappa = 10^2$, $\beta = 0.95$ with $\gamma R^2 = \frac{1}{6}$ at $\kappa \in \{3 \cdot 10^2, 10^3, 10^4\}$, and $\beta = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ with $\gamma R^2 = \frac{1}{6}$ at $3 \cdot 10^3$ and with $\gamma R^2 = \frac{1}{20}$ at $\kappa \in \{3 \cdot 10^4, 10^5\}$ (on (R) one seed selects $\beta = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$, with $\gamma R^2 = \frac{1}{6}$, already at $\kappa = 3 \cdot 10^2$). Its local slopes between consecutive values of κ are 0.82, 0.28, 0.73, 1.30, 0.79, 0.58 on (H) and 0.65, 0.45, 0.75, 1.29, 0.75, 0.50 on (R): close to $\frac{1}{2}$ from $3 \cdot 10^2$ to $3 \cdot 10^3$ (0.50 and 0.59; $\beta = 0.95$ is the momentum $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ of $\kappa \approx 1.5 \cdot 10^3$), a jump of slope about 1.3 between $3 \cdot 10^3$ and 10^4, then 0.5 to 0.8. L-Katyusha without negative momentum converges at $\kappa = 10^2$ and $3 \cdot 10^2$ and diverges for every $\kappa \geq 10^3$ on both instances.

Constants, L_F and finite dimension. On the block-separable instances the full-gradient methods share the exponents of the variance-reduced ones with smaller constants. At $\kappa = 10^5$, reaching the target costs, on (H) and (R), $1.47 \cdot 10^7$ and $1.49 \cdot 10^7$ gradients to L-SVRG, $3.27 \cdot 10^6$ and $3.31 \cdot 10^6$ to SAGA, $8.2 \cdot 10^5$ and $8.3 \cdot 10^5$ to GD, $3.1 \cdot 10^5$ to the naive scheme, $7.85 \cdot 10^4$ to L-Katyusha and $1.3 \cdot 10^4$ to AGD. L-SVRG costs 4.5 times SAGA at every $\kappa \geq 10^3$ (4.5–4.6 overall): its step $1/(6R^2)$ instead of $1/(4R^2)$ and the 3 gradients per iteration charged to it give 1.5×3 (the factor would be 3 with stored reference gradients). The ratio $T_{\text{GD}}/T_{\text{SAGA}}$ is 0.25 on (H), (R) and (S) for every $\kappa \geq 10^3$ and 4.0 on (A) for every $\kappa \geq 10^4$; both values equal $nL_F/(4R^2)$ up to rounding, the ratio expected if the error of SAGA decays like that of its mean iterate, which follows gradient descent with step $1/(4R^2)$ and one gradient per iteration by Proposition A1.22 (mean iterate of SAGA on quadratics), since $L_F \approx R^2/n$ on the block-separable instances and $L_F = R^2$ on (A). The counts of SAGA and of L-Katyusha agree within 6% between (S) and (A) at every common nominal κ, whereas GD costs 14 to 16 times more and AGD 3.9 to 4.4 times more on (A): about n and $\sqrt{n}$, the factors by which nL_F/μ and $n\sqrt{L_F/\mu}$ grow when L_F goes from about R^2/n to R^2. The ratio $T_{\text{SAGA}}/T_{\text{LK}}$ grows like $\sqrt{\kappa/n}$, with a constant between 0.52 and 0.54 over the whole sweep on (H)

and (R) (41.7 and 42.1 at $\kappa = 10^5$, where $\sqrt{\kappa/n} = 79$), between 0.51 and 0.55 on (A), and of 0.54 and 0.48 at $\kappa = 10^4$ and 10^5 on (S) (13 and 38); with the counting convention of Proposition A1.26 (testable prediction of hypothesis A4), which stores the reference gradients, these constants would be about $\frac{3}{2}$ times larger. On (S), $\kappa_{\text{eff}}/\kappa$ is 0.98 at the nominal $\kappa = 10^4$, 0.85 at 10^5 and 0.44 at 10^6; the slopes of SAGA and GD drop from 0.99 inside the window to 0.95 and 0.94 over $\kappa \geq 10n$ against the nominal κ, and are both 1.03 against κ_{eff}: plotting against κ_{eff} largely removes the bend, as part (iii) of Proposition A1.26 and Remark A1.27 (what to expect in a plot) anticipate, with a slight overshoot (local slopes against κ_{eff} up to 1.07 between $3 \cdot 10^4$ and $3 \cdot 10^5$). L-Katyusha and AGD, whose parameters depend on $\lambda_{\min}(F'')$, i.e. on κ_{eff}, rather than on μ, bend less (0.50 to 0.49 and 0.53 to 0.50), and their slopes against κ_{eff}, 0.53 and 0.55, are slightly above $\frac{1}{2}$. The theory does not fix these values: inside the window its bounds leave constant and logarithmic factors, such as the $\log(1 + \frac{5}{8}Q)$ of (A1.29), and beyond $\kappa_{\max}$ the lower bound behind the exponent $\frac{1}{2}$ no longer applies, while the upper bounds involve $\lambda_{\min}(F'')$ rather than μ because of this tuning (for L-Katyusha, part (iii) of Proposition A1.26, with κ_{eff} in place of κ).

Interpretation. The exponents predicted by hypothesis A4 are observed on the four instances with 2 or 3 seeds, up to the constant and logarithmic factors of Proposition A1.26 (testable prediction of hypothesis A4): over $\kappa \geq 10n$ and against the nominal κ, the slopes are 0.95–1.02 for SAGA (0.95 on (S), whose chain is exhausted beyond $\kappa_{\max}$; 0.99 inside its window and 1.03 against κ_{eff}), 1.00–1.02 for L-SVRG (on (H) and (R), the two instances where it was run) and 0.49–0.52 for L-Katyusha. As orders of growth, the exponent 1 of SAGA is a theorem for constant steps, by Proposition B3.17 (lower bound for SAGA with any constant step), and, on (S) and (A), inside their windows and for the step $1/(4R^2)$ of the experiments, by Corollary A1.23 (SAGA sandwich) with Lemma A1.24 (slow-mode mass of the initial error); on (S) the exponent $\frac{1}{2}$ of L-Katyusha is optimal inside the window, by Proposition A1.18 (the $\sqrt{n\kappa}$ term for any index sequence) with part (ii) of Proposition A1.26. The measured slopes agree with these orders, which do not by themselves fix the value of a slope fitted over a finite window: the two sides of (A1.28) and of (A1.29), with ρ and C_0 replaced by their bounds, differ there by factors of about 8 and 10^2 (Section 4.4 and Remark A1.27). The expectation of the proposal that AGD costs $n\sqrt{\kappa}$ and GD $n\kappa$ holds only when $L_F \approx R^2$, as on (A): on block-separable instances, the worst case for incremental methods, the full-gradient methods do at least as well as their variance-reduced counterparts, and what variance reduction buys is the uniform guarantee over all instances with given (n, R^2, μ) (Section A1.4). The re-tuned naive scheme, a variant of the scheme of Definition A3.15 (naive accelerated SVRG) (Section 5.8), has fitted slopes between the two exponents on this grid. This is consistent with, but not predicted by, the obstruction that Proposition A3.19 (closing by smoothness and strong convexity costs κ) proves for the scheme of Definition A3.15 itself, a limit of the proof, and with the absence of an acceleration guarantee for such schemes with one sample per iteration [67]; it does not show that no tuning can accelerate the scheme.

6 Discussion

B3: where the proof of Proposition 5.9 loses. The diagnosis was carried out and locates the loss, which depends on the regime. On ℓ_2-regularized logistic regression ($n = 1000$, $d = 50$, 5 seeds) the exact one-step contraction exponent of SAGA, averaged over passes 2–20, exceeds the certified one by a factor 3.2–3.7 for $\kappa \leq 10^2$ and 19–220 at $\kappa = 10^5$. For $\kappa \gg n$ the steps that cost are those that replace a quantity depending on the iterate by a global constant: for $\kappa \geq 10^4$, cocoercivity with $R^2 = \max_i L_i$ (I2) and strong convexity with $\mu = \lambda$ (I4) carry 0.77–0.96 of the logarithmic

gap in proof order. For $\kappa \le 10^2$, where the memory term is active, the gap is carried in proof order by the square-norm step I1 (0.39–0.44) and by I2 (0.32–0.37), and in order A by I5 and I1. The variance step (I3) is nearly tight ($r_{\text{var}} \ge 0.958$; at most 0.6% of the gap in proof order, 6.3% in order A) and the minimum of the two rates (I5) matters only when $\kappa \lesssim n$. The additive shares hide a multiplicative structure: the κ-dependent part of the gap is the factor μ_{eff}/μ of I4, whereas, by Proposition B3.10 (exact contraction in terms of the ratios), I1 and I2 together can contribute at most $\frac{8}{3}$ at the fixed step (measured 2.45–2.62), and the product of the two reproduces 91–96% of the speed-ups for $\kappa \ge 10^4$. By Proposition B3.16 (optimal constants of the proof structure) and Remark B3.18, optimizing (γ, Δ) within the proof gains 17% at $\kappa = 10^3$ (at most $\frac{4}{3}$ whenever $\kappa \ge \frac{9n}{16}$), but up to a factor 2.4–2.9 for $\kappa \le 10^2$, comparable to the whole gap there; and, by Proposition B3.17 (lower bound for SAGA with any constant step), no analysis of SAGA with a constant step can remove the exponent 1 of κ. The theorem has the right order in the worst case; on these instances it is pessimistic by a factor 3.2–3.7 already for $\kappa \le 10^2$, and its pessimism grows with κ because the curvature met along the trajectory exceeds μ by a factor that grows with κ (up to 81 here).

B2: trajectory-dependent constants. Confirmed in its main claim, with one refuted prediction. By Corollary B2.15 (effective curvature versus curvature at the optimum), near η_*, μ_{eff}/μ is bounded below by about $\lambda_{\min}(F''(\eta_*))/\mu$ (up to 70 in our problems), a value that SAGA reaches when its error aligns with the flattest direction, and R^2 overestimates L_{eff} by a factor 2.6 to 86; the a-posteriori bound with μ_{eff} and L_{eff} (Proposition B3.19) recovers 0.59–0.96 of the logarithmic gap for $\kappa \ge 10^4$, and nothing for $\kappa \le 10^2$, where the bound is saturated by the memory term. Our prediction that the error of SAGA stays spread over many directions holds only for $\kappa \lesssim n$, and not in every run with $\kappa \approx n$: at $\kappa = 10^4 = 10n$ and for $\alpha \in \{0, 1\}$ (one run each) SAGA keeps and sharpens, as gradient descent does, the alignment of the error with the flattest direction of $F''(\eta_*)$, on which the initial error already carries 89–98% of its squared norm, a behaviour that Remark B2.22 (outcome of prediction (P2)) explains only heuristically; for $\alpha = 2$, whose spectrum is nearly degenerate at the bottom, neither method aligns within our runs. The effective constants are a-posteriori quantities: they explain an observed trajectory, they do not certify one in advance, and, by Remark B2.24 (what each constant can explain), at a fixed step the gain from L_{eff} is capped at $\frac{8}{3}$.

B1: preconditioned SAGA. Nuanced. For logistic regression no fixed metric improves the global condition number: $\kappa_P \ge \kappa$ by Proposition B1.21 (no free lunch for global constants), so that Corollary B1.12 (Proposition 5.9 transferred to P-SAGA) never favours preconditioning. Yet at $\kappa = 10^3$ the statistical and Hessian metrics raise the observed exponent per pass (in component gradients; a dense metric costs $O(d^2)$ instead of $O(d)$ per iteration, and $P = F''(\eta_*)$ requires η_*, an oracle benchmark) from 1.00 to 1.11 ($\alpha = 1$) and from 0.61 to 1.24–1.26 ($\alpha = 2$). The gain is local: the trajectory bound of Corollary B1.23 (leverage-score bound along trajectories), which pays the smallest per-sample curvature, does not explain it, whereas with the local constant at the optimum, a heuristic substitution that Lemma B1.27 (limits of the effective constants at η_*) supports only as $\theta \to \eta_*$, the bound reproduces the observed ordering but saturates at the memory term, $\frac{1}{3}$ per pass, 3.3–3.8 times below the observed exponents. Along the trajectories, the exact memory coefficient accounts for a factor of about 2 of this gap (about $1 - \frac{0.65}{n}$ instead of $1 - \frac{1}{3n}$), and the parameter part, which contracts faster but is charged at the memory rate by I5, for most of the rest; but no bound on $\mathbb{E}V_t^P$ can gain more than a factor 3 over $\frac{1}{3}$, since the memory slots that are never refreshed keep $\mathbb{E}V_t^P$ from decreasing faster than $(1 - \frac{1}{n})^n \approx e^{-1}$ per pass: a factor 1.1–1.26 separates

the typical trajectories, whose slopes are measured, from the expectation that the bound controls. The metric shifts the slack of the proof from cocoercivity to strong convexity, as predicted, and the Jacobi metric does not help when the features are rotated.

A1: lower bounds. This hypothesis is theoretical; Appendix A1 (Section 4.4) assesses it against the known bounds, restated with their exact hypotheses. The first clause holds on instances whose dimension is of the order of the number of queries certified: span-based methods sampling i.i.d. from a fixed distribution need $\Omega((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon})$, by Theorem A1.8 (Lan–Zhou lower bound; Theorem 3 and Corollary 3 of [50]) when $\kappa \geq n$ and, on the instance (S), for every $\kappa > 1$ by Corollary A1.20 (iteration count and dimension requirement). The second clause holds only in a restricted sense: for oblivious methods (up to logarithmic factors) and for loopless methods with a full-gradient coin, full-gradient steps can divide the $n\log\frac{1}{\varepsilon}$ term by at most $1+(\log(n/\kappa))_+$, a gain only when $\kappa \ll n$, and leave the $\sqrt{n\kappa}\log\frac{1}{\varepsilon}$ term untouched, by part (c) of Theorem A1.11 (Hannah–Liu–O'Connor–Yin bounds; [38]) and by Theorem A1.12 (Han–Xie–Zhang lower bound; Theorem E.6 of [37]). Deterministic methods, cyclic ones included, face instead $\Omega(n\sqrt{\kappa}\log\frac{1}{\varepsilon})$ for small enough ε when $\kappa > 73$, by Theorem A1.9 (Woodworth–Srebro lower bounds; Theorems 4 and 8 of [87]). Two questions remain open: for randomized methods with fully adaptive queries only an additive n is proved (randomized part of Theorem A1.9), and whether randomly reshuffled span-based methods, or deterministic ones when $\kappa \leq 73$, can beat $n\log\frac{1}{\varepsilon}$ is not settled. Our experiments use the lower bound only through its instance: the exponent $\frac{1}{2}$ is certified there inside the window of κ of Proposition A1.26 (testable prediction of hypothesis A4), which the dimension limits ($\kappa_{\max} \approx 4.9\cdot 10^4$ for $m = 256$ and $\varepsilon = 10^{-8}$), and the regime $\kappa \ll n$ of the second clause was not explored numerically.

A3: corrected acceleration. Confirmed. The one-step contraction of the Lyapunov proof of L-Katyusha, Theorem A3.4 (contraction and rate of L-Katyusha; Lemma 10 and Theorem 11 of [47]), holds at every recorded state and is pessimistic in exponent by a factor 2.8–4.7 on our three problems (3.7–9.3 for its closed-form rate). On the two logistic problems this is less than for Proposition 5.9 (2.8 and 4.1 against 5.2 and 15.7) and grows less from $\kappa = 10^3$ to 10^4 ($\times 1.5$ against $\times 3.1$); two values of κ, in different regimes of the certificate, do not show how it scales. Its variance bound loses, as I2 does, the gap between the worst-case curvature $\max_i L_i$ and the curvature met along the trajectory (tightness 0.125–0.15). The term that the negative momentum cancels in Proposition A3.14 (Bregman-gap cancellation by the negative momentum) exceeds the whole certified decrease in absolute value on 63% and 65% of the steps of the two ill-conditioned problems, but mostly with the sign that helps a bound without the anchor: it is positive and exceeds this decrease on 9.9% and 19.3% of the steps (5.2% on logistic regression with $\kappa = 10^3$), and only there could it not have been absorbed, within this proof, into the certified decrease instead of cancelled.

A2: naive acceleration. The failure of naive acceleration is confirmed on block quadratics but is not universal. On (H) and (R) with $m = 256$ (Section 5.9), L-Katyusha without its negative momentum diverges for every $\kappa \geq 10^3$ ($n = 16$), and Nesterov's momentum on the SVRG estimator, even re-tuned at every κ over six settings, has the log–log slope 0.81 on both instances instead of $\frac{1}{2}$ (95% intervals $[0.62, 0.99]$ and $[0.65, 0.98]$, wide because its profile is irregular) and needs 3.0 times the gradients of L-Katyusha at $\kappa = 10^4$. In Section 5.8 (one seed, $\kappa = 10^4$), the best of fifteen settings needs 2.9 times the gradients of L-Katyusha to reach 10^{-6} on (H) with $m = 64$, a chain too short for the lower bound to be certified, whereas on logistic regression both anchor-free schemes

converge and need about half the gradients of L-Katyusha with its worst-case parameters. The proof-level obstruction of Proposition A3.19 (closing by smoothness and strong convexity costs κ) is a limit of the argument, not a lower bound, as Remark A3.20 (other routes, and scope of the obstruction) details, and our evidence comes from small grids.

A4: exponents on hard instances. Confirmed inside the window of κ where Proposition A1.26 (testable prediction of hypothesis A4) certifies the exponents, with two corrections to the expectations of the proposal. On four quadratic instances ($n = 16$, 2 or 3 seeds) the log–log slopes of the gradient counts over $\kappa \geq 10n$ are 0.95–1.02 for SAGA (0.99 inside the window on (S)), 1.00–1.02 for L-SVRG (run on (H) and (R) only) and 0.49–0.52 for L-Katyusha. First, on block-separable instances GD and AGD share these exponents with smaller constants, because $L_F \approx R^2/n$; the orders $n\kappa$ and $n\sqrt{\kappa}$ of the proposal appear only on the aligned instance, where GD costs 14–16 times and AGD 3.9–4.4 times more than on the block instance (S). Second, the exponents are tied to a finite window: beyond $\kappa_{\max}$ the chain of (S) is exhausted ($\kappa_{\text{eff}}/\kappa = 0.44$ at the nominal $\kappa = 10^6$), and the slopes of SAGA and GD against the nominal κ drop from 0.99 inside the window to 0.95 and 0.94 (95% intervals $[0.90, 0.99]$ for both; 1.03 against κ_{eff}).

Limitations.

- *Data.* All logistic problems are synthetic: Gaussian features with power-law spectra, $n = 1000$, $d = 50$, one data set per α shared by all κ (which only set λ), $\theta_0 = 0$, and isotropic features with $d = 40$ and $n \in \{10^3, 10^4\}$ for the validation of Section 5.2. Real data, other ratios d/n and nearly separable data were not tested, and the reproduction of the experiment of the book is only qualitative.
- *Seeds and intervals.* We used 5 seeds (Sections 5.3 and 5.6), 3 (Figure C, Section 5.7, (H) and (R) in Section 5.9), 2 ((S) and (A)) or 1 (Sections 5.2, 5.5 and 5.8); bands are min–max ranges over the seeds, not confidence intervals. For the stochastic methods the seeds are independent replicates (on (R) each seed also draws its own instance), whereas GD and AGD are deterministic, so that on (H), (S) and (A) their counts coincide for every seed. The slopes are least-squares fits of the per-κ geometric means over the seeds, and their 95% t-intervals ($K - 2$ degrees of freedom, K values of κ) measure the departure of these means from a single power law over a finite window of κ, the dominant uncertainty on an exponent, rather than sampling noise: additional seeds would not reduce it, no coverage is claimed, and an exponent remains a local, window-dependent quantity.
- *Attribution.* The shares depend on the order in which the inequalities are undone, and the two orders of Proposition B3.11 (attribution of the gap) differ in the memory regime; the shares are additive in contraction exponents and not multiplicative factors, and the product decomposition used for $\kappa \geq 10^4$ is approximate (91–96% of the speed-ups).
- *Exactness.* Conditional expectations are exact only along the realized trajectories; products of conditional factors are realized quantities, and the trajectory-dependent bounds (Proposition B3.19) are a-posteriori, not guarantees.
- *Precision floors.* Nothing is analysed below $V_t = 10^{-18}$ for SAGA ($\|\theta_t - \eta_*\|^2 = 10^{-18}$ in Section 5.5) or $\Psi_t = 10^{-10}$ for L-Katyusha, so the late phase of the well-conditioned runs is truncated; conversely, the runs with $\kappa = 10^5$ and $\alpha = 2$ do not reach the neighbourhood of η_* in 100 passes.
- *Hard instances.* One n (16), one accuracy ($\varepsilon = 10^{-8}$) and a finite window of κ; the dependence

on n and the regime $\kappa \ll n$ were not explored, and the instance (H) with $m = 64$ used in Section 5.8 is too short for the lower bound to be certified at the accuracies used there.

- *Tuning and counting.* The naive scheme was tuned on small grids (fifteen settings in Section 5.8; six per κ in Section 5.9, selected on seed 0 on (H) and on each seed on (R)), while L-Katyusha kept its theoretical parameters; a finer grid, or one in which β and γ scale with κ beyond the single rule $\beta = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$, could change the measured exponent of the naive scheme. The reference gradients $f_i'(w)$ of the SVRG-type methods are charged as if recomputed at every iteration (the code stores them, as SAGA stores its table), which raises their counts by about one half compared with the convention with storage used by the bounds of Appendix A1 (3 instead of 2 gradients per iteration on average); this changes constants, not exponents, but it decides the order of SAGA and L-Katyusha on logistic regression in Section 5.8 (with storage L-Katyusha needs $7.1 \cdot 10^4$ gradients to 10^{-10}, against $8.8 \cdot 10^4$ for SAGA).

Open questions. (i) The exact worst-case one-step contraction of the Lyapunov function of Proposition 5.9, computable by performance estimation when interpolation conditions are known for its class of functions [84, 31] and bounded from above by the linear matrix inequality of [41], which only uses quadratic constraints between the iterate and η_* (see also [82]), would separate the slack of the proof from the benign character of our instances, and its dual multipliers would say which inequalities are active in the worst case, to be compared with our trajectory-wise attribution. (ii) An a-priori version of the trajectory-dependent bound can be obtained on a region (Proposition B3.21), with the local strong convexity of logistic regression on a ball around η_* (Proposition B3.22); how much of the gap such a computable certificate recovers is open. (iii) Once the curvature is handled, the remaining slack of Proposition 5.9 is in its memory coefficient and in the step I5 (Section 5.6). At the textbook step a larger weight Δ already certifies up to twice the memory rate $\frac{1}{3n}$ when $n \gg \kappa$, by Corollary B3.15 (parametrization $\Delta = c\gamma^2$), and Proposition B3.16 (optimal constants of the proof structure) allows at most a factor 3 over all (γ, Δ). For $\kappa \leq \frac{9n}{16}$, where the textbook rate is $\frac{1}{3n}$, this factor 3 is also the most that any analysis of $\mathbb{E}[V_t]$, whatever Δ, can gain, since, by Remark B3.18 (optimizing (γ, Δ) gains at most a factor 3), no bound on $\mathbb{E}[V_t]$ certifies more than $\frac{1}{n}$ per iteration: the open question is whether another Lyapunov function certifies more at the textbook step. (iv) The alignment of SAGA with the flattest direction for $\kappa \gg n$, seen in two runs, should be tested on more configurations, for instance by recording the spectrum of the noise of the estimator. (v) No lower bound shows that Nesterov's momentum on the SVRG estimator with one sample per iteration cannot accelerate; measuring its exponent as a function of the mini-batch size would connect our observations with the mini-batch analyses of [68, 21]. (vi) The largest additive share for $\kappa \geq 10^4$ belongs to cocoercivity with $\max_i L_i$, which non-uniform sampling replaces by an average of the L_i [88, 75, 63], removing the part of this slack due to the spread of the L_i; instrumenting such a variant, and exploiting μ_{eff} in accelerated methods through restarts [24, 73] within the limits set by [36, 4], are natural next steps.

Use of generative AI

A large language model (Claude, Anthropic) was used as a research and engineering assistant for this project, under the supervision of the authors: to extract and transcribe the proof of Proposition 5.9 from the book, to implement and test the numerical library and the experiment scripts, to search and verify the bibliography (every reference was checked against the arXiv, Crossref or Semantic Scholar records), to draft the theory notes (each note was reviewed by independent adversarial passes checking the constants and the consistency with the cited sources, and then corrected), and

to format this report. All mathematical statements were re-derived and all numerical claims come from the scripts in the repository, which are deterministic given the seeds reported in Appendix R, except that a run of the κ sweeps of Section 5.9 that exceeds its CPU-time budget (`VRLAB_MAX_TIME`, Appendix R) is dropped from the fits, so that which points enter could depend on the machine and on the budget; in the reported sweeps every run reached its target within the budget, except the runs of L-Katyusha without negative momentum, which diverge. Representative prompts: *"Transcribe the proof of Proposition 5.9 as a chain of labelled inequalities"*, *"Try to refute this note: check every constant"*, *"Verify that this reference exists and that its main result is stated correctly"*.

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B3 Proposition 5.9 as a chain of inequalities: exact one-step identity, tightness ratios, optimal (γ, Δ), trajectory-dependent version

This note takes the proof of Proposition 5.9 of [11] (linear convergence of SAGA [16]) and splits it into an *exact* one-step identity followed by six inequalities (I1)–(I6). Each inequality receives a tightness ratio in $[0,1]$ (in $(0,1]$ for all of them except r_{var}) that is computable from the state of the algorithm, so that the experiments of hypothesis (B3) can measure, on ℓ_2-regularized logistic regression, which inequality is responsible for the gap between the worst-case contraction factor $1-\min\{1/(3n),\, 3\mu/(16R^2)\}$ and the contraction actually observed. We then prove that, within the structure of this proof (the book's Lyapunov function with a free weight Δ and the inequalities (I1)–(I5)), no choice of the two constants (γ, Δ) can improve the certified one-step rate $1-\rho_{\mathrm{bound}}$ by more than a factor 3 (a factor $4/3$ when $\kappa \geq 9n/16$), the optimal constants being exactly those of [40], so that a larger gap between the certified and the exact contraction must come from the slack of these inequalities (other Lyapunov functions or analyses are not covered by this optimality statement and could certify more); we show by an elementary argument, valid for every n and every constant step, that the exponent 1 of κ in the complexity of SAGA cannot be removed by a better analysis; and we derive the trajectory-dependent version of the proposition that hypothesis (B2) builds on.

B3.1 Setting, the SAGA iteration and Proposition 5.9

Assumption B3.1 (convex smooth components, strongly convex sum)**.** *$F(\theta) = \frac{1}{n}\sum_{i=1}^n f_i(\theta)$ on $\mathbb{R}^d$, where each f_i is convex, differentiable and R^2-smooth, i.e. (Definition 5.3 of the book, Lipschitz-gradient form) $\|f_i'(\theta) - f_i'(\eta)\| \leq R^2\,\|\theta-\eta\|$ for all $\theta,\eta \in \mathbb{R}^d$; and F is μ-strongly convex, $\mu > 0$ (Definition 5.2 of the book: $F(\eta) \geq F(\theta) + F'(\theta)^\top(\eta-\theta) + \frac{\mu}{2}\|\eta-\theta\|^2$ for all θ,η). We write $\kappa = R^2/\mu$.*

Lemma B3.2 (consequences of the assumptions)**.** *Under Assumption B3.1:*

(a) F has a unique minimizer η_*, characterized by $F'(\eta_*) = 0$, i.e. $\sum_{i=1}^n f_i'(\eta_*) = 0$;

(b) for all θ, $\langle F'(\theta), \theta - \eta_*\rangle \geq \mu\,\|\theta-\eta_*\|^2$;

(c) F is R^2-smooth and $\mu \leq R^2$, i.e. $\kappa \geq 1$.

Proof. (a) is Proposition 5.2 and Exercise 5.6 of the book: a strongly convex function is coercive and strictly convex, hence has exactly one minimizer, and for a differentiable convex function the minimizers are exactly the stationary points. (b) Write the strong convexity inequality at the pairs $(\theta,\eta) = (\theta,\eta_*)$ and (η_*,θ) and add: $0 \geq (F'(\theta)-F'(\eta_*))^\top(\eta_*-\theta) + \mu\,\|\theta-\eta_*\|^2$; conclude with $F'(\eta_*) = 0$. (c) $\|F'(\theta)-F'(\eta)\| \leq \frac{1}{n}\sum_i \|f_i'(\theta)-f_i'(\eta)\| \leq R^2\,\|\theta-\eta\|$ by the triangle inequality. For $\theta \neq \eta_*$, (b) and Cauchy–Schwarz give $\mu\,\|\theta-\eta_*\|^2 \leq \|F'(\theta)\|\;\|\theta-\eta_*\| \leq R^2\,\|\theta-\eta_*\|^2$. □

SAGA. Let $i(1), i(2), \ldots$ be i.i.d. uniform on $\{1,\ldots,n\}$, let $\mathcal{F}_t = \sigma(i(1),\ldots,i(t))$ ($\mathcal{F}_0$ trivial) and $\mathbb{E}_{i(t)}[\,\cdot\,] := \mathbb{E}[\,\cdot \mid \mathcal{F}_{t-1}]$. Given $\theta_0 \in \mathbb{R}^d$ and stored gradients $z_1^{(0)},\ldots,z_n^{(0)} \in \mathbb{R}^d$, the iteration of [11, Sec. 5.4.4] is

$$\begin{aligned} \theta_t &= \theta_{t-1} - \gamma\,\omega_t, \qquad \omega_t := f_{i(t)}'(\theta_{t-1}) + \frac{1}{n}\sum_{j=1}^n z_j^{(t-1)} - z_{i(t)}^{(t-1)}, \\ z_{i(t)}^{(t)} &= f_{i(t)}'(\theta_{t-1}), \qquad z_j^{(t)} = z_j^{(t-1)} \quad (j \neq i(t)). \end{aligned} \tag{B3.1}$$

We write $\bar{z}_{t-1} := \frac{1}{n}\sum_j z_j^{(t-1)}$. The variables θ_t, $z^{(t)}$ are $\mathcal{F}_t$-measurable, and $i(t)$ is independent of $\mathcal{F}_{t-1}$ and uniform, so that for any $\mathcal{F}_{t-1}$-measurable vectors $(v_i)_{i\le n}$, $\mathbb{E}_{i(t)}[v_{i(t)}] = \frac{1}{n}\sum_i v_i$. The Lyapunov function of the book is, for a constant $\Delta > 0$,

$$V_t := \|\theta_t - \eta_*\|^2 + \Delta \sum_{i=1}^n \left\| z_i^{(t)} - f_i'(\eta_*) \right\|^2, \qquad \text{with } \gamma = \frac{1}{4R^2},\ \Delta = 3\gamma^2 = \frac{3}{16R^4} \text{ in the book.} \tag{B3.2}$$

Proposition B3.3 (convergence of SAGA; Proposition 5.9 of [11]). *Under Assumption B3.1, if $z_i^{(0)} = f_i'(\theta_0)$ for all i and $\gamma = \frac{1}{4R^2}$, then for all $t \ge 0$,*

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \le \left(1 - \min\left\{\frac{1}{3n}, \frac{3\mu}{16R^2}\right\}\right)^t \left(1 + \frac{n}{4}\right) \|\theta_0 - \eta_*\|^2 . \tag{B3.3}$$

The book's proof (pp. 147–149) is reproduced in Section B3.3 as the exact identity of Section B3.2 followed by the six inequalities (I1)–(I6).

B3.2 The exact one-step identity

Fix $t \ge 1$ and write $(\theta, z) = (\theta_{t-1}, z^{(t-1)})$ for the state before the t-th draw; everything below is $\mathcal{F}_{t-1}$-measurable. Define

$$D_{t-1} := \|\theta - \eta_*\|^2, \qquad M_{t-1} := \sum_{i=1}^n \left\| z_i - f_i'(\eta_*) \right\|^2, \qquad G_{t-1} := (\theta - \eta_*)^\top F'(\theta),$$

$$S_{t-1} := \sum_{i=1}^n \left\| f_i'(\theta) - f_i'(\eta_*) \right\|^2, \qquad a_i := f_i'(\theta) - f_i'(\eta_*), \qquad b_i := f_i'(\eta_*) - z_i + \bar{z}_{t-1},$$

so that $\omega_t = a_{i(t)} + b_{i(t)}$ (indeed $a_i + b_i = f_i'(\theta) - z_i + \bar{z}_{t-1}$), and

$$\Omega_{t-1} := \mathbb{E}_{i(t)}\,\|\omega_t\|^2 = \frac{1}{n}\sum_{i=1}^n \|a_i + b_i\|^2, \qquad C_{t-1} := \frac{1}{n}\sum_{i=1}^n \langle a_i, b_i \rangle ,$$

$$B_{t-1} := \mathbb{E}_{i(t)}\left\|b_{i(t)}\right\|^2 = \frac{1}{n}\sum_{i=1}^n \left\| f_i'(\eta_*) - z_i + \frac{1}{n}\sum_{j=1}^n z_j \right\|^2 .$$

Thus $V_{t-1} = D_{t-1} + \Delta M_{t-1}$. We drop the index $t-1$ when no confusion is possible.

Lemma B3.4 (exact one-step identities). *For every $t \ge 1$, almost surely:*

(a) $\mathbb{E}_{i(t)}[\omega_t] = F'(\theta_{t-1})$ *and*

$$\mathbb{E}_{i(t)}\big[D_t\big] = D_{t-1} - 2\gamma\, G_{t-1} + \gamma^2\, \Omega_{t-1}; \tag{B3.4}$$

(b)

$$\mathbb{E}_{i(t)}\big[M_t\big] = \left(1 - \frac{1}{n}\right) M_{t-1} + \frac{1}{n} S_{t-1}; \tag{B3.5}$$

(c)

$$B_{t-1} = \frac{1}{n} M_{t-1} - \|\bar{z}_{t-1}\|^2 ; \tag{B3.6}$$

(d) $\Omega_{t-1} = \frac{1}{n}S_{t-1} + 2C_{t-1} + B_{t-1}$, *and* $\Omega_{t-1} = \|F'(\theta_{t-1})\|^2 + \mathbb{E}_{i(t)}\|\omega_t - F'(\theta_{t-1})\|^2$;

(e) $\mathbb{E}_{i(t)}\left\|a_{i(t)}\right\|^2 = S_{t-1}/n$, $\mathbb{E}_{i(t)}[b_{i(t)}] = 0$ *and* $|C_{t-1}| \le \sqrt{(S_{t-1}/n)\,B_{t-1}}$.

Proof. (a) Since θ_{t-1} and $z^{(t-1)}$ are $\mathcal{F}_{t-1}$-measurable and $i(t)$ is uniform and independent of $\mathcal{F}_{t-1}$, $\mathbb{E}_{i(t)}[f'_{i(t)}(\theta_{t-1})] = \frac{1}{n}\sum_i f'_i(\theta_{t-1}) = F'(\theta_{t-1})$ and $\mathbb{E}_{i(t)}[z^{(t-1)}_{i(t)}] = \bar{z}_{t-1}$, hence $\mathbb{E}_{i(t)}[\omega_t] = F'(\theta_{t-1}) + \bar{z}_{t-1} - \bar{z}_{t-1} = F'(\theta_{t-1})$. Expanding the square in $\theta_t - \eta_* = (\theta_{t-1} - \eta_*) - \gamma\omega_t$, $D_t = D_{t-1} - 2\gamma\langle\theta_{t-1} - \eta_*, \omega_t\rangle + \gamma^2\|\omega_t\|^2$; taking $\mathbb{E}_{i(t)}$ and pulling the $\mathcal{F}_{t-1}$-measurable vector $\theta_{t-1} - \eta_*$ out of the inner product gives (B3.4). (b) Only the coordinate $i(t)$ of the memory changes, so $M_t = M_{t-1} - \left\|z^{(t-1)}_{i(t)} - f'_{i(t)}(\eta_*)\right\|^2 + \left\|f'_{i(t)}(\theta_{t-1}) - f'_{i(t)}(\eta_*)\right\|^2$, and $\mathbb{E}_{i(t)}$ of the last two terms equals M_{t-1}/n and S_{t-1}/n. (c) Let $Z := f'_{i(t)}(\eta_*) - z^{(t-1)}_{i(t)}$. Then $\mathbb{E}_{i(t)}[Z] = \frac{1}{n}\sum_i f'_i(\eta_*) - \bar{z}_{t-1} = -\bar{z}_{t-1}$ by Lemma B3.2(a), so $b_{i(t)} = Z - \mathbb{E}_{i(t)}[Z]$ and $B_{t-1} = \mathbb{E}_{i(t)}\left\|Z - \mathbb{E}_{i(t)}Z\right\|^2 = \mathbb{E}_{i(t)}\|Z\|^2 - \left\|\mathbb{E}_{i(t)}Z\right\|^2 = M_{t-1}/n - \|\bar{z}_{t-1}\|^2$. (d) Expand $\|a_i + b_i\|^2 = \|a_i\|^2 + 2\langle a_i, b_i\rangle + \|b_i\|^2$ and average over i. The second identity is $\mathbb{E}_{i(t)}\|\omega_t\|^2 = \left\|\mathbb{E}_{i(t)}\omega_t\right\|^2 + \mathbb{E}_{i(t)}\left\|\omega_t - \mathbb{E}_{i(t)}\omega_t\right\|^2$ with (a). (e) The first two claims are the computations in (a) and (c). For the third, Cauchy–Schwarz in $\mathbb{R}^d$ and then in $\mathbb{R}^n$: $|C| \le \frac{1}{n}\sum_i \|a_i\|\,\|b_i\| \le \left(\frac{1}{n}\sum_i \|a_i\|^2\right)^{1/2}\left(\frac{1}{n}\sum_i \|b_i\|^2\right)^{1/2}$. □

Proposition B3.5 (exact conditional contraction)**.** *For every* $t \ge 1$ *and every* $\gamma, \Delta > 0$, *almost surely,*

$$\mathbb{E}_{i(t)}[V_t] = D_{t-1} - 2\gamma G_{t-1} + \gamma^2\Omega_{t-1} + \Delta\Big[\Big(1 - \frac{1}{n}\Big)M_{t-1} + \frac{1}{n}S_{t-1}\Big], \tag{B3.7}$$

so that, on $\{V_{t-1} > 0\}$, *the exact conditional contraction factor is*

$$\rho_{\text{exact}}(t) := \frac{\mathbb{E}_{i(t)}[V_t]}{V_{t-1}} = 1 - \frac{2\gamma G_{t-1} - \gamma^2\Omega_{t-1} + \Delta\,(M_{t-1} - S_{t-1})/n}{D_{t-1} + \Delta M_{t-1}}. \tag{B3.8}$$

Proof. Add (B3.4) and Δ times (B3.5), then subtract $V_{t-1} = D_{t-1} + \Delta M_{t-1}$. □

Remark B3.6 (reading the identity). The numerator of (B3.8) is the one-step *progress*: $2\gamma G - \gamma^2\Omega$ is the progress of the parameter (as for gradient descent, with the second moment $\Omega = \|F'(\theta)\|^2 + \mathrm{Var}_{i(t)}(\omega_t)$ of the direction in place of $\|F'(\theta)\|^2$), and $\Delta(M-S)/n$ is the progress of the memory: one stored error $\|z_i - f'_i(\eta_*)\|^2$ (of average size M/n) is replaced by the current error $\|f'_i(\theta) - f'_i(\eta_*)\|^2$ (of average size S/n). For admissible constants (Proposition B3.14 below, in particular the textbook ones), the expected progress is bounded below by $(1 - \rho_{\text{bound}})V_{t-1} > 0$ whenever $V_{t-1} > 0$, by the chain (I1)–(I5) of Section B3.3: the memory progress $\Delta(M - S)/n$ alone can be negative (when $S > M$), but it is then compensated by the parameter term. Only the realised one-step change $V_t - V_{t-1}$ can be positive. In the experiments, a negative numerator in (B3.8) with the textbook constants would therefore signal a bug, not a feature. Note that $V_{t-1} = 0$ iff $\theta_{t-1} = \eta_*$ and $z_i = f'_i(\eta_*)$ for all i; then $\omega_t = 0$ and the state is absorbing, so ρ_{exact} is only needed on $\{V_{t-1} > 0\}$.

B3.3 The proof of Proposition 5.9 as a chain of six inequalities

We now follow the book's proof line by line; apart from the two final steps $\mathbb{E}\|\theta_t - \eta_*\|^2 \le \mathbb{E}[V_t]$ (dropping $\Delta M_t \ge 0$) and $1 + \frac{3n}{16} \le 1 + \frac{n}{4}$, which affect the constant of (B3.3) but not its rate, the only steps that are not identities are the six inequalities below (named as in the project description).

(I1) Square-norm inequality. For $a, b \in \mathbb{R}^d$, $\|a+b\|^2 = 2\|a\|^2 + 2\|b\|^2 - \|a-b\|^2 \le 2\|a\|^2 + 2\|b\|^2$. Applied to $\omega_t = a_{i(t)} + b_{i(t)}$ and averaged with Lemma B3.4(e):

$$\Omega_{t-1} \;\le\; \frac{2}{n} S_{t-1} + 2B_{t-1}, \qquad \text{with} \quad \frac{2}{n} S_{t-1} + 2B_{t-1} - \Omega_{t-1} = \frac{1}{n}\sum_{i=1}^{n} \|a_i - b_i\|^2 . \tag{I1}$$

(I2) Co-coercivity (Proposition 5.4 of the book, eq. (5.11), as applied in the book's proof at eq. (5.31)). Proposition 5.4 states that for a convex L-smooth function f and all θ, η, $\frac{1}{L}\|f'(\theta) - f'(\eta)\|^2 \le (f'(\theta) - f'(\eta))^\top(\theta - \eta)$. Applied to each f_i with $L = R^2$, $\eta = \eta_*$, summed over i, divided by n, and using $\frac{1}{n}\sum_i f_i'(\eta_*) = F'(\eta_*) = 0$:

$$\frac{1}{n} S_{t-1} = \frac{1}{n}\sum_{i=1}^{n} \left\| f_i'(\theta_{t-1}) - f_i'(\eta_*) \right\|^2 \;\le\; \frac{R^2}{n}\sum_{i=1}^{n} \left(f_i'(\theta_{t-1}) - f_i'(\eta_*) \right)^\top (\theta_{t-1} - \eta_*) = R^2\, G_{t-1}. \tag{I2}$$

This inequality is used *twice*: once in the recursion for D (through (I1)) and once in the recursion for M (through (B3.5)).

(I3) Variance $\le$ second moment. By Lemma B3.4(c) and $\|\bar{z}_{t-1}\|^2 \ge 0$:

$$B_{t-1} = \mathbb{E}_{i(t)} \left\| Z - \mathbb{E}_{i(t)} Z \right\|^2 \;\le\; \mathbb{E}_{i(t)} \|Z\|^2 = \frac{1}{n} M_{t-1}, \qquad Z = f_{i(t)}'(\eta_*) - z_{i(t)}^{(t-1)}. \tag{I3}$$

Step 1 of the book (parameter recursion). Inserting (I1), then (I2), then (I3) into (B3.4):

$$\mathbb{E}_{i(t)}[D_t] \le D_{t-1} - 2\gamma G_{t-1} + 2\gamma^2 \Big(\frac{1}{n} S_{t-1} + B_{t-1} \Big) \le D_{t-1} - 2\gamma(1 - \gamma R^2)\, G_{t-1} + \frac{2\gamma^2}{n} M_{t-1}. \tag{B3.9}$$

Step 2 of the book (memory recursion). Inserting (I2) into (B3.5):

$$\mathbb{E}_{i(t)}[M_t] \le \Big(1 - \frac{1}{n}\Big) M_{t-1} + R^2\, G_{t-1}. \tag{B3.10}$$

Combination. Adding (B3.9) and Δ times (B3.10), for any $\gamma, \Delta > 0$,

$$\mathbb{E}_{i(t)}[V_t] \le D_{t-1} - 2\gamma \Big(1 - \gamma R^2 - \frac{R^2 \Delta}{2\gamma}\Big) G_{t-1} + \Big[\frac{2\gamma^2}{n\Delta} + 1 - \frac{1}{n}\Big] \Delta M_{t-1}. \tag{B3.11}$$

With $\Delta = 3\gamma^2$ and $\gamma = \frac{1}{4R^2}$: $1 - \gamma R^2 - \frac{R^2\Delta}{2\gamma} = 1 - \frac{1}{4} - \frac{3}{8} = \frac{3}{8}$ and $\frac{2\gamma^2}{n\Delta} = \frac{2}{3n}$, so the memory coefficient is $1 - \frac{1}{n} + \frac{2}{3n} = 1 - \frac{1}{3n}$ and

$$\mathbb{E}_{i(t)}[V_t] \le D_{t-1} - \frac{3\gamma}{4}\, G_{t-1} + \Big(1 - \frac{1}{3n}\Big) \Delta M_{t-1}. \tag{B3.12}$$

(I4) Strong convexity. By Lemma B3.2(b),

$$G_{t-1} = (\theta_{t-1} - \eta_*)^\top F'(\theta_{t-1}) \;\ge\; \mu\, D_{t-1}. \tag{I4}$$

Since the coefficient $\frac{3\gamma}{4}$ of G_{t-1} in (B3.12) is positive, $-\frac{3\gamma}{4} G_{t-1} \le -\frac{3\gamma\mu}{4} D_{t-1} = -\frac{3\mu}{16R^2} D_{t-1}$, hence

$$\mathbb{E}_{i(t)}[V_t] \le \Big(1 - \frac{3\mu}{16R^2}\Big) D_{t-1} + \Big(1 - \frac{1}{3n}\Big) \Delta M_{t-1}. \tag{B3.13}$$

(I5) The min step. For $p, q \geq 0$ and $x, y \geq 0$, $px+qy \leq \max\{p,q\}(x+y)$. With $\rho_D := 1-\frac{3\mu}{16R^2} \in (0,1)$ (recall $\mu \leq R^2$) and $\rho_M := 1-\frac{1}{3n} \in (0,1)$:

$$\mathbb{E}_{i(t)}[V_t] \;\leq\; \Big(1-\min\Big\{\frac{1}{3n}, \frac{3\mu}{16R^2}\Big\}\Big)V_{t-1} =: \rho_{\text{bound}}\, V_{t-1}. \tag{I5}$$

Taking expectations and iterating (tower property; V_0 is deterministic): $\mathbb{E}[V_t] \leq \rho_{\text{bound}}^t V_0$, and $\mathbb{E}\,\|\theta_t - \eta_*\|^2 \leq \mathbb{E}[V_t]$ because $\Delta M_t \geq 0$.

(I6) Initialisation (Lipschitz continuity of each f_i'). With $z_i^{(0)} = f_i'(\theta_0)$ one has $M_0 = S_0$ and

$$M_0 = \sum_{i=1}^{n} \big\|f_i'(\theta_0) - f_i'(\eta_*)\big\|^2 \;\leq\; n\,R^4\,\|\theta_0 - \eta_*\|^2\,, \tag{I6}$$

so that $V_0 = D_0 + \frac{3}{16R^4}M_0 \leq \big(1+\frac{3n}{16}\big)D_0 \leq \big(1+\frac{n}{4}\big)D_0$, which is (B3.3). (The last step, $1+\frac{3n}{16} \leq 1+\frac{n}{4}$, is a cosmetic loss by a factor at most $4/3$ on the constant, not on the rate.) □

Remark B3.7 (the constant R^2 for logistic regression). For $f_i(\theta) = \ell(y_i x_i^\top \theta) + \frac{\lambda}{2}\|\theta\|^2$ with $\ell(u) = \log(1+e^{-u})$, $y_i \in \{-1,1\}$, one has $\ell''(u) = \sigma(u)\sigma(-u) \in (0, \frac{1}{4}]$ where $\sigma(u) = 1/(1+e^{-u})$, so $f_i''(\theta) = \ell''(y_i x_i^\top\theta)\, x_i x_i^\top + \lambda I \preceq \big(\frac{\|x_i\|^2}{4} + \lambda\big)I$. Hence f_i is L_i-smooth with $L_i = \|x_i\|^2/4 + \lambda$, and the smallest constant valid for all i is

$$R^2 := \max_i \Big(\frac{\|x_i\|^2}{4} + \lambda\Big).$$

With features bounded by R_x in ℓ_2-norm, the book's constant $R^2 := R_x^2$ ("features bounded by R in ℓ_2-norm") is therefore a valid smoothness constant whenever $\lambda \leq \frac{3}{4}R_x^2$, but a loose one: it ignores the factor $\frac{1}{4}$ of the logistic second derivative (and λ), so that it exceeds $\max_i L_i$ by a factor close to 4 when $\lambda \ll R_x^2$. Since every constant of the proof is expressed through R^2 ($\gamma = 1/(4R^2)$, $\Delta = 3/(16R^4)$, the rate $3\mu/(16R^2)$), using the loose value would divide the step and the guaranteed rate by about 4; the experiments must use $R^2 = \max_i(\|x_i\|^2/4 + \lambda)$. The global strong convexity constant is $\mu = \lambda$ (the logistic part is convex), which is much smaller than the curvature actually met along the trajectory when λ is small compared with the curvature of the data term: in the experiments of Section B3.7, the ratio μ_{eff}/μ of Definition B3.8 (geometric means over passes 2–20) lies between 1.1 and 2.3 for $\kappa \leq 10^2$ and reaches 81 at $\kappa = 10^5$; see Section B3.6.

B3.4 Tightness ratios and attribution of the gap

Definition B3.8 (tightness ratios). *For a state $(\theta, z) = (\theta_{t-1}, z^{(t-1)})$ with $\theta \neq \eta_*$, define, with the quantities of Section B3.2,*

$$r_{\text{sq}} := \frac{\Omega}{\frac{2}{n}S + 2B} \;\; \textit{(I1)}, \qquad r_{\text{coco}} := \frac{S/n}{R^2 G} = \frac{L_{\text{eff}}}{R^2}, \quad L_{\text{eff}} := \frac{S}{nG} \;\; \textit{(I2)},$$

$$r_{\text{var}} := \frac{B}{M/n} = 1 - \frac{n\,\|\bar z\|^2}{M} \;\; \textit{(I3)}, \qquad r_{\text{sc}} := \frac{\mu D}{G} = \frac{\mu}{\mu_{\text{eff}}}, \quad \mu_{\text{eff}} := \frac{G}{D} \;\; \textit{(I4)},$$

$$r_{\min} := \frac{\big(1-\frac{3\mu}{16R^2}\big)D + \big(1-\frac{1}{3n}\big)\Delta M}{\big(1-\min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}\big)(D+\Delta M)} \;\; \textit{(I5)}, \qquad r_{\text{init}} := \frac{\sum_i \|f_i'(\theta_0) - f_i'(\eta_*)\|^2}{nR^4\,\|\theta_0 - \eta_*\|^2} \;\; \textit{(I6)},$$

with the convention $r_{\text{var}} := 1$ if $M = 0$ (then (I3) is the equality $0 \leq 0$; no convention is needed for r_{sq}, since $S > 0$ when $\theta \neq \eta_*$, Lemma B3.9(i)). Each ratio is the smaller side divided by the larger side of the corresponding inequality of Section B3.3 (for (I4), $G \geq \mu D$, this is $\mu D/G$); $r_{\text{sq}}, r_{\text{coco}}, r_{\text{var}}, r_{\text{sc}}$ depend on the state only (not on γ, Δ), $r_{\min}$ also depends on (Δ, n, μ, R^2).

Lemma B3.9 (ranges of the tightness ratios). *Let $\theta \neq \eta_*$. Then:*

(i) $G > 0$, $S > 0$ *and* $0 < \mu \leq \mu_{\text{eff}} \leq L_{\text{eff}} \leq R^2$; *hence* $r_{\text{sc}} \in (0, 1]$ *and* $r_{\text{coco}} \in (0, 1]$.

(ii) $0 \leq r_{\text{var}} \leq 1$; *for* $M > 0$, $r_{\text{var}} = 0$ *iff the memory errors* $z_i - f_i'(\eta_*)$ *are the same (nonzero) vector for all* i; *if* $M = 0$ *then* $B = 0$ *and* $r_{\text{var}} = 1$ *by convention.*

(iii) $S/n + B > 0$, $\Omega > 0$ *and* $r_{\text{sq}} \in (0, 1]$; *moreover*

$$r_{\text{sq}} = \frac{1}{2} + \frac{C}{S/n + B}, \qquad 1 - r_{\text{sq}} = \frac{\frac{1}{n}\sum_i \|a_i - b_i\|^2}{2(S/n + B)},$$

so $r_{\text{sq}} = 1$ *iff* $a_i = b_i$ *for all* i, *and* $r_{\text{sq}} = \frac{1}{2}$ *iff* $C = 0$ *(gradient changes* a_i *and memory errors* b_i *uncorrelated under* $i(t)$*).*

(iv) $r_{\min} \in (0, 1]$, *with* $r_{\min} = 1$ *iff* $\kappa = 9n/16$ *(the two coefficients coincide) or* $M = 0$ *and* $\kappa \geq 9n/16$ *(the Lyapunov mass sits on the parameter component, which is then the slower one). Since* $D > 0$, *the mass never sits on the memory component alone.*

(v) If $\theta_0 \neq \eta_*$ *then* $r_{\text{init}} \in (0, 1]$.

Proof. (i) $G \geq \mu D > 0$ by (I4), which is $\mu_{\text{eff}} \geq \mu$. By Jensen, $\|F'(\theta)\|^2 = \left\|\frac{1}{n}\sum_i a_i\right\|^2 \leq \frac{1}{n}\sum_i \|a_i\|^2 = S/n$, and $F'(\theta) \neq 0$ by Lemma B3.2(a), so $S > 0$. By Cauchy–Schwarz $G \leq \|F'(\theta)\|\sqrt{D}$, hence $S/n \geq \|F'(\theta)\|^2 \geq G^2/D$, i.e. $L_{\text{eff}} = S/(nG) \geq G/D = \mu_{\text{eff}}$. Finally $L_{\text{eff}} \leq R^2$ is (I2). (ii) By (B3.6), $0 \leq B \leq M/n$, and $B = 0$ iff $Z = f_{i(t)}'(\eta_*) - z_{i(t)}^{(t-1)}$ is almost surely equal to its mean, i.e. $f_i'(\eta_*) - z_i$ does not depend on i. If $M = 0$ then $z_i = f_i'(\eta_*)$ for all i and $B = 0$. (iii) $S/n + B \geq S/n > 0$ by (i). The two formulas follow from Lemma B3.4(d) and from the identity in (I1); $r_{\text{sq}} \leq 1$ is (I1) and $r_{\text{sq}} \geq 0$ is $\Omega \geq 0$ (equivalently $|C| \leq \sqrt{(S/n)B} \leq \frac{1}{2}(S/n + B)$ by Lemma B3.4(e)). Suppose $\Omega = 0$; then $a_i + b_i = f_i'(\theta) - z_i + \bar{z} = 0$ for all i; averaging over i gives $F'(\theta) = 0$, so $\theta = \eta_*$, which is excluded. Hence $\Omega > 0$ and $r_{\text{sq}} > 0$. (iv) The numerator is $\rho_D D + \rho_M \Delta M$ with the coefficients $\rho_D = 1 - \frac{3\mu}{16R^2}$ and $\rho_M = 1 - \frac{1}{3n}$ of (I5), both in $(0, 1)$, and the denominator is $\max\{\rho_D, \rho_M\}(D + \Delta M)$; the ratio is in $(0, 1]$, with equality iff $\rho_D = \rho_M$ (i.e. $16R^2 = 9n\mu$, $\kappa = 9n/16$) or the component with the strictly smaller coefficient has zero mass. Since $D > 0$ (because $\theta \neq \eta_*$), the latter can only be the memory component: $M = 0$ and $\rho_M \leq \rho_D$, i.e. $\frac{1}{3n} \geq \frac{3\mu}{16R^2}$, i.e. $\kappa \geq 9n/16$. (v) $S_0 > 0$ by (i) and $S_0 \leq nR^4 D_0$ is (I6). □

Proposition B3.10 (exact contraction in terms of the ratios). *Let* $\theta_{t-1} \neq \eta_*$ *and* $M_{t-1} > 0$, *and let* $\gamma, \Delta > 0$. *Then*

$$\mathbb{E}_{i(t)}[V_t] = c_D\, D_{t-1} + c_M\, \Delta M_{t-1}, \qquad \begin{aligned} c_D &= 1 - 2\gamma\,\mu_{\text{eff}}\Big(1 - \gamma\, r_{\text{sq}} L_{\text{eff}} - \frac{L_{\text{eff}}\Delta}{2\gamma}\Big), \\ c_M &= 1 - \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\, r_{\text{sq}}\, r_{\text{var}}\Big), \end{aligned} \tag{B3.14}$$

where $\mu_{\text{eff}}, L_{\text{eff}}, r_{\text{sq}}, r_{\text{var}}$ *are evaluated at the state* $(\theta_{t-1}, z^{(t-1)})$. *For the textbook constants* $\gamma = \frac{1}{4R^2}$, $\Delta = 3\gamma^2$,

$$c_D = 1 - \frac{\mu_{\text{eff}}}{2R^2}\Big(1 - \frac{L_{\text{eff}}}{R^2}\Big(\frac{r_{\text{sq}}}{4} + \frac{3}{8}\Big)\Big), \qquad c_M = 1 - \frac{1}{n}\Big(1 - \frac{2}{3}\, r_{\text{sq}}\, r_{\text{var}}\Big). \tag{B3.15}$$

Substituting the worst-case values $r_{\text{sq}} = r_{\text{var}} = 1$, $L_{\text{eff}} = R^2$, $\mu_{\text{eff}} = \mu$ *gives exactly the two coefficients* $1 - \frac{3\mu}{16R^2}$ *and* $1 - \frac{1}{3n}$ *of* (B3.13).

Proof. By definition of the ratios, $S/n = L_{\text{eff}}G = L_{\text{eff}}\mu_{\text{eff}}D$, $B = r_{\text{var}}M/n$ and $\Omega = r_{\text{sq}}(2S/n + 2B) = 2r_{\text{sq}}(L_{\text{eff}}\mu_{\text{eff}}D + r_{\text{var}}M/n)$. Inserting into (B3.7):

$$\mathbb{E}_{i(t)}[V_t] = D\big[1 - 2\gamma\mu_{\text{eff}} + 2\gamma^2 r_{\text{sq}}L_{\text{eff}}\mu_{\text{eff}} + \Delta L_{\text{eff}}\mu_{\text{eff}}\big] + \Delta M\Big[1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}r_{\text{sq}}r_{\text{var}}\Big],$$

and factoring $2\gamma\mu_{\text{eff}}$ in the first bracket gives c_D. With $\gamma = 1/(4R^2)$, $\Delta = 3\gamma^2$: $2\gamma\mu_{\text{eff}} = \mu_{\text{eff}}/(2R^2)$, $\gamma L_{\text{eff}} = L_{\text{eff}}/(4R^2)$, $L_{\text{eff}}\Delta/(2\gamma) = 3\gamma L_{\text{eff}}/2 = 3L_{\text{eff}}/(8R^2)$ and $2\gamma^2/\Delta = 2/3$. □

Proposition B3.10 says that the book's proof *is* the substitution of worst-case values into an exact formula: the two coefficients of (B3.13) are c_D and c_M at $(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}}, r_{\text{var}}) = (1, R^2, \mu, 1)$, and (I5) replaces the weighted mean $(c_D D + c_M \Delta M)/V$ by $\max\{c_D, c_M\}$. This gives a canonical way of attributing the gap $\rho_{\text{bound}} - \rho_{\text{exact}}$ to the individual inequalities: undo them one at a time.

Proposition B3.11 (attribution of the gap)**.** *Let $\theta_{t-1} \neq \eta_*$, $M_{t-1} > 0$, $\gamma, \Delta > 0$ with $\alpha := 1 - \gamma R^2 - \frac{R^2\Delta}{2\gamma} > 0$, and write D, M, G, S, B, Ω, V for the state quantities. Define the coefficient functions*

$$c_D(r, L, m) := 1 - 2\gamma m\Big(1 - \gamma r L - \frac{L\Delta}{2\gamma}\Big), \qquad c_M(r, v) := 1 - \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}rv\Big),$$

and

$$\rho_{\text{bound}}(\gamma, \Delta) := \max\{c_D(1, R^2, \mu),\, c_M(1, 1)\} = 1 - \min\Big\{2\gamma\mu\alpha,\ \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\Big)\Big\},$$

which is the worst-case coefficient of Proposition B3.14 (it is < 1 iff $2\gamma^2 < \Delta$) and equals the ρ_{bound} of (I5) for the textbook constants; in the rest of this proposition we write ρ_{bound} for $\rho_{\text{bound}}(\gamma, \Delta)$.

Order A (I4 → I2 → I3 → I1 → I5; the min is kept until the end).

$$\begin{aligned}
\rho_0^A &:= \max\{c_D(1, R^2, \mu),\, c_M(1, 1)\} = \rho_{\text{bound}},\\
\rho_1^A &:= \max\{c_D(1, R^2, \mu_{\text{eff}}),\, c_M(1, 1)\} = \max\Big\{1 - 2\gamma\alpha\frac{G}{D},\ 1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\Big\},\\
\rho_2^A &:= \max\{c_D(1, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(1, 1)\} = \max\Big\{1 - 2\gamma\frac{G}{D} + (2\gamma^2 + \Delta)\frac{S}{nD},\ 1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\Big\},\\
\rho_3^A &:= \max\{c_D(1, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(1, r_{\text{var}})\} = \max\Big\{1 - 2\gamma\frac{G}{D} + (2\gamma^2 + \Delta)\frac{S}{nD},\ 1 - \frac{1}{n} + \frac{2\gamma^2 B}{\Delta M}\Big\},\\
\rho_4^A &:= \max\{c_D(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(r_{\text{sq}}, r_{\text{var}})\}\\
&\quad = \max\Big\{1 - 2\gamma\frac{G}{D} + \frac{2\gamma^2 r_{\text{sq}}S}{nD} + \frac{\Delta S}{nD},\ 1 - \frac{1}{n} + \frac{2\gamma^2 r_{\text{sq}}B}{\Delta M}\Big\},\\
\rho_5^A &:= \frac{c_D(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}})\, D + c_M(r_{\text{sq}}, r_{\text{var}})\, \Delta M}{D + \Delta M} = \rho_{\text{exact}}(t),
\end{aligned}$$

with $r_{\text{sq}} = \Omega/(2S/n + 2B)$.

Order B (I5 $\to$ I4 $\to$ I2 $\to$ I3 $\to$ I1; reverse proof order, the min is undone first).

$$\rho_0^B := \rho_{\text{bound}},$$
$$\rho_1^B := \frac{(1-2\gamma\mu\alpha)D + \big(1-\frac{1}{n}+\frac{2\gamma^2}{n\Delta}\big)\Delta M}{V} \quad \big(= r_{\min}\,\rho_{\text{bound}} \text{ for the textbook constants}\big),$$
$$\rho_2^B := \frac{D - 2\gamma\alpha G + \big(1-\frac{1}{n}+\frac{2\gamma^2}{n\Delta}\big)\Delta M}{V},$$
$$\rho_3^B := \frac{D - 2\gamma G + (2\gamma^2+\Delta)\frac{S}{n} + \frac{2\gamma^2}{n}M + \big(1-\frac{1}{n}\big)\Delta M}{V},$$
$$\rho_4^B := \frac{D - 2\gamma G + (2\gamma^2+\Delta)\frac{S}{n} + 2\gamma^2 B + \big(1-\frac{1}{n}\big)\Delta M}{V},$$
$$\rho_5^B := \frac{D - 2\gamma G + \gamma^2\Omega + \Delta\frac{S}{n} + \big(1-\frac{1}{n}\big)\Delta M}{V} = \rho_{\text{exact}}(t).$$

Then both sequences are nonincreasing:

$$\rho_{\text{bound}} = \rho_0^A \ge \rho_1^A \ge \rho_2^A \ge \rho_3^A \ge \rho_4^A \ge \rho_5^A = \rho_{\text{exact}}(t), \qquad \rho_{\text{bound}} = \rho_0^B \ge \rho_1^B \ge \cdots \ge \rho_5^B = \rho_{\text{exact}}(t),$$

and consequently the nonnegative increments $\delta_k := \rho_{k-1} - \rho_k$ *(additive attribution) and, when* $\rho_{\text{exact}}(t) > 0$, $\nu_k := \log(\rho_{k-1}/\rho_k)$ *(attribution in nats per iteration) satisfy*

$$\sum_{k=1}^{5}\delta_k = \rho_{\text{bound}} - \rho_{\text{exact}}(t), \qquad \sum_{k=1}^{5}\nu_k = \log\frac{1}{\rho_{\text{exact}}(t)} - \log\frac{1}{\rho_{\text{bound}}}, \qquad \prod_{k=1}^{5}\frac{\rho_k}{\rho_{k-1}} = \frac{\rho_{\text{exact}}(t)}{\rho_{\text{bound}}}.$$

In order A the increments are attributed to (I4), (I2), (I3), (I1), (I5) in this order; in order B to (I5), (I4), (I2), (I3), (I1).

Proof. The closed forms follow from $\mu_{\text{eff}} = G/D$, $L_{\text{eff}}\mu_{\text{eff}} = S/(nD)$, $r_{\text{var}}M/n = B$ and $r_{\text{sq}}(2S/n + 2B) = \Omega$, exactly as in the proof of Proposition B3.10; e.g. $c_D(1, L_{\text{eff}}, \mu_{\text{eff}}) = 1 - 2\gamma G/D + 2\gamma^2 L_{\text{eff}}\mu_{\text{eff}} + \Delta L_{\text{eff}}\mu_{\text{eff}} = 1 - 2\gamma G/D + (2\gamma^2+\Delta)S/(nD)$, and $c_D(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}})D + c_M(r_{\text{sq}}, r_{\text{var}})\Delta M$ equals the right-hand side of (B3.7).

Monotonicity. For $r \le 1$ and $L \le R^2$ the bracket in c_D satisfies $1 - \gamma r L - \frac{L\Delta}{2\gamma} \ge 1 - \gamma R^2 - \frac{R^2\Delta}{2\gamma} = \alpha > 0$, so $m \mapsto c_D(r, L, m)$ is nonincreasing; moreover c_D is nondecreasing in L and in r, and c_M is nondecreasing in r and v. Lemma B3.9 gives $\mu \le \mu_{\text{eff}}$, $L_{\text{eff}} \le R^2$, $r_{\text{var}} \le 1$, $r_{\text{sq}} \le 1$, so each substitution lowers (weakly) the coefficient it touches; max and the weighted mean $(x, y) \mapsto (xD + y\Delta M)/V$ are monotone in both arguments; and finally $\max\{x, y\} \ge (xD + y\Delta M)/V$ since $D, \Delta M \ge 0$. This proves both chains; the telescoping identities are immediate. □

Remark B3.12 (reading the two attribution orders). The attribution is path dependent, as for any decomposition of a difference of a nonlinear function. Order B (undo from the end) attributes to each inequality its full effect on the weighted contraction; order A keeps the max structure, so that undoing (I4) and (I2), which touch c_D alone, has *no* effect as long as the memory coefficient c_M is the larger one (regime $\kappa < 9n/16$ for the textbook constants), and undoing (I3), which touches c_M alone, has no effect as long as c_D is the larger one. Undoing (I1) is different: r_{sq} enters both coefficients (through $2\gamma^2 r_{\text{sq}} L_{\text{eff}}\mu_{\text{eff}}$ in c_D and $2\gamma^2 r_{\text{sq}} r_{\text{var}}/\Delta$ in c_M), so the step $\rho_3^A \to \rho_4^A$ lowers both of them and has an effect in both regimes, except in the degenerate cases $r_{\text{sq}} = 1$, or $r_{\text{var}} = 0$ with c_M the larger coefficient. Reporting both orders therefore shows, in addition to the individual contributions, which component (D or M) is the bottleneck of the bound. Since ρ is close to 1, $\nu_k \approx \delta_k$; per effective pass the natural unit is $n\nu_k$.

Remark B3.13 (the square-norm step discards the control variate). Suppose the memory is fresh, $z_i = f_i'(\theta)$ for all i (e.g. at $t = 1$ with the book's initialization, or approximately when the iterates move slowly). Then $\omega_t = F'(\theta)$ is deterministic, $\Omega = \|F'(\theta)\|^2$, $M = S$, $B = S/n - \|F'(\theta)\|^2$ and

$$r_{\mathrm{sq}} = \frac{\|F'(\theta)\|^2}{4S/n - 2\|F'(\theta)\|^2} \le \frac{1}{2}, \qquad \text{with } r_{\mathrm{sq}} \ll \frac{1}{2} \text{ when } \frac{S}{n} \gg \left\|F'(\theta)\right\|^2,$$

i.e. when the per-sample gradient changes a_i are diverse. In general, if index i was last drawn at time $u_i \le t-1$, then $z_i = z_i^{(u_i)} = f_i'(\theta_{u_i-1})$ by (B3.1); writing $s_i := u_i - 1 \in \{0, \dots, t-2\}$, and $s_i := 0$ for the indices not yet drawn (with the book's initialization $z_i^{(0)} = f_i'(\theta_0)$), one has $z_i = f_i'(\theta_{s_i})$ in all cases, and

$$C = \left\langle F'(\theta), \bar{z} \right\rangle - \frac{1}{n}\sum_{i=1}^{n} \left\langle a_i(\theta), a_i(\theta_{s_i}) \right\rangle, \qquad a_i(\theta') := f_i'(\theta') - f_i'(\eta_*) :$$

the correlation between the current gradient change and the stored one is exactly what the control variate $-z_{i(t)} + \bar{z}$ exploits, and (I1) discards it. Since $F'(\theta) = \frac{1}{n}\sum_i a_i(\theta)$ and $\bar{z} = \frac{1}{n}\sum_i a_i(\theta_{s_i})$ (Lemma B3.2(a)), this is

$$C = -\frac{1}{n}\sum_{i=1}^{n} \left\langle a_i(\theta) - F'(\theta), a_i(\theta_{s_i}) - \bar{z} \right\rangle,$$

minus the empirical covariance over i of the current and the stored gradient changes, whose sign is not determined in general: $r_{\mathrm{sq}} < \frac{1}{2}$ iff this covariance is positive. Moreover $\left|r_{\mathrm{sq}} - \frac{1}{2}\right| = |C| / (S/n + B) \le \sqrt{x}/(1+x)$ with $x := B/(S/n)$, by Lemma B3.4(e), so that $r_{\mathrm{sq}} \ll \frac{1}{2}$ requires both a strong positive correlation and B comparable with S/n. One therefore expects two regimes. When the memory is nearly fresh, i.e. when the iterates move little during the $\approx n$ iterations that refresh it (for the textbook step, roughly when $2n\gamma\mu_{\mathrm{eff}} \lesssim 1$, which requires $\kappa \gtrsim n/2$), the correlation is strong, $r_{\mathrm{sq}} \le \frac{1}{2}$ (possibly $\ll \frac{1}{2}$) and (I1) loses at least a factor 2 in the γ^2 terms. When the iterates move fast compared with this refresh, the stored errors dominate ($x \gg 1$), the covariance may have either sign, and r_{sq} stays close to $\frac{1}{2}$ on either side, since $\sqrt{x}/(1+x) \le 1/\sqrt{x}$: (I1) still costs a factor close to 2. The experiments show both regimes (Section B3.7).

B3.5 The best constants of the proof: general (γ, Δ)

The inequalities (I1)–(I3) and the identities do not depend on γ or Δ, so (B3.11) holds for all $\gamma, \Delta > 0$. We now optimize the two constants.

Proposition B3.14 (certified rate for general (γ, Δ)). *Let $\gamma, \Delta > 0$ satisfy*

$$\alpha(\gamma, \Delta) := 1 - \gamma R^2 - \frac{R^2 \Delta}{2\gamma} > 0 \qquad \text{and} \qquad \frac{2\gamma^2}{\Delta} < 1.$$

Then for all $t \ge 1$, almost surely,

$$\mathbb{E}_{i(t)}[V_t] \le \left(1 - \min\left\{2\gamma\mu\,\alpha(\gamma, \Delta),\ \frac{1}{n}\left(1 - \frac{2\gamma^2}{\Delta}\right)\right\}\right) V_{t-1}, \tag{B3.16}$$

and both arguments of the min lie in $(0, \frac{1}{2}]$ and $(0, \frac{1}{n})$ respectively.

Proof. In (B3.11) the coefficient $2\gamma\alpha$ of G_{t-1} is positive, so (I4) gives $-2\gamma\alpha G_{t-1} \le -2\gamma\mu\alpha D_{t-1}$ and $\mathbb{E}_{i(t)}[V_t] \le (1-2\gamma\mu\alpha)D_{t-1} + (1-\frac{1}{n}(1-\frac{2\gamma^2}{\Delta}))\Delta M_{t-1}$. Since $\alpha > 0$ forces $\gamma R^2 < 1$, we have $2\gamma\mu\alpha \le 2\gamma\mu(1-\gamma R^2) \le 2x(1-x) \le \frac{1}{2}$ with $x = \gamma R^2$ (using $\mu \le R^2$), and $0 < \frac{1}{n}(1-\frac{2\gamma^2}{\Delta}) < \frac{1}{n}$. Both coefficients are thus in $[\frac{1}{2}, 1)$ and $(1-\frac{1}{n}, 1)$, and (I5) concludes. □

Corollary B3.15 (parametrization $\Delta = c\gamma^2$)**.** *Write $\Delta = c\gamma^2$ with $c > 2$. Then $\alpha = 1-\gamma R^2(1+\frac{c}{2})$, the condition $\alpha > 0$ reads $\gamma < \frac{2}{R^2(2+c)}$, and (B3.16) becomes*

$$\begin{aligned} &\mathbb{E}_{i(t)}[V_t] \le \big(1 - \min\{\tau_D(\gamma, c), \tau_M(c)\}\big)V_{t-1}, \\ &\tau_D(\gamma, c) := 2\gamma\mu\Big(1 - \gamma R^2\Big(1 + \frac{c}{2}\Big)\Big), \qquad \tau_M(c) := \frac{1 - 2/c}{n}. \end{aligned} \tag{B3.17}$$

For fixed c, $\gamma \mapsto \tau_D(\gamma, c)$ is a concave parabola, maximal at

$$\gamma^\star(c) = \frac{1}{R^2(2+c)}, \qquad \tau_D(\gamma^\star(c), c) = \frac{\mu}{R^2(2+c)} = \frac{1}{\kappa(2+c)},$$

while τ_M does not depend on γ. Hence the best rate that (B3.17) certifies for a given c is

$$\tau(c) := \min\Big\{\frac{1}{\kappa(2+c)},\ \frac{c-2}{cn}\Big\}.$$

Proof. $\alpha = 1 - \gamma R^2 - \frac{R^2 c\gamma^2}{2\gamma} = 1 - \gamma R^2(1+\frac{c}{2})$ and $\frac{2\gamma^2}{\Delta} = \frac{2}{c}$. Then $\partial_\gamma \tau_D = 2\mu(1 - 2\gamma R^2(1+\frac{c}{2}))$ vanishes at $\gamma^\star = \frac{1}{2R^2(1+c/2)} = \frac{1}{R^2(2+c)}$, where $1 - \gamma^\star R^2(1+\frac{c}{2}) = \frac{1}{2}$. For fixed c the min of a function of γ and a constant is maximized by maximizing the function. □

Proposition B3.16 (optimal constants of the proof structure)**.** *Let $s := n/\kappa = n\mu/R^2$. The maximum over $c > 2$ and admissible γ of the rate $\min\{\tau_D, \tau_M\}$ in (B3.17) is attained at*

$$\begin{aligned} c^\star &= \frac{s + \sqrt{s^2+16}}{2}, \qquad \Delta^\star = c^\star(\gamma^\star)^2, \\ \gamma^\star &= \frac{1}{R^2(2+c^\star)} = \frac{2}{R^2\big(4 + s + \sqrt{s^2+16}\big)} = \frac{2}{4R^2 + n\mu + \sqrt{n^2\mu^2 + 16R^4}}, \end{aligned}$$

and equals

$$\tau^\star = \frac{1}{\kappa(2+c^\star)} = \frac{2}{4\kappa + n + \sqrt{n^2 + 16\kappa^2}}, \qquad \frac{1}{n+4\kappa} \ \le\ \tau^\star \ \le\ \frac{4-2\sqrt{2}}{n+4\kappa} \approx \frac{1.172}{n+4\kappa}.$$

In particular $\tau^\star \sim \frac{1}{4\kappa}$, $\gamma^\star \sim \frac{1}{4R^2}$ and $c^\star \to 2$ when $n/\kappa \to 0$, and $\tau^\star \sim \frac{1}{n}$, $\gamma^\star \sim \frac{1}{n\mu}$ and $c^\star \sim n/\kappa$ when $n/\kappa \to \infty$.

Proof. By Corollary B3.15 we must maximize $\tau(c) = \min\{\frac{1}{\kappa(2+c)}, \frac{c-2}{cn}\}$ over $c > 2$. The first argument is decreasing in c, the second is increasing from 0 (at $c = 2$) to $\frac{1}{n}$; the maximum of the min is therefore attained where they cross: $cn = \kappa(c+2)(c-2)$, i.e. $\kappa c^2 - nc - 4\kappa = 0$, whose positive root is $c^\star = \frac{n + \sqrt{n^2+16\kappa^2}}{2\kappa} = \frac{s + \sqrt{s^2+16}}{2}$; it satisfies $c^\star > 2$ for $s > 0$ because $\sqrt{s^2+16} > 4 - s$. Then $2 + c^\star = \frac{4 + s + \sqrt{s^2+16}}{2}$, which gives $\tau^\star$ and $\gamma^\star$; the last expression for $\gamma^\star$ follows from $R^2 s = n\mu$ and $R^2\sqrt{s^2+16} = \sqrt{n^2\mu^2 + 16R^4}$. For the bounds, $\sqrt{n^2+16\kappa^2} \le n + 4\kappa$ gives $\tau^\star \ge \frac{2}{2(n+4\kappa)}$, and $\sqrt{n^2+16\kappa^2} \ge \frac{n+4\kappa}{\sqrt{2}}$ (since $(a+b)^2 \le 2(a^2+b^2)$) gives $\tau^\star \le \frac{2}{(1+1/\sqrt{2})(n+4\kappa)} = \frac{4-2\sqrt{2}}{n+4\kappa}$. The limits are read off the formulas ($c^\star = 2 + \frac{s}{2} + O(s^2)$ as $s \to 0$; $c^\star = s + \frac{4}{s} + O(s^{-3})$ as $s \to \infty$). □

These are the constants of Hofmann et al. Proposition B3.16 is not new: $\tau^\star$ and $\gamma^\star$ are exactly the optimal rate and step of [40, Theorem 1 and Corollary 1] for SAGA. *Dictionary* (their symbol $\to$ ours): $w, w^* \to \theta, \eta_*$; memory $\alpha_i \to z_i$, $\bar\alpha \to \bar z$; direction $g_i(w) = f_i'(w) - \alpha_i + \bar\alpha \to \omega_t$; $f \to F$ and $f^\delta(w) = f(w) - f(w^*) \to F(\theta) - F(\eta_*)$; step $\gamma \to \gamma$; L (Lipschitz constant of every f_i') $\to R^2$; $\mu \to \mu$ and $\kappa := L/\mu \to \kappa$; $q \to 1$ (SAGA is the case $q = 1$ of their Definition 1 of a "uniform q-memorization algorithm", in which every memory slot is refreshed with probability q/n); $a := 4L\gamma \to 4R^2\gamma$; $K := \frac{4qL}{n\mu} \to \frac{4\kappa}{n} = \frac{4}{s}$; the weight $\sigma \in [0,1]$ of their Lyapunov function (see below) $\to R^2\Delta/\gamma$; their rate ρ, whose contraction factor is $1-\rho$, $\to$ our rate τ (their ρ is *not* a contraction factor such as our ρ_{bound}). Their S, their α_i and the constant $c \in (0,1]$ of their Lemma 3 (where $\rho = c\mu\gamma$) are unrelated to our S_{t-1}, $\alpha(\gamma,\Delta)$ and c. Their standing assumption (each f_i is a convex loss plus a μ-strongly convex regularizer and has an L-Lipschitz gradient) is stronger than Assumption B3.1, but the proofs of their Lemmas 1–3 and Theorem 1 use only the μ-strong convexity of f (their eq. (5)), the convexity and L-smoothness of each f_i (their eqs. (7), (8), (10)) and $f'(w^*) = 0$, so that their Theorem 1 holds under Assumption B3.1. It states, for a uniform q-memorization algorithm and any step $\gamma = a/(4L)$ with $a < 1$, a geometric rate of at least $1 - \rho(\gamma)$ with $\rho(\gamma) = \frac{q}{n} \cdot \frac{1-a}{1-a/2}$ if $\gamma \geq \gamma^\star(K)$ and $\rho(\gamma) = \mu\gamma$ otherwise, where $\gamma^\star(K) := a^\star(K)/(4L)$, $a^\star(K) := \frac{2K}{1+K+\sqrt{1+K^2}}$ and $K := \frac{4qL}{n\mu}$; their Corollary 1 states that ρ is maximized at $\gamma = \gamma^\star(K)$, with $\rho^\star(K) = \frac{q}{n}\big[\frac{2}{1+K+\sqrt{1+K^2}}\big]$. With $q = 1$ and $K = 4/s$,

$$a^\star(K) = \frac{8}{4+s+\sqrt{s^2+16}}, \qquad \gamma^\star(K) = \frac{a^\star(K)}{4R^2} = \frac{2}{R^2(4+s+\sqrt{s^2+16})} = \gamma^\star,$$
$$\rho^\star(K) = \frac{1}{n} \cdot \frac{2}{1+\frac{4}{s}+\sqrt{1+\frac{16}{s^2}}} = \frac{2s}{n(4+s+\sqrt{s^2+16})} = \frac{2}{\kappa(4+s+\sqrt{s^2+16})} = \tau^\star,$$

and moreover $c^\star = 4/a^\star(K) - 2$, while their best weight for a given step, $\sigma^* = 1 - 2L\gamma$ (eq. (33) of their appendix), takes at $\gamma = \gamma^\star(K)$ the value $1 - a^\star(K)/2 = c^\star/(2+c^\star) = R^2\Delta^\star/\gamma^\star$: the optimal pairs (γ,σ) and $(\gamma^\star,\Delta^\star)$ correspond exactly under the dictionary. Their Corollary 3 ("Choosing $\gamma = \frac{a}{4L}$ with $a < 1$ yields $\rho = \min\{\frac{1-a}{1-\frac{1}{2}a}\frac{q}{n}, \frac{a}{4}\frac{\mu}{L}\}$. In particular, we have for the choice $\gamma = \frac{1}{5L}$ that $\rho = \min\{\frac{1}{3}\frac{q}{n}, \frac{1}{5}\frac{\mu}{L}\}$") is Corollary B3.15 at $\gamma = \gamma^\star(c)$: with $a = 4/(2+c)$ one has $\frac{1-a}{1-a/2} = 1 - \frac{2}{c}$ and $\frac{a}{4}\frac{\mu}{L} = \frac{1}{\kappa(2+c)}$, i.e. $\tau(c)$, and $\gamma = \frac{1}{5L}$ is $c = 3$. The coincidence is structural. Their Lyapunov function $\mathcal{L}_\sigma(w,H) = \|w - w^*\|^2 + S\sigma\bar H$, with $S = \gamma n/(Lq)$ and $\bar H = \frac{1}{n}\sum_i H_i$, has the two-term form (B3.2) with $\Delta = \gamma\sigma/(Lq)$, which is the entry $\sigma \to R^2\Delta/\gamma$ of the dictionary (their H_i are upper bounds on $\|\alpha_i - f_i'(w^*)\|^2$ obtained from the Bregman divergences of f_i at the stored points, their eqs. (10)–(12)); and their Lemma 1 rests on their eqs. (6) and (9), which are our (I1) and (I3), the former with the comment that $\|x \pm y\|^2 \leq (1+\beta)\|x\|^2 + (1+\beta^{-1})\|y\|^2$ could be used but "for the sake of simplicity, we sacrifice tightness and choose $\beta = 1$". The difference is that they apply strong convexity and co-coercivity through function values (their eqs. (5), (7)–(8)) instead of through G, and that they require the resulting coefficient of $f(w) - f(w^*)$ to be nonnegative and drop it (their eq. (31): $\gamma \leq \frac{1-\sigma}{2L}$); this caps their parameter rate at $\mu\gamma$ ("we will have $\rho \leq \gamma\mu$ based on the $\|w - w^*\|^2$ part", proof of their Lemma 3) and restricts their constants to $\Delta \leq \gamma(1 - 2\gamma R^2)/R^2$, i.e. to $\gamma \leq \gamma^\star(c)$ in our parametrization $\Delta = c\gamma^2$. On the curve $\gamma = \gamma^\star(c)$, where the bracket $1 - \gamma R^2(1 + \frac{c}{2})$ equals $\frac{1}{2}$, the two one-step bounds coincide, and the joint maximizer lies on this curve: this is why the optima agree. Off the curve the bounds differ: for $\gamma < \gamma^\star(c)$ the book's argument certifies the parameter rate $2\gamma\mu(1 - \gamma R^2(1 + \frac{c}{2})) > \mu\gamma$, and the textbook constants $\gamma = \frac{1}{4R^2}$, $c = 3$ (their $a = 1$, excluded in their Theorem 1) are admissible in Proposition B3.14 but not in their Lemma 3.

Comparison with the textbook choice. The book takes $c = 3$ and $\gamma = \frac{1}{4R^2}$, i.e. $\tau_{\text{book}} = \min\{\frac{1}{3n}, \frac{3}{16\kappa}\}$. (For $c = 3$ the best step would be $\gamma^\star(3) = \frac{1}{5R^2}$, with $\tau_D = \frac{1}{5\kappa}$ instead of $\frac{3}{16\kappa}$: the step $\frac{1}{4R^2}$ is slightly too large for $c = 3$, by 6.7% on the rate; this is the choice of Corollary 3 of [40] quoted above.) The ratio $Q(s) := \tau^\star/\tau_{\text{book}}$ between the best rate this proof structure can certify and the textbook rate is, with $s = n/\kappa$,

$$Q(s) = \begin{cases} \dfrac{32}{3\big(4 + s + \sqrt{s^2+16}\big)}, & s \le \frac{16}{9} \quad (\text{i.e. } \kappa \ge \frac{9n}{16},\ \text{parameter term is the bottleneck}), \\ \dfrac{6s}{4 + s + \sqrt{s^2+16}}, & s \ge \frac{16}{9} \quad (\text{memory term is the bottleneck}), \end{cases}$$

which decreases from $Q(0) = \frac{4}{3}$ to $Q(\frac{16}{9}) \approx 1.050$ and then increases to $Q(\infty) = 3$; numerically $Q(0.5) \approx 1.25$, $Q(1) \approx 1.17$, $Q(5) \approx 1.95$, $Q(20) \approx 2.70$, $Q(100) \approx 2.94$. Thus:

- in the ill-conditioned regime $\kappa \ge \frac{9n}{16}$ (the regime of hypothesis (A4), $\kappa \gg n$), the textbook constants are within a factor $\frac{4}{3}$ of the best this proof can give, and the optimal step is $\gamma^\star = \frac{\tau^\star \kappa}{R^2} \in [\frac{0.1969}{R^2}, \frac{1}{4R^2})$;
- in the big-data regime $n \gg \kappa$ the textbook rate $\frac{1}{3n}$ can be improved, within this proof structure, to at most $\frac{1}{n}$ (factor 3). A factor up to 2 of this gain requires no change of step: at $\gamma = \frac{1}{4R^2}$, $\Delta = c\gamma^2$ is admissible for $c \in (2, 6)$ and certifies the rate $\min\{\frac{6-c}{16\kappa}, \frac{c-2}{cn}\}$ (Corollary B3.15), maximal at $c_\Delta = \frac{6s-16+\sqrt{(6s-16)^2+128s}}{2s}$ (so that $c_\Delta = 3$ at $s = \frac{16}{9}$) with value $\frac{6-c_\Delta}{16\kappa}$, which is asymptotically equivalent to $\frac{2}{3n}$ as $s \to \infty$ and to $\frac{1}{4\kappa}$ as $s \to 0$; this is the certificate "with Δ optimized" at the textbook step used in Section B3.7. Only the remaining factor $\frac{3}{2}$ requires the much smaller step $\gamma^\star \approx \frac{1}{n\mu}$ (this is the step $\frac{1}{2(\mu n + L)}$ of [16] up to a factor 2).

In both regimes the certified complexity is $\Theta\big((n+\kappa)\log(1/\varepsilon)\big)$: by Proposition B3.16 the best iteration complexity of the proof structure is $(n + 4\kappa)\log(V_0/\varepsilon)$ up to a factor 1.172, and the textbook's is $\max\{3n, \frac{16}{3}\kappa\}\log(V_0/\varepsilon)$.

Proposition B3.17 (lower bound for SAGA with any constant step)**.** *Let $n \ge 1$, $0 < \mu < R^2$, $d = 2$, and $f_i(\theta) = \frac{1}{2}(\mu\theta_1^2 + R^2\theta_2^2)$ for all $i \le n$. Assumption B3.1 holds with these constants, $F = f_1$, $F'(\theta) = Q\theta$ with $Q = \operatorname{diag}(\mu, R^2)$, and $\eta_* = 0$. Run SAGA (B3.1) with any constant step $\gamma > 0$ (which may depend on μ) and any deterministic initial memory $z^{(0)}$. Then for all $t \ge 0$,*

$$\mathbb{E}[\theta_t] = (I - \gamma Q)^t \theta_0 \qquad \text{and} \qquad \mathbb{E}\,\|\theta_t - \eta_*\|^2 \ \ge\ \|\mathbb{E}[\theta_t] - \eta_*\|^2 \ \ge\ \Big(\frac{\kappa-1}{\kappa+1}\Big)^{2t} \min\{\theta_{0,1}^2, \theta_{0,2}^2\}. \tag{B3.18}$$

In particular, for $\theta_0 = (r, r)$ with $r \ne 0$ and $\varepsilon < \frac{1}{2}$, reaching $\mathbb{E}\,\|\theta_t - \eta_*\|^2 \le \varepsilon\,\|\theta_0 - \eta_*\|^2$ requires $t \ge \frac{\kappa-1}{4}\log\frac{1}{2\varepsilon}$, whatever the step.

Proof. By Lemma B3.4(a), $\mathbb{E}[\omega_t \mid \mathcal{F}_{t-1}] = F'(\theta_{t-1}) = Q\theta_{t-1}$ whatever the content of the memory, so $\mathbb{E}[\theta_t \mid \mathcal{F}_{t-1}] = (I - \gamma Q)\theta_{t-1}$; taking expectations (the map is linear) and iterating from the deterministic θ_0 gives the first identity. Next, $\mathbb{E}\,\|X\|^2 = \|\mathbb{E}X\|^2 + \mathbb{E}\,\|X - \mathbb{E}X\|^2 \ge \|\mathbb{E}X\|^2$ with $X = \theta_t$. Finally, Q is diagonal, so $\big\|(I - \gamma Q)^t\theta_0\big\|^2 = (1-\gamma\mu)^{2t}\theta_{0,1}^2 + (1-\gamma R^2)^{2t}\theta_{0,2}^2 \ge \max\{|1-\gamma\mu|, |1-\gamma R^2|\}^{2t}\min\{\theta_{0,1}^2, \theta_{0,2}^2\}$, and $\max\{|1-\gamma\mu|, |1-\gamma R^2|\} \ge \frac{\kappa-1}{\kappa+1}$ for every $\gamma > 0$: if $\gamma \le \frac{2}{\mu+R^2}$ then $1 - \gamma\mu \ge 1 - \frac{2\mu}{\mu+R^2} = \frac{R^2-\mu}{R^2+\mu}$, and if $\gamma \ge \frac{2}{\mu+R^2}$ then $\gamma R^2 - 1 \ge \frac{2R^2}{\mu+R^2} - 1 = \frac{R^2-\mu}{R^2+\mu}$. For the last claim, $\min\{\theta_{0,1}^2, \theta_{0,2}^2\} = r^2 = \frac{1}{2}\,\|\theta_0 - \eta_*\|^2$, so the target forces $(\frac{\kappa-1}{\kappa+1})^{2t} \le 2\varepsilon$, i.e. $t \ge \log\frac{1}{2\varepsilon}\big/\big(2\log\frac{\kappa+1}{\kappa-1}\big) \ge \frac{\kappa-1}{4}\log\frac{1}{2\varepsilon}$, using $\log\frac{\kappa+1}{\kappa-1} = \log(1 + \frac{2}{\kappa-1}) \le \frac{2}{\kappa-1}$. □

The argument uses only that the SAGA direction is unbiased for $F'(\theta_{t-1})$ and that F' is affine: on any finite sum with a quadratic F (least squares, for instance), the *mean* SAGA iterate is the gradient-descent iterate with the same step, and the variance $\mathbb{E}\,\|\theta_t - \mathbb{E}\theta_t\|^2$ only adds to the error. Hence no analysis of SAGA with a constant step can certify, uniformly over Assumption B3.1, a per-iteration contraction of $\mathbb{E}\,\|\theta_t - \eta_*\|^2$ better than $(\frac{\kappa-1}{\kappa+1})^2 \geq 1 - \frac{4}{\kappa+1}$. For $n = 1$ the memory is useless and (B3.1) *is* gradient descent; the proposition shows that the obstruction persists for every n.

Remark B3.18 (optimizing (γ, Δ) gains at most a factor 3). Within the proof structure of Proposition 5.9 (Lyapunov function $D + \Delta M$, inequalities (I1)–(I5)), optimizing (γ, Δ) changes the certified one-step rate by the factor $Q(s)$, between 1.05 and 3, and by at most 4/3 when $\kappa \geq 9n/16$. This factor is not negligible when $n \gg \kappa$: for $n = 1000$ it is $Q(10) \approx 2.42$ at $\kappa = 10^2$ and $Q(100) \approx 2.94$ at $\kappa = 10$, of the same size as the speed-ups 3.2–3.7 measured there (Section B3.7). When $\kappa \leq 9n/16$, where the textbook rate is $\frac{1}{3n}$, a factor 3 is in fact the most that any analysis of $\mathbb{E}[V_t]$, for any $\Delta > 0$, can gain over it: by (B3.5) and $S_{t-1} \geq 0$, $\mathbb{E}_{i(t)}[M_t] \geq (1 - \frac{1}{n})M_{t-1}$, hence $\mathbb{E}[V_t] \geq \Delta(1 - \frac{1}{n})^t M_0$, and a bound $\mathbb{E}[V_t] \leq c_0 \rho^t V_0$ valid for all t, with c_0 independent of t and $M_0 > 0$, forces $1 - \rho \leq \frac{1}{n}$. On the hard instances of hypothesis (A4) ($\kappa \gg n$), the rate per effective pass that this proof structure certifies, whatever the constants, is at most $n\tau^\star \leq \frac{1.172\,n}{4\kappa} \ll 1$. Whenever the exponent of the contraction of V_t observed along SAGA exceeds the certified exponent $-\log \rho_{\text{bound}}$ by more than the factor $Q(s)$, the excess must therefore be located in the state-dependent ratios $r_{\text{sq}}, r_{\text{coco}}, r_{\text{var}}, r_{\text{sc}}$ (inequalities (I1)–(I4)) and in $r_{\min}$, which is exactly what the chains of Proposition B3.11 measure. The gap itself depends on the instance: on the worst-case instance of caveat (ii) below, for $\kappa \gg n$, it is at most the factor $\frac{8}{3}$ on the leading term of the complexity, and on the logistic-regression instances of Section B3.7 it comes mainly from (I4) ($\mu_{\text{eff}} \gg \mu$) and (I2) ($L_{\text{eff}} < R^2$) where it is large ($\kappa \geq 10^4$). Two caveats delimit the claim. (i) The optimality of $(\gamma^\star, \Delta^\star)$ is relative to this proof structure: it does not show that $\tau^\star$ is the best rate that an analysis of SAGA can certify under Assumption B3.1, and another Lyapunov function or another chain of inequalities could certify more. The closed-form analyses we checked give rates of the same order. [40] give exactly $\tau^\star$ and $\gamma^\star$ (see the paragraph following Proposition B3.16); [16] (Theorem 1, Corollary 1 and the "adaptivity to strong convexity" comment at the end of the proof of Theorem 1) prove, with a Lyapunov function containing the Bregman divergences $f_i(\phi_i) - f_i(x^*) - \langle f_i'(x^*), \phi_i - x^*\rangle$ and assuming every f_i to be μ-strongly convex, the rate $1 - \frac{\mu}{2(\mu n + L)}$ for $\gamma = \frac{1}{2(\mu n + L)}$ and the rate $1 - \min\{\frac{1}{4n}, \frac{\mu}{3L}\}$ for the μ-free step $\gamma = \frac{1}{3L}$ (this stronger assumption implies Assumption B3.1 with $R^2 = L$, and on this subclass the first rate exceeds $\tau^\star$ as soon as $\kappa > n/\sqrt{2}$, by a factor tending to 2 as $\kappa/n \to \infty$: an analysis that exploits the stronger assumption does certify more there); when strong convexity only holds on average (our Assumption B3.1) they state the rate $1 - \frac{\mu}{6(\mu n + L)}$ for $\gamma = \frac{1}{3(\mu n + L)}$; the LMI-based analysis of [41] (Corollary 3, eq. (42)) certifies $1 - \min\{\frac{1}{3n}, \frac{\mu}{10L}\}$ for $\gamma = \frac{1}{3L}$ under assumptions implied by Assumption B3.1: each f_i convex and L-smooth ($f_i \in \mathcal{F}(0, L)$ in their notation) and F in their class $\mathcal{S}(m, L)$ of functions with sector-bounded gradients, which contains the m-strongly convex L-smooth functions but, as they stress, also non-convex ones. The rates quoted here that hold under Assumption B3.1 itself are all at most $\tau^\star$, and all of the above have the form $1 - \min\{\Theta(\frac{1}{n}), \Theta(\frac{1}{\kappa})\}$. (ii) The exponent 1 of κ is intrinsic to SAGA with a constant step, not to its analysis. Proposition B3.17 shows that on an instance of Assumption B3.1 with n identical quadratic components, for every n and every constant step (whether or not it depends on μ), $\mathbb{E}\,\|\theta_t - \eta_*\|^2$ decays no faster than $(1 - \frac{2}{\kappa+1})^{2t}$, so that at least $\frac{\kappa-1}{4}\log\frac{1}{2\varepsilon}$ iterations are needed, against $O((n + \sqrt{n\kappa})\log\frac{1}{\varepsilon})$ for accelerated methods (hypotheses (A2)–(A4)). For $\kappa \gg n$ the certified complexity $\Theta(\kappa \log\frac{1}{\varepsilon})$ is therefore optimal for SAGA up to a constant factor. The textbook bound guarantees the target after $\frac{16\kappa}{3}\log\frac{1+n/4}{\varepsilon}$ iterations; for the textbook step itself, the identity and the first inequality in

(B3.18) give on the same instance, with $\theta_0 = (r, r)$, $\mathbb{E}\,\|\theta_t - \eta_*\|^2 \geq (1 - \frac{1}{4\kappa})^{2t} \frac{1}{2} \|\theta_0 - \eta_*\|^2$, hence at least $(2\kappa - \frac{1}{2}) \log \frac{1}{2\varepsilon}$ iterations (using $\log \frac{1}{1-x} \leq \frac{x}{1-x}$), which is the bound of part (a) of the SAGA sandwich of the A1 note (Corollary A1.23) on this instance. On the leading term $\kappa \log \frac{1}{\varepsilon}$ the textbook analysis is thus tight up to the factor $\frac{8}{3}$, and within $\frac{64}{3} \approx 21$ of the lower bound valid for every constant step. For oblivious but non-constant step schedules the same conclusion follows from [6, Theorem 2]: "Suppose that only L the smoothness parameter is known, i.e. $\mathcal{I}(\cdot) = \{L\}$. If the iteration complexity of a given oblivious, possibly stochastic, p-CLI optimization algorithm is $\tilde{O}(\kappa(f)^\alpha \ln(1/\epsilon))$, then $\alpha \geq 1$"; SAGA (with SAG, SVRG and SDCA without duality) is listed there (Section 2.2) as a stationary stochastic p-CLI, and the book's step $\gamma = \frac{1}{4R^2}$, or any schedule depending on R^2 only, is oblivious in their sense (coefficients depending on the side-information $\{L\}$ and on time only). When μ is also known, their Theorem 1 only gives the lower bound $\tilde{\Omega}(\sqrt{\kappa} \ln(1/\epsilon))$ for oblivious schedules, which accelerated gradient descent attains with full gradients, i.e. with n component gradients per iteration (for finite sums the oblivious lower bound counted in component gradients is $\tilde{\Omega}(n + \sqrt{n(\kappa - 1)} \ln(1/\epsilon))$ by [5, Theorem 2], recalled in the A1 note as Theorem A1.10, and accelerated variance-reduced methods match it for $\kappa \gtrsim n$); this does not contradict Proposition B3.17, which concerns constant steps. Improving κ to $\sqrt{n\kappa}$ therefore requires changing the algorithm (acceleration, hypotheses (A2)–(A3)), consistently with the lower bound of hypothesis (A1). The computer-aided approach of [82] automates this kind of search over potential (Lyapunov) functions and parameters by semidefinite programming, with a tightness guarantee for the chosen potential class; it is developed there mainly for sublinear rates of stochastic-gradient-type methods without strong convexity, the authors noting that the same tools apply under strong convexity and that "minor adaptations" cover finite-sum methods such as SAGA, and the LMI analysis of [41] quoted in (i) is a search of this kind for SAGA with linear rates.

B3.6 Trajectory-dependent version (hook for hypothesis (B2))

Strong convexity (I4) and co-coercivity (I2) are only used at the current point θ_{t-1}. Define, as functions of $\theta \neq \eta_*$ (they do not depend on the memory),

$$\mu_{\mathrm{eff}}(\theta) := \frac{(\theta - \eta_*)^\top F'(\theta)}{\|\theta - \eta_*\|^2}, \qquad L_{\mathrm{eff}}(\theta) := \frac{\frac{1}{n} \sum_{i=1}^n \|f_i'(\theta) - f_i'(\eta_*)\|^2}{(\theta - \eta_*)^\top F'(\theta)},$$

and $\mu_{\mathrm{eff}}(t-1) := \mu_{\mathrm{eff}}(\theta_{t-1})$, $L_{\mathrm{eff}}(t-1) := L_{\mathrm{eff}}(\theta_{t-1})$. By Lemma B3.9(i), $\mu \leq \mu_{\mathrm{eff}}(\theta) \leq L_{\mathrm{eff}}(\theta) \leq R^2$. For quadratic F, $\mu_{\mathrm{eff}}(\theta) = \hat{u}^\top \nabla^2 F\, \hat{u}$ with $\hat{u} := (\theta - \eta_*)/\|\theta - \eta_*\|$, so that $\inf_{\theta \neq \eta_*} \mu_{\mathrm{eff}}(\theta) = \lambda_{\min}(\nabla^2 F)$: the constant $\mu_F = \lambda_{\min}(\nabla^2 F)$ in which the A1 note states its SAGA bounds for quadratic instances (Corollary A1.23) is this infimum, not the trajectory-dependent value $\mu_{\mathrm{eff}}(\theta_{t-1})$ used here.

Proposition B3.19 (a-posteriori rates with effective constants). *Let $\gamma = \frac{1}{4R^2}$ and $\Delta = 3\gamma^2$. For every $t \geq 1$, almost surely on $\{\theta_{t-1} \neq \eta_*\}$:*

$$\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \Big(1 - \min\Big\{\frac{1}{3n},\ \frac{3\,\mu_{\mathrm{eff}}(t-1)}{16R^2}\Big\}\Big) V_{t-1}, \tag{B3.19}$$

$$\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \Big(1 - \min\Big\{\frac{1}{3n},\ \frac{\mu_{\mathrm{eff}}(t-1)}{2R^2}\Big(1 - \frac{5\,L_{\mathrm{eff}}(t-1)}{8R^2}\Big)\Big\}\Big) V_{t-1}, \tag{B3.20}$$

and the coefficient in (B3.20) satisfies $\frac{3\mu_{\mathrm{eff}}(t-1)}{16R^2} \leq \frac{\mu_{\mathrm{eff}}(t-1)}{2R^2}\big(1 - \frac{5L_{\mathrm{eff}}(t-1)}{8R^2}\big) \leq \frac{\mu_{\mathrm{eff}}(t-1)}{2R^2} \leq \frac{1}{2}$, so that both contraction factors lie in $[\frac{1}{2}, 1)$. More generally, for γ, Δ as in Proposition B3.14,

$$\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \Big(1 - \min\Big\{2\gamma\,\mu_{\mathrm{eff}}(t-1)\Big(1 - \gamma L_{\mathrm{eff}}(t-1) - \frac{L_{\mathrm{eff}}(t-1)\Delta}{2\gamma}\Big),\ \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\Big)\Big\}\Big) V_{t-1}. \tag{B3.21}$$

On $\{\theta_{t-1}=\eta_*\}$ *one has* $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \le \big(1-\frac{1}{n}+\frac{2\gamma^2}{n\Delta}\big)V_{t-1}$, *so that* (B3.19)–(B3.21) *hold there with any value assigned to* $\mu_{\text{eff}}(t-1)$ *and* $L_{\text{eff}}(t-1)$.

Proof. Apply (I1) and (I3) to (B3.4) and keep (I2) as the identity $S/n = L_{\text{eff}}(t-1)\,G$: with $G=\mu_{\text{eff}}(t-1)D$,

$$\mathbb{E}_{i(t)}[D_t] \le D - 2\gamma G + 2\gamma^2 L_{\text{eff}} G + \frac{2\gamma^2}{n}M, \qquad \mathbb{E}_{i(t)}[M_t] = \Big(1-\frac{1}{n}\Big)M + L_{\text{eff}}G,$$

hence $\mathbb{E}_{i(t)}[V_t] \le \big[1-2\gamma\mu_{\text{eff}}\,\alpha_{\text{eff}}\big]D + \big[1-\frac{1}{n}(1-\frac{2\gamma^2}{\Delta})\big]\Delta M$ with $\alpha_{\text{eff}} := 1-\gamma L_{\text{eff}} - \frac{L_{\text{eff}}\Delta}{2\gamma}$ (this is (B3.14) with $r_{\text{sq}}=r_{\text{var}}=1$, and it follows from it since c_D, c_M are nondecreasing in $r_{\text{sq}}, r_{\text{var}}$). Since $0 < L_{\text{eff}} \le R^2$, we have $\alpha(\gamma,\Delta) \le \alpha_{\text{eff}} \le 1$, so $2\gamma\mu_{\text{eff}}\alpha_{\text{eff}} > 0$; and since $\mu_{\text{eff}} \le L_{\text{eff}}$ (Lemma B3.9(i)) and $\gamma L_{\text{eff}} \le \gamma R^2 < 1$, we have $2\gamma\mu_{\text{eff}}\alpha_{\text{eff}} \le 2\gamma L_{\text{eff}}(1-\gamma L_{\text{eff}}) \le \frac{1}{2}$. Hence the first coefficient lies in $[\frac{1}{2},1)$ and the second in $(1-\frac{1}{n},1)$, and the step (I5) gives (B3.21). For the textbook constants, $2\gamma\mu_{\text{eff}} = \frac{\mu_{\text{eff}}}{2R^2}$, $\gamma L_{\text{eff}} + \frac{L_{\text{eff}}\Delta}{2\gamma} = \frac{5L_{\text{eff}}}{8R^2}$ and $\frac{1}{n}(1-\frac{2\gamma^2}{\Delta}) = \frac{1}{3n}$, which is (B3.20); replacing $L_{\text{eff}}(t-1)$ by its upper bound R^2 (the bracket is decreasing in L_{eff}) gives $\frac{\mu_{\text{eff}}}{2R^2}\cdot\frac{3}{8} = \frac{3\mu_{\text{eff}}}{16R^2}$, which is (B3.19); the chain of inequalities between the coefficients follows from $\mu \le \mu_{\text{eff}} \le R^2$ and $0 \le L_{\text{eff}} \le R^2$. On $\{\theta_{t-1}=\eta_*\}$: $S=G=D=0$, $\omega_t = b_{i(t)}$, so $\mathbb{E}_{i(t)}[D_t] = \gamma^2 B \le \gamma^2 M/n$ by (I3) and $\mathbb{E}_{i(t)}[M_t] = (1-\frac{1}{n})M$; thus $\mathbb{E}_{i(t)}[V_t] \le (1-\frac{1}{n}+\frac{\gamma^2}{n\Delta})\Delta M \le (1-\frac{1}{n}+\frac{2\gamma^2}{n\Delta})V_{t-1}$. □

Remark B3.20 (the step is fixed in advance). One may not replace R^2 by $L_{\text{eff}}(t-1)$ inside γ: the step was chosen before the trajectory was observed, and $\Delta = 3\gamma^2$ with it. With $\gamma = \frac{1}{4R^2}$ fixed, a smaller effective smoothness $L_{\text{eff}} < R^2$ enters only through the bracket $1-\frac{5L_{\text{eff}}}{8R^2} \in [\frac{3}{8},1)$: the gain from (I2) being loose is at most a factor $\frac{8}{3}$ on the parameter rate. By contrast $\mu_{\text{eff}}(t-1)$ enters linearly and may exceed μ by orders of magnitude (for logistic regression, $\mu=\lambda$ while $\mu_{\text{eff}}(\theta)$ is λ plus the curvature of the data term in the direction $\theta-\eta_*$, averaged along the segment $[\eta_*,\theta]$; see the proof of Proposition B3.22), which is why hypothesis (B2) focuses on μ_{eff}. A step adapted to L_{eff} would be a different algorithm, and Δ would have to follow γ; see also the discussion of step-size selection in [35, Sec. 3.A].

A posteriori versus a priori. The factors in Proposition B3.19 are random and $\mathcal{F}_{t-1}$-measurable: writing $\rho_{\text{eff}}(t)$ for any of them, we have $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \le \rho_{\text{eff}}(t)V_{t-1}$, but *not* $\mathbb{E}[V_T] \le \mathbb{E}\big[\prod_{t\le T}\rho_{\text{eff}}(t)\big]V_0$ in general: $\rho_{\text{eff}}(t)$ and V_{t-1} are correlated, and $\rho_{\text{eff}}(t)$ is not $\mathcal{F}_{t-2}$-measurable, so the conditional expectations cannot be iterated. The experiments can still report the realised products $\prod_{t\le T}\rho_{\text{eff}}(t)$ (and $\prod_t \rho_{\text{exact}}(t)$) next to the realised V_T/V_0; the difference is a martingale-type fluctuation. To obtain a rigorous a-priori statement one needs a deterministic bound on the effective constants over a region.

Proposition B3.21 (a-priori rate on a region). *Let* $\Theta \subset \mathbb{R}^d$ *and*

$$\mu_\Theta := \inf_{\theta\in\Theta\setminus\{\eta_*\}} \mu_{\text{eff}}(\theta) \ \ge \mu, \qquad L_\Theta := \sup_{\theta\in\Theta\setminus\{\eta_*\}} L_{\text{eff}}(\theta) \ \le R^2,$$
$$\rho_\Theta := 1-\min\Big\{\frac{1}{3n},\ \frac{\mu_\Theta}{2R^2}\Big(1-\frac{5L_\Theta}{8R^2}\Big)\Big\} \ \le\ 1-\min\Big\{\frac{1}{3n},\frac{3\mu_\Theta}{16R^2}\Big\},$$

and let $\tau := \inf\{t\ge 0 : \theta_t \notin \Theta\}$ *(with* $\inf\emptyset = +\infty$*). For* $\gamma = \frac{1}{4R^2}$, $\Delta = 3\gamma^2$ *and every* $T \ge 0$,

$$\mathbb{E}\big[V_T\,\mathbf{1}_{\{\tau>T\}}\big] \le \rho_\Theta^T\, V_0. \tag{B3.22}$$

In particular, if $\theta_t \in \Theta$ *for all* t *almost surely, then* $\mathbb{E}[V_T] \le \rho_\Theta^T V_0$ *and* $\mathbb{E}\,\|\theta_T-\eta_*\|^2 \le \rho_\Theta^T(1+\frac{n}{4})\,\|\theta_0-\eta_*\|^2$ *for the book's initialization.*

Proof. Since $\theta_0, \dots, \theta_t$ are $\mathcal{F}_t$-measurable, $\{\tau > t\} \in \mathcal{F}_t$, and τ is a stopping time. Let $W_t := \rho_\Theta^{-(t\wedge\tau)} V_{t\wedge\tau}$. On $\{\tau \le t\}$, $W_{t+1} = W_t$. On $\{\tau > t\}$, $\theta_t \in \Theta$, so Proposition B3.19 (inequality (B3.20) at time $t+1$, whose coefficient is nondecreasing in μ_{eff} and nonincreasing in L_{eff}, together with the case $\theta_t = \eta_*$) gives $\mathbb{E}[V_{t+1} \mid \mathcal{F}_t] \le \rho_\Theta V_t$, hence $\mathbb{E}[W_{t+1}\mathbf{1}_{\{\tau>t\}} \mid \mathcal{F}_t] = \mathbf{1}_{\{\tau>t\}}\rho_\Theta^{-(t+1)}\mathbb{E}[V_{t+1} \mid \mathcal{F}_t] \le \mathbf{1}_{\{\tau>t\}}\rho_\Theta^{-t} V_t = \mathbf{1}_{\{\tau>t\}} W_t$. Thus (W_t) is a supermartingale (all variables are integrable since $i(1), \dots, i(T)$ take finitely many values), $\mathbb{E}[W_T] \le W_0 = V_0$, and $W_T \ge \mathbf{1}_{\{\tau>T\}}\rho_\Theta^{-T} V_T$ gives (B3.22). The last claim uses $\tau = \infty$ a.s., $D_T \le V_T$ and (I6). □

For ℓ_2-regularized logistic regression the infimum μ_Θ over a ball can be bounded below explicitly, thanks to the self-concordance-type property of the logistic loss that underlies the adaptivity results of [10] (Assumption (A2) there: along any segment, $|\varphi'''| \le R_x \|\theta_1 - \theta_2\| \varphi''$, with $R_x = \max_i \|x_i\|$, denoted R in [10], not our R).

Proposition B3.22 (local strong convexity of logistic regression). *Let $f_i(\theta) = \ell(y_i x_i^\top \theta) + \frac{\lambda}{2}\|\theta\|^2$ as in Remark B3.7, $\lambda > 0$, $R_x := \max_i \|x_i\|$, and $\psi(x) := \frac{1-e^{-x}}{x}$ for $x > 0$, $\psi(0) := 1$ (ψ is decreasing on $[0,\infty)$, with values in $(0,1]$). Then for every $\theta \ne \eta_*$,*

$$\mu_{\text{eff}}(\theta) \;\ge\; \lambda + \psi\big(R_x\|\theta-\eta_*\|\big)\big(\lambda_{\min}(F''(\eta_*)) - \lambda\big) \;\ge\; \max\big\{\lambda,\; \psi\big(R_x\|\theta-\eta_*\|\big)\lambda_{\min}(F''(\eta_*))\big\},$$

where $F''(\eta_*) = \lambda I + \frac{1}{n}\sum_i \ell''(y_i x_i^\top \eta_*) x_i x_i^\top \succeq \lambda I$. Consequently, on the ball $\Theta_r = \{\theta : \|\theta - \eta_*\| \le r\}$, the constant μ_{Θ_r} of Proposition B3.21 satisfies

$$\mu_{\Theta_r} \ge \lambda + \psi(R_x r)(\lambda_{\min}(F''(\eta_*)) - \lambda) \ge \max\{\lambda, \psi(R_x r)\lambda_{\min}(F''(\eta_*))\};$$

in particular $\mu_{\Theta_r} \ge \frac{1}{2}(\lambda + \lambda_{\min}(F''(\eta_*))) \ge \max\{\lambda, \frac{1}{2}\lambda_{\min}(F''(\eta_*))\}$ as soon as $R_x r \le \frac{3}{2}$ (since $\psi(3/2) \approx 0.518$).

Proof. Let $u = \theta - \eta_*$ and $\varphi(s) := F(\eta_* + su)$, $s \in [0,1]$. With $m_i(s) = y_i x_i^\top(\eta_* + su)$ and $y_i^2 = 1$,

$$\varphi''(s) = \lambda\|u\|^2 + \chi(s), \qquad \chi(s) := \frac{1}{n}\sum_{i=1}^n \ell''(m_i(s))(x_i^\top u)^2 \ge 0,$$

and $\chi'(s) = \frac{1}{n}\sum_{i=1}^n \ell'''(m_i(s))\, y_i (x_i^\top u)^3$. Since $\ell'''(u) = \ell''(u)(1 - 2\sigma(u))$, $|\ell'''| \le \ell''$, hence

$$|\chi'(s)| \le \frac{1}{n}\sum_i \ell''(m_i(s))\left|x_i^\top u\right|^3 \le R_x\|u\| \cdot \frac{1}{n}\sum_i \ell''(m_i(s))(x_i^\top u)^2 = R_x\|u\|\,\chi(s).$$

If $\chi(0) = 0$, then $x_i^\top u = 0$ for all i (because $\ell'' > 0$) and $\chi \equiv 0$; otherwise $\chi > 0$ on $[0,1]$ and $(\log\chi)'(s) \ge -R_x\|u\|$. In both cases $\chi(s) \ge \chi(0)e^{-R_x\|u\|s}$ for $s \in [0,1]$. Now

$$G = \langle u, F'(\eta_* + u) - F'(\eta_*)\rangle = \int_0^1 \langle u, F''(\eta_* + su)\,u\rangle\, ds = \int_0^1 \varphi''(s)\, ds \ge \lambda\|u\|^2 + \chi(0)\,\psi(R_x\|u\|),$$

and $\chi(0) = u^\top F''(\eta_*)u - \lambda\|u\|^2 \ge (\lambda_{\min}(F''(\eta_*)) - \lambda)\|u\|^2 \ge 0$. Dividing by $D = \|u\|^2$ gives the first inequality. With $\psi = \psi(R_x\|u\|) \in (0,1]$, its right-hand side is $(1-\psi)\lambda + \psi\,\lambda_{\min}(F''(\eta_*))$, which is at least λ (as $\lambda_{\min}(F''(\eta_*)) \ge \lambda$) and at least $\psi\,\lambda_{\min}(F''(\eta_*))$ (as $(1-\psi)\lambda \ge 0$): this is the second inequality. Finally, ψ is decreasing on $(0,\infty)$ (its derivative is $\frac{(1+x)e^{-x}-1}{x^2} \le 0$ because $e^x \ge 1 + x$), $\psi \le 1$ (because $1 - e^{-x} \le x$), and both lower bounds are nondecreasing in ψ, which gives the statement on balls; for $R_x r \le \frac{3}{2}$, $\psi(R_x r) \ge \psi(\frac{3}{2}) > \frac{1}{2}$. (Before $\chi(0)$ is bounded below, the argument gives $\mu_{\text{eff}}(\theta) \ge \lambda + \psi(R_x\|u\|)\,(\hat{u}^\top F''(\eta_*)\hat{u} - \lambda)$ with $\hat{u} := u/\|u\|$: this is the pointwise bound (B2.24) of the B2 note (Corollary B2.15) with its factor $e^{-R_x\|u\|}$ replaced by $\psi(R_x\|u\|) \ge e^{-R_x\|u\|}$, as $e^x - 1 \ge x$.) □

Combining Propositions B3.21 and B3.22 with $\Theta = \Theta_r$, $r = \frac{3}{2R_x}$, and using that the upper bound $1 - \min\{\frac{1}{3n}, \frac{3\mu_\Theta}{16R^2}\}$ on ρ_Θ is nonincreasing in μ_Θ: inequality (B3.22) holds with ρ_Θ replaced by $1 - \min\{\frac{1}{3n}, \frac{3}{16R^2}\max\{\lambda, \frac{1}{2}\lambda_{\min}(F''(\eta_*))\}\}$, and even with $\frac{1}{2}(\lambda + \lambda_{\min}(F''(\eta_*)))$ in place of the maximum. Up to the exit time from this ball, the Lyapunov function thus contracts in expectation at this rate instead of $1 - \min\{\frac{1}{3n}, \frac{3\lambda}{16R^2}\}$: the global constant λ is replaced by $\max\{\lambda, \frac{1}{2}\lambda_{\min}(F''(\eta_*))\}$ (or by $\frac{1}{2}(\lambda + \lambda_{\min}(F''(\eta_*)))$), a substantial gain only when $\lambda_{\min}(F''(\eta_*)) \gg \lambda$. The maximum with λ matters: in the fifteen configurations of the experiments of hypothesis (B3) ($n = 1000$, $d = 50$, $\kappa \in \{10, \dots, 10^5\}$, covariance spectrum $j^{-\alpha}$ with exponent $\alpha \in \{0, 1, 2\}$, unrelated to $\alpha(\gamma, \Delta)$), the ratio $\lambda_{\min}(F''(\eta_*))/\lambda$ ranges from 1.0003 to 69.9 and is below 2 in nine of them, where $\frac{1}{2}\lambda_{\min}(F''(\eta_*)) < \lambda$. The smallest eigenvalue $\lambda_{\min}(F''(\eta_*))$ of the Hessian at the optimum is the quantity to which averaged SGD is shown to adapt in [10]. The experiments of (B2) measure $\mu_{\text{eff}}(t)$ directly and compare it with λ and with the extreme eigenvalues $\lambda_{\min}(F''(\eta_*))$ and $\lambda_{\max}(F''(\eta_*))$ (Figure B2 of the report); they do not evaluate the bound of Proposition B3.22, which is weak at the start of those runs ($R_x \|\theta_0 - \eta_*\|$ between 5.9 and 51, so $\psi \le 0.17$) and tends to $\lambda_{\min}(F''(\eta_*))$ as $\theta_t \to \eta_*$.

B3.7 What to measure

At a measurement time (once per effective pass keeps the $O(nd)$ cost of a measurement of the same order as that of the n SAGA steps; the experiments reported below measure every $n/10$ iterations, which costs more but does not change the trajectory), the state is $(\theta, z) = (\theta_{t-1}, z^{(t-1)})$. The code needs η_* (computed beforehand to a precision far below the smallest V_t that will be reported, e.g. by a second-order method or by SAGA itself run to machine precision), the stored vectors $g_i^* := f_i'(\eta_*)$ ($n \times d$, computed once), and, at each measurement, one pass computing $g_i := f_i'(\theta)$ for all i. For logistic regression, $f_i'(\theta) = -y_i\sigma(-y_i x_i^\top \theta)x_i + \lambda\theta$, $R^2 = \max_i(\|x_i\|^2/4 + \lambda)$, $\mu = \lambda$, $R_x = \max_i \|x_i\|$.

Ineq.	Ratio	Closed form	State quantities	Code
(I1)	r_{sq}	$\Omega/(2S/n + 2B) = \frac{1}{2} + C/(S/n + B)$	Ω, S, B (or C)	$a_i = g_i - g_i^*$, $b_i = g_i^* - z_i + \bar{z}$, $\Omega = \frac{1}{n}\sum_i \|a_i + b_i\|^2$, $C = \frac{1}{n}\sum_i \langle a_i, b_i\rangle$
(I2)	$r_{\text{coco}} = L_{\text{eff}}/R^2$	$(S/n)/(R^2 G)$	S, G	$S = \sum_i \|a_i\|^2$, $F'(\theta) = \frac{1}{n}\sum_i g_i$, $G = (\theta - \eta_*)^\top F'(\theta)$
(I3)	r_{var}	$B/(M/n) = 1 - n\|\bar{z}\|^2/M$	$B, M, \bar{z}$	$M = \sum_i \|z_i - g_i^*\|^2$; compute $B = \frac{1}{n}\sum_i \|b_i\|^2$ directly (avoids cancellation)
(I4)	$r_{\text{sc}} = \mu/\mu_{\text{eff}}$	$\mu D/G$	D, G	$D = \|\theta - \eta_*\|^2$; also report $\mu_{\text{eff}} = G/D$ (B2)
(I5)	$r_{\min}$	Definition B3.8	D, M	needs Δ, n, μ, R^2; equals $\rho_1^B/\rho_{\text{bound}}$
(I6)	r_{init}	$S_0/(nR^4 D_0)$	S_0, D_0	at $t = 0$ only
exact	$\rho_{\text{exact}}(t)$	(B3.8)	D, M, G, S, Ω	compare with the realised V_t/V_{t-1} and with ρ_{bound}
chains	ρ_k^A, ρ_k^B	Proposition B3.11	all of the above	report δ_k, ν_k and $n\nu_k$ (per pass) for both orders
(B2)	$\rho_{\text{eff}}(t)$	(B3.19), (B3.20)	$\mu_{\text{eff}}, L_{\text{eff}}$	compare $\mu_{\text{eff}}(t)$ with λ, with $\lambda_{\min}(F''(\eta_*))$ and with Proposition B3.22

Practical remarks. (a) The four ratios of (I1)–(I4) are independent of (γ, Δ). Since Δ does not affect the iterates, the same measured states can be re-used to evaluate the chains of Proposition B3.11 for any Δ; for a step other than the one that generated the trajectory (e.g. the $\gamma^\star$ of Proposition B3.16) they only describe a single hypothetical step from these states, and SAGA must be rerun with that step (as for Figure C of the report). (b) Stop measuring when D or M reaches

the precision of η_* (the ratios become numerical noise). (c) Expect $r_{\mathrm{sq}} \approx \frac{1}{2}$ when the iterates move fast compared with the refresh of the memory (stale memory) and $r_{\mathrm{sq}} \leq \frac{1}{2}$, possibly $\ll \frac{1}{2}$, when they move slowly (nearly fresh memory; Remark B3.13), r_{var} close to 1 once the memory errors have no common direction, $r_{\mathrm{coco}} = L_{\mathrm{eff}}/R^2$ well below 1 (the per-sample curvature along $\theta - \eta_*$ is typically much smaller than the maximal one), and $r_{\mathrm{sc}} = \lambda/\mu_{\mathrm{eff}}$ very small when λ is small compared with the curvature of the data term. With (γ, Δ) fixed to the textbook values, (B3.15) shows that r_{sq}, r_{var} and r_{coco} each change the corresponding certified rate by a bounded factor only: the parameter rate $1 - c_D = \frac{\mu_{\mathrm{eff}}}{2R^2}\big(1 - r_{\mathrm{coco}}(\frac{r_{\mathrm{sq}}}{4} + \frac{3}{8})\big)$ varies by at most a factor $\frac{8}{3}$ with r_{coco} and by at most $\frac{5}{3}$ with r_{sq} (the bracket lies in $[\frac{3}{8}, 1)$, Remark B3.20), and the memory rate $1 - c_M = \frac{1}{n}(1 - \frac{2}{3} r_{\mathrm{sq}} r_{\mathrm{var}})$ varies by at most a factor 3 with $r_{\mathrm{sq}} r_{\mathrm{var}}$. Only r_{sc} (through μ_{eff}, which is unbounded relative to μ) and r_{min} (when the Lyapunov mass sits on the faster component: for $\kappa \geq 9n/16$ and $\Delta M \gg D$, $(1 - \rho_1^B)/(1 - \rho_{\mathrm{bound}}) \approx \frac{1/(3n)}{3/(16\kappa)} = \frac{16\kappa}{9n}$, e.g. ≈ 1778 for $n = 10$, $\kappa = 10^4$) can explain a gap larger than a constant factor.

Outcome of the experiments. On ℓ_2-regularized logistic regression ($n = 1000$, $d = 50$, covariance spectrum $j^{-\alpha}$, $\alpha \in \{0, 1, 2\}$, $\kappa \in \{10, 10^2, \ldots, 10^5\}$, textbook constants, 5 seeds, averages over passes 2–20; Figures 1–3 of the report), $-\log \rho_{\mathrm{exact}}$ exceeds $-\log \rho_{\mathrm{bound}}$ by a factor 3.2–3.7 for $\kappa \leq 10^2$ and 19–220 at $\kappa = 10^5$; as (c) predicts, large factors go with small r_{sc} (0.012–0.14 at $\kappa = 10^5$). For $\kappa \leq 10^2$, order B attributes the log-gap to (I1) (0.39–0.44) and (I2) (0.32–0.37), order A to (I5) (0.56–0.61) and (I1) (0.38–0.44), with zero share for (I4) and (I2), as Remark B3.12 predicts; for $\kappa \geq 10^4$, (I2) and (I4) together carry 0.77–0.96 (order B) and 0.59–0.96 (order A). (I3) carries below 0.006 (order B) and at most 0.063 (order A), consistently with $r_{\mathrm{var}} \geq 0.958$. r_{sq} is 0.50–0.52 for $\kappa \leq 10^2$ and at $\kappa = 10^3$, $\alpha = 0$, where the memory is stale (median $2n\gamma\mu_{\mathrm{eff}}$ at least 3.5, median $B/(S/n)$ between 13 and 154) and r_{sq} exceeds $\frac{1}{2}$ ($C > 0$) on 60–99% of the records, and 0.0035–0.075 at $\kappa = 10^5$, where the memory is nearly fresh (median $2n\gamma\mu_{\mathrm{eff}}$ at most 0.38, median $B/(S/n)$ between 1.07 and 1.8): these are the two regimes of Remark B3.13. Optimizing (γ, Δ) at $\kappa = 10^3$, $\alpha = 1$ ($s = 1$) multiplies the certified rate by only $Q(1) \approx 1.17$, whereas the exponent observed at the textbook step is 4.5 times the certificate at that step with Δ optimized (3 seeds; Figure C of the report). For $\kappa \leq 10^2$ ($s \geq 10$) the constants matter much more: at the textbook step, the weight $\Delta = c\gamma^2$ with $c \approx 5.04$ ($\kappa = 10^2$) or $c \approx 5.89$ ($\kappa = 10$) instead of $c = 3$ multiplies the certificate by 1.81 or 1.98, and the joint optimum of Proposition B3.16 (at a smaller step) by $Q(10) \approx 2.42$ or $Q(100) \approx 2.94$, leaving factors 1.75–1.95 and 1.22–1.45 of the measured speed-ups 3.2–3.7. The measured exponents there even exceed the ceiling $1 - \rho \leq \frac{1}{n}$ of Remark B3.18: the means of $-\log \rho_{\mathrm{exact}}$ are 1.06–1.24 times $-\log(1 - \frac{1}{n})$, and the realised slopes of $-\log V_t$ (passes 2–20, mean over the seeds) 1.17–1.33 times. Along a typical trajectory V_t thus decreases faster than $\mathbb{E}[V_t]$ can, the lower bound $\Delta(1 - \frac{1}{n})^t M_0$ on the latter coming from the memory slots not yet refreshed at time t (see also the paragraph "A posteriori versus a priori" of Section B3.6).

B2 Effective curvature and effective smoothness along the trajectory (hypothesis B2)

The proof of Proposition 5.9 of [11] (convergence of SAGA, transcribed in the B3 note) uses exactly two problem constants. The per-sample smoothness constant R^2 enters through the co-coercivity inequality (equation (5.31) of the book),

$$\frac{1}{n}\sum_{i=1}^{n}\left\|f_i'(\theta_{t-1})-f_i'(\eta_*)\right\|^2 \le R^2\, F'(\theta_{t-1})^\top(\theta_{t-1}-\eta_*),$$

and the strong-convexity constant μ enters through $(\theta_{t-1}-\eta_*)^\top F'(\theta_{t-1}) \ge \mu\,\|\theta_{t-1}-\eta_*\|^2$. Both inequalities are applied at the *current iterate* θ_{t-1}, always with the same second point η_*. Hypothesis (B2) of the project is that the slack of these two inequalities along the trajectory actually followed by the algorithm explains a large part of the gap between the rate $1-\min\{1/(3n), 3\mu/(16R^2)\}$ and the observed contraction. This note defines the two trajectory-dependent constants that make the two inequalities equalities, proves their basic properties, makes them explicit for ℓ_2-regularized logistic regression, analyses their dependence on the direction of the error $\theta_t-\eta_*$ (which can differ between gradient descent and SAGA, and which is constrained by an exact invariance property: a trajectory started in the span of the features never leaves it, Lemma B2.16), and finally explains how they enter the SAGA bound and how to measure the fraction of the theory–practice gap they explain.

B2.1 Setting and definitions

Assumption B2.1 (convex R^2-smooth f_i, μ-strongly convex F). *(i) Each $f_i:\mathbb{R}^d\to\mathbb{R}$ is convex, differentiable and R^2-smooth, i.e. f_i' is R^2-Lipschitz-continuous.*

(ii) $F=\frac{1}{n}\sum_{i=1}^n f_i$ is μ-strongly convex with $\mu>0$; its (unique) minimizer is η_*, so that $F'(\eta_*)=0$ (Proposition 5.2 of [11]). We denote by $L\le R^2$ the smoothness constant of F (F' is L-Lipschitz-continuous; $L\le R^2$ because an average of R^2-Lipschitz maps is R^2-Lipschitz), and by $\kappa=R^2/\mu$ the condition number used in the book.

(iii) Whenever second derivatives appear (Lemma B2.5, Proposition B2.8, Section B2.3 and after), each f_i is twice continuously differentiable.

Throughout, for $\theta\neq\eta_*$ we write

$$v=\theta-\eta_*, \qquad u=v/\|v\|,$$

and we use repeatedly that $F'(\theta)=F'(\theta)-F'(\eta_*)=\frac{1}{n}\sum_i[f_i'(\theta)-f_i'(\eta_*)]$.

Definition B2.2 (effective constants μ_{eff} and L_{eff}). *For $\theta\neq\eta_*$, define*

$$\mu_{\text{eff}}(\theta) \;=\; \frac{(\theta-\eta_*)^\top F'(\theta)}{\|\theta-\eta_*\|^2}, \tag{B2.1}$$

$$L_{\text{eff}}(\theta) \;=\; \frac{\frac{1}{n}\sum_{i=1}^n\|f_i'(\theta)-f_i'(\eta_*)\|^2}{(\theta-\eta_*)^\top F'(\theta)}. \tag{B2.2}$$

Along a trajectory $(\theta_t)_{t\ge0}$ we abbreviate $\mu_{\text{eff}}(t)=\mu_{\text{eff}}(\theta_t)$ and $L_{\text{eff}}(t)=L_{\text{eff}}(\theta_t)$.

Lemma B2.4 below shows that $(\theta-\eta_*)^\top F'(\theta) \geq \mu\|\theta-\eta_*\|^2 > 0$, so both quantities are well defined and positive. By construction the two inequalities quoted above become *identities* when μ and R^2 are replaced by $\mu_{\text{eff}}(\theta)$ and $L_{\text{eff}}(\theta)$:

$$(\theta-\eta_*)^\top F'(\theta) = \mu_{\text{eff}}(\theta)\,\|\theta-\eta_*\|^2, \qquad \frac{1}{n}\sum_{i=1}^n \left\|f_i'(\theta)-f_i'(\eta_*)\right\|^2 = L_{\text{eff}}(\theta)\,(\theta-\eta_*)^\top F'(\theta).$$

We call $\mu_{\text{eff}}(\theta)$ the *effective curvature* and $L_{\text{eff}}(\theta)$ the *effective co-coercivity (or effective smoothness) constant* at θ.

B2.2 General properties

We first record the two-point inequalities that we use, with proofs, since the whole note rests on them.

Lemma B2.3 (two-point inequalities)**.** *Let $f:\mathbb{R}^d\to\mathbb{R}$ be convex and differentiable.*

(a) (Monotonicity) For all θ,η: $(f'(\theta)-f'(\eta))^\top(\theta-\eta)\geq 0$.

(b) (Strong monotonicity) If f is μ-strongly convex, then for all θ,η: $(f'(\theta)-f'(\eta))^\top(\theta-\eta)\geq \mu\|\theta-\eta\|^2$.

(c) (Co-coercivity, Proposition 5.4 of [11]) If f is L_f-smooth, then for all θ,η: $\|f'(\theta)-f'(\eta)\|^2 \leq L_f\,(f'(\theta)-f'(\eta))^\top(\theta-\eta)$, and consequently $(f'(\theta)-f'(\eta))^\top(\theta-\eta)\leq L_f\|\theta-\eta\|^2$.

Proof. (b) By definition of μ-strong convexity (Definition 5.2 of the book; the same computation with $\mu=0$, i.e. with the definition of convexity, Definition 5.1, eq. (5.6), gives (a)), $f(\eta)\geq f(\theta)+f'(\theta)^\top(\eta-\theta)+\frac{\mu}{2}\|\eta-\theta\|^2$ and $f(\theta)\geq f(\eta)+f'(\eta)^\top(\theta-\eta)+\frac{\mu}{2}\|\eta-\theta\|^2$. Summing the two inequalities and cancelling $f(\theta)+f(\eta)$ gives $0\geq -(f'(\theta)-f'(\eta))^\top(\theta-\eta)+\mu\|\theta-\eta\|^2$, which is the claim.

(c) Fix θ and let $\varphi(\eta)=f(\eta)-f'(\theta)^\top\eta$. The function φ is convex, L_f-smooth, and $\varphi'(\theta)=0$, so θ minimizes φ. The descent lemma for L_f-smooth functions (Definition 5.3 of the book: $\varphi(\eta')\leq\varphi(\eta)+\varphi'(\eta)^\top(\eta'-\eta)+\frac{L_f}{2}\|\eta'-\eta\|^2$) applied with $\eta'=\eta-\frac{1}{L_f}\varphi'(\eta)$ gives $\varphi(\theta)\leq\varphi(\eta')\leq\varphi(\eta)-\frac{1}{2L_f}\|\varphi'(\eta)\|^2$, that is,

$$f(\theta)\;\leq\; f(\eta)+f'(\theta)^\top(\theta-\eta)-\frac{1}{2L_f}\left\|f'(\eta)-f'(\theta)\right\|^2.$$

This is equation (5.12) of the book. Writing it also with θ and η exchanged and adding the two inequalities yields $0\leq(f'(\theta)-f'(\eta))^\top(\theta-\eta)-\frac{1}{L_f}\|f'(\theta)-f'(\eta)\|^2$, the co-coercivity inequality. The consequence follows from the Cauchy–Schwarz inequality $(f'(\theta)-f'(\eta))^\top(\theta-\eta)\leq\|f'(\theta)-f'(\eta)\|\,\|\theta-\eta\|$ together with the Lipschitz property $\|f'(\theta)-f'(\eta)\|\leq L_f\|\theta-\eta\|$. □

Lemma B2.4 (μ_{eff} lies between μ and L)**.** *Under Assumption B2.1, for all $\theta\neq\eta_*$,*

$$\mu\|\theta-\eta_*\|^2\;\leq\;(\theta-\eta_*)^\top F'(\theta) \qquad \textit{(strong convexity at the iterate; i.e. } \mu\leq\mu_{\text{eff}}(\theta)\textit{)}, \tag{B2.3}$$

$$(\theta-\eta_*)^\top F'(\theta)\;\leq\;L\|\theta-\eta_*\|^2 \qquad \textit{(smoothness at the iterate; i.e. } \mu_{\text{eff}}(\theta)\leq L\textit{)}. \tag{B2.4}$$

Proof. Apply Lemma B2.3(b) and (c) to $f=F$ with $\eta=\eta_*$ and use $F'(\eta_*)=0$. □

The tightness ratio of (B2.3) at θ is exactly $\mu/\mu_{\text{eff}}(\theta) \in [\mu/L, 1]$; this is the quantity the experiments of the B3 note record under the name r_{sc} (Definition B3.8 of that note, which likewise calls $r_{\text{coco}} = L_{\text{eff}}/R^2$ the tightness ratio of the co-coercivity step, see Lemma B2.6); we use the same two names below.

Lemma B2.5 (integral form of the effective curvature). *Assume F is twice continuously differentiable and let, for $\theta \neq \eta_*$,*

$$\bar{H}(\theta) \;=\; \int_0^1 F''(\eta_* + s\,v)\,ds \qquad (v = \theta - \eta_*,\; u = v/\|v\|).$$

Then $\bar{H}(\theta)$ is symmetric, $\mu I \preccurlyeq \bar{H}(\theta) \preccurlyeq LI$, $F'(\theta) = \bar{H}(\theta)\,v$, and

$$\mu_{\text{eff}}(\theta) \;=\; \frac{v^\top \bar{H}(\theta)\,v}{\|v\|^2} \;=\; u^\top \bar{H}(\theta)\,u, \tag{B2.5}$$

i.e. $\mu_{\text{eff}}(\theta)$ is the Rayleigh quotient, in the direction of the error, of the Hessian averaged along the segment $[\eta_*, \theta]$.

Proof. Let $g(s) = F'(\eta_* + sv)$ for $s \in [0,1]$. Since F is C^2, g is C^1 with $g'(s) = F''(\eta_* + sv)\,v$, and the fundamental theorem of calculus gives $F'(\theta) - F'(\eta_*) = g(1) - g(0) = \int_0^1 F''(\eta_* + sv)v\,ds = \bar{H}(\theta)v$; as $F'(\eta_*) = 0$, this is $F'(\theta) = \bar{H}(\theta)v$. The integral of a continuous function with values in the set $\{A \text{ symmetric} : \mu I \preccurlyeq A \preccurlyeq LI\}$ (a closed convex set, since it is defined by linear inequalities $\mu\|w\|^2 \leq w^\top A w \leq L\|w\|^2$ for all w) stays in that set, because Riemann sums are convex combinations; and for a twice differentiable function, μ-strong convexity and L-smoothness are equivalent to $\mu I \preccurlyeq F''(\theta) \preccurlyeq LI$ for all θ (Section 5.2.3 of [11]). Finally $\mu_{\text{eff}}(\theta) = v^\top F'(\theta)/\|v\|^2 = v^\top \bar{H}(\theta)v/\|v\|^2$. □

Lemma B2.6 (L_{eff} lies between μ_{eff} and R^2). *Under Assumption B2.1, for all $\theta \neq \eta_*$, with $v = \theta - \eta_*$,*

$$\big\|F'(\theta)\big\|^2 \;\leq\; \frac{1}{n}\sum_{i=1}^n \big\|f_i'(\theta) - f_i'(\eta_*)\big\|^2 \qquad \textit{(Jensen / gradient diversity)}, \tag{B2.6}$$

$$\frac{1}{n}\sum_{i=1}^n \big\|f_i'(\theta) - f_i'(\eta_*)\big\|^2 \;\leq\; R^2\,v^\top F'(\theta) \qquad \textit{(averaged co-coercivity, eq. (5.31) of the book)}, \tag{B2.7}$$

$$\mu \;\leq\; \mu_{\text{eff}}(\theta) \;\leq\; \frac{\|F'(\theta)\|}{\|v\|} \;\leq\; \frac{\|F'(\theta)\|^2}{v^\top F'(\theta)}. \tag{B2.8}$$

Consequently

$$\mu \;\leq\; \mu_{\text{eff}}(\theta) \;\leq\; \frac{\|F'(\theta)\|^2}{v^\top F'(\theta)} \;\leq\; L_{\text{eff}}(\theta) \;\leq\; R^2, \tag{B2.9}$$

and, more precisely, with $h_i = f_i'(\theta) - f_i'(\eta_*)$ (the vectors a_i of Section B3.2) and $\bar{h} = \frac{1}{n}\sum_i h_i = F'(\theta)$,

$$L_{\text{eff}}(\theta) \;=\; \frac{\|F'(\theta)\|^2 + \frac{1}{n}\sum_{i=1}^n \big\|h_i - \bar{h}\big\|^2}{v^\top F'(\theta)}. \tag{B2.10}$$

If moreover each f_i is L_i-smooth, then $L_{\text{eff}}(\theta) \leq \max_i L_i$ (so one may take $R^2 = \max_i L_i$).

Proof. (B2.10) is the variance decomposition $\frac{1}{n}\sum_i \|h_i\|^2 = \|\bar{h}\|^2 + \frac{1}{n}\sum_i \|h_i - \bar{h}\|^2$ (expand the squares and use $\sum_i (h_i - \bar{h}) = 0$), divided by $v^\top F'(\theta) > 0$; dropping the nonnegative variance term gives (B2.6). For (B2.7), apply Lemma B2.3(c) to each f_i (convex, R^2-smooth) with $\eta = \eta_*$: $\|h_i\|^2 \le R^2 h_i^\top v$; averaging over i gives $\frac{1}{n}\sum_i \|h_i\|^2 \le R^2 \bar{h}^\top v = R^2 v^\top F'(\theta)$. If f_i is L_i-smooth, the same argument gives $\frac{1}{n}\sum_i \|h_i\|^2 \le \frac{1}{n}\sum_i L_i h_i^\top v \le (\max_i L_i)\, \bar{h}^\top v$, using $h_i^\top v \ge 0$ (Lemma B2.3(a)). For (B2.8): the first inequality is Lemma B2.4; the second is Cauchy–Schwarz, $v^\top F'(\theta) \le \|v\|\,\|F'(\theta)\|$, divided by $\|v\|^2$; the third is the same Cauchy–Schwarz inequality in the form $\|F'(\theta)\| / \|v\| \le \|F'(\theta)\|^2 / (v^\top F'(\theta))$. Combining (B2.6)–(B2.8) with the definition of L_{eff} gives (B2.9). □

Remark B2.7. The chain (B2.9) shows that the "effective condition number" $L_{\text{eff}}(\theta)/\mu_{\text{eff}}(\theta)$ is always at least 1, and that two effects are separated in $L_{\text{eff}}(\theta)$, whose ratio to R^2 is the tightness ratio of the co-coercivity step: (B2.6) measures the *diversity* of the individual gradient differences (its tightness ratio $\|F'(\theta)\|^2 / \frac{1}{n}\sum_i \|h_i\|^2$ equals 1 iff all h_i coincide), while the remaining factor $\|F'(\theta)\|^2 / (v^\top F'(\theta)) \le L_{\text{eff}}(\theta) \le R^2$ measures how much of the bound R^2 is actually used by the full gradient in the direction of the error. Both ratios are cheap to compute once η_* is known (one pass over the data).

Proposition B2.8 (directional limits at the optimum)**.** *Assume each f_i is twice continuously differentiable, and write $H_* = F''(\eta_*)$ and $M_* = \frac{1}{n}\sum_{i=1}^n f_i''(\eta_*)^2$. For a unit vector u define*

$$\Lambda(u) \;=\; \frac{u^\top M_* u}{u^\top H_* u}.$$

Then, as $\theta \to \eta_*$ (with $u = (\theta - \eta_*)/\|\theta - \eta_*\|$ allowed to vary),

$$\mu_{\text{eff}}(\theta) - u^\top H_* u \;\to\; 0, \qquad L_{\text{eff}}(\theta) - \Lambda(u) \;\to\; 0,$$

uniformly in the direction. In particular, if $\theta \to \eta_*$ along a fixed direction u, then $\mu_{\text{eff}}(\theta) \to u^\top H_* u$ (the Rayleigh quotient of the Hessian at the optimum in the direction of approach) and $L_{\text{eff}}(\theta) \to \Lambda(u)$. Moreover, for every unit u,

$$u^\top H_* u \;\le\; \frac{u^\top H_*^2 u}{u^\top H_* u} \;\le\; \Lambda(u) \;\le\; R^2, \tag{B2.11}$$

consistently with (B2.9).

Proof. Let $r = \|v\|$ and $\bar{H}_i(\theta) = \int_0^1 f_i''(\eta_* + sv)\, ds$, so that, exactly as in Lemma B2.5, $f_i'(\theta) - f_i'(\eta_*) = \bar{H}_i(\theta) v$, $0 \preccurlyeq \bar{H}_i(\theta) \preccurlyeq R^2 I$, and $\bar{H}(\theta) = \frac{1}{n}\sum_i \bar{H}_i(\theta)$. Set $\varepsilon_i(r) = \sup_{\|w\| \le r} \|f_i''(\eta_* + w) - f_i''(\eta_*)\|_{\text{op}}$ and $\varepsilon(r) = \frac{1}{n}\sum_i \varepsilon_i(r)$; by continuity of f_i'' at η_*, $\varepsilon(r) \to 0$ as $r \to 0$. Since every point $\eta_* + sv$ of the segment is within distance r of η_*, $\|\bar{H}_i(\theta) - f_i''(\eta_*)\|_{\text{op}} \le \varepsilon_i(r)$, hence $\|\bar{H}(\theta) - H_*\|_{\text{op}} \le \varepsilon(r)$ and, by (B2.5), $|\mu_{\text{eff}}(\theta) - u^\top H_* u| = |u^\top (\bar{H}(\theta) - H_*) u| \le \varepsilon(r)$, which proves the first limit.

For the second, write $L_{\text{eff}}(\theta) = \mathcal{N}(\theta)/\mathcal{D}(\theta)$ with $\mathcal{N}(\theta) = \frac{1}{r^2} \cdot \frac{1}{n}\sum_i \|\bar{H}_i(\theta) v\|^2 = \frac{1}{n}\sum_i u^\top \bar{H}_i(\theta)^2 u$ and $\mathcal{D}(\theta) = \frac{1}{r^2} v^\top F'(\theta) = u^\top \bar{H}(\theta) u = \mu_{\text{eff}}(\theta)$. For symmetric matrices $0 \preccurlyeq P, Q \preccurlyeq R^2 I$ one has $P^2 - Q^2 = P(P-Q) + (P-Q)Q$, so $\|P^2 - Q^2\|_{\text{op}} \le 2R^2 \|P - Q\|_{\text{op}}$; therefore $|\mathcal{N}(\theta) - u^\top M_* u| \le 2R^2 \varepsilon(r)$, while $|\mathcal{D}(\theta) - u^\top H_* u| \le \varepsilon(r)$. Both denominators are at least μ (Lemma B2.4 and $H_* \succcurlyeq \mu I$) and both numerators are at most R^4, so

$$|L_{\text{eff}}(\theta) - \Lambda(u)| = \left|\frac{\mathcal{N}}{\mathcal{D}} - \frac{\mathcal{N}_*}{\mathcal{D}_*}\right| \le \frac{|\mathcal{N} - \mathcal{N}_*|}{\mathcal{D}} + \mathcal{N}_* \frac{|\mathcal{D} - \mathcal{D}_*|}{\mathcal{D}\,\mathcal{D}_*} \le \frac{2R^2 \varepsilon(r)}{\mu} + \frac{R^4 \varepsilon(r)}{\mu^2} \;\xrightarrow[r \to 0]{}\; 0,$$

uniformly in u (here $\mathcal{N}_* = u^\top M_* u$, $\mathcal{D}_* = u^\top H_* u$).

For (B2.11): with $A_i = f_i''(\eta_*)$ and $\bar{A} = H_*$, the matrix $\frac{1}{n}\sum_i (A_i - \bar{A})^2 = \frac{1}{n}\sum_i A_i^2 - \bar{A}^2 = M_* - H_*^2$ is positive semidefinite (each $(A_i - \bar{A})^2 = (A_i - \bar{A})^\top (A_i - \bar{A}) \succcurlyeq 0$), which gives the middle inequality; the first one is Cauchy–Schwarz, $(u^\top H_* u)^2 \le \|H_* u\|^2 \|u\|^2 = u^\top H_*^2 u$; the last one uses $A_i^2 \preccurlyeq R^2 A_i$ for $0 \preccurlyeq A_i \preccurlyeq R^2 I$ (indeed $R^2 A_i - A_i^2 = A_i^{1/2}(R^2 I - A_i) A_i^{1/2} \succcurlyeq 0$), so that $M_* \preccurlyeq R^2 H_*$ and $\Lambda(u) \le R^2$. □

Remark B2.9. Proposition B2.8 says that, close to the optimum, μ_{eff} and L_{eff} are not numbers but functions of the direction of the error. This is the reason why the direction in which an algorithm approaches η_* (Section B2.4) matters for the observed contraction rate. The "local strong convexity constant" $\lambda_{\min}(H_*)$ of [10] is the worst case of $u^\top H_* u$ over directions, and the quantity $u^\top H_*^2 u / (u^\top H_* u)$ in (B2.11) is the Rayleigh-type quotient that governs gradient descent on a quadratic.

B2.3 Logistic regression

Setting and constants. Let $\ell(a) = \log(1 + e^{-a})$ and $\sigma(a) = 1/(1 + e^{-a})$. Then

$$\ell'(a) = \sigma(a) - 1 = -\sigma(-a), \qquad \ell''(a) = \sigma(a)\sigma(-a) = \sigma(a)\big(1 - \sigma(a)\big) = \sigma'(a) \in (0, \tfrac{1}{4}],$$

$$\ell'''(a) = \ell''(a)\big(1 - 2\sigma(a)\big),$$

so that ℓ'' is even, maximal at 0 with $\ell''(0) = 1/4$, and non-increasing on $[0, \infty)$ (because $1 - 2\sigma(a) \le 0$ for $a \ge 0$). The data are $(x_i, y_i) \in \mathbb{R}^d \times \{-1, 1\}$, $i = 1, \dots, n$; we write $R_x = \max_i \|x_i\|$, $\hat{\Sigma} = \frac{1}{n}\sum_i x_i x_i^\top$ (empirical second-moment matrix of the features), $a_i(\theta) = y_i x_i^\top \theta$, and

$$f_i(\theta) = \ell(a_i(\theta)) + \frac{\lambda}{2}\|\theta\|^2, \qquad f_i'(\theta) = \ell'(a_i(\theta))\, y_i x_i + \lambda\theta, \qquad f_i''(\theta) = \ell''(a_i(\theta))\, x_i x_i^\top + \lambda I,$$

using $y_i^2 = 1$. Hence

$$F''(\theta) = \frac{1}{n}\sum_{i=1}^n \ell''\big(y_i x_i^\top \theta\big)\, x_i x_i^\top + \lambda I. \tag{B2.12}$$

The function f_i is L_i-smooth with $L_i = \|x_i\|^2/4 + \lambda$ (the largest eigenvalue of $f_i''(\theta)$, $\ell''(a_i(\theta))\|x_i\|^2 + \lambda$, is at most L_i, with equality when $a_i(\theta) = 0$), so the per-sample smoothness constant of the project is $R^2 = \max_i L_i = R_x^2/4 + \lambda$. The smoothness constant of F is $L = \lambda + \lambda_{\max}(\hat{\Sigma})/4$: indeed $F''(\theta) \preccurlyeq \frac{1}{4}\hat{\Sigma} + \lambda I$ for all θ, with equality at $\theta = 0$. Finally $F''(\theta) \succcurlyeq \lambda I$ for all θ, so F is λ-strongly convex; Proposition B2.12 shows that $\mu = \lambda$ is the exact global constant.

Lemma B2.10 (mean-value weights and explicit formulas)**.** *Let $\theta \ne \eta_*$, $v = \theta - \eta_*$, $u = v/\|v\|$, $c_i = x_i^\top u$, $a_i = a_i(\theta)$, $a_i^* = a_i(\eta_*)$, and define the mean-value weights*

$$w_i(\theta) \;=\; \int_0^1 \ell''\big(a_i^* + s(a_i - a_i^*)\big)\, ds \;\in\; (0, \tfrac{1}{4}]. \tag{B2.13}$$

By the fundamental theorem of calculus (see the proof), the weights have the closed form

$$w_i(\theta) = \frac{\ell'(a_i) - \ell'(a_i^*)}{a_i - a_i^*} = \frac{\sigma(a_i) - \sigma(a_i^*)}{a_i - a_i^*} \quad \text{if } a_i \ne a_i^*, \qquad w_i(\theta) = \ell''(a_i^*) \quad \text{if } a_i = a_i^*, \tag{B2.14}$$

so that the formulas below are exact and computable in $O(nd)$ operations without any quadrature (in floating point, when $|a_i - a_i^*|$ is below the square root of the machine precision, use $\ell''(\frac{1}{2}(a_i + a_i^*))$,

which agrees with $w_i(\theta)$ up to $O(|a_i - a_i^*|^2)$, to avoid cancellation; alternatively μ_{eff} and L_{eff} can be computed directly from Definition B2.2). Then

$$f_i'(\theta) - f_i'(\eta_*) = w_i(\theta)\,(x_i^\top v)\,x_i + \lambda v, \qquad \bar{H}(\theta) = \frac{1}{n}\sum_{i=1}^n w_i(\theta)\,x_i x_i^\top + \lambda I, \tag{B2.15}$$

and, with $A(\theta) = \frac{1}{n}\sum_i w_i(\theta)c_i^2$ and $B(\theta) = \frac{1}{n}\sum_i w_i(\theta)^2 c_i^2\,\|x_i\|^2$,

$$\mu_{\text{eff}}(\theta) = \lambda + A(\theta) = \lambda + \frac{1}{n}\sum_{i=1}^n w_i(\theta)\,(x_i^\top u)^2, \tag{B2.16}$$

$$L_{\text{eff}}(\theta) = \frac{\lambda^2 + 2\lambda A(\theta) + B(\theta)}{\lambda + A(\theta)} = \mu_{\text{eff}}(\theta) + \frac{B(\theta) - A(\theta)^2}{\mu_{\text{eff}}(\theta)}, \qquad B(\theta) \geq A(\theta)^2. \tag{B2.17}$$

Proof. The map $s \mapsto \ell'(a_i^* + s(a_i - a_i^*))$ is C^1 with derivative $\ell''(a_i^* + s(a_i - a_i^*))(a_i - a_i^*)$, so the fundamental theorem of calculus gives $\ell'(a_i) - \ell'(a_i^*) = w_i(\theta)(a_i - a_i^*) = w_i(\theta)\,y_i x_i^\top v$, which is (B2.14) (when $a_i = a_i^*$ the integrand of (B2.13) is the constant $\ell''(a_i^*)$; $\ell' = \sigma - 1$ gives the second expression). Therefore $f_i'(\theta) - f_i'(\eta_*) = [\ell'(a_i) - \ell'(a_i^*)]y_i x_i + \lambda v = w_i(\theta)(x_i^\top v)x_i + \lambda v$ (using $y_i^2 = 1$). Averaging over i gives $F'(\theta) = [\frac{1}{n}\sum_i w_i(\theta)x_i x_i^\top + \lambda I]v$; by linearity of the integral this matrix is exactly $\int_0^1 F''(\eta_* + sv)ds$ computed from (B2.12), i.e. the matrix $\bar{H}(\theta)$ of Lemma B2.5. Formula (B2.16) is the Rayleigh quotient (B2.5). For (B2.17), $f_i'(\theta) - f_i'(\eta_*) = \|v\|\,(w_i c_i x_i + \lambda u)$ and $\|w_i c_i x_i + \lambda u\|^2 = w_i^2 c_i^2\,\|x_i\|^2 + 2\lambda w_i c_i^2 + \lambda^2$ (because $x_i^\top u = c_i$ and $\|u\| = 1$), so $\frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|^2 = \|v\|^2\,(B + 2\lambda A + \lambda^2)$, while $v^\top F'(\theta) = \|v\|^2\,(\lambda + A)$; dividing gives the first expression, and $\lambda^2 + 2\lambda A + B = (\lambda + A)^2 + (B - A^2)$ gives the second. Finally $\|x_i\|^2 \geq c_i^2$ (Cauchy–Schwarz) gives $B \geq \frac{1}{n}\sum_i (w_i c_i^2)^2 \geq (\frac{1}{n}\sum_i w_i c_i^2)^2 = A^2$ (Jensen). □

Lemma B2.11 (two-sided bounds on the effective curvature)**.** *Let $\theta \neq \eta_*$. For each i, let $m_i(\theta)$ be the smaller of $\ell''(a_i^*)$ and $\ell''(a_i)$, and let $m(\theta) = \min_i m_i(\theta)$. Then $m_i(\theta) \leq w_i(\theta) \leq \frac{1}{4}$, and*

$$\lambda + m(\theta)\,u^\top \hat{\Sigma} u \;\leq\; \lambda + \frac{1}{n}\sum_{i=1}^n m_i(\theta)(x_i^\top u)^2 \;\leq\; \mu_{\text{eff}}(\theta), \tag{B2.18}$$

$$\mu_{\text{eff}}(\theta) \;\leq\; \lambda + \frac{1}{4}\,u^\top \hat{\Sigma} u \;\leq\; \lambda + \frac{1}{4}\lambda_{\max}(\hat{\Sigma}) = L. \tag{B2.19}$$

In particular $\mu_{\text{eff}}(\theta) \geq \lambda + m(\theta)\lambda_{\min}(\hat{\Sigma})$, where $m(\theta) = \min_i \min\{\sigma'(y_i x_i^\top \eta_*), \sigma'(y_i x_i^\top \theta)\}$ is the minimum of $\ell'' = \sigma'$ over the n segments $[a_i^*, a_i]$ and is computable in $O(nd)$ operations.

Proof. Every point $a = a_i^* + s(a_i - a_i^*)$, $s \in [0,1]$, of the segment satisfies $|a| \leq \max\{|a_i^*|, |a_i|\}$; since ℓ'' is even and non-increasing in $|a|$, ℓ'' on the segment is at least $\min\{\ell''(a_i^*), \ell''(a_i)\} = m_i(\theta)$ and at most $1/4$. Integrating over s gives $m_i(\theta) \leq w_i(\theta) \leq 1/4$, and inserting these bounds into (B2.16), with $\frac{1}{n}\sum_i (x_i^\top u)^2 = u^\top \hat{\Sigma} u \in [\lambda_{\min}(\hat{\Sigma}), \lambda_{\max}(\hat{\Sigma})]$, gives (B2.18)–(B2.19). □

Global versus local strong convexity. The next proposition states precisely in which sense "the global strong convexity constant is exactly λ" for logistic regression. Recall that, for a twice continuously differentiable F, F is μ-strongly convex if and only if $\theta \mapsto F(\theta) - \frac{\mu}{2}\|\theta\|^2$ is convex (Exercise 5.5 of [11]), if and only if $F''(\theta) \succcurlyeq \mu I$ for all θ, so that the best (largest) strong convexity constant is

$$\mu_{\text{glob}} \;=\; \sup\{\mu \geq 0 :\; F \text{ is } \mu\text{-strongly convex}\} \;=\; \inf_{\theta \in \mathbb{R}^d} \lambda_{\min}\big(F''(\theta)\big).$$

Proposition B2.12 (the global strong convexity constant is λ)**.** *For ℓ_2-regularized logistic regression, let $I_0 = \{i : x_i \neq 0\}$ (if $I_0 = \emptyset$ then $F'' \equiv \lambda I$ and there is nothing to prove).*

(a) There exists a unit vector $\hat{d} \in \mathbb{R}^d$ with $x_i^\top \hat{d} \neq 0$ for every $i \in I_0$; in fact Lebesgue-almost every direction has this property.

(b) For every such $\hat{d}$, with $\delta(\hat{d}) = \min_{i \in I_0} \left| x_i^\top \hat{d} \right| > 0$, and every $t \geq 0$,

$$\lambda \;\leq\; \lambda_{\min}\big(F''(t\hat{d})\big) \;\leq\; \lambda + R_x^2\, e^{-t\,\delta(\hat{d})}. \tag{B2.20}$$

(c) Consequently $\mu_{\mathrm{glob}} = \inf_\theta \lambda_{\min}(F''(\theta)) = \lambda$: the global strong convexity constant of F is exactly λ, whatever the data.

(d) The infimum is attained at some θ if and only if $\operatorname{span}\{x_1, \dots, x_n\} \neq \mathbb{R}^d$; in that case $\lambda_{\min}(F''(\theta)) = \lambda$ for every θ. If the features span $\mathbb{R}^d$, then $\lambda_{\min}(F''(\theta)) > \lambda$ for every θ and the value λ is only approached as $\|\theta\| \to \infty$; it is in fact approached along every *ray: for every unit vector $\hat{d}$ and every $t \geq 0$,*

$$\lambda_{\min}\big(F''(t\hat{d})\big) \;\leq\; \lambda + \frac{1}{n}\sum_{i=1}^n \ell''\big(t\, y_i x_i^\top \hat{d}\big)(x_i^\top \hat{d})^2 \;\leq\; \lambda + R_x^2\, e^{-t\,\delta'(\hat{d})}, \tag{B2.21}$$

where $\delta'(\hat{d}) = \min\{\left| x_i^\top \hat{d} \right| : x_i^\top \hat{d} \neq 0\}$, with the conventions $\delta'(\hat{d}) = +\infty$ and $e^{-\infty} = 0$ when $x_i^\top \hat{d} = 0$ for all i; hence $\lim_{t\to\infty} \lambda_{\min}(F''(t\hat{d})) = \lambda$ for every direction $\hat{d}$. Directions as in (a) are those along which the whole matrix $F''(t\hat{d}) - \lambda I$ tends to 0; along a general ray,

$$F''(t\hat{d}) - \lambda I \;\xrightarrow[t\to\infty]{}\; \frac{1}{4n} \sum_{i:\, x_i^\top \hat{d} = 0} x_i x_i^\top,$$

a matrix which is singular because $\hat{d}$ lies in its kernel.

Proof. (a) For $i \in I_0$ the set $\{\hat{d} : x_i^\top \hat{d} = 0\}$ is a hyperplane, hence has Lebesgue measure zero; a finite union of null sets is null, so its complement is nonempty (indeed of full measure); normalize any element.

(b) The lower bound is $F'' \succcurlyeq \lambda I$. For the upper bound, $F''(t\hat{d}) - \lambda I = \frac{1}{n}\sum_{i\in I_0} \ell''(t\, y_i x_i^\top \hat{d})\, x_i x_i^\top$ (the terms with $x_i = 0$ vanish), and this matrix is positive semidefinite with

$$\Big\| F''(t\hat{d}) - \lambda I \Big\|_{\mathrm{op}} \leq \frac{1}{n}\sum_{i\in I_0} \ell''(t\, y_i x_i^\top \hat{d})\, \|x_i\|^2 \leq R_x^2 \max_{i\in I_0} \ell''(t\, y_i x_i^\top \hat{d}) \leq R_x^2 e^{-t\delta(\hat{d})},$$

where we used $\ell''(a) = \sigma(a)\sigma(-a) \leq \sigma(-|a|) = \frac{1}{1+e^{|a|}} \leq e^{-|a|}$ (because $\sigma \leq 1$ and ℓ'' is even) and $\left| t\, y_i x_i^\top \hat{d} \right| = t \left| x_i^\top \hat{d} \right| \geq t\delta(\hat{d})$. Hence $\lambda_{\min}(F''(t\hat{d})) \leq \lambda + \Big\| F''(t\hat{d}) - \lambda I \Big\|_{\mathrm{op}} \leq \lambda + R_x^2 e^{-t\delta(\hat{d})}$.

(c) Letting $t \to \infty$ in (B2.20) gives $\inf_\theta \lambda_{\min}(F''(\theta)) \leq \lambda$; the reverse inequality is $F'' \succcurlyeq \lambda I$.

(d) $\lambda_{\min}(F''(\theta)) = \lambda$ iff the positive semidefinite matrix $\frac{1}{n}\sum_i \ell''(a_i(\theta)) x_i x_i^\top$ is singular, iff there is $w \neq 0$ with $\frac{1}{n}\sum_i \ell''(a_i(\theta))(x_i^\top w)^2 = 0$, iff $x_i^\top w = 0$ for all i (since $\ell'' > 0$), iff the x_i do not span $\mathbb{R}^d$. This condition does not depend on θ. For (B2.21): $\lambda_{\min}(F''(t\hat{d})) \leq \hat{d}^\top F''(t\hat{d})\, \hat{d} = \lambda + \frac{1}{n}\sum_i \ell''(t\, y_i x_i^\top \hat{d})(x_i^\top \hat{d})^2$ (Rayleigh quotient with $\left\| \hat{d} \right\| = 1$, using $y_i^2 = 1$); the terms with $x_i^\top \hat{d} = 0$ vanish, and for the others $\ell''(t\, y_i x_i^\top \hat{d})(x_i^\top \hat{d})^2 \leq e^{-t\left| x_i^\top \hat{d} \right|}\, \|x_i\|^2 \leq R_x^2 e^{-t\delta'(\hat{d})}$ by the bound $\ell''(a) \leq e^{-|a|}$ used in (b). The limit of $F''(t\hat{d}) - \lambda I$ follows termwise from $\ell''(t\, y_i x_i^\top \hat{d}) \to 0$ when $x_i^\top \hat{d} \neq 0$ and $\ell''(0) = \frac{1}{4}$ when $x_i^\top \hat{d} = 0$. $\square$

Corollary B2.13 (strong convexity on a bounded region). *In ℓ_2-regularized logistic regression, for every $D > 0$ and every θ with $\|\theta\| \le D$,*

$$\lambda_{\min}\big(F''(\theta)\big) \;\ge\; \lambda + \ell''(R_x D)\,\lambda_{\min}(\hat\Sigma) \;\ge\; \lambda + \tfrac{1}{4}e^{-R_x D}\lambda_{\min}(\hat\Sigma). \tag{B2.22}$$

Moreover $\|\eta_*\| \le \sqrt{2\log 2/\lambda}$, so that the ball $\{\|\theta\| \le 2\sqrt{2\log 2/\lambda}\}$ contains the ball of radius $\sqrt{2\log 2/\lambda}$ around η_*.

Proof. If $\|\theta\| \le D$ then $|a_i(\theta)| \le \|x_i\|\,\|\theta\| \le R_x D$, hence $\ell''(a_i(\theta)) \ge \ell''(R_x D)$ (ℓ'' even, non-increasing in $|a|$) and $F''(\theta) \succcurlyeq \lambda I + \ell''(R_x D)\hat\Sigma$. The second inequality uses $\ell''(a) = \sigma(|a|)\sigma(-|a|) \ge \frac{1}{2}\cdot\frac{1}{1+e^{|a|}} \ge \frac{1}{4}e^{-|a|}$. For the norm of η_*: $\frac{\lambda}{2}\|\eta_*\|^2 \le F(\eta_*) \le F(0) = \log 2$. □

Thus the global constant λ is only "seen" at infinity: on any bounded region the Hessian is uniformly larger than λI by a data-dependent amount, and an algorithm whose iterates stay in a bounded region never experiences the curvature λ unless $\lambda_{\min}(\hat\Sigma) = 0$ (for instance when $d > n$; Lemma B2.16 below identifies which trajectories then actually see it) or in directions where $\hat\Sigma$ is nearly singular. Gradient descent with $\gamma \le 1/L$ stays in the ball of radius $\|\theta_0 - \eta_*\|$ around η_*: with $v = \theta_{t-1} - \eta_*$, $\|\theta_t - \eta_*\|^2 = \|v\|^2 - 2\gamma v^\top F'(\theta_{t-1}) + \gamma^2\|F'(\theta_{t-1})\|^2 \le \|v\|^2 - \gamma(2-\gamma L)\,v^\top F'(\theta_{t-1}) \le \|v\|^2$ by co-coercivity of F (Lemma B2.3(c) with $\eta = \eta_*$) and $v^\top F'(\theta_{t-1}) \ge 0$. For SAGA the statement is only probabilistic: under Proposition 5.9, $V_t \ge 0$ and $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \le V_{t-1}$, so $(V_t)_{t\ge 0}$ is a nonnegative supermartingale; it converges almost surely, hence $\sup_t V_t < \infty$ almost surely, and Ville's maximal inequality gives $\mathbb{P}(\sup_{t\ge 0} V_t \ge c) \le V_0/c$ for every $c > 0$. Since $\|\theta_t - \eta_*\|^2 \le V_t$, the SAGA iterates are almost surely bounded; they are not, in general, confined to a deterministic ball (for $n = 1$ SAGA is gradient descent with $\gamma = 1/(4R^2) \le 1/L$, which stays in the ball of radius $\|\theta_0 - \eta_*\|$). For instance, take $n = 2$, $d = 1$, $x_1 = 1$, $x_2 = 0.1$, $y = (1, -1)$, $\lambda = 10^{-3}$, $\theta_0 = 0$, the step and initialization of Proposition 5.9, and cycles of $2\cdot 10^4$ draws of the index 2 followed by one draw of the index 1. While the index 2 is repeated, the iterate converges to the point where $f_2'(\theta) = -z_1^{(t)}$, the stored $z_1^{(t)}$ being stale, and $|\theta_t - \eta_*|$, initially 2.7, equals about 396, 1294, 2192 and 3090 after the cycles 1, 3, 5 and 7: it grows by about $(1-x_2)/\lambda = 900$ every two cycles (a draw of the index 1 at a point $\theta \ll 0$ stores $z_1 \approx -1 + \lambda\theta$, and the next phase brings the iterate to about $|\theta| + (1-x_2)/\lambda$), and keeps growing if the phases are long enough. Since every finite sequence of indices has positive probability, $\mathbb{P}(\sup_t \|\theta_t - \eta_*\| \ge c) > 0$ for every $c > 0$. The next lemma, which is the scalar case of the generalized self-concordance property of the logistic loss ([8], Lemma 1 and Proposition 1, eq. (6); Assumption (A2) in [10]), turns this into a quantitative comparison between $\mu_{\text{eff}}(\theta)$ and the curvature at the optimum.

Lemma B2.14 (generalized self-concordance of the logistic loss). *For all $a, b \in \mathbb{R}$,*

$$e^{-|b-a|}\,\ell''(a) \;\le\; \ell''(b) \;\le\; e^{|b-a|}\,\ell''(a).$$

Consequently, for $\theta \ne \eta_*$ and all i, with $p_i = \ell''(a_i^*) = \sigma'(y_i x_i^\top \eta_*)$,

$$e^{-|x_i^\top v|}\,p_i \;\le\; w_i(\theta) \;\le\; e^{|x_i^\top v|}\,p_i, \qquad \text{so that} \qquad e^{-R_x\|v\|}p_i \le w_i(\theta) \le e^{R_x\|v\|}p_i. \tag{B2.23}$$

Proof. Since $\ell'' > 0$ and $\ell''' = \ell''(1-2\sigma)$, the function $\log \ell''$ is differentiable with $|(\log\ell'')'(a)| = |1 - 2\sigma(a)| \le 1$; integrating between a and b gives $|\log\ell''(b) - \log\ell''(a)| \le |b-a|$, which is the claim. For (B2.23), every point of the segment $[a_i^*, a_i]$ is at distance at most $|a_i - a_i^*| = |y_i x_i^\top v| = |x_i^\top v| \le R_x\|v\|$ from a_i^*; apply the first claim with $a = a_i^*$ and integrate over $s \in [0,1]$ in (B2.13). □

Corollary B2.15 (effective curvature versus curvature at the optimum). *For ℓ_2-regularized logistic regression, let $p_i = \sigma'(y_i x_i^\top \eta_*)$, $H_* = F''(\eta_*) = \frac{1}{n}\sum_i p_i x_i x_i^\top + \lambda I$ and $p_{\min} = \min_i p_i$. For every $\theta \neq \eta_*$, with $u = (\theta - \eta_*)/\|\theta - \eta_*\|$,*

$$\lambda + e^{-R_x\|\theta-\eta_*\|}\left(u^\top H_* u - \lambda\right) \;\le\; \mu_{\text{eff}}(\theta), \tag{B2.24}$$

$$\mu_{\text{eff}}(\theta) \;\le\; \lambda + e^{R_x\|\theta-\eta_*\|}\left(u^\top H_* u - \lambda\right), \tag{B2.25}$$

and

$$u^\top H_* u \;\ge\; \lambda + p_{\min}\, u^\top \hat\Sigma u \;\ge\; \lambda + p_{\min}\lambda_{\min}(\hat\Sigma). \tag{B2.26}$$

In particular, $\liminf_{\theta\to\eta_*} \mu_{\text{eff}}(\theta) \ge \lambda + p_{\min}\lambda_{\min}(\hat\Sigma)$, and for $\|\theta - \eta_*\| \le 1/R_x$ one has $\mu_{\text{eff}}(\theta) \ge \lambda + e^{-1}p_{\min}\lambda_{\min}(\hat\Sigma)$. If moreover $\theta \in S = \operatorname{span}\{x_1, \dots, x_n\}$ (which holds along the whole trajectory of gradient descent and of SAGA when $\theta_0 \in S$, Lemma B2.16), then $u \in S$, and $\lambda_{\min}(\hat\Sigma)$ may be replaced, in (B2.26) and in the two consequences, by $\lambda^+_{\min}(\hat\Sigma)$, the smallest nonzero eigenvalue of $\hat\Sigma$ (which is positive even when $d > n$).

Proof. By (B2.16) and (B2.23), $\mu_{\text{eff}}(\theta) - \lambda = \frac{1}{n}\sum_i w_i(\theta)c_i^2$ lies between $e^{\mp R_x\|v\|}\frac{1}{n}\sum_i p_i c_i^2$, and $\frac{1}{n}\sum_i p_i c_i^2 = u^\top H_* u - \lambda$. Inequality (B2.26) is $p_i \ge p_{\min}$ and $\frac{1}{n}\sum_i c_i^2 = u^\top \hat\Sigma u \ge \lambda_{\min}(\hat\Sigma)$. For the last claim, $\eta_* \in S$ (Lemma B2.16(a)), so $v = \theta - \eta_* \in S$ and $u \in S$; the kernel of the positive semidefinite matrix $\hat\Sigma$ is $\{w : x_i^\top w = 0\ \forall i\} = S^\perp$, so its range is S and its restriction to S has exactly the nonzero eigenvalues of $\hat\Sigma$, whence $u^\top\hat\Sigma u \ge \lambda^+_{\min}(\hat\Sigma)$ for unit $u \in S$. □

How the data spectrum and λ control the gap μ_{eff}/μ. Since $\mu = \lambda$ exactly (Proposition B2.12), the ratio measuring the slack of the strong-convexity step along a trajectory is $\mu_{\text{eff}}(\theta_t)/\lambda$. Corollary B2.15 shows that, once $R_x\|\theta_t - \eta_*\| \lesssim 1$ (for SAGA this takes a number of passes that grows with κ in the runs of Remark B2.22, where $n = 1000$: the first recorded state with $R_x\|\theta_t - \eta_*\| \le 1$ comes after 0.6 to 7.1 passes for $\kappa \le 10^3$, after 10.6 and 24.7 passes for $\kappa = 10^4$ with $\alpha \in \{0, 1\}$, and not within the 60 passes of the run for $\kappa = 10^4$, $\alpha = 2$; key `saga_first_pass_Rx_dist_le_1` of `results/B2_summary.json`), this ratio is within a factor $e^{\pm 1}$ of

$$\frac{u_t^\top H_* u_t}{\lambda} \;=\; 1 + \frac{1}{\lambda}\cdot\frac{1}{n}\sum_{i=1}^n p_i\,(x_i^\top u_t)^2, \qquad u_t = \frac{\theta_t - \eta_*}{\|\theta_t - \eta_*\|}, \tag{B2.27}$$

which, as u_t ranges over the unit sphere, spans the interval $[1 + \lambda_{\min}(H_* - \lambda I)/\lambda,\ 1 + \lambda_{\max}(H_* - \lambda I)/\lambda]$, where the "data Hessian" $H_* - \lambda I = \frac{1}{n}\sum_i p_i x_i x_i^\top$ satisfies $p_{\min}\hat\Sigma \preccurlyeq H_* - \lambda I \preccurlyeq \frac{1}{4}\hat\Sigma$. Three regimes follow.

- *Strong regularization, $\lambda \gg \lambda_{\max}(\hat\Sigma)/4$:* the ratio is at most $1 + \lambda_{\max}(\hat\Sigma)/(4\lambda) \approx 1$ in every direction, the strong-convexity step is essentially tight, but then $\kappa - 1 = R_x^2/(4\lambda) \ll R_x^2/\lambda_{\max}(\hat\Sigma) \le n$ (as $\hat\Sigma \succcurlyeq \frac{1}{n}x_i x_i^\top$ for every i), so that $\kappa \ll n$ and the problem lies outside the regime $\kappa \gg n$ of (A4).

- *Intermediate regularization, $\lambda_{\min}(H_* - \lambda I) \ll \lambda \ll \lambda_{\max}(H_* - \lambda I)$* (the typical situation with anisotropic features: a decaying spectrum of $\hat\Sigma$, or $d > n$, in which case $\lambda_{\min}(\hat\Sigma) = 0$ and $\lambda_{\min}(H_*) = \lambda$ exactly, the λ-eigenspace of H_* being the orthogonal complement $S^\perp$ of the feature span $S = \operatorname{span}\{x_1, \dots, x_n\}$; by Lemma B2.16 below this value λ is experienced only by trajectories having a component outside S, hence never when $\theta_0 \in S$, e.g. $\theta_0 = 0$, in which case the relevant bottom eigenvalue is $\lambda_{\min}(H_*|_S) = \lambda + \lambda^+_{\min}(H_* - \lambda I)$, with $\lambda^+_{\min}$ the smallest nonzero eigenvalue): the gap is *direction dependent*, close to 1 along the bottom eigenvectors of H_* and of order $\lambda_{\max}(H_*)/\lambda \gg 1$ along the top ones. Whether the slack is observed is then decided by the direction of the error u_t, which is the object of Section B2.4.

- *Weak regularization with well-conditioned features,* $\lambda \ll p_{\min}\lambda_{\min}(\hat{\Sigma})$: the gap is at least $1 + p_{\min}\lambda_{\min}(\hat{\Sigma})/\lambda \gg 1$ in *every* direction, and the μ-term of Proposition 5.9 is pessimistic, whatever the algorithm, by at least the factor $1+e^{-1}p_{\min}\lambda_{\min}(\hat{\Sigma})/\lambda$ as soon as $R_x \|\theta_t - \eta_*\| \leq 1$ and by at least $1 + p_{\min}\lambda_{\min}(\hat{\Sigma})/\lambda$ in the limit $t \to \infty$ (Corollary B2.15); far from η_* the data term carries the weight $e^{-R_x\|\theta_t-\eta_*\|}$ (or $m(\theta_t)$ in Lemma B2.11), which can be tiny, so the factor is not uniform along the whole trajectory. Here $p_{\min}$ is controlled by the margins at the optimum: $p_i \geq \frac{1}{4}e^{-|a_i^*|}$ with $|a_i^*| \leq R_x \|\eta_*\| \leq R_x\sqrt{2\log 2/\lambda}$ (Corollary B2.13), so $p_{\min}$ is of order $1/4$ when $\lambda \gtrsim R_x^2$ but may be exponentially small when λ is small and the data are nearly separable; in the latter case $H_* - \lambda I$ is dominated by the points with $|a_i^*| = O(1)$ (the points near the decision boundary), and the relevant quantity is the spectrum of $\frac{1}{n}\sum_i p_i x_i x_i^\top$ rather than that of $\hat{\Sigma}$.

In all cases, the statement "the gap is small when $\lambda \gg \lambda_{\min}(H_* - \lambda I)$" is correct only for the worst direction (the bottom eigenvector of H_*); in a typical (e.g. isotropically random) direction the ratio (B2.27) is $1 + \operatorname{tr}(H_* - \lambda I)/(d\lambda)$, which can be large even when $\lambda \gg \lambda_{\min}(H_* - \lambda I)$. The tightness ratios to record are therefore $(\lambda + m(\theta_t)u_t^\top \hat{\Sigma} u_t)/\mu_{\text{eff}}(\theta_t)$ for (B2.18), $\mu_{\text{eff}}(\theta_t)/(\lambda + \frac{1}{4}u_t^\top \hat{\Sigma} u_t)$ for (B2.19), and the two self-concordance ratios of (B2.24)–(B2.25), all of which only require η_* (from a high-precision solver), one pass over the data and the eigen-decomposition of H_*.

B2.4 Direction dependence: gradient descent versus SAGA

Proposition B2.8 shows that near the optimum $\mu_{\text{eff}}(\theta_t)$ is the Rayleigh quotient of $H_* = F''(\eta_*)$ in the direction u_t of the error. Which direction an algorithm selects is therefore part of the answer to (B2). We treat the quadratic case exactly and use it as a model for the logistic case.

An exact invariant subspace: the feature span. Before the quadratic model, we record an exact constraint on the directions that gradient descent and SAGA can explore for logistic regression (the argument only uses $f_i'(\theta) - \lambda\theta \in \operatorname{span}\{x_i\}$, so it applies verbatim to ridge regression and to every ℓ_2-regularized generalized linear model $f_i(\theta) = \varphi_i(x_i^\top\theta) + \frac{\lambda}{2}\|\theta\|^2$). Below, $z_j^{(t)}$ is the SAGA memory, ω_t the SAGA direction, and, for $i = 1, \dots, n$,

$$g_i^{(t+1)} \;=\; F'(\theta_t) - f_i'(\theta_t) + z_i^{(t)} - \frac{1}{n}\sum_{j=1}^{n} z_j^{(t)}$$

are the n possible values of the estimator error $\xi_{t+1} = F'(\theta_t) - \omega_{t+1}$ at the state $(\theta_t, z^{(t)})$ (for the quadratic model of Lemma B2.21 below these are the vectors $g_i^{(t+1)}$ defined there), and $C_{t+1} = \frac{1}{n}\sum_i g_i^{(t+1)} g_i^{(t+1)\top}$ is the conditional covariance of ω_{t+1} given the state.

Lemma B2.16 (invariance of the feature span)**.** *Let* $S = \operatorname{span}\{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$*, let* $S^\perp$ *be its orthogonal complement, and write* $\theta = \theta_S + \theta_\perp$ *for the orthogonal decomposition of a vector* θ *(similarly* $v_S, v_\perp$*). For* ℓ_2*-regularized logistic regression:*

(a) $\eta_* \in S$*, and* $f_i'(\theta) - \lambda\theta \in S$ *for every* θ *and every* i*; in particular* $f_i'(\theta)_\perp = \lambda\theta_\perp$ *for all* i*, and* $F'(\theta)_\perp = \lambda\theta_\perp$*.*

(b) (Invariance) If $\theta_0 \in S$*, then, for gradient descent and for SAGA with the initialization* $z_i^{(0)} = f_i'(\theta_0)$ *of Proposition 5.9 (or with* $z_i^{(0)} = 0$*), all iterates* θ_t*, all memories* $z_i^{(t)}$*, all directions* ω_t *and all error vectors* $g_i^{(t)}$ *lie in* S*, for every* t *and every realization of the indices. The*

same holds for the quadratic model of Lemma B2.21 whenever $H_i S \subseteq S$ for every i and $v_0, e_1^{(0)}, \dots, e_n^{(0)} \in S$.

(c) (The complementary component is autonomous) For an arbitrary θ_0, the $S^\perp$-component of the error $v_t = \theta_t - \eta_*$ evolves as if F were the quadratic $\frac{\lambda}{2}\|\theta_\perp\|^2$. For gradient descent, $v_{t,\perp} = (1-\gamma\lambda)^t v_{0,\perp}$. For SAGA, let $s_j(t)$ be the time at which the current memory was computed, $z_j^{(t)} = f_j'(\theta_{s_j(t)})$ (so $s_j(0) = 0$, $s_{i(t)}(t) = t-1$ and $s_j(t) = s_j(t-1)$ for $j \neq i(t)$); then

$$v_{t,\perp} = (1-\gamma\lambda)\, v_{t-1,\perp} - \gamma\lambda\Big(\frac{1}{n}\sum_{j=1}^n v_{s_j(t-1),\perp} - v_{s_{i(t)}(t-1),\perp}\Big),$$

and the error vectors satisfy

$$g_{i,\perp}^{(t+1)} = \lambda\Big(v_{s_i(t),\perp} - \frac{1}{n}\sum_{j=1}^n v_{s_j(t),\perp}\Big),$$

so that $\mathbb{E}[v_{t,\perp}] = (1-\gamma\lambda)^t v_{0,\perp}$, and the noise injected by SAGA in a unit direction $q \in S^\perp$ is

$$q^\top C_{t+1}\, q \;=\; \lambda^2\, \mathrm{Var}_j\left(q^\top v_{s_j(t)}\right) \;:=\; \lambda^2 \cdot \frac{1}{n}\sum_{i=1}^n \Big(q^\top v_{s_i(t)} - \frac{1}{n}\sum_{j=1}^n q^\top v_{s_j(t)}\Big)^2,$$

the empirical variance, over the memory, of the q-component of the stored errors.

Proof. (a) $f_i'(\theta) = \ell'(a_i(\theta))\, y_i x_i + \lambda\theta$ and $\ell'(a_i(\theta))\, y_i x_i \in S$; averaging over i, $F'(\theta) - \lambda\theta \in S$. At $\theta = \eta_*$, $F'(\eta_*) = 0$ gives $\lambda\eta_* = -\frac{1}{n}\sum_i \ell'(a_i^*)\, y_i x_i \in S$, hence $\eta_* \in S$ because $\lambda > 0$. Projecting $f_i'(\theta) = [f_i'(\theta) - \lambda\theta] + \lambda\theta$ onto $S^\perp$ gives $f_i'(\theta)_\perp = \lambda\theta_\perp$.

(b) Induction on t. At $t = 0$, $\theta_0 \in S$ and $z_i^{(0)} = f_i'(\theta_0) \in S$ by (a), since $\lambda\theta_0 \in S$ (and trivially if $z_i^{(0)} = 0$). If $\theta_{t-1} \in S$ and $z_j^{(t-1)} \in S$ for all j, then $f_i'(\theta_{t-1}) \in S$ for all i by (a), hence $\omega_t = f_{i(t)}'(\theta_{t-1}) + \frac{1}{n}\sum_j z_j^{(t-1)} - z_{i(t)}^{(t-1)} \in S$ (for gradient descent, $F'(\theta_{t-1}) \in S$), $\theta_t = \theta_{t-1} - \gamma\omega_t \in S$, the updated memory $z_{i(t)}^{(t)} = f_{i(t)}'(\theta_{t-1})$ lies in S and the other memories are unchanged. Finally each $g_i^{(t+1)}$ is a linear combination of vectors of S. For the quadratic model, $f_i'(\theta) - f_i'(\eta_*) = H_i v$ with $H_i S \subseteq S$, and the same induction applies to v_t, $e_i^{(t)}$ and $g_i^{(t+1)}$ (Lemma B2.21(a)).

(c) By (a) and $\eta_* \in S$, $f_i'(\theta)_\perp = \lambda\theta_\perp = \lambda(\theta - \eta_*)_\perp$ for every θ and i. For gradient descent, $v_{t,\perp} = v_{t-1,\perp} - \gamma F'(\theta_{t-1})_\perp = (1-\gamma\lambda)v_{t-1,\perp}$. For SAGA, $z_j^{(t-1)} = f_j'(\theta_{s_j(t-1)})$ has $S^\perp$-component $\lambda v_{s_j(t-1),\perp}$, so the $S^\perp$-component of ω_t is $\lambda v_{t-1,\perp} + \lambda\frac{1}{n}\sum_j v_{s_j(t-1),\perp} - \lambda v_{s_{i(t)}(t-1),\perp}$, and $v_{t,\perp} = v_{t-1,\perp} - \gamma\,\omega_{t,\perp}$ is the displayed recursion. Since $i(t)$ is uniform and independent of the past, the bracket has conditional mean zero, so $\mathbb{E}[v_{t,\perp} \mid \mathcal{F}_{t-1}] = (1-\gamma\lambda)v_{t-1,\perp}$ and $\mathbb{E}[v_{t,\perp}] = (1-\gamma\lambda)^t v_{0,\perp}$. In $g_i^{(t+1)}$ the components $F'(\theta_t)_\perp$ and $f_i'(\theta_t)_\perp$ are both equal to $\lambda v_{t,\perp}$ and cancel, leaving $z_{i,\perp}^{(t)} - \frac{1}{n}\sum_j z_{j,\perp}^{(t)} = \lambda(v_{s_i(t),\perp} - \frac{1}{n}\sum_j v_{s_j(t),\perp})$. For $q \in S^\perp$, $q^\top g_i^{(t+1)} = q^\top g_{i,\perp}^{(t+1)}$, and $q^\top C_{t+1} q = \frac{1}{n}\sum_i (q^\top g_i^{(t+1)})^2$ gives the last formula. □

The lemma has two consequences for the measurements of this section. First, with the standard initialization $\theta_0 = 0$ (or any $\theta_0 \in S$) the error v_t never leaves S, so only the restriction $H_*|_S$ of the Hessian at the optimum to S is explored; S is H_*-invariant (as $H_* = \frac{1}{n}\sum_i p_i x_i x_i^\top + \lambda I$), and when $d > n$ the λ-eigenspace $S^\perp$ of H_* is exactly the part that is never explored. Second, for a generic $\theta_0 \notin S$ the $S^\perp$-component of the error is governed by the smallest possible curvature λ, exactly as for the quadratic $\frac{\lambda}{2}\|\theta\|^2$, for both algorithms.

Gradient descent on a quadratic. Let $F(\theta) = F(\eta_*) + \frac{1}{2}(\theta-\eta_*)^\top H(\theta-\eta_*)$ with H symmetric positive definite, eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d = \lambda_{\min} > 0$ and an orthonormal basis of eigenvectors $q_1, \ldots, q_d$, and let $K_{\min} = \{k : \lambda_k = \lambda_{\min}\}$. For a quadratic, $\bar{H}(\theta) = H$ for all θ, so by Lemma B2.5 $\mu_{\text{eff}}(\theta) = v^\top H v / \|v\|^2$ exactly, and, along a trajectory (θ_t) with $v_t = \theta_t - \eta_* \neq 0$, the spectral profile of the error

$$\pi_k(t) \;=\; \frac{(q_k^\top v_t)^2}{\|v_t\|^2}, \qquad k = 1, \ldots, d, \qquad \sum_k \pi_k(t) = 1, \tag{B2.28}$$

determines it: $\mu_{\text{eff}}(\theta_t) = \sum_k \lambda_k \pi_k(t)$.

Proposition B2.17 (alignment of gradient descent with the flattest direction)**.** *Let $F(\theta) = F(\eta_*) + \frac{1}{2}(\theta-\eta_*)^\top H(\theta-\eta_*)$, where H is symmetric positive definite with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d = \lambda_{\min} > 0$ and orthonormal eigenvectors $q_1, \ldots, q_d$; let $E_{\min}$ be the eigenspace of $\lambda_{\min}$, P the orthogonal projector onto $E_{\min}$, and $\lambda' = \min\{\lambda_k : \lambda_k > \lambda_{\min}\}$ the second smallest distinct eigenvalue (if $H = \lambda_{\min} I$, then $P = I$ and the statement is trivial). Let $\theta_t = \theta_{t-1} - \gamma F'(\theta_{t-1})$ be gradient descent, $v_t = \theta_t - \eta_*$, and $\pi_k(t)$ the spectral profile (B2.28) of the error. Let $0 < \gamma \leq 1/\lambda_1$ and $v_0 \neq 0$; if $\gamma = 1/\lambda_1$, assume moreover that v_0 does not lie in the eigenspace of λ_1 (otherwise $\theta_t = \eta_*$ for all $t \geq 1$). Then $v_t \neq 0$ for every t, and:*

(a) $v_t = (I - \gamma H)^t v_0$, i.e. $q_k^\top v_t = (1-\gamma\lambda_k)^t q_k^\top v_0$ for all k, and $\mu_{\text{eff}}(\theta_t) = \sum_k \lambda_k \pi_k(t)$ with $\pi_k(t) \propto (1-\gamma\lambda_k)^{2t}(q_k^\top v_0)^2$.

(b) (Monotonicity) $t \mapsto \mu_{\text{eff}}(\theta_t)$ is non-increasing.

(c) (Alignment) If $Pv_0 \neq 0$ (true for Lebesgue-almost every v_0), then with $\rho_{\text{gap}} = (1-\gamma\lambda')/(1-\gamma\lambda_{\min}) \in [0,1)$,

$$\frac{\|v_t - Pv_t\|}{\|Pv_t\|} \;\leq\; \rho_{\text{gap}}^t \, \frac{\|v_0 - Pv_0\|}{\|Pv_0\|},$$

so that $v_t/\|v_t\|$ converges to $E_{\min}$ at the geometric rate ρ_{gap}, and

$$0 \;\leq\; \mu_{\text{eff}}(\theta_t) - \lambda_{\min} \;\leq\; (\lambda_1 - \lambda_{\min})\, \rho_{\text{gap}}^{2t} \, \frac{\|v_0 - Pv_0\|^2}{\|Pv_0\|^2} \;\xrightarrow[t\to\infty]{}\; 0. \tag{B2.29}$$

(d) (Invariant subspaces) If S is a linear subspace with $HS \subseteq S$ and $v_0 \in S$, then $v_t \in S$ for all t, and (a)–(c) hold with H replaced by its restriction $H|_S$: the sums run over the eigenvectors $q_k \in S$ only, $\lambda_{\min}$ is replaced by $\lambda_{\min}(H|_S) = \min\{\lambda_k : q_k \in S\}$, $E_{\min}$ and P by the corresponding eigenspace within S and its orthogonal projector, λ' by the second smallest distinct eigenvalue of $H|_S$, and the hypothesis $Pv_0 \neq 0$ by its analogue for the restricted projector.

Proof. (a) $F'(\theta) = H(\theta - \eta_*)$, so $v_t = v_{t-1} - \gamma H v_{t-1} = (I - \gamma H) v_{t-1}$, and $q_k^\top (I-\gamma H)^t v_0 = (1-\gamma\lambda_k)^t q_k^\top v_0$ since $Hq_k = \lambda_k q_k$. The formula for μ_{eff} follows from $v_t^\top H v_t = \sum_k \lambda_k (q_k^\top v_t)^2$. Since $1 - \gamma\lambda_k > 0$ unless $\lambda_k = \lambda_1 = 1/\gamma$, the assumptions give $v_t \neq 0$ for every t.

(b) Put $w_k(t) = (q_k^\top v_t)^2 = (1-\gamma\lambda_k)^{2t}(q_k^\top v_0)^2 \geq 0$, $W(t) = \sum_k w_k(t) = \|v_t\|^2 > 0$, and $\varphi(\lambda) = (1-\gamma\lambda)^2$, so that $w_k(t+1) = \varphi(\lambda_k) w_k(t)$. On $[0, 1/\gamma] \supset [\lambda_{\min}, \lambda_1]$ the function φ is non-increasing, hence $(\lambda_k - \lambda_j)(\varphi(\lambda_k) - \varphi(\lambda_j)) \leq 0$ for all k, j. Writing $\varphi_k = \varphi(\lambda_k)$,

$$\begin{aligned} W(t)W(t+1)\big[\mu_{\text{eff}}(\theta_{t+1}) - \mu_{\text{eff}}(\theta_t)\big] &= \sum_{k,j} w_k w_j \lambda_k \varphi_k - \sum_{k,j} w_k w_j \lambda_k \varphi_j \\ &= \frac{1}{2} \sum_{k,j} w_k w_j (\lambda_k - \lambda_j)(\varphi_k - \varphi_j) \;\leq\; 0, \end{aligned}$$

where the middle equality is obtained by symmetrizing the sum in (k, j) (Chebyshev's sum inequality).

(c) $\|Pv_t\|^2 = \sum_{k\in K_{\min}}(1-\gamma\lambda_{\min})^{2t}(q_k^\top v_0)^2 = (1-\gamma\lambda_{\min})^{2t}\|Pv_0\|^2$, where $1-\gamma\lambda_{\min} > 0$ because $\gamma \le 1/\lambda_1 < 1/\lambda_{\min}$. For $k \notin K_{\min}$, $\lambda' \le \lambda_k \le \lambda_1 \le 1/\gamma$ gives $0 \le 1-\gamma\lambda_k \le 1-\gamma\lambda'$, hence $\|v_t - Pv_t\|^2 = \sum_{k\notin K_{\min}}(1-\gamma\lambda_k)^{2t}(q_k^\top v_0)^2 \le (1-\gamma\lambda')^{2t}\|v_0 - Pv_0\|^2$. Dividing gives the first claim. Then $\mu_{\text{eff}}(\theta_t) - \lambda_{\min} = \sum_k(\lambda_k - \lambda_{\min})\pi_k(t) \le (\lambda_1 - \lambda_{\min})\sum_{k\notin K_{\min}}\pi_k(t) = (\lambda_1 - \lambda_{\min})\|v_t - Pv_t\|^2 / \|v_t\|^2 \le (\lambda_1 - \lambda_{\min})\|v_t - Pv_t\|^2 / \|Pv_t\|^2$, using $\|v_t\| \ge \|Pv_t\|$; the lower bound 0 is $\lambda_k \ge \lambda_{\min}$. The set $\{v_0 : Pv_0 = 0\}$ is a proper linear subspace, hence Lebesgue-null.

(d) Since H is symmetric and $HS \subseteq S$, also $HS^\perp \subseteq S^\perp$ (for $w \in S^\perp$ and $s \in S$, $s^\top Hw = (Hs)^\top w = 0$), so an orthonormal eigenbasis (q_k) of H can be chosen with every q_k in S or in $S^\perp$. Then $q_k^\top v_0 = 0$ for $q_k \in S^\perp$, hence by (a) $q_k^\top v_t = 0$ for all t and $v_t \in S$; all the sums in (a)–(c) run over the k with $q_k \in S$ only, and the proofs apply verbatim to this restricted spectrum. □

Remark B2.18 (scope of Proposition B2.17). (i) The alignment time is of order $1/\log(1/\rho_{\text{gap}}) \approx 1/(\gamma(\lambda' - \lambda_{\min}))$ iterations; when the spectrum of H is dense near $\lambda_{\min}$ this can exceed the horizon of an experiment, in which case $\mu_{\text{eff}}(\theta_t)$ decreases monotonically towards $\lambda_{\min}$ without reaching it: monotonicity (b) is the robust prediction, the limit (c) the asymptotic one. (ii) The case $Pv_0 = 0$ is not exceptional: by Lemma B2.16, for logistic regression with $d > n$ and $\theta_0 \in S$ the bottom eigenspace $E_{\min} = S^\perp$ of H_* is orthogonal to the whole trajectory, and in the quadratic model (d) then gives the alignment with the flattest direction of $H|_S$, not with $E_{\min}$. Both statements hold in exact arithmetic. Rounding errors re-inject a component along $E_{\min}$; that this component remains invisible, except possibly close to the precision floor, is *not proved* here: it is the heuristic of Remark B2.19. (iii) For $\gamma \in (1/\lambda_1, 2/\lambda_1)$ the contraction factors are $|1-\gamma\lambda_k|$ and the largest one may be attained at λ_1 (for $\gamma = 2/(\lambda_1 + \lambda_{\min})$ both extreme components contract by the same factor $(\lambda_1 - \lambda_{\min})/(\lambda_1 + \lambda_{\min})$, the top one with alternating sign), so (b)–(c) fail; the project uses $\gamma_{\text{GD}} \le 1/L = 1/\lambda_1$, as in the book.

Remark B2.19 (rounding errors and the feature span: a heuristic). This remark is a heuristic, not a proved statement. It concerns gradient descent in the situation of Remark B2.18(ii): logistic regression with $d > n$, $\theta_0 \in S$ and $E_{\min} = S^\perp$. *What is proved* is the exact-arithmetic invariance of Lemma B2.16(b), the alignment of Proposition B2.17(d) for a quadratic, and the following bound. Write the computed iterates as $\theta_t = \theta_{t-1} - \gamma F'(\theta_{t-1}) + \delta_t$, where δ_t collects all the rounding errors of step t (including those made in computing the gradient). Since $F'(\theta)_\perp = \lambda\theta_\perp$ for every θ and $\eta_* \in S$ (Lemma B2.16(a)), the $S^\perp$-component of the error satisfies exactly $v_{t,\perp} = (1-\gamma\lambda)v_{t-1,\perp} + \delta_{t,\perp}$; hence, for $\theta_0 \in S$ and $0 < \gamma\lambda \le 1$ (true for $\gamma \le 1/L$),

$$\|v_{t,\perp}\| \;\le\; \sum_{s=1}^{t}(1-\gamma\lambda)^{t-s}\,\|\delta_s\| \;\le\; \frac{\max_{s\le t}\|\delta_s\|}{\gamma\lambda}.$$

What is not proved is the size of δ_t. In the standard model of floating-point arithmetic it is of order $\varepsilon_{\text{mach}} \approx 10^{-16}$ times the size of the quantities that are added ($\|\theta_{t-1}\|$, and γ times the individual terms of the gradient), up to factors depending on n and d; in particular it does not decrease with the error $\|v_{t-1}\|$. Under this model the component of the error along $E_{\min}$ stays below a level of order $\varepsilon_{\text{mach}}\max_s\|\theta_s\|/(\gamma\lambda)$ (and is typically much smaller, since rounding errors do not add up coherently), while the component in S keeps decreasing geometrically. The former can therefore become comparable with the error only when the error is within a factor of order $1/(\gamma\lambda)$ of the rounding level; above that level one observes the behaviour inside S, that is, the alignment of Proposition B2.17(d) with the bottom eigenspace of $H|_S$ in the quadratic model (for logistic regression, the limit described, itself heuristically, in Remark B2.20). Consistently, in the sanity

check of Remark B2.20 the $S^\perp$-component of the error stays at the rounding level 10^{-14}. No proved statement of this note depends on this remark; prediction (P1) below uses it only when $S \neq \mathbb{R}^d$ (e.g. $d > n$) and $\theta_0 \in S$, a situation that does not occur in the experiments of the project, where the features span $\mathbb{R}^d$ (Remark B2.22).

Remark B2.20 (gradient descent beyond quadratics: a heuristic). For a general C^2 function, Lemma B2.5 gives $v_t = (I - \gamma H_*)v_{t-1} - \gamma(\bar{H}(\theta_{t-1}) - H_*)v_{t-1}$, and $\left\|\bar{H}(\theta) - H_*\right\|_{\mathrm{op}} \to 0$ as $\theta \to \eta_*$. Gradient descent is thus asymptotically the linear recursion of Proposition B2.17 with $H = H_*$, and we expect, for a generic θ_0, $v_t/\left\|v_t\right\| \to E_{\min}(H_*)$ and $\mu_{\mathrm{eff}}(\theta_t) \to \lambda_{\min}(H_*)$ (for logistic regression, $\lambda_{\min}(H_*) = \lambda + \lambda_{\min}(\frac{1}{n}\sum_i p_i x_i x_i^\top)$, which equals λ when $d > n$). When $\theta_0 \in S = \mathrm{span}\{x_1, \dots, x_n\}$, e.g. $\theta_0 = 0$, Lemma B2.16 confines the whole trajectory to S, which is H_*-invariant, and the expectation becomes, as in Proposition B2.17(d),

$$\mu_{\mathrm{eff}}(\theta_t) \;\to\; \lambda_{\min}(H_*|_S) \;=\; \lambda + \lambda^+_{\min}\Big(\frac{1}{n}\sum_{i=1}^n p_i x_i x_i^\top\Big),$$

where $\lambda^+_{\min}$ denotes the smallest nonzero eigenvalue: the matrix $\frac{1}{n}\sum_i p_i x_i x_i^\top$ has range S (all $p_i > 0$) and its restriction to S is positive definite. For $d > n$ this limit is strictly larger than $\lambda_{\min}(H_*) = \lambda$, and the value λ is approached only from a $\theta_0 \notin S$, through the component $v_{t,\perp} = (1 - \gamma\lambda)^t v_{0,\perp}$ of Lemma B2.16(c), which then dominates the error for large t. (Sanity check, reproduced by the script `experiments/check_B2_fp_heuristic.py`, whose output is `results/B2_fp_check_summary.json`. Instance: $n = 20$, $d = 40$, $\lambda = 10^{-4}$; with NumPy's generator `default_rng(0)` (seed 0) one draws, in this order, a 20×40 standard Gaussian matrix whose rows, normalized to unit norm, are the features x_i, a standard Gaussian $\beta \in \mathbb{R}^{40}$ and a standard Gaussian noise $\epsilon \in \mathbb{R}^{20}$, and one sets $y_i = \mathrm{sign}(x_i^\top\beta + 0.5\,\epsilon_i)$. Gradient descent with $\gamma = 1/L$, $L = \lambda + \lambda_{\max}(\hat{\Sigma})/4$, is run for 3000 iterations. From $\theta_0 = 0$, $\mu_{\mathrm{eff}}(\theta_t)$ decreases monotonically to $1.356{\cdot}10^{-4}$, against $\lambda + \lambda^+_{\min} = 1.346{\cdot}10^{-4}$, while the $S^\perp$-component of $\theta_t - \eta_*$ stays at the rounding level (at most $1.6 \cdot 10^{-14}$); from a standard Gaussian θ_0, the next draw of the same generator, $\mu_{\mathrm{eff}}(\theta_t)$ reaches λ to within $2.6 \cdot 10^{-5}$ in relative terms. The seeds 1 and 2 behave alike: monotone decrease, $\mu_{\mathrm{eff}}(\theta_{3000})$ within 1% of $\lambda + \lambda^+_{\min}$ and $S^\perp$-component below $2 \cdot 10^{-14}$ from $\theta_0 = 0$, relative gap to λ below $3 \cdot 10^{-5}$ from the Gaussian θ_0.) We state this as a heuristic: a rigorous proof requires the stable-manifold theory of the C^1 map $T(\theta) = \theta - \gamma F'(\theta)$ at its fixed point η_*, whose Jacobian $I - \gamma H_*$ has eigenvalues in $[0, 1)$, and is not needed here. For logistic regression the linearization error is explicit: by (B2.23), $|w_i(\theta) - p_i| \le (e^{R_x\|v\|} - 1)p_i$, hence for every unit vector ψ, $\left|\psi^\top(\bar{H}(\theta) - H_*)\psi\right| \le \frac{1}{n}\sum_i |w_i(\theta) - p_i|\,(x_i^\top\psi)^2 \le (e^{R_x\|v\|} - 1)\,\psi^\top(H_* - \lambda I)\psi$, i.e. $\left\|\bar{H}(\theta) - H_*\right\|_{\mathrm{op}} \le (e^{R_x\|\theta - \eta_*\|} - 1)\left\|H_* - \lambda I\right\|_{\mathrm{op}}$, and likewise $\left|\mu_{\mathrm{eff}}(\theta) - u^\top H_* u\right| \le (e^{R_x\|\theta-\eta_*\|} - 1)(u^\top H_* u - \lambda)$ (Corollary B2.15). Once $R_x\left\|\theta_t - \eta_*\right\| \le 0.1$, the quadratic model is accurate to about 10% in relative terms, and the measured $\mu_{\mathrm{eff}}(\theta_t)$ can be compared with the "quadratic proxy" $\sum_k \lambda_k \pi_k(t)$ built from the eigenpairs (λ_k, q_k) of H_*.

SAGA on a quadratic: a linear random recursion. For quadratic components f_i with Hessians H_i (for instance ridge regression, $H_i = x_i x_i^\top + \lambda I$; logistic regression is of this form to first order near η_* with $H_i = f_i''(\eta_*)$), $f_i'(\theta) - f_i'(\eta_*) = H_i v$ exactly, and SAGA is a linear recursion with random coefficients.

Lemma B2.21 (SAGA on a quadratic: exact moment recursions). *Let $f_i(\theta) = \frac{1}{2}\theta^\top H_i\theta - b_i^\top\theta + c_i$ with H_i symmetric, $0 \preccurlyeq H_i \preccurlyeq R^2 I$, and $H = \frac{1}{n}\sum_i H_i \succcurlyeq \mu I$. Consider SAGA with step γ and memory $z_i^{(t)}$ as in the book, write $v_t = \theta_t - \eta_*$, and define the memory errors $e_i^{(t)} = z_i^{(t)} - f_i'(\eta_*)$ and $\bar{e}^{(t)} = \frac{1}{n}\sum_i e_i^{(t)}$. Let $i(t)$ be uniform on $\{1, \dots, n\}$ and independent of the past $\mathcal{F}_{t-1} = \sigma(i(1), \dots, i(t-1))$.*

(a) (State recursion) $v_t = (I - \gamma H)v_{t-1} + \gamma\xi_t$ *with*

$$\xi_t = \big(e_{i(t)}^{(t-1)} - \bar{e}^{(t-1)}\big) - \big(H_{i(t)} - H\big)v_{t-1}, \qquad \mathbb{E}_{i(t)}[\xi_t] = 0,$$

and $e_{i(t)}^{(t)} = H_{i(t)}v_{t-1}$, $e_j^{(t)} = e_j^{(t-1)}$ *for* $j \neq i(t)$. *Moreover* $\xi_t = F'(\theta_{t-1}) - \omega_t$ *is minus the error of the SAGA gradient estimator* ω_t.

(b) (Mean) $\mathbb{E}[v_t] = (I - \gamma H)^t v_0$*: the mean error follows exactly the gradient descent recursion.*

(c) (Second moment) With $C_t = \mathbb{E}_{i(t)}[\xi_t\xi_t^\top] = \frac{1}{n}\sum_{i=1}^n g_i^{(t)} g_i^{(t)\top}$, *where* $g_i^{(t)} = (e_i^{(t-1)} - \bar{e}^{(t-1)}) - (H_i - H)v_{t-1}$ *(so that* C_t *is the conditional covariance of* ω_t *given* $\mathcal{F}_{t-1}$*),*

$$\mathbb{E}[v_t v_t^\top] = (I - \gamma H)\,\mathbb{E}[v_{t-1}v_{t-1}^\top]\,(I - \gamma H) + \gamma^2\mathbb{E}[C_t],$$

and therefore, for every eigenpair (λ_k, q_k) *of* H,

$$\mathbb{E}\big[(q_k^\top v_t)^2\big] = (1 - \gamma\lambda_k)^{2t}(q_k^\top v_0)^2 + \gamma^2 \sum_{s=1}^{t} (1 - \gamma\lambda_k)^{2(t-s)}\, q_k^\top \mathbb{E}[C_s]\, q_k. \tag{B2.30}$$

Proof. (a) By definition $\omega_t = f'_{i(t)}(\theta_{t-1}) + \frac{1}{n}\sum_j z_j^{(t-1)} - z_{i(t)}^{(t-1)}$. Write $f'_{i(t)}(\theta_{t-1}) = f'_{i(t)}(\eta_*) + H_{i(t)}v_{t-1}$, $z_j^{(t-1)} = f'_j(\eta_*) + e_j^{(t-1)}$ and $\frac{1}{n}\sum_j f'_j(\eta_*) = F'(\eta_*) = 0$. Then $\omega_t = H_{i(t)}v_{t-1} + \bar{e}^{(t-1)} - e_{i(t)}^{(t-1)} = Hv_{t-1} - \xi_t$, and $v_t = v_{t-1} - \gamma\omega_t$ gives the recursion. Since $i(t)$ is uniform and independent of $\mathcal{F}_{t-1}$, $\mathbb{E}_{i(t)}[e_{i(t)}^{(t-1)}] = \bar{e}^{(t-1)}$ and $\mathbb{E}_{i(t)}[H_{i(t)}] = H$, so $\mathbb{E}_{i(t)}[\xi_t] = 0$. The memory update is $z_{i(t)}^{(t)} = f'_{i(t)}(\theta_{t-1})$, i.e. $e_{i(t)}^{(t)} = f'_{i(t)}(\theta_{t-1}) - f'_{i(t)}(\eta_*) = H_{i(t)}v_{t-1}$.

(b) Take expectations in (a): $\mathbb{E}[v_t] = (I - \gamma H)\mathbb{E}[v_{t-1}] + \gamma\mathbb{E}[\mathbb{E}_{i(t)}[\xi_t]] = (I - \gamma H)\mathbb{E}[v_{t-1}]$.

(c) Expand $v_t v_t^\top = (I - \gamma H)v_{t-1}v_{t-1}^\top(I - \gamma H) + \gamma\big[(I - \gamma H)v_{t-1}\xi_t^\top + \xi_t v_{t-1}^\top(I - \gamma H)\big] + \gamma^2\xi_t\xi_t^\top$. Since v_{t-1} is $\mathcal{F}_{t-1}$-measurable and $\mathbb{E}_{i(t)}[\xi_t] = 0$, the conditional expectation of the bracket vanishes; $\mathbb{E}_{i(t)}[\xi_t\xi_t^\top] = \frac{1}{n}\sum_i g_i^{(t)} g_i^{(t)\top} = C_t$ because $\xi_t = g_{i(t)}^{(t)}$. Taking total expectations gives the matrix recursion; multiplying it on both sides by q_k and unrolling from t down to 0 gives (B2.30). □

Reading of (B2.30) and a quasi-stationary heuristic. The first term of (B2.30) is the gradient-descent term of Proposition B2.17, which concentrates the error on $E_{\min}$. The second term is the gradient-estimator noise accumulated along the trajectory: the noise injected at time s in direction q_k is damped by $(1 - \gamma\lambda_k)^{2(t-s)}$, i.e. it is remembered for about $1/(2\gamma\lambda_k)$ iterations, longer in flat directions. The noise covariance C_s is itself a quadratic function of the state $(v_{s-1}, e^{(s-1)})$, with $\operatorname{tr} C_s = \frac{1}{n}\sum_i \big\|g_i^{(s)}\big\|^2 \le \frac{2}{n}\sum_i \big\|e_i^{(s-1)}\big\|^2 + \frac{2}{n}\sum_i \|H_i v_{s-1}\|^2$ (the two quantities bounded in Step 1 of the proof of Proposition 5.9), so it decays at the rate of the Lyapunov function. To extract a prediction, assume (quasi-stationarity) that in the linear-convergence regime $\mathbb{E}[C_s] \approx \rho^{s-t}\mathbb{E}[C_t]$ for $s \le t$, where ρ is the contraction factor of the second moments. For the proved factor $\rho = 1 - \min\{1/(3n), 3\mu/(16R^2)\}$, in the regime $\kappa \gg n$ with $\gamma = 1/(4R^2)$ one has $1 - \rho = \frac{3}{4}\gamma\mu$ while $1 - (1 - \gamma\lambda_k)^2 = \gamma\lambda_k(2 - \gamma\lambda_k) \ge \gamma\lambda_k \ge \gamma\mu$, so $\rho > (1 - \gamma\lambda_k)^2$ for every k: the noise term decays more slowly than the gradient-descent term in *every* direction and dominates for large t (the experiments contradict this conclusion for $\kappa = 10^4 = 10n$ and $\alpha \in \{0, 1\}$, see Remark B2.22). For the observed factor ρ_{obs} this strict comparison is not established; what is guaranteed is the weaker inequality $\rho_{\text{obs}} \ge (1 - \gamma\lambda_k)^2$ for every k with $q_k^\top v_0 \neq 0$, because by (b) and Jensen's inequality $\mathbb{E}[(q_k^\top v_t)^2] \ge (q_k^\top \mathbb{E}v_t)^2 = (1 - \gamma\lambda_k)^{2t}(q_k^\top v_0)^2$, so that the second moment along q_k cannot

contract faster than $(1-\gamma\lambda_k)^2$ per step; in particular, if $Pv_0 \neq 0$, $1-\rho_{\mathrm{obs}} \leq 2\gamma\lambda_{\min} - \gamma^2\lambda_{\min}^2$, which keeps the denominators $2\lambda_k - (1-\rho_{\mathrm{obs}})/\gamma \geq 2(\lambda_k - \lambda_{\min}) + \gamma\lambda_{\min}^2 > 0$ appearing below positive (with $\lambda_{\min}$ read as $\lambda_{\min}(H|_S)$ in the situation of Proposition B2.17(d)). Summing the geometric series,

$$\mathbb{E}\big[(q_k^\top v_t)^2\big] \;\approx\; \gamma^2\,\frac{c_k(t)}{1-(1-\gamma\lambda_k)^2/\rho} \;\approx\; \frac{\gamma\, c_k(t)}{2\lambda_k - (1-\rho)/\gamma}, \qquad c_k(t) = q_k^\top \mathbb{E}[C_t] q_k, \tag{B2.31}$$

using $(1-\gamma\lambda_k)^2 \approx 1 - 2\gamma\lambda_k$ and $\rho \approx 1$. The predicted spectral profile of SAGA is therefore

$$\pi_k^{\mathrm{SAGA}}(t) \;\propto\; \frac{c_k(t)}{2\lambda_k - (1-\rho)/\gamma}, \qquad \mu_{\mathrm{eff}}(\theta_t) \;\approx\; \frac{\sum_k \lambda_k\, c_k(t)/(2\lambda_k - (1-\rho)/\gamma)}{\sum_k c_k(t)/(2\lambda_k - (1-\rho)/\gamma)}, \tag{B2.32}$$

a weighted mean of the eigenvalues in which direction k receives the weight $c_k/(2\lambda_k - (1-\rho)/\gamma)$: the noise re-injects *all* directions it feeds, and the weight of a flat direction is larger only by the factor $1/\lambda_k$, not by the exponentially growing factor $((1-\gamma\lambda_{\min})/(1-\gamma\lambda_k))^{2t}$ of gradient descent. Two illustrative noise shapes: (i) isotropic noise, $c_k \equiv c$, gives $\mu_{\mathrm{eff}} \approx \big[\sum_k \lambda_k/(2\lambda_k - \frac{3}{4}\mu)\big]/\big[\sum_k 1/(2\lambda_k - \frac{3}{4}\mu)\big]$; with one eigenvalue equal to μ and $d-1$ eigenvalues $L \gg d\mu$ this is $\approx \mu\,(0.8 + (d-1)/2)/0.8$, i.e. larger than $\lambda_{\min} = \mu$ by a factor of order d; (ii) "data-shaped" noise, $c_k \propto \lambda_k - \lambda$ (for logistic or ridge regression started at $\theta_0 \in S = \mathrm{span}\{x_1, \dots, x_n\}$, e.g. $\theta_0 = 0$, all vectors $g_i^{(t)}$ lie in S by Lemma B2.16(b), so the noise has no energy in the pure-regularization directions $S^\perp$; for a generic θ_0 these directions carry, by Lemma B2.16(c), the noise $c_q = \lambda^2\,\mathrm{Var}_j(q^\top v_{s_j})$, $q \in S^\perp$, i.e. λ^2 times the spread of the $S^\perp$-errors stored in the memory, which is negligible compared with the data-direction noise unless λ is comparable to the data curvature), gives weights $\propto (\lambda_k - \lambda)/(2\lambda_k - \frac{3}{4}\mu)$, which vanish on the directions with $\lambda_k \approx \lambda$ and are $\approx \frac{1}{2}$ on all directions with $\lambda_k \gg \lambda$: $\mu_{\mathrm{eff}}(\theta_t)$ is then of the order of the *average* curvature of the data directions, far above $\lambda_{\min}(H_*) = \lambda + \lambda_{\min}(\frac{1}{n}\sum_i p_i x_i x_i^\top)$ (and above $\lambda_{\min}(H_*|_S)$ when $\theta_0 \in S$).

Testable predictions and what to measure. Let (λ_k, q_k) be the eigenpairs of $H_* = F''(\eta_*)$ (with η_* from a high-precision solver), and record along each run: the profile $\pi_k(t)$ of (B2.28), its effective number of directions $d_{\mathrm{eff}}(t) = 1/\sum_k \pi_k(t)^2 \in [1, d]$, the exact $\mu_{\mathrm{eff}}(\theta_t)$ (formula (B2.16) with the closed-form weights (B2.14), or directly Definition B2.2) and its quadratic proxy $\sum_k \lambda_k \pi_k(t)$, and, for SAGA, the noise spectrum $c_k(t) = q_k^\top C_t q_k$ where, for a general (non-quadratic) F,

$$C_{t+1} = \frac{1}{n}\sum_{i=1}^n g_i g_i^\top, \qquad g_i = F'(\theta_t) - f_i'(\theta_t) + z_i^{(t)} - \frac{1}{n}\sum_j z_j^{(t)},$$

is the conditional covariance of the SAGA estimator ω_{t+1} over the n possible draws of $i(t+1)$, given the state $(\theta_t, z^{(t)})$ (Lemma B2.21(c), Lemma B2.16). Forming all the $c_k(t)$ costs $O(nd^2)$ per evaluation (the product $Q^\top G$ with $G = [g_1, \dots, g_n] \in \mathbb{R}^{d\times n}$ and $Q = [q_1, \dots, q_d]$, or n rank-one updates of a $d \times d$ matrix); it is $O(nd + d^2)$ only if a single direction q_k is monitored. Done once per pass, this is affordable for d up to a few thousands. The initial point must be part of the protocol, since it selects the subspace explored (Lemma B2.16): run every experiment both from $\theta_0 = 0 \in S$ and from a generic θ_0 (e.g. standard Gaussian), and compare $\mu_{\mathrm{eff}}(\theta_t)$ with $\lambda_{\min}(H_*|_S) = \lambda + \lambda^+_{\min}(H_* - \lambda I)$ in the first case and with $\lambda_{\min}(H_*)$ in the second (the two coincide when the features span $\mathbb{R}^d$).

(P1) *Gradient descent*: $\mu_{\mathrm{eff}}(\theta_t)$ is non-increasing (exactly for a quadratic, Proposition B2.17(b)) and converges to $\lambda_{\min}(H_*)$ for a generic θ_0, and to $\lambda_{\min}(H_*|_S) = \lambda + \lambda^+_{\min}(H_* - \lambda I)$ when

$\theta_0 \in S$, e.g. $\theta_0 = 0$ (Proposition B2.17(d) and Lemma B2.16, completed by the heuristic Remarks B2.19 and B2.20); under alignment, $\pi(t)$ concentrates on the corresponding bottom eigenspace, so that $d_{\text{eff}}(t)$ converges to a value in $[1, m]$, m being the multiplicity of that bottom eigenvalue: the limit is $1/\sum_{k\in K_{\min}}(\pi_k^\infty)^2$ with $\pi_k^\infty = (q_k^\top P v_0)^2/\|Pv_0\|^2$, which equals m only when Pv_0 is equidistributed over the m eigenvectors (generically $m = 1$ and $d_{\text{eff}}(t) \to 1$); the convergence speed is governed by the spectral gap $\lambda' - \lambda_{\min}$ of the explored spectrum.

(P2) *SAGA*: after the initial transient, $\pi_k(t)$ remains spread over the directions fed by the noise, with $\pi_k(t) \propto c_k(t)/(2\lambda_k - (1-\rho)/\gamma)$ up to fluctuations (B2.32); $d_{\text{eff}}(t)$ stays of the order of the number of data directions; $\mu_{\text{eff}}(\theta_t)$ fluctuates around a level strictly above $\lambda_{\min}(H_*)$ (resp. $\lambda_{\min}(H_*|_S)$ when $\theta_0 \in S$) and does not decrease monotonically; for the same problem and large t, $\mu_{\text{eff}}^{\text{SAGA}}(t) > \mu_{\text{eff}}^{\text{GD}}(t)$. The experiments confirm (P2) only for κ at most of the order of n, and not in every such run (Remark B2.22).

(P3) *Consequence for (B2)*: the slack of the strong-convexity step (B2.3), measured by $\mu_{\text{eff}}(\theta_t)/\mu$, is larger for SAGA than the asymptotic slack $\lambda_{\min}(H_*)/\mu$ (resp. $\lambda_{\min}(H_*|_S)/\mu$ for $\theta_0 \in S$) of gradient descent; in the "intermediate regularization" regime of Section B2.3 this is precisely the situation where the data anisotropy makes the bottom eigenvalue of the explored spectrum close to λ while the trajectory-averaged curvature is much larger. A violation of (P2) (e.g. $\pi(t)$ concentrating on $E_{\min}$ for SAGA as well) would mean that the noise term in (B2.30) decays faster than assumed, which can be checked directly on $c_k(t)$.

Remark B2.22 (outcome of prediction (P2)). *Measurements* (Figure B2 of the report shows the case $\alpha = 1$). ℓ_2-regularized logistic regression with $n = 1000$, $d = 50$, Gaussian features with covariance spectrum $j^{-\alpha}$ (randomly rotated, rescaled so that $R_x = \max_i \|x_i\| = 1$), $\alpha \in \{0, 1, 2\}$, $\kappa = R^2/\mu \in \{10^2, 10^3, 10^4\}$, a single seed (seed 0 for the data and for the sampling); $\theta_0 = 0$; SAGA with $\gamma = 1/(4R^2)$ and $z_i^{(0)} = f_i'(\theta_0)$ for 60 passes, recorded every $n/10$ iterations; gradient descent with $\gamma = 1/L$ for 400 iterations, recorded at every iteration. States with $\|\theta_t - \eta_*\|^2 < 10^{-18}$ are discarded and "end" denotes the last recorded state above this floor (SAGA reaches it after 34 to 49 passes for $\kappa = 10^2$, and for $\kappa = 10^3$ with $\alpha \in \{0, 1\}$; the other SAGA runs end after 60 passes). Since $\lambda_{\min}(H_*) > \lambda$ in every configuration (table below), the features span $\mathbb{R}^d$ (Proposition B2.12(d)), so that $S = \mathbb{R}^d$ and $\lambda_{\min}(H_*|_S) = \lambda_{\min}(H_*)$. We write $q_{\min}$ for a unit eigenvector of H_* for its smallest eigenvalue (which is simple in all nine configurations), $\pi_{\min}(t) = (q_{\min}^\top(\theta_t - \eta_*))^2/\|\theta_t - \eta_*\|^2$ for its share in the profile (B2.28), and λ' for the second smallest eigenvalue of H_*; "g. mean" is the geometric mean over the recorded states. Gradient descent reaches the floor after 23 to 171 iterations for $\kappa = 10^2$ and for $(\alpha, \kappa) = (0, 10^3)$, and otherwise ends at iteration 400, with $\|\theta_t - \eta_*\|^2 = 1.5 \cdot 10^{-18}$, $4.7 \cdot 10^{-9}$, 1.0, $5.7 \cdot 10^{-4}$ and 202 for $(\alpha, \kappa) = (0, 10^4)$, $(1, 10^3)$, $(1, 10^4)$, $(2, 10^3)$, $(2, 10^4)$ (from $\|\theta_0 - \eta_*\|^2 = 486, 1954, 389$ and 2565 for the last four). Its end values coincide with the limit $\lambda_{\min}(H_*)$ of (P1) only for $\alpha = 0$, $\kappa \geq 10^3$: for $\kappa = 10^2$ it reaches the floor before its error has aligned with $q_{\min}$ ($\pi_{\min} = 0.53$, 0.074 and 0.001 at the end), for $\kappa = 10^3$, $\alpha \in \{1, 2\}$ the error is not yet aligned (μ_{eff}/μ ends 9% and 11% above $\lambda_{\min}(H_*)/\mu$), and for $\kappa = 10^4$ it ends at the edge of the local regime ($\alpha = 1$) or far from it ($\alpha = 2$). The table and the other numbers of this remark are read from, or computed from the values stored in, `results/B2_summary.json`, written by `experiments/exp_B2_mu_eff.py` from the cached trajectories.

				SAGA: μ_{eff}/μ			GD: μ_{eff}/μ	$\pi_{\min}$ at the end		d_{eff} at the end	
α	κ	$\lambda_{\min}(H_*)/\mu$	$9n\lambda_{\min}(H_*)/(16R^2)$	start	g. mean	end	end	SAGA	GD	SAGA	GD
0	10^2	1.609	9.05	2.086	1.996	2.088	1.635	0.0008	0.530	23.7	2.51
0	10^3	3.275	1.84	7.917	7.137	7.341	3.275	0.015	1.000	23.7	1.000
0	10^4	9.863	0.555	50.05	10.59	9.942	9.863	0.998	1.000	1.005	1.000
1	10^2	1.100	6.19	1.793	1.901	2.147	1.155	0.0008	0.074	15.2	10.1
1	10^3	1.737	0.977	4.632	2.198	1.904	1.900	0.308	0.148	3.79	3.89
1	10^4	4.970	0.280	21.02	5.386	4.974	5.058	0.9996	0.996	1.001	1.008
2	10^2	1.003	5.64	2.497	2.167	4.980	1.029	$8 \cdot 10^{-6}$	0.0010	2.71	11.6
2	10^3	1.020	0.573	5.340	1.163	1.062	1.129	0.0025	0.0010	12.1	14.6
2	10^4	1.142	0.0643	13.79	1.809	1.468	1.787	0.0017	0.0007	11.9	15.3

Outcome. (P2) holds only for κ at most of the order of n, and not in every such run: it is confirmed in the four runs with $9n\lambda_{\min}(H_*)/(16R^2) > 1$ ($\kappa = 10^2$, all α; $\kappa = 10^3$, $\alpha = 0$), refuted for $\kappa = 10^4$ with $\alpha \in \{0, 1\}$ (one seed each; the initial error already carries 89–98% of its energy on $q_{\min}$), and the three remaining runs ($\kappa = 10^3$ with $\alpha \in \{1, 2\}$; $\kappa = 10^4$ with $\alpha = 2$) are intermediate, see (c).

(a) For $\kappa = 10^2$ (all α) and for $\kappa = 10^3$ with $\alpha = 0$, the error of SAGA does not concentrate on $q_{\min}$ ($\pi_{\min} \le 0.05$ at every state of the last five recorded passes), and over these passes the geometric mean of μ_{eff}/μ is between 1.25 and 2.6 times $\lambda_{\min}(H_*)/\mu$ (single states fluctuate between 1.2 and 6.0 times; the "end" columns of the table are single states), whereas gradient descent ends closer to $\lambda_{\min}(H_*)/\mu$. For $\alpha \in \{0, 1\}$ the error stays spread over many directions (d_{eff} averages 14 to 18 over the last five passes). The run $\kappa = 10^3$, $\alpha = 0$ is the clearest: the initial error $-\eta_*$ is concentrated on $q_{\min}$ ($\pi_{\min}(0) = 0.96$), gradient descent keeps it there and SAGA spreads it out, although the spectral gap is large ($\lambda'/\lambda_{\min}(H_*) = 1.46$). For $\alpha = 2$, $\kappa = 10^2$, d_{eff} is smaller (mean 5.4 over the last five passes), but the five lowest eigenvectors of H_* carry only 0.005 of the error and the geometric mean of μ_{eff}/μ is 2.6: the error sits on intermediate- and high-curvature directions, consistently with the "data-shaped" noise (ii) above, whose weights vanish on the directions with $\lambda_k \approx \lambda$.

(b) For $\kappa = 10^4$ and $\alpha \in \{0, 1\}$, the error of SAGA concentrates on $q_{\min}$ as that of gradient descent does ($\pi_{\min} = 0.998$ and 0.9996, $d_{\text{eff}} = 1.005$ and 1.001), and μ_{eff}/μ ends within 1% of $\lambda_{\min}(H_*)/\mu$. The initial error is already largely aligned ($\pi_{\min}(0) = 0.98$ and 0.89); SAGA keeps and sharpens this alignment instead of spreading the error. The large initial values $\mu_{\text{eff}}(\theta_0) = 50.05\,\mu$ and $21.02\,\mu$ are far from the Rayleigh quotients of H_* in the initial direction ($10.38\,\mu$ and $5.885\,\mu$); they are Rayleigh quotients of the averaged Hessian $\bar{H}(\theta_0)$ of Lemma B2.5: the point $\theta_0 = 0$ is far from η_* ($R_x \|\theta_0 - \eta_*\| = 32$ and 44), where the comparison of Corollary B2.15 is uninformative. (P3) fails accordingly: at the end of the run the slack μ_{eff}/μ of SAGA exceeds $\lambda_{\min}(H_*)/\mu$ by less than 1%.

(c) Intermediate cases. For $\kappa = 10^3$, $\alpha = 1$, the alignment with $q_{\min}$ is partial ($\pi_{\min} = 0.31$, $d_{\text{eff}} = 3.79$), but the five lowest eigenvectors of H_* carry 0.88 of the error at the end, and $\lambda'/\lambda_{\min}(H_*) = 1.005$, so that alignment with $q_{\min}$ itself is slow (Remark B2.18(i)). For $\alpha = 2$ and $\kappa \ge 10^3$ the bottom of the spectrum is nearly degenerate (29 and 11 eigenvalues of H_* lie within 10% of $\lambda_{\min}(H_*)$ for $\kappa = 10^3$ and 10^4) and neither algorithm aligns with $q_{\min}$ (for gradient descent, alignment would take of the order of $L/(\lambda' - \lambda_{\min}(H_*)) \approx 9 \cdot 10^4$ iterations, far more than the 400 of the protocol, and at $\kappa = 10^4$ it is moreover still far from η_*); at $\kappa = 10^3$ SAGA nevertheless ends at $\mu_{\text{eff}}/\mu = 1.062$, 4.2% above $\lambda_{\min}(H_*)/\mu$, while at $\kappa = 10^4$ the run is not finished after 60 passes: $\|\theta_t - \eta_*\|^2 = 8.6$ at the end ($R_x \|\theta_t - \eta_*\| \approx 2.9$, outside the region where the bounds of Corollary B2.15 are informative, although the quadratic proxy $\sum_k \lambda_k \pi_k(t)$ already matches $\mu_{\text{eff}}(\theta_t)$ to 0.2%) and μ_{eff}/μ is still decreasing.

Explanation (a heuristic). Write $\lambda_{\min} = \lambda_{\min}(H_*)$. Formula (B2.32) involves the actual decay factor ρ of the noise covariance, and the spread predicted in (P2) came from inserting the certified factor of Proposition 5.9, which exceeds $(1-\gamma\lambda_k)^2$ for every k (its distance to 1 is at most $\frac{3}{4}\gamma\mu < \gamma\lambda_k(2-\gamma\lambda_k)$). If instead the noise decays as fast as the slowest mode of the mean dynamics, i.e. if $1-\rho$ is close to its largest admissible value $2\gamma\lambda_{\min} - \gamma^2\lambda_{\min}^2$ (discussion before (B2.31)), the denominator $2\lambda_{\min} - (1-\rho)/\gamma$ of the weight of $q_{\min}$ becomes small compared with those of the other directions, which are at least $2(\lambda_k - \lambda_{\min})$, and the same formula predicts concentration on $q_{\min}$ (unless $q_{\min}^\top \mathbb{E}[C_t] q_{\min}$ is comparatively negligible). Which case occurs is decided by the memory. For SAGA with $z_i^{(0)} = f_i'(\theta_0)$ one has $z_j^{(t-1)} = f_j'(\theta_{s_j(t-1)})$, with s_j as in Lemma B2.16(c). Writing $\zeta_j = f_j'(\theta_{s_j(t-1)}) - f_j'(\theta_{t-1})$ and inserting $z_j^{(t-1)} = f_j'(\theta_{t-1}) + \zeta_j$ in the definition of ω_t gives, exactly and for any F,

$$\xi_t = F'(\theta_{t-1}) - \omega_t = \zeta_{i(t)} - \frac{1}{n}\sum_{j=1}^n \zeta_j,$$

$$\mathbb{E}_{i(t)} \|\xi_t\|^2 \le \frac{1}{n}\sum_{j=1}^n \|\zeta_j\|^2 \le \frac{R^4}{n}\sum_{j=1}^n \left\|\theta_{s_j(t-1)} - \theta_{t-1}\right\|^2$$

(a variance is at most the second moment, and the f_i' are R^2-Lipschitz): the noise is produced by the displacement of the iterate since each memory slot was last refreshed, about n iterations earlier on average. Along $q_{\min}$ the mean error decays by the factor $(1-\gamma\lambda_{\min})^n \approx e^{-n\gamma\lambda_{\min}}$ per pass (Lemma B2.21(b)), and faster along the other eigenvectors. If $n\gamma\lambda_{\min}$ is small, then, once the faster components have decayed, the iterate moves little during one refresh cycle (in relative terms, by about $n\gamma\lambda_{\min}$ along $q_{\min}$), the noise is a small perturbation of the gradient-descent dynamics and decays with the slow component, and the error aligns with $q_{\min}$. If $n\gamma\lambda_{\min} \gtrsim 1$, every mode decays by a constant factor within one refresh cycle, the stored gradients were computed much farther from η_* than θ_{t-1}, the noise decays only as fast as the memory is refreshed, more slowly than every mode, and the error stays spread. For $\gamma = 1/(4R^2)$ the condition $\frac{9}{4}n\gamma\lambda_{\min} < 1$ is exactly $\frac{3\lambda_{\min}}{16R^2} < \frac{1}{3n}$: up to a constant factor, the dividing line is the condition that the parameter term be the active one in the minimum of Proposition 5.9 with μ replaced by the local constant $\lambda_{\min}(H_*)$ (Proposition B2.23 with $\mu_{\text{eff}} = \lambda_{\min}(H_*)$, the value reached once the error is aligned). For $\alpha \in \{0,1\}$ the column $9n\lambda_{\min}(H_*)/(16R^2)$ of the table is below 1 for the two runs that align, equal to 0.977 for the partially aligned run, and between 1.84 and 9.05 for the three runs that stay spread; with the global μ in place of $\lambda_{\min}(H_*)$, the value $9n/(16\kappa) = 0.56$ at $\kappa = 10^3$ would wrongly predict alignment for $\alpha = 0$. For $\alpha = 2$ the column is below 1 for $\kappa \ge 10^3$, but the near-degeneracy of the bottom of the spectrum makes alignment with a single eigenvector slow for both algorithms. One recorded quantity is consistent with this dichotomy: the least-squares slope of $\log\|\theta_t - \eta_*\|^2$ against the number of passes, over the second half of each run, is -0.495 and -0.249 in the two runs that align, i.e. the rate $2n\gamma\lambda_{\min}(H_*) = 0.493$ and 0.249 per pass of the slowest mode of the mean dynamics (to within 0.5%), whereas in the four runs with $9n\lambda_{\min}(H_*)/(16R^2) > 1$ it lies between -1.22 and -1.10: a decay slower than the values 1.64 to 8.05 of $2n\gamma\lambda_{\min}(H_*)$ in these runs and close to the refresh rate of the memory (about 1 per pass, a slot being refreshed with probability $\frac{1}{n}$ at each iteration). This explanation is nevertheless only a heuristic: it relies on the quadratic model, and the noise spectrum $c_k(t)$ was not recorded in these runs, so that the mechanism is inferred, not measured; recording $q_{\min}^\top C_t q_{\min}$ and $\operatorname{tr} C_t$ (one pass over the data per evaluation) would turn the consistency check above into a measurement.

B2.5 How μ_{eff} and L_{eff} enter the SAGA bound

We use the setting of Proposition 5.9 of [11] (SAGA [16] with uniform sampling): step $\gamma = 1/(4R^2)$, $\Delta = 3\gamma^2$, and the Lyapunov function

$$V_t = \|\theta_t - \eta_*\|^2 + \Delta \sum_{i=1}^{n} \left\| z_i^{(t)} - f_i'(\eta_*) \right\|^2 .$$

Proposition B3.19 of the B3 note is the following trajectory-dependent bound, obtained there by keeping the co-coercivity step (I2) and the strong-convexity step (I4) of the book's proof as identities; we include the short derivation because it shows exactly where μ_{eff} and L_{eff} enter.

Proposition B2.23 (trajectory-dependent one-step contraction)**.** *Under Assumption B2.1, for every $t \geq 1$, conditionally on the past and on the event $\theta_{t-1} \neq \eta_*$,*

$$\mathbb{E}_{i(t)}[V_t] \;\leq\; \Big(1 - \min\Big\{\frac{1}{3n},\, a(\theta_{t-1})\Big\}\Big) V_{t-1}, \qquad a(\theta) = \frac{\mu_{\text{eff}}(\theta)\,\big(8R^2 - 5L_{\text{eff}}(\theta)\big)}{16R^4}, \tag{B2.33}$$

and, since $L_{\text{eff}}(\theta) \leq R^2$ implies $a(\theta) \geq 3\mu_{\text{eff}}(\theta)/(16R^2)$,

$$\mathbb{E}_{i(t)}[V_t] \;\leq\; \Big(1 - \min\Big\{\frac{1}{3n},\, \frac{3\mu_{\text{eff}}(\theta_{t-1})}{16R^2}\Big\}\Big) V_{t-1}. \tag{B2.34}$$

On the event $\theta_{t-1} = \eta_*$ one has $\mathbb{E}_{i(t)}[V_t] \leq (1 - \frac{1}{3n})V_{t-1}$. The bound of Proposition 5.9, $\mathbb{E}_{i(t)}[V_t] \leq (1 - \min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\})V_{t-1}$, follows from (B2.34) and $\mu_{\text{eff}} \geq \mu$.

Proof. Write $v = \theta_{t-1} - \eta_*$, $e_i = z_i^{(t-1)} - f_i'(\eta_*)$, and ω_t for the SAGA direction, so that $\theta_t - \eta_* = v - \gamma\omega_t$ and $\mathbb{E}_{i(t)}[\omega_t] = F'(\theta_{t-1})$. Step 1 of the book's proof gives, using only the expansion of the square, unbiasedness, $\|a+b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ and $\mathbb{E}_{i(t)}\|Z - \mathbb{E}_{i(t)}Z\|^2 \leq \mathbb{E}_{i(t)}\|Z\|^2$,

$$\mathbb{E}_{i(t)}\|\theta_t - \eta_*\|^2 \leq \|v\|^2 - 2\gamma\, v^\top F'(\theta_{t-1}) + 2\gamma^2 \frac{1}{n}\sum_i \left\|f_i'(\theta_{t-1}) - f_i'(\eta_*)\right\|^2 + \frac{2\gamma^2}{n}\sum_i \|e_i\|^2 ,$$

and Step 2 gives the *identity* $\mathbb{E}_{i(t)}\sum_i \left\|z_i^{(t)} - f_i'(\eta_*)\right\|^2 = (1-\frac{1}{n})\sum_i \|e_i\|^2 + \frac{1}{n}\sum_i \|f_i'(\theta_{t-1}) - f_i'(\eta_*)\|^2$. Instead of bounding $\frac{1}{n}\sum_i \|f_i'(\theta_{t-1}) - f_i'(\eta_*)\|^2$ by $R^2 v^\top F'(\theta_{t-1})$ (co-coercivity), we use its exact value $L_{\text{eff}}(\theta_{t-1})\, v^\top F'(\theta_{t-1})$ (Definition B2.2). Adding Δ times the second relation to the first,

$$\mathbb{E}_{i(t)}[V_t] \leq \|v\|^2 - 2\gamma\Big(1 - \gamma L_{\text{eff}} - \frac{L_{\text{eff}}\Delta}{2\gamma}\Big) v^\top F'(\theta_{t-1}) + \Big(\frac{2\gamma^2}{n\Delta} + 1 - \frac{1}{n}\Big)\Delta \sum_i \|e_i\|^2 ,$$

with $L_{\text{eff}} = L_{\text{eff}}(\theta_{t-1})$. With $\Delta = 3\gamma^2$ and $\gamma = 1/(4R^2)$: $1 - \gamma L_{\text{eff}} - \frac{3}{2}\gamma L_{\text{eff}} = 1 - \frac{5}{2}\gamma L_{\text{eff}} = 1 - \frac{5L_{\text{eff}}}{8R^2} \geq \frac{3}{8} > 0$ (as $L_{\text{eff}} \leq R^2$), and $\frac{2\gamma^2}{n\Delta} + 1 - \frac{1}{n} = 1 - \frac{1}{3n}$. Now, instead of $v^\top F'(\theta_{t-1}) \geq \mu\|v\|^2$, we use the exact value $v^\top F'(\theta_{t-1}) = \mu_{\text{eff}}(\theta_{t-1})\|v\|^2$, and the coefficient of $\|v\|^2$ becomes $1 - 2\gamma(1 - \frac{5}{2}\gamma L_{\text{eff}})\mu_{\text{eff}} = 1 - \frac{\mu_{\text{eff}}(8R^2 - 5L_{\text{eff}})}{16R^4} = 1 - a(\theta_{t-1})$, which is nonnegative because $a \leq \mu_{\text{eff}}/(2R^2) \leq L/(2R^2) \leq \frac{1}{2}$. Hence $\mathbb{E}_{i(t)}[V_t] \leq (1-a)\|v\|^2 + (1-\frac{1}{3n})\Delta\sum_i \|e_i\|^2 \leq (1 - \min\{\frac{1}{3n}, a\})V_{t-1}$. If $v = 0$, the gradient-difference term vanishes and the same computation gives the factor $1 - \frac{1}{3n}$. □

Remark B2.24 (what each constant can explain). With the step size fixed at $\gamma = 1/(4R^2)$, the gain coming from $L_{\text{eff}} < R^2$ is the factor $(8R^2 - 5L_{\text{eff}})/(3R^2) \in [1, 8/3]$ on the parameter-term

exponent: the co-coercivity slack can explain at most a factor 8/3 of the gap, however small L_{eff} is (a smaller L_{eff} would allow a larger step, but the algorithm does not take it). In contrast, the gain from $\mu_{\text{eff}} > \mu$ is the factor μ_{eff}/μ, which can be as large as L/μ (Lemma B2.4) and is typically of the order described in Sections B2.3–B2.4. The parameter term is the active one in the minimum as long as $a(\theta_{t-1}) < 1/(3n)$, i.e. (with $L_{\text{eff}} = R^2$) $\mu_{\text{eff}}(\theta_{t-1}) < 16R^2/(9n)$; in the regime $\kappa \gg n$ of hypothesis (A4) this holds for μ and for moderate values of μ_{eff}, and the bound saturates at $1 - \frac{1}{3n}$ once μ_{eff} exceeds $16R^2/(9n)$: beyond that point no improvement of the curvature term can be "explained", and the residual gap belongs to the memory coefficient $1 - \frac{1}{3n}$, which is set by the square-norm inequality (I1), the inequality $\mathbb{E}_{i(t)} \left\| Z - \mathbb{E}_{i(t)} Z \right\|^2 \le \mathbb{E}_{i(t)} \|Z\|^2$ (I3) and the choice of Δ, and to the min step (I5) between the two rates, all instrumented in the B3 note. In its experiments with $\kappa \le 10^2$, where the memory term is active from the start, order A of Proposition B3.11 (which, as here, keeps the minimum until the end) assigns 0.56–0.61 of the logarithmic gap to (I5), 0.38–0.44 to (I1) and almost nothing to (I3).

Explained fraction of the theory–practice gap. Fix a window $W = \{t_0, \dots, t_1 - 1\}$ of $T = t_1 - t_0$ iterations inside the linear-convergence regime (e.g. one effective pass, $T = n$), and let `V[t]`, `mu_eff[t]`, `L_eff[t]` be the recorded values of V_t, $\mu_{\text{eff}}(\theta_t)$, $L_{\text{eff}}(\theta_t)$ (computed with η_* from a high-precision solver). Define the per-iteration exponents (written ν, to distinguish them from the tightness ratios r_{sc} and r_{coco} recalled at the end of this subsection)

$$\nu_{\text{obs}} = \frac{1}{T} \log \frac{V_{t_0}}{V_{t_1}}, \tag{B2.35}$$

$$\nu_{\text{th}} = -\log\left(1 - \min\left\{\frac{1}{3n}, \frac{3\mu}{16R^2}\right\}\right), \tag{B2.36}$$

$$\nu_{\mu} = -\frac{1}{T} \sum_{t \in W} \log\left(1 - \min\left\{\frac{1}{3n}, \frac{3\mu_{\text{eff}}(\theta_t)}{16R^2}\right\}\right), \tag{B2.37}$$

$$\nu_{\mu L} = -\frac{1}{T} \sum_{t \in W} \log\left(1 - \min\left\{\frac{1}{3n}, \frac{\mu_{\text{eff}}(\theta_t)\,(8R^2 - 5L_{\text{eff}}(\theta_t))}{16R^4}\right\}\right), \tag{B2.38}$$

and the explained fractions

$$\phi_\mu = \frac{\nu_\mu - \nu_{\text{th}}}{\nu_{\text{obs}} - \nu_{\text{th}}}, \qquad \phi_L = \frac{\nu_{\mu L} - \nu_\mu}{\nu_{\text{obs}} - \nu_{\text{th}}}, \qquad \phi_{\text{rest}} = 1 - \phi_\mu - \phi_L = \frac{\nu_{\text{obs}} - \nu_{\mu L}}{\nu_{\text{obs}} - \nu_{\text{th}}}. \tag{B2.39}$$

By Lemmas B2.4 and B2.6, $a(\theta) \ge 3\mu_{\text{eff}}(\theta)/(16R^2) \ge 3\mu/(16R^2)$ and the monotonicity of $x \mapsto -\log(1 - \min\{\frac{1}{3n}, x\})$, one has $\nu_{\text{th}} \le \nu_\mu \le \nu_{\mu L}$, so $\phi_\mu, \phi_L \ge 0$ whenever $\nu_{\text{obs}} > \nu_{\text{th}}$ (the fractions are meaningful only when there is a gap to explain). The quantity $e^{-T\nu_{\mu L}} = \prod_{t \in W}(1 - \min\{\frac{1}{3n}, a(\theta_t)\})$ is the product of the one-step factors of Proposition B2.23 along the realized trajectory. It does not bound $\mathbb{E}[V_{t_1} \mid \mathcal{F}_{t_0}]/V_{t_0}$ in general, because the factors are random, but the comparison holds in expectation. Let $c_s = 1 - \min\{\frac{1}{3n}, a(\theta_s)\}$, and $c_s = 1 - \frac{1}{3n}$ if $\theta_s = \eta_*$; each c_s is $\mathcal{F}_s$-measurable ($\mathcal{F}_s = \sigma(i(1), \dots, i(s))$ as in Lemma B2.21), and Proposition B2.23 gives $\mathbb{E}[V_{s+1} \mid \mathcal{F}_s] \le c_s V_s$, so that $Y_t = V_t / \prod_{s=t_0}^{t-1} c_s$, $t \ge t_0$, is a nonnegative supermartingale. Hence $\mathbb{E}[e^{T(\nu_{\mu L} - \nu_{\text{obs}})} \mid \mathcal{F}_{t_0}] = \mathbb{E}[Y_{t_1} \mid \mathcal{F}_{t_0}]/V_{t_0} \le 1$ and, by Jensen's inequality, $\mathbb{E}[\nu_{\text{obs}} - \nu_{\mu L} \mid \mathcal{F}_{t_0}] \ge 0$: the numerator of ϕ_{rest} is nonnegative in conditional expectation, and for a single realization only up to fluctuations (average over seeds, or use windows of several passes). Thus ϕ_μ is the fraction of the gap (in exponent) explained by the trajectory-dependent curvature alone, ϕ_L the additional fraction explained by the effective co-coercivity constant (at most a factor 8/3 by Remark B2.24), and ϕ_{rest} the fraction left to the other inequalities of the proof (square-norm inequality, variance versus second moment, the

minimum of the two rates, the choice of Δ), each of which the B3 note measures separately. The same decomposition can be reported instantaneously through the gain factors

$$g_\mu(t) = \frac{\min\{\frac{1}{3n}, \frac{3\mu_{\text{eff}}(\theta_t)}{16R^2}\}}{\min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}} \geq 1, \qquad g_L(t) = \frac{\min\{\frac{1}{3n}, a(\theta_t)\}}{\min\{\frac{1}{3n}, \frac{3\mu_{\text{eff}}(\theta_t)}{16R^2}\}} \in \Big[1, \frac{8}{3}\Big],$$

whose product is the ratio between the guaranteed per-step exponent along the trajectory and the one of Proposition 5.9. Finally, the per-step tightness ratios of the two inequalities themselves are, in the notation of Definition B3.8 of the B3 note,

$$r_{\text{sc}}(t) = \frac{\mu \left\|\theta_t - \eta_*\right\|^2}{(\theta_t - \eta_*)^\top F'(\theta_t)} = \frac{\mu}{\mu_{\text{eff}}(\theta_t)} \in \Big[\frac{\mu}{L}, 1\Big],$$
$$r_{\text{coco}}(t) = \frac{\frac{1}{n}\sum_i \left\|f_i'(\theta_t) - f_i'(\eta_*)\right\|^2}{R^2(\theta_t - \eta_*)^\top F'(\theta_t)} = \frac{L_{\text{eff}}(\theta_t)}{R^2} \in \Big[\frac{\mu}{R^2}, 1\Big],$$

for (B2.3) and (B2.7) respectively.

B2.6 Related work

Restricted secant inequality. Since the minimizer is unique, $\mu_{\text{eff}}(\theta)$ is the constant, at the point θ, of the restricted secant inequality of [89] (Definition 2 there: $\langle F'(x), x - x_{\text{prj}}\rangle \geq \nu \left\|x - x_{\text{prj}}\right\|^2$, with x_{prj} the projection of x on the solution set), whose constant "can be viewed as a lower bound of the average curvature" of the function between x and x_{prj}; Lemma B2.5 makes this exact. The best global constant of this inequality is $\inf_{\theta \neq \eta_*} \mu_{\text{eff}}(\theta)$, which equals λ for logistic regression: along $\theta = \eta_* + t\hat{d}$, $\hat{d}$ a unit vector, (B2.16) gives $\mu_{\text{eff}}(\theta) - \lambda = \frac{1}{n}\sum_i w_i(\theta)(x_i^\top \hat{d})^2$, where $w_i(\theta) \leq 1/\left|a_i - a_i^*\right| = 1/(t\left|x_i^\top \hat{d}\right|)$ by (B2.14) when $x_i^\top \hat{d} \neq 0$, so that $\mu_{\text{eff}}(\eta_* + t\hat{d}) \to \lambda$ as $t \to \infty$.

Local strong convexity and self-concordance. The observation that the curvature actually experienced by an algorithm is the local one is classical; the book itself notes that "locally around the global optimum, the strong convexity constant converges to the minimal eigenvalue of the Hessian at η_*, which can be significantly larger than μ (the global constant)" ([11], Section 5.2.3, paragraph "Adaptivity"). For logistic regression, [10] proves that averaged SGD with constant step size proportional to $1/(R^2\sqrt{N})$ has an $O(1/\sqrt{N})$ rate which "improves to $O(R^2/(\mu N))$ where μ is the lowest eigenvalue of the Hessian at the global optimum" (when this eigenvalue is greater than $R^2/\sqrt{N}$), without knowledge of μ (adaptivity to unknown local strong convexity); the tool is the generalized self-concordance of the logistic loss, Assumption (A2) there: along every line, $|\varphi'''(t)| \leq R\left\|\theta_1 - \theta_2\right\| \varphi''(t)$, a notion introduced in [8], whose Proposition 1, eq. (6), gives the Hessian comparison $e^{-R\|v\|}F''(w) \preccurlyeq F''(w+v) \preccurlyeq e^{R\|v\|}F''(w)$. Lemma B2.14 and Corollary B2.15 are the per-sample, directional version of this comparison, and $\mu_{\text{eff}}(\theta_t)$ is the trajectory-wise counterpart of the local constant $\lambda_{\min}(H_*)$: by Proposition B2.8 it converges to the Rayleigh quotient $u_t^\top H_* u_t \geq \lambda_{\min}(H_*)$, with equality only when the error aligns with the flattest direction, which Section B2.4 predicts for gradient descent but not for SAGA (the experiments confirm the latter only for κ at most of the order of n, and not in every such run, Remark B2.22).

Expected smoothness. The constant L_{eff} is a trajectory-wise version of the *expected smoothness* constant of [34] (Assumption 3.1 there: there is $\mathcal{L}_1 > 0$ with $\mathbb{E}_{\mathcal{D}}\left\|\nabla f_S(x) - \nabla f_S(x^*)\right\|^2 \leq 2\mathcal{L}_1(f(x) - f(x^*))$ for all x), which governs the rates of SAGA and of its JacSketch generalizations; for minibatch sketches with τ-nice sampling, Theorem 2 there (Theorem 4.16 of the arXiv version 1805.02632v1) shows that Assumption 3.1 is satisfied with $\mathcal{L}_1 = L_{\max}^{\mathcal{G}}$ (eq. (85) there), which for single-element uniform sampling ($\tau = 1$) is $\max_i L_i$, a valid constant rather

than the smallest one; the same assumption is the basis of the general SGD analysis of [33] (Assumption 2.1: $\mathbb{E}_{\mathcal{D}} \|\nabla f_v(x) - \nabla f_v(x^*)\|^2 \leq 2\mathcal{L}(f(x) - f(x^*))$). Defining the trajectory-wise expected smoothness $\mathcal{L}_{\mathrm{ES}}(\theta) = \frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|^2 / \big(2(F(\theta) - F(\eta_*))\big)$, the convexity inequality $F(\theta) - F(\eta_*) \leq (\theta - \eta_*)^\top F'(\theta)$ (eq. (5.7) of the book) gives $\mathcal{L}_{\mathrm{ES}}(\theta) \geq L_{\mathrm{eff}}(\theta)/2$, with $\mathcal{L}_{\mathrm{ES}} = L_{\mathrm{eff}}$ for quadratics (where $2(F(\theta) - F(\eta_*)) = (\theta - \eta_*)^\top F'(\theta)$). We work with L_{eff} because the proof of Proposition 5.9 is written with the inner product $(\theta - \eta_*)^\top F'(\theta)$ rather than with function values; both constants measure how much of the worst-case $R^2 = \max_i L_i$ is used along the trajectory, and both can be recorded at the same cost.

B1 Preconditioned (quadratic-metric) SAGA (hypothesis B1)

Throughout, $F(\theta) = \frac{1}{n}\sum_{i=1}^{n} f_i(\theta)$ on $\mathbb{R}^d$, each f_i is convex and R^2-smooth, F is μ-strongly convex with minimizer η_*, $\kappa = R^2/\mu$, and $P \in \mathbb{R}^{d\times d}$ is a *fixed* symmetric positive definite (SPD) matrix; $P^{1/2}$ denotes its SPD square root and $P^{-1/2} = (P^{1/2})^{-1}$. Whenever a Hessian is written we assume the corresponding function is twice continuously differentiable (this is the case for logistic regression); every statement that only involves gradients holds without this assumption (Remark B1.10). We write $\mathbb{E}_{i(t)}$ for the expectation with respect to the index $i(t)$ drawn at iteration t, conditionally on the past.

The content of this note is the following chain of facts. (1) SAGA run in the metric of P ("P-SAGA") is *exactly* Euclidean SAGA on a linearly transformed problem (Proposition B1.5), so Proposition 5.9 of [11] transfers verbatim, with (R^2, μ) replaced by the relative constants (R_P^2, μ_P) of (B1.6) (Corollary B1.12). (2) Redoing the proof in the P-metric shows which of the six proof inequalities change (Section B1.3.1); this defines the six tightness ratios to be instrumented in the P-geometry (Definition B1.14). (3) On ℓ_2-regularized logistic regression, *no* quadratic metric improves the certified global condition number, $\kappa_P \geq \kappa$ for every P (Proposition B1.21): the benefit of preconditioning is entirely a trajectory-dependent effect, captured by the effective constants of the B2 note. For the statistical preconditioner $P = \hat{\Sigma} + \lambda I$ the trajectory-dependent condition number is at most $(d_\infty(\lambda)/4 + 1)/m(\theta)$, where $d_\infty(\lambda)$ (n times the largest ridge leverage score, at most $C\, d_{\text{eff}}(\lambda)$ under a coherence assumption) is the only trace of the spectrum of $\hat{\Sigma}$ and $m(\theta)$ is the *smallest* per-sample logistic curvature averaged on the segment $[\eta_*, \theta]$ (Corollary B1.23). Because of the factor $1/m(\theta)$ this rigorous bound need not improve on the Euclidean one, and on the instances of our experiments it does not: it exceeds κ, i.e. it predicts no gain. The observed gain is accounted for only by the local constants at the optimum (Lemma B1.27), whose use along the whole run is heuristic (Section B1.6). (4) Cost and the case of a non-constant metric (Section B1.5), and a prediction table for the experiments (Section B1.6).

B1.1 The P-geometry and the algorithm

Definition B1.1 (P-norm and its dual norm)**.** *For $v, g \in \mathbb{R}^d$ let $\|v\|_P^2 := v^\top P v$ (norm on parameters) and $\|g\|_{P^{-1}}^2 := g^\top P^{-1} g$ (norm on gradients and memory vectors).*

Lemma B1.2 (elementary facts on the P-geometry)**.** *Let P be SPD. Then:*

(i) (generalized Cauchy–Schwarz) $|\langle g, v\rangle| \leq \|g\|_{P^{-1}} \|v\|_P$ for all $g, v \in \mathbb{R}^d$, with equality if and only if g and Pv are collinear; consequently $\|g\|_{P^{-1}} = \sup_{\|v\|_P \leq 1} \langle g, v\rangle$, i.e. $\|\cdot\|_{P^{-1}}$ is the dual norm of $\|\cdot\|_P$;

(ii) $\left\|P^{1/2} v\right\| = \|v\|_P$, $\left\|P^{-1/2} g\right\| = \|g\|_{P^{-1}}$ and $\left\|P^{-1} g\right\|_P = \|g\|_{P^{-1}}$;

(iii) (square-norm inequality) $\|a + b\|_{P^{-1}}^2 \leq 2\|a\|_{P^{-1}}^2 + 2\|b\|_{P^{-1}}^2$, with equality iff $a = b$;

(iv) (variance $\leq$ second moment) for a random vector Z with $\mathbb{E}\|Z\|^2 < \infty$, $\mathbb{E}\|Z - \mathbb{E}Z\|_{P^{-1}}^2 = \mathbb{E}\|Z\|_{P^{-1}}^2 - \|\mathbb{E}Z\|_{P^{-1}}^2 \leq \mathbb{E}\|Z\|_{P^{-1}}^2$;

(v) $\lambda_{\min}(P)\|v\|^2 \leq \|v\|_P^2 \leq \lambda_{\max}(P)\|v\|^2$ and $\lambda_{\max}(P)^{-1}\|g\|^2 \leq \|g\|_{P^{-1}}^2 \leq \lambda_{\min}(P)^{-1}\|g\|^2$.

Proof. (i) $\langle g, v\rangle = \langle P^{-1/2} g, P^{1/2} v\rangle$, and the Euclidean Cauchy–Schwarz inequality gives $|\langle g, v\rangle| \leq \left\|P^{-1/2} g\right\| \left\|P^{1/2} v\right\| = \|g\|_{P^{-1}} \|v\|_P$, with equality iff $P^{-1/2} g$ and $P^{1/2} v$ are collinear, i.e. iff g and Pv are; the choice $v = P^{-1} g / \|g\|_{P^{-1}}$ (for $g \neq 0$) has $\|v\|_P = 1$ and attains the supremum.

(ii) $\left\|P^{1/2}v\right\|^2 = v^\top P v$, $\left\|P^{-1/2}g\right\|^2 = g^\top P^{-1} g$ and $(P^{-1}g)^\top P (P^{-1}g) = g^\top P^{-1} g$. (iii) $\|a+b\|_{P^{-1}}^2 = \|a\|_{P^{-1}}^2 + 2a^\top P^{-1} b + \|b\|_{P^{-1}}^2$ and $2a^\top P^{-1} b \le \|a\|_{P^{-1}}^2 + \|b\|_{P^{-1}}^2$ because $\|a-b\|_{P^{-1}}^2 \ge 0$, with equality iff $a = b$. (iv) Expand $\mathbb{E}\,\|Z - \mathbb{E}Z\|_{P^{-1}}^2 = \mathbb{E}\,\|Z\|_{P^{-1}}^2 - 2(\mathbb{E}Z)^\top P^{-1}\mathbb{E}Z + \|\mathbb{E}Z\|_{P^{-1}}^2$. (v) These are the extremal values of the Rayleigh quotients of P and of P^{-1}. □

Definition B1.3 (P-SAGA: SAGA in the metric of P)**.** *Fix $\gamma > 0$, $\theta_0 \in \mathbb{R}^d$ and memory vectors $z_i^{(0)} \in \mathbb{R}^d$, $i = 1, \dots, n$. At iteration $t \ge 1$, draw $i(t)$ uniformly in $\{1, \dots, n\}$, independently of the past, and set*

$$g_t := f'_{i(t)}(\theta_{t-1}) + \frac{1}{n}\sum_{i=1}^n z_i^{(t-1)} - z_{i(t)}^{(t-1)}, \tag{B1.1}$$

$$\theta_t := \arg\min_{\theta \in \mathbb{R}^d} \Big\{ \langle g_t, \theta \rangle + \frac{1}{2\gamma}(\theta - \theta_{t-1})^\top P (\theta - \theta_{t-1}) \Big\}, \tag{B1.2}$$

$$z_{i(t)}^{(t)} := f'_{i(t)}(\theta_{t-1}), \qquad z_j^{(t)} := z_j^{(t-1)} \quad (j \ne i(t)). \tag{B1.3}$$

The memory update (B1.3) is the one of SAGA [16]; only the step (B1.2) depends on P.

Lemma B1.4 (closed form, unbiasedness, Bregman form)**.** *The minimization (B1.2) has the unique solution*

$$\theta_t = \theta_{t-1} - \gamma P^{-1} g_t, \tag{B1.4}$$

which for $P = I$ is the SAGA iteration of [11, Sec. 5.4.4]. Moreover $\mathbb{E}_{i(t)}[g_t] = F'(\theta_{t-1})$. Finally, (B1.2) is the Bregman step $\arg\min_\theta \{\gamma \langle g_t, \theta \rangle + D_h(\theta, \theta_{t-1})\}$ for the quadratic reference function $h = \frac{1}{2}\|\cdot\|_P^2$, whose Bregman divergence is $D_h(\theta, \theta') = \frac{1}{2}\|\theta - \theta'\|_P^2$; i.e. P-SAGA is Algorithm 1 (Bregman-SAGA) of [19] with a quadratic h.

Proof. The objective in (B1.2) is a strictly convex quadratic (its Hessian is $P/\gamma \succ 0$), so it has a unique minimizer, characterized by a vanishing gradient: $g_t + \gamma^{-1} P(\theta - \theta_{t-1}) = 0$, i.e. $\theta = \theta_{t-1} - \gamma P^{-1} g_t$. Since $i(t)$ is uniform and independent of the past, $\mathbb{E}_{i(t)}[f'_{i(t)}(\theta_{t-1})] = \frac{1}{n}\sum_i f'_i(\theta_{t-1}) = F'(\theta_{t-1})$ and $\mathbb{E}_{i(t)}[z_{i(t)}^{(t-1)}] = \frac{1}{n}\sum_i z_i^{(t-1)}$, so the two memory terms of (B1.1) cancel in expectation. For $h = \frac{1}{2}\|\cdot\|_P^2$ one has $\nabla h(\theta') = P\theta'$ and $D_h(\theta, \theta') = h(\theta) - h(\theta') - \langle \nabla h(\theta'), \theta - \theta' \rangle = \frac{1}{2}\|\theta - \theta'\|_P^2$; multiplying the objective of (B1.2) by γ gives the Bregman form. □

B1.2 Change of variables: P-SAGA is SAGA on a transformed problem

Proposition B1.5 (P-SAGA as SAGA on a transformed problem)**.** *Let $u := P^{1/2}\theta$, $\tilde f_i(u) := f_i(P^{-1/2}u)$ and $\tilde F := \frac{1}{n}\sum_i \tilde f_i = F \circ P^{-1/2}$. Then:*

(i) $\nabla \tilde f_i(u) = P^{-1/2} f'_i(P^{-1/2}u)$ and, when f_i is C^2, $\nabla^2 \tilde f_i(u) = P^{-1/2} f''_i(P^{-1/2}u) P^{-1/2}$; the same formulas hold for $(\tilde F, F)$.

(ii) $\tilde F$ is strongly convex and $\tilde u_* := P^{1/2}\eta_*$ is its unique minimizer.

(iii) Let $(\theta_t, z^{(t)})_{t \ge 0}$ be generated by P-SAGA (Definition B1.3) with step γ and index sequence $(i(t))_{t \ge 1}$, and let $u_t := P^{1/2}\theta_t$, $\tilde z_i^{(t)} := P^{-1/2} z_i^{(t)}$. Then $(u_t, \tilde z^{(t)})_{t \ge 0}$ is generated by Euclidean SAGA ($P = I$) on $(\tilde f_i)_{i \le n}$ with the same *step γ and the* same *index sequence, started from $(u_0, \tilde z^{(0)})$.*

(iv) For all t and i, $\|u_t-\tilde{u}_*\|^2=\|\theta_t-\eta_*\|_P^2$ and $\left\|\tilde{z}_i^{(t)}-\nabla\tilde{f}_i(\tilde{u}_*)\right\|^2=\left\|z_i^{(t)}-f_i'(\eta_*)\right\|_{P^{-1}}^2$; in particular $z_i^{(0)}=f_i'(\theta_0)$ for all i iff $\tilde{z}_i^{(0)}=\nabla\tilde{f}_i(u_0)$ for all i.

Proof. (i) $\tilde{f}_i=f_i\circ A$ with the linear map $A=P^{-1/2}$ (symmetric), so by the chain rule $\nabla\tilde{f}_i(u)=A^\top f_i'(Au)=P^{-1/2}f_i'(P^{-1/2}u)$ and $\nabla^2\tilde{f}_i(u)=A^\top f_i''(Au)A$. Averaging over i gives the statements for $\tilde{F}$.

(ii) By (i), $\nabla\tilde{F}(\tilde{u}_*)=P^{-1/2}F'(\eta_*)=0$. For $u,u'\in\mathbb{R}^d$ write $\theta=P^{-1/2}u$, $\theta'=P^{-1/2}u'$; by (i) and μ-strong convexity of F,

$$\begin{aligned}\tilde{F}(u')-\tilde{F}(u)-\left\langle\nabla\tilde{F}(u),u'-u\right\rangle&=F(\theta')-F(\theta)-\left\langle F'(\theta),\theta'-\theta\right\rangle\\&\geq\frac{\mu}{2}\left\|\theta'-\theta\right\|^2\geq\frac{\mu}{2\lambda_{\max}(P)}\left\|u'-u\right\|^2,\end{aligned}$$

where we used $\left\langle P^{-1/2}F'(\theta),u'-u\right\rangle=\left\langle F'(\theta),\theta'-\theta\right\rangle$ and Lemma B1.2(v), which gives $\|\theta'-\theta\|^2=\left\|P^{-1/2}(u'-u)\right\|^2\geq\|u'-u\|^2/\lambda_{\max}(P)$. Hence $\tilde{F}$ is $(\mu/\lambda_{\max}(P))$-strongly convex and its stationary point $\tilde{u}_*$ is its unique minimizer.

(iii) By induction on t; the initialization is the definition of $(u_0,\tilde{z}^{(0)})$. Let $\tilde{g}_t:=\nabla\tilde{f}_{i(t)}(u_{t-1})+\frac{1}{n}\sum_i\tilde{z}_i^{(t-1)}-\tilde{z}_{i(t)}^{(t-1)}$ be the Euclidean SAGA direction built from $(u_{t-1},\tilde{z}^{(t-1)})$ and the index $i(t)$. By (i) and the induction hypothesis, $\tilde{g}_t=P^{-1/2}f_{i(t)}'(\theta_{t-1})+P^{-1/2}\big(\frac{1}{n}\sum_i z_i^{(t-1)}-z_{i(t)}^{(t-1)}\big)=P^{-1/2}g_t$. Hence, by (B1.4), $u_t=P^{1/2}\theta_t=P^{1/2}\theta_{t-1}-\gamma P^{1/2}P^{-1}g_t=u_{t-1}-\gamma P^{-1/2}g_t=u_{t-1}-\gamma\tilde{g}_t$, which is the Euclidean SAGA step. For the memory, $\tilde{z}_{i(t)}^{(t)}=P^{-1/2}f_{i(t)}'(\theta_{t-1})=\nabla\tilde{f}_{i(t)}(u_{t-1})$ and $\tilde{z}_j^{(t)}=\tilde{z}_j^{(t-1)}$ for $j\neq i(t)$: this is the SAGA memory update.

(iv) By Lemma B1.2(ii), $\|u_t-\tilde{u}_*\|^2=\left\|P^{1/2}(\theta_t-\eta_*)\right\|^2=\|\theta_t-\eta_*\|_P^2$ and $\left\|\tilde{z}_i^{(t)}-\nabla\tilde{f}_i(\tilde{u}_*)\right\|^2=\left\|P^{-1/2}(z_i^{(t)}-f_i'(\eta_*))\right\|^2=\left\|z_i^{(t)}-f_i'(\eta_*)\right\|_{P^{-1}}^2$, using (i) at $\tilde{u}_*$. The last claim is (i) at u_0, $P^{-1/2}$ being invertible. □

Lemma B1.6 (relative smoothness and strong convexity constants). *(i) $\tilde{f}_i$ is L-smooth (i.e. $\nabla\tilde{f}_i$ is L-Lipschitz) if and only if*

$$\left\|f_i'(\theta)-f_i'(\theta')\right\|_{P^{-1}}\leq L\left\|\theta-\theta'\right\|_P\qquad\forall\theta,\theta'\in\mathbb{R}^d;\tag{B1.5}$$

when f_i is C^2 and convex, this is equivalent to $\sup_\theta\lambda_{\max}(P^{-1/2}f_i''(\theta)P^{-1/2})\leq L$.

(ii) $\tilde{F}$ is m-strongly convex if and only if $F(\theta')\geq F(\theta)+\langle F'(\theta),\theta'-\theta\rangle+\frac{m}{2}\|\theta'-\theta\|_P^2$ for all θ,θ'; when F is C^2, this is equivalent to $\inf_\theta\lambda_{\min}(P^{-1/2}F''(\theta)P^{-1/2})\geq m$.

(iii) If each f_i is R^2-smooth and F is μ-strongly convex (Euclidean constants), then (B1.5) holds with $L=R^2/\lambda_{\min}(P)$ and (ii) holds with $m=\mu/\lambda_{\max}(P)$. Conversely, if (B1.5) holds with $L=R_P^2$ and (ii) with $m=\mu_P$, then each f_i is $\lambda_{\max}(P)R_P^2$-smooth and F is $\lambda_{\min}(P)\mu_P$-strongly convex.

Proof. (i) With $\theta=P^{-1/2}u$, $\theta'=P^{-1/2}u'$, Proposition B1.5(i) and Lemma B1.2(ii) give

$$\left\|\nabla\tilde{f}_i(u)-\nabla\tilde{f}_i(u')\right\|=\left\|P^{-1/2}(f_i'(\theta)-f_i'(\theta'))\right\|=\left\|f_i'(\theta)-f_i'(\theta')\right\|_{P^{-1}},\qquad\|u-u'\|=\left\|\theta-\theta'\right\|_P;$$

as u ranges over $\mathbb{R}^d$ so does θ. For a C^2 convex function, L-smoothness is equivalent to all Hessians being $\preceq LI$ [11, Def. 5.3 and Sec. 5.2]; apply this to $\tilde{f}_i$, whose Hessian at u is $P^{-1/2}f_i''(P^{-1/2}u)P^{-1/2}$.

(ii) Same substitution in the definition of strong convexity, using $\left\langle \nabla \tilde{F}(u), u'-u \right\rangle = \langle F'(\theta), \theta'-\theta\rangle$ and $\|u'-u\|^2 = \|\theta'-\theta\|_P^2$; for C^2 functions m-strong convexity is equivalent to all Hessians being $\succeq mI$ [11, Sec. 5.2]. (iii) By Lemma B1.2(v),

$$\begin{aligned}\left\|f_i'(\theta)-f_i'(\theta')\right\|_{P^{-1}} &\le \lambda_{\min}(P)^{-1/2}\left\|f_i'(\theta)-f_i'(\theta')\right\| \le \lambda_{\min}(P)^{-1/2}R^2\left\|\theta-\theta'\right\| \\ &\le \lambda_{\min}(P)^{-1}R^2\left\|\theta-\theta'\right\|_P,\end{aligned}$$

and $\frac{\mu}{2}\|\theta'-\theta\|^2 \ge \frac{\mu}{2\lambda_{\max}(P)}\|\theta'-\theta\|_P^2$. The converse uses the reverse inequalities of Lemma B1.2(v) in the same way. □

Definition B1.7 (relative constants and relative condition number)**.** *We set*

$$\begin{aligned} R_P^2 &:= \max_{1\le i\le n}\ \sup_{\theta\in\mathbb{R}^d} \lambda_{\max}\big(P^{-1/2}f_i''(\theta)P^{-1/2}\big), \\ \mu_P &:= \inf_{\theta\in\mathbb{R}^d} \lambda_{\min}\big(P^{-1/2}F''(\theta)P^{-1/2}\big), \qquad \kappa_P := \frac{R_P^2}{\mu_P}. \end{aligned} \tag{B1.6}$$

By Lemma B1.6, every $\tilde{f}_i$ of Proposition B1.5 is R_P^2-smooth and $\tilde{F}$ is μ_P-strongly convex; in the terminology of [39, Def. 1] and [58], f_i is R_P^2-smooth and F is μ_P-strongly convex relative to *the reference function $h=\frac{1}{2}\|\cdot\|_P^2$ (whose Hessian is P). For $P=I$ these are the usual constants, and R^2,μ,κ always denote the Euclidean ($P=I$) constants.*

Lemma B1.8 (comparison with the Euclidean constants)**.** *Let* $\operatorname{cond}(P) := \lambda_{\max}(P)/\lambda_{\min}(P)$ *be the condition number of P. Then*

$$\begin{aligned} \mu_P \le \sup_\theta \lambda_{\max}\big(P^{-1/2}F''(\theta)P^{-1/2}\big) \le R_P^2, \qquad R_P^2 \le \frac{R^2}{\lambda_{\min}(P)}, \qquad \mu_P \ge \frac{\mu}{\lambda_{\max}(P)}, \\ \frac{\kappa}{\operatorname{cond}(P)} \le \kappa_P \le \kappa\ \operatorname{cond}(P). \end{aligned} \tag{B1.7}$$

In particular $\kappa_P \ge 1$, $\kappa_{cP}=\kappa_P$ for every $c>0$, and a badly chosen metric can degrade the condition number by at most $\operatorname{cond}(P)$.

Proof. $F''=\frac{1}{n}\sum_i f_i''$ and $\lambda_{\max}$ is subadditive on symmetric matrices, so $\lambda_{\max}(P^{-1/2}F''(\theta)P^{-1/2}) \le \frac{1}{n}\sum_i \lambda_{\max}(P^{-1/2}f_i''(\theta)P^{-1/2}) \le R_P^2$ for every θ, while $\lambda_{\min}\le\lambda_{\max}$ at every θ gives the first inequality. The two middle inequalities are Lemma B1.6(iii) combined with the Hessian characterizations; dividing them gives the upper bound on κ_P, and the converse part of Lemma B1.6(iii) gives $\kappa \le \kappa_P \operatorname{cond}(P)$. Finally $R_{cP}^2 = R_P^2/c$ and $\mu_{cP}=\mu_P/c$ by (B1.6). □

Remark B1.9 (scaling of P and of the step). (cP)-SAGA with step $c\gamma$ produces exactly the same iterates as P-SAGA with step γ, since $c\gamma(cP)^{-1}=\gamma P^{-1}$; consistently, the normalized step $\gamma = 1/(4R_P^2)$ of Corollary B1.12 is multiplied by c when P is. Only κ_P is scale free; the values of R_P^2 and μ_P quoted below for $P=\hat{\Sigma}+\lambda I$ depend on the normalization $\hat{\Sigma}=\frac{1}{n}X^\top X$ (with $P'=nP=X^\top X+n\lambda I$ one would get $R_{P'}^2=R_P^2/n$).

Remark B1.10 (C^1 functions). If the f_i are only C^1, define R_P^2 as the smallest constant L in (B1.5) (over all i) and μ_P as the largest m in Lemma B1.6(ii). Everything below that does not display a Hessian remains valid, because the transfer argument and the proof of Proposition 5.9 only use the gradient forms (co-coercivity, Proposition 5.4 of [11], is proved from L-smoothness and convexity alone).

Lemma B1.11 (relative constants of logistic regression). *Let $f_i(\theta) = \ell(y_i x_i^\top \theta) + \frac{\lambda}{2}\|\theta\|^2$ with $\ell(u) = \log(1+e^{-u})$, $y_i \in \{-1,1\}$, $x_i \in \mathbb{R}^d$, $\lambda > 0$, and let $\sigma(u) = (1+e^{-u})^{-1}$. Then $\ell'(u) = -\sigma(-u)$ and $\ell''(u) = \sigma'(u) = \sigma(u)\sigma(-u) = \left(2 + e^u + e^{-u}\right)^{-1} \in (0, \frac{1}{4}]$, which equals $\frac{1}{4}$ at $u = 0$ and is strictly decreasing in $|u|$. Writing $\sigma_i'(\theta) := \sigma'(y_i x_i^\top \theta)$,*

$$f_i'(\theta) = \ell'(y_i x_i^\top \theta)\, y_i x_i + \lambda\theta, \qquad f_i''(\theta) = \sigma_i'(\theta)\, x_i x_i^\top + \lambda I, \qquad F''(\theta) = \frac{1}{n}\sum_{i=1}^n \sigma_i'(\theta)\, x_i x_i^\top + \lambda I. \tag{B1.8}$$

For every SPD P,

$$\max\left\{\frac{1}{4}\max_i \|x_i\|_{P^{-1}}^2,\ \frac{\max_i \|x_i\|^2/4 + \lambda}{\lambda_{\max}(P)}\right\} \ \le\ R_P^2 \ \le\ \frac{1}{4}\max_i \|x_i\|_{P^{-1}}^2 + \frac{\lambda}{\lambda_{\min}(P)}, \tag{B1.9}$$

and more precisely $R_P^2 \ge \frac{1}{4}\|x_i\|_{P^{-1}}^2 + \lambda \frac{x_i^\top P^{-2} x_i}{x_i^\top P^{-1} x_i}$ for every i with $x_i \neq 0$;

$$\mu_P = \lambda\, \lambda_{\min}(P^{-1}) = \frac{\lambda}{\lambda_{\max}(P)}. \tag{B1.10}$$

In particular, for $P = I$: $R^2 := \frac{1}{4}\max_i \|x_i\|^2 + \lambda$ is a valid smoothness constant of every f_i, $\mu = \lambda$, and $\kappa = R^2/\mu = \max_i \|x_i\|^2/(4\lambda) + 1$; these are the Euclidean constants used in the whole project.

Proof. $\ell'(u) = -e^{-u}/(1+e^{-u}) = -\sigma(-u)$ and $\ell''(u) = \sigma'(u) = \sigma(u)(1-\sigma(u)) = \sigma(u)\sigma(-u) = 1/(2+e^u+e^{-u})$, which is maximal at $u = 0$ (value $\frac{1}{4}$) and decreasing in $|u|$ since $e^u + e^{-u} = 2\cosh u$ is increasing in $|u|$. The gradient and Hessians follow from the chain rule and $y_i^2 = 1$. Upper bound in (B1.9): by Weyl's inequality $\lambda_{\max}(A+B) \le \lambda_{\max}(A) + \lambda_{\max}(B)$ with $A = \sigma_i'(\theta) P^{-1/2} x_i x_i^\top P^{-1/2}$ (rank one, with nonzero eigenvalue $\sigma_i'(\theta)\left\|P^{-1/2}x_i\right\|^2 \le \frac{1}{4}\|x_i\|_{P^{-1}}^2$) and $B = \lambda P^{-1}$. Lower bounds: at $\theta = 0$, $\sigma_i'(0) = \frac{1}{4}$; for $x_i \neq 0$ the Rayleigh quotient of $P^{-1/2} f_i''(0) P^{-1/2}$ along $v = P^{-1/2}x_i$ equals $\frac{1}{4}(x_i^\top P^{-1} x_i)^2 / \left\|P^{-1/2}x_i\right\|^2 + \lambda \left\|P^{-1}x_i\right\|^2 / \left\|P^{-1/2}x_i\right\|^2 = \frac{1}{4}\|x_i\|_{P^{-1}}^2 + \lambda\,(x_i^\top P^{-2} x_i)/(x_i^\top P^{-1} x_i)$, which is the "more precise" bound; it is $\ge \frac{1}{4}\|x_i\|_{P^{-1}}^2$, and also $\ge (\|x_i\|^2/4 + \lambda)/\lambda_{\max}(P)$ because, with $y := P^{-1/2}x_i$, the fraction $(x_i^\top P^{-2} x_i)/(x_i^\top P^{-1} x_i) = y^\top P^{-1} y / \|y\|^2$ is a Rayleigh quotient of P^{-1}, hence $\ge \lambda_{\min}(P^{-1}) = 1/\lambda_{\max}(P)$, while $\|x_i\|_{P^{-1}}^2 \ge \|x_i\|^2/\lambda_{\max}(P)$ by Lemma B1.2(v). If $x_i = 0$ then $f_i'' \equiv \lambda I$ and $\sup_\theta \lambda_{\max}(P^{-1/2} f_i''(\theta) P^{-1/2}) = \lambda/\lambda_{\min}(P) \ge (\|x_i\|^2/4 + \lambda)/\lambda_{\max}(P)$, the first lower bound being trivial. Taking the maximum over i gives the two lower bounds in (B1.9). For (B1.10): $F''(\theta) \succeq \lambda I$ for all θ gives $P^{-1/2} F''(\theta) P^{-1/2} \succeq \lambda P^{-1} \succeq \lambda\lambda_{\min}(P^{-1}) I$, hence $\mu_P \ge \lambda\lambda_{\min}(P^{-1})$. Conversely, let $I_0 = \{i : x_i \neq 0\}$ and pick v with $v^\top x_i \neq 0$ for all $i \in I_0$ (the excluded set is a finite union of hyperplanes). For $s \to \infty$, $|y_i x_i^\top (sv)| \to \infty$ hence $\sigma_i'(sv) \to 0$ for $i \in I_0$, while the terms $i \notin I_0$ vanish identically; thus $F''(sv) \to \lambda I$ and, by continuity of $\lambda_{\min}$, $\lambda_{\min}(P^{-1/2} F''(sv) P^{-1/2}) \to \lambda\lambda_{\min}(P^{-1})$, so the infimum in (B1.6) is $\le \lambda\lambda_{\min}(P^{-1})$. The $P = I$ statements are the special case $\lambda_{\min}(P) = \lambda_{\max}(P) = 1$. □

B1.3 Convergence of P-SAGA: Proposition 5.9 in the P-metric

Corollary B1.12 (Proposition 5.9 transferred to P-SAGA). *Let P be SPD and let R_P^2, μ_P be the constants of (B1.6) (or any constants such that each $\tilde{f}_i$ of Proposition B1.5 is R_P^2-smooth and $\tilde{F}$ is μ_P-strongly convex). Run P-SAGA (Definition B1.3) with $\gamma = 1/(4R_P^2)$ and define*

$$V_t^P := \|\theta_t - \eta_*\|_P^2 + 3\gamma^2 \sum_{i=1}^n \left\|z_i^{(t)} - f_i'(\eta_*)\right\|_{P^{-1}}^2,$$

$$\rho_P := 1 - \min\left\{\frac{1}{3n}, \frac{3\mu_P}{16 R_P^2}\right\} = 1 - \min\left\{\frac{1}{3n}, \frac{3}{16\kappa_P}\right\}. \tag{B1.11}$$

Then for every $t \geq 1$,

$$\mathbb{E}_{i(t)}\big[V_t^P\big] \leq \rho_P\, V_{t-1}^P, \tag{B1.12}$$

and if $z_i^{(0)} = f_i'(\theta_0)$ for all i,

$$\mathbb{E}\,\|\theta_t - \eta_*\|_P^2 \leq \rho_P^t\Big(1 + \frac{n}{4}\Big)\,\|\theta_0 - \eta_*\|_P^2\,. \tag{B1.13}$$

Consequently

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \leq \mathrm{cond}(P)\,\rho_P^t\Big(1 + \frac{n}{4}\Big)\,\|\theta_0 - \eta_*\|^2$$

(the condition number of P enters the constant, not the rate), and $t \geq \max\{3n, \frac{16}{3}\kappa_P\}\log\frac{2n}{\varepsilon}$ iterations guarantee $\mathbb{E}\,\|\theta_t - \eta_*\|_P^2 \leq \varepsilon\,\|\theta_0 - \eta_*\|_P^2$, i.e. a running time of order $c_P\,(3n + \frac{16}{3}\kappa_P)\log\frac{2n}{\varepsilon}$, where c_P is the cost of one iteration (Section B1.5).

Proof by transfer. By Proposition B1.5(iii), $(u_t, \tilde{z}^{(t)}) = (P^{1/2}\theta_t, P^{-1/2}z^{(t)})$ is the Euclidean SAGA sequence on $(\tilde{f}_i)$ with step γ; the $\tilde{f}_i$ are convex and R_P^2-smooth and $\tilde{F}$ is μ_P-strongly convex with minimizer $\tilde{u}_*$ (Lemma B1.6, Proposition B1.5(ii)). The proof of Proposition 5.9 in [11] (transcribed in the B3 note) establishes, for Euclidean SAGA with $\gamma = 1/(4L)$ on any family of convex L-smooth functions whose average is m-strongly convex, the one-step contraction $\mathbb{E}_{i(t)}[\tilde{V}_t] \leq (1 - \min\{\frac{1}{3n}, \frac{3m}{16L}\})\tilde{V}_{t-1}$ of the Lyapunov function $\tilde{V}_t = \|u_t - \tilde{u}_*\|^2 + 3\gamma^2\sum_i \Big\|\tilde{z}_i^{(t)} - \nabla\tilde{f}_i(\tilde{u}_*)\Big\|^2$, and then the bound $\mathbb{E}\,\|u_t - \tilde{u}_*\|^2 \leq (1 - \min\{\frac{1}{3n}, \frac{3m}{16L}\})^t(1 + \frac{n}{4})\,\|u_0 - \tilde{u}_*\|^2$ when $\tilde{z}_i^{(0)} = \nabla\tilde{f}_i(u_0)$. By Proposition B1.5(iv), $\tilde{V}_t = V_t^P$ and $\|u_t - \tilde{u}_*\|^2 = \|\theta_t - \eta_*\|_P^2$ identically, and the initialization conditions correspond. The Euclidean consequence follows from Lemma B1.2(v) applied on both sides; the iteration count follows from $\rho_P^t \leq \exp(-t\min\{\frac{1}{3n}, \frac{3}{16\kappa_P}\})$ and $1 + \frac{n}{4} \leq 2n$. □

B1.3.1 The same proof written in the P-metric: which inequalities change

To see exactly where P enters, we rewrite the proof of Proposition 5.9 in the original coordinates; every inequality is labelled, and these labels are the ones instrumented by the experiments (Definition B1.14). Fix $t \geq 1$ and write $\delta := \theta_{t-1} - \eta_*$, $\omega_t := g_t$, $\bar{z} := \frac{1}{n}\sum_i z_i^{(t-1)}$ and, for each i,

$$a_{t,i} := f_i'(\theta_{t-1}) - f_i'(\eta_*), \qquad b_{t,i} := f_i'(\eta_*) - z_i^{(t-1)} + \bar{z}, \qquad \text{so that} \quad \omega_t = a_{t,i(t)} + b_{t,i(t)}.$$

Step 1. Expanding the square in the P-inner product and using (B1.4),

$$\begin{aligned}\|\theta_t - \eta_*\|_P^2 = \big\|\delta - \gamma P^{-1}\omega_t\big\|_P^2 &= \|\delta\|_P^2 - 2\gamma\,\delta^\top P P^{-1}\omega_t + \gamma^2\omega_t^\top P^{-1}PP^{-1}\omega_t \\ &= \|\delta\|_P^2 - 2\gamma\,\langle\delta, \omega_t\rangle + \gamma^2\,\|\omega_t\|_{P^{-1}}^2\,.\end{aligned} \tag{B1.14}$$

The pairing $\langle\delta, \omega_t\rangle$ is the same number for every P; taking $\mathbb{E}_{i(t)}$ and using Lemma B1.4 (unbiasedness, also metric independent), $\mathbb{E}_{i(t)}\,\|\theta_t - \eta_*\|_P^2 = \|\delta\|_P^2 - 2\gamma\,\langle\delta, F'(\theta_{t-1})\rangle + \gamma^2\mathbb{E}_{i(t)}\,\|\omega_t\|_{P^{-1}}^2$. The three inequalities of Step 1 are

$$\text{(I1}_P\text{, square norm)} \quad \mathbb{E}_{i(t)}\,\|\omega_t\|_{P^{-1}}^2 \leq 2\,\mathbb{E}_{i(t)}\,\big\|a_{t,i(t)}\big\|_{P^{-1}}^2 + 2\,\mathbb{E}_{i(t)}\,\big\|b_{t,i(t)}\big\|_{P^{-1}}^2\,, \tag{B1.15}$$

$$\begin{aligned}\text{(I2}_P\text{, co-coercivity)} \quad \mathbb{E}_{i(t)}\,\big\|a_{t,i(t)}\big\|_{P^{-1}}^2 &= \frac{1}{n}\sum_{i=1}^n \big\|f_i'(\theta_{t-1}) - f_i'(\eta_*)\big\|_{P^{-1}}^2 \\ &\leq R_P^2\,\big\langle F'(\theta_{t-1}), \theta_{t-1} - \eta_*\big\rangle\,,\end{aligned} \tag{B1.16}$$

$$\text{(I3}_P\text{, variance} \leq \text{2nd moment)} \quad \mathbb{E}_{i(t)}\,\big\|b_{t,i(t)}\big\|_{P^{-1}}^2 \leq \frac{1}{n}\sum_{i=1}^n \Big\|f_i'(\eta_*) - z_i^{(t-1)}\Big\|_{P^{-1}}^2\,. \tag{B1.17}$$

(B1.15) is Lemma B1.2(iii) applied for each value of $i(t)$. For (B1.16): each $\tilde f_i$ is convex and R_P^2-smooth, so co-coercivity (Proposition 5.4 of [11]) gives $\left\|\nabla\tilde f_i(u)-\nabla\tilde f_i(u')\right\|^2 \le R_P^2\left\langle\nabla\tilde f_i(u)-\nabla\tilde f_i(u'),u-u'\right\rangle$; with $u=P^{1/2}\theta_{t-1}$, $u'=P^{1/2}\eta_*$ and Proposition B1.5(i), this reads

$$\left\|f_i'(\theta_{t-1})-f_i'(\eta_*)\right\|_{P^{-1}}^2 \le R_P^2\left\langle f_i'(\theta_{t-1})-f_i'(\eta_*),\theta_{t-1}-\eta_*\right\rangle \qquad (i=1,\dots,n), \tag{B1.18}$$

because $\left\langle P^{-1/2}(f_i'(\theta)-f_i'(\eta_*)),P^{1/2}(\theta-\eta_*)\right\rangle=\langle f_i'(\theta)-f_i'(\eta_*),\theta-\eta_*\rangle$; averaging over i and using $\sum_i f_i'(\eta_*)=nF'(\eta_*)=0$ gives (B1.16), which is (5.31) of the book in the P-metric. For (B1.17): with $Z:=f_{i(t)}'(\eta_*)-z_{i(t)}^{(t-1)}$ one has $\mathbb{E}_{i(t)}Z=-\bar z$ (again $\sum_i f_i'(\eta_*)=0$), so $Z-\mathbb{E}_{i(t)}Z=b_{t,i(t)}$ and Lemma B1.2(iv) applies. Combining, Step 1 yields

$$\mathbb{E}_{i(t)}\left\|\theta_t-\eta_*\right\|_P^2 \le \|\delta\|_P^2-2\gamma(1-\gamma R_P^2)\left\langle F'(\theta_{t-1}),\delta\right\rangle+\frac{2\gamma^2}{n}\sum_{i=1}^n\left\|f_i'(\eta_*)-z_i^{(t-1)}\right\|_{P^{-1}}^2. \tag{B1.19}$$

Step 2. Only the $i(t)$-th memory vector changes, so, exactly,

$$\begin{aligned}\sum_{i=1}^n\left\|f_i'(\eta_*)-z_i^{(t)}\right\|_{P^{-1}}^2 &= \sum_{i=1}^n\left\|f_i'(\eta_*)-z_i^{(t-1)}\right\|_{P^{-1}}^2-\left\|f_{i(t)}'(\eta_*)-z_{i(t)}^{(t-1)}\right\|_{P^{-1}}^2\\ &\quad+\left\|f_{i(t)}'(\eta_*)-f_{i(t)}'(\theta_{t-1})\right\|_{P^{-1}}^2;\end{aligned}$$

taking $\mathbb{E}_{i(t)}$ (each index has probability $1/n$) and using (B1.16) once more,

$$\begin{aligned}\mathbb{E}_{i(t)}\sum_{i=1}^n\left\|f_i'(\eta_*)-z_i^{(t)}\right\|_{P^{-1}}^2 &= \Big(1-\frac1n\Big)\sum_{i=1}^n\left\|f_i'(\eta_*)-z_i^{(t-1)}\right\|_{P^{-1}}^2+\frac1n\sum_{i=1}^n\left\|f_i'(\eta_*)-f_i'(\theta_{t-1})\right\|_{P^{-1}}^2\\ &\le \Big(1-\frac1n\Big)\sum_{i=1}^n\left\|f_i'(\eta_*)-z_i^{(t-1)}\right\|_{P^{-1}}^2+R_P^2\left\langle F'(\theta_{t-1}),\delta\right\rangle.\end{aligned} \tag{B1.20}$$

Combination. Multiply (B1.20) by $\Delta=3\gamma^2$ and add to (B1.19). With $\gamma=1/(4R_P^2)$: $1-\gamma R_P^2-R_P^2\Delta/(2\gamma)=1-\frac14-\frac38=\frac38$ and $\frac{2\gamma^2}{n\Delta}+1-\frac1n=1-\frac{1}{3n}$, so

$$\mathbb{E}_{i(t)}[V_t^P]\le\|\delta\|_P^2-\frac{3\gamma}{4}\left\langle F'(\theta_{t-1}),\delta\right\rangle+\Big(1-\frac{1}{3n}\Big)\Delta\sum_{i=1}^n\left\|f_i'(\eta_*)-z_i^{(t-1)}\right\|_{P^{-1}}^2. \tag{B1.21}$$

The last two inequalities are

$$\text{(I4}_P\text{, strong convexity)}\quad \left\langle F'(\theta_{t-1}),\theta_{t-1}-\eta_*\right\rangle \ge \mu_P\left\|\theta_{t-1}-\eta_*\right\|_P^2, \tag{B1.22}$$

$$\text{(I5}_P\text{, min)}\quad \Big(1-\frac{3\mu_P}{16R_P^2}\Big)\|\delta\|_P^2+\Big(1-\frac{1}{3n}\Big)\Delta\sum_{i=1}^n\left\|f_i'(\eta_*)-z_i^{(t-1)}\right\|_{P^{-1}}^2 \le \rho_P V_{t-1}^P. \tag{B1.23}$$

For (B1.22): $\tilde F$ is μ_P-strongly convex, so adding the inequalities of Lemma B1.6(ii) written at $(\theta,\theta')=(\theta_{t-1},\eta_*)$ and at (η_*,θ_{t-1}), and using $F'(\eta_*)=0$, gives $\langle F'(\theta_{t-1}),\theta_{t-1}-\eta_*\rangle\ge\mu_P\left\|\theta_{t-1}-\eta_*\right\|_P^2$; inserting it in (B1.21) (the coefficient $-\frac{3\gamma}{4}$ is negative, so the inequality direction is preserved) produces the left-hand side of (B1.23), since $\frac{3\gamma}{4}\mu_P=\frac{3\mu_P}{16R_P^2}$. (B1.23) is the scalar

inequality $(1-\alpha)x+(1-\beta)y \le (1-\min\{\alpha,\beta\})(x+y)$ for $x,y\ge 0$, with $\alpha=\frac{3\mu_P}{16R_P^2}$, $\beta=\frac{1}{3n}$. This proves (B1.12). Iterating with the tower property, $\mathbb{E}V_t^P \le \rho_P^t V_0^P$, and

$$\text{(I6}_P\text{, initialization)} \qquad \sum_{i=1}^n \left\|f_i'(\eta_*)-f_i'(\theta_0)\right\|_{P^{-1}}^2 \le n\,R_P^4\,\|\theta_0-\eta_*\|_P^2 \tag{B1.24}$$

(each term is bounded by the square of (B1.5)) gives, when $z_i^{(0)}=f_i'(\theta_0)$,

$$V_0^P \le (1+3\gamma^2 nR_P^4)\,\|\theta_0-\eta_*\|_P^2 = \left(1+\frac{3n}{16}\right)\|\theta_0-\eta_*\|_P^2 \le \left(1+\frac{n}{4}\right)\|\theta_0-\eta_*\|_P^2,$$

which is (B1.13) because $\|\theta_t-\eta_*\|_P^2 \le V_t^P$. This completes the second proof of Corollary B1.12. □

Remark B1.13 (comparison with Bregman-SAGA). Corollary 1 of [19] states, for Bregman-SAGA with a quadratic reference function ($\nabla^2 h$ constant, which is exactly P-SAGA by Lemma B1.4) and step $1/(8L_{f/h})$, the contraction $\mathbb{E}[\psi_t] \le (1-\min\{\frac{1}{8\kappa_{f/h}},\frac{1}{2n}\})^t\psi_0$ of their potential $\psi_t=\frac{1}{\eta}D_h(x^\star,x_t)+\frac{n}{2}\cdot\frac{1}{n}\sum_i D_{f_i}(\phi_i^t,x^\star)$, where $L_{f/h}$ is the relative smoothness of every f_i (their Assumption 3) and $\kappa_{f/h}=L_{f/h}/\mu_{f/h}$ the relative condition number: these are our R_P^2 and κ_P. The rate has the same structure $1-\min\{c_1/n,\,c_2/\kappa_P\}$ as (B1.11) (with $(c_1,c_2)=(\frac{1}{2},\frac{1}{8})$ instead of $(\frac{1}{3},\frac{3}{16})$), the difference coming from the step and from a potential expressed with Bregman distances of the f_i instead of squared norms of memory vectors).

B1.3.2 Metric dependence of each step, and the six tightness ratios in the P-geometry

Table 2 summarizes where P enters. Three kinds of objects appear. (a) Quantities that are the same numbers for every P, given the current state $(\theta_{t-1},z^{(t-1)})$: the pairing $\langle\theta_{t-1}-\eta_*,\omega_t\rangle$ and its conditional expectation $\langle\theta_{t-1}-\eta_*,F'(\theta_{t-1})\rangle$, the unbiasedness, the memory bookkeeping identity of Step 2 (an identity, valid for any norm), the scalar inequality (I5), and the algebra of the combination (the constants $\frac{3}{8}$, $\frac{2}{3n}$, $\frac{3}{16}$ are unchanged because γ and Δ are normalized by R_P^2). (b) Inequalities whose *form* is metric independent but which are instantiated in the P^{-1}-inner product: (I1) and (I3), both valid in every Hilbert norm (Lemma B1.2). (c) Inequalities whose norm *and* constant change: (I2) (left side in $\|\cdot\|_{P^{-1}}$, right side metric free, constant R_P^2), (I4) (left side metric free, right side in $\|\cdot\|_P$, constant μ_P), (I6) (both norms and R_P^4). Of course the iterates themselves depend on P, so the numerical values of all ratios depend on P; the point of the classification is that the *same six ratios* can be computed, by Proposition B1.5(iv), either in θ-coordinates with P-norms (one linear solve with P per vector), or in u-coordinates with Euclidean norms on the transformed problem (whose regularizer is the non-isotropic quadratic $\frac{\lambda}{2}u^\top P^{-1}u$ for logistic regression, so the B3 code must accept a general quadratic regularizer if the second route is taken).

Definition B1.14 (tightness ratios in the P-geometry)**.** *The effective (trajectory-dependent) constants of the B2 note, written in the P-metric, are, for $\theta\neq\eta_*$,*

$$\mu_{\text{eff}}^P(\theta) := \frac{\langle F'(\theta),\theta-\eta_*\rangle}{\|\theta-\eta_*\|_P^2}, \qquad L_{\text{eff}}^P(\theta) := \frac{\frac{1}{n}\sum_i\|f_i'(\theta)-f_i'(\eta_*)\|_{P^{-1}}^2}{\langle F'(\theta),\theta-\eta_*\rangle}. \tag{B1.25}$$

Convention on indices. *As in the B3 note, the ratios of iteration t are evaluated at the state $(\theta_{t-1},z^{(t-1)})$ seen before $i(t)$ is drawn, so they involve $\mu_{\text{eff}}^P(\theta_{t-1})$ and $L_{\text{eff}}^P(\theta_{t-1})$; the B2 note abbreviates $\mu_{\text{eff}}(t):=\mu_{\text{eff}}(\theta_t)$, which for $P=I$ is the value entering the ratios of iteration $t+1$. To avoid any off-by-one error in code shared by the three notes, we never write $\mu_{\text{eff}}^P(t)$ but always display*

Step	Inequality	What changes with P	Metric-free part
(I1) (B1.15)	square norm	norm $\|\cdot\|_{P^{-1}}$ only	form of the inequality
(I2) (B1.16)	co-coercivity	norm $\|\cdot\|_{P^{-1}}$, constant R_P^2	pairing $\langle F'(\theta_{t-1}), \theta_{t-1} - \eta_* \rangle$
(I3) (B1.17)	variance $\leq$ 2nd moment	norm $\|\cdot\|_{P^{-1}}$ only	form of the inequality
(I4) (B1.22)	strong convexity	norm $\|\cdot\|_P$, constant μ_P	pairing $\langle F'(\theta_{t-1}), \theta_{t-1} - \eta_* \rangle$
(I5) (B1.23)	min	only through the value of $3\mu_P/(16R_P^2)$	scalar inequality
(I6) (B1.24)	initialization	both norms, constant R_P^4	—
exact steps	(B1.14), Step 2 identity, combination	norms only	unbiasedness, pairing, algebra

Table 2: Metric dependence of the steps of the proof of Proposition 5.9 (Section B1.3.1).

the point, $\mu_{\text{eff}}^P(\theta_{t-1})$. With the notation of Section B1.3.1 ($\delta = \theta_{t-1} - \eta_*$, $a_{t,i}$, $b_{t,i}$, $\bar{z}$), $\alpha = \frac{3\mu_P}{16R_P^2}$, $\beta = \frac{1}{3n}$ and $\Delta = 3\gamma^2$, define, for $\theta_{t-1} \neq \eta_*$,

$$
\begin{aligned}
r_{\text{sq}}^P(t) &:= \frac{\frac{1}{n}\sum_i \|a_{t,i} + b_{t,i}\|_{P^{-1}}^2}{\frac{2}{n}\sum_i \left(\|a_{t,i}\|_{P^{-1}}^2 + \|b_{t,i}\|_{P^{-1}}^2 \right)}, \\
r_{\text{coco}}^P(t) &:= \frac{\frac{1}{n}\sum_i \|f_i'(\theta_{t-1}) - f_i'(\eta_*)\|_{P^{-1}}^2}{R_P^2 \, \langle F'(\theta_{t-1}), \delta \rangle} = \frac{L_{\text{eff}}^P(\theta_{t-1})}{R_P^2}, \\
r_{\text{var}}^P(t) &:= \frac{\frac{1}{n}\sum_i \|b_{t,i}\|_{P^{-1}}^2}{\frac{1}{n}\sum_i \left\| f_i'(\eta_*) - z_i^{(t-1)} \right\|_{P^{-1}}^2} = 1 - \frac{\|\bar{z}\|_{P^{-1}}^2}{\frac{1}{n}\sum_i \left\| f_i'(\eta_*) - z_i^{(t-1)} \right\|_{P^{-1}}^2}, \\
r_{\text{sc}}^P(t) &:= \frac{\mu_P \|\delta\|_P^2}{\langle F'(\theta_{t-1}), \delta \rangle} = \frac{\mu_P}{\mu_{\text{eff}}^P(\theta_{t-1})}, \\
r_{\text{min}}^P(t) &:= \frac{(1-\alpha)\|\delta\|_P^2 + (1-\beta)\Delta \sum_i \left\| f_i'(\eta_*) - z_i^{(t-1)} \right\|_{P^{-1}}^2}{\rho_P \, V_{t-1}^P},
\end{aligned}
$$

with the convention $r_{\text{var}}^P(t) := 1$ when the memory is exact ($z_i^{(t-1)} = f_i'(\eta_*)$ for all i, the only case in which its denominator vanishes; (B1.17) is then the equality $0 \leq 0$), and, for $\theta_0 \neq \eta_*$,

$$
r_{\text{init}}^P := \frac{\sum_i \|f_i'(\eta_*) - f_i'(\theta_0)\|_{P^{-1}}^2}{n R_P^4 \|\theta_0 - \eta_*\|_P^2} = \frac{L_{\text{eff}}^P(\theta_0)\, \mu_{\text{eff}}^P(\theta_0)}{R_P^4} = r_{\text{coco}}^P(1)\, \frac{\mu_{\text{eff}}^P(\theta_0)}{R_P^2}. \tag{B1.26}
$$

Each ratio is the smaller side of one of (B1.15)-(B1.24) divided by the larger side, the expectations over $i(t)$ being exact averages over i. The identities in (B1.26) are immediate from (B1.25), the pairing $\langle F'(\theta_0), \theta_0 - \eta_* \rangle$ cancelling in the product $L_{\text{eff}}^P(\theta_0)\mu_{\text{eff}}^P(\theta_0)$: r_{init}^P is a derived ratio, not an independent one. The one-step ratio $r_{\text{tot}}^P(t) := \mathbb{E}_{i(t)}[V_t^P]/(\rho_P V_{t-1}^P)$, defined when $V_{t-1}^P > 0$, is also computable exactly (average of V_t^P over the n possible draws). For $P = I$ all of them reduce to the ratios of the B3 note.

Lemma B1.15 (ranges and degenerate cases)**.** *Let $\theta_{t-1} \neq \eta_*$. Then*

$$
0 < \mu_P \leq \mu_{\text{eff}}^P(\theta_{t-1}) \leq L_{\text{eff}}^P(\theta_{t-1}) \leq R_P^2, \tag{B1.27}
$$

all the denominators in Definition B1.14 are positive (that of r^P_{var} when the memory is not exact), and $r^P_{\mathrm{sc}}(t), r^P_{\mathrm{coco}}(t), r^P_{\mathrm{sq}}(t), r^P_{\min}(t) \in (0,1]$, $r^P_{\mathrm{var}}(t) \in [0,1]$; if $\theta_0 \neq \eta_*$ then $r^P_{\mathrm{init}} \in (0,1]$. At $\theta_{t-1} = \eta_*$ the ratios r^P_{sc} and r^P_{coco} are of the form $0/0$ (both numerator and denominator vanish since $F'(\eta_*) = 0$), as are r^P_{sq} when in addition all memory errors $z_i^{(t-1)} - f_i'(\eta_*)$ coincide, and $r^P_{\min}$, r^P_{tot} when in addition the memory is exact; they are left undefined there, as in the B3 note.

Proof. By Proposition B1.5(i),(iv), every quantity above is the corresponding Euclidean quantity of the B3 note computed on the transformed problem $(\tilde{f}_i)$ at the state $(u_{t-1}, \tilde{z}^{(t-1)})$, for which the hypotheses of the B3 note hold (convex R_P^2-smooth $\tilde{f}_i$, μ_P-strongly convex $\tilde{F}$, $u_{t-1} \neq \tilde{u}_*$); the inequalities of the statement are Lemma B3.9 of the B3 note (ranges of the tightness ratios), transferred. For completeness, directly: (B1.22) gives $\langle F'(\theta_{t-1}), \delta\rangle \geq \mu_P \|\delta\|_P^2 > 0$, i.e. $\mu^P_{\mathrm{eff}}(\theta_{t-1}) \geq \mu_P > 0$, so $F'(\theta_{t-1}) \neq 0$; by convexity of the squared norm (Jensen) and Lemma B1.2(i), $\frac{1}{n}\sum_i \|f_i'(\theta_{t-1}) - f_i'(\eta_*)\|^2_{P^{-1}} \geq \|F'(\theta_{t-1})\|^2_{P^{-1}} \geq \langle F'(\theta_{t-1}), \delta\rangle^2 / \|\delta\|_P^2 > 0$, which is $L^P_{\mathrm{eff}} \geq \mu^P_{\mathrm{eff}}$ together with the positivity of the numerator of r^P_{coco}; $L^P_{\mathrm{eff}} \leq R_P^2$ is (B1.16). The numerator of r^P_{sq} is $\geq \left\|\frac{1}{n}\sum_i (a_{t,i} + b_{t,i})\right\|^2_{P^{-1}} = \|F'(\theta_{t-1})\|^2_{P^{-1}} > 0$ (Jensen again, and $\sum_i b_{t,i} = 0$ because $\sum_i f_i'(\eta_*) = 0$); the numerator of $r^P_{\min}$ is $\geq (1-\alpha)\|\delta\|_P^2 > 0$ as $\alpha \leq \frac{3}{16}$, and its denominator is $\rho_P V^P_{t-1} \geq \rho_P \|\delta\|_P^2 > 0$. The upper bounds are the inequalities (B1.15)–(B1.23) themselves, and $r^P_{\mathrm{var}} \geq 0$ is clear. For r^P_{init}, apply (B1.27) at θ_0 in (B1.26). □

Remark B1.16 (effective one-step contraction). Keep $\gamma = 1/(4R_P^2)$ and $\Delta = 3\gamma^2$ but, in Steps 1–2, replace the two uses of (B1.16) and the use of (B1.22) by the identities

$$\frac{1}{n}\sum_i \left\|f_i'(\theta_{t-1}) - f_i'(\eta_*)\right\|^2_{P^{-1}} = L^P_{\mathrm{eff}}(\theta_{t-1})\left\langle F'(\theta_{t-1}), \delta\right\rangle, \qquad \left\langle F'(\theta_{t-1}), \delta\right\rangle = \mu^P_{\mathrm{eff}}(\theta_{t-1})\|\delta\|_P^2 .$$

The coefficient of $\langle F'(\theta_{t-1}), \delta\rangle$ in the combination becomes (writing L^P_{eff} for $L^P_{\mathrm{eff}}(\theta_{t-1})$)

$$-2\gamma\big(1 - \gamma L^P_{\mathrm{eff}} - L^P_{\mathrm{eff}}\Delta/(2\gamma)\big) = -2\gamma\big(1 - \tfrac{5}{2}\gamma L^P_{\mathrm{eff}}\big) = -2\gamma\big(1 - \tfrac{5}{8} r^P_{\mathrm{coco}}(t)\big),$$

whence, with only (B1.15) and (B1.17) still used,

$$\mathbb{E}_{i(t)}[V_t^P] \leq \left(1 - \left(1 - \frac{5}{8} r^P_{\mathrm{coco}}(t)\right)\frac{\mu^P_{\mathrm{eff}}(\theta_{t-1})}{2R_P^2}\right)\|\theta_{t-1} - \eta_*\|_P^2 + \left(1 - \frac{1}{3n}\right)\Delta \sum_{i=1}^n \left\|f_i'(\eta_*) - z_i^{(t-1)}\right\|^2_{P^{-1}} . \tag{B1.28}$$

Since $r^P_{\mathrm{coco}} \in (0,1]$ (Lemma B1.15), the contraction factor of the parameter part lies between $1 - \frac{\mu^P_{\mathrm{eff}}(\theta_{t-1})}{2R_P^2}$ and $1 - \frac{3\mu^P_{\mathrm{eff}}(\theta_{t-1})}{16R_P^2}$: the rate "seen by the proof" is governed by the effective relative condition number $R_P^2/\mu^P_{\mathrm{eff}}(\theta_{t-1})$ (the B2 quantity in the P-metric), the looseness of co-coercivity affecting only the constant, between $\frac{3}{16}$ and $\frac{1}{2}$. This is the quantity whose behaviour under the four metrics is predicted in Section B1.6.

B1.4 Choice of P on ℓ_2-regularized logistic regression

Setting and notation. $X \in \mathbb{R}^{n\times d}$ has rows $x_i^\top$, $\hat{\Sigma} := \frac{1}{n}X^\top X$ has eigenvalues $\sigma_1 \geq \cdots \geq \sigma_d \geq 0$ and eigenvectors $v_1, \ldots, v_d$, and f_i is as in Lemma B1.11. We use the ridge leverage scores and the effective dimension ([23, Def. 1] with the linear kernel $K = XX^\top$; [9, Sec. 4.1]):

$$\mathrm{lev}_i(\lambda) := x_i^\top (X^\top X + n\lambda I)^{-1} x_i = \left[X(X^\top X + n\lambda I)^{-1} X^\top\right]_{ii}, \tag{B1.29}$$

$$d_{\mathrm{eff}}(\lambda) := \mathrm{tr}\left(\hat{\Sigma}(\hat{\Sigma} + \lambda I)^{-1}\right) = \sum_{k=1}^d \frac{\sigma_k}{\sigma_k + \lambda}, \qquad d_\infty(\lambda) := n \max_i \mathrm{lev}_i(\lambda),$$

d_∞ being the "maximal marginal degrees of freedom" $n\,\|\operatorname{diag}(K(K+n\lambda I)^{-1})\|_\infty$ of [9, Sec. 4.1, eq. (5)].

Notation shared with the other notes. The segment average $\bar\sigma_i'(\theta)$ of Proposition B1.22 below is the mean-value weight $w_i(\theta)$ of the B2 note, and $w_i := \sigma_i'(\eta_*)$ (paragraph (d)) is its value $w_i(\eta_*)$ at the optimum, denoted p_i in the B2 note; our $m(\theta) = \min_i \bar\sigma_i'(\theta)$ is at least the $m(\theta)$ of the B2 note (the minimum of σ' over the segments, attained at an endpoint), so every bound below stated with $m(\theta)$ also holds with the latter. The coherence constants C (Assumption B1.18) and C_w (paragraph (d)) are unrelated to the cross term C of the B3 note, and the relative Hessian M_P of Section B1.6 is unrelated to the memory error M of the B3 note and to the matrix M_* of the B2 note. Inside this note, m also denotes a generic strong-convexity constant (Lemma B1.6) and the inner-loop length in [61, 32].

Lemma B1.17 (leverage scores). *(i)* $\|x_i\|^2_{(\hat\Sigma+\lambda I)^{-1}} = n\,\mathrm{lev}_i(\lambda)$ *for every* i.

(ii) $\sum_{i=1}^n \mathrm{lev}_i(\lambda) = d_{\mathrm{eff}}(\lambda)$; *hence* $d_{\mathrm{eff}}(\lambda) \le d_\infty(\lambda)$.

(iii) $0 \le \mathrm{lev}_i(\lambda) < 1$ *for every* i, *hence* $d_\infty(\lambda) < n$; *for every* i *the map* $\lambda \mapsto \mathrm{lev}_i(\lambda)$ *is nonincreasing on* $(0,\infty)$, *hence so are* d_∞ *and* d_{eff}; *moreover* $d_{\mathrm{eff}}(\lambda) \le \min\{\mathrm{rank}(X), \mathrm{tr}(\hat\Sigma)/\lambda\}$.

(iv) $X(X^\top X + n\lambda I)^{-1}X^\top = K(K+n\lambda I)^{-1}$ *with* $K = XX^\top$, *so* $\mathrm{lev}_i(\lambda)$ *is the* λ*-ridge leverage score of [23, Def. 1].*

Proof. (i) $(\hat\Sigma+\lambda I)^{-1} = \big(\frac1n(X^\top X + n\lambda I)\big)^{-1} = n(X^\top X+n\lambda I)^{-1}$. (ii) By cyclicity of the trace, $\sum_i \mathrm{lev}_i = \mathrm{tr}\big(X(X^\top X+n\lambda I)^{-1}X^\top\big) = \mathrm{tr}\big((X^\top X+n\lambda I)^{-1}X^\top X\big) = \mathrm{tr}\big((\hat\Sigma+\lambda I)^{-1}\hat\Sigma\big)$, which equals $\sum_k \sigma_k/(\sigma_k+\lambda)$ in the eigenbasis of $\hat\Sigma$; the inequality is $\max_i \mathrm{lev}_i \ge \frac1n\sum_i \mathrm{lev}_i$. (iii) Let $X = \sum_{k=1}^r s_k u_k w_k^\top$ be a thin SVD ($r = \mathrm{rank}\,X$, $s_k>0$, $u_k \in \mathbb{R}^n$ orthonormal). Then $X(X^\top X + n\lambda I)^{-1}X^\top = \sum_k \frac{s_k^2}{s_k^2+n\lambda} u_k u_k^\top$, whose i-th diagonal entry is $\mathrm{lev}_i = \sum_k \frac{s_k^2}{s_k^2+n\lambda}u_{ki}^2$. If $\sum_k u_{ki}^2 = 0$ then $\mathrm{lev}_i = 0$; otherwise $\frac{s_k^2}{s_k^2+n\lambda} < 1$ gives $\mathrm{lev}_i < \sum_k u_{ki}^2 \le 1$, the last inequality because the u_k are orthonormal columns of an $n\times r$ matrix. Each coefficient $s_k^2/(s_k^2+n\lambda)$ is decreasing in λ and $u_{ki}^2 \ge 0$, so $\mathrm{lev}_i(\lambda)$ is nonincreasing in λ, and so are $d_\infty = n\max_i \mathrm{lev}_i$ and $d_{\mathrm{eff}} = \sum_i \mathrm{lev}_i$. The bound on d_{eff} follows from $\sigma_k/(\sigma_k+\lambda) \le \min\{\mathbf{1}_{\sigma_k>0}, \sigma_k/\lambda\}$. (iv) Push-through identity: $X(X^\top X+n\lambda I) = (XX^\top+n\lambda I)X$, hence $(XX^\top+n\lambda I)^{-1}X = X(X^\top X+n\lambda I)^{-1}$; multiply by $X^\top$ on the right. □

Assumption B1.18 (coherence of the leverage scores). *There is* $C \ge 1$ *such that the ridge leverage scores* (B1.29) *satisfy* $\max_i \mathrm{lev}_i(\lambda) \le C\,d_{\mathrm{eff}}(\lambda)/n$, *i.e.* $d_\infty(\lambda) \le C\,d_{\mathrm{eff}}(\lambda)$: *no data point has a leverage score larger than* C *times the average leverage score.*

(a) $P = I$ **(SAGA).** By Lemma B1.11, $R^2 = \frac14\max_i\|x_i\|^2 + \lambda$, $\mu = \lambda$ and

$$\kappa = \frac{\max_i\|x_i\|^2}{4\lambda} + 1 \;\ge\; \frac{\mathrm{tr}(\hat\Sigma)}{4\lambda} + 1 = \frac{\sum_{k=1}^d \sigma_k}{4\lambda} + 1, \tag{B1.30}$$

since $\max_i\|x_i\|^2 \ge \frac1n\sum_i\|x_i\|^2 = \mathrm{tr}\,\hat\Sigma$. Thus κ contains the whole spectrum of $\hat\Sigma$ through its trace (not only σ_1, which is what the condition number L/μ of F alone would contain, since $L \le \sigma_1/4+\lambda$), divided by λ.

(b) $P = P_{\mathrm{J}} := \operatorname{diag}(\hat{\Sigma}) + \lambda I$ **(feature scaling, Jacobi preconditioning).** Write $\hat{\Sigma}_{jj} = \frac{1}{n}\sum_i x_{ij}^2$ for the diagonal entries of $\hat{\Sigma}$, assume no feature is identically zero (otherwise drop it), and let $C_{\mathrm{corr}} := \operatorname{diag}(\hat{\Sigma})^{-1/2}\,\hat{\Sigma}\,\operatorname{diag}(\hat{\Sigma})^{-1/2}$ be the correlation matrix (unit diagonal, so $\lambda_{\min}(C_{\mathrm{corr}}) \le 1 \le \lambda_{\max}(C_{\mathrm{corr}}) \le d$).

Proposition B1.19 (constants of the Jacobi preconditioner). *(i)* $\|x_i\|^2_{P_{\mathrm{J}}^{-1}} = \sum_{j=1}^d \frac{x_{ij}^2}{\hat{\Sigma}_{jj}+\lambda}$ *and*

$$\frac{1}{n}\sum_i \|x_i\|^2_{P_{\mathrm{J}}^{-1}} = \sum_{j=1}^d \frac{\hat{\Sigma}_{jj}}{\hat{\Sigma}_{jj}+\lambda} =: d_{\mathrm{eff}}^{\mathrm{diag}}(\lambda) \;\ge\; d_{\mathrm{eff}}(\lambda).$$

(ii) $R^2_{P_{\mathrm{J}}} \le \frac{1}{4}\max_i \|x_i\|^2_{P_{\mathrm{J}}^{-1}} + \frac{\lambda}{\min_j \hat{\Sigma}_{jj}+\lambda}$ *and* $\mu_{P_{\mathrm{J}}} = \frac{\lambda}{\max_j \hat{\Sigma}_{jj}+\lambda}$.

(iii) $\lambda_{\min}(C_{\mathrm{corr}})\,P_{\mathrm{J}} \preceq \hat{\Sigma} + \lambda I \preceq \lambda_{\max}(C_{\mathrm{corr}})\,P_{\mathrm{J}}$; *consequently* $\lambda_{\min}(C_{\mathrm{corr}})\,n\,\mathrm{lev}_i(\lambda) \le \|x_i\|^2_{P_{\mathrm{J}}^{-1}} \le \lambda_{\max}(C_{\mathrm{corr}})\,n\,\mathrm{lev}_i(\lambda)$ *for all* i.

Proof. (i) P_{J} is diagonal, so $x_i^\top P_{\mathrm{J}}^{-1} x_i = \sum_j x_{ij}^2/(\hat{\Sigma}_{jj}+\lambda)$; averaging over i and using $\frac{1}{n}\sum_i x_{ij}^2 = \hat{\Sigma}_{jj}$ gives $d_{\mathrm{eff}}^{\mathrm{diag}}$. For the inequality, $g(s) = s/(s+\lambda)$ is concave on $[0,\infty)$, the vector $(\hat{\Sigma}_{jj})_j$ of diagonal entries of $\hat{\Sigma}$ is majorized by the vector of its eigenvalues (Schur–Horn theorem), and $x \mapsto \sum_j g(x_j)$ is Schur-concave when g is concave [60]; hence $\sum_j g(\hat{\Sigma}_{jj}) \ge \sum_k g(\sigma_k)$. (ii) Lemma B1.11 with $\lambda_{\min}(P_{\mathrm{J}}) = \min_j \hat{\Sigma}_{jj} + \lambda$ and $\lambda_{\max}(P_{\mathrm{J}}) = \max_j \hat{\Sigma}_{jj} + \lambda$. (iii) $\hat{\Sigma} = \operatorname{diag}(\hat{\Sigma})^{1/2} C_{\mathrm{corr}} \operatorname{diag}(\hat{\Sigma})^{1/2}$, so $\lambda_{\min}(C_{\mathrm{corr}})\operatorname{diag}(\hat{\Sigma}) \preceq \hat{\Sigma} \preceq \lambda_{\max}(C_{\mathrm{corr}})\operatorname{diag}(\hat{\Sigma})$; adding λI and using $\lambda_{\min}(C_{\mathrm{corr}}) \le 1 \le \lambda_{\max}(C_{\mathrm{corr}})$ gives the sandwich. If $\lambda_{\min}(C_{\mathrm{corr}}) > 0$, inverting it (the map $A \mapsto A^{-1}$ is order reversing on SPD matrices) gives

$$\lambda_{\max}(C_{\mathrm{corr}})^{-1} P_{\mathrm{J}}^{-1} \preceq (\hat{\Sigma}+\lambda I)^{-1} \preceq \lambda_{\min}(C_{\mathrm{corr}})^{-1} P_{\mathrm{J}}^{-1},$$

which with Lemma B1.17(i) yields the bounds on $\|x_i\|^2_{P_{\mathrm{J}}^{-1}}$; if $\lambda_{\min}(C_{\mathrm{corr}}) = 0$ (singular $\hat{\Sigma}$), the lower bound is trivial and the upper bound follows by inverting the right-hand inequality alone. □

Thus the Jacobi metric removes the *scale* anisotropy of the features (up to the λ terms, it is invariant under a rescaling of each feature) but keeps their *correlation* anisotropy: every constant of the statistical preconditioner $\hat{\Sigma} + \lambda I$ derived below transfers to P_{J} at the price of the factor $\mathrm{cond}(C_{\mathrm{corr}}) = \lambda_{\max}(C_{\mathrm{corr}})/\lambda_{\min}(C_{\mathrm{corr}})$ (the argument of Lemma B1.8, which applies verbatim to any pair of metrics, applied to $(P_{\mathrm{J}}, \hat{\Sigma}+\lambda I)$ through (iii)), and $P_{\mathrm{J}} = \hat{\Sigma}+\lambda I$ exactly when the features are uncorrelated ($C_{\mathrm{corr}} = I$).

(c) $P = \hat{\Sigma} + \lambda I$ **(statistical preconditioning).** Here $\lambda_{\max}(P) = \sigma_1 + \lambda$, $\lambda_{\min}(P) = \sigma_d + \lambda$ and, by Lemma B1.17(i), $\|x_i\|^2_{P^{-1}} = n\,\mathrm{lev}_i(\lambda)$.

Proposition B1.20 (global constants of the statistical preconditioner). *For* $P = \hat{\Sigma} + \lambda I$,

$$\frac{d_\infty(\lambda)}{4} \;\le\; R_P^2 \;\le\; \frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda} \;\le\; \frac{d_\infty(\lambda)}{4} + 1 \;\le\; \frac{C\,d_{\mathrm{eff}}(\lambda)}{4} + 1, \tag{B1.31}$$

the last inequality under Assumption B1.18, and

$$\mu_P = \frac{\lambda}{\sigma_1+\lambda}, \qquad \kappa_P = \frac{R_P^2}{\mu_P} \in \Big[\frac{d_\infty(\lambda)}{4}\cdot\frac{\sigma_1+\lambda}{\lambda},\; \Big(\frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda}\Big)\frac{\sigma_1+\lambda}{\lambda}\Big]. \tag{B1.32}$$

Proof. Insert $\max_i \|x_i\|^2_{P^{-1}} = n \max_i \mathrm{lev}_i(\lambda) = d_\infty(\lambda)$ and $\lambda_{\min}(P) = \sigma_d + \lambda$ in (B1.9), and $\lambda_{\max}(P) = \sigma_1 + \lambda$ in (B1.10); $\lambda/(\sigma_d + \lambda) \le 1$ and Assumption B1.18 give the last two bounds. □

The smoothness side is now controlled by leverage scores, i.e. by an effective dimension, with no trace of the spectrum of $\hat\Sigma$; but the *global* strong convexity side has become $\lambda/(\sigma_1 + \lambda)$, and the following statement shows that this is not an artefact of the proof: with global constants, no quadratic metric can beat $P = I$ on this problem.

Proposition B1.21 (no free lunch for global constants)**.** *For ℓ_2-regularized logistic regression and* every *SPD matrix P,*

$$\kappa_P \;\ge\; \frac{\max_i \|x_i\|^2}{4\lambda} + 1 = \kappa, \tag{B1.33}$$

with κ the Euclidean condition number (B1.30)*; consequently $\rho_P \ge \rho_I$ for every P. In particular, for $P = \hat\Sigma + \lambda I$, $\kappa \le \kappa_P \le \kappa\, \frac{\sigma_1+\lambda}{\sigma_d+\lambda}$.*

Proof. By the second lower bound in (B1.9), $R^2_P \ge (\max_i \|x_i\|^2/4 + \lambda)/\lambda_{\max}(P) = R^2/\lambda_{\max}(P)$, while $\mu_P = \lambda/\lambda_{\max}(P)$ by (B1.10); divide. Since $\kappa_P \mapsto \rho_P = 1 - \min\{\frac{1}{3n}, \frac{3}{16\kappa_P}\}$ is nondecreasing, $\rho_P \ge \rho_I$. The upper bound for $P = \hat\Sigma + \lambda I$ is (B1.7). □

The mechanism is transparent: the supremum over θ in R^2_P is attained on the decision boundary of sample i ($x_i^\top \theta = 0$, curvature $\frac14 x_i x_i^\top$), whereas the infimum over θ in μ_P is approached far from all decision boundaries, where $F'' \to \lambda I$ is isotropic; no single metric can be adapted both to a rank-one curvature $\frac14 x_i x_i^\top$ and to an isotropic one. (For the square loss the curvature is constant, $F'' \equiv \hat\Sigma + \lambda I = P$, every $f_i'' = x_i x_i^\top + \lambda I$ has relative smoothness in $[n\,\mathrm{lev}_i, n\,\mathrm{lev}_i + 1]$, and the global κ_P lies in $[d_\infty(\lambda), d_\infty(\lambda) + 1]$: a genuine gain. This is the ridge-regression setting of [28], whose Theorem 4 bounds the *average* condition number of the preconditioned problem by $\mathrm{tr}(P^{-1/2}(\hat\Sigma + \lambda I)P^{-1/2})/\lambda_{\min}(P^{-1/2}(\hat\Sigma + \lambda I)P^{-1/2})$, equal to d for $P = \hat\Sigma + \lambda I$; with uniform sampling and the maximum in (B1.6), d is replaced by $d_\infty(\lambda)$.) Hence, for logistic regression, any benefit of the metric must come from the *trajectory-dependent* constants, which we now quantify; this is exactly the point of view of the B2 note.

Proposition B1.22 (effective strong convexity from segment curvatures)**.** *For the logistic regression of Lemma B1.11, every $\theta \ne \eta_*$, $\delta := \theta - \eta_*$, and $\bar\sigma_i'(\theta) := \int_0^1 \sigma'\big(y_i x_i^\top(\eta_* + s\delta)\big)\, ds \in (0, \frac14]$,*

$$f_i'(\theta) - f_i'(\eta_*) = \bar\sigma_i'(\theta)\,(x_i^\top\delta)\,x_i + \lambda\delta, \qquad \big\langle F'(\theta), \theta - \eta_*\big\rangle = \frac1n \sum_{i=1}^n \bar\sigma_i'(\theta)\,(x_i^\top\delta)^2 + \lambda\,\|\delta\|^2. \tag{B1.34}$$

Consequently, with $m(\theta) := \min_i \bar\sigma_i'(\theta) \;\ge\; \min_i \sigma'\big(\max\{|x_i^\top\eta_*|, |x_i^\top\theta|\}\big) > 0$:

(i) for $P = \hat\Sigma + \lambda I$, the effective strong convexity $\mu^P_{\mathrm{eff}}(\theta) := \langle F'(\theta), \delta\rangle / \|\delta\|^2_P$ is the convex combination

$$\mu^P_{\mathrm{eff}}(\theta) = \frac{\frac1n\sum_i \bar\sigma_i'(\theta)(x_i^\top\delta)^2 + \lambda\,\|\delta\|^2}{\frac1n\sum_i (x_i^\top\delta)^2 + \lambda\,\|\delta\|^2} \;\in\; [\,m(\theta),\,1\,] \tag{B1.35}$$

of the numbers $\bar\sigma_1'(\theta), \dots, \bar\sigma_n'(\theta), 1$, and more precisely

$$\mu^P_{\mathrm{eff}}(\theta) \;\ge\; \frac{m(\theta)\,\sigma_1 + \lambda}{\sigma_1 + \lambda} \;\ge\; \max\Big\{m(\theta),\; \frac{\lambda}{\sigma_1+\lambda}\Big\} = \max\{m(\theta), \mu_P\}; \tag{B1.36}$$

(ii) for $P = I$, $\mu_{\mathrm{eff}}(\theta) = \frac1n\sum_i \bar\sigma_i'(\theta)\,(x_i^\top\delta/\|\delta\|)^2 + \lambda \in [\,m(\theta)\sigma_d + \lambda,\; \sigma_1/4 + \lambda\,]$.

Proof. By the fundamental theorem of calculus applied to $s \mapsto \ell'(y_i x_i^\top(\eta_* + s\delta))$, $\ell'(y_i x_i^\top \theta) - \ell'(y_i x_i^\top \eta_*) = \int_0^1 \ell''(y_i x_i^\top(\eta_* + s\delta))\, ds \cdot y_i x_i^\top \delta = \bar{\sigma}_i'(\theta)\, y_i x_i^\top \delta$; multiplying by $y_i x_i$ ($y_i^2 = 1$) and adding $\lambda\delta$ gives the first identity in (B1.34) by (B1.8). Pairing with δ, averaging over i and using $F'(\eta_*) = 0$ gives the second. Lower bound on $m(\theta)$: $s \mapsto x_i^\top(\eta_* + s\delta)$ is affine, so $|x_i^\top(\eta_* + s\delta)| \le \max\{|x_i^\top \eta_*|, |x_i^\top \theta|\}$ on $[0,1]$, and σ' is decreasing in $|u|$ (Lemma B1.11). (i) $\|\delta\|_P^2 = \delta^\top \hat{\Sigma}\delta + \lambda \|\delta\|^2 = \frac{1}{n}\sum_i (x_i^\top \delta)^2 + \lambda \|\delta\|^2$, which gives (B1.35); it is a convex combination with (normalized) weights $(x_i^\top \delta)^2/n$ and $\lambda \|\delta\|^2$, whence $m(\theta) \le \mu_{\text{eff}}^P \le 1$ because $\bar{\sigma}_i' \le \frac{1}{4} < 1$. For (B1.36): the numerator is $\ge \delta^\top(m\hat{\Sigma} + \lambda I)\delta$ with $m = m(\theta)$, and since $\hat{\Sigma}$ and P are simultaneously diagonalizable, $\delta^\top(m\hat{\Sigma} + \lambda I)\delta \ge \min_k \frac{m\sigma_k + \lambda}{\sigma_k + \lambda}\, \delta^\top P \delta = \frac{m\sigma_1 + \lambda}{\sigma_1 + \lambda} \|\delta\|_P^2$, because $s \mapsto \frac{ms + \lambda}{s + \lambda}$ is nonincreasing on $[0, \infty)$ when $m \le 1$ (its derivative is $\lambda(m-1)/(s+\lambda)^2$). Finally $\frac{m\sigma_1 + \lambda}{\sigma_1 + \lambda}$ is a convex combination of m and 1 with weights $\frac{\sigma_1}{\sigma_1 + \lambda}$ and $\frac{\lambda}{\sigma_1 + \lambda}$, hence it is $\ge m$ and $\ge \frac{\lambda}{\sigma_1 + \lambda}$. (ii) Divide (B1.34) by $\|\delta\|^2$; the bounds use $m \le \bar{\sigma}_i' \le \frac{1}{4}$ and $\sigma_d \|\delta\|^2 \le \delta^\top \hat{\Sigma}\delta \le \sigma_1 \|\delta\|^2$. □

Note that (B1.36) is the segment-averaged, *per-sample* version of the pointwise inequality $F''(\theta') \succeq \min_i \sigma_i'(\theta')\, \hat{\Sigma} + \lambda I$: integrating the latter along the segment $[\eta_*, \theta]$ would only produce the constant $\int_0^1 \min_i \sigma_i'(\eta_* + s\delta)\, ds$, which is $\le \min_j \int_0^1 \sigma_j'(\eta_* + s\delta)\, ds = m(\theta)$ (for each j, $\min_i \sigma_i' \le \sigma_j'$ pointwise), so (B1.36), which uses the minimum of the per-sample segment averages, is at least as strong as the integrated pointwise bound. Its global version only gives $\mu_P = \lambda/(\sigma_1 + \lambda)$ because $\inf_\theta \min_i \sigma_i'(\theta) = 0$. The experiments can test (B1.36) directly (its tightness ratio is $\frac{m(\theta_{t-1})\sigma_1 + \lambda}{\sigma_1 + \lambda} / \mu_{\text{eff}}^P(\theta_{t-1})$), and (B1.35) shows that $\mu_{\text{eff}}^P(\theta_{t-1})$ is a direction-weighted *average* of the per-sample curvatures along the segment, typically much larger than the minimum m: a single sample with a huge margin makes $m(\theta)$ tiny but has a negligible weight in (B1.35) unless the error δ is aligned with it.

Corollary B1.23 (leverage-score bound along trajectories)**.** *Let $\kappa_{\text{eff}}^P(\theta) := R_P^2 / \mu_{\text{eff}}^P(\theta)$ be the effective relative condition number at θ (the quantity governing the effective contraction (B1.28)). Then, for $P = \hat{\Sigma} + \lambda I$ and $m(\theta)$ as in Proposition B1.22,*

$$\kappa_{\text{eff}}^P(\theta) \;\le\; \Big(\frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d + \lambda}\Big) \frac{\sigma_1 + \lambda}{m(\theta)\sigma_1 + \lambda} \;\le\; \frac{d_\infty(\lambda)/4 + 1}{m(\theta)} \;\le\; \frac{C\, d_{\text{eff}}(\lambda)/4 + 1}{m(\theta)}, \tag{B1.37}$$

the last inequality under Assumption B1.18, whereas for $P = I$ the same argument only gives

$$\frac{\max_i \|x_i\|^2/4 + \lambda}{\sigma_1/4 + \lambda} \;\le\; \kappa_{\text{eff}}(\theta) \;\le\; \frac{\max_i \|x_i\|^2/4 + \lambda}{m(\theta)\sigma_d + \lambda}, \qquad \textit{with} \qquad \frac{\max_i \|x_i\|^2/4 + \lambda}{\sigma_1/4 + \lambda} \;\ge\; \frac{\operatorname{tr}\hat{\Sigma} + 4\lambda}{\sigma_1 + 4\lambda}. \tag{B1.38}$$

Proof. Divide (B1.31) by (B1.36), then use $\lambda/(\sigma_d + \lambda) \le 1$ and $\frac{\sigma_1 + \lambda}{m\sigma_1 + \lambda} \le \frac{1}{m}$ (as $m \le 1$). For $P = I$, divide R^2 by the bounds of Proposition B1.22(ii), and use $\max_i \|x_i\|^2 \ge \operatorname{tr}\hat{\Sigma}$. □

In words: along the trajectory, the effective condition number in the statistical metric is at most a leverage count, $d_\infty(\lambda)/4 + 1 \le C\, d_{\text{eff}}(\lambda)/4 + 1$, divided by the *smallest* of the per-sample curvatures of the logistic loss averaged on the segment $[\eta_*, \theta]$ (the typical, direction-weighted curvature of (B1.35) does not enter the bound); the eigenvalues $\sigma_1, \ldots, \sigma_d$ of $\hat{\Sigma}$ have disappeared, except through $d_\infty(\lambda) \le C\, d_{\text{eff}}(\lambda)$, with $d_{\text{eff}}(\lambda) = \sum_k \sigma_k/(\sigma_k + \lambda) \le d$, which is a *count* of the directions not killed by the regularization and is insensitive to the dynamic range of the spectrum. In the Euclidean metric, by contrast, even the trajectory-dependent bound keeps the factor $\max_i \|x_i\|^2/(m\sigma_d + \lambda) \ge \operatorname{tr}\hat{\Sigma}/(\sigma_d/4 + \lambda)$, the trace of $\hat{\Sigma}$ divided by its smallest eigenvalue (or by λ when $\lambda \gg m\sigma_d$); and the lower bound in (B1.38) shows that no trajectory can make the Euclidean κ_{eff} smaller than $(\operatorname{tr}\hat{\Sigma})/\sigma_1$

(up to the λ terms), the stable rank $\|X\|_F^2 / \|X\|_{\mathrm{op}}^2$ of X. The price paid in the P-metric is the coherence constant C and the curvature $m(\theta)$, which is small when some margins $|x_i^\top \theta|$ are large ($\sigma'(u) \approx e^{-|u|}$); the Newton metric (d) addresses this second issue.

On the instances of our experiments this price dominates, and Corollary B1.23 predicts no gain. In the experiment of Figure B1 of the report ($n = 1000$, $d = 50$, $\kappa = 10^3$, covariance spectrum $j^{-\alpha}$; Section B1.6), $m(\theta) \to \min_i w_i = 7.4 \cdot 10^{-3}$ ($\alpha = 1$) and $6.1 \cdot 10^{-3}$ ($\alpha = 2$) as $\theta \to \eta_*$, with $w_i = \sigma_i'(\eta_*)$ as in paragraph (d), so that the first bound in (B1.37) tends to $1.7 \cdot 10^3$ and $1.2 \cdot 10^3$: above $\kappa = 10^3$, and above the upper bound in (B1.38) for $P = I$ with the same m (968 and 999). Along the run of seed 0 it exceeds κ from pass 0.6 on, and lies between $1.5 \cdot 10^3$ and $1.8 \cdot 10^3$ ($\alpha = 1$), resp. $1.1 \cdot 10^3$ and $1.3 \cdot 10^3$ ($\alpha = 2$), over passes 2–20; the condition (B1.39) below, even with $m = \min_i w_i$ (no lower bound on $m(\theta_t)$ along a trajectory converging to η_* can be larger), would require $n \geq 4859$ and $n \geq 3288$. By contrast, $R_P^2 / \lambda_{\min}(P^{-1/2} F''(\eta_*) P^{-1/2})$, which bounds $\limsup_{\theta \to \eta_*} \kappa_{\mathrm{eff}}^P(\theta)$ by Lemma B1.27, equals 166 and 103 (against $R^2/\lambda_{\min}(F''(\eta_*)) = 576$ and 981 for $P = I$), and the values of $\kappa_{\mathrm{eff}}^P(\theta_{t-1})$ measured over passes 15–20 are about 84 and 31 (against 460 and 878 for $P = I$; geometric means over states and seeds), R_P^2 being here the upper bound of (B1.9), as in the experiments. The reason is that $\lambda_{\min}(P^{-1/2} F''(\eta_*) P^{-1/2}) = 0.117$ and 0.109 is the smallest over directions v of a weighted *average* of $w_1, \dots, w_n, 1$ (weights $(x_i^\top v)^2/n$ and $\lambda \|v\|^2$), 16 and 18 times $\min_i w_i$. The observed gain of $\hat{\Sigma} + \lambda I$ is therefore not certified by Corollary B1.23; it is described only by the local constants at the optimum, which Lemma B1.27 justifies as $\theta \to \eta_*$ and which Section B1.6 uses heuristically along the whole run, with the weighted average $\bar{w}$ in place of the minimum $m(\theta)$.

Remark B1.24 (memory-bottleneck regime). The one-step contraction factor is at most $1 - \min\{\frac{1}{3n}, \frac{3}{16\kappa'}\}$, with $\kappa' = \kappa_P$ for the certified bound (B1.12) and $\kappa' = R_P^2 / \mu_{\mathrm{eff}}^P(\theta_{t-1})$ for the state-dependent bound (B1.28) (a rate only under a deterministic bound on κ' along the trajectory); the memory term $\frac{1}{3n}$ is the binding one iff $\kappa' \leq \frac{9n}{16}$. For $P = \hat{\Sigma} + \lambda I$ and the trajectory constants of Corollary B1.23, a sufficient condition is therefore

$$n \;\geq\; \frac{16}{9} \cdot \frac{d_\infty(\lambda)/4 + 1}{m} = \frac{4 d_\infty(\lambda) + 16}{9m} \qquad \Big(\leq \frac{4C\, d_{\mathrm{eff}}(\lambda) + 16}{9m} \;\text{ under Assumption B1.18} \Big), \tag{B1.39}$$

with m a deterministic constant such that $m(\theta_t) \geq m$ for all t almost surely: a trajectory-dependent m cannot be pulled out of the expectations when the one-step bounds are iterated (see the case of a non-constant metric in Section B1.5), and without an a priori confinement of the iterates Proposition B3.21 of the B3 note, transferred, gives the bound only up to the exit time from a region (end of Section B1.4). In this regime ($n \gg d_{\mathrm{eff}}/m$) the bound on the contraction of P-SAGA is $1 - \frac{1}{3n}$ whatever λ and the spectrum of $\hat{\Sigma}$: it needs $3 \log(2n/\varepsilon)$ effective passes (the $\log(2n)$ coming from the prefactor $1 + \frac{n}{4}$ of (B1.13)), which is the $n \log(1/\varepsilon)$ term of the lower bound of hypothesis (A1). That bound is a worst case over the problem class for span-based methods with i.i.d. sampling, which cover P-SAGA only through Proposition B1.5 and for P fixed independently of the data; with full-gradient steps this term becomes of order $\frac{n}{1+\ln(n/\kappa)} \log \frac{1}{\varepsilon}$ when $\kappa \ll n$ (A1 note), and the accelerated complexity $O((n + \sqrt{n\kappa}) \log(1/\varepsilon))$ of (A3) keeps it. For $P = I$ the corresponding condition $n \geq \frac{16}{9} \frac{\max_i \|x_i\|^2/4 + \lambda}{m\sigma_d + \lambda}$ fails by a factor of order $\max_i \|x_i\|^2 / (d_\infty(\lambda)\, \sigma_d)$ when $\lambda \ll m\sigma_d$; and with global constants both conditions read $n \geq \frac{16}{9}\kappa_P \geq \frac{16}{9}\kappa$ by Proposition B1.21. This fails, by at least the factor $\frac{16}{9}$ and for every P, when $\lambda = R^2/n$, i.e. $\kappa = n$, which is the setting of the experiment of Section B1.6. The experiments of [11, Sec. 5.4.4] take $\mu = R_x^2/n$ with R_x the bound on $\|x_i\|$: this is $\kappa = n$ in the book's normalization (smoothness constant R_x^2), but $\kappa = \frac{n}{4} + 1 \leq \frac{9n}{16}$ (for $n \geq 4$) with the exact constant $R^2 = R_x^2/4 + \lambda$ used here, so that there the condition holds for $P = I$ and the memory term is already the binding one with global constants (in

our reproduction of that setting, Figure 0 of the report with $d = 40$, it also holds for $P = \hat{\Sigma} + \lambda I$: $\kappa_P \le 347$ for $n = 10^3$ and $\kappa_P \le 2814$ for $n = 10^4$).

Remark B1.25 (the variant $\frac{1}{4}\hat{\Sigma} + \lambda I$). $P' = \frac{1}{4}\hat{\Sigma} + \lambda I = \frac{1}{4}(\hat{\Sigma} + 4\lambda I)$ is the exact global upper envelope of F'' ($F''(\theta) \preceq P'$ for all θ by (B1.8)), so that F is 1-smooth relative to P'. It is the statistical preconditioner $P_{4\lambda} := \hat{\Sigma} + 4\lambda I$ at regularization 4λ, rescaled by $\frac{1}{4}$ (Remark B1.9; the loss keeps its regularization λ), so that $\kappa_{P'} = \kappa_{P_{4\lambda}}$ and $\kappa_{\mathrm{eff}}^{P'}(\theta) = \kappa_{\mathrm{eff}}^{P_{4\lambda}}(\theta)$ (both R_P^2 and $\mu_{\mathrm{eff}}^P(\theta)$ are multiplied by $1/c$ under $P \mapsto cP$). By Lemma B1.17(i) at regularization 4λ, $\|x_i\|_{P_{4\lambda}^{-1}}^2 = n\,\mathrm{lev}_i(4\lambda)$, so Lemma B1.11 gives $R_{P_{4\lambda}}^2 \le \frac{d_\infty(4\lambda)}{4} + \frac{\lambda}{\sigma_d + 4\lambda}$ and $\mu_{P_{4\lambda}} = \frac{\lambda}{\sigma_1 + 4\lambda}$; moreover $\mu_{\mathrm{eff}}^{P'}(\theta)$ is a convex combination of $4\bar{\sigma}_1', \dots, 4\bar{\sigma}_n', 1$ (weights $(x_i^\top \delta)^2/(4n)$ and $\lambda\|\delta\|^2$), hence lies in $[4m(\theta), 1]$, and the argument of (B1.36), now with the function $s \mapsto \frac{ms+\lambda}{s+4\lambda}$ (nonincreasing since $m \le \frac{1}{4}$), gives $\mu_{\mathrm{eff}}^{P_{4\lambda}}(\theta) \ge \frac{m(\theta)\sigma_1 + \lambda}{\sigma_1 + 4\lambda}$. Hence

$$\begin{aligned} \kappa_{P'} &\le \Big(\frac{d_\infty(4\lambda)}{4} + \frac{\lambda}{\sigma_d + 4\lambda}\Big)\frac{\sigma_1 + 4\lambda}{\lambda}, \\ \kappa_{\mathrm{eff}}^{P'}(\theta) &\le \Big(\frac{d_\infty(4\lambda)}{4} + \frac{\lambda}{\sigma_d + 4\lambda}\Big)\frac{\sigma_1 + 4\lambda}{m(\theta)\sigma_1 + \lambda} \le \frac{d_\infty(4\lambda)/4 + 1}{m(\theta)}, \end{aligned} \tag{B1.40}$$

the last step because $m(\sigma_1 + 4\lambda) \le m\sigma_1 + \lambda$ when $m \le \frac{1}{4}$. Since $\lambda \mapsto \mathrm{lev}_i(\lambda)$ is nonincreasing (Lemma B1.17(iii)), $d_\infty(4\lambda) \le d_\infty(\lambda)$, and $\frac{\lambda}{\sigma_d + 4\lambda} \le \frac{\lambda}{\sigma_d + \lambda}$: the brackets in (B1.40) are never larger than those of (B1.32) and (B1.37), but the factor $\sigma_1 + 4\lambda$ replaces $\sigma_1 + \lambda$. Thus the variant is never worse than $P = \hat{\Sigma} + \lambda I$ in the global and in the finer trajectory bounds *up to the factor* $(\sigma_1 + 4\lambda)/(\sigma_1 + \lambda) \le 4$, which is negligible when $\lambda \ll \sigma_1$, and never worse at all in the crude bound $(d_\infty/4 + 1)/m$; the experiments may use either.

(d) $P = H := F''(\eta_*)$ (oracle Newton metric). Let $w_i := \sigma_i'(\eta_*) = \sigma'(y_i x_i^\top \eta_*) \in (0, \frac{1}{4}]$, $W := \mathrm{diag}(w_1, \dots, w_n)$ and $\hat{\Sigma}_w := \frac{1}{n} X^\top W X$ (eigenvalues $\sigma_1^w \ge \cdots \ge \sigma_d^w \ge 0$), so that $H = \hat{\Sigma}_w + \lambda I$ by (B1.8). Define the w-weighted ridge leverage scores and effective dimension

$$\begin{aligned} \mathrm{lev}_i^w(\lambda) &:= w_i\, x_i^\top (X^\top W X + n\lambda I)^{-1} x_i = \big[W^{1/2} X (X^\top W X + n\lambda I)^{-1} X^\top W^{1/2}\big]_{ii}, \\ d_{\mathrm{eff}}^w(\lambda) &:= \mathrm{tr}\,\big(\hat{\Sigma}_w (\hat{\Sigma}_w + \lambda I)^{-1}\big), \qquad d_\infty^w(\lambda) := n \max_i \mathrm{lev}_i^w(\lambda), \end{aligned} \tag{B1.41}$$

i.e. the ridge leverage scores of the weighted design $W^{1/2}X$ (rows $\sqrt{w_i}\, x_i^\top$). Lemma B1.17 applied to $W^{1/2}X$ gives $\big\|\sqrt{w_i}\, x_i\big\|_{H^{-1}}^2 = n\,\mathrm{lev}_i^w(\lambda)$, $\sum_i \mathrm{lev}_i^w = d_{\mathrm{eff}}^w$ and $0 \le \mathrm{lev}_i^w < 1$.

Proposition B1.26 (constants in the Newton metric)**.** *For $P = H = F''(\eta_*)$:*

(i) $\|x_i\|_{H^{-1}}^2 = n\,\mathrm{lev}_i^w(\lambda)/w_i$, *hence*

$$\max_i \frac{n\,\mathrm{lev}_i^w(\lambda)}{4w_i} \;\le\; R_H^2 \;\le\; \max_i \frac{n\,\mathrm{lev}_i^w(\lambda)}{4w_i} + \frac{\lambda}{\sigma_d^w + \lambda} \;\le\; \frac{d_\infty^w(\lambda)}{4\min_i w_i} + 1, \qquad \mu_H = \frac{\lambda}{\sigma_1^w + \lambda}. \tag{B1.42}$$

(ii) For $\theta \ne \eta_*$, $\delta := \theta - \eta_*$ and $\bar{\sigma}_i'(\theta)$ as in Proposition B1.22,

$$\mu_{\mathrm{eff}}^H(\theta) = \frac{\frac{1}{n}\sum_i \bar{\sigma}_i'(\theta)(x_i^\top \delta)^2 + \lambda\|\delta\|^2}{\frac{1}{n}\sum_i w_i (x_i^\top \delta)^2 + \lambda\|\delta\|^2} \;\in\; \Big[\min\Big\{1, \min_i \frac{\bar{\sigma}_i'(\theta)}{w_i}\Big\},\; \max\Big\{1, \max_i \frac{\bar{\sigma}_i'(\theta)}{w_i}\Big\}\Big], \tag{B1.43}$$

a convex combination of $\bar{\sigma}_1'/w_1, \dots, \bar{\sigma}_n'/w_n, 1$; in particular $\mu_{\mathrm{eff}}^H(\theta) \to 1$ as $\theta \to \eta_*$, and $\mu_{\mathrm{eff}}^H(\theta) \ge \mu_H$ always.

Proof. (i) $x_i^\top H^{-1} x_i = w_i^{-1}(\sqrt{w_i}x_i)^\top H^{-1}(\sqrt{w_i}x_i) = n\,\mathrm{lev}_i^w/w_i$ by Lemma B1.17(i) for the design $W^{1/2}X$ (whose empirical covariance is $\hat\Sigma_w$); then (B1.9)–(B1.10) with $\lambda_{\min}(H) = \sigma_d^w + \lambda$, $\lambda_{\max}(H) = \sigma_1^w + \lambda$, and $\lambda/(\sigma_d^w+\lambda) \le 1$. (ii) Divide (B1.34) by $\|\delta\|_H^2 = \frac{1}{n}\sum_i w_i (x_i^\top\delta)^2 + \lambda\|\delta\|^2$; the (normalized) weights are $w_i(x_i^\top\delta)^2/n$ for the values $\bar\sigma_i'/w_i$ and $\lambda\|\delta\|^2$ for the value 1. As $\theta \to \eta_*$, $\bar\sigma_i'(\theta) \to \sigma_i'(\eta_*) = w_i$ by continuity of σ', so every value tends to 1. The last claim is (B1.22). □

Thus the Newton metric makes the strong convexity side perfect near the optimum ($\mu_{\text{eff}}^H \to 1$), but the smoothness side still pays the supremum over θ of the per-sample curvature: sample i enters R_H^2 with the weight $\sup_\theta \sigma_i'(\theta) = \frac{1}{4}$ in the numerator and with its curvature at the optimum w_i in the metric, hence the factor $1/w_i$ in (B1.42); a sample with a large margin at η_* (small w_i) but a non-negligible weighted leverage is the worst case. Under a *homogeneous curvature* assumption $\min_i w_i \ge w_{\min}$ and coherence of the weighted leverage scores ($d_\infty^w \le C_w\, d_{\text{eff}}^w$ for a constant $C_w \ge 1$), one gets $R_H^2 \le C_w\, d_{\text{eff}}^w(\lambda)/(4w_{\min}) + 1$ and $\kappa_{\text{eff}}^H(\theta) = R_H^2/\mu_{\text{eff}}^H(\theta) \to R_H^2 \le C_w\, d_{\text{eff}}^w/(4w_{\min}) + 1$ as $\theta \to \eta_*$. The next lemma shows that this $1/w_{\min}$ is again an artefact of the global supremum: the *realized* smoothness constant L_{eff}^H is leverage-controlled, with no curvature factor.

Lemma B1.27 (limits of the effective constants at η_*)**.** *Let the f_i be C^2, $H_i := f_i''(\eta_*)$, $H := F''(\eta_*) = \frac{1}{n}\sum_i H_i \succ 0$, and let $\theta \to \eta_*$ with $\delta := \theta - \eta_*$ satisfying $\delta/\|\delta\| \to v$, $\|v\| = 1$. Then, for every SPD P,*

$$\begin{aligned} L_{\text{eff}}^P(\theta) &\longrightarrow \frac{\frac{1}{n}\sum_i v^\top H_i P^{-1} H_i v}{v^\top H v}, \\ \mu_{\text{eff}}^P(\theta) &\longrightarrow \frac{v^\top H v}{v^\top P v} \in \big[\lambda_{\min}(P^{-1/2}HP^{-1/2}), \lambda_{\max}(P^{-1/2}HP^{-1/2})\big]. \end{aligned} \tag{B1.44}$$

Moreover $\frac{1}{n}\sum_i H_i P^{-1} H_i \succeq H P^{-1} H$, so the limit of L_{eff}^P is at least $\lambda_{\min}(P^{-1/2}HP^{-1/2})$, and for $P = H$

$$\lim L_{\text{eff}}^H(\theta) = 1 + \frac{\frac{1}{n}\sum_i \left\|H^{-1/2}(H_i - H)v\right\|^2}{\left\|H^{1/2}v\right\|^2} \;\ge\; 1, \tag{B1.45}$$

with equality iff $H_i v = Hv$ for all i (homogeneous per-sample curvature in the direction v). For the logistic regression of Lemma B1.11, in addition, $\lim L_{\text{eff}}^H(\theta) \le \max\{d_\infty^w(\lambda), 2\}$ for $P = H$, $\lim L_{\text{eff}}^P(\theta) \le d_\infty(\lambda)/4 + \frac{3}{2}$ for $P = \hat\Sigma + \lambda I$, and, for $P = I$, with $c_i := x_i^\top v$ and $B_v := \frac{1}{n}\sum_i w_i^2 c_i^2 \|x_i\|^2$,

$$\lim L_{\text{eff}}(\theta) = \frac{\lambda^2 + 2\lambda\, v^\top\hat\Sigma_w v + B_v}{\lambda + v^\top\hat\Sigma_w v}, \qquad (v^\top\hat\Sigma_w v)^2 \le B_v \le \frac{\max_i\|x_i\|^2}{4}\, v^\top\hat\Sigma_w v = (\kappa-1)\lambda\, v^\top\hat\Sigma_w v \tag{B1.46}$$

(this is the exact formula $L_{\text{eff}} = (\lambda^2 + 2\lambda A + B)/(\lambda + A)$ of Lemma B2.10, in the limit, where its A and B become $v^\top\hat\Sigma_w v$ and B_v); in particular $\lambda + v^\top\hat\Sigma_w v \le \lim L_{\text{eff}}(\theta) \le \lambda + (\kappa+1)\, v^\top\hat\Sigma_w v$.

Proof. Since f_i is C^2, $f_i'(\theta) - f_i'(\eta_*) = H_i\delta + o(\|\delta\|)$ and $F'(\theta) = H\delta + o(\|\delta\|)$ (as $F'(\eta_*) = 0$). Hence $\frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|_{P^{-1}}^2 = \frac{1}{n}\sum_i \delta^\top H_i P^{-1} H_i \delta + o(\|\delta\|^2)$, $\langle F'(\theta), \delta\rangle = \delta^\top H\delta + o(\|\delta\|^2)$ and $\|\delta\|_P^2 = \delta^\top P\delta$; dividing numerators and denominators by $\|\delta\|^2$ and letting $\delta/\|\delta\| \to v$ gives (B1.44), because $v^\top Hv > 0$ and $v^\top Pv > 0$; the bracket is the range of the Rayleigh quotient of $P^{-1/2}HP^{-1/2}$. The Loewner inequality: $0 \preceq \frac{1}{n}\sum_i (H_i - H)P^{-1}(H_i - H) = \frac{1}{n}\sum_i H_i P^{-1} H_i - HP^{-1}H$, since the cross terms give $-\frac{1}{n}\sum_i (H_i P^{-1} H + H P^{-1} H_i) = -2HP^{-1}H$. Then $v^\top HP^{-1}Hv = \left\|M\, P^{1/2}v\right\|^2$ with $M := P^{-1/2}HP^{-1/2}$, and for an SPD matrix M and any w, $\|Mw\|^2 = w^\top M^2 w \ge \lambda_{\min}(M)\, w^\top M w$ (because $M^2 - \lambda_{\min}(M)M = M^{1/2}(M - \lambda_{\min}(M)I)M^{1/2} \succeq 0$), with $w^\top Mw = v^\top Hv$ here; with

$P = H$ the same decomposition gives (B1.45). For logistic regression, $H_i = w_i x_i x_i^\top + \lambda I$ and $v^\top H_i P^{-1} H_i v = w_i^2 (x_i^\top v)^2 \|x_i\|_{P^{-1}}^2 + 2\lambda w_i (x_i^\top v)(x_i^\top P^{-1} v) + \lambda^2 v^\top P^{-1} v$. For $P = H$: $w_i \|x_i\|_{H^{-1}}^2 = n \operatorname{lev}_i^w$, so the first terms average to $\frac{1}{n}\sum_i n \operatorname{lev}_i^w w_i (x_i^\top v)^2 \le d_\infty^w \, v^\top \hat\Sigma_w v$; the remaining terms average to $2\lambda v^\top \hat\Sigma_w H^{-1} v + \lambda^2 v^\top H^{-1} v = 2\lambda v^\top (H - \lambda I) H^{-1} v + \lambda^2 v^\top H^{-1} v = 2\lambda \|v\|^2 - \lambda^2 v^\top H^{-1} v \le 2\lambda \|v\|^2$; hence the numerator is $\le d_\infty^w \, v^\top \hat\Sigma_w v + 2\lambda \|v\|^2 \le \max\{d_\infty^w, 2\} \, v^\top H v$. For $P = \hat\Sigma + \lambda I$: $\|x_i\|_{P^{-1}}^2 = n \operatorname{lev}_i$ and $w_i \le \frac{1}{4}$ give $\frac{1}{n}\sum_i w_i^2 (x_i^\top v)^2 \, n \operatorname{lev}_i \le \frac{d_\infty}{4} \, v^\top \hat\Sigma_w v$; next $P^{-1} \preceq \lambda^{-1} I$ gives $\lambda^2 v^\top P^{-1} v \le \lambda \|v\|^2$; finally, by Cauchy–Schwarz, $\hat\Sigma_w \preceq \frac{1}{4}\hat\Sigma$ and $\hat\Sigma \preceq P$,

$$
\begin{aligned}
2\lambda \, v^\top \hat\Sigma_w P^{-1} v &\le 2\lambda \left\|\hat\Sigma_w^{1/2} v\right\| \left\|\hat\Sigma_w^{1/2} P^{-1} v\right\| \le \lambda \left\|\hat\Sigma_w^{1/2} v\right\| \left\|\hat\Sigma^{1/2} P^{-1} v\right\| \le \lambda \left\|\hat\Sigma_w^{1/2} v\right\| \left\|P^{-1/2} v\right\| \\
&\le \sqrt{\lambda} \left\|\hat\Sigma_w^{1/2} v\right\| \|v\| \le \tfrac{1}{2}\big(v^\top \hat\Sigma_w v + \lambda \|v\|^2\big),
\end{aligned}
$$

where we used $\left\|\hat\Sigma^{1/2} P^{-1} v\right\|^2 = v^\top P^{-1} \hat\Sigma P^{-1} v \le v^\top P^{-1} v = \left\|P^{-1/2} v\right\|^2 \le \|v\|^2 / \lambda$ and $ab \le \frac{1}{2}(a^2 + b^2)$. Summing the three bounds, the numerator is $\le (\frac{d_\infty}{4} + \frac{1}{2}) v^\top \hat\Sigma_w v + \frac{3}{2}\lambda \|v\|^2 \le (\frac{d_\infty}{4} + \frac{3}{2}) \, v^\top H v$. For $P = I$: $v^\top H_i^2 v = \|H_i v\|^2 = \|w_i c_i x_i + \lambda v\|^2 = w_i^2 c_i^2 \|x_i\|^2 + 2\lambda w_i c_i^2 + \lambda^2$ (using $x_i^\top v = c_i$ and $\|v\| = 1$); averaging over i gives $B_v + 2\lambda v^\top \hat\Sigma_w v + \lambda^2$, and $v^\top H v = v^\top \hat\Sigma_w v + \lambda$, which is (B1.46). For the bounds on B_v: $\|x_i\|^2 \ge c_i^2$ (Cauchy–Schwarz, $\|v\| = 1$) and the convexity of the square give $B_v \ge \frac{1}{n}\sum_i w_i^2 c_i^4 \ge \big(\frac{1}{n}\sum_i w_i c_i^2\big)^2 = (v^\top \hat\Sigma_w v)^2$, while $w_i \le \frac{1}{4}$ and $\|x_i\|^2 \le \max_j \|x_j\|^2 = 4(\kappa - 1)\lambda$ (by (B1.30)) give $B_v \le \frac{\max_j \|x_j\|^2}{4} \cdot \frac{1}{n}\sum_i w_i c_i^2$. The two-sided bound on the limit follows, with $A := v^\top \hat\Sigma_w v$, from $(\lambda^2 + 2\lambda A + B_v)/(\lambda + A) \ge (\lambda + A)^2/(\lambda + A)$ and $(\lambda^2 + 2\lambda A + B_v)/(\lambda + A) \le (\lambda^2 + 2\lambda A + (\kappa - 1)\lambda A)/\lambda$. □

Global versus trajectory-dependent constants: where the supremum over θ is needed. Proposition 5.9 uses (I2) for each i at the pair (θ_{t-1}, η_*) for every t, and (I6) at (θ_0, η_*). The proof of co-coercivity in Proposition 5.4 of [11] uses smoothness at an auxiliary point, hence the global R_P^2; for C^2 convex f_i, however, the segment suffices: with $\delta = \theta_{t-1} - \eta_*$ and $\bar A_i := \int_0^1 f_i''(\eta_* + s\delta)\, ds \succeq 0$ one has $f_i'(\theta_{t-1}) - f_i'(\eta_*) = \bar A_i \delta$ and, since $M^2 \preceq \lambda_{\max}(M) M$ for every $M \succeq 0$,

$$
\left\|\bar A_i \delta\right\|_{P^{-1}}^2 \le \lambda_{\max}\big(P^{-1/2} \bar A_i P^{-1/2}\big) \, \big\langle f_i'(\theta_{t-1}) - f_i'(\eta_*), \delta \big\rangle ,
$$

where for logistic regression $\bar A_i = \bar\sigma_i'(\theta_{t-1}) x_i x_i^\top + \lambda I$ (as in (B1.34)), so that $\lambda_{\max}(P^{-1/2} \bar A_i P^{-1/2}) \le \bar\sigma_i'(\theta_{t-1}) \|x_i\|_{P^{-1}}^2 + \lambda / \lambda_{\min}(P)$. The value $\sup_\theta \sigma_i' = \frac{1}{4}$ thus enters only as the worst case over trajectories (the same holds for (I6)): the supremum over θ in R_P^2 is needed only because the step $\gamma = 1/(4R_P^2)$ is fixed before the trajectory is known. In the experiment of Section B1.6 the largest of these per-sample segment constants at η_* is still $0.75R^2$ for $P = I$ and $0.85R_P^2$ for $P = \hat\Sigma + \lambda I$ at $\alpha = 1$ ($0.62R^2$ and $0.94R_P^2$ at $\alpha = 2$, with R_P^2 the supremum of (B1.6)), some samples lying near their decision boundary: the slack of (I2) lies mostly in the average over samples and directions that defines L_{eff}^P, not in the supremum over θ. Strong convexity (I4) is also needed only at the pairs (θ_{t-1}, η_*), and the exact formula (B1.34) shows that it is a direction-weighted average of curvatures on the segment $[\eta_*, \theta_{t-1}]$, which depends on the trajectory. The global constants (R_P^2, μ_P) are therefore pessimistic in two different ways: μ_P by the gap between $\inf_\theta$ and the segment averages (Propositions B1.22 and B1.26), and R_P^2 by the gap between $\max_i \sup_\theta$ and the realized L_{eff}^P (Lemma B1.27). The effective constants of the B2 note, written in the P-metric (Definition B1.14), measure both gaps: $\mu_{\text{eff}}^P(\theta_{t-1})$ and $L_{\text{eff}}^P(\theta_{t-1})$; the statements above predict that preconditioning moves the gap from one inequality to the other (Section B1.6). A rigorous way to use trajectory-dependent constants is Proposition B3.21 of the B3 note, transferred by Proposition B1.5:

on a set $\Theta \ni \eta_*$, with $\inf_\Theta \mu_{\text{eff}}^P$ and $\sup_\Theta L_{\text{eff}}^P$ in place of μ_P and of the co-coercivity constant (the step staying $1/(4R_P^2)$), it bounds $\mathbb{E}[V_T^P \mathbf{1}_{\{\tau>T\}}]$, τ being the exit time from Θ, and it gives a rate if the iterates stay in Θ almost surely, which is not known a priori for SAGA (it can be enforced by a projection onto Θ, a proximal variant whose analysis differs from Proposition 5.9). For non-quadratic losses [39, Sec. 4.2] constrain their problem to a ball $B(0, D)$ through an indicator function in the composite term and prove a uniform concentration of Hessians over it, the ball being needed because the iterates depend on the sample that defines the preconditioner.

B1.5 Cost, and the case of a non-constant metric

Per iteration and per pass. One iteration of P-SAGA computes one gradient $f'_{i(t)}(\theta_{t-1})$ ($O(d)$, or $O(\text{nnz}(x_i))$ for sparse data), updates the memory vector $z_{i(t)}$ and the running sum $\sum_i z_i$ ($O(d)$), and applies P^{-1} to g_t: $O(d)$ if P is diagonal, $O(d^2)$ if P is dense and a Cholesky factor $P = LL^\top$ (or P^{-1} itself) has been precomputed (two triangular solves). Hence the cost of one iteration is $c_I = O(d)$, $c_{P_\text{J}} = O(d)$, $c_{\hat\Sigma+\lambda I} = c_H = O(d^2)$, and one effective pass (n iterations) costs $O(nd)$ for $P \in \{I, P_\text{J}\}$ and $O(nd^2)$ for a dense P. Precomputation: $\text{diag}(\hat\Sigma)$ costs $O(nd)$; $\hat\Sigma$ costs $O(nd^2)$ (one matrix product) and its Cholesky factorization $O(d^3)$; $F''(\eta_*)$ needs η_* (an oracle; in practice a Hessian at a reference point, same cost $O(nd^2)$). Memory: $O(nd)$ for the z_i as in Proposition 5.9 (the usual implementation for linear models stores one scalar $\ell'(y_i x_i^\top \theta_{\tau_i})$ per sample and keeps the regularizer outside the memory, $O(n)$ memory; this is a composite variant of SAGA [16] whose analysis differs slightly from Proposition 5.9), plus $O(d^2)$ for the factor of a dense P. A dense P also destroys sparsity ($P^{-1}x_i$ is dense), whereas a diagonal P preserves it.

Total complexity. By Corollary B1.12, the time to reach accuracy ε is

$$T_P \asymp c_P\Big(3n + \frac{16}{3}\kappa_P\Big)\log\frac{2n}{\varepsilon} + (\text{precomputation}), \tag{B1.47}$$

to be compared with $T_I \asymp d\,(3n + \frac{16}{3}\kappa)\log\frac{2n}{\varepsilon}$ for SAGA [11, Sec. 5.4.4]. The diagonal metric is free. A dense metric multiplies the per-iteration cost by d and adds $O(nd^2 + d^3)$, so it pays off only if the number of iterations decreases by more than a factor d: dividing (B1.47) by $d\log\frac{2n}{\varepsilon}$, the condition is

$$d\Big(3n + \frac{16}{3}\kappa_P\Big) + O\Big(\frac{nd + d^2}{\log(2n/\varepsilon)}\Big) \lesssim 3n + \frac{16}{3}\kappa, \qquad \text{i.e. roughly} \qquad \kappa \gtrsim d\,(n + \kappa_P),$$

the very ill-conditioned regime $\kappa \gg nd$ (the regime $\kappa \gg n$ of hypothesis (A4), with an extra factor d). With global constants this never happens for logistic regression ($\kappa_P \geq \kappa$, Proposition B1.21). With trajectory constants, *both* condition numbers must be replaced by their effective values: inserting the upper bounds (B1.37) for $P = \hat\Sigma + \lambda I$ and (B1.38) for $P = I$ (with the same lower bound m on the segment curvatures along both trajectories), the condition reads

$$\frac{\max_i \|x_i\|^2/4 + \lambda}{m\sigma_d + \lambda} \gtrsim d\left(n + \frac{d_\infty(\lambda)/4 + 1}{m}\right) \tag{B1.48}$$

if both bounds are tight. Since the left-hand side of (B1.48) is at most the global $\kappa = \max_i \|x_i\|^2/(4\lambda) + 1$, replacing it by κ (global-constant SAGA against effective-constant P-SAGA, an asymmetric comparison) gives a condition, $\kappa \gtrsim d(n + (d_\infty(\lambda)/4 + 1)/m)$, which is only *necessary*: it is a lower bound on the ill-conditioning required for a dense metric to pay off, and it overstates the benefit of the metric when $\lambda \ll m\sigma_d$ (then the left-hand side of (B1.48) is of order $\max_i \|x_i\|^2/(4m\sigma_d) \ll \kappa$). In the memory-bottleneck regime of Remark B1.24,

$T_P \asymp 3c_P n \log(2n/\varepsilon)$: $3\log(2n/\varepsilon)$ passes, each d times more expensive than a SAGA pass when P is dense.

A non-constant metric breaks the transfer argument. Suppose the step uses P_t at iteration t: $\theta_t = \theta_{t-1} - \gamma_t P_t^{-1} g_t$ (variable-metric or quasi-Newton variance reduction). Proposition B1.5 no longer applies: the transformed functions $\tilde{f}_i^{(t)} = f_i \circ P_t^{-1/2}$ and the transformed memory $P_t^{-1/2} z_i$ change with t, so there is no single Euclidean SAGA run to which Proposition 5.9 could be applied. In the P-metric proof of Section B1.3.1, the one-step inequality (B1.12) still holds at iteration t for the Lyapunov function $V_t^{P_t}$ built with P_t and $\gamma_t = 1/(4R_{P_t}^2)$, *provided* P_t (hence γ_t, $R_{P_t}^2$, ρ_{P_t}) is measurable with respect to the past $\mathcal{F}_{t-1} := \sigma(i(1), \dots, i(t-1))$, i.e. is computed before $i(t)$ is drawn: conditionally on $\mathcal{F}_{t-1}$ the metric is then fixed, the unbiasedness $\mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = F'(\theta_{t-1})$ can be used inside the P_t-norms, and the proof applies verbatim, giving $\mathbb{E}[V_t^{P_t} \mid \mathcal{F}_{t-1}] \le \rho_{P_t} V_{t-1}^{P_t}$ (if P_t depended on $i(t)$, then $\mathbb{E}[P_t^{-1} g_t \mid \mathcal{F}_{t-1}] \neq P_t^{-1} F'(\theta_{t-1})$ in general and the argument already breaks at (B1.14)). But this compares $\mathbb{E}[V_t^{P_t} \mid \mathcal{F}_{t-1}]$ with $V_{t-1}^{P_t}$, not with $V_{t-1}^{P_{t-1}}$; one must pay the change of metric

$$V_{t-1}^{P_t} \le c_t \, V_{t-1}^{P_{t-1}}, \qquad c_t := \max\left\{ \lambda_{\max}\big(P_{t-1}^{-1/2} P_t P_{t-1}^{-1/2}\big), \ \frac{\gamma_t^2}{\gamma_{t-1}^2} \lambda_{\max}\big(P_t^{-1/2} P_{t-1} P_t^{-1/2}\big) \right\} \tag{B1.49}$$

(Lemma B1.2(v) applied to the pairs (P_{t-1}, P_t) and (P_{t-1}^{-1}, P_t^{-1}); with the convention $P_0 := P_1$, $\gamma_0 := \gamma_1$, so that $c_1 = 1$). Both ρ_{P_t} and c_t are $\mathcal{F}_{t-1}$-measurable but random for an adaptive metric, and correlated with $V_{t-1}^{P_{t-1}}$, so they cannot be pulled out of the expectation; if however $\rho_{P_s} c_s \le \bar{\rho}_s$ almost surely for deterministic constants $\bar{\rho}_s$, then taking expectations and iterating gives $\mathbb{E} V_t^{P_t} \le \prod_{s=1}^t \bar{\rho}_s \, V_0^{P_0}$. Within this Lyapunov argument, linear convergence is thus certified only if the metric drift is controlled by deterministic bounds, e.g. $c_t \le \bar{c}_t$ with $\sum_t \log \bar{c}_t < \infty$ (a stationary limit metric, reached fast enough), or $c_t \le 1 + \frac{1}{2}\min\{\frac{1}{3n}, \frac{3}{16\bar{\kappa}}\}$ for a deterministic $\bar{\kappa} \ge \sup_t \kappa_{P_t}$ (then $\rho_{P_t} c_t \le 1 - \frac{1}{2}\min\{\frac{1}{3n}, \frac{3}{16\bar{\kappa}}\}$), which is the analogue of the condition "η_t does not vary too much (not more than a factor $1 - O(1/n)$)" stated after Theorem 4 of [19] (there the gains G_t must be nonincreasing for the Bregman potential to be a supermartingale). A uniform sandwich $\gamma I \preceq P_t^{-1} \preceq \Gamma I$ alone gives only $c_t \le \Gamma/\gamma$ for a constant step (from here on, $\gamma, \Gamma, \eta, \lambda, \Lambda, m, \mu$ in the quoted results are the symbols of the cited papers), useless per step; this is why the stochastic quasi-Newton variance-reduced methods of [61] (stochastic L-BFGS with SVRG gradients; Lemma 4: $\gamma I \preceq H_r \preceq \Gamma I$) and [32] (stochastic block BFGS with SVRG gradients; Assumption 1 and Lemma 1: $\gamma I \preceq H_t \preceq \Gamma I$) are analysed in the Euclidean norm, the metric entering only through (γ, Γ): Theorem 7 of [61] gives, for $\lambda I \preceq \nabla^2 f_T \preceq \Lambda I$, step η and inner-loop length m, the rate $\alpha = \big(\frac{1}{2m\eta} + \eta\Gamma^2\Lambda^2\big)/\big(\gamma\lambda - \eta\Gamma^2\Lambda^2\big)$, and Theorem 1 of [32] the rate $\rho = \big(\frac{1}{2m\eta} + \eta\Gamma^2\Lambda(\Lambda-\lambda)\big)/\big(\gamma\lambda - \eta\Gamma^2\Lambda^2\big)$. In the first formula, $\alpha < 1$ iff $\frac{1}{2m\eta} + 2\eta\Gamma^2\Lambda^2 < \gamma\lambda$; minimizing the left-hand side over η (AM–GM: its minimum is $2\Gamma\Lambda/\sqrt{m}$, attained at $\eta = 1/(2\Gamma\Lambda\sqrt{m})$) shows that $\alpha < 1$ for some η if and only if $m > 4\big(\Gamma\Lambda/(\gamma\lambda)\big)^2 \ge 4(\Lambda/\lambda)^2$, since $\Gamma \ge \gamma$: the certified epoch length is at least $4(\Lambda/\lambda)^2$. Since both papers require $\lambda I \preceq \nabla^2 f_T \preceq \Lambda I$ for *every* nonempty subset T of indices (single components and the full sum included, f_T being the average of the f_i, $i \in T$), $\Lambda \ge R^2$ and $\lambda \le \mu$ with our R^2 and μ, so this epoch length is at least $4\kappa^2$, whereas the epoch length of Euclidean SVRG is linear in the condition number; the benefit of the metric is not captured at all. Capturing it requires a relative-smoothness analysis with a fixed (or asymptotically fixed) reference function, as here ([19]: Corollary 1 for quadratic h, Theorem 4 with gains G_t otherwise), or a preconditioner fixed in advance from statistics of the data: a sub-sampled Hessian plus ridge as in [39], whose Theorem 2 shows that for quadratic losses with $\|a_i\| \le R$, a sub-sample of size $n > \frac{28}{3}\log\frac{2d}{\delta}$ and the ridge

$\mu = \frac{1}{2}\big(\frac{28R^2}{3n}\log\frac{2d}{\delta} - \lambda\big)^+$ give, with probability $1-\delta$, $(\frac{3}{2} + \frac{2\mu}{\lambda})^{-1}(H_f + \mu I) \preceq H_F \preceq 2(H_f + \mu I)$, i.e. a relative condition number $O\big(1 + \frac{\kappa_\ell}{n}\log\frac{d}{\delta}\big)$ with $\kappa_\ell = R^2/\lambda$; or a sketched preconditioner as in [28] (rank-k sketch of the covariance for ridge regression, computed in time $O(ndk\log n)$; their Theorem 1 replaces the average condition number $\sum_i(\sigma_i+\lambda)/(\sigma_d+\lambda)$ of SVRG by one of order $(k\sigma_k + \sum_{i>k}\sigma_i)/\lambda$). In all these cases the reference function is fixed in advance; the metric itself is constant only for a quadratic h (quadratic losses in [39, 28], Corollary 1 of [19]), which is exactly what Proposition B1.5 needs. (Non-uniform sampling of $i(t)$ is outside the uniform-sampling setting of Proposition 5.9 and of this note. For the record: Gower et al. [34, Thm. 5.2 and Table 1, row 8] proved (see also [71]) that SAGA with the reweighted unbiased estimator $\frac{1}{n}\sum_i z_i^{(t-1)} + \frac{1}{np_{i(t)}}\big(f'_{i(t)}(\theta_{t-1}) - z_{i(t)}^{(t-1)}\big)$ and the probabilities $p_i \propto \mu n + 4L_i$ that minimize their bound reaches accuracy ε in $O\big((n + 4\bar{L}/\mu)\log\frac{1}{\varepsilon}\big)$ iterations, $\bar{L} = \frac{1}{n}\sum_i L_i$ being the *average* of the per-sample smoothness constants; this resolves what they call a conjecture of Schmidt et al. [75], who had obtained a rate depending on $\bar{L}$ only for a variant drawing two indices per iteration (one with $p_i \propto L_i$ for the gradient, one uniform for the memory update) [75, Prop. 1]; a non-rigorous motivation of Lipschitz sampling for SAG is given in [77, Sec. 4.8]. The change of variables of Proposition B1.5 applies verbatim to this variant, and for $P = \hat{\Sigma} + \lambda I$ the per-sample relative smoothness constants satisfy $L_i \le \frac{n\,\mathrm{lev}_i(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda}$ by the Weyl bound in the proof of Lemma B1.11, so that $d_\infty(\lambda)$ would be replaced by $d_{\mathrm{eff}}(\lambda) = \sum_i \mathrm{lev}_i(\lambda)$ in the *global* bound (B1.32); whether the same substitution holds in the trajectory bounds of Corollary B1.23 is not covered by those analyses and is left open.)

B1.6 Predictions for the experiments

Tables 3 and 4 collect, for the four metrics, the certified constants (Lemma B1.11, Propositions B1.19, B1.20 and B1.26), the certified rate $\rho_P = 1 - \min\{\frac{1}{3n}, \frac{3}{16\kappa_P}\}$, and the predicted late-phase behaviour of the effective constants (Lemma B1.27). The late-phase entries are evaluated for the extreme direction v = bottom eigenvector of the relative Hessian $M_P := P^{-1/2}F''(\eta_*)P^{-1/2}$ (the most slowly contracted direction), as if the late-phase error δ aligned with it, as it does for gradient descent on a quadratic; $\bar{w}$ denotes a leverage-weighted average of the curvatures $w_i = \sigma_i'(\eta_*)$ in that direction and $\overline{n\,\mathrm{lev}}$ a weighted average of the $n\,\mathrm{lev}_i(\lambda)$ (equal to $d_{\mathrm{eff}}(\lambda)$ when leverage scores are homogeneous). Some entries do not depend on this alignment. By Lemma B1.27 and the compactness of the unit sphere, whatever the direction of the error, $\liminf_{\theta\to\eta_*}\mu_{\mathrm{eff}}^P(\theta) \ge \lambda_{\min}(M_P)$, hence $\limsup r_{\mathrm{sc}} \le \mu_P/\lambda_{\min}(M_P)$ and $\limsup R_P^2/\mu_{\mathrm{eff}}^P \le R_P^2/\lambda_{\min}(M_P)$; the bounds $\lim L_{\mathrm{eff}}^P \le d_\infty(\lambda)/4 + \frac{3}{2}$ ($P = \hat{\Sigma} + \lambda I$) and $\lim L_{\mathrm{eff}}^H \in [1, \max\{d_\infty^w(\lambda), 2\}]$ hold along every limiting direction; and for $P = I$ the formulas for L_{eff} and r_{coco} hold with σ_d^w replaced by $v^\top\hat{\Sigma}_w v \in [\sigma_d^w, \sigma_1^w]$. The other entries, in particular the approximate values ($\approx$), assume the alignment, a heuristic borrowed from gradient descent. For SAGA, the (heuristic) criterion of Remark B2.22 of the B2 note, transferred by Proposition B1.5, expects alignment only when the parameter term is locally the active one, $9n\lambda_{\min}(M_P)/(16R_P^2) < 1$. On the instances of Figure B1 of the report this quantity is 0.98, 0.88, 3.4, 6.8 ($\alpha = 1$) and 0.57, 0.23, 5.5, 13.6 ($\alpha = 2$) for I, P_{J}, $\hat{\Sigma} + \lambda I$, H (with R_P^2 the upper bound of (B1.9) used in the experiments): the dense metrics are in the memory-bottleneck regime of (P3), where the error stays spread (over passes 15–20, $\mu_{\mathrm{eff}}^P/\lambda_{\min}(M_P) \approx 2.0$ and 3.3 for $\hat{\Sigma} + \lambda I$). Even for $P = I$ alignment is not observed at $\kappa = 10^3$: in the B2 runs ($P = I$, same problems, one seed) the final error has 0.31 ($\alpha = 1$) and 0.0025 ($\alpha = 2$) of its energy on the bottom eigenvector of H (participation ratios 3.8 and 12), the bottom of the spectrum of H being nearly degenerate (5 and 29 eigenvalues within 10% of $\lambda_{\min}(H)$); alignment occurs there only at $\kappa = 10^4$ with $\alpha \in \{0, 1\}$. For $P = I$ and $\alpha = 2$,

for instance, the limit of r_{coco} along the bottom eigenvector is 0.0028, against 0.013 measured over passes 15–20.

	$P = I$	$P = P_{\text{J}} = \text{diag}(\hat{\Sigma}) + \lambda I$	$P = \hat{\Sigma} + \lambda I$	$P = H = F''(\eta_*)$
R_P^2 (certified)	$\frac{1}{4}\max_i \|x_i\|^2 + \lambda$	$\frac{1}{4}\max_i \|x_i\|^2_{P_{\text{J}}^{-1}} + \frac{\lambda}{\min_j \hat{\Sigma}_{jj} + \lambda}$	$\frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d + \lambda}$ $\le \frac{C d_{\text{eff}}(\lambda)}{4} + 1$	$\max_i \frac{n \operatorname{lev}_i^w(\lambda)}{4 w_i} + \frac{\lambda}{\sigma_d^w + \lambda}$
μ_P (global, exact)	λ	$\frac{\lambda}{\max_j \hat{\Sigma}_{jj} + \lambda}$	$\frac{\lambda}{\sigma_1 + \lambda}$	$\frac{\lambda}{\sigma_1^w + \lambda}$
κ_P (global)	κ	$\ge \kappa$	$\in [\kappa,\ \kappa\,\text{cond}(P)]$	$\ge \kappa$
certified ρ_P	$1 - \min\{\frac{1}{3n}, \frac{3}{16\kappa}\}$	$\ge \rho_I$: not better than $P = I$	$\ge \rho_I$: not better than $P = I$	$\ge \rho_I$: not better than $P = I$
cost per pass	$O(nd)$	$O(nd)$	$O(nd^2)$ $+O(nd^2 + d^3)$ once	$O(nd^2)$ $+O(nd^2 + d^3)$ once, needs η_*

Table 3: Certified (global) constants for ℓ_2-regularized logistic regression.

late phase	$P = I$	$P = P_{\text{J}}$	$P = \hat{\Sigma} + \lambda I$	$P = H = F''(\eta_*)$
$\mu_{\text{eff}}^P \to$	$\lambda_{\min}(H) = \sigma_d^w + \lambda$	$\lambda_{\min}(M_{P_{\text{J}}})$	$\lambda_{\min}(M_P) \approx \bar{w}$, $\in \left[\frac{w_{\min}\sigma_1 + \lambda}{\sigma_1 + \lambda}, 1\right]$	1
$L_{\text{eff}}^P \to$	$\frac{\lambda^2 + 2\lambda\sigma_d^w + B_v}{\lambda + \sigma_d^w}$ $\in [\sigma_d^w + \lambda,\ \lambda + (\kappa+1)\sigma_d^w]$	same with $\|\cdot\|_{P_{\text{J}}^{-1}}$	$\le \frac{d_\infty(\lambda)}{4} + \frac{3}{2}$, $\approx \bar{w}\,\overline{n \operatorname{lev}}$	$\in [1, \max\{d_\infty^w(\lambda), 2\}]$
$r_{\text{sc}} \to$	$\frac{\lambda}{\sigma_d^w + \lambda}$ (≈ 1 if $\sigma_d^w \ll \lambda$)	$\mu_{P_{\text{J}}}/\lambda_{\min}(M_{P_{\text{J}}})$	$\approx \frac{\lambda}{(\sigma_1+\lambda)\bar{w}} \ll 1$	$\frac{\lambda}{\sigma_1^w + \lambda} \ll 1$
$r_{\text{coco}} \to$	$\frac{\lambda^2 + 2\lambda\sigma_d^w + B_v}{(\lambda + \sigma_d^w)R^2}$ $\in \left[\frac{1}{\kappa},\ \frac{1}{\kappa} + \frac{2\sigma_d^w}{\lambda}\right]$ ($\approx \frac{1}{\kappa}$ if $\sigma_d^w \ll \lambda/\kappa$)	no prediction	$\approx 4\bar{w}\,\overline{n \operatorname{lev}}/d_\infty(\lambda) \le 1$	$\frac{1 + \text{heterogeneity}}{R_H^2}$
loosest step	(I2) co-coercivity	(I2) and (I4)	(I4) strong convexity	(I4) strong convexity
$R_P^2/\mu_{\text{eff}}^P \to$	$\frac{\max_i \|x_i\|^2/4 + \lambda}{\sigma_d^w + \lambda}$	within a factor $\text{cond}(C_{\text{corr}})$ of the $\hat{\Sigma} + \lambda I$ value (either direction)	$\lesssim \frac{C d_{\text{eff}}(\lambda)/4 + 1}{\bar{w}}$	$R_H^2 \le \frac{C_w d_{\text{eff}}^w(\lambda)}{4 w_{\min}} + 1$

Table 4: Predicted late-phase behaviour ($M_P = P^{-1/2} F''(\eta_*) P^{-1/2}$, v the bottom eigenvector of M_P, taken heuristically as the late-phase error direction (see the caveat before Table 3); the μ_{eff}^P row is also a lower bound for the lim inf, and the r_{sc} and R_P^2/μ_{eff}^P rows upper bounds for the lim sup, whatever the direction of the error; for $P = I$, $v^\top \hat{\Sigma}_w v = \sigma_d^w$ and $B_v = \frac{1}{n}\sum_i w_i^2 (x_i^\top v)^2 \|x_i\|^2 \le (\kappa - 1)\lambda\sigma_d^w$ as in (B1.46); "heterogeneity" is the second term of (B1.45)).

In words, with the statements they follow from:

(P1) *Certified rates.* With global constants, none of the three preconditioners improves ρ_P for logistic regression ($\rho_P \ge \rho_I$ since $\kappa_P \ge \kappa$, Proposition B1.21); any observed speed-up must show up in the effective constants, not in ρ_P. For the square loss the statistical preconditioner does improve ρ_P, to at most $1 - \min\{\frac{1}{3n}, \frac{3}{16(d_\infty(\lambda)+1)}\}$.

(P2) *Looseness moves from (I2) to (I4).* Under $P = I$ the late-phase error lives in the low-curvature

directions of H, where (I4) is nearly tight ($r_{\mathrm{sc}} \to \lambda/\lambda_{\min}(H) = \lambda/(\sigma_d^w + \lambda)$ along the bottom eigenvector of H, and $\limsup r_{\mathrm{sc}} \le \lambda/\lambda_{\min}(H)$ in general; close to 1 when $\sigma_d^w \ll \lambda$) and (I2) is loose: by (B1.46) with v the bottom eigenvector of H (so $v^\top \hat{\Sigma}_w v = \sigma_d^w$),

$$r_{\mathrm{coco}} \;\to\; \frac{\lambda^2 + 2\lambda\sigma_d^w + B_v}{(\lambda + \sigma_d^w)R^2} \;\in\; \Big[\frac{\sigma_d^w + \lambda}{R^2},\; \frac{1}{\kappa} + \Big(1 + \frac{1}{\kappa}\Big)\frac{\sigma_d^w}{\lambda}\Big],$$

which is $\approx \lambda/R^2 = 1/\kappa$ when $\sigma_d^w \ll \lambda$ *and* $B_v \ll \lambda^2$ (sufficient: $\sigma_d^w \ll \lambda/\kappa$, since $B_v \le (\kappa - 1)\lambda\sigma_d^w$), and in any case $\ll 1$ as soon as $\sigma_d^w \ll \lambda$. The condition $\sigma_d^w \ll \lambda$ alone does not give $L_{\mathrm{eff}} \approx \lambda$: it kills the term $2\lambda\sigma_d^w$ but not B_v, which may be as large as $(\kappa - 1)\lambda\sigma_d^w \gg \lambda^2$ on ill-conditioned logistic instances ($\kappa \gg n$); the experiments should therefore record B_v (or directly the limit value) rather than test $r_{\mathrm{coco}} \approx 1/\kappa$. Under $P = \hat{\Sigma} + \lambda I$ and $P = H$ the bottom eigenspace of M_P is a high-variance data direction with relative curvature $\approx \bar{w}$ (resp. 1), so (I4) with the global μ_P becomes the loosest step ($r_{\mathrm{sc}} \approx \lambda/((\sigma_1 + \lambda)\bar{w})$, resp. $\lambda/(\sigma_1^w + \lambda)$), while r_{coco} rises to $\approx 4\bar{w}\,\overline{n\,\mathrm{lev}}/d_\infty(\lambda)$ (Lemma B1.27): it moves closer to 1 when the per-sample curvature is homogeneous and close to its maximum $\frac{1}{4}$ (data near the decision boundary) *and* the leverage scores are homogeneous ($\overline{n\,\mathrm{lev}} \approx d_{\mathrm{eff}} \approx d_\infty$); it remains bounded by $4\bar{w} \le 1$.

(P3) *Effective rate.* By (B1.28) the parameter part contracts at $1 - (1 - \frac{5}{8}r_{\mathrm{coco}})\mu_{\mathrm{eff}}^P/(2R_P^2)$; the predicted late-phase effective condition numbers $R_P^2/\mu_{\mathrm{eff}}^P$ are the last row of Table 4: equal to $(\max_i \|x_i\|^2/4 + \lambda)/(\sigma_d^w + \lambda) = R^2/\lambda_{\min}(F''(\eta_*)) \;\ge\; (\mathrm{tr}\,\hat{\Sigma}/4 + \lambda)/(\sigma_d^w + \lambda)$ for $P = I$ (the anisotropy of $\hat{\Sigma}$), $\lesssim (Cd_{\mathrm{eff}}/4 + 1)/\bar{w}$ for the statistical metric (no anisotropy), $\to R_H^2$ for the Newton metric; for the Jacobi metric Proposition B1.19(iii) only gives $\mathrm{cond}(C_{\mathrm{corr}})^{-1} \le \kappa_{\mathrm{eff}}^{P_{\mathrm{J}}}/\kappa_{\mathrm{eff}}^{\hat{\Sigma}+\lambda I} \le \mathrm{cond}(C_{\mathrm{corr}})$, hence no ordering with respect to $P = I$ (with randomly rotated features it may be worse than $P = I$). These values are *local*: they rest on the weighted average $\bar{w}$ of the curvatures at η_* (Lemma B1.27), whereas the rigorous trajectory bound (B1.37) keeps the smallest per-sample curvature, $m(\theta) \to \min_i w_i$, and need not improve on $P = I$. On the instances of Figure B1 it does not (comment after Corollary B1.23): any predicted gain of $\hat{\Sigma} + \lambda I$ over $P = I$ rests on the local constants alone, and their use as a rate over a finite run is heuristic. On ill-conditioned logistic instances ($\kappa \gg n$) with incoherent leverage scores ($d_\infty(\lambda) \ll n$), preconditioning should thus bring the *effective* (local) regime back to $\kappa_{\mathrm{eff}} \lesssim \frac{9n}{16}$, where the memory term $\frac{1}{3n}$ is the bottleneck (Remark B1.24). This does not extend to the block-separable quadratic instances of (A4), on which each f_i depends only on its own block of d/n coordinates: there each f_i'' has rank d/n, so that $\lambda_{\min}(P^{-1/2}F''P^{-1/2}) \le \mathrm{tr}(P^{-1/2}F''P^{-1/2})/d \le R_P^2/n$, i.e. $\kappa_P \ge n > \frac{9n}{16}$ for every P, with equality for $P = F''$ ($R_P^2 = n$, $\mu_P = 1$): the parameter term still binds.

(P4) *The other ratios.* When $\frac{3}{16\kappa_P} < \frac{1}{3n}$ (for logistic regression as soon as $\kappa > \frac{9n}{16}$, by Proposition B1.21; on the block-separable instances of (A4) for every P, by (P3)), $r_{\min} = s + (1 - s)\frac{1 - 1/(3n)}{1 - 3/(16\kappa_P)}$ exactly, with $s := \|\delta\|_P^2/V_{t-1}^P$ the parameter share of V^P: $r_{\min}$ always lies within $\frac{1}{3n}$ of 1, tends to 1 iff V^P is dominated by the parameter term, and the min step costs $-n\log r_{\min} \approx (1 - s)\big(\frac{1}{3} - \frac{3n}{16\kappa_P}\big)$ in the exponent per pass. If the memory errors $z_i^{(t-1)} - f_i'(\eta_*) \approx H_i\delta'$ were built from a single past error δ' aligned with δ, one would get $r_{\mathrm{var}} \approx 1 - \mu_{\mathrm{eff}}^P/L_{\mathrm{eff}}^P$ (close to 1 when the effective condition number is large, hence smaller under preconditioning) and a negative correlation of $a_{t,i}$ and $b_{t,i}$, i.e. $r_{\mathrm{sq}} \le \frac{1}{2}$, since $r_{\mathrm{sq}} = \frac{1}{2} + \sum_i a_{t,i}^\top P^{-1} b_{t,i} / \sum_i \big(\|a_{t,i}\|_{P^{-1}}^2 + \|b_{t,i}\|_{P^{-1}}^2\big)$. Exactly, $r_{\mathrm{init}} = r_{\mathrm{coco}}(1)\,\mu_{\mathrm{eff}}^P(\theta_0)/R_P^2$ by (B1.26), which is expected to increase under preconditioning.

(P5) *Check of the transfer.* Running Euclidean SAGA on $(\tilde{f}_i)$ with the same seed must reproduce P-SAGA exactly, up to rounding (Proposition B1.5(iii)); this is a sanity check of the implementation, and the six ratios computed either way must coincide.

Outcome of the experiments. In the experiment of Figure B1 of the report ($n = 1000$, $d = 50$, $\kappa = 10^3$, Gaussian features with covariance spectrum $j^{-\alpha}$, i.e. covariance condition number d^α, for $\alpha \in \{1, 2\}$; this α is not the decrement $\frac{3\mu_P}{16R_P^2}$ of (B1.23); 5 sampling seeds, $\gamma = 1/(4R_P^2)$ with R_P^2 the upper bound in (B1.9)), the contraction exponents per pass of $\|\theta_t - \eta_*\|_P^2$ (least-squares slopes over passes 2–20, seed-averaged) for $P = I$, P_{J}, $\hat{\Sigma} + \lambda I$, $F''(\eta_*)$ are 1.000, 0.9414, 1.114, 1.106 ($\alpha = 1$) and 0.6104, 0.2687, 1.257, 1.242 ($\alpha = 2$). As (P1) predicts, the global constants get the ordering wrong: $\kappa_P = 1000$, 1899, 5083, 3623 and 1000, 2852, 3639, 2089, computed like the step with the upper bound of (B1.9) (the exact values of (B1.6) are within 5.2%: 1000, 1898, 5057, 3611 and 1000, 2841, 3459, 2051; all $\geq \kappa$), so $P = I$ gets the largest certified exponent (0.1875).

The local constants give the observed ordering (up to the near tie of the dense metrics): $\lambda_{\max}(M_P)/\lambda_{\min}(M_P) = 25.17$, 25.67, 2.801, 1 and 49.5, 116.2, 8.42, 1, and the bound with $\lambda_{\min}(M_P)$ in place of μ_P (heuristic: Lemma B1.27 supports it only as $\theta \to \eta_*$) gives 0.3256, 0.2924, 0.3333, 0.3333 and 0.1912, 0.07705, 0.3333, 0.3333. Corollary B1.23 does not account for this ordering: its bound for $\hat{\Sigma} + \lambda I$ exceeds κ over the whole fitting window and tends to a value above its bound for $P = I$ (comment after the corollary). What accounts for it is Lemma B1.27, rigorous as $\theta \to \eta_*$ and used heuristically as a rate over passes 2–20: along every direction, $\mu_{\mathrm{eff}}^P(\theta)$ tends to a value $\geq \lambda_{\min}(M_P)$. For $\hat{\Sigma} + \lambda I$, $\lambda_{\min}(M_P) = 0.117$ and 0.109 is the smallest over directions of a weighted average of $w_1, \ldots, w_n, 1$, not their minimum: it is 16 and 18 times $\min_i w_i$, so that $R_P^2/\lambda_{\min}(M_P) = 166$ and 103, against $R^2/\lambda_{\min}(F''(\eta_*)) = 576$ and 981 for $P = I$.

The dense metrics saturate the memory term $\frac{1}{3}$, as in (P3); their observed exponents are 3.3–3.8 times larger. Along the measured trajectories, the exact one-step contraction of Proposition B3.10 of the B3 note, transferred by Proposition B1.5 and evaluated at the instrumented states of passes 2–20, splits this factor as follows when the inequalities are undone in order A of Proposition B3.11 ((I4), (I2), (I3), (I1), then the min step (I5); logarithms of mean exponents, all seeds). The memory coefficient, $1 - \frac{1}{n}(1 - \frac{2}{3} r_{\mathrm{sq}} r_{\mathrm{var}}) \approx 1 - \frac{0.65}{n}$ here instead of $1 - \frac{1}{3n}$ (through $r_{\mathrm{sq}} \approx 0.51$–$0.53$, i.e. the square-norm step (I1), since $r_{\mathrm{var}} \approx 1$), accounts for about 55% ($\alpha = 1$) and 52% ($\alpha = 2$) of the logarithm, a factor 1.95–1.99. The min step (I5), which replaces by the slower memory rate a weighted mean in which the parameter term (median share 2–7% of V^P) contracts 7–21 times faster, accounts for about 36% and 39–40%, a factor 1.55–1.70. The remaining 8–9%, a factor 1.10–1.12, separates the measured slopes from the exact one-step contraction (mean exponents 1.01 and 1.11–1.12 per pass) and is pathwise: by Jensen's inequality, $\mathbb{E}_{i(t)}[\log(V_t^P/V_{t-1}^P)] \leq \log(\mathbb{E}_{i(t)}[V_t^P]/V_{t-1}^P)$. Relative to the exact one-step contraction, the memory coefficient and (I5) thus share the gap about 60/40 ($\alpha = 1$) and 57/43 ($\alpha = 2$). None of this is a statement about $\mathbb{E}V_t^P$: the identity of Step 2 gives $\mathbb{E}\sum_i \left\| z_i^{(t)} - f_i'(\eta_*) \right\|_{P^{-1}}^2 \geq (1 - \frac{1}{n})^t \sum_i \left\| z_i^{(0)} - f_i'(\eta_*) \right\|_{P^{-1}}^2 > 0$, the contribution of the slots never refreshed (an event of probability $(1 - \frac{1}{n})^t \approx e^{-t/n}$ per slot; in the five runs every slot has been refreshed after 5.7 to 9.1 passes), so that no bound of the form $\mathbb{E}V_t^P \leq C\rho^t V_0^P$ holds with $\rho < 1 - \frac{1}{n}$. Such a bound certifies at most $n \log \frac{n}{n-1} \approx 1.0005$ per pass, 3 times the local bound, whereas along the measured trajectories the memory error itself decays with exponents 1.12–1.25 per pass, faster than its expectation can (the exact one-step contraction is faster than $1 - \frac{1}{n}$ at the typical states, while $\mathbb{E}V_t^P$ is carried by the rare states with stale slots): a factor 1.10–1.26 of the observed exponents is invisible in the expectation $\mathbb{E}V_t^P$ that the analysis controls.

The Jacobi metric is the slowest, which (P3) leaves open and Proposition B1.19 allows: here

$\mathrm{cond}(C_{\mathrm{corr}}) = 52$ ($\alpha = 1$) and 2866 ($\alpha = 2$), and $\lambda_{\max}(M_{P_{\mathrm{J}}})/\lambda_{\min}(M_{P_{\mathrm{J}}})$ is 9.2 and 13.8 times that of $\hat{\Sigma} + \lambda I$. As in (P2), r_{coco} rises from 0.1289 ($P = I$) to 0.3288 and 0.3377, and r_{sc} falls from 0.4233 to 0.01692 and 0.02288 ($\alpha = 1$, geometric means over passes 2–20).

The heuristics of (P4) fail for the dense metrics. r_{var} *increases* under preconditioning (geometric means 0.9936, 0.9956, 0.9990, 0.9990 for $\alpha = 1$ and 0.958, 0.970, 0.9991, 0.9984 for $\alpha = 2$): $1 - r_{\mathrm{var}}$ is 36–173 times smaller than $\mu^P_{\mathrm{eff}}/L^P_{\mathrm{eff}}$ for the dense metrics ($(0.9\text{–}1.6) \cdot 10^{-3}$ against 0.035–0.036 for $\alpha = 1$ and 0.14–0.16 for $\alpha = 2$; geometric means), and within a factor 3.5 of it for $P = I$ and P_{J}. r_{sq} exceeds $\frac{1}{2}$ (positive correlation of $a_{t,i}$ and $b_{t,i}$) in 86–93% of the instrumented states for $\hat{\Sigma} + \lambda I$ and $F''(\eta_*)$ (geometric means 0.525, 0.526 and 0.507, 0.509). Both failures occur in the memory-bottleneck regime, where the stored gradients were computed at past iterates much farther from η_* than θ_{t-1}, contrary to the single aligned past error assumed in (P4). r_{init} increases as predicted ($8.5 \cdot 10^{-4}$, $1.2 \cdot 10^{-3}$, $3.7 \cdot 10^{-3}$, $7.3 \cdot 10^{-3}$ for $\alpha = 1$), and $r_{\min}$, an exact function of the parameter share s of V^P here (median $s = 0.64$, 0.50, 0.066, 0.063 for $\alpha = 1$), shows that with global constants the min step costs 0.23–0.27 per pass for the memory-dominated dense metrics. (P5) was not run as a separate experiment; the instrumentation in the P-metric is checked instead against an exact enumeration of the n possible steps, for a dense and a diagonal P (`tests/test_vrlab.py`).

A1 Lower bounds for finite-sum optimization and the hard instance (hypotheses A1, A4)

Throughout, as in [11], Section 5.4.4, we consider

$$F(\theta) = \frac{1}{n}\sum_{i=1}^{n} f_i(\theta), \qquad \theta \in \mathbb{R}^d,$$

where every f_i is convex and R^2-smooth (its gradient $f_i' = \nabla f_i$ is R^2-Lipschitz), F is μ-strongly convex with minimizer η_*, and $\kappa = R^2/\mu$. We write $F' = \nabla F$ and L_F for the smoothness constant of F; since $L_F \le R^2$ one always has $L_F/\mu \le \kappa$, and the gap between these two numbers is the main theme of this note. The cost of a method is the number of individual gradients $f_i'(\theta)$ it evaluates (one per SAGA iteration, n per full-gradient step).

The note has four parts. Section A1.1 fixes the oracle models and the span assumption and gives Nesterov's single-function construction with a complete proof. Section A1.2 quotes, with their exact hypotheses, the finite-sum lower bounds of Agarwal–Bottou [1], Lan–Zhou [50], Woodworth–Srebro [87], Arjevani–Shamir [5], Hannah–Liu–O'Connor–Yin [38] and Han–Xie–Zhang [37], and settles the meaning of hypothesis (A1). Section A1.3 builds the shared-regularizer block-separable hard instance used for (A4) in Figure A4b of the report, with every constant derived in the notation above, and proves the lower bound on it (the instances of Figure A4 of the report are instead the block-local normalization of [50] with Nesterov's chain $\zeta = 2$, discussed after Proposition A1.13, for which the per-block condition number is κ/n and $\kappa = R^2/\lambda_{\min}(\nabla^2 F)$ by construction, and block quadratics with random spectra; the results of Section A1.3, stated for $\zeta = \zeta_Q$ and the shared regularizer, do not apply verbatim to them). Section A1.4 derives the predictions for the numerics (exponents κ^1 versus $\kappa^{1/2}$, crossover, finite-dimension effects) and explains why logistic regression is not a hard instance.

A1.1 Oracle models, the span assumption, and Nesterov's chain

A1.1.1 Oracle models and classes of algorithms

Definition A1.1 (incremental first-order oracle)**.** *Following [1] (Definition 2), the* incremental first-order oracle *(IFO) takes as input a point $\theta \in \mathbb{R}^d$ and an index $i \in \{1,\dots,n\}$ and returns the pair $(f_i(\theta), f_i'(\theta))$. An IFO algorithm is an algorithm whose specification depends on F only through such calls; its cost is the number of calls. The* proximal *IFO (PIFO) of [87, 37] additionally returns $\operatorname{prox}_{f_i}(\theta,\beta) = \arg\min_u\{f_i(u) + \frac{\beta}{2}\|u-\theta\|^2\}$: for any $\beta > 0$ in [37], who write $\frac{1}{2\gamma}$, $\gamma > 0$, for $\frac{\beta}{2}$; in display (3) of [87] the* arg min *is over the domain $\mathcal{X}$ and $\beta = 0$ is allowed ("one prox oracle call (with $\beta = 0$) is sufficient for exact optimization" of a single function). The oracle of [87] is thus the stronger one, and lower bounds proved for it hold a fortiori for the other.*

A full gradient $F'(\theta) = \frac{1}{n}\sum_i f_i'(\theta)$ costs n IFO calls at the same point. The lower bounds below differ in *how the algorithm may choose the sequence of indices* and in *how it may combine the answers.*

Definition A1.2 (linear-span protocol and classes (S1)–(S3))**.** *An algorithm follows the* linear-span protocol *(Nesterov's "black-box" assumption, [65] Section 2.1, relation (3.41) of [50], relation (1.2) of [38]) if, when it has queried the oracle at the pairs (θ_{s-1}, i_s), $s = 1,\dots,t$, its iterate satisfies*

$$\theta_t \in \theta_0 + \operatorname{span}\big\{f_{i_1}'(\theta_0),\ f_{i_2}'(\theta_1),\ \dots,\ f_{i_t}'(\theta_{t-1})\big\}. \tag{A1.1}$$

We distinguish three classes according to the index sequence (i_s):

(S1) span-based with i.i.d. sampling*: (i_s) is i.i.d. from a fixed distribution p on $\{1,\dots,n\}$ (uniform in the project), independent of the oracle answers [50, 38]. This contains SGD, SAG, Finito/MISO, SDCA-without-duality, SAGA started from the zero memory $z^{(0)}=0$, and, on block-separable problems, randomized block-coordinate methods. With the book's initialization $z_i^{(0)}=f_i'(\theta_0)$, SAGA is an (S1) method* preceded by one full gradient at θ_0 *(n deterministic queries); Lemma A1.16(ii) and Corollary A1.21 treat this variant.*

(S2) span-based with an arbitrary index sequence*: (i_s) is any sequence, deterministic (cyclic, or n consecutive queries at the same point, which is a full gradient), random, or adaptive. This contains GD, Nesterov's accelerated gradient (AGD), cyclic incremental methods, and the hybrid methods SVRG, SARAH, L-SVRG, Katyusha and L-Katyusha, which mix i.i.d. single queries with full gradients.*

(S3) no span restriction*: arbitrary deterministic algorithms [1, 87] or arbitrary randomized algorithms [87], and the* oblivious canonical linear iterative *(CLI) class of [5], in which new iterates are linear combinations $\sum_j O_F(\theta^{(j)};\alpha^{(j)})$ of oracle outputs (gradients possibly multiplied by fixed matrices, coordinate steps, full gradients) whose coefficients $\alpha^{(j)}$ are chosen* without looking at the oracle answers *(they may be random). Fixed step sizes, fixed momentum and fixed sampling schemes are allowed, line searches and quasi-Newton updates are not.*

SAGA's update direction $f_{i(t)}'(\theta_{t-1})-z_{i(t)}^{(t-1)}+\frac{1}{n}\sum_j z_j^{(t-1)}$ is a linear combination of the current gradient and of the stored gradients $z_j^{(t-1)}$, each of which is either its initial value or a gradient $f_j'(\theta_s)$ computed at an earlier iterate. Hence (A1.1) holds when $z^{(0)}=0$, whereas with the book's initialization $z_i^{(0)}=f_i'(\theta_0)$ the span must be enlarged by the n initial gradients $f_1'(\theta_0),\dots,f_n'(\theta_0)$, which are not sampled i.i.d.: this is the only point where SAGA, as analysed in Proposition 5.9 of [11], leaves class (S1). [37] make the same observation (SAGA and Point-SAGA "only calculate the full gradient at the first iteration") and let the full gradient at θ_0 be free in their class. L-Katyusha [47] is *not* in (S1) because of its full gradients at the reference points w_k (each costs n queries at one point); it is in (S2), in the oblivious CLI class (its coin flips and parameters do not depend on the answers) and in the class of [37].

Dimension-free versus span-based bounds. A lower bound is *dimension-free* if it holds in every dimension d [5]. Span-based constructions are not: they reveal at most one new coordinate per query, so they can only certify as many iterations as there are unrevealed coordinates. This is Nesterov's caveat $t<d$ (in [11], Section 15.2.1, the nonsmooth lower bounds "are valid for $k<d$ only" and, for the smooth convex construction F_{2k+1}, "the number of iterations has to be less than half the dimension for the lower bound to hold"; the strongly convex construction of that section lives in ℓ_2 and carries no such caveat), the condition $n\geq \underline{n}(m,k)$ in [50], and the condition $m\gtrsim\sqrt{\kappa/n}\log(1/\varepsilon)$ of [37]; in the words of [87], "if we do assume that the queries lie in that span, the necessary dimension is only on the order of the number of oracle queries needed". Removing the span assumption costs, for randomized algorithms, a much larger dimension (polynomial in $1/\varepsilon$ in [87]) or an obliviousness assumption [5]; for deterministic algorithms a resisting oracle needs only a dimension of the order of the number K of queries ($2K+2$ in [1], see also Theorem A1.9). Quantitatively, Section A1.3 shows that the block-separable instance certifies the bound at accuracy ε when each block has at least $\frac{\log(1/\varepsilon)}{2\log(1/q)}$ coordinates, where $q=\frac{\sqrt{Q}-1}{\sqrt{Q}+1}$ and $Q=1+\frac{\kappa-1}{n}$ (Corollary A1.20), a number between $\frac{1}{4}(\sqrt{Q}-1)\log\frac{1}{\varepsilon}$ and $\frac{1}{4}\sqrt{Q}\log\frac{1}{\varepsilon}$.

A1.1.2 The chain matrix

For $m \geq 1$ and $\zeta \in [1,3]$ let $A_m(\zeta) \in \mathbb{R}^{m\times m}$ be the symmetric tridiagonal matrix with entries 2 on the diagonal except $(A_m(\zeta))_{mm} = \zeta$, and -1 on the two off-diagonals:

$$A_m(\zeta) = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & \zeta \end{pmatrix}. \tag{A1.2}$$

The value $\zeta = 2$ is Nesterov's original finite-dimensional chain ([65], Section 2.1.2; [11], Section 15.2.1); the value $\zeta = \zeta_Q$ chosen in (A1.11) below makes the minimizer an exact geometric sequence in finite dimension [50].

Lemma A1.3 (quadratic form of the chain)**.** *For every $s \in \mathbb{R}^m$ and $\zeta \in [1,3]$,*

$$\langle A_m(\zeta)s, s\rangle = s_1^2 + \sum_{j=1}^{m-1}(s_j - s_{j+1})^2 + (\zeta - 1)s_m^2, \tag{A1.3}$$

so that $0 \preceq A_m(\zeta) \preceq 4I$; more precisely

$$\langle A_m(\zeta)s, s\rangle \leq 4\,\|s\|^2. \tag{A1.4}$$

Moreover, for $\zeta \in [1,2]$, $\lambda_{\min}(A_m(\zeta)) \in \left[\,4\sin^2\frac{\pi}{2(2m+1)},\ 4\sin^2\frac{\pi}{2(m+1)}\,\right]$, which is of order $1/m^2$.

Proof. Since

$$\sum_{j=1}^{m-1}(s_j - s_{j+1})^2 = s_1^2 + 2\sum_{j=2}^{m-1} s_j^2 + s_m^2 - 2\sum_{j=1}^{m-1} s_j s_{j+1},$$

the right-hand side of (A1.3) equals $2s_1^2 + 2\sum_{j=2}^{m-1} s_j^2 + \zeta s_m^2 - 2\sum_{j=1}^{m-1} s_j s_{j+1} = \langle A_m(\zeta)s, s\rangle$ (for $m = 1$ both sides equal ζs_1^2). Nonnegativity follows from $\zeta \geq 1$. For the upper bound, $(s_j - s_{j+1})^2 \leq 2s_j^2 + 2s_{j+1}^2$ gives $\langle A_m(\zeta)s, s\rangle \leq 3s_1^2 + 4\sum_{j=2}^{m-1} s_j^2 + (\zeta+1)s_m^2 \leq 4\,\|s\|^2$ because $\zeta \leq 3$. For the eigenvalue bounds: $A_m(\zeta) - A_m(1) = (\zeta - 1)e_m e_m^\top \succeq 0$ and $A_m(2) - A_m(\zeta) = (2-\zeta)e_m e_m^\top \succeq 0$, so $\lambda_{\min}(A_m(1)) \leq \lambda_{\min}(A_m(\zeta)) \leq \lambda_{\min}(A_m(2))$ by the Courant–Fischer theorem; the eigenvalues of $A_m(2)$ are $2 - 2\cos\frac{j\pi}{m+1} = 4\sin^2\frac{j\pi}{2(m+1)}$ and those of $A_m(1)$ are $2 - 2\cos\frac{(2j-1)\pi}{2m+1} = 4\sin^2\frac{(2j-1)\pi}{2(2m+1)}$, $j = 1,\dots,m$ (the vectors $(\sin\frac{jk\pi}{m+1})_k$, resp. $(\sin\frac{(2j-1)k\pi}{2m+1})_k$, are eigenvectors, as one checks row by row; the last row is where the two boundary values $\zeta = 2$, resp. $\zeta = 1$, enter). □

A1.1.3 Nesterov's construction for a single function

Fix $L > \mu > 0$, $Q = L/\mu$, and define on ℓ_2 (square-summable sequences) the function of [65], Section 2.1.4 (also [11], Section 15.2.1):

$$f(\theta) = \frac{L-\mu}{4}\Big\{\frac{1}{2}\Big[\theta_1^2 + \sum_{j\geq 1}(\theta_j - \theta_{j+1})^2\Big] - \theta_1\Big\} + \frac{\mu}{2}\,\|\theta\|^2. \tag{A1.5}$$

In the notation of (A1.3) the bracket is $\langle A\theta, \theta\rangle$ for the infinite tridiagonal matrix A with 2 on the diagonal and -1 off it, so $f(\theta) = \frac{L-\mu}{4}\{\frac{1}{2}\langle A\theta,\theta\rangle - \theta_1\} + \frac{\mu}{2}\|\theta\|^2$, $f'(\theta) = \frac{L-\mu}{4}(A\theta - e_1) + \mu\theta$, and $f'' = \frac{L-\mu}{4}A + \mu I$ has spectrum in $[\mu, \mu + (L-\mu)] = [\mu, L]$ by (A1.4): f is L-smooth and μ-strongly convex, with condition number exactly Q.

Proposition A1.4 (minimizer of the chain). *Let* $q = \dfrac{\sqrt{Q}-1}{\sqrt{Q}+1} \in (0,1)$. *The minimizer of* (A1.5) *is* $\theta_j^* = q^j$, $j \geq 1$, *with* $\|\theta^*\|^2 = \dfrac{q^2}{1-q^2}$ *and* $f(\theta^*) = -\dfrac{(L-\mu)q}{8}$.

Proof. The optimality condition $f'(\theta) = 0$ reads $(A + \frac{4}{Q-1}I)\theta = e_1$, i.e.

$$\Big(2 + \tfrac{4}{Q-1}\Big)\theta_1 - \theta_2 = 1, \qquad -\theta_{j-1} + \Big(2 + \tfrac{4}{Q-1}\Big)\theta_j - \theta_{j+1} = 0 \quad (j \geq 2).$$

Since $q + q^{-1} = \frac{(\sqrt{Q}-1)^2+(\sqrt{Q}+1)^2}{Q-1} = \frac{2(Q+1)}{Q-1} = 2 + \frac{4}{Q-1}$, the sequence $\theta_j = q^j$ satisfies the j-th equation, $q^{j-1}[-1 + (q+q^{-1})q - q^2] = 0$, and the first one, $(q+q^{-1})q - q^2 = 1$. It belongs to ℓ_2 and the strongly convex f has a unique minimizer, hence $\theta_j^* = q^j$. Then $\|\theta^*\|^2 = \sum_{j\geq 1} q^{2j} = q^2/(1-q^2)$ and, using $A\theta^* = e_1 - \frac{4}{Q-1}\theta^*$, $f(\theta^*) = \frac{L-\mu}{4}\{\frac{1}{2}\theta_1^* - \frac{2}{Q-1}\|\theta^*\|^2 - \theta_1^*\} + \frac{\mu}{2}\|\theta^*\|^2 = -\frac{(L-\mu)q}{8}$ because $\frac{L-\mu}{4} \cdot \frac{2}{Q-1} = \frac{\mu}{2}$. □

Lemma A1.5 (one coordinate per gradient call). *For* $\theta \in \ell_2$ *(or* $\mathbb{R}^m$*) let* $N(\theta) = \max\{j : \theta_j \neq 0\}$ *(*$N(0) = 0$*). Then* $N(f'(\theta)) \leq N(\theta) + 1$ *for the function* (A1.5)*, and the same holds for* $\theta \mapsto c(A_m(\zeta)\theta - e_1) + \mu\theta$ *on* $\mathbb{R}^m$ *for any* $c > 0$ *(with* $N \leq m$*).*

Proof. $(A\theta)_j = -\theta_{j-1} + 2\theta_j - \theta_{j+1}$ (with the obvious boundary modifications) vanishes as soon as $j - 1 > N(\theta)$, so $N(A\theta) \leq N(\theta) + 1$; $-e_1$ only touches coordinate 1, and $\mu\theta$ does not enlarge the support. □

Theorem A1.6 (Nesterov's chain lower bound; Theorem 2.1.13 of [65]). *Let* $\theta_0 = 0$ *and let* (θ_t) *be produced by any method following the span protocol* (A1.1) *for the single function* (A1.5) *(here all* $i_s = 1$*). Then for every* $t \geq 0$

$$\|\theta_t - \theta^*\|^2 \;\geq\; q^{2t}\,\|\theta_0 - \theta^*\|^2\,, \qquad f(\theta_t) - f(\theta^*) \;\geq\; \frac{\mu}{2}\,q^{2t}\,\|\theta_0 - \theta^*\|^2\,. \tag{A1.6}$$

Consequently, reaching $\|\theta_t - \theta^*\|^2 \leq \varepsilon\,\|\theta_0 - \theta^*\|^2$ *requires* $t \geq \dfrac{\log(1/\varepsilon)}{2\log(1/q)} \geq \dfrac{\sqrt{Q}-1}{4}\log\dfrac{1}{\varepsilon}$ *gradient evaluations, i.e.* $\Omega(\sqrt{Q}\log(1/\varepsilon))$*, which is the rate* $(1-\sqrt{\mu/L})^t$ *of accelerated gradient descent ([11], Section 5.2.5) up to constants.*

Proof. By induction, $N(\theta_t) \leq t$: this is true for $t = 0$, and if $N(\theta_{s-1}) \leq s-1$ for all $s \leq t$ then each $f'(\theta_{s-1})$ in (A1.1) has $N \leq s \leq t$ by Lemma A1.5, so $N(\theta_t) \leq t$. Hence $\|\theta_t - \theta^*\|^2 \geq \sum_{j>t}(\theta_j^*)^2 = \sum_{j>t} q^{2j} = q^{2t}\frac{q^2}{1-q^2} = q^{2t}\,\|\theta^*\|^2 = q^{2t}\,\|\theta_0 - \theta^*\|^2$. The function-value bound follows from μ-strong convexity, $f(\theta) - f(\theta^*) \geq \frac{\mu}{2}\|\theta - \theta^*\|^2$ (since $f'(\theta^*) = 0$). For the iteration count, $q^{2t} \leq \varepsilon$ means $t \geq \log(1/\varepsilon)/(2\log(1/q))$ and $\log(1/q) = \log\big(1 + \frac{2}{\sqrt{Q}-1}\big) \leq \frac{2}{\sqrt{Q}-1}$. □

The mechanism is the whole story of this note: a gradient of the chain reveals one new coordinate, the minimizer has geometrically decaying coordinates with ratio $q \approx 1 - 2/\sqrt{Q}$, and the energy in the unrevealed coordinates is q^{2t} of the total. The finite-sum constructions below simply run n chains in parallel, one per component, so that a query to component i reveals a coordinate of chain i only.

A1.2 Precise statements of the finite-sum lower bounds

We quote the theorems in the notation of their authors, then translate them into the book's notation (n components, per-component smoothness R^2, strong convexity μ of the average, $\kappa = R^2/\mu$). All sources were downloaded and read (arXiv 1410.0723, 1507.02000, 1605.08003, 1606.09333, 1805.07786, 2103.08280, and 1901.08689 for L-Katyusha); the arXiv identifier 1809.09608 given in the project description for Hannah et al. is wrong (it is an unrelated logic paper): the correct one is 1805.07786, and the paper has four authors (Hannah, Liu, O'Connor, Yin).

Deterministic IFO algorithms (Agarwal–Bottou). In [1] the problem is $f(x) = \frac{\mu}{2}\|x\|^2 + \frac{1}{n}\sum_{i=1}^n g_i(x)$ with each g_i convex and $(L-\mu)$-smooth on ℓ_2 (class $\mathcal{F}_n^{\mu,L}$), $\kappa = L/\mu$; so each component $\frac{\mu}{2}\|x\|^2 + g_i$ is L-smooth and f is μ-strongly convex: in our notation $R^2 = L$ and their κ is ours. The algorithms are deterministic IFO algorithms (Definition A1.1); the span assumption is *not* made: their Appendix A removes it in two steps, first by showing that for the worst case one may assume that the algorithm's final point lies in the span of its previous queries and of the oracle's answers (their Section A.1), then by a resisting oracle which builds, call after call, orthonormal families S_k of dimension at most $2k+2$ (Gram–Schmidt, their Definition 11) and applies Nesterov's function to the coordinates in these bases, so that the final point is confined to a $(2K+2)$-dimensional subspace while the minimizer of the constructed function stays far from it (their Sections A.2–A.3).

Theorem A1.7 (Agarwal–Bottou lower bound; Theorem 1 and Corollary 1 of [1])**.** *Consider a deterministic IFO algorithm that makes $K \geq 0$ calls and outputs x_K. For any $\varepsilon > 0$ there is $f \in \mathcal{F}_n^{\mu,L}(\ell_2)$ with $\left\|x_f^*\right\| = \varepsilon$ and*

$$\left\|x_f^* - x_K\right\| \;\geq\; \varepsilon\, q^{2t}, \qquad q = \frac{\sqrt{1+\frac{\kappa-1}{n}}-1}{\sqrt{1+\frac{\kappa-1}{n}}+1}, \qquad t = \begin{cases} 0 & \text{if } K < n, \\ K/n & \text{otherwise.} \end{cases}$$

Hence an algorithm guaranteeing $\left\|x_f^* - x_K\right\| \leq \varepsilon \left\|x_f^*\right\|$ for all f in the class, with $\varepsilon < 1$, must perform $K = \Omega\big(n + \sqrt{n(\kappa-1)}\log(1/\varepsilon)\big)$ IFO calls.

The number $1+\frac{\kappa-1}{n}$ is the condition number of f restricted to one of n disjoint coordinate blocks (their construction is block separable with the regularizer shared by all components, exactly as ours in Section A1.3); the bound is attained for $\kappa \gtrsim n$, on their own instance, by accelerated gradient descent, and [87] show that it is loose for deterministic algorithms in general (Theorem A1.9 below).

Span-based randomized incremental methods (Lan–Zhou). In [50], Section 3.3 (displays and constants are quoted from arXiv:1507.02000v3, which writes the objective as a sum, with $L = \sum_i L_i$; the journal version writes it as an average of components m times larger, with $L = \frac{1}{m}\sum_i L_i$, so that the function, $\mathcal{C}$ and the bounds are the same), the hard problem is

$$\min_{x_i \in \mathbb{R}^{\tilde{n}},\, i=1,\dots,m} \Psi(x) := \sum_{i=1}^m \Big[f_i(x_i) + \frac{\mu}{2}\|x_i\|^2 \Big], \qquad f_i(x_i) = \frac{\mu(\mathcal{Q}-1)}{4}\Big[\frac{1}{2}\langle A x_i, x_i\rangle - \langle e_1, x_i\rangle\Big], \tag{A1.7}$$

with $A = A_{\tilde{n}}(\zeta)$ as in (A1.2) and $\zeta = \frac{\sqrt{\mathcal{Q}}+3}{\sqrt{\mathcal{Q}}+1}$ (their display (3.39), where the boundary entry is called κ); each f_i is convex with Lipschitz constant $L_i = \mu(\mathcal{Q}-1)$. The class is: i_t i.i.d. with $\mathbb{P}\{i_t = i\} = p_i$ for a fixed distribution p (their (3.40)), and $x^k \in x^0 + \mathrm{Lin}\{\nabla f_{i_1}(x^0), \dots, \nabla f_{i_k}(x^{k-1})\}$ (their (3.41)), i.e. class (S1).

Theorem A1.8 (Lan–Zhou lower bound; Theorem 3 and Corollary 3 of [50]). *Let $\mathcal{Q}$ be the parameter of the instance (A1.7) and $q = \frac{\sqrt{\mathcal{Q}}-1}{\sqrt{\mathcal{Q}}+1}$. The iterates of any randomized incremental gradient method of class (S1) (Definition A1.2) applied to (A1.7) satisfy*

$$\frac{\mathbb{E}\left\|x^k - x^*\right\|^2}{\left\|x^0 - x^*\right\|^2} \;\geq\; \frac{1}{2}\exp\Big(-\frac{4k\sqrt{\mathcal{Q}}}{m(\sqrt{\mathcal{Q}}+1)^2 - 4\sqrt{\mathcal{Q}}}\Big) \qquad \text{for all } \; n \geq \underline{n}(m,k),$$

$$\underline{n}(m,k) := \frac{m\log\left[(1-(1-q^2)/m)^k/2\right]}{2\log q},$$

where $n = m\tilde{n}$ is the total dimension. Consequently the number of gradient evaluations needed by such a method to find $\bar{x}$ with $\mathbb{E}\left\|\bar{x} - x^*\right\|^2 \leq \varepsilon$ cannot be smaller than

$$\Omega\Big\{\big(m + \sqrt{m\,\mathcal{C}}\big)\log\frac{\left\|x^0 - x^*\right\|^2}{\varepsilon}\Big\}, \qquad \mathcal{C} = \frac{L}{\mu},\; L = \sum_{i=1}^m L_i,$$

if n is sufficiently large; precisely (their display (3.45) in arXiv:1507.02000v3) $k \geq \big[\frac{m}{2}\big(\frac{\sqrt{\mathcal{Q}}}{2}+1\big) - 1\big]\log\frac{\left\|x^0-x^*\right\|^2}{2\varepsilon}$.

Translation. Write $\Psi = \frac{1}{m}\sum_i g_i$ with $g_i(x) = m[f_i(x_i) + \frac{\mu}{2}\left\|x_i\right\|^2]$. Then g_i is $m\mu\mathcal{Q}$-smooth (Lemma A1.3), Ψ is μ-strongly convex, so in the book's notation $R^2 = m\mu\mathcal{Q}$, $n_{\text{book}} = m$, $\kappa = R^2/\mu = m\mathcal{Q}$, and $m + \sqrt{m\mathcal{C}} = m + m\sqrt{\mathcal{Q}-1} \asymp n_{\text{book}} + \sqrt{n_{\text{book}}\,\kappa}$. Theorem A1.8 is therefore the bound

$$\Omega\big((n + \sqrt{n\kappa})\log(1/\varepsilon)\big) \qquad \text{for class (S1)},$$

which is the first half of hypothesis (A1). Note that the proof minimizes the bound over the sampling distribution p and that the minimum is the uniform distribution: the bound holds for every fixed p. Note also that on these block-separable instances $\kappa = m\mathcal{Q} \geq m$ always (the per-block condition number $\mathcal{Q} = \kappa/n_{\text{book}}$ is what the chain mechanism sees), so the regime $\kappa \ll n$ cannot be probed with (A1.7); our instance in Section A1.3 shares the regularizer, as in [1], which gives $\mathcal{Q} = 1 + (\kappa-1)/n$ and allows every $\kappa \geq 1$.

Arbitrary deterministic and randomized algorithms (Woodworth–Srebro). In [87] the components are γ-smooth and F is λ-strongly convex ($R^2 = \gamma$, $\mu = \lambda$, $m = n_{\text{book}}$ components), the oracle is the gradient-and-prox oracle h_F, and no span assumption is made.

Theorem A1.9 (Woodworth–Srebro lower bounds; Theorems 4 and 8 of [87]). (Deterministic.) *For any $\gamma, \lambda > 0$ with $\gamma/\lambda > 73$, any $\varepsilon > 0$, any $\varepsilon_0 > 3\gamma\varepsilon/\lambda$, any $m \geq 2$ and any deterministic algorithm A with access to the gradient-and-prox oracle h_F (Definition A1.1), there exist a dimension $d = O\big(m\sqrt{\gamma/\lambda}\log\frac{\lambda\varepsilon_0}{\gamma\varepsilon}\big)$ and m functions f_i on $\mathcal{X} \subseteq \mathbb{R}^d$, γ-smooth and λ-strongly convex with $F(0) - F(x^*) = \varepsilon_0$, such that finding $\hat{x}$ with $F(\hat{x}) - F(x^*) < \varepsilon$ requires $\Omega\big(m\sqrt{\gamma/\lambda}\log\frac{\lambda\varepsilon_0}{\gamma\varepsilon}\big)$ queries.*

(Randomized.) *For any $m \geq 2$, any $\gamma, \lambda > 0$ with $\gamma/\lambda > 161m$, any $\varepsilon > 0$, any $\varepsilon_0 > 60\varepsilon\sqrt{\gamma/(\lambda m)}$ and any randomized algorithm A, there exist a dimension $d = O\big(\frac{\gamma m}{\lambda\varepsilon}\log^4\frac{\varepsilon_0^2\lambda m}{\gamma\varepsilon^2}\log\frac{\gamma m^2}{\lambda\varepsilon}\big)$, a domain $\mathcal{X} \subseteq \mathbb{R}^d$, $x_0 \in \mathcal{X}$ and m functions f_i, γ-smooth and λ-strongly convex with $F(x_0) - F(x^*) = \varepsilon_0$, such that finding $\hat{x}$ with $\mathbb{E}[F(\hat{x}) - F(x^*)] < \varepsilon$ requires $\Omega\big(m + \sqrt{m\gamma/\lambda}\log\big(\frac{\varepsilon_0}{\varepsilon}\sqrt{\frac{m\lambda}{\gamma}}\big)\big)$ queries.*

In our notation: deterministic algorithms need $\Omega\big(n\sqrt{\kappa}\log\frac{\varepsilon_0}{\kappa\varepsilon}\big)$ queries (the argument of the logarithm is $\frac{\lambda\varepsilon_0}{\gamma\varepsilon}=\frac{\varepsilon_0}{\kappa\varepsilon}$, so the bound is informative only for relative accuracies $\varepsilon/\varepsilon_0\ll 1/\kappa$; up to the constant inside the logarithm it is attained by AGD on F, since $L_F\le R^2$), and randomized algorithms with arbitrary adaptive queries need $\Omega\big(n+\sqrt{n\kappa}\log\big(\frac{\varepsilon_0}{\varepsilon}\sqrt{n/\kappa}\big)\big)$ in the regime $\kappa>161n$. Both theorems use components that are individually λ-strongly convex; this is a subclass of the book's class (where only F is μ-strongly convex), so the lower bounds apply a fortiori to the book's class. Two facts matter for (A1): the $\sqrt{n\kappa}\log(1/\varepsilon)$ term survives without any span or i.i.d. assumption, and the $n\log(1/\varepsilon)$ term of Theorem A1.8 is replaced by a bare n (no accuracy dependence).

Oblivious CLI algorithms, dimension-free (Arjevani–Shamir).

Theorem A1.10 (Arjevani–Shamir dimension-free bound; Theorem 2 of [5]). *The iteration complexity of oblivious (possibly stochastic) CLI algorithms for finite sums $\frac{1}{n}\sum_{i=1}^n f_i$ with L-smooth and μ-strongly convex components (their problem (1); the sum is then μ-strongly convex, and the bound transfers a fortiori to the book's class, which contains this one), $\kappa=L/\mu$, equipped with first-order and coordinate-descent oracles, is bounded from below by $\tilde\Omega\big(n+\sqrt{n(\kappa-1)}\ln(1/\varepsilon)\big)$, in every dimension d.*

The proof is not a chain argument: on a parametrized family of quadratics, the coordinates of the k-th iterate of an oblivious CLI are polynomials of degree $\le k$ in the parameters, and approximating the minimizer well requires a high degree (Chebyshev-type approximation bounds). The "n" term holds for any algorithm using an incremental oracle (their Theorem 5), and the class covers deterministic schedules (cyclic coordinate/component orders), mini-batches and full-gradient steps.

Breaking the span assumption (Hannah–Liu–O'Connor–Yin). In [38], $F=\frac{1}{n}\sum f_i+\psi$ with f_i L-smooth (or L_i-smooth, $L_Q=\frac{1}{n}\sum L_i$) and F μ-strongly convex, $\kappa=L/\mu$ (so $R^2=L$ and their κ is ours). Their span condition is relation (A1.1) "for some fixed IID random variable i_k over the indices", i.e. class (S1); a full gradient step violates it because it uses all n indices at the same point. The hard functions of their Theorems 2 and 4 are individually L-smooth and μ-strongly convex, a subclass of the book's class to which the lower bounds therefore apply a fortiori.

Theorem A1.11 (Hannah–Liu–O'Connor–Yin bounds; [38]). *Let the components be L-smooth and F be μ-strongly convex, $\kappa=L/\mu$.*

(a) (Their Corollary 3, a rewording of Theorem A1.8.) For every ε and every randomized algorithm following the span assumption there are a dimension d and L-smooth, μ-strongly convex f_i on $\mathbb{R}^d$ such that reaching $\mathbb{E}f(x^k)-f(x^*)<\varepsilon$ takes the algorithm at least $\Omega((n+\sqrt{\kappa n})\ln(1/\varepsilon))$ steps.

(b) (Upper bound; their Corollary 2.) With epoch length $m=n+121\kappa_Q$ and step size $\eta=\kappa_Q^{1/2}m^{-1/2}/(2L_Q)$, Prox-SVRG needs

$$K(\varepsilon)=O\Big(\Big(\frac{n}{1+(\ln(n/\kappa_Q))_+}+\kappa_Q\Big)\ln\frac{1}{\varepsilon}+n+\kappa_Q\Big)$$

gradient evaluations to reach $\mathbb{E}f(x^K)-f(x^*)<\varepsilon$ (here $(a)_+=\max\{a,0\}$ and $\kappa_Q=L_Q/\mu$).

(c) (Lower bound; their Theorem 2, stated in full as their Theorem 4.) For any oblivious p-CLI algorithm (their Appendix B lists "SDCA, SAG, SAGA, SVRG, SARAH, etc. without proximal terms", and they state before their Theorem 2 that the class "contains all VR algorithms

mentioned in this paper", Katyusha included; L-Katyusha with fixed parameters, which is posterior to [38], is an oblivious CLI as well, since its coin flips and parameters do not depend on the oracle answers: this last inclusion is ours) and all μ, L there exist L-smooth, μ-strongly convex f_i such that at least

$$K(\varepsilon) = \tilde{\Omega}\Big(\Big(\frac{n}{1+(\ln(n/\kappa))_+} + \sqrt{n\kappa}\Big)\ln\frac{1}{\varepsilon} + n\Big)$$

iterations are needed; equivalently (their (1.3)) $K(\varepsilon) = \Omega(n + \sqrt{n\kappa}\ln(1/\varepsilon))$ for $n = O(\kappa)$ and $K(\varepsilon) = \Omega\big(n + \frac{n}{1+(\ln(n/\kappa))_+}\ln(1/\varepsilon)\big)$ for $\kappa = O(n)$.

(d) (Example; their Corollary 1.) For $\varepsilon = 1/n^\alpha$ and $\kappa = n^\beta$ with $\alpha, \beta \in (0,1)$, modified SVRG needs $O(n)$ iterations whereas algorithms following the span assumption need $\Omega(n\ln n)$ (the source writes "require $O(n\ln(n))$ iterations", quoting Theorem A1.8; the content is a lower bound).

Their Section 3 explains the mechanism on a shared-regularizer variant of (A1.7) (in their words, "a variation of the adversarial function example from" [50] and Nesterov), $f(x) = \sum_{i=1}^n[\psi(x_i) + \frac{\mu}{2}\|x\|^2]$ on ℓ_2 with $\psi(x) = \frac{L-\mu}{4}(\frac{1}{2}\langle Ax, x\rangle - x_1)$: the infinite-chain version of (A1.8), with the regularizer shared by all components as in [1] and the per-block condition number $1 + \frac{\kappa-1}{n}$, which can be close to 1 (whereas (A1.7) has $\kappa \geq n$). After k i.i.d. uniform queries, block i has been touched a Binomial$(k, 1/n)$ number of times, and the *average* over blocks of the unrevealed energy is $(1-(1-q^2)/n)^k \geq (1-1/n)^k$ whatever the conditioning, which forces $n\ln(1/\varepsilon)$ queries; one full-gradient pass (n queries) touches *every* block once, so with full gradients the unrevealed energy of every block decreases by the factor q^2 per n queries and the cost $n\ln(1/\varepsilon)$ is divided by $2\ln(1/q) \approx 2\ln\frac{4n}{\kappa-1}$, a logarithmic gain that matters only when $\kappa \ll n$. Part (c) shows that the gain is at most the factor $1 + (\ln(n/\kappa))_+$, and part (b) that SVRG with a well-chosen epoch length achieves it.

Loopless methods and small dimension (Han–Xie–Zhang). The class $\mathcal{A}$ of [37] (their Definition 3.1) consists of randomized algorithms which at each step t sample i_t i.i.d. from a fixed distribution $\mathcal{D}$ and may query the PIFO of f_{i_t} *at all previous iterates* $x_0, \ldots, x_{t-1}$ ("simultaneous queries"; earlier answers may be reused), and with an independent probability $\bar{q} \leq c_0/n$ also query the full gradient at the current point; the set $\mathcal{H}$ of points where the full gradient is available is initialized with x_0, so the full gradient at x_0 is free. The iterate is then any point of the linear span of the previous iterates, of the oracle answers and of the full gradients at the points of $\mathcal{H}$, projected on the domain. By the authors' own account, this covers SAGA and Point-SAGA (whose only full gradient is the free one at x_0) and the loopless methods L-SVRG, L-Katyusha, KatyushaX, L2S, PAGE and Anita, but *not* the two-loop methods SVRG and Katyusha, whose full gradients are computed at deterministic times.

Theorem A1.12 (Han–Xie–Zhang lower bound; Theorem E.6 of [37]). *Let $n \geq 2$ and consider the class $\mathcal{F}^{\mathrm{C}}(R, L, \mu)$ of finite sums $f = \frac{1}{n}\sum_i f_i$ on a closed convex domain $\mathcal{X}$ of diameter at most $2R$, with L-smooth components and μ-strongly convex f, $\kappa = L/\mu \geq 2$, and let $\varepsilon \leq LR^2/4$ (the hard instance is constrained to the ball $\{\|x\| \leq R\}$, which contains $x_0 = 0$ and x^*; its components are individually μ-strongly convex, their Proposition E.13). The number of PIFO queries that algorithms of class $\mathcal{A}$ (their Definition 3.1, which by their account contains SAGA, L-SVRG and L-Katyusha) need to find $\hat{x} \in \mathcal{X}$ with $\mathbb{E}f(\hat{x}) - \min_{\mathcal{X}} f < \varepsilon$ is*

$$\mathfrak{m}_*^{\mathrm{C}}(\varepsilon, R, L, \mu) = \begin{cases} \Omega\big((n + \sqrt{\kappa n})\log(1/\varepsilon)\big) & \text{for } \kappa = \Omega(n), \\ \Omega\big(n + \frac{n}{1+(\log(n/\kappa))_+}\log(1/\varepsilon)\big) & \text{for } \kappa = O(n), \end{cases}$$

and the construction only requires the dimension to be $O\big(1+\sqrt{\kappa/n}\,\log(1/\varepsilon)\big)$.

Their instance is a single chain of length $m = O(1+\sqrt{\kappa/n}\log(1/\varepsilon))$ whose rows are distributed among the n components (component i owns the rows $l \equiv i-1 \pmod n$, scaled by n): only the component owning the next row can extend the support, so the time to extend it is geometric with mean n, which gives the n factor without needing n separate chains.

Assessment of hypothesis (A1). Hypothesis (A1) reads: "Any span-based method with i.i.d. uniform sampling needs $\Omega((n+\sqrt{n\kappa})\log(1/\varepsilon))$ gradients. For an arbitrary index sequence, the $\sqrt{n\kappa}$ term survives but the $n\log(1/\varepsilon)$ term weakens by a factor $\log(n/\kappa)$ when $\kappa \ll n$."

(i) The first clause is exactly Theorem A1.8 (class (S1), any fixed sampling distribution, dimension $\gtrsim$ number of iterations) and is correct.

(ii) The second clause is correct *if* "arbitrary index sequence" means "index sequences that are not i.i.d. single queries, in particular full-gradient sweeps (SVRG, SARAH, L-SVRG, L-Katyusha)", and if it is understood within the oblivious CLI class and with the precise factor $1+(\ln(n/\kappa))_+$: by Theorem A1.11(c) and Theorem A1.12 the bound becomes $\Omega\big(n+\big(\frac{n}{1+(\ln(n/\kappa))_+}+\sqrt{n\kappa}\big)\ln\frac{1}{\varepsilon}\big)$, the $\sqrt{n\kappa}\ln(1/\varepsilon)$ term is untouched, and the $n\ln(1/\varepsilon)$ term is divided by $1+(\ln(n/\kappa))_+$ but never falls below the bare n (one query per component is unavoidable, Theorem A1.10). This is tight: Theorem A1.11(b) attains it when $\kappa = O(n)$, and Katyusha/L-Katyusha attain $(n+\sqrt{n\kappa})\log(1/\varepsilon)$ when $n = O(\kappa)$ [2, 47].

(iii) The clause is *not* correct for other readings. For deterministic (adaptive) index sequences the worst case is $\Omega(n\sqrt{\kappa}\log(1/\varepsilon))$ (Theorem A1.9, deterministic part): the $\sqrt{n\kappa}$ term does not merely survive, it deteriorates by $\sqrt{n}$, and the $\sqrt{n}$ saving of randomized methods is genuinely due to randomization. For randomized algorithms with fully adaptive queries and no span restriction, the known bound is $\Omega(n+\sqrt{n\kappa}\log(1/\varepsilon))$ (Theorem A1.9, randomized part, stated for $\kappa > 161n$), which says nothing about an $n\log(1/\varepsilon)$ term; for deterministic index sequences, cyclic ones included, the deterministic part of Theorem A1.9 already excludes, when $\kappa > 73$, any gain of the type $1+\ln(n/\kappa)$ over $n\log(1/\varepsilon)$ (its bound $\Omega(n\sqrt{\kappa}\log\frac{\varepsilon_0}{\kappa\varepsilon})$ is at least of order $n\log\frac{1}{\varepsilon}$ for small ε); whether randomly reshuffled (without-replacement) span-based methods, or deterministic ones when $\kappa \le 73$, can beat $n\log(1/\varepsilon)$ is not settled by these sources.

(iv) Finally, the factor is $1+(\ln(n/\kappa))_+$ rather than $\log(n/\kappa)$ (which is <1 for $\kappa > n/e$); the gain is only logarithmic and only in the big-data regime $\kappa \ll n$; in the regime $\kappa \gtrsim n$ of hypothesis (A4) the two bounds coincide, $\Theta((n+\sqrt{n\kappa})\log(1/\varepsilon))$.

On the hard instance of the next section, Proposition A1.18 gives a self-contained proof that the $\sqrt{n\kappa}\log(1/\varepsilon)$ term is unavoidable for *every* index sequence (class (S2)), and Theorem A1.19 proves the full $(n+\sqrt{n\kappa})\log(1/\varepsilon)$ bound for class (S1).

A1.3 The block-separable hard instance (hypothesis A4)

A1.3.1 Definition and constants

Fix $n \ge 1$ components, a block length $m \ge 2$, $\mu > 0$, a target condition number $\kappa > 1$ and $\zeta \in [1,3]$. The variable is $\theta = (\theta^{(1)},\dots,\theta^{(n)}) \in \mathbb{R}^{nm}$ with blocks $\theta^{(i)} \in \mathbb{R}^m$; $E_i : \mathbb{R}^m \to \mathbb{R}^{nm}$ denotes the embedding of block i (so $E_i^\top \theta = \theta^{(i)}$). Let

$$\begin{aligned} &c = \frac{\mu(\kappa-1)}{4}, \qquad \phi(x) = \frac{1}{2}\langle A_m(\zeta)x, x\rangle - x_1 \quad (x \in \mathbb{R}^m), \\ &f_i(\theta) = c\,\phi(\theta^{(i)}) + \frac{\mu}{2}\|\theta\|^2, \quad i = 1,\dots,n. \end{aligned} \tag{A1.8}$$

Then

$$F(\theta) = \frac{1}{n}\sum_{i=1}^{n} f_i(\theta) = \frac{c}{n}\sum_{i=1}^{n}\phi(\theta^{(i)}) + \frac{\mu}{2}\|\theta\|^2,$$
$$f_i'(\theta) = c\,E_i\big(A_m(\zeta)\theta^{(i)} - e_1\big) + \mu\theta, \qquad F'(\theta) = \frac{c}{n}\sum_{i=1}^{n} E_i\big(A_m(\zeta)\theta^{(i)} - e_1\big) + \mu\theta. \tag{A1.9}$$

In numpy, with θ stored as an $n \times m$ array Θ and $A = A_m(\zeta)$: $f_i'(\theta)$ is $\mu\Theta$ plus, on row i only, $c\,(A\Theta_{i,:} - e_1)$; $F'(\theta) = \frac{c}{n}(\Theta A - \mathbf{1}e_1^\top) + \mu\Theta$ (row-wise matrix product, $\mathbf{1}$ the all-ones column); the products with the tridiagonal A cost $O(m)$ per row.

Proposition A1.13 (constants of the instance)**.** *Let $Q := 1 + \frac{\kappa-1}{n}$. For the instance (A1.8):*

(a) each f_i is μ-strongly convex and R^2-smooth with $R^2 = \mu\kappa$ (its exact Lipschitz constant is $\mu + c\,\lambda_{\max}(A_m(\zeta)) \le \mu\kappa$, with equality only in the limit $m \to \infty$);

(b) $\nabla^2 F = \operatorname{diag}\big(\frac{c}{n}A_m(\zeta) + \mu I, \dots, \frac{c}{n}A_m(\zeta) + \mu I\big)$ is block diagonal; F is μ-strongly convex, its smallest Hessian eigenvalue is $\mu_F := \mu + \frac{c}{n}\lambda_{\min}(A_m(\zeta)) = \mu\big(1 + \frac{Q-1}{4}\lambda_{\min}(A_m(\zeta))\big) \ge \mu$ and its smoothness constant is $L_F = \mu + \frac{c}{n}\lambda_{\max}(A_m(\zeta)) \le \mu Q$;

(c) hence $\kappa = R^2/\mu = 1 + n(Q-1)$ and

$$\frac{L_F}{\mu} \;\le\; Q = 1 + \frac{\kappa-1}{n} \;\le\; \kappa, \tag{A1.10}$$

so that $(L_F/\mu)/\kappa \le \frac{1}{n} + \frac{1}{\kappa}$: the condition number of F is about n times smaller than κ as soon as $\kappa \gg n$.

Proof. (a) $\nabla^2 f_i = cE_iA_m(\zeta)E_i^\top + \mu I$ has eigenvalues μ (on the other blocks) and $\mu + c\lambda$ for λ an eigenvalue of $A_m(\zeta)$; by Lemma A1.3 these lie in $[\mu, \mu + 4c] = [\mu, \mu\kappa]$. (b) $\nabla^2 F = \frac{1}{n}\sum_i \nabla^2 f_i$ is block diagonal with blocks $\frac{c}{n}A_m(\zeta) + \mu I$, whose eigenvalues are $\mu + \frac{c}{n}\lambda \in [\mu, \mu + \frac{4c}{n}]$ and $\mu + \frac{4c}{n} = \mu(1 + \frac{\kappa-1}{n}) = \mu Q$. (c) is arithmetic. □

Why the chain of f_i carries the factor n. The restriction of F to block i is $\frac{c}{n}\phi + \frac{\mu}{2}\|\cdot\|^2$: the chain that the i-th block of F sees is the chain of f_i divided by n. Nesterov's per-block function with condition number Q is $\frac{\mu(Q-1)}{4}\phi + \frac{\mu}{2}\|\cdot\|^2$ (compare (A1.5) with $L = \mu Q$), so one needs $\frac{c}{n} = \frac{\mu(Q-1)}{4}$, i.e. $c = \frac{n\mu(Q-1)}{4} = \frac{\mu(\kappa-1)}{4}$: the component must carry n times the chain of the block. Conversely, if one naively takes $f_i(\theta) = \psi(\theta^{(i)})$ with a single μ-strongly convex, L-smooth ψ (block-only regularizer, as in (A1.7)), then $F = \frac{1}{n}\sum_i \psi(\theta^{(i)})$ is only μ/n-strongly convex, $R^2 = L$, and the book's κ equals nL/μ while the per-block condition number is $L/\mu = \kappa/n$. In both normalizations the quantity that drives the chain mechanism is the *per-block* condition number $Q \approx \kappa/n$, and $\sqrt{n\kappa} \approx n\sqrt{Q}$ is "n blocks times $\sqrt{Q}$ touches per block".

Proposition A1.14 (closed-form minimizer)**.** *For the instance (A1.8), let $Q = 1 + \frac{\kappa-1}{n}$ and $q = \frac{\sqrt{Q}-1}{\sqrt{Q}+1}$.*

(a) (Geometric chain.) If

$$\zeta = \zeta_Q := \frac{\sqrt{Q}+3}{\sqrt{Q}+1} \in (1, 2), \tag{A1.11}$$

then the minimizer is $\eta_* = (x^*, \dots, x^*)$ with $x_j^* = q^j$, $j = 1, \dots, m$, so that

$$\|\eta_*\|^2 = n\,\frac{q^2(1-q^{2m})}{1-q^2}, \qquad F(\eta_*) = -\frac{c\,q}{2} = -\frac{\mu(\kappa-1)\,q}{8}.$$

(b) (Nesterov's chain, $\zeta = 2$.) The minimizer is $\eta_* = (x^*, \dots, x^*)$ with

$$x_j^* = \frac{q^{\,j} - q^{\,2(m+1)-j}}{1 - q^{\,2(m+1)}} = \frac{\sinh\big((m+1-j)\tau\big)}{\sinh\big((m+1)\tau\big)}, \qquad \tau = \log\frac{1}{q} = 2\,\mathrm{artanh}\,\frac{1}{\sqrt{Q}}, \quad j = 1, \dots, m.$$

(c) (Any $\zeta \in [1, 3]$.) x^* is the solution of the tridiagonal system $\big(A_m(\zeta) + \frac{4}{Q-1} I\big)x = e_1$, computable in $O(m)$ by the Thomas algorithm (forward elimination, backward substitution), which is numerically stable here because the matrix is symmetric positive definite and diagonally dominant.

In all cases $F(\theta) - F(\eta_*) = \frac{1}{2}\langle \nabla^2 F(\theta - \eta_*), \theta - \eta_* \rangle$ lies between $\frac{\mu_F}{2}\|\theta - \eta_*\|^2$ and $\frac{L_F}{2}\|\theta - \eta_*\|^2$, with μ_F and L_F as in Proposition A1.13(b).

Proof. Since F is block separable and the blocks play symmetric roles, $\eta_* = (x^*, \dots, x^*)$ where x^* minimizes $\frac{c}{n}\phi(x) + \frac{\mu}{2}\|x\|^2$, i.e. solves $\frac{c}{n}(A_m(\zeta)x - e_1) + \mu x = 0$, or $(A_m(\zeta) + \frac{n\mu}{c} I)x = e_1$; and $\frac{n\mu}{c} = \frac{4n}{\kappa - 1} = \frac{4}{Q-1}$, which proves (c). Written row by row, with $a := 2 + \frac{4}{Q-1} = q + q^{-1}$ (see the proof of Proposition A1.4):

$$a\,x_1 - x_2 = 1, \qquad -x_{j-1} + a\,x_j - x_{j+1} = 0 \ (2 \le j \le m-1), \qquad -x_{m-1} + \Big(\zeta + \frac{4}{Q-1}\Big)x_m = 0.$$

(a) The first $m-1$ equations are satisfied by $x_j = q^j$ exactly as in Proposition A1.4. The last one holds iff $\zeta + \frac{4}{Q-1} = q^{-1}$; since $q^{-1} = \frac{\sqrt{Q}+1}{\sqrt{Q}-1} = \frac{(\sqrt{Q}+1)^2}{Q-1}$, this means $\zeta = \frac{(\sqrt{Q}+1)^2 - 4}{Q-1} = \frac{(\sqrt{Q}-1)(\sqrt{Q}+3)}{(\sqrt{Q}-1)(\sqrt{Q}+1)} = \zeta_Q$, and $1 < \zeta_Q < 2$ for $Q > 1$. Then $\|x^*\|^2 = \sum_{j=1}^m q^{2j} = q^2(1 - q^{2m})/(1 - q^2)$. For the value, $A_m(\zeta_Q)x^* = e_1 - \frac{4}{Q-1}x^*$ gives $\phi(x^*) = -\frac{x_1^*}{2} - \frac{2}{Q-1}\|x^*\|^2$ and $F(\eta_*) = c\,\phi(x^*) + \frac{n\mu}{2}\|x^*\|^2 = -\frac{cq}{2}$ because $\frac{2c}{Q-1} = \frac{\mu(\kappa-1)}{2(Q-1)} = \frac{n\mu}{2}$. (b) The general solution of the three-term recursion $x_{j+1} = a\,x_j - x_{j-1}$ is $x_j = \alpha q^j + \beta q^{-j}$ (q and q^{-1} are the roots of $z^2 - az + 1 = 0$). The first equation says that the recursion extends to $j = 1$ with $x_0 := 1$, and the last one, with $\zeta = 2$, that it extends to $j = m$ with $x_{m+1} := 0$. Hence $\alpha + \beta = 1$ and $\alpha q^{m+1} + \beta q^{-(m+1)} = 0$, i.e. $\beta = -\alpha q^{2(m+1)}$ and $\alpha = 1/(1 - q^{2(m+1)})$, which is the stated formula; the sinh form follows by multiplying numerator and denominator by $q^{-(m+1)}$, and $\tau = \log\frac{\sqrt{Q}+1}{\sqrt{Q}-1} = 2\,\mathrm{artanh}(Q^{-1/2})$. The formula with q is the one to implement (all powers are in $(0, 1]$). The last claim is Taylor's formula for the quadratic F with $F'(\eta_*) = 0$. □

A1.3.2 The zero-chain property and the sequence-free bound

For $\theta \in \mathbb{R}^{nm}$ and $j \in \{1, \dots, n\}$ let $N_j(\theta) = \max\{l : \theta_l^{(j)} \neq 0\}$ ($N_j = 0$ if the block vanishes). Consider a method following the linear-span protocol (A1.1) with $\theta_0 = 0$ and an *arbitrary* index sequence $(i_s)_{s \ge 1}$ (class (S2); full gradients are n consecutive queries at the same point, and the indices may be chosen adaptively). Write $\tau_j(t) = \#\{s \le t : i_s = j\}$ for the number of queries to component j among the first t queries, so that $\sum_j \tau_j(t) = t$. (For $\theta_0 \neq 0$ the bounds below apply to the translated instance $f_i(\cdot - \theta_0)$, whose minimizer is $\eta_* + \theta_0$: for it, $\theta_t - \theta_0$ follows the span protocol on (A1.8) started at 0, so the bounds hold for $\|\theta_t - (\eta_* + \theta_0)\|^2$, with $\|\theta_0 - (\eta_* + \theta_0)\| = \|\eta_*\|$.)

Lemma A1.15 (zero chain, block version)**.** *For every θ and i, j: $N_j(f_i'(\theta)) \le N_j(\theta) + \mathbf{1}[i = j]$.*

Proof. By (A1.9), the j-th block of $f_i'(\theta)$ is $\mu\theta^{(j)}$ if $j \neq i$ and $c(A_m(\zeta)\theta^{(i)} - e_1) + \mu\theta^{(i)}$ if $j = i$; apply Lemma A1.5. □

Lemma A1.16 (support under the span protocol)**.** *Consider a method of class (S2) (Definition A1.2) on the instance (A1.8), started at $\theta_0 = 0$; let $N_j(\theta) = \max\{l : \theta_l^{(j)} \neq 0\}$ ($N_j = 0$ if*

the block vanishes) and let $\tau_j(t)$ be the number of queries to component j among the first t. (i) For every $t \geq 0$ and j: $N_j(\theta_t) \leq \tau_j(t)$. (ii) If the span in (A1.1) is enlarged by the n initial gradients $f_1'(\theta_0), \ldots, f_n'(\theta_0)$ (an initial full gradient, as for SAGA with $z_i^{(0)} = f_i'(\theta_0)$), then $N_j(\theta_t) \leq 1 + \tau_j(t)$, where $\tau_j(t)$ counts the queries made after the initial full gradient; equivalently, $N_j(\theta_t) \leq \tau_j(t)$ still holds if the initial full gradient is counted as one query to every component.

Proof. (i) Induction on t; the case $t = 0$ is $\theta_0 = 0$. If the claim holds up to $t-1$, each generator $f_{i_s}'(\theta_{s-1})$, $s \leq t$, of the span in (A1.1) satisfies $N_j \leq \tau_j(s-1) + \mathbf{1}[i_s = j] = \tau_j(s) \leq \tau_j(t)$ by Lemma A1.15, and the support of a linear combination is contained in the union of the supports. (ii) Same induction: the n additional generators $f_l'(\theta_0) = f_l'(0)$ satisfy $N_j(f_l'(0)) \leq \mathbf{1}[l = j] \leq 1$ by Lemma A1.15, and if $N_j(\theta_{s-1}) \leq 1 + \tau_j(s-1)$ for all $s \leq t$ then each generator $f_{i_s}'(\theta_{s-1})$ has $N_j \leq 1 + \tau_j(s-1) + \mathbf{1}[i_s = j] = 1 + \tau_j(s) \leq 1 + \tau_j(t)$. □

Proposition A1.17 (tail bound)**.** *On the instance (A1.8) with $\zeta = \zeta_Q$ and q as in Proposition A1.14, let $\theta \in \mathbb{R}^{nm}$ and let $\tau_1, \ldots, \tau_n$ be integers. If $N_j(\theta) \leq \tau_j \leq m$ for all j (N_j as in Lemma A1.16), then*

$$\frac{\|\theta - \eta_*\|^2}{\|\eta_*\|^2} \;\geq\; \frac{1}{n} \sum_{j=1}^{n} \frac{q^{2\tau_j} - q^{2m}}{1 - q^{2m}}. \tag{A1.12}$$

Proof. $\|\theta - \eta_*\|^2 = \sum_j \left\|\theta^{(j)} - x^*\right\|^2 \geq \sum_j \sum_{l > \tau_j} (x_l^*)^2 = \sum_j \sum_{l=\tau_j+1}^{m} q^{2l} = \frac{q^2}{1-q^2} \sum_j \left(q^{2\tau_j} - q^{2m}\right)$, and $\|\eta_*\|^2 = n \frac{q^2(1-q^{2m})}{1-q^2}$ by Proposition A1.14. □

Proposition A1.18 (the $\sqrt{n\kappa}$ term for any index sequence)**.** *On the instance (A1.8) with $\zeta = \zeta_Q$, and Q, q as in Proposition A1.14, consider any method of class (S2) (Definition A1.2) started at $\theta_0 = 0$. After G gradient evaluations in total (full gradients counted as n), of which τ_j are queries to component j, every iterate θ in the span of the answers satisfies*

$$\frac{1}{n} \sum_{j=1}^{n} q^{2\tau_j} \;\geq\; q^{2G/n}, \qquad \textit{hence} \qquad \frac{\|\theta - \eta_*\|^2}{\|\eta_*\|^2} \;\geq\; \frac{q^{2G/n} - q^{2m}}{1 - q^{2m}}. \tag{A1.13}$$

Consequently, if $q^{2m} \leq \varepsilon < \frac{1}{2}$, reaching $\|\theta - \eta_*\|^2 \leq \varepsilon \|\eta_*\|^2$ requires

$$G \;\geq\; \frac{n \log \frac{1}{2\varepsilon}}{2 \log(1/q)} \;\geq\; \frac{n(\sqrt{Q} - 1)}{4} \log \frac{1}{2\varepsilon} \;\geq\; \frac{\sqrt{n(\kappa - 1)}}{8} \log \frac{1}{2\varepsilon} \quad \textit{(the last step for } \kappa - 1 \geq 2n\textit{)}. \tag{A1.14}$$

Proof. Let $\bar{\tau}_j = \min(\tau_j(G), m)$. By Lemma A1.16 and $N_j \leq m$, the hypothesis of Proposition A1.17 holds with $\tau_j = \bar{\tau}_j$, and $q^{2\bar{\tau}_j} \geq q^{2\tau_j(G)}$. The map $\tau \mapsto q^{2\tau}$ is convex and $\sum_j \tau_j(G) = G$, so Jensen's inequality gives $\frac{1}{n}\sum_j q^{2\bar{\tau}_j} \geq \frac{1}{n}\sum_j q^{2\tau_j(G)} \geq q^{2G/n}$, and (A1.12) yields the second inequality in (A1.13). If $\|\theta - \eta_*\|^2 \leq \varepsilon \|\eta_*\|^2$ then $q^{2G/n} \leq \varepsilon(1 - q^{2m}) + q^{2m} \leq \varepsilon + q^{2m} \leq 2\varepsilon$, i.e. $\frac{2G}{n} \log \frac{1}{q} \geq \log \frac{1}{2\varepsilon}$. Finally $\log \frac{1}{q} = \log\left(1 + \frac{2}{\sqrt{Q}-1}\right) \leq \frac{2}{\sqrt{Q}-1}$, and for $y = \frac{\kappa - 1}{n} \geq 2$ one has $\sqrt{1+y} - 1 \geq \frac{\sqrt{y}}{2}$ (the function $y \mapsto \sqrt{1+y} - 1 - \frac{\sqrt{y}}{2}$ is positive at $y = 2$ and increasing for $y > \frac{1}{3}$), so $n(\sqrt{Q} - 1) \geq \frac{1}{2}\sqrt{n(\kappa - 1)}$. □

If the number G_k of gradient evaluations after k iterations is random (L-SVRG, L-Katyusha), (A1.13) holds pathwise with $G = G_k$. Since $x \mapsto q^{2x/n}$ is convex, $\mathbb{E}\|\theta_k - \eta_*\|^2 / \|\eta_*\|^2 \geq (q^{2\mathbb{E}G_k/n} - q^{2m})/(1 - q^{2m})$, so (A1.14) holds for the expected number of gradient evaluations $\mathbb{E}G_k$ and the criterion $\mathbb{E}\|\theta_k - \eta_*\|^2 \leq \varepsilon \|\eta_*\|^2$.

Proposition A1.18 therefore applies (in expected gradient evaluations for the methods with coin flips) to GD, AGD, cyclic incremental methods, SAGA, SVRG, SARAH, Katyusha and L-Katyusha, whatever their randomization: on this instance none of them can beat $\Omega(\sqrt{n\kappa}\log(1/\varepsilon))$ gradient evaluations, and accelerated gradient descent on F attains this order (it needs $O(\sqrt{L_F/\mu}\log(1/\varepsilon)) \le O(\sqrt{Q}\log(1/\varepsilon))$ iterations of n gradients, i.e. $O(n\sqrt{Q}\log(1/\varepsilon))$ gradient evaluations, where $n\sqrt{Q} = \sqrt{n^2+n(\kappa-1)}$). What the sequence-free argument cannot produce is the $n\log(1/\varepsilon)$ term: a full-gradient sweep advances all n chains at once, and this is exactly the point of [38].

A1.3.3 The i.i.d. bound

Theorem A1.19 (Lan–Zhou bound on the instance)**.** *On the instance (A1.8) with $\zeta = \zeta_Q$, and Q, q as in Proposition A1.14, let $\theta_0 = 0$ and let the method belong to class (S1) (Definition A1.2) with a fixed sampling distribution p (uniform or not). Then for every $t \ge 0$,*

$$\frac{\mathbb{E}\,\|\theta_t-\eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{\big(1-\frac{1-q^2}{n}\big)^t - q^{2m}}{1-q^{2m}}, \tag{A1.15}$$

and, whenever $q^{2m} \le \frac12\big(1-\frac{1-q^2}{n}\big)^t$,

$$\frac{\mathbb{E}\,\|\theta_t-\eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac12\Big(1-\frac{1-q^2}{n}\Big)^t \;\ge\; \frac12\exp\Big(-\frac{4\sqrt{Q}\,t}{n(\sqrt{Q}+1)^2-4\sqrt{Q}}\Big). \tag{A1.16}$$

Proof. Under (S1), $\tau_j(t) \sim \mathrm{Binomial}(t,p_j)$, so $\mathbb{E}q^{2\tau_j(t)} = \sum_{k=0}^t \binom{t}{k}p_j^k(1-p_j)^{t-k}q^{2k} = \big(1-(1-q^2)p_j\big)^t$. Taking expectations in (A1.12) (with $\tau_j = \min(\tau_j(t),m)$ as in the previous proof),

$$\frac{\mathbb{E}\,\|\theta_t-\eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{\frac1n\sum_j\big(1-(1-q^2)p_j\big)^t - q^{2m}}{1-q^{2m}} \;\ge\; \frac{\big(1-\frac{1-q^2}{n}\big)^t - q^{2m}}{1-q^{2m}},$$

where the last step is Jensen's inequality for the convex map $p \mapsto (1-(1-q^2)p)^t$ on $[0,1]$ and $\frac1n\sum_j p_j = \frac1n$: the uniform distribution is the most favourable one for the algorithm, and the bound is the same for all p. For (A1.16), write $a = (1-\frac{1-q^2}{n})^t$ and $b = q^{2m}$; if $b \le a/2$ then $\frac{a-b}{1-b} \ge a-b \ge \frac{a}{2}$. Next, $1-q^2 = \frac{(\sqrt{Q}+1)^2-(\sqrt{Q}-1)^2}{(\sqrt{Q}+1)^2} = \frac{4\sqrt{Q}}{(\sqrt{Q}+1)^2}$, so $a = (1-x)^t$ with $x = \frac{4\sqrt{Q}}{n(\sqrt{Q}+1)^2} \in (0,1)$, and $\log(1-x) = -\sum_{k\ge1}x^k/k \ge -\sum_{k\ge1}x^k = -\frac{x}{1-x}$ gives $(1-x)^t \ge \exp(-\frac{tx}{1-x})$ with $\frac{x}{1-x} = \frac{4\sqrt{Q}}{n(\sqrt{Q}+1)^2-4\sqrt{Q}}$. □

Corollary A1.20 (iteration count and dimension requirement)**.** *In the setting of Theorem A1.19, let $\varepsilon \in (0,\frac12)$ and assume the block length m satisfies*

$$m \;\ge\; \frac{\log(1/\varepsilon)}{2\log(1/q)}, \qquad \textit{for which it suffices that} \qquad m \;\ge\; \frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon} = \frac14\sqrt{1+\frac{\kappa-1}{n}}\,\log\frac{1}{\varepsilon}. \tag{A1.17}$$

Then any method of class (S1) needs

$$t \;\ge\; \Big(\frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}}-1\Big)\log\frac{1}{2\varepsilon} \;\ge\; \Big(\frac{n+\sqrt{n(\kappa-1)}}{8}+\frac{n}{2}-1\Big)\log\frac{1}{2\varepsilon} = \Omega\Big((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon}\Big) \tag{A1.18}$$

gradient evaluations to reach $\mathbb{E}\,\|\theta_t-\eta_*\|^2 \le \varepsilon\,\|\eta_*\|^2$. In particular $t \ge (n-1)\log\frac{1}{2\varepsilon}$ for every $\kappa > 1$, because $(\sqrt{Q}+1)^2 \ge 4\sqrt{Q}$.

Proof. Let $x = \frac{4\sqrt{Q}}{n(\sqrt{Q}+1)^2}$ and $t^* = (\frac{1}{x}-1)\log\frac{1}{2\varepsilon}$. For $t \le t^*$ the condition $q^{2m} \le \frac{1}{2}(1-x)^t$ of (A1.16) holds: indeed $t\log\frac{1}{1-x} \le t^*\frac{x}{1-x} = \log\frac{1}{2\varepsilon}$, so $\frac{1}{2}(1-x)^t \ge \varepsilon \ge q^{2m}$ by (A1.17) (the sufficient form uses $\log\frac{1}{q} = 2\operatorname{artanh}(Q^{-1/2}) \ge 2Q^{-1/2}$). Hence for $t \le t^*$, (A1.16) gives $\mathbb{E}\left\|\theta_t - \eta_*\right\|^2 / \left\|\eta_*\right\|^2 \ge \frac{1}{2}\exp(-\frac{tx}{1-x})$, which is $\le \varepsilon$ only if $t \ge t^*$: so at least t^* iterations are needed. Finally $\frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}} = \frac{n}{4}(\sqrt{Q}+2+Q^{-1/2}) \ge \frac{n\sqrt{Q}}{4} + \frac{n}{2}$ and $n\sqrt{Q} = \sqrt{n^2+n(\kappa-1)} \ge \max\{n, \sqrt{n(\kappa-1)}\} \ge \frac{1}{2}(n + \sqrt{n(\kappa-1)})$. □

Corollary A1.21 (initial full gradient)**.** *On the instance (A1.8) with $\zeta = \zeta_Q$, and Q, q as in Proposition A1.14, let $\theta_0 = 0$ and let the method be of class (S1) (Definition A1.2) preceded by one full gradient at θ_0 (Lemma A1.16(ii)), t denoting the number of i.i.d. queries made after this initial full gradient. Then for every $t \ge 0$*

$$\frac{\mathbb{E}\left\|\theta_t - \eta_*\right\|^2}{\left\|\eta_*\right\|^2} \;\ge\; \frac{q^2\big(1-\frac{1-q^2}{n}\big)^t - q^{2m}}{1-q^{2m}}, \tag{A1.19}$$

and, under (A1.17) and for $\varepsilon < q^2/2$, reaching $\mathbb{E}\left\|\theta_t - \eta_*\right\|^2 \le \varepsilon\left\|\eta_*\right\|^2$ requires

$$t \;\ge\; \Big(\frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}} - 1\Big)\log\frac{q^2}{2\varepsilon} \;\ge\; (n-1)\log\frac{q^2}{2\varepsilon}$$

i.i.d. queries, i.e. $G = n + t$ gradient evaluations in total. Compared with (A1.18), ε is replaced by ε/q^2: the initial full gradient reveals the first coordinate of every block, which carries the fraction $1-q^2$ of the energy of each block, and this is worth about n i.i.d. queries since $\big(\frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}}-1\big)\cdot 2\log\frac{1}{q} \le \frac{n(\sqrt{Q}+1)^2}{\sqrt{Q}(\sqrt{Q}-1)}$, which is $n(1+O(Q^{-1/2}))$ for large Q (for $\kappa \le n$, however, $q^2 \le 0.03$ and the shift $(n-1)\log(1/q^2)$ is a noticeable part of the count).

Proof. By Lemma A1.16(ii) and $N_j \le m$, Proposition A1.17 applies with $\tau_j = \min(1+\tau_j(t), m)$, and $q^{2\min(1+\tau_j(t),m)} \ge q^{2(1+\tau_j(t))}$. Under (S1), $\tau_j(t) \sim \mathrm{Binomial}(t, p_j)$, so $\mathbb{E}q^{2(1+\tau_j(t))} = q^2(1-(1-q^2)p_j)^t$, and Jensen's inequality in p gives (A1.19) exactly as in the proof of Theorem A1.19. For the count, let $x = \frac{1-q^2}{n}$ and $t_1^* = (\frac{1}{x}-1)\log\frac{q^2}{2\varepsilon} > 0$. For $t \le t_1^*$, $\frac{q^2}{2}(1-x)^t \ge \frac{q^2}{2}\exp(-\frac{tx}{1-x}) \ge \varepsilon \ge q^{2m}$ (the middle inequality is the definition of t_1^*, the last one is (A1.17)), so the right-hand side of (A1.19) is at least $q^2(1-x)^t - q^{2m} \ge \frac{q^2}{2}(1-x)^t \ge \varepsilon$; hence at least t_1^* i.i.d. queries are needed, and $\frac{1}{x}-1 \ge n-1$ as in Corollary A1.20. For the last claim, $\log\frac{1}{q} \le \frac{2}{\sqrt{Q}-1}$ and $\frac{1}{x} = \frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}}$. □

The dimension requirement (A1.17) is the precise form of "m must exceed the number of touches an accelerated method needs per block": $\frac{\log(1/\varepsilon)}{2\log(1/q)}$ is exactly the number of coordinates of a single chain that must be revealed before the unrevealed energy $q^{2\tau}$ drops below ε (Theorem A1.6), and it is of order $\frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon} = \frac{1}{4}\sqrt{\kappa/n}\log\frac{1}{\varepsilon}$ for $\kappa \gg n$ (the project's "$m \gtrsim \sqrt{\kappa}\log(1/\varepsilon)$" should read $\sqrt{\kappa/n}$, since each block only sees the per-block condition number Q). It is also necessary for the argument: since $\log(1/q) \le 2/(\sqrt{Q}-1)$, the condition (A1.17) forces $m \ge \frac{\sqrt{Q}-1}{4}\log\frac{1}{\varepsilon}$. The total dimension is $nm \gtrsim \frac{1}{4}\sqrt{n\kappa}\log(1/\varepsilon)$, of the order of the number of iterations certified, in agreement with the discussion of Section A1.1.

A1.4 Predictions for the numerics, finite-dimension effects, and logistic regression

A1.4.1 Gradient counts on the instance

Write $T_{\mathrm{alg}}(\varepsilon)$ for the number of gradient evaluations (its expectation for the methods with coin flips) an algorithm needs on the instance (A1.8) to reach $\mathbb{E}\,\|\theta-\eta_*\|^2 \le \varepsilon\,\|\theta_0-\eta_*\|^2$. For SAGA with the book's initialization $z_i^{(0)}=f_i'(\theta_0)$ the n gradients of the initial sweep are counted. For L-Katyusha, T_{LK} counts the n gradients of $F'(w^0)$ and, per iteration, $1+n\,\mathbf{1}[\mathrm{coin}]$ gradients with $\mathbb{P}[\mathrm{coin}]=p=\frac{1}{n}$ (the gradients $f_i'(w^k)$ of each full pass are stored, so that $f_i'(w^k)$ is a look-up), i.e. $n+2k$ in expectation after k iterations; Proposition A1.18 applies to this expectation (see the paragraph after its proof). The experiments of the report count differently: their code charges L-SVRG, L-Katyusha and the naive accelerated schemes n gradients initially, 2 per iteration ($f_i'(x)$ and $f_i'(w)$, the latter as if it were recomputed, although the table of the $f_i'(w)$ is stored) and n per refresh of the reference point, i.e. $n+3k=\frac{3}{2}(n+2k)-\frac{n}{2}$ in expectation after k iterations; their counts are thus about $\frac{3}{2}$ times the stored-gradient counts used in the bounds below, which changes constants, not log-log slopes. They also measure, for each seed, the first recorded count at which $\|\theta-\eta_*\|^2 \le \varepsilon\,\|\theta_0-\eta_*\|^2$ (log-linearly interpolated between records), averaged geometrically over the seeds. The pathwise bound of Proposition A1.18 applies to this hitting count; the in-expectation statements (Theorem A1.19, Corollaries A1.20, A1.21 and A1.23, (A1.22) and the time t^* of Remark A1.27) concern T_{alg} and are compared with the measurements only up to the fluctuations of the hitting time (a single run may reach the accuracy slightly before an in-expectation lower bound allows). For the function-value criterion divide and multiply by $\mu_F/2$ and $L_F/2$ (Proposition A1.14); for a guarantee stated for a Lyapunov function, such as Theorem 11 of [47], which follows from the one-step Lyapunov inequality of its Lemma 10 (Lemma 4.5 and Theorem 4.6 in the arXiv version 1901.08689), the conversion to the distance criterion puts the ratio $\Psi^0/\,\|\theta_0-\eta_*\|^2$ inside the logarithm, see (A1.21) below. The mechanism is the one of Lemma A1.16: the support of a block only grows when its component is queried, and a block with per-block condition number $Q\approx\kappa/n$ needs about $\sqrt{Q}$ touches with an accelerated scheme and about Q touches with a plain gradient scheme (in mean, the slow mode of a block contracts by $1-\gamma\mu_F$ per iteration for a gradient-type step of size $\gamma\le 1/R^2$, i.e. by about $1-n\gamma\mu_F\approx 1-\frac{1}{4Q}$ per touch for $\gamma=\frac{1}{4R^2}$, and by $1-O(Q^{-1/2})$ per touch for accelerated schemes). With i.i.d. uniform sampling each block is touched once per n queries on average, and $n\log(1/\varepsilon)$ queries are needed before the *average* unrevealed energy $(1-(1-q^2)/n)^t$ is small (Theorem A1.19), whatever Q. This gives:

method	upper bound on T (gradient evaluations)	lower bound on T
SAGA, $\gamma=\frac{1}{4R^2}$, $z_i^{(0)}=f_i'(\theta_0)$	$n+\max\{3n,\frac{16\kappa_{\mathrm{eff}}}{3}\}\log\frac{1+n/4}{\varepsilon}$ (Cor. A1.23)	$(\kappa-\frac{1}{2})\log\frac{\rho}{\varepsilon}$ with $\rho\ge\frac{1}{105}$ (Cor. A1.23, Lem. A1.24); $n+(n-1)\log\frac{q^2}{2\varepsilon}$ (Cor. A1.21)
L-Katyusha	$n+2\max\{8n,2\sqrt{6}\sqrt{n\kappa}\}\log\frac{C_0}{\varepsilon}$, (A1.22)	$\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon}$ for $\kappa-1\ge 2n$, and $\frac{n\log(1/(2\varepsilon))}{2\log(1/q)}$ for all κ (Prop. A1.18)
GD on F, $\gamma=1/L_F$	$\frac{nQ}{2}\log\frac{1}{\varepsilon}$ ($Q\ge L_F/\mu$)	$n(\frac{L_F}{4\mu}-\frac{1}{2})\log\frac{\rho}{\varepsilon}$ (slow modes, Lem. A1.24)
AGD on F	$O(n\sqrt{Q}\log\frac{1}{\varepsilon})$ ([11], Sec. 5.2.5)	$\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon}$ (Prop. A1.18)
any method of class (S1)	–	$(\frac{n+\sqrt{n(\kappa-1)}}{8}+\frac{n}{2}-1)\log\frac{1}{2\varepsilon}$ (Cor. A1.20)
any method of class (S2)	–	$\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon}$ (Prop. A1.18)

The first lower bounds of the first four rows are stated for $4n+1\le\kappa\le\kappa_{\max}(m,\varepsilon)$, the window

(A1.27) in which m is large enough both for the dimension condition (A1.17) and for the slow-mode mass of Lemma A1.24, and for $\varepsilon < \frac{1}{105}$; the second lower bound of the first row (Corollary A1.21) only needs (A1.17) and $\varepsilon < q^2/2$; the sequence-free bound $\frac{n\log(1/(2\varepsilon))}{2\log(1/q)}$ only needs $q^{2m} \le \varepsilon < \frac{1}{2}$; the slow-mode bound for GD uses the exact identity $e_t = (I-\gamma\nabla^2 F)^t e_0$ and $\nu' = 2\mu \le L_F$ (Proposition A1.26). In orders of magnitude, in the window $4n+1 \le \kappa \le \kappa_{\max}(m,\varepsilon)$ (beyond it, see Proposition A1.26(iii)): SAGA and GD cost $\Theta((n+\kappa)\log\frac{1}{\varepsilon})$, L-Katyusha and AGD $\Theta((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon})$, up to the logarithmic factors inside the bounds (Proposition A1.26).

Two corrections to the counts announced in the project description ("AGD $\sim n\sqrt{\kappa}\log(1/\varepsilon)$, GD $\sim n\kappa\log(1/\varepsilon)$") are necessary. Those figures are the worst-case costs over all instances with the same (n, R^2, μ), obtained from $L_F \le R^2$; they are attained only when $L_F \approx R^2$, i.e. when the stiff directions of the components are aligned (for instance identical components). On the block-separable instance $L_F \le \mu Q \approx R^2/n$ (Proposition A1.13), so GD costs about as much as SAGA and AGD about as much as L-Katyusha; AGD even attains the lower bound of Proposition A1.18. This is not a defect of the instance: the lower bound $\Omega((n+\sqrt{n\kappa})\log(1/\varepsilon))$ is a statement about the worst case over instances, and on the worst-case instance for variance-reduced methods full-gradient methods happen to be as good. What variance reduction buys is the *uniform* guarantee $(n+\sqrt{n\kappa})\log(1/\varepsilon)$ over all instances, including the aligned ones where AGD needs $n\sqrt{\kappa}\log(1/\varepsilon)$. To exhibit the four distinct orders κ, $\sqrt{n\kappa}$, $n\kappa$, $n\sqrt{\kappa}$ on one problem one can use the aligned variant $f_i(\theta) = c\,\phi(\theta) + \frac{\mu}{2}\|\theta\|^2$ for all i (one chain in $\mathbb{R}^m$, $L_F = R^2 = \mu\kappa$): there GD needs $\Theta(n\kappa\log(1/\varepsilon))$, AGD $\Theta(n\sqrt{\kappa}\log(1/\varepsilon))$, SAGA $\Theta(\kappa\log(1/\varepsilon))$ (Corollary A1.23 applies to every convex quadratic instance, and Lemma A1.24 with $n=1$, $Q=\kappa$ provides the slow-mode mass ρ for this aligned instance, which has the block structure of (A1.8) with a single block), and L-Katyusha $O((n+\sqrt{n\kappa})\log(\kappa/\varepsilon))$ by (A1.22)–(A1.21) with $L_F/\mu = \kappa$; but on that instance no chain argument certifies more than $\Omega(\sqrt{\kappa}\log(1/\varepsilon))$ for randomized methods, so only the block instance certifies the exponent $1/2$ as optimal.

A1.4.2 SAGA on quadratic instances: the exponent 1 is a theorem

Proposition A1.22 (mean iterate of SAGA on quadratics)**.** *Let the f_i be convex quadratics with Hessians $H_i \succeq 0$, let $\bar H = \frac{1}{n}\sum_i H_i = \nabla^2 F$ have eigenpairs $(\nu_l, w_l)_{l=1}^d$ (w_l orthonormal) and $\mu_F = \lambda_{\min}(\bar H)$, and for $\nu' \ge \mu_F$ let $P_{\nu'} = \sum_{l:\nu_l\le\nu'} w_l w_l^\top$ be the orthogonal projector onto the sum of the eigenspaces of $\bar H$ with eigenvalues at most ν'. For SAGA with step $\gamma > 0$, uniform sampling and any initial memory $z^{(0)}$ (deterministic, or random and independent of the sampling), $e_t = \theta_t - \eta_*$ satisfies $\mathbb{E}[e_t] = (I-\gamma\bar H)^t e_0$; hence, for every $\nu' \in [\mu_F, 1/\gamma]$,*

$$\mathbb{E}\,\|\theta_t-\eta_*\|^2 \;\ge\; \big\|(I-\gamma\bar H)^t(\theta_0-\eta_*)\big\|^2 = \sum_{l=1}^{d}(1-\gamma\nu_l)^{2t}\,\langle\theta_0-\eta_*, w_l\rangle^2 \;\ge\; (1-\gamma\nu')^{2t}\,\|P_{\nu'}(\theta_0-\eta_*)\|^2 . \tag{A1.20}$$

With $\nu' = \mu_F$ and a simple smallest eigenvalue with unit eigenvector $v_{\min}$, the last term is $(1-\gamma\mu_F)^{2t}\,\langle\theta_0-\eta_*, v_{\min}\rangle^2$.

Proof. Since $f_i'(\theta) = f_i'(\eta_*) + H_i(\theta-\eta_*)$ and $\sum_i f_i'(\eta_*) = nF'(\eta_*) = 0$, the SAGA direction (Section 5.4.4 of [11]) at time t, with $i = i(t)$ and $u_j^{(t-1)} = z_j^{(t-1)} - f_j'(\eta_*)$, equals

$$\omega_t = f_i'(\theta_{t-1}) - z_i^{(t-1)} + \frac{1}{n}\sum_j z_j^{(t-1)} = H_i e_{t-1} - u_i^{(t-1)} + \frac{1}{n}\sum_j u_j^{(t-1)}.$$

Conditionally on the past (which determines θ_{t-1} and $z^{(t-1)}$), $i(t)$ is uniform and independent, so $\mathbb{E}_{i(t)}[H_i e_{t-1}] = \bar H e_{t-1}$ and $\mathbb{E}_{i(t)}[u_i^{(t-1)}] = \frac{1}{n}\sum_j u_j^{(t-1)}$: the memory terms cancel in expectation,

whatever the memory, and $\mathbb{E}_{i(t)}[e_t] = (I - \gamma\bar{H})e_{t-1}$. Taking expectations and iterating gives $\mathbb{E}[e_t] = (I-\gamma\bar{H})^t e_0$. Then $\mathbb{E}\,\|e_t\|^2 \geq \|\mathbb{E}e_t\|^2$ (Jensen), and expanding e_0 in the eigenbasis of $\bar{H}$ gives the equality in (A1.20). Finally, for $\nu_l \leq \nu' \leq 1/\gamma$ one has $0 \leq 1-\gamma\nu' \leq 1-\gamma\nu_l$, so $(1-\gamma\nu_l)^{2t} \geq (1-\gamma\nu')^{2t}$ for every l in the range of $P_{\nu'}$, and the remaining terms of the sum are nonnegative. □

Corollary A1.23 (SAGA sandwich)**.** *Consider an instance with convex quadratic components such that $\lambda_{\max}(H_i) \leq R^2$ for all i (each f_i is then convex and R^2-smooth), with μ_F and the projectors $P_{\nu'}$ of Proposition A1.22. Let $\kappa_{\text{eff}} = R^2/\mu_F$, $\gamma = 1/(4R^2)$, and let $T_{\text{SAGA}}(\varepsilon)$ be the number of gradient evaluations (the n of an initial sweep $z_i^{(0)} = f_i'(\theta_0)$ included) that SAGA with step γ and uniform sampling needs to reach $\mathbb{E}\,\|\theta_t - \eta_*\|^2 \leq \varepsilon\,\|\theta_0 - \eta_*\|^2$.*

(a) (Lower bound, any initial memory.) If $\|P_{\nu'}(\theta_0 - \eta_*)\|^2 \geq \rho\,\|\theta_0 - \eta_*\|^2$ for some $\nu' \in [\mu_F, 4R^2]$ and $\rho \in (0,1]$, then for $\varepsilon < \rho$

$$T_{\text{SAGA}}(\varepsilon) \;\geq\; \Big(\frac{2R^2}{\nu'} - \frac{1}{2}\Big)\log\frac{\rho}{\varepsilon},$$

i.e. $(\kappa - \frac{1}{2})\log\frac{\rho}{\varepsilon}$ for $\nu' = 2\mu$ and $(2\kappa_{\text{eff}} - \frac{1}{2})\log\frac{\rho}{\varepsilon}$ for $\nu' = \mu_F$.

(b) (Upper bound.) With the initialization $z_i^{(0)} = f_i'(\theta_0)$ of Proposition 5.9 of [11], the number of iterations needed is at most

$$\max\Big\{3n, \frac{16\kappa_{\text{eff}}}{3}\Big\}\log\frac{1+n/4}{\varepsilon} \qquad (\textit{add } n \textit{ gradient evaluations for the initial sweep}),$$

and with $z^{(0)} = 0$ the same bound holds with $1 + n/4$ replaced by $1 + \frac{3\sum_i \|f_i'(\eta_*)\|^2}{16R^4\|\theta_0 - \eta_*\|^2}$.

Proof. (a) By (A1.20) with $x = \gamma\nu' = \nu'/(4R^2) \in (0,1]$, $\mathbb{E}\,\|e_t\|^2 \leq \varepsilon\,\|e_0\|^2$ forces $(1-x)^{2t} \leq \varepsilon/\rho$, i.e. $2t\log\frac{1}{1-x} \geq \log\frac{\rho}{\varepsilon}$, and $\log\frac{1}{1-x} \leq \frac{x}{1-x}$ gives $t \geq \frac{1-x}{2x}\log\frac{\rho}{\varepsilon} = (\frac{2R^2}{\nu'} - \frac{1}{2})\log\frac{\rho}{\varepsilon}$; for $\nu' = 2\mu$, $\frac{2R^2}{\nu'} = \frac{R^2}{\mu} = \kappa$. (b) The proof of Proposition 5.9 of [11] (transcribed in the SAGA note) uses the convexity and R^2-smoothness of each f_i through cocoercivity (Proposition 5.4 of the book, which requires each f_i to be convex), and strong convexity only through $\langle \theta_{t-1} - \eta_*, F'(\theta_{t-1})\rangle \geq \mu\,\|\theta_{t-1} - \eta_*\|^2$; for a quadratic F the latter holds with $\mu_F = \lambda_{\min}(\nabla^2 F)$ in place of μ, and the proof gives, for any initial memory,

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \;\leq\; \Big(1 - \min\Big\{\frac{1}{3n}, \frac{3\mu_F}{16R^2}\Big\}\Big)^t\Big[\,\|\theta_0 - \eta_*\|^2 + \frac{3}{16R^4}\sum_i \Big\|f_i'(\eta_*) - z_i^{(0)}\Big\|^2\Big].$$

For $z_i^{(0)} = f_i'(\theta_0)$ the bracket is at most $(1 + \frac{3n}{16})\,\|\theta_0 - \eta_*\|^2 \leq (1 + \frac{n}{4})\,\|\theta_0 - \eta_*\|^2$ by the R^2-Lipschitz continuity of each f_i', and for $z^{(0)} = 0$ it is the stated constant times $\|\theta_0 - \eta_*\|^2$; in both cases the right-hand side is $\leq \varepsilon\,\|\theta_0 - \eta_*\|^2$ as soon as $t \geq \max\{3n, \frac{16\kappa_{\text{eff}}}{3}\}\log\frac{C}{\varepsilon}$, C the constant in the bracket (using $(1-y)^t \leq e^{-ty}$). □

On the instance (A1.8), Proposition A1.13 gives $\mu_F = \mu\big(1 + \frac{Q-1}{4}\lambda_{\min}(A_m(\zeta))\big)$, so that $\kappa_{\text{eff}} \approx \kappa$ for large m (as $\lambda_{\min}(A_m(\zeta)) \leq \pi^2/(m+1)^2$ by Lemma A1.3) and the upper bound of Corollary A1.23 is $O(\kappa\log(1/\varepsilon))$ for $\kappa \gtrsim n$. The matching lower bound requires the initial error to carry a fixed fraction of its energy on the eigenspaces of $\nabla^2 F$ with curvature at most 2μ; this is the object of Lemma A1.24 below, and the two together make the log-log slope 1 of hypothesis (A4) a theorem for SAGA with the book's step size. The exponent is *not* a consequence of the lower bound of Section A1.3 (which only says $\sqrt{n\kappa}$), but of the fact that the step must be $O(1/R^2)$ while a fixed fraction of the initial error sits on modes of curvature between μ and 2μ. Note also that on the

instance, since $A_m(\zeta_Q)x^* = e_1 - \frac{4}{Q-1}x^*$ and $c \cdot \frac{4}{Q-1} = n\mu$, one has $f_i'(\eta_*) = \mu(\eta_* - nE_ix^*)$: its block i equals $-\mu(n-1)x^*$ and its other blocks equal μx^*, so $\|f_i'(\eta_*)\|^2 = \mu^2 n(n-1)\|x^*\|^2$ and $\sum_i \|f_i'(\eta_*)\|^2 = \mu^2 n(n-1)\|\eta_*\|^2$. The constant of Corollary A1.23(b) for the zero memory is therefore $1 + \frac{3n(n-1)}{16\kappa^2} \le 1 + \frac{3}{16}$ when $\kappa \ge n$: on this instance the zero memory, which keeps SAGA literally in class (S1) and dispenses with the initial sweep, has a guarantee (Corollary A1.23(b)) at least as good as that of the book's initialization.

For L-Katyusha, Theorem 11 of [47] (parameters $\theta_1 = \min\{\sqrt{2n\sigma/3}, \frac{1}{2}\}$, $\theta_2 = \frac{1}{2}$, $p = \frac{1}{n}$, $\sigma = \mu/L$, $\eta = \theta_2/((1+\theta_2)\theta_1) = 1/(3\theta_1)$, with $L = R^2$ the component smoothness of their Assumption 1 and μ the strong convexity of F of their Assumption 2, Assumptions 1.1 and 1.2 in the arXiv version) states that the Lyapunov function

$$\Psi^k = \frac{L(1+\eta\sigma)}{2\eta}\left\|z^k - \eta_*\right\|^2 + \frac{1}{\theta_1}\big(F(y^k) - F(\eta_*)\big) + \frac{\theta_2(1+\theta_1)}{p\,\theta_1}\big(F(w^k) - F(\eta_*)\big)$$

satisfies $\mathbb{E}\Psi^{k+1} \le (1-\theta)\mathbb{E}\Psi^k$ with $\theta = \min\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\}$ (their proof of Theorem 11; the second term is $\theta_1/(2n)$), whence $\mathbb{E}\Psi^k \le \varepsilon\Psi^0$ after $k = O((n + \sqrt{nL/\mu})\log(1/\varepsilon))$ iterations. Explicitly, $\frac{1}{\theta} = \max\{\frac{6\theta_1}{\sigma}, \frac{2n}{\theta_1}\}$ equals $\max\{3\kappa, 4n\} \le 8n$ when $\kappa \le \frac{8n}{3}$ (then $\theta_1 = \frac{1}{2}$), and $2\sqrt{6}\sqrt{n\kappa}$ when $\kappa > \frac{8n}{3}$ (then $\theta_1 = \sqrt{2n\sigma/3} < \frac{1}{2}$, $\frac{6\theta_1}{\sigma} = 2\sqrt{6}\sqrt{n\kappa}$ and $\frac{2n}{\theta_1} = \sqrt{6}\sqrt{n\kappa}$). For the distance criterion on the sequence z^k, started from $y^0 = z^0 = w^0 = \theta_0$, one has $\mathbb{E}\left\|z^k - \eta_*\right\|^2 \le \frac{2\eta}{L(1+\eta\sigma)}\mathbb{E}\Psi^k \le (1-\theta)^k C_0\|\theta_0 - \eta_*\|^2$ with, using $F(\theta_0) - F(\eta_*) \le \frac{L_F}{2}\|\theta_0 - \eta_*\|^2$ and $\frac{\eta}{\theta_1} = \frac{1}{3\theta_1^2}$,

$$C_0 = 1 + \frac{L_F}{3L\theta_1^2}\Big(1 + \frac{n(1+\theta_1)}{2}\Big) \;\le\; \begin{cases} 1 + \frac{5}{8}\frac{L_F}{\mu} \;\le\; 1 + \frac{5}{8}Q & \text{if } \kappa \ge \frac{8n}{3} \text{ and } n \ge 2, \\ n + \frac{7}{3} & \text{if } \kappa < \frac{8n}{3}, \end{cases} \tag{A1.21}$$

because in the first case $\frac{1}{3\theta_1^2} = \frac{1}{2\sigma n} = \frac{\kappa}{2n}$, so that $\frac{L_F}{3L\theta_1^2}(1 + \frac{n(1+\theta_1)}{2}) = \frac{L_F}{\mu}(\frac{1}{2n} + \frac{1+\theta_1}{4}) \le \frac{L_F}{\mu}(\frac{1}{4} + \frac{3}{8})$, and in the second case $\frac{1}{3\theta_1^2} = \frac{4}{3}$ and $L_F \le L$. Hence, in expected gradient evaluations,

$$T_{\mathrm{LK}}(\varepsilon) \;\le\; n + \frac{2}{\theta}\log\frac{C_0}{\varepsilon} \;\le\; n + 2\max\{8n,\, 2\sqrt{6}\sqrt{n\kappa}\}\log\frac{C_0}{\varepsilon}, \tag{A1.22}$$

where the number of iterations is treated as a real number, as for SAGA (rounding it up adds 2), and the logarithm carries the extra term $\log(1 + \frac{5}{8}Q) \approx \log(\kappa/n)$ in the regime $\kappa \gg n$ (for the sequence y^k, bounded through $F(y^k) - F(\eta_*) \le \theta_1\Psi^k$ and μ_F-strong convexity, the constant is $C_y = \frac{3L\theta_1^2(1+\eta\sigma)}{\mu_F} + \frac{L_F}{\mu_F}\big(1 + \frac{n(1+\theta_1)}{2}\big) \le 3n + \big(1 + \frac{3n}{4}\big)Q = O(\kappa)$ for $\kappa \ge \frac{8n}{3}$, instead of C_0); this must be kept in mind when reading slopes. L-Katyusha is not in class (S1) (its full gradients are not i.i.d. single queries, Section A1.1), so Theorem A1.19 does not apply to it; it is in class (S2), and the lower bound (A1.14) of Proposition A1.18, which holds for every index sequence, together with (A1.22) gives $T_{\mathrm{LK}}(\varepsilon) = \Theta(\sqrt{n\kappa}\log(1/\varepsilon))$ for $\kappa \ge 3n$, up to the constants and the $\log(\kappa/n)$ just mentioned, as long as (A1.17) holds: this is the slope 1/2. This exponent is certified optimal within class (S2), hence within its subclass (S1), by Proposition A1.18 on this instance, and within the class of [37] by Theorem A1.12 on their own instance.

A1.4.3 The testable prediction (A4) and the finite-dimension effect

Corollary A1.23(a) needs the initial error to carry a fixed fraction of its energy on the slow modes of $\nabla^2 F$. On the instance, the eigenvectors of the chain are explicit enough for this fraction to be bounded below uniformly in κ, as long as the chain is long compared with the decay length

$1/\log(1/q) \approx \sqrt{Q}/2$ of the minimizer (the slow modes are the sine waves whose wavelength exceeds this decay length; the slowest mode alone carries a fraction of order m^{-3} only when $m \gg \sqrt{Q}$, which is why the full slow subspace, and not a single eigenvector, must be used).

Lemma A1.24 (slow-mode mass of the initial error)**.** *Let $m \geq 2$, $\zeta \in [1,2]$ and $s \in (0,1]$. Let $0 < \lambda_1 < \cdots < \lambda_m < 4$ be the eigenvalues of the chain matrix $A_m(\zeta)$ of (A1.2) with orthonormal eigenvectors $v_1, \dots, v_m$, and let $\Pi_s = \sum_{k:\lambda_k \leq s} v_k v_k^\top$ be the orthogonal projector onto the eigenspaces with eigenvalue at most s. For $x^* = (A_m(\zeta) + sI)^{-1} e_1$ and $K_0 := \lfloor (m+1)\sqrt{s}/\pi \rfloor$,*

$$\|\Pi_s x^*\|^2 \;\geq\; \frac{7\pi^2(1 - s/4)}{96\, m\, s^2 (2m+1)^2}\, K_0\big(4K_0^2 - 1\big). \tag{A1.23}$$

If moreover $\zeta = \zeta_Q$ and $s = \frac{4}{Q-1}$ (so that $x^* = (q^j)_{j \leq m}$ is the block of η_*, Proposition A1.14(a)), the slow-mode fraction $\rho := \|\Pi_s x^*\|^2 / \|x^*\|^2$ satisfies, with $y := (m+1)\sqrt{s}/\pi$,

$$\rho \;\geq\; \frac{7(1 - s/4)}{192\pi} \cdot \frac{K_0(4K_0^2 - 1)}{y^3} \;\geq\; \begin{cases} 1/105 & \text{if } y \geq 2, \\ 1/310 & \text{if } y \geq 1. \end{cases} \tag{A1.24}$$

On the instance (A1.8) with $\theta_0 = 0$, $\|P_{2\mu}(\theta_0 - \eta_*)\|^2 = \rho\, \|\theta_0 - \eta_*\|^2$, where $P_{2\mu}$ is the projector of Proposition A1.22 for $\nu' = 2\mu$.

Proof. Step 1: the eigenvectors are sampled sines. Write $A = A_m(\zeta)$. By Lemma A1.3, $A_m(1) \preceq A \preceq A_m(2)$, so Weyl's monotonicity theorem (the Courant–Fischer characterization of the k-th eigenvalue) gives, with the eigenvalues of $A_m(1)$ and $A_m(2)$ computed in Lemma A1.3,

$$4\sin^2 \frac{(2k-1)\pi}{2(2m+1)} \;\leq\; \lambda_k \;\leq\; 4\sin^2 \frac{k\pi}{2(m+1)}, \qquad k = 1, \dots, m. \tag{A1.25}$$

In particular $0 < \lambda_k < 4$, and we may write $\lambda_k = 2 - 2\cos\theta_k = 4\sin^2(\theta_k/2)$ with $\theta_k \in (0, \pi)$. Let $Av = \lambda v$, $v \neq 0$, $\lambda = 2 - 2\cos\theta$. Rows 1 to $m-1$ of this system read $v_2 = 2\cos\theta\, v_1$ and $v_{j+1} = 2\cos\theta\, v_j - v_{j-1}$ for $2 \leq j \leq m-1$; the sequence $u_j = \sin(j\theta)$ satisfies the same recursion with $u_0 = 0$, since $\sin((j+1)\theta) + \sin((j-1)\theta) = 2\cos\theta \sin(j\theta)$, and a solution of this second-order recursion with prescribed $v_0 = 0$ is determined by v_1. Hence $v_j = (v_1/\sin\theta)\sin(j\theta)$ for $j = 1, \dots, m$; in particular $v_1 \neq 0$ and every eigenvalue is simple (the last row of the system is the equation that selects θ). Normalizing, $1 = \|v\|^2 = (v_1^2/\sin^2\theta) S(\theta)$ with $S(\theta) = \sum_{j=1}^m \sin^2(j\theta) \leq m$, hence

$$(v_k)_1^2 = \frac{\sin^2\theta_k}{S(\theta_k)} \;\geq\; \frac{\sin^2\theta_k}{m} = \frac{\lambda_k(1 - \lambda_k/4)}{m}, \tag{A1.26}$$

because $\sin^2\theta = 4\sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2} = \lambda(1 - \frac{\lambda}{4})$.

Step 2: Fourier coefficients of x^*. With $B = A + sI$, $Bv_k = (\lambda_k + s)v_k$ and B symmetric, $\langle x^*, v_k \rangle = \langle e_1, B^{-1} v_k \rangle = (v_k)_1/(\lambda_k + s)$, so that

$$\|\Pi_s x^*\|^2 = \sum_{k:\lambda_k \leq s} \frac{(v_k)_1^2}{(\lambda_k + s)^2}, \qquad \|x^*\|^2 = \sum_{k=1}^m \frac{(v_k)_1^2}{(\lambda_k + s)^2}.$$

By (A1.25) and $\sin x \leq x$, $\lambda_k \leq (k\pi/(m+1))^2 \leq s$ for $k \leq K_0$: the slow set $\{k : \lambda_k \leq s\}$ contains $\{1, \dots, K_0\}$. For such k, $(\lambda_k + s)^2 \leq 4s^2$ and $1 - \lambda_k/4 \geq 1 - s/4$, so (A1.26) gives

$$\|\Pi_s x^*\|^2 \;\geq\; \frac{1 - s/4}{4ms^2} \sum_{k=1}^{K_0} \lambda_k.$$

Step 3: lower bound on the slow eigenvalues. For $k \le K_0$ put $x_k = \frac{(2k-1)\pi}{2(2m+1)}$. Then $x_k \le \frac{K_0\pi}{2m+1} \le \frac{(m+1)\sqrt{s}}{2m+1} \le \frac{3}{5}\sqrt{s} \le \frac{3}{5}$ (as $m \ge 2$ and $s \le 1$), and $\sin x \ge x - \frac{x^3}{6} \ge x\big(1 - \frac{9}{25\cdot 6}\big) = \frac{47}{50}x$ on $[0, \frac{3}{5}]$, so $\sin^2 x_k \ge \frac{2209}{2500}x_k^2 \ge \frac{7}{8}x_k^2$. By (A1.25), $\lambda_k \ge 4\sin^2 x_k \ge \frac{7}{2}x_k^2 = \frac{7\pi^2(2k-1)^2}{8(2m+1)^2}$, and $\sum_{k=1}^{K_0}(2k-1)^2 = \frac{K_0(4K_0^2-1)}{3}$. Inserting this in Step 2 gives (A1.23).

Step 4: the fraction for $\zeta = \zeta_Q$. By Proposition A1.14(a), $\|x^*\|^2 = \frac{q^2(1-q^{2m})}{1-q^2} \le \frac{q^2}{1-q^2} = \frac{(\sqrt{Q}-1)^2}{4\sqrt{Q}} \le \frac{Q-1}{4\sqrt{Q}} = \frac{1}{s\sqrt{Q}} \le \frac{1}{2\sqrt{s}}$, using $(\sqrt{Q}-1)^2 \le (\sqrt{Q}-1)(\sqrt{Q}+1) = Q-1 = \frac{4}{s}$ and $\sqrt{Q} \ge \sqrt{4/s} = \frac{2}{\sqrt{s}}$. Hence $\rho \ge 2\sqrt{s}\cdot\frac{7\pi^2(1-s/4)K_0(4K_0^2-1)}{96ms^2(2m+1)^2}$; substituting $s = \pi^2y^2/(m+1)^2$, i.e. $s^{-3/2} = (m+1)^3/(\pi^3y^3)$, and using $\frac{(m+1)^3}{m(2m+1)^2} = \frac{m+1}{m}\big(\frac{m+1}{2m+1}\big)^2 \ge \frac{1}{4}$, we obtain the first inequality in (A1.24). For $y \in [K, K+1)$ one has $K_0 = K$ and $\frac{K_0(4K_0^2-1)}{y^3} \ge g(K) := \frac{K(4K^2-1)}{(K+1)^3}$; writing $u = K+1$, $g = 4 - \frac{12}{u} + \frac{11}{u^2} - \frac{3}{u^3}$ has derivative $\frac{12u^2-22u+9}{u^4} > 0$ for $u \ge 2$ (the larger root of $12u^2-22u+9$ is $\frac{22+\sqrt{52}}{24} < 2$), so g is increasing on $K \ge 1$, with $g(1) = \frac{3}{8}$ and $g(2) = \frac{30}{27}$. With $1 - \frac{s}{4} \ge \frac{3}{4}$: for $y \ge 2$, $\rho \ge \frac{7}{192\pi}\cdot\frac{3}{4}\cdot\frac{30}{27} = \frac{35}{1152\pi} > \frac{1}{105}$, and for $y \ge 1$, $\rho \ge \frac{7}{192\pi}\cdot\frac{3}{4}\cdot\frac{3}{8} = \frac{63}{6144\pi} > \frac{1}{310}$.

Step 5: the instance. $\nabla^2 F = \operatorname{diag}(H_b, \dots, H_b)$ with $H_b = \frac{c}{n}A_m(\zeta_Q) + \mu I = \frac{\mu}{s}\big(A_m(\zeta_Q) + sI\big)$, because $\frac{c}{n} = \frac{\mu(Q-1)}{4} = \frac{\mu}{s}$; its eigenvalues are $\mu(1+\lambda_k/s)$, each with the n eigenvectors $E_1v_k, \dots, E_nv_k$, and $\mu(1+\lambda_k/s) \le 2\mu$ iff $\lambda_k \le s$. Since $\theta_0 - \eta_* = -(x^*, \dots, x^*)$, $\|P_{2\mu}(\theta_0-\eta_*)\|^2 = \sum_{i=1}^n \|\Pi_s x^*\|^2 = \rho\, n\, \|x^*\|^2 = \rho\, \|\theta_0 - \eta_*\|^2$. □

Remark A1.25 (sharpness of the slow-mode bound, and the exhausted chain). The constants in (A1.24) are not sharp: the proof loses a factor $m/S(\theta_k)$ in $S(\theta) \le m$ (for every mode, the first one included, $S(\theta_k) = \frac{m}{2} - \frac{\sin(m\theta_k)\cos((m+1)\theta_k)}{2\sin\theta_k} \approx \frac{m}{2}$, exactly $\frac{m+1}{2}$ for $\zeta = 2$ and $\frac{2m+1}{4}$ for $\zeta = 1$; for $m = 200$, $S(\theta_k)/m \in [0.50, 0.61]$ along the instance, the largest values being those of the first mode), a factor up to 4 in $(\lambda_k + s)^2 \le 4s^2$, and the rounding in K_0. The exact ρ is computable from the $m \times m$ tridiagonal eigenproblem and is the quantity to report in the experiments; for $n = 100$, $m = 200$ and $\kappa \in [10^3, 3\cdot 10^5]$ (so that $y \in [2.3, 40.5]$; 4001 log-spaced values of κ, completed by the jump points of both quantities) the exact ρ ranges from 0.13 to 0.35, against 0.013–0.046 for the bound (A1.24), and the ratio of the bound to the exact ρ lies between 0.06 and 0.21. The exact ρ is not monotone in κ: it drops each time an eigenvalue λ_k crosses s (one mode leaves the slow set); its largest values occur near $\kappa = 10^3$ and its smallest ones (≈ 0.13) just above $\kappa \approx 2.2\cdot 10^5$, where the third mode leaves the slow set. For the infinite chain ($m = \infty$) the spectral measure of e_1 for A is $d\nu(\lambda) = \frac{1}{2\pi}\sqrt{\lambda(4-\lambda)}\,d\lambda$ on $[0,4]$ (the orthogonal polynomials of the recursion of Step 1 are Chebyshev polynomials of the second kind), $\|x^*\|^2 = \int_0^4 (\lambda+s)^{-2}d\nu(\lambda)$, and the slow fraction $\int_0^s (\lambda+s)^{-2}d\nu \big/ \int_0^4 (\lambda+s)^{-2}d\nu$ tends, as $s \to 0$, to $\frac{\int_0^1 \sqrt{u}\,(1+u)^{-2}du}{\int_0^\infty \sqrt{u}\,(1+u)^{-2}du} = \frac{\pi/4 - 1/2}{\pi/2} = \frac{1}{2} - \frac{1}{\pi} \approx 0.18$; this is the value approached by the exact ρ when $m\sqrt{s} \to \infty$ and then $s \to 0$, and it is not used in the proofs. At the other extreme, when $\kappa \to \infty$ with m fixed ($s \to 0$, $y \to 0$), eventually no eigenvalue of $A_m(\zeta_Q)$ lies below s and the slow set is empty; the relevant quantity is then the single-mode overlap $\delta := \langle x^*, v_1\rangle^2 / \|x^*\|^2$ of Corollary A1.23(a) with $\nu' = \mu_F$ (the eigenspace of μ_F is spanned by $E_1v_1, \dots, E_nv_1$, so $\|P_{\mu_F}(\theta_0-\eta_*)\|^2 = n\,\langle x^*, v_1\rangle^2 = \delta\,\|\theta_0-\eta_*\|^2$). Since $\zeta_Q \to 1$, $x^* = (A_m(\zeta_Q) + sI)^{-1}e_1 \to A_m(1)^{-1}e_1$ and $v_1(\zeta_Q) \to v_1(1)$ (simple eigenvalue, eigenvectors normalized with positive first coordinate), so $\delta \to \delta_\infty(m) := \big\langle A_m(1)^{-1}e_1, v_1(1)\big\rangle^2 / \big\|A_m(1)^{-1}e_1\big\|^2$. In fact $A_m(1)\mathbf{1} = e_1$ (row sums), so $A_m(1)^{-1}e_1 = \mathbf{1}$, and $v_1(1)$ is the normalized vector $(\sin\frac{j\pi}{2m+1})_{j\le m}$ (Lemma A1.3), whose entries are positive; hence $\delta_\infty(m) = \langle \mathbf{1}, v_1(1)\rangle^2/m = \cot^2\big(\frac{\pi}{2(2m+1)}\big)/(m(2m+1)) > 0$, which is 0.81 for $m = 200$ and tends to $8/\pi^2$. The overlap δ approaches this value only once $s \ll \lambda_1$: for $m = 200$

and $n = 100$, $\delta \approx 0.18$ at $\kappa = 10^6$ (one mode is still slow there, so $\delta = \rho$), 0.56 at 10^7 and 0.74 at 10^8.

Proposition A1.26 (testable prediction of hypothesis A4). *Fix $n \geq 2$, $m \geq 2$, μ and a relative accuracy $\varepsilon \in (0, \frac{1}{105})$, start at $\theta_0 = 0$, and let κ vary. On the instance (A1.8) with $\zeta = \zeta_Q$ (Q, q as in Proposition A1.14; μ_F, L_F as in Proposition A1.13), let $T_{\mathrm{alg}}(\varepsilon)$ be the number of gradient evaluations (its expectation for L-Katyusha) that an algorithm needs to reach $\mathbb{E}\,\|\theta - \eta_*\|^2 \leq \varepsilon\,\|\theta_0 - \eta_*\|^2$: SAGA with $\gamma = \frac{1}{4R^2}$ and $z_i^{(0)} = f_i'(0)$, the n gradients of this sweep being counted; L-Katyusha with the parameters of Theorem 11 of [47], θ being its mirror sequence z^k, charged n gradients for $F'(w^0)$ and, per iteration, one gradient plus n when the reference point is refreshed (probability $p = \frac{1}{n}$; the gradients $f_i'(w^k)$ of each full pass are stored), i.e. 2 per iteration in expectation; GD and AGD on F, n per iteration. Let C_0 be the constant of (A1.21).*

(i) (Plateau, $\kappa \leq n$.) For SAGA, under (A1.17) and $\varepsilon < q^2/2$,

$$n + (n-1)\log\frac{q^2}{2\varepsilon} \;\leq\; T_{\mathrm{SAGA}}(\varepsilon) \;\leq\; n + \max\Big\{3n, \frac{16\kappa}{3}\Big\}\log\frac{1+n/4}{\varepsilon} \;\leq\; n + \frac{16n}{3}\log\frac{1+n/4}{\varepsilon},$$

both sides being $\Theta(n\log(1/\varepsilon))$ once $\varepsilon \ll q^2$ and $\varepsilon \ll 1/n$ (the upper bound is $\Theta(n\log(n/\varepsilon))$ in general): the plateau with slope ≈ 0 is a theorem for SAGA. For L-Katyusha only

$$\frac{n\log(1/(2\varepsilon))}{2\log(1/q)} \;\leq\; T_{\mathrm{LK}}(\varepsilon) \;\leq\; n + 16n\log\frac{C_0}{\varepsilon}, \qquad C_0 \leq n + \tfrac{7}{3},$$

is proved (the lower bound under (A1.17)); the two sides are of the same order $n\log(1/\varepsilon)$ only when $\log(1/q) = O(1)$, i.e. when κ is of order n (for $\kappa = n$, $\log(1/q) = 2\,\mathrm{artanh}\,\big((2-\frac{1}{n})^{-1/2}\big)$, which decreases to $\log(3+2\sqrt{2}) \approx 1.763$ as $n \to \infty$: 1.81 for $n = 16$, 1.77 for $n = 100$). For $\kappa \ll n$, $\log(1/q) \approx \log\frac{4n}{\kappa-1}$, and a gain of this logarithmic order over $n\log(1/\varepsilon)$ is not excluded for L-Katyusha; it is realized by GD on F with step $1/L_F$, which contracts the error by $1 - \frac{\mu_F}{L_F} \leq 1 - \frac{1}{Q} = \frac{\kappa-1}{n+\kappa-1} \approx \frac{\kappa-1}{n}$ per full gradient, in agreement with Theorem A1.11(c) and with item (ii) of the assessment of hypothesis (A1) in Section A1.2. In the upper bounds, the second term takes over at $\kappa = \frac{9}{16}n$ for SAGA ($\frac{16\kappa}{3} = 3n$) and at $\kappa = \frac{8}{3}n$ for L-Katyusha (its momentum parameter θ_1 reaches $\frac{1}{2}$).

(ii) (Power laws.) Let

$$\kappa_{\max}(m,\varepsilon) := 1 + n\min\Big\{\frac{16m^2}{\log^2(1/\varepsilon)} - 1,\; \frac{(m+1)^2}{\pi^2}\Big\}, \tag{A1.27}$$

where the first term is the sufficient form $m \geq \frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon}$ of the dimension condition (A1.17) solved for κ (so that $q^{2m} \leq \varepsilon$), and the second is the condition $y \geq 2$ of Lemma A1.24 solved for κ. In the window $4n+1 \leq \kappa \leq \kappa_{\max}(m,\varepsilon)$, with $\rho = \rho(m,\kappa)$ the slow-mode fraction of Lemma A1.24,

$$\Big(\kappa - \frac{1}{2}\Big)\log\frac{\rho}{\varepsilon} \;\leq\; T_{\mathrm{SAGA}}(\varepsilon) \;\leq\; n + \frac{16\kappa}{3}\log\frac{1+n/4}{\varepsilon}, \qquad \rho = \rho(m,\kappa) \geq \frac{1}{105}, \tag{A1.28}$$

$$\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon} \;\leq\; T_{\mathrm{LK}}(\varepsilon) \;\leq\; n + 4\sqrt{6}\,\sqrt{n\kappa}\,\log\frac{C_0}{\varepsilon}, \qquad C_0 \leq 1 + \frac{5}{8}Q, \tag{A1.29}$$

so that $T_{\mathrm{SAGA}} = \Theta(\kappa\log(1/\varepsilon))$ and $T_{\mathrm{LK}} = \Theta(\sqrt{n\kappa}\log(1/\varepsilon))$ up to the displayed constants and the $\log(1+\frac{5}{8}Q)$ of (A1.29): κ enters T_{SAGA} and T_{LK} with the exponents 1 and $\frac{1}{2}$ as orders of growth (not as the slope of a line fitted over the window, see Remark A1.27), and

$T_{\mathrm{SAGA}}/T_{\mathrm{LK}} \asymp \sqrt{\kappa/n}$. *GD on F (step $1/L_F$) and AGD on F have the same two orders of growth:* $n(\frac{L_F}{4\mu}-\frac{1}{2})\log\frac{\rho}{\varepsilon} \le T_{\mathrm{GD}} \le \frac{nQ}{2}\log\frac{1}{\varepsilon}$ *with* $L_F \ge \mu(1+\frac{13}{20}(Q-1))$, *and* $\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon} \le T_{\mathrm{AGD}} = O(n\sqrt{Q}\log\frac{1}{\varepsilon})$.

(iii) *(Saturation.) For $\kappa \gg nm^2$ the chain is exhausted: $\mu_F/\mu = 1+\frac{\kappa-1}{4n}\lambda_{\min}(A_m(\zeta_Q))$ grows linearly in κ, so*

$$\kappa_{\mathrm{eff}} = \frac{R^2}{\mu_F} = \frac{\kappa}{1+\frac{\kappa-1}{4n}\lambda_{\min}(A_m(\zeta_Q))} \xrightarrow[\kappa\to\infty]{} \frac{4n}{\lambda_{\min}(A_m(1))} = \frac{n}{\sin^2\frac{\pi}{2(2m+1)}} \approx \frac{4n(2m+1)^2}{\pi^2}, \tag{A1.30}$$

because $\zeta_Q \to 1$ and $\lambda_{\min}$ is continuous in the matrix; for every finite κ, Lemma A1.3 gives the two-sided bound $\frac{\kappa}{1+\frac{\kappa-1}{n}\sin^2\frac{\pi}{2(m+1)}} \le \kappa_{\mathrm{eff}} \le \frac{\kappa}{1+\frac{\kappa-1}{n}\sin^2\frac{\pi}{2(2m+1)}}$. *Then* $(2\kappa_{\mathrm{eff}}-\frac{1}{2})\log\frac{\delta}{\varepsilon} \le T_{\mathrm{SAGA}}(\varepsilon) \le n+\frac{16\kappa_{\mathrm{eff}}}{3}\log\frac{1+n/4}{\varepsilon}$ *for $\varepsilon<\delta$, with the single-mode overlap $\delta = \langle x^*, v_1\rangle^2/\|x^*\|^2$ (x^*, v_1 as in Lemma A1.24), which tends to $\delta_\infty(m)>0$ (Remark A1.25): T_{SAGA} saturates at $\Theta(\kappa_{\mathrm{eff}}\log(1/\varepsilon)) = \Theta(nm^2\log(1/\varepsilon))$, while T_{LK} is no longer bounded below by $\sqrt{n\kappa}$ (the hypothesis $q^{2m}\le\varepsilon$ of Proposition A1.18 fails) and its behaviour depends on the strong-convexity constant used to set the parameter σ of L-Katyusha. Since F is μ_F-strongly convex, Theorem 11 of [47] also applies with $\sigma=\mu_F/R^2$ (the tuning used in the experiments, where $\mu=\lambda_{\min}(\nabla^2 F)$ is passed to the method): (A1.21)–(A1.22) then hold with μ_F and κ_{eff} in place of μ and κ, so that $T_{\mathrm{LK}} = O\big((n+\sqrt{n\kappa_{\mathrm{eff}}})\log\frac{C_0}{\varepsilon}\big)$ with C_0 bounded in κ, and T_{LK} saturates at $O(nm\log\frac{C_0}{\varepsilon})$. With the nominal $\sigma=\mu/R^2$ of (A1.22) no saturation is proved: the upper bound keeps growing like $\sqrt{n\kappa}$, and the momentum $\theta_1=\sqrt{2n\mu/(3R^2)}$ tends to 0, whereas the value matched to the curvature, $\sqrt{2n\mu_F/(3R^2)}$, does not. Thus T_{SAGA}, and T_{LK} with $\sigma=\mu_F/R^2$, stay bounded as $\kappa\to\infty$, and their log-log slopes against κ tend to 0.*

Proof. (i) For SAGA, the lower bound is Corollary A1.21 (the sweep is one query to every component, Lemma A1.16(ii)) plus the n gradients of the sweep, and the upper bound is Corollary A1.23(b) with $\max\{3n, \frac{16\kappa_{\mathrm{eff}}}{3}\} \le \max\{3n, \frac{16\kappa}{3}\} \le \frac{16n}{3}$, since $\kappa_{\mathrm{eff}}\le\kappa\le n$; the maximum equals $3n$ only when $\frac{16\kappa_{\mathrm{eff}}}{3}\le 3n$, i.e. $\kappa_{\mathrm{eff}}\le\frac{9}{16}n$. For L-Katyusha, the lower bound is (A1.14), valid for every index sequence under (A1.17), and the upper bound is (A1.22) with $2\max\{8n, 2\sqrt{6}\sqrt{n\kappa}\} = 16n$ for $\kappa\le n$ and $C_0 \le n+\frac{7}{3}$ from (A1.21). For $\kappa \ll n$: $Q-1=\frac{\kappa-1}{n}$, $\sqrt{Q}-1\approx\frac{Q-1}{2}$ and $q=\frac{\sqrt{Q}-1}{\sqrt{Q}+1}\approx\frac{Q-1}{4}=\frac{\kappa-1}{4n}$. The GD contraction follows from $\left\|(I-\frac{1}{L_F}\nabla^2 F)e\right\| \le (1-\frac{\mu_F}{L_F})\|e\|$ and $\frac{L_F}{\mu_F}\le\frac{L_F}{\mu}\le Q$ (Proposition A1.13).

(ii) The window conditions: $\kappa\ge 4n+1$ means $Q\ge 5$, i.e. $s=\frac{4}{Q-1}\le 1$; $\kappa\le 1+\frac{n(m+1)^2}{\pi^2}$ means $Q-1\le\frac{(m+1)^2}{\pi^2}$, i.e. $(m+1)\sqrt{s}\ge 2\pi$, i.e. $y\ge 2$; and $\kappa\le 1+n(\frac{16m^2}{\log^2(1/\varepsilon)}-1)$ is $m\ge\frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon}$, the sufficient form of (A1.17). For SAGA, Lemma A1.24 gives $\|P_{2\mu}(\theta_0-\eta_*)\|^2 = \rho\|\theta_0-\eta_*\|^2$ with $\rho\ge\frac{1}{105}>\varepsilon$, so Corollary A1.23(a) with $\nu'=2\mu\le 4R^2$ gives the lower bound, for any initial memory and in particular after the sweep (the mean-iterate identity of Proposition A1.22 holds whatever the memory, and the sweep only adds n to the count); the upper bound is Corollary A1.23(b) with $\kappa_{\mathrm{eff}}\le\kappa$ and $\frac{16\kappa}{3}\ge 3n$ for $\kappa\ge\frac{9}{16}n$. For L-Katyusha, the lower bound is (A1.14), which needs $q^{2m}\le\varepsilon$ and $\kappa-1\ge 2n$, both true in the window, and the upper bound is (A1.22)–(A1.21) with $\kappa>\frac{8}{3}n$; it also holds when σ is set to $\mu_F/R^2\ge\mu/R^2$ (the tuning of the experiments), since θ does not decrease and C_0 does not increase when σ increases. For GD, $e_t=(I-\frac{1}{L_F}\nabla^2 F)^t e_0$ exactly, so the second inequality of (A1.20) applies with $\gamma=1/L_F$ and $\nu'=2\mu$, which is $\le 1/\gamma = L_F$ because $\lambda_{\max}(A_m(\zeta_Q)) \ge \lambda_{\max}(A_m(1)) = 4\sin^2\frac{(2m-1)\pi}{2(2m+1)} \ge 4\sin^2\frac{3\pi}{10} > \frac{13}{5}$ for $m\ge 2$ (Lemma A1.3 and Weyl's inequality), whence $L_F = \mu(1+\frac{Q-1}{4}\lambda_{\max}(A_m(\zeta_Q))) \ge \mu(1+\frac{13}{20}(Q-1)) \ge \frac{18}{5}\mu$ for $Q\ge 5$;

the argument of Corollary A1.23(a) then gives $t \geq (\frac{L_F}{4\mu} - \frac{1}{2}) \log \frac{\rho}{\varepsilon}$ iterations of n gradients, while $(1 - \frac{\mu_F}{L_F})^{2t} \leq \varepsilon$ for $t \geq \frac{L_F}{2\mu_F} \log \frac{1}{\varepsilon}$ with $\frac{L_F}{\mu_F} \leq Q$. For AGD, Proposition A1.18 and Section 5.2.5 of [11] ($O(\sqrt{L_F/\mu_F} \log \frac{1}{\varepsilon})$ iterations of n gradients, and $L_F/\mu_F \leq Q$) give the two sides. Since $\rho \geq \frac{1}{105}$ and $C_0 \leq 1 + \frac{5}{8}Q$ depend on κ only through the displayed logarithm, all four pairs of bounds have the announced orders.

(iii) As $\kappa \to \infty$ with m fixed, $\zeta_Q = \frac{\sqrt{Q}+3}{\sqrt{Q}+1} \to 1$, the eigenvalues of $A_m(\zeta)$ depend continuously on ζ, and $\frac{\kappa}{1+\frac{\kappa-1}{4n}\lambda} \to \frac{4n}{\lambda}$; the two-sided bound is Lemma A1.3 inserted in Proposition A1.13(b). The bounds on T_{SAGA} are Corollary A1.23 with $\nu' = \mu_F$ and the value of $\|P_{\mu_F}(\theta_0 - \eta_*)\|^2$ computed in Remark A1.25, which also gives the limit of δ; for the upper bound, $\frac{16\kappa_{\mathrm{eff}}}{3} \geq 3n$ for κ large. For T_{LK}, (A1.22) holds for all κ (Theorem 11 of [47] has no dimension hypothesis); applied with the strong-convexity constant μ_F and $\sigma = \mu_F/R^2$, the same theorem gives (A1.21)–(A1.22) with μ_F, κ_{eff} in place of μ, κ, and κ_{eff} and L_F/μ_F converge as $\kappa \to \infty$ by (A1.30) and the same continuity argument (uniformly, $\frac{L_F}{\mu_F} \leq \frac{\lambda_{\max}(A_m(\zeta_Q))}{\lambda_{\min}(A_m(\zeta_Q))} \leq 1/\sin^2 \frac{\pi}{2(2m+1)}$ by Lemma A1.3). On the other hand (A1.14) requires $q^{2m} \leq \varepsilon$, i.e. $\kappa \leq \kappa_{\mathrm{dim}}(m, \varepsilon) := 1 + n\big[\big(\frac{1+e^{-\ell/(2m)}}{1-e^{-\ell/(2m)}}\big)^2 - 1\big]$ with $\ell = \log \frac{1}{\varepsilon}$ (a threshold $\geq \kappa_{\max}(m, \varepsilon)$, since (A1.27) uses the sufficient form of (A1.17)), and fails for $\kappa > \kappa_{\mathrm{dim}}(m, \varepsilon)$, in particular for $\kappa \gg nm^2$. □

Numerically (for $n = 8$, $m = 16$, $\varepsilon = 10^{-6}$), with the nominal $\sigma = \mu/R^2$ the cost of L-Katyusha keeps growing long after κ_{eff} has saturated and levels off only at a non-accelerated level, about $\frac{3}{2}\kappa_{\mathrm{eff}} \log \frac{1}{\varepsilon}$ iterations.

Remark A1.27 (what to expect in a plot). For $\varepsilon = 10^{-6}$ ($\log(1/\varepsilon) \approx 13.8$), $n = 100$ and $m = 200$: the dimension condition gives $1 + n(\frac{16m^2}{\log^2(1/\varepsilon)} - 1) \approx 3.4 \cdot 10^5$ and the slow-mode condition $1 + \frac{n(m+1)^2}{\pi^2} \approx 4.1 \cdot 10^5$, so $\kappa_{\max} \approx 3.4 \cdot 10^5$ and the window $401 \leq \kappa \leq 3.4 \cdot 10^5$ of Proposition A1.26(ii) spans almost three decades, enough to read the slopes 1 and $\frac{1}{2}$, with the plateau below $\kappa \approx n$ and the bend above $\kappa_{\max}$. Two caveats. First, a sandwich $c_1\kappa \leq T \leq c_2\kappa$ pins the slope of a line fitted over D decades only up to $\pm \log_{10}(c_2/c_1)/D$; for SAGA, $c_2/c_1 = \frac{16}{3} \cdot \frac{\log((1+n/4)/\varepsilon)}{\log(\rho/\varepsilon)} \approx 7.7$ with the smallest value of the exact ρ over the window, $\rho \approx 0.13$ (reached just above $\kappa \approx 2.2 \cdot 10^5$, Remark A1.25), and for L-Katyusha the ratio of the two sides of (A1.29) is between 91 (bottom of the window, C_0 bounded by $1 + \frac{5}{8}Q$ at that point) and about 128 (C_0 replaced by its bound $1 + \frac{5}{8}Q$ at the top of the window, which gives constants valid over the whole window). The order of growth is a theorem; the numerical value of the slope must be read from exact quantities: the mean-iterate time $t^*(\kappa) = \min\{t : \|(I - \gamma\nabla^2 F)^t(\theta_0 - \eta_*)\|^2 \leq \varepsilon \|\theta_0 - \eta_*\|^2\}$, computable from the m eigenpairs of $A_m(\zeta_Q)$, is a lower bound on T_{SAGA} (first inequality of (A1.20)), and for these parameters we found $t^* \approx 1.4\,\kappa \log(1/\varepsilon)$ with log-log slope 0.99 over $\kappa \in [10^3, 3 \cdot 10^5]$. Second, with $m = 20$ the window collapses ($\kappa_{\max} \approx 3.3 \cdot 10^3$, less than a decade above $4n + 1$): the slopes measured over $\kappa \in [10^3, 10^6]$ would be far below 1 and $\frac{1}{2}$. This finite-dimension effect is quantified by (A1.30): plotting T_{SAGA} against κ_{eff} instead of κ restores a linear growth (Corollary A1.23 with $\nu' = \mu_F$), which is the cleanest test of the "effective constants" idea of hypothesis (B2) on an instance where the infimum over θ of the effective curvature $\mu_{\mathrm{eff}}(\theta) = \langle \theta - \eta_*, F'(\theta) \rangle / \|\theta - \eta_*\|^2$ of (B2) is known exactly: it is $\mu_F = \lambda_{\min}(\nabla^2 F)$. For the configuration of the experiments of the report ($n = 16$, $m = 256$, $\varepsilon = 10^{-8}$), the dimension condition gives $\kappa_{\max} \approx 4.9 \cdot 10^4$ (the slow-mode condition gives $1.1 \cdot 10^5$), so the window $65 \leq \kappa \leq 4.9 \cdot 10^4$ spans 2.9 decades, the smallest exact ρ in it is 0.15 and $c_2/c_1 \approx 6.5$ for SAGA; the swept values $\kappa \geq 10^5$ lie outside the window, and slopes should be fitted over $10n \leq \kappa \leq \kappa_{\max}$.

A1.4.4 Logistic regression is not a hard instance

The lower bounds of Section A1.2 are statements about the worst case over all instances with given (n, R^2, μ); for a given instance only the upper bounds are informative, and the ℓ_2-regularized logistic problems of the project, $f_i(\theta) = \log(1 + e^{-y_i x_i^\top \theta}) + \frac{\lambda}{2}\|\theta\|^2$ with $R^2 = \max_i \|x_i\|^2/4 + \lambda$ and $\mu = \lambda$, are far from the worst case, for three reasons.

(a) *No chain.* Every component acts on the whole vector θ through x_i; after one pass the span of the gradients is typically all of $\mathbb{R}^d$ (generic x_i), so nothing prevents the iterates from moving in every direction. The "one coordinate per query" mechanism, which is the only source of the $\sqrt{n\kappa}$ term, is absent.

(b) *Slack in the constants.* The certified smoothness R^2 is reached only at $\sigma'(y_i x_i^\top \theta) = 1/4$, i.e. on the decision boundary, and only for the sample with the largest $\|x_i\|$; along the trajectory the per-sample curvature is $\sigma'(y_i x_i^\top \theta)\|x_i\|^2 + \lambda$, often much smaller (this is the cocoercivity slack measured in (B3)). Likewise the strong convexity $\mu = \lambda$ ignores the data term: the trajectory-dependent constant of (B2),

$$\mu_{\text{eff}}(t) = \frac{\langle \theta_t - \eta_*, F'(\theta_t)\rangle}{\|\theta_t - \eta_*\|^2} \;\geq\; \lambda + \lambda_{\min}\Big(\frac{1}{n}\sum_i \bar{\sigma}'_{i,t}\, x_i x_i^\top\Big) \;\geq\; \lambda, \tag{A1.31}$$

where $\bar{\sigma}'_{i,t} \in (0, \frac{1}{4}]$ is the average of σ' on the segment $[\eta_*, \theta_t]$ (mean-value form of $F'(\theta_t) - F'(\eta_*)$). When the features span $\mathbb{R}^d$ (which requires $n \geq d$) and the data are not separable, the data Hessian is uniformly positive definite on every bounded set, so that $\mu_{\text{eff}}(t) \geq \lambda + \mu_{\text{data}}$ with $\mu_{\text{data}} > 0$ independent of λ as long as θ_t stays in a fixed ball $\{\|\theta\| \leq D\}$ containing η_* (one may take $\mu_{\text{data}} = \frac{1}{4}e^{-D\max_i\|x_i\|}\lambda_{\min}(\frac{1}{n}\sum_i x_i x_i^\top)$ by Corollary B2.13; the SAGA iterates are not confined to a deterministic ball in general, see the discussion after that corollary). The one-step inequality of the proof of Proposition 5.9 of [11] holds at every step with $\mu_{\text{eff}}(t)$ in place of μ (Proposition B3.19), but these random one-step factors cannot in general be multiplied in expectation; on a region Θ where $\mu_{\text{eff}} \geq \mu_\Theta$, the Lyapunov function of SAGA contracts in expectation by a factor at most $1 - \min\{\frac{1}{3n}, \frac{3\mu_\Theta}{16R^2}\}$ per step up to the exit time from Θ (Proposition B3.21), instead of the factor $1 - \min\{\frac{1}{3n}, \frac{3\lambda}{16R^2}\}$ dictated by $\kappa = R^2/\lambda$.

(c) $L_F \ll R^2$ *by isotropy rather than by block structure.* For isotropic features (x_i with i.i.d. unit-variance coordinates), $\|x_i\|^2 \approx d$ so $R^2 \approx d/4$, while $L_F \leq \lambda + \frac{1}{4}\lambda_{\max}(\frac{1}{n}\sum_i x_i x_i^\top) \approx \lambda + \frac{1}{4}(1 + \sqrt{d/n})^2$; thus $L_F/R^2 \approx 1/d$ for $n \gg d$ (compare with $1/n$ on the block instance). Full-gradient methods are therefore much faster on these problems than the worst-case counts $n\kappa$, $n\sqrt{\kappa}$ suggest, by a factor about d for GD (nL_F/μ against $n\kappa$) and about $\sqrt{d}$ for AGD ($n\sqrt{L_F/\mu}$ against $n\sqrt{\kappa}$).

Consequently, when $\kappa = R^2/\lambda$ is increased by decreasing λ at fixed data, the number of gradient evaluations of SAGA, whose step $\frac{1}{4R^2}$ is essentially independent of λ, should grow like κ_{eff}^1 at most, with $\kappa_{\text{eff}} = R^2/\min_t \mu_{\text{eff}}(t)$ saturating as soon as $\lambda \ll \mu_{\text{data}}$: its empirical log-log slope against the nominal κ is expected to be strictly smaller than 1 and to decrease towards 0 for small λ (for $d > n$ or separable data $\mu_{\text{data}} = 0$ and the slope can stay close to its hard-instance value, with large constants). For L-Katyusha the outcome depends on the constant used to set σ: with the nominal $\sigma = \lambda/R^2$ (the tuning of the experiments on logistic data, where $\mu = \lambda$ is passed to the method), Theorem 11 of [47] only gives $O((n + \sqrt{n\kappa})\log\frac{C_0}{\varepsilon})$, the momentum $\theta_1 = \sqrt{2n\lambda/(3R^2)}$ keeps decreasing with λ, and nothing prevents the cost from growing after κ_{eff} has saturated (in small checks on non-separable data, with $n = 64$ and $d = 4$, it does); a growth like $\kappa_{\text{eff}}^{1/2}$ can only be expected when σ is set with a strong-convexity constant valid along the trajectory, such as a lower bound on $\mu_{\text{eff}}(t)$. Measuring the slopes against R^2/μ_{eff} (with μ_{eff} the time average or the

terminal value of (A1.31)) is the natural way to connect (A4) with (B2); the hard instance, where $\inf_\theta \mu_{\text{eff}}(\theta) = \mu_F = \lambda_{\min}(\nabla^2 F)$ is known in closed form, serves as the calibration of that procedure.

A1.4.5 Inequalities to instrument

For a lower-bound inequality LHS $\geq$ RHS the tightness ratio is RHS/LHS, which lies in $(0, 1]$ whenever RHS > 0; for an upper bound LHS $\leq$ RHS it is LHS/RHS. On the instance (A1.8) every quantity below is computable exactly from the iterates (η_* is known in closed form, and the eigenpairs of the $m \times m$ tridiagonal matrix $A_m(\zeta_Q)$ are cheap). Every gradient query is counted, including the initial sweep of SAGA ($\tau_j = 1$ for all j and $G = n$ right after the sweep) and the full gradients of L-Katyusha (n queries each).

- (A1.4) *chain smoothness*: ratio $\langle A_m(\zeta)s, s\rangle / (4\|s\|^2)$ at $s = \theta_t^{(i)} - x^*$ (how much of the error sits in the stiff directions; also certifies the slack between $R^2 = \mu\kappa$ and the true Lipschitz constant of f_i').
- (A1.10) *condition number of F versus κ*: ratio $(L_F/\mu)/\kappa$ ($\approx 1/n$ on the instance; on logistic data compare with $1/d$).
- Lemma A1.16 *support growth*: ratio $N_j(\theta_t)/\min(\tau_j(t), m)$ per block, with τ_j counting the sweep (generically equal to 1 for SAGA, and for L-Katyusha when its stored reference gradients are not counted again, which validates the zero-chain mechanism; about $\frac{2}{3}$ for L-Katyusha if every use of $f_i'(w^k)$ is counted as a query, as in the gradient counts of the experiments).
- (A1.12) *tail bound*: ratio $\left[\frac{1}{n}\sum_j \frac{q^{2\min(\tau_j(t),m)} - q^{2m}}{1-q^{2m}}\right] / \left[\|\theta_t - \eta_*\|^2 / \|\eta_*\|^2\right]$ (how much of the error is the unrevealed energy; a ratio near 1 means the method is optimal on the revealed coordinates; the cap at m keeps the right-hand side nonnegative once a block has been touched more than m times).
- (A1.13) *Jensen over blocks*: ratio $q^{2G/n} / \frac{1}{n}\sum_j q^{2\tau_j}$ (equal to 1 for cyclic or full-gradient schedules, smaller for i.i.d. sampling: this is the price of i.i.d. sampling isolated by [38]).
- (A1.15) and (A1.19) *i.i.d. bound*: ratio of the right-hand side to the empirical mean over seeds of the relative error $\|\theta_t - \eta_*\|^2 / \|\eta_*\|^2$, using (A1.19) for SAGA with $z_i^{(0)} = f_i'(\theta_0)$ (t = number of i.i.d. iterations after the sweep) and (A1.15) for $z^{(0)} = 0$; report the ratio only while the right-hand side is positive, i.e. while $t \log \frac{1}{1-x} < 2m \log \frac{1}{q}$ (resp. $< 2(m-1)\log\frac{1}{q}$ for (A1.19)), $x = \frac{1-q^2}{n}$, since beyond that point the bound is vacuous; under (A1.17) these ranges contain $[0, t^*]$ of Corollary A1.20, resp. $[0, t_1^*]$ of the proof of Corollary A1.21.
- (A1.16) *exponential bound*: ratio $\exp(-\frac{tx}{1-x})/(1-x)^t$ with $x = \frac{1-q^2}{n}$.
- (A1.17) *dimension condition*: the scalar q^{2m}/ε (≤ 1 required; report it next to each measured slope), together with $y = (m+1)\sqrt{s}/\pi$ (≥ 2 required for (A1.28)).
- (A1.24) *slow-mode mass*: the exact $\rho = \sum_{k:\lambda_k \leq s} \langle x^*, v_k\rangle^2 / \|x^*\|^2$ from the eigenpairs of $A_m(\zeta_Q)$, and the ratio of the bound (A1.24) to it (between 0.06 and 0.21 for $n = 100$, $m = 200$, $\kappa \in [10^3, 3\cdot 10^5]$, Remark A1.25); also the single-mode overlap $\delta = \langle x^*, v_1\rangle^2 / \|x^*\|^2$ for the saturation regime.
- (A1.20) *SAGA mean iterate*: ratio $\left\|(I - \gamma\nabla^2 F)^t(\theta_0 - \eta_*)\right\|^2 / \mathbb{E}\|\theta_t - \eta_*\|^2$ (exact mean iterate versus the empirical mean square error over seeds) and, for the second inequality, $(1-2\gamma\mu)^{2t}\rho\|\theta_0 - \eta_*\|^2 / \left\|(I - \gamma\nabla^2 F)^t(\theta_0 - \eta_*)\right\|^2$ with $\nu' = 2\mu$ (or $(1-\gamma\mu_F)^{2t}\delta\|\theta_0 - \eta_*\|^2$ in the numerator with $\nu' = \mu_F$); the mean-iterate time $t^*(\kappa)$ of Remark A1.27 is the natural reference curve for $T_{\text{SAGA}}(\kappa)$.
- (A1.22) *L-Katyusha upper bound*: ratio of the measured count to $n + \frac{c}{\theta}\log\frac{C}{\varepsilon}$, with $c = 2$ (stored reference gradients, the convention of T_{LK}) or $c = 3$ (reference gradients charged as recomputed, the convention of the experiments) and C the constant of the recorded sequence

(C_0 for z^k; C_y, given after (A1.22), for y^k, which the experiments record), and the one-step ratio $\mathbb{E}[\Psi^{k+1} \mid y^k, z^k, w^k]/((1-\theta)\Psi^k)$, computed exactly along the run (the pathwise ratio is not bounded by 1; the individual steps of that proof are instrumented in the L-Katyusha note).

- (A1.31) *effective strong convexity*: the ratio $\mu/\mu_{\text{eff}}(t)$, on the instance (for gradient descent with step at most $1/L_F$ it converges there to $\mu/\lambda_{\min}(\nabla^2 F)$ when the initial error has a component on the bottom eigenspace of $\nabla^2 F$, Proposition B2.17(c); for SAGA this alignment is not proved, and the measurements of Remark B2.22, on logistic data, find it in some runs only) and on logistic data.
- (A1.6) *single chain* ($n = 1$ version of the instance, finite m): ratio of $\frac{q^{2t}-q^{2m}}{1-q^{2m}}\,\|\eta_*\|^2$, i.e. the right-hand side of (A1.12) with $n = 1$, to $\|\theta_t - \eta_*\|^2$, for AGD and for GD; the infinite-chain form $q^{2t}\,\|\eta_*\|^2$ of (A1.6) is not a valid lower bound on the finite instance (at $t = m$ an optimal span method has zero error), so the ratio defined with it could exceed 1.

Outcome of the experiments. In Figures A4b and A4 of the report ($n = 16$, $\varepsilon = 10^{-8}$, 2 or 3 seeds), exponents are least-squares slopes of the per-κ geometric means over the seeds, with 95% t-intervals with $K-2$ degrees of freedom for K values of κ. The counts charge L-SVRG, L-Katyusha and the naive scheme 3 gradients per iteration in expectation, about $\frac{3}{2}$ times the stored-gradient convention of T_{LK}, which leaves slopes unchanged. On the instance (A1.8) ($m = 256$; (S) in the report), inside the window $10n \le \kappa \le \kappa_{\max} \approx 4.9 \cdot 10^4$, the slopes are 0.994 [0.990, 0.998] for SAGA, 0.500 [0.479, 0.520] for L-Katyusha, 0.990 for GD and 0.530 for AGD; on the aligned instance ($m = 2048$; (A)), 0.999, 0.502, 0.999 and 0.511. Beyond $\kappa_{\max}$ the nominal slopes bend (SAGA: 0.945 [0.897, 0.993] over $\kappa \ge 10n$), whereas against κ_{eff} they are 1.03 for SAGA and GD, 0.53 for L-Katyusha and 0.55 for AGD: since L-Katyusha and AGD are tuned with μ_F, the upper bounds of all four methods involve μ_F rather than μ. On the block-local instance with Nesterov's chain and on block quadratics with random spectra ((H) and (R), $m = 256$, $10n \le \kappa \le 10^5$), the slopes are 1.022 and 1.003 for SAGA, 1.021 and 1.002 for L-SVRG, 0.525 and 0.500 for L-Katyusha, 0.806 and 0.814 for the re-tuned naive scheme (irregular profile, wide intervals), 0.550 and 0.526 for AGD, 1.026 and 1.007 for GD; L-Katyusha without negative momentum diverges for $\kappa \ge 10^3$. On (H) the analogue of the dimension condition, $q^{2m} \le \varepsilon$ with $q = \frac{\sqrt{\mathcal{Q}}-1}{\sqrt{\mathcal{Q}}+1}$ the chain ratio of a block ($\mathcal{Q}$ the parameter of a block, as in (A1.7)), holds up to $\kappa = 3 \cdot 10^4$ ($q^{2m} = 1.2 \cdot 10^{-10}$) but not at $\kappa = 10^5$ ($q^{2m} = 1.2 \cdot 10^{-5} > \varepsilon$): the top point of these fits lies outside the regime in which a span lower bound of the type of Proposition A1.18 can certify the exponent $\frac{1}{2}$ of L-Katyusha (the upper bounds hold at every κ). As predicted, on these block-separable instances ($L_F \approx R^2/n$) GD and AGD have the exponents of SAGA and L-Katyusha, with smaller constants.

A3 Loopless Katyusha: a Lyapunov proof in the style of the book (hypotheses A2, A3)

This note reproduces, in the notation and in the style of Section 5.4.4 of the book [11] (one Lyapunov function, a short list of elementary inequalities, explicit constants), the convergence proof of the *loopless Katyusha* method (L-Katyusha) of Kovalev, Horváth and Richtárik [47], i.e. the accelerated variance-reduced method of Allen-Zhu [2] in which the outer loop (a full gradient every m iterations) is replaced by a coin flip. The goal is twofold. (A3) We want to see *exactly* which term of the proof is paid by the "negative momentum", that is the pull of the extrapolation point towards the reference point w_t at which the full gradient is known. (A2) We want to see why the same proof does not close when Nesterov's momentum of Section 5.2.5 of the book is combined with the variance-reduced estimator *without* this pull. Every inequality is labelled so that its tightness can be measured in the experiments (Section A3.5). The constants are those of [47]: Algorithm 2, Lemmas 6–10 and Theorem 11 there (Lemmas 4.1–4.5 and Theorem 4.6 in the arXiv version 1901.08689), with proofs in their Appendix C; we quote them where we use them. We number the results of [47] as in this ALT 2020 version, the one listed in the bibliography: its Assumptions 1–2 and Theorem 5 are Assumptions 1.1–1.2 and Theorem 3.5 of the arXiv version, and the algorithms, equations, sections and appendices are numbered identically in both.

A3.1 Setting, estimator and algorithm in the notation of the book

Assumption A3.1 (convex R^2-smooth components, μ-strongly convex F)**.** *$F(\theta) = \frac{1}{n}\sum_{i=1}^n f_i(\theta)$ on $\mathbb{R}^d$, where each f_i is convex and R^2-smooth; we write $L := R^2$ so that L is the* per-sample *smoothness constant, as in Assumption 1 of [47]. F is μ-strongly convex, with minimizer η_* and optimal value $F_* := F(\eta_*)$. We write $\kappa := R^2/\mu = L/\mu \geq 1$ and $\sigma := \mu/L = 1/\kappa \in (0,1]$ (the quantity called σ in [47]).*

Since $F' = \frac{1}{n}\sum_i f_i'$, the function F is itself L-smooth (its own constant may be smaller, but we only use L). For ℓ_2-regularized logistic regression, $f_i(\theta) = \log(1+\exp(-y_i x_i^\top \theta)) + \frac{\lambda}{2}\|\theta\|^2$, one can take $L = \max_i \|x_i\|^2/4 + \lambda$ and $\mu = \lambda$ (hypothesis (B3)).

Toolbox. The proof uses only the following facts, all of which appear in Chapter 5 of the book. For a convex L-smooth f and all $\theta, \eta \in \mathbb{R}^d$:

(T1) convexity (eq. (5.7) of the book): $f(\theta) \geq f(\eta) + \langle f'(\eta), \theta - \eta\rangle$;

(T2) quadratic upper bound (eq. (5.10)): $f(\theta) \leq f(\eta) + \langle f'(\eta), \theta - \eta\rangle + \frac{L}{2}\|\theta - \eta\|^2$;

(T3) co-coercivity in its "Bregman" form (Prop. 5.4, eq. (5.12)): $f(\theta) \geq f(\eta) + \langle f'(\eta), \theta - \eta\rangle + \frac{1}{2L}\|f'(\theta) - f'(\eta)\|^2$;

(T4) strong convexity of F (eq. (5.9)): $F(\theta) \geq F(\eta) + \langle F'(\eta), \theta - \eta\rangle + \frac{\mu}{2}\|\theta - \eta\|^2$;

(T5) three-point identity: $2\langle a - b, b - c\rangle = \|a - c\|^2 - \|a - b\|^2 - \|b - c\|^2$;

(T6) Young's inequality: for $\beta > 0$, $\langle a, b\rangle \geq -\frac{1}{2\beta}\|a\|^2 - \frac{\beta}{2}\|b\|^2$;

(T7) variance is at most the second moment (inequality (I3) of the proof of Prop. 5.9 in the B3 note): for a random vector X, $\mathbb{E}\|X - \mathbb{E}X\|^2 = \mathbb{E}\|X\|^2 - \|\mathbb{E}X\|^2 \leq \mathbb{E}\|X\|^2$.

In (T1)–(T4), θ and η denote arbitrary points of $\mathbb{R}^d$, as in the book; this η is unrelated to the step η of (A3.2). We write $D_F(w,x) := F(w) - F(x) - \langle F'(x), w - x\rangle \geq 0$ for the Bregman divergence of F; by (T2), $D_F(w,x) \leq \frac{L}{2}\|w-x\|^2$.

Definition A3.2 (loopless SVRG estimator)**.** *Let $p \in (0,1]$. At iteration t the method holds a query point x_t and a* reference point *w_t at which the full gradient $F'(w_t)$ has been computed. An index $i(t)$ is drawn uniformly in $\{1,\dots,n\}$ and the estimator is*

$$g_t := f'_{i(t)}(x_t) - f'_{i(t)}(w_t) + F'(w_t). \tag{A3.1}$$

Independently of $i(t)$ and of the past, a coin $b_t \sim$ Bernoulli(p) decides whether the reference point is refreshed: $w_{t+1} = y_t$ if $b_t = 1$ and $w_{t+1} = w_t$ otherwise (with y_t the "gradient-step" sequence defined below). Refreshing costs one pass (n gradients f'_i) and happens with probability p; with $p = 1/n$ it costs one gradient per iteration on average.

Remark A3.3 (conditional expectation and comparison with SAGA). We write $\mathbb{E}_{i(t)}$ for the conditional expectation given the state (y_t, z_t, w_t) of the algorithm, i.e. over $i(t)$ and b_t (this is the notation $\mathbb{E}[\cdot \mid y^k, z^k, w^k]$ of [47]). Since $i(t)$ is uniform, $\mathbb{E}_{i(t)}[g_t] = F'(x_t) - F'(w_t) + F'(w_t) = F'(x_t)$: the estimator is unbiased, exactly as the SAGA direction ω_t of Prop. 5.9. The difference with SAGA is that the memory is a single point w_t and a single vector $F'(w_t)$ instead of the table $(z_i^{(t)})_{i\leq n}$, and that each iteration then evaluates two gradients $f'_{i(t)}(x_t), f'_{i(t)}(w_t)$ instead of one. If, at the price of $\mathcal{O}(nd)$ memory as for the table of SAGA, the n gradients $f'_i(w_t)$ computed with $F'(w_t)$ are stored, $f'_{i(t)}(w_t)$ is a look-up; this is the convention of the gradient counts of the A1 note, and Corollary A3.12 gives both counts.

The algorithm. L-Katyusha (Algorithm 2 of [47]) keeps three sequences x_t, y_t, z_t and the reference point w_t. Parameters: $\theta_1, \theta_2 > 0$ with $\theta_1 + \theta_2 \leq 1$, $p \in (0,1]$, $\sigma = \mu/L$ and the step

$$\eta := \frac{\theta_2}{(1+\theta_2)\theta_1}. \tag{A3.2}$$

Initialization $y_0 = w_0 = z_0 = \theta_0 \in \mathbb{R}^d$ (with $F'(w_0)$ computed). For $t = 0, 1, 2, \dots$:

$$x_t = \theta_1 z_t + \theta_2 w_t + (1 - \theta_1 - \theta_2) y_t, \tag{K1}$$

$$g_t = f'_{i(t)}(x_t) - f'_{i(t)}(w_t) + F'(w_t), \tag{K2}$$

$$z_{t+1} = \frac{1}{1+\eta\sigma}\Big(\eta\sigma\, x_t + z_t - \frac{\eta}{L}\, g_t\Big), \tag{K3}$$

$$y_{t+1} = x_t + \theta_1(z_{t+1} - z_t), \tag{K4}$$

$$w_{t+1} = \begin{cases} y_t & \text{with probability } p, \\ w_t & \text{with probability } 1-p. \end{cases} \tag{K5}$$

The reading of the three steps is the one of "linear coupling" [3]: (K3) is a *mirror* (or dual-averaging) step with step size η/L on a μ-strongly convex surrogate, namely $z_{t+1} = \arg\min_z\{\langle g_t, z\rangle + \frac{L}{2\eta}\|z - z_t\|^2 + \frac{\mu}{2}\|z - x_t\|^2\}$ (set the gradient to zero: $g_t + \frac{L}{\eta}(z - z_t) + \mu(z - x_t) = 0$, multiply by η/L and use $\eta\mu/L = \eta\sigma$); (K4) is a *gradient* step from x_t with step $\theta_1\eta/L$ up to a term of order σ: indeed by (K3), $y_{t+1} = x_t - \frac{\theta_1\eta}{L(1+\eta\sigma)} g_t + \frac{\theta_1\eta\sigma}{1+\eta\sigma}(x_t - z_t)$; and (K1) is the coupling. The term $\theta_2 w_t$ in (K1) is the *negative momentum* ("Katyusha momentum") of [2]: the query point is pulled towards the point where the estimator has zero variance.

Parameters of the paper. Theorem 11 of [47] uses

$$\theta_2 = \tfrac{1}{2}, \qquad \theta_1 = \min\Big\{\sqrt{\tfrac{2\sigma n}{3}},\, \tfrac{1}{2}\Big\} = \min\Big\{\sqrt{\tfrac{2n}{3\kappa}},\, \tfrac{1}{2}\Big\}, \qquad p = \tfrac{1}{n}, \qquad \eta = \frac{\theta_2}{(1+\theta_2)\theta_1} = \frac{1}{3\theta_1}. \tag{A3.3}$$

Here θ_1, θ_2 are the coupling weights of [47], not iterates; the initial point is θ_0. With the parameters (A3.3), the gradient step in (K4) has length $\theta_1\eta/L = 1/(3L)$ (up to the factor $1/(1+\eta\sigma)$), the same as in Katyusha and close to the $1/(4L)$ of SAGA in Prop. 5.9, while the mirror step $\eta/L = 1/(3\theta_1 L)$ is longer by the factor $1/\theta_1 = \max\{\sqrt{3\kappa/(2n)}, 2\}$: this is where the momentum lives.

The Lyapunov function. Equation (13) of [47] defines, verbatim in their notation, $\Psi^k = \mathcal{Z}^k + \mathcal{Y}^k + \mathcal{W}^k$ with $\mathcal{Z}^k = \frac{L(1+\eta\sigma)}{2\eta}\|z^k - x^*\|^2$, $\mathcal{Y}^k = \frac{1}{\theta_1}(f(y^k) - f(x^*))$, $\mathcal{W}^k = \frac{\theta_2(1+\theta_1)}{p\theta_1}(f(w^k) - f(x^*))$. In our notation:

$$\begin{aligned} &\Psi_t := Z_t + Y_t + W_t, \\ &Z_t := \frac{L(1+\eta\sigma)}{2\eta}\,\|z_t - \eta_*\|^2, \qquad Y_t := \frac{1}{\theta_1}\big(F(y_t) - F_*\big), \qquad W_t := \frac{\theta_2(1+\theta_1)}{p\theta_1}\big(F(w_t) - F_*\big). \end{aligned} \tag{A3.4}$$

Compare with Prop. 5.9: $V_t = \|\theta_t - \eta_*\|^2 + \Delta\sum_i \Big\|z_i^{(t)} - f_i'(\eta_*)\Big\|^2$ is a distance plus a memory term; here Z_t is the distance (of the *mirror* sequence), Y_t a function value (of the *gradient* sequence, as in the potential of Exercise 5.14 of the book for Nesterov's method) and W_t the memory term: it measures how good the reference point is, and it is the loopless substitute for the "aggregation across the inner loop" of the original Katyusha proof.

Theorem A3.4 (contraction and rate of L-Katyusha; Lemma 10 and Theorem 11 of [47])**.** *Under Assumption A3.1, for any $\theta_1, \theta_2 > 0$ with $\theta_1 + \theta_2 \le 1$, any $p \in (0,1]$, $\sigma = \mu/L$ and η as in (A3.2), the iterates of (K1)–(K5) and the potential (A3.4) satisfy, for all $t \ge 0$,*

$$\mathbb{E}_{i(t)}\big[\Psi_{t+1}\big] \;\le\; \frac{1}{1+\eta\sigma}\,Z_t + \big(1 - \theta_1(1-\theta_2)\big)\,Y_t + \Big(1 - \frac{p\theta_1}{1+\theta_1}\Big)\,W_t \qquad \textit{(one-step contraction)}. \tag{A3.5}$$

With the parameters (A3.3), for all $t \ge 0$,

$$\mathbb{E}\big[\Psi_t\big] \le (1-\rho)^t\,\Psi_0, \qquad \rho := \min\Big\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\Big\} = \begin{cases} \dfrac{1}{\sqrt{24\,n\kappa}} & \text{if } \kappa \ge \frac{8n}{3}, \\ \min\Big\{\dfrac{1}{3\kappa}, \dfrac{1}{4n}\Big\} \;\Big(\ge \dfrac{1}{8n}\Big) & \text{if } \kappa < \frac{8n}{3}, \end{cases} \tag{A3.6}$$

*so that in all cases $\rho \ge \min\{1/(4n), 1/\sqrt{24n\kappa}\}$ (*global rate*).*

The proof occupies Section A3.2. The complexity in gradient evaluations is derived in Corollary A3.12: $\mathcal{O}\big((n+\sqrt{n\kappa})\log(1/\varepsilon)\big)$, to be compared with $(3n+\frac{16}{3}\kappa)\log(2n/\varepsilon)$ for SAGA (Prop. 5.9) and with $n\sqrt{\kappa}\log(1/\varepsilon)$ for Nesterov's method of Section 5.2.5 applied to the full gradient: the condition number enters through $\sqrt{n\kappa}$ instead of κ (SAGA) or $n\sqrt{\kappa}$ (accelerated gradient descent).

Remark A3.5 (relation with the original Katyusha and with the book). Theorem 2.1 of [2] states (their notation: σ is our μ, m the epoch length, and the objective of their Problem (1.1) is $F = \frac{1}{n}\sum_i f_i + \psi$ with a proximal term ψ) that "if each $f_i(x)$ is convex, L-smooth, and $\psi(x)$ is σ-strongly convex in Problem (1.1)", then Katyusha satisfies $\mathbb{E}[F(\tilde{x}^S)] - F(x^*) \le \mathcal{O}\big((1 + \sqrt{\sigma/(3Lm)})^{-Sm}\big)(F(\tilde{x}^0) - F(x^*))$ when $m\sigma/L \le 3/4$ and $\mathcal{O}(1.5^{-S})(F(\tilde{x}^0) - F(x^*))$ otherwise, "choosing $m = \Theta(n)$, Katyusha achieves an ε-additive error using at most $\mathcal{O}\big((n +$

$\sqrt{nL/\sigma}) \log \frac{F(\tilde{x}^0)-F(x^*)}{\varepsilon}\big)$ iterations". The strong convexity is thus assumed on the proximal term ψ (for ℓ_2-regularized logistic regression one can take $\psi = \frac{\lambda}{2}\|\cdot\|^2$), whereas L-Katyusha has no proximal term and assumes it on F (Assumption 2 of [47]), as in Assumption A3.1. The loopless version has the same complexity with a proof that needs a single iteration (no telescoping over an epoch). Allen-Zhu also explains the role of the three sequences: "In a classical accelerated gradient method, x^{k+1} is only a convex combination of y^k and z^k (or equivalently, $\tau_2 = 0$ in our formulation)"; we make this precise in Section A3.4 where we show that (K1)–(K4) with $\theta_2 = 0$ and exact gradients is an accelerated gradient method with the rate $(1 - \frac{1}{2}\sqrt{\mu/L})^t$, i.e. a variant of the method of eq. (5.16)–(5.17) of the book, from which it differs by the choice of constants and by an $\mathcal{O}(\sqrt{\mu/L})$ correction of the momentum term (Remark A3.16).

A3.2 The Lyapunov proof

Lemma A3.6 (L1: unbiasedness and variance of the estimator; Lemma 6 of [47]). *For all t, $\mathbb{E}_{i(t)}[g_t] = F'(x_t)$ and*

$$\mathbb{E}_{i(t)}\left\|g_t - F'(x_t)\right\|^2 = \frac{1}{n}\sum_{i=1}^n \left\|f_i'(x_t) - f_i'(w_t)\right\|^2 - \left\|F'(x_t) - F'(w_t)\right\|^2 \;\le\; \frac{1}{n}\sum_{i=1}^n \left\|f_i'(x_t) - f_i'(w_t)\right\|^2 \tag{A3.7}$$

$$\le\; 2L\left(F(w_t) - F(x_t) - \left\langle F'(x_t), w_t - x_t\right\rangle\right) \;=\; 2L\, D_F(w_t, x_t). \tag{A3.8}$$

We call (A3.7) *the* centering inequality *and* (A3.8) *the* co-coercivity bound*; their product is the* variance bound

$$\mathbb{E}_{i(t)}\left\|g_t - F'(x_t)\right\|^2 \le 2L\, D_F(w_t, x_t). \tag{A3.9}$$

Proof. Given the state, x_t and w_t are fixed and $X := f_{i(t)}'(x_t) - f_{i(t)}'(w_t)$ is a random vector with $\mathbb{E}_{i(t)}X = F'(x_t) - F'(w_t)$, so that $g_t - F'(x_t) = X - \mathbb{E}_{i(t)}X$ (unbiasedness). By (T7), $\mathbb{E}_{i(t)}\left\|X - \mathbb{E}_{i(t)}X\right\|^2 = \mathbb{E}_{i(t)}\|X\|^2 - \left\|\mathbb{E}_{i(t)}X\right\|^2$, and $\mathbb{E}_{i(t)}\|X\|^2 = \frac{1}{n}\sum_i \|f_i'(x_t) - f_i'(w_t)\|^2$ because $i(t)$ is uniform: this is (A3.7). For (A3.8), apply (T3) to each convex L-smooth f_i with $\theta = w_t$, $\eta = x_t$: $\|f_i'(w_t) - f_i'(x_t)\|^2 \le 2L\big(f_i(w_t) - f_i(x_t) - \langle f_i'(x_t), w_t - x_t\rangle\big)$. Average over i and use $\frac{1}{n}\sum_i f_i = F$, $\frac{1}{n}\sum_i f_i' = F'$. This is the same use of co-coercivity as in eq. (5.31) of the book (there with $\eta = \eta_*$, which gives the inner-product form $R^2\, F'(\theta)^\top(\theta - \eta_*)$). Here we keep the one-sided (Bregman) form (5.12) rather than the symmetric form (5.11) used in (5.31): since $\langle F'(w_t) - F'(x_t), w_t - x_t\rangle = D_F(w_t, x_t) + D_F(x_t, w_t)$, the bound $L\langle F'(w_t) - F'(x_t), w_t - x_t\rangle$ given by (5.11) is the average of the one-sided bounds $2L\, D_F(w_t, x_t)$ and $2L\, D_F(x_t, w_t)$, so neither is uniformly sharper; (5.12) is used because it produces the divergence $D_F(w_t, x_t)$, which is exactly what the anchor term cancels in Step 6 of Lemma A3.11. □

Lemma A3.7 (L2: the mirror step; Lemma 7 of [47]). *For every realization of $i(t)$ (no expectation is needed),*

$$\langle g_t, \eta_* - z_{t+1}\rangle + \frac{\mu}{2}\|x_t - \eta_*\|^2 \;\ge\; \frac{L}{2\eta}\|z_t - z_{t+1}\|^2 + Z_{t+1} - \frac{1}{1+\eta\sigma}Z_t \qquad \text{(mirror-step inequality)}, \tag{A3.10}$$

and the slack is exactly $\frac{\mu}{2}\|x_t - z_{t+1}\|^2$ (i.e. (A3.10) becomes an identity if this term is added to the right-hand side).

Proof. Multiplying (K3) by $1 + \eta\sigma$ gives $(1+\eta\sigma)z_{t+1} = \eta\sigma x_t + z_t - \frac{\eta}{L}g_t$, i.e.

$$\frac{\eta}{L}\, g_t = \eta\sigma\,(x_t - z_{t+1}) + (z_t - z_{t+1}).$$

Take the inner product with $z_{t+1}-\eta_*$ and multiply by L/η, using $L\sigma=\mu$:

$$\langle g_t, z_{t+1}-\eta_*\rangle = \mu\,\langle x_t - z_{t+1}, z_{t+1}-\eta_*\rangle + \frac{L}{\eta}\,\langle z_t - z_{t+1}, z_{t+1}-\eta_*\rangle\,.$$

Apply (T5) twice, with $(a,b,c)=(x_t, z_{t+1}, \eta_*)$ and $(a,b,c)=(z_t, z_{t+1}, \eta_*)$:

$$\begin{aligned}\langle g_t, z_{t+1}-\eta_*\rangle &= \frac{\mu}{2}\Big(\|x_t-\eta_*\|^2 - \|x_t-z_{t+1}\|^2 - \|z_{t+1}-\eta_*\|^2\Big)\\ &\quad + \frac{L}{2\eta}\Big(\|z_t-\eta_*\|^2 - \|z_t-z_{t+1}\|^2 - \|z_{t+1}-\eta_*\|^2\Big)\\ &= \frac{\mu}{2}\|x_t-\eta_*\|^2 - \frac{\mu}{2}\|x_t-z_{t+1}\|^2 + \frac{L}{2\eta}\|z_t-\eta_*\|^2\\ &\quad - \frac{L}{2\eta}(1+\eta\sigma)\|z_{t+1}-\eta_*\|^2 - \frac{L}{2\eta}\|z_t-z_{t+1}\|^2\,,\end{aligned}$$

where we grouped $\frac{\mu}{2}+\frac{L}{2\eta}=\frac{L}{2\eta}(\eta\sigma+1)$. By definition (A3.4), $\frac{L}{2\eta}(1+\eta\sigma)\|z_{t+1}-\eta_*\|^2 = Z_{t+1}$ and $\frac{L}{2\eta}\|z_t-\eta_*\|^2 = \frac{Z_t}{1+\eta\sigma}$. Moving $\langle g_t, z_{t+1}-\eta_*\rangle$ to the right and $\frac{\mu}{2}\|x_t-\eta_*\|^2$ to the left gives the claimed identity with the slack $\frac{\mu}{2}\|x_t-z_{t+1}\|^2\ge 0$; dropping it gives (A3.10). □

Lemma A3.8 (L3: the gradient step; Lemma 8 of [47])**.** *For every realization of $i(t)$, the* gradient-step inequality *holds:*

$$\frac{1}{\theta_1}\Big(F(y_{t+1})-F(x_t)\Big) - \frac{\theta_2}{2L\theta_1}\big\|g_t-F'(x_t)\big\|^2 \;\le\; \frac{L}{2\eta}\|z_{t+1}-z_t\|^2 + \langle g_t, z_{t+1}-z_t\rangle\,. \tag{A3.11}$$

It is the product of two inequalities, the smoothness bound

$$F(y_{t+1}) \le F(x_t) + \big\langle F'(x_t), y_{t+1}-x_t\big\rangle + \frac{L}{2}\|y_{t+1}-x_t\|^2\,, \tag{A3.12}$$

and the Young bound *on the cross term between the noise of the estimator and the step,*

$$\big\langle g_t - F'(x_t), y_{t+1}-x_t\big\rangle \;\ge\; -\frac{\theta_2}{2L}\big\|g_t-F'(x_t)\big\|^2 - \frac{L}{2\theta_2}\|y_{t+1}-x_t\|^2\,. \tag{A3.13}$$

Proof. By (K4), $y_{t+1}-x_t=\theta_1(z_{t+1}-z_t)$, hence

$$\frac{L}{2\eta}\|z_{t+1}-z_t\|^2 + \langle g_t, z_{t+1}-z_t\rangle = \frac{1}{\theta_1}\Big(\frac{L}{2\eta\theta_1}\|y_{t+1}-x_t\|^2 + \langle g_t, y_{t+1}-x_t\rangle\Big).$$

Write $g_t = F'(x_t) + (g_t-F'(x_t))$ and split the quadratic term as $\frac{L}{2\eta\theta_1} = \frac{L}{2} + \frac{L}{2}\big(\frac{1}{\eta\theta_1}-1\big)$. By the choice (A3.2), $\eta\theta_1 = \frac{\theta_2}{1+\theta_2}<1$ and $\frac{1}{\eta\theta_1}-1=\frac{1}{\theta_2}$. Therefore the bracket equals

$$\Big[\frac{L}{2}\|y_{t+1}-x_t\|^2 + \big\langle F'(x_t), y_{t+1}-x_t\big\rangle\Big] + \frac{L}{2\theta_2}\|y_{t+1}-x_t\|^2 + \big\langle g_t-F'(x_t), y_{t+1}-x_t\big\rangle\,.$$

The first bracket is at least $F(y_{t+1})-F(x_t)$ by (T2) applied to F ($\theta=y_{t+1}$, $\eta=x_t$), which is (A3.12). The last two terms are at least $-\frac{\theta_2}{2L}\|g_t-F'(x_t)\|^2$ by (T6) with $\beta=L/\theta_2$, $a=g_t-F'(x_t)$, $b=y_{t+1}-x_t$, which is (A3.13). Dividing by θ_1 gives (A3.11). □

Remark A3.9 (the step size couples the two momenta). In the proof of L3 the coefficient of the variance term is $\frac{1}{2L}\cdot\frac{\eta\theta_1}{1-\eta\theta_1}$, a function of the step only. The choice (A3.2) is precisely the one which makes this coefficient equal to $\frac{\theta_2}{2L}$, with the *same* θ_2 as the weight of w_t in (K1). This is what allows the exact cancellation (A3.19) in the proof of L5 (Lemma A3.11). For a general step we write $\nu := \frac{\eta\theta_1}{1-\eta\theta_1}$ (so $\nu=\theta_2$ for L-Katyusha); L3 then holds with θ_2 replaced by ν in the variance term. We use this in Section A3.4.

Lemma A3.10 (L4: the reference point; Lemma 9 of [47]). *For all t,*

$$\mathbb{E}_{i(t)}\big[W_{t+1}\big] = (1-p)\,W_t + \theta_2(1+\theta_1)\,Y_t. \tag{A3.14}$$

Proof. By (K5) and independence of the coin from $i(t)$, $\mathbb{E}_{i(t)}[F(w_{t+1})] = (1-p)F(w_t)+pF(y_t)$. Subtract F_* and multiply by $\frac{\theta_2(1+\theta_1)}{p\theta_1}$: the first term gives $(1-p)W_t$ and the second $\frac{\theta_2(1+\theta_1)}{\theta_1}(F(y_t)-F_*)=\theta_2(1+\theta_1)Y_t$. □

Lemma A3.11 (L5: combination of L1–L4; Lemma 10 of [47]). *Under the assumptions of Theorem A3.4,*

$$\mathbb{E}_{i(t)}\big[Z_{t+1}+Y_{t+1}\big] \le \frac{1}{1+\eta\sigma}Z_t + (1-\theta_1-\theta_2)\,Y_t + \frac{p}{1+\theta_1}\,W_t, \tag{A3.15}$$

and consequently the one-step contraction (A3.5) holds.

Proof. Step 1 (strong convexity at the query point). By (T4) with $\theta=\eta_*$ and $\eta=x_t$,

$$F_* \;\ge\; F(x_t) + \big\langle F'(x_t), \eta_* - x_t\big\rangle + \frac{\mu}{2}\,\|x_t-\eta_*\|^2 \qquad (\textit{strong convexity at } x_t). \tag{A3.16}$$

Step 2 (decomposition through the coupling). By (K1), $\theta_2 w_t + (1-\theta_1-\theta_2)y_t = x_t - \theta_1 z_t$, hence

$$\theta_2(x_t-w_t)+(1-\theta_1-\theta_2)(x_t-y_t) = (1-\theta_1)x_t - (x_t-\theta_1 z_t) = \theta_1(z_t-x_t),$$

that is

$$z_t - x_t = \frac{\theta_2}{\theta_1}(x_t-w_t) + \frac{1-\theta_1-\theta_2}{\theta_1}(x_t-y_t). \tag{A3.17}$$

Writing $\eta_*-x_t=(\eta_*-z_t)+(z_t-x_t)$ in (A3.16) and using (A3.17),

$$F_* \;\ge\; F(x_t)+\frac{\mu}{2}\,\|x_t-\eta_*\|^2+\big\langle F'(x_t),\eta_*-z_t\big\rangle+\frac{\theta_2}{\theta_1}\big\langle F'(x_t),x_t-w_t\big\rangle+\frac{1-\theta_1-\theta_2}{\theta_1}\big\langle F'(x_t),x_t-y_t\big\rangle.$$

Step 3 (convexity towards y_t). Since $1-\theta_1-\theta_2\ge 0$, we may use (T1) with $\theta=y_t$, $\eta=x_t$:

$$\big\langle F'(x_t), x_t-y_t\big\rangle \;\ge\; F(x_t)-F(y_t) \qquad (\textit{convexity towards } y_t). \tag{A3.18}$$

Step 4 (unbiasedness). Since z_t is fixed given the state and $\mathbb{E}_{i(t)}g_t = F'(x_t)$,

$$\big\langle F'(x_t),\eta_*-z_t\big\rangle = \mathbb{E}_{i(t)}\,\langle g_t,\eta_*-z_t\rangle = \mathbb{E}_{i(t)}\Big[\,\langle g_t,\eta_*-z_{t+1}\rangle + \langle g_t, z_{t+1}-z_t\rangle\,\Big].$$

(The deterministic quantity $\frac{\mu}{2}\|x_t-\eta_*\|^2$ may be put inside $\mathbb{E}_{i(t)}$ as well.) Steps 1–4 give

$$\begin{aligned} F_* \;\ge\; & F(x_t)+\frac{\theta_2}{\theta_1}\big\langle F'(x_t),x_t-w_t\big\rangle+\frac{1-\theta_1-\theta_2}{\theta_1}\big(F(x_t)-F(y_t)\big)\\ & +\mathbb{E}_{i(t)}\Big[\frac{\mu}{2}\,\|x_t-\eta_*\|^2+\langle g_t,\eta_*-z_{t+1}\rangle+\langle g_t,z_{t+1}-z_t\rangle\Big]. \end{aligned}$$

Step 5 (the three lemmas). Inside the expectation we use, in this order, L2 (A3.10), L3 (A3.11) and, after taking $\mathbb{E}_{i(t)}$, L1 (A3.9):

$$\begin{aligned}
\tfrac{\mu}{2}\|x_t-\eta_*\|^2+\langle g_t,\eta_*-z_{t+1}\rangle &\geq \tfrac{L}{2\eta}\|z_t-z_{t+1}\|^2+Z_{t+1}-\tfrac{Z_t}{1+\eta\sigma},\\
\tfrac{L}{2\eta}\|z_{t+1}-z_t\|^2+\langle g_t,z_{t+1}-z_t\rangle &\geq \tfrac{1}{\theta_1}\big(F(y_{t+1})-F(x_t)\big)-\tfrac{\theta_2}{2L\theta_1}\big\|g_t-F'(x_t)\big\|^2,\\
-\tfrac{\theta_2}{2L\theta_1}\mathbb{E}_{i(t)}\big\|g_t-F'(x_t)\big\|^2 &\geq -\tfrac{\theta_2}{\theta_1}D_F(w_t,x_t).
\end{aligned}$$

Altogether

$$\begin{aligned}
F_* \;\geq\; & F(x_t)+\frac{\theta_2}{\theta_1}\big\langle F'(x_t),x_t-w_t\big\rangle+\frac{1-\theta_1-\theta_2}{\theta_1}\big(F(x_t)-F(y_t)\big)+\mathbb{E}_{i(t)}[Z_{t+1}]-\frac{Z_t}{1+\eta\sigma}\\
&+\frac{1}{\theta_1}\mathbb{E}_{i(t)}\big[F(y_{t+1})-F(x_t)\big]-\frac{\theta_2}{\theta_1}\Big(F(w_t)-F(x_t)-\big\langle F'(x_t),w_t-x_t\big\rangle\Big).
\end{aligned}$$

Step 6 (the cancellation: this is where the negative momentum acts). The inner products in w_t cancel *exactly*:

$$\underbrace{\frac{\theta_2}{\theta_1}\big\langle F'(x_t),x_t-w_t\big\rangle}_{\text{from }\theta_2 w_t\text{ in (K1), Step 2}}\;-\;\underbrace{\frac{\theta_2}{\theta_1}\Big(F(w_t)-F(x_t)-\big\langle F'(x_t),w_t-x_t\big\rangle\Big)}_{\text{variance term, L1 and L3}}\;=\;-\frac{\theta_2}{\theta_1}\big(F(w_t)-F(x_t)\big),\quad\text{(A3.19)}$$

because $-\langle F'(x_t),w_t-x_t\rangle=\langle F'(x_t),x_t-w_t\rangle$ enters with the sign $-\frac{\theta_2}{\theta_1}$. What survives of the variance term is a *difference of function values* only.

Step 7 (bookkeeping). Collect the coefficients of $F(x_t)$ after (A3.19): they sum to $1+\frac{\theta_2}{\theta_1}+\frac{1-\theta_1-\theta_2}{\theta_1}-\frac{1}{\theta_1}=0$. Hence

$$F_*\;\geq\;-\frac{\theta_2}{\theta_1}F(w_t)-\frac{1-\theta_1-\theta_2}{\theta_1}F(y_t)+\mathbb{E}_{i(t)}[Z_{t+1}]-\frac{Z_t}{1+\eta\sigma}+\frac{1}{\theta_1}\mathbb{E}_{i(t)}[F(y_{t+1})].$$

Since $\frac{1}{\theta_1}-\frac{\theta_2}{\theta_1}-\frac{1-\theta_1-\theta_2}{\theta_1}=1$, we may write $F_*=\frac{1}{\theta_1}F_*-\frac{\theta_2}{\theta_1}F_*-\frac{1-\theta_1-\theta_2}{\theta_1}F_*$ and rearrange:

$$\mathbb{E}_{i(t)}[Z_{t+1}]+\frac{1}{\theta_1}\mathbb{E}_{i(t)}\big[F(y_{t+1})-F_*\big]\leq\frac{Z_t}{1+\eta\sigma}+\frac{1-\theta_1-\theta_2}{\theta_1}\big(F(y_t)-F_*\big)+\frac{\theta_2}{\theta_1}\big(F(w_t)-F_*\big).$$

By (A3.4), $\frac{1-\theta_1-\theta_2}{\theta_1}(F(y_t)-F_*)=(1-\theta_1-\theta_2)Y_t$ and $\frac{\theta_2}{\theta_1}(F(w_t)-F_*)=\frac{p}{1+\theta_1}W_t$: this is (A3.15).

Step 8 (add the reference-point term). Adding (A3.14) to (A3.15): the coefficient of Y_t becomes $1-\theta_1-\theta_2+\theta_2(1+\theta_1)=1-\theta_1(1-\theta_2)$ and the coefficient of W_t becomes $\frac{p}{1+\theta_1}+1-p=1-\frac{p\theta_1}{1+\theta_1}$, which is (A3.5). □

Proof of Theorem A3.4. (A3.5) is Lemma A3.11. For the rate, (A3.5) gives

$$\mathbb{E}_{i(t)}[\Psi_{t+1}]\leq\max\{c_Z,c_Y,c_W\}\,\Psi_t,\qquad c_Z=\tfrac{1}{1+\eta\sigma},\quad c_Y=1-\theta_1(1-\theta_2),\quad c_W=1-\tfrac{p\theta_1}{1+\theta_1},$$

and by the tower property $\mathbb{E}[\Psi_{t+1}]\leq\max\{c_Z,c_Y,c_W\}\mathbb{E}[\Psi_t]$. With the parameters (A3.3):

- $c_Z\leq 1-\frac{\sigma}{6\theta_1}$. Indeed $\eta\sigma=\frac{\sigma}{3\theta_1}\leq 1$: if $\theta_1=\frac{1}{2}$ then $\eta\sigma=\frac{2\sigma}{3}\leq\frac{2}{3}$; if $\theta_1=\sqrt{2\sigma n/3}$ then $\eta\sigma=\frac{\sigma}{3\sqrt{2\sigma n/3}}=\sqrt{\frac{\sigma}{6n}}\leq 1$. And for $x\in[0,1]$, $\frac{1}{1+x}\leq 1-\frac{x}{2}$ because $(1-\frac{x}{2})(1+x)=1+\frac{x}{2}-\frac{x^2}{2}\geq 1$.
- $c_Y=1-\frac{\theta_1}{2}$ since $\theta_2=\frac{1}{2}$.

- $c_W \le 1-\frac{\theta_1}{2n}$: with $p=\frac{1}{n}$ and $\theta_1 \le \frac12$, $\frac{p\theta_1}{1+\theta_1} = \frac{\theta_1}{n(1+\theta_1)} \ge \frac{2\theta_1}{3n} \ge \frac{\theta_1}{2n}$.

Hence $\max\{c_Z,c_Y,c_W\} \le 1-\min\{\frac{\sigma}{6\theta_1},\frac{\theta_1}{2},\frac{\theta_1}{2n}\} = 1-\rho$ with $\rho=\min\{\frac{\sigma}{6\theta_1},\frac{\theta_1}{2n}\}$, which is the θ of the proof of Theorem 11 of [47] ("$\theta=\min\{\sigma/6\theta_1,\theta_1/2n\}$"). Explicitly, with $\sigma=1/\kappa$: if $\kappa \ge \frac{8n}{3}$ then $\frac{2n}{3\kappa}\le\frac14$, $\theta_1=\sqrt{2n/(3\kappa)}$, and $\frac{\sigma}{6\theta_1}=\frac{1}{6\kappa}\sqrt{\frac{3\kappa}{2n}}=\frac{1}{\sqrt{24n\kappa}}$, $\frac{\theta_1}{2n}=\frac{1}{2n}\sqrt{\frac{2n}{3\kappa}}=\frac{1}{\sqrt{6n\kappa}}$, so $\rho=\frac{1}{\sqrt{24n\kappa}}$; if $\kappa<\frac{8n}{3}$ then $\theta_1=\frac12$ and $\rho=\min\{\frac{1}{3\kappa},\frac{1}{4n}\}\ge\frac{1}{8n}$. Finally $\frac{1}{3\kappa}\ge\frac{1}{\sqrt{24n\kappa}}$ iff $9\kappa^2\le 24n\kappa$ iff $\kappa\le\frac{8n}{3}$, which gives the unified bound $\rho\ge\min\{\frac{1}{4n},\frac{1}{\sqrt{24n\kappa}}\}$. Iterating, $\mathbb{E}[\Psi_t]\le(1-\rho)^t\Psi_0$. □

Corollary A3.12 (gradient complexity of L-Katyusha). *With the parameters (A3.3), ρ as in (A3.6) and $y_0=z_0=w_0=\theta_0$, for all $t\ge0$,*

$$\begin{aligned}\mathbb{E}\big[F(y_t)-F_*\big] &\le \theta_1(1-\rho)^t\Psi_0, \qquad \mathbb{E}\,\|z_t-\eta_*\|^2 \le \frac{2\eta}{L(1+\eta\sigma)}(1-\rho)^t\Psi_0,\\ \theta_1\Psi_0 &\le \frac{3L(n+3)}{8}\,\|\theta_0-\eta_*\|^2.\end{aligned} \tag{A3.20}$$

Consequently $\mathbb{E}[F(y_t)-F_*]\le\varepsilon$ as soon as $t\ge\big(4n+\sqrt{24n\kappa}\big)\log\frac{3L(n+3)\|\theta_0-\eta_*\|^2}{8\varepsilon}$. An iteration evaluates $f'_{i(t)}(x_t)$ and $f'_{i(t)}(w_t)$ in (K2) and, with probability $p=1/n$, the n gradients of a full pass: $2+pn=3$ evaluations of some f'_i in expectation. If the n gradients $f'_i(w)$ computed during the pass are stored ($\mathcal{O}(nd)$ memory, as for the table of SAGA), $f'_{i(t)}(w_t)$ is a look-up and the count is $1+pn=2$ (this is the "$\mathcal{O}(1+pn)$ times per iteration" of [47]). Adding the n evaluations of $F'(w_0)$ at initialization, for $\varepsilon\le\frac{3L(n+3)}{16}\|\theta_0-\eta_*\|^2$ the expected number of gradient evaluations is at most

$$\begin{aligned}&n+\big(12\,n+3\sqrt{24}\,\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon}\\ &\quad\le\ n+\big(12\,n+15\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon}\quad\textit{(without storage)},\\ &n+\big(8\,n+2\sqrt{24}\,\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon}\\ &\quad\le\ n+\big(8\,n+10\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon}\quad\textit{(with storage)},\end{aligned} \tag{A3.21}$$

and in both cases $\mathcal{O}\big((n+\sqrt{n\kappa})\log\frac{nL\|\theta_0-\eta_*\|^2}{\varepsilon}\big)$.

Proof. The first two bounds follow from $Y_t\le\Psi_t$, $Z_t\le\Psi_t$ and (A3.6). For Ψ_0, since $1/\eta=3\theta_1$ and by (T2) at the point η_* ($F'(\eta_*)=0$),

$$\begin{aligned}Z_0 &= \frac{3\theta_1L(1+\eta\sigma)}{2}\,\|\theta_0-\eta_*\|^2,\\ Y_0+W_0 &= \frac{2+n(1+\theta_1)}{2\theta_1}\big(F(\theta_0)-F_*\big) \le \frac{2+n(1+\theta_1)}{2\theta_1}\cdot\frac{L}{2}\,\|\theta_0-\eta_*\|^2.\end{aligned}$$

Using $\eta\sigma\le\frac23$, $\theta_1\le\frac12$ and $1\le\frac{1}{2\theta_1}$: $\Psi_0\le\frac{L}{2}\big(\frac52+\frac{2+3n/2}{2\theta_1}\big)\|\theta_0-\eta_*\|^2\le\frac{L}{2}\cdot\frac{9/2+3n/2}{2\theta_1}\|\theta_0-\eta_*\|^2=\frac{3L(n+3)}{8\theta_1}\|\theta_0-\eta_*\|^2$. For the iteration count use $(1-\rho)^t\le e^{-\rho t}$ and $1/\rho\le\max\{4n,\sqrt{24n\kappa}\}\le 4n+\sqrt{24n\kappa}$. For the gradient count let $A:=\frac{3L(n+3)}{8}\|\theta_0-\eta_*\|^2$: the smallest integer $t_\varepsilon\ge\frac1\rho\log\frac{A}{\varepsilon}$ suffices, and since $4n+\sqrt{24n\kappa}-\frac1\rho\ge\min\{4n,\sqrt{24n\kappa}\}\ge4$ and $\log\frac{A}{\varepsilon}\ge\log2>\frac14$, $t_\varepsilon\le\frac1\rho\log\frac{A}{\varepsilon}+1\le(4n+\sqrt{24n\kappa})\log\frac{A}{\varepsilon}$; t_ε iterations cost $3t_\varepsilon$ (resp. $2t_\varepsilon$) evaluations in expectation. □

Remark A3.13 (reading of the rate). Per iteration, Ψ contracts by $1-\rho$ with $\rho = \min\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\}$: the first term is the contraction of the mirror sequence (a strongly convex mirror step with the long step $\eta/L = 1/(3\theta_1 L)$ contracts the distance by $\frac{1}{1+\eta\sigma} \approx 1 - \frac{\sigma}{3\theta_1}$), the second the contraction of the reference term (w is refreshed with probability $p = 1/n$ and Y decreases by the fraction $\theta_1(1-\theta_2) = \theta_1/2$). Balancing the two gives $\theta_1 \asymp \sqrt{\sigma n}$ and $\rho \asymp \sqrt{\sigma/n} = 1/\sqrt{n\kappa}$: this is the $\sqrt{\kappa}$ scaling of hypothesis (A4). For SAGA (Prop. 5.9) the analogous balance is $\min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}$: the "parameter" contraction is $\asymp \sigma$ because the step is $\asymp 1/L$; here the mirror step is $\asymp 1/(\theta_1 L) \gg 1/L$ and the price is paid by the variance, which is cancelled by the negative momentum (Step 6) instead of being absorbed by the progress term $-2\gamma(1-\gamma R^2)F'(\theta)^\top(\theta - \eta_*)$ as in Step 1 of the proof of Prop. 5.9.

A3.3 Where the negative momentum acts (A3)

The proof of Lemma A3.11 contains a single place where θ_2 is used for something other than bookkeeping: Step 6, equation (A3.19). Let us isolate it.

Proposition A3.14 (Bregman-gap cancellation by the negative momentum). *In the chain of Lemma A3.11, the variance of the estimator enters through L3 and L1 (Lemmas A3.8 and A3.6) as the term $-\frac{\theta_2}{\theta_1} D_F(w_t, x_t)$ with $D_F(w_t, x_t) = \big(F(w_t) - F(x_t)\big) + \langle F'(x_t), x_t - w_t\rangle$, and the weight θ_2 of w_t in (K1) enters, through the decomposition (A3.17), as the term $+\frac{\theta_2}{\theta_1}\langle F'(x_t), x_t - w_t\rangle$. The two inner products cancel identically, and the remainder $-\frac{\theta_2}{\theta_1}(F(w_t) - F(x_t))$ is absorbed as follows: $F(x_t)$ disappears because the coefficients of $F(x_t)$ sum to zero (Step 7 of the proof of Lemma A3.11), and $\frac{\theta_2}{\theta_1}(F(w_t) - F_*) = \frac{p}{1+\theta_1} W_t$ is paid by the reference-point potential, which contracts by (A3.14) because w is refreshed with probability p.*

Proof. This is Steps 5–8 of the proof of Lemma A3.11; the identity $D_F(w_t, x_t) = (F(w_t) - F(x_t)) + \langle F'(x_t), x_t - w_t\rangle$ is the definition of D_F. □

Three comments make the mechanism explicit.

1. *Why a cancellation is needed.* The variance term is not small: by (T2), $D_F(w_t, x_t) \le \frac{L}{2}\|w_t - x_t\|^2$ is a squared distance at the scale L, whereas the certified decrease of the potential per step is of order $\rho\Psi_t$ with $\rho \asymp 1/\sqrt{n\kappa}$. To absorb a term $\asymp L\|w_t - x_t\|^2$ into $\rho\Psi_t$ one would have to convert a squared distance into function values, which by strong convexity costs the factor κ (we quantify this in Proposition A3.19). The cancellation avoids any such conversion: the inner-product part of D_F is removed identically, and the function-value part (a signed difference of function values; see the discussion of $c(t)$ in Table 6) is of the same nature as Y_t and W_t.

2. *What the negative momentum costs.* Because $\theta_2 w_t$ takes a fraction θ_2 of the weight in (K1), the function-value potential Y_t contracts by $\theta_1(1-\theta_2)$ instead of θ_1 (compare (A3.5) with the deterministic Proposition A3.17 below), and the reference-point potential must carry the weight $\frac{\theta_2(1+\theta_1)}{p\theta_1} \asymp n/\theta_1$ in front of $F(w_t) - F_*$. With $\theta_2 = \frac{1}{2}$ this halves the momentum contraction, a constant factor.

3. *Why the step size is tied to θ_2.* By Remark A3.9, the coefficient of $\|g_t - F'(x_t)\|^2$ produced by Young's inequality in L3 is $\frac{\nu}{2L\theta_1}$ with $\nu = \frac{\eta\theta_1}{1-\eta\theta_1}$. For the cancellation (A3.19) to be exact one needs $\nu = \theta_2$, i.e. (A3.2). A larger step ($\nu > \theta_2$) leaves a residual $+\frac{\nu-\theta_2}{\theta_1} D_F(w_t, x_t)$ on the right-hand side of (A3.15), which is exactly the uncontrolled term of the naive scheme below (with $\nu - \theta_2$ in place of ν). A smaller step ($\nu < \theta_2$) wastes part of the pull.

A3.4 (A2) Naive acceleration: Nesterov's momentum with the SVRG estimator

Definition A3.15 (naive accelerated SVRG). *The* naive scheme $\mathrm{N}(\theta_1,\eta,p)$ *is* (K1)–(K5) *with* $\theta_2=0$ *and the step* $\eta>0$ *taken as a free parameter with* $\eta\theta_1\le 1$*:*

$$\begin{aligned} &x_t=\theta_1 z_t+(1-\theta_1)y_t, \qquad g_t \text{ as in (K2)}, \qquad z_{t+1}=\frac{\eta\sigma x_t+z_t-\frac{\eta}{L}g_t}{1+\eta\sigma},\\ &y_{t+1}=x_t+\theta_1(z_{t+1}-z_t), \qquad w_{t+1}=\begin{cases} y_t & w.p.\ p,\\ w_t & w.p.\ 1-p.\end{cases}\end{aligned} \tag{A3.22}$$

We set $\nu:=\frac{\eta\theta_1}{1-\eta\theta_1}\in(0,\infty]$*, with* $\nu=\infty$ *in the limiting case* $\eta\theta_1=1$*; this case is admissible only with exact gradients* $g_t=F'(x_t)$ *(Proposition A3.17), because the Young step of L3 (Lemma A3.8) requires* $\nu<\infty$ *(Remark A3.9). The potential is the same* Ψ_t *as in* (A3.4) *(with* W_t *defined with an arbitrary constant* $a_W\ge 0$ *in place of* $\frac{\theta_2(1+\theta_1)}{p\theta_1}$*, which is* 0 *when* $\theta_2=0$*).*

Remark A3.16 (relation with Nesterov's method of the book). Eliminating z from (A3.22): $y_{t+1}=\theta_1 z_{t+1}+(1-\theta_1)y_t$ and $x_{t+1}=\theta_1 z_{t+1}+(1-\theta_1)y_{t+1}$ give $x_{t+1}=y_{t+1}+(1-\theta_1)(y_{t+1}-y_t)$, which is the extrapolation (5.17) of the book, $\eta_t=\theta_t+\beta(\theta_t-\theta_{t-1})$ (a point, in the book's notation; it is written v_t in the outcome of the experiments, Section A3.5), with $\beta=1-\theta_1$. The book takes $\beta=\frac{1-\sqrt{\mu/L}}{1+\sqrt{\mu/L}}=1-2\sqrt{\mu/L}+\mathcal{O}(\mu/L)$: with $\theta_1=\sqrt{\mu/L}$ the two momentum coefficients agree at zeroth order (both are $1-\Theta(\sqrt{\mu/L})$), but the momentum deficits differ by a factor 2 ($1-\beta=\sqrt{\mu/L}$ here, against $2\sqrt{\mu/L}+\mathcal{O}(\mu/L)$ in the book). The gradient step reads $y_{t+1}=x_t-\frac{\theta_1\eta}{L(1+\eta\sigma)}g_t+\frac{\theta_1\eta\sigma}{1+\eta\sigma}(x_t-z_t)$: it is (5.16) with $F'(\eta_{t-1})$ replaced by g_t, step $\frac{\theta_1\eta}{L(1+\eta\sigma)}$, and a drift which comes from putting the strong convexity in the mirror step. Since $x_t-z_t=-\frac{1-\theta_1}{\theta_1}(x_t-y_t)$, the drift equals $-\frac{\eta\sigma(1-\theta_1)}{1+\eta\sigma}(x_t-y_t)$, i.e. it is of relative size $\eta\sigma$ with respect to the momentum term $x_t-y_t=(1-\theta_1)(y_t-y_{t-1})$, where $\eta\sigma=\sqrt{\sigma}$ for the parameters of Proposition A3.17 and $\eta\sigma=\frac{\sigma}{3\theta_1}$ for (A3.3) ($=\sqrt{\sigma/(6n)}$ when $\kappa\ge 8n/3$). The drift is therefore small compared with the momentum term, but of the same order $\mathcal{O}(\sqrt{\sigma})$ as the discrepancy between the two momentum coefficients, which is the order that governs the accelerated rate: the two-sequence form of the book and (A3.22) coincide only up to $\mathcal{O}(\sqrt{\sigma})$ corrections of the momentum term. (Carrying the strong convexity by a proximal term does not remove the drift. If $f_i=\ell_i+\frac{\mu}{2}\|\cdot\|^2$ with ℓ_i convex, as for ℓ_2-regularized logistic regression with $\mu=\lambda$, then $g_t=\tilde{g}_t+\mu x_t$ with $\tilde{g}_t$ the SVRG estimator of $\frac{1}{n}\sum_i\ell_i$, and (K3) is exactly $z_{t+1}=\mathrm{prox}_{\gamma g}(z_t-\gamma\tilde{g}_t)=\frac{z_t-\gamma\tilde{g}_t}{1+\gamma\mu}$ with $g=\frac{\mu}{2}\|\cdot\|^2$ and $\gamma=\eta/L$, the z-step of Algorithm 1 of [21]: the drift is the gradient of g taken at z_t instead of x_t. On such problems (A3.22) is therefore their Algorithm 1 with $\tau=\theta_1$ and $\gamma=\eta/L$, except that their SVRG variant refreshes the reference point at the query point.) The two-sequence form of the book with F' replaced by a mini-batch SVRG estimator, a proximal step, a reference point refreshed every m iterations and the momentum restarted at each refresh ($x_1=y_1=\tilde{x}$ in their notation) is the method of Nitanda [67]; Allen-Zhu writes that it "corresponds to this paper but *without* Katyusha momentum (i.e., $\tau_2=0$)" [2, Appendix A], and Driggs et al. write their scheme in the same two-sequence form ("$x_{k+1}=y_k+(1-\tau_k)(y_k-y_{k-1})$") [21, Section 2].

The deterministic case closes. First we check that the proof machinery of Section A3.2 does prove acceleration for the scheme with $\theta_2=0$ when the gradient is exact; this is the analogue of Exercise 5.14 of the book in the linear-coupling form of [3].

Proposition A3.17 (acceleration with exact gradients and $\theta_2 = 0$)**.** *Let $g_t = F'(x_t)$ in (A3.22) (full gradient), $\theta_2 = 0$, $\eta = 1/\theta_1$ (so $\eta\theta_1 = 1$ and $\nu = \infty$: the limiting case of Definition A3.15) and $\theta_1 = \sqrt{\sigma} = \sqrt{\mu/L}$. Then for all t,*

$$Z_{t+1} + Y_{t+1} \le \frac{1}{1+\sqrt{\sigma}} Z_t + (1-\sqrt{\sigma})\, Y_t \le \Big(1 - \frac{1}{2}\sqrt{\frac{\mu}{L}}\Big)(Z_t + Y_t), \tag{A3.23}$$

hence $F(y_t) - F_* \le \sqrt{\mu/L}\,(1 - \frac{1}{2}\sqrt{\mu/L})^t (Z_0 + Y_0)$: the accelerated rate, as for (5.16)–(5.17) in the book.

Proof. L2 holds verbatim (it does not involve x_t's composition nor the nature of g_t). For L3 with $g_t = F'(x_t)$ and $\eta\theta_1 = 1$, the computation of Lemma A3.8 gives $\frac{L}{2\eta}\|z_{t+1} - z_t\|^2 + \langle F'(x_t), z_{t+1} - z_t\rangle = \frac{1}{\theta_1}\big[\frac{L}{2}\|y_{t+1} - x_t\|^2 + \langle F'(x_t), y_{t+1} - x_t\rangle\big] \ge \frac{1}{\theta_1}(F(y_{t+1}) - F(x_t))$ by (T2): no Young step and no variance term. Steps 1–5 and 7 of Lemma A3.11 with $\theta_2 = 0$ (so (A3.17) reads $z_t - x_t = \frac{1-\theta_1}{\theta_1}(x_t - y_t)$ and no w-terms appear) give $Z_{t+1} + Y_{t+1} \le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1) Y_t$ exactly as in (A3.15) with $\theta_2 = 0$, $p = 0$. With $\eta = 1/\theta_1$ and $\theta_1 = \sqrt{\sigma}$: $\eta\sigma = \sqrt{\sigma} \le 1$, hence $\frac{1}{1+\sqrt{\sigma}} \le 1 - \frac{\sqrt{\sigma}}{2}$ and $1 - \sqrt{\sigma} \le 1 - \frac{\sqrt{\sigma}}{2}$. □

The stochastic case does not close. Now replace $F'(x_t)$ by the loopless SVRG estimator.

Proposition A3.18 (one-step inequality of the naive scheme)**.** *For the scheme $\mathrm{N}(\theta_1, \eta, p)$ of Definition A3.15 with $\eta\theta_1 < 1$ (so that $\nu < \infty$), for all t,*

$$\begin{aligned} \mathbb{E}_{i(t)}\big[Z_{t+1} + Y_{t+1}\big] &\le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1)\, Y_t + \frac{\nu}{2L\theta_1}\mathbb{E}_{i(t)}\big\|g_t - F'(x_t)\big\|^2 \\ &\le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1)\, Y_t + \frac{\nu}{\theta_1} D_F(w_t, x_t), \end{aligned} \tag{A3.24}$$

where the last inequality is the variance bound (A3.9). The extra term $\frac{\nu}{\theta_1} D_F(w_t, x_t)$ is nonnegative; under Assumption A3.1 it vanishes only when $w_t = x_t$, since $D_F(w_t, x_t) \ge \frac{\mu}{2}\|w_t - x_t\|^2$ by the μ-strong convexity of F.

Proof. L1 and L2 hold verbatim. L3 holds with θ_2 replaced by ν (Remark A3.9; the proof of Lemma A3.8 only used $\frac{1}{\eta\theta_1} - 1 = \frac{1}{\nu} > 0$ in Young's inequality with $\beta = L/\nu$). In the proof of Lemma A3.11, Step 2 now reads $z_t - x_t = \frac{1-\theta_1}{\theta_1}(x_t - y_t)$, so Steps 1–5 give

$$\begin{aligned} F_* \;\ge\; & F(x_t) + \frac{1-\theta_1}{\theta_1}\big(F(x_t) - F(y_t)\big) + \mathbb{E}_{i(t)}[Z_{t+1}] - \frac{Z_t}{1+\eta\sigma} \\ & + \frac{1}{\theta_1}\mathbb{E}_{i(t)}\big[F(y_{t+1}) - F(x_t)\big] - \frac{\nu}{2L\theta_1}\mathbb{E}_{i(t)}\big\|g_t - F'(x_t)\big\|^2. \end{aligned}$$

The coefficients of $F(x_t)$ sum to $1 + \frac{1-\theta_1}{\theta_1} - \frac{1}{\theta_1} = 0$, and $F_* = \frac{1}{\theta_1}F_* - \frac{1-\theta_1}{\theta_1}F_*$; rearranging as in Step 7 gives the first inequality of (A3.24). There is no Step 6: nothing in the chain produces a term $+\frac{\nu}{\theta_1}\langle F'(x_t), x_t - w_t\rangle$, because w_t does not appear in x_t. Nonnegativity of D_F is (T1), and $D_F(w_t, x_t) \ge \frac{\mu}{2}\|w_t - x_t\|^2$ by (T4) with $\theta = w_t$, $\eta = x_t$. □

Compared with (A3.15), the right-hand side of (A3.24) contains the *uncontrolled term* $\frac{\nu}{\theta_1} D_F(w_t, x_t)$, where ν is tied to the step: for the Katyusha step ($\eta\theta_1 = \frac{1}{3}$) $\nu = \frac{1}{2}$, and for the deterministic step of Proposition A3.17 ($\eta\theta_1 \to 1$) $\nu \to \infty$. The decrease of the potential Ψ that (A3.24) can certify per step is at most $\frac{\eta\sigma}{1+\eta\sigma} Z_t$ (distance of the mirror sequence to η_*) plus $\theta_1 Y_t$

(function value at y_t), plus, if one adds a reference term $a_W(F(w_t)-F_*)$, the expected decrease $pa_W(F(w_t)-F_*)-pa_W(F(y_t)-F_*)$ of that term; none of these involves the query point x_t or $F'(x_t)$. The next statement quantifies what it costs to express $D_F(w_t,x_t)$ in terms of the potential.

Proposition A3.19 (closing by smoothness and strong convexity costs κ)**.** *Under Assumption A3.1, for the scheme* $\mathrm{N}(\theta_1,\eta,p)$ *of Definition A3.15 with* $\eta\theta_1<1$,

$$\frac{\nu}{\theta_1}D_F(w_t,x_t)\ \le\ 2\kappa\nu\Big[\frac{1}{\theta_1}\big(F(w_t)-F_*\big)+\frac{\eta}{1+\eta\sigma}Z_t+(1-\theta_1)\,Y_t\Big], \tag{A3.25}$$

so that, writing $W_t=a_W(F(w_t)-F_*)$ *with* $a_W>0$ *and using* (A3.14) *with* $\theta_2(1+\theta_1)/(p\theta_1)$ *replaced by* a_W,

$$\mathbb{E}_{i(t)}[\Psi_{t+1}]\le\Big(\frac{1+2\kappa\nu\eta}{1+\eta\sigma}\Big)Z_t+\Big(1-\theta_1+2\kappa\nu(1-\theta_1)+pa_W\theta_1\Big)Y_t+\Big(1-p+\frac{2\kappa\nu}{\theta_1 a_W}\Big)W_t. \tag{A3.26}$$

The coefficient of Z_t *is* <1 *iff* $2\kappa\nu\eta<\eta\sigma$, *i.e.* $\nu<\frac{1}{2\kappa^2}$; *the coefficient of* Y_t *is* <1 *only if* $2\kappa\nu(1-\theta_1)<\theta_1$, *i.e.* $\nu<\frac{\theta_1}{2\kappa(1-\theta_1)}$; *the coefficient of* W_t *is* <1 *iff* $a_W>\frac{2\kappa\nu}{p\theta_1}$ *(substituting this bound in the* Y*-coefficient sharpens the second condition to* $\nu<\frac{\theta_1}{2\kappa(2-\theta_1)}$, *which we do not need). No choice of* (θ_1,η,p,a_W) *makes* (A3.26) *certify a rate better than* $1-\kappa^{-2}$ *per iteration: worse than gradient descent with step* $1/L$, *which contracts by* $1-1/\kappa$ *per iteration (per gradient evaluation, with* $p=1/n$, *the comparison is* $1-\frac{1}{3\kappa^2}$ *against* $1-\frac{1}{n\kappa}$, *so the naive certificate is worse as soon as* $\kappa>n/3$*), let alone the accelerated* $1-\Theta(1/\sqrt{n\kappa})$.

Proof. By (T2), $D_F(w_t,x_t)\le\frac{L}{2}\|w_t-x_t\|^2\le L\|w_t-\eta_*\|^2+L\|x_t-\eta_*\|^2$ (triangle and $(a+b)^2\le 2a^2+2b^2$). By (T4) at $\eta=\eta_*$, $\|\theta-\eta_*\|^2\le\frac{2}{\mu}(F(\theta)-F_*)$ for all θ, so $L\|w_t-\eta_*\|^2\le 2\kappa(F(w_t)-F_*)$ and $L\|x_t-\eta_*\|^2\le 2\kappa(F(x_t)-F_*)$. By convexity of F and (A3.22), $F(x_t)-F_*\le\theta_1(F(z_t)-F_*)+(1-\theta_1)(F(y_t)-F_*)$, and by (T2) at η_*, $F(z_t)-F_*\le\frac{L}{2}\|z_t-\eta_*\|^2=\frac{\eta}{1+\eta\sigma}Z_t$ by (A3.4); also $F(y_t)-F_*=\theta_1Y_t$. Multiplying by ν/θ_1 gives (A3.25). Adding (A3.25) to (A3.24) and then $\mathbb{E}_{i(t)}[W_{t+1}]=(1-p)W_t+pa_W\theta_1Y_t$ gives (A3.26); the stated conditions are read off the coefficients, using $\sigma=1/\kappa$. Suppose all three coefficients are <1. Since $\nu\ge\eta\theta_1$, two cases: if $\theta_1\le\frac12$, the Y-condition gives $\eta\le\frac{\nu}{\theta_1}<\frac{1}{2\kappa(1-\theta_1)}\le\frac{1}{\kappa}$, hence $\eta\sigma<\kappa^{-2}$; if $\theta_1>\frac12$, the Z-condition gives $\eta\le\frac{\nu}{\theta_1}<\frac{1}{2\kappa^2\theta_1}<\kappa^{-2}$, hence $\eta\sigma<\kappa^{-3}$. In all cases $\eta\sigma<\kappa^{-2}$, and the certified contraction factor is at least the Z-coefficient, $\frac{1+2\kappa\nu\eta}{1+\eta\sigma}\ge\frac{1}{1+\eta\sigma}>\frac{1}{1+\kappa^{-2}}\ge 1-\kappa^{-2}$, which proves the last claim. □

Remark A3.20 (other routes, and scope of the obstruction). (i) In the proof of Proposition A3.19, the route through Z_t instead of μ is no better: $L\|x_t-\eta_*\|^2\le L\theta_1\|z_t-\eta_*\|^2+L(1-\theta_1)\|y_t-\eta_*\|^2$ and $L\theta_1\|z_t-\eta_*\|^2=\frac{2\eta\theta_1}{1+\eta\sigma}Z_t$, so the Z-coefficient receives $\frac{2\nu\eta}{1+\eta\sigma}$ against the available $\frac{\eta\sigma}{1+\eta\sigma}$: again $\nu<\sigma/2=\frac{1}{2\kappa}$, a factor κ away from the $\nu=\Theta(1)$ needed for a step $\asymp 1/L$. (ii) Paying D_F by the reference term alone is impossible: the reference potential pays differences of function values, and $D_F(w_t,x_t)-(F(w_t)-F(x_t))=\langle F'(x_t),x_t-w_t\rangle$ is not sign-definite and not a difference of function values; this inner product is precisely what the anchor term produces in (A3.19). (iii) The comparison with the *non-accelerated* proof of Prop. 5.9 is instructive: there the variance term $2\gamma^2R^2F'(\theta_{t-1})^\top(\theta_{t-1}-\eta_*)$ (eq. (5.31)) is absorbed by the progress term $-2\gamma F'(\theta_{t-1})^\top(\theta_{t-1}-\eta_*)$, which is possible because both scale with the *same* quantity and the step is $\gamma\le 1/(4R^2)$; the price is that the contraction is then $\asymp\gamma\mu\asymp\sigma$, not $\sqrt{\sigma}$. In the accelerated proof the progress is measured by Y_t and Z_t, which are not comparable with $D_F(w_t,x_t)$ without the factor κ. (iv) Propositions A3.18 and A3.19 show that *this* Lyapunov argument does not certify acceleration for the naive scheme; they do not prove that the scheme is not accelerated (no lower bound specific to the scheme is known to us).

What is known. We summarize the literature on the anchor-free scheme, quoting the statements we rely on.

- *Nitanda (2014, 2016): acceleration needs large mini-batches.* Corollary 1 and Table 1 of [67] give, for Acc-Prox-SVRG with mini-batch size b, the overall complexity $\mathcal{O}\big((n+\frac{n-b}{n-1}\kappa)\log\frac{1}{\varepsilon}\big)$ when $b<b_0$ and $\mathcal{O}\big((n+b\sqrt{\kappa})\log\frac{1}{\varepsilon}\big)$ otherwise (with a threshold b_0 depending on n and κ), and the paper comments: "if $b=1$, then Acc-Prox-SVRG has the same complexity as that of Prox-SVRG, while if $b=n$ then the complexity of this method is equal to that of APG". For $b=1$ this is $\mathcal{O}((n+\kappa)\log\frac{1}{\varepsilon})$: no acceleration. Allen-Zhu [2, Remark 5.7] summarizes the same bound as follows: "In the simple case $\bar{L}=L$, Nitanda [46] obtained a total work $\tilde{\mathcal{O}}\big(n+\frac{n-b}{n-1}\frac{L}{\sigma}+b\sqrt{L/\sigma}\big)$ (see their page 7), which also implies a phase transition for b. However, this result is no better than ours for all parameters $b\in\{1,2,\dots,n\}$. In fact, in terms of total work, Nitanda's result is even slower than SVRG when $b\le n/2$, and slower than Nesterov's method when $b>n/2$." (Here σ is our μ.) Allen-Zhu adds (footnote 21): "Nitanda's method is usually not considered as an accelerated method, since it requires mini-batch size to be very large in order to be accelerated. If mini-batch is large then one can use full-gradient method directly and acceleration is trivial." Nitanda's later AMSVRG [68, Theorem 3] with *increasing* mini-batches has, under the assumptions stated there, an overall complexity "$\mathcal{O}\big((n+\frac{n\kappa}{n+\sqrt{\kappa}})\log\frac{1}{\varepsilon}\big)$"; since $\frac{n\kappa}{n+\sqrt{\kappa}}=\frac{1}{2}H(\kappa,n\sqrt{\kappa})$ with H the harmonic mean, this is $\tilde{\mathcal{O}}(n+\min\{\kappa,n\sqrt{\kappa}\})$, the entry of Table 1 there for both Acc-SVRG and AMSVRG in the strongly convex column, while the optimal methods (Acc-SDCA, APCG, SPDC) are listed with $\tilde{\mathcal{O}}(n+\min\{\kappa,\sqrt{n\kappa}\})$; the paper states that "for strongly convex problems, AMSVRG is slower than optimal methods". Neither bound is better than the best of SVRG and accelerated gradient descent.

- *Allen-Zhu's discussion and experiment.* Footnote 1 of [2]: "experimentalists have observed that momentums could sometimes help if stochastic gradient iterations are used. However, the so-obtained methods (1) sometimes fail to converge in an accelerated rate, (2) become unstable and hard to tune, and (3) have no support theory behind them." Section 7.1 ("Effectiveness of Katyusha momentum") compares Katyusha with $\tau_2=\frac{1}{2}$ and $\tau_2=0$ on ridge regression and concludes: "the old heuristics (i.e., $\tau_2=0$) sometimes indeed make the method faster after careful parameter tuning. However, for certain tasks such as Figure 1(c), without Katyusha momentum the algorithm does not even enjoy an accelerated convergence rate."

- *Hu, Wright and Lessard (2018): the SDP certificate needs the anchor.* In [42] the one-iteration inequality of Katyusha is recovered as a dissipation inequality certified by a 6×6 linear matrix inequality (their Section 4.3 and Lemma 11). Generalizing the parameters they obtain: "the one-iteration convergence result (27) holds for any $\frac{1}{5}\le\tau_2<1$ and $\tau_1\le\min\{\frac{5\tau_2-1}{9\alpha L\tau_2},1-\tau_2\}$" (in their notation, α is Katyusha's mirror step size, σ the strong-convexity modulus, i.e. our μ, and ρ^2 the contraction factor). This is derived for one explicit storage function (their matrix P of eq. (32)) and fixed multipliers ($\rho^2=\frac{1}{1+\alpha\sigma}$, $\lambda_1=\alpha/\tau_1$): with this certificate the admissible momentum weight τ_1 shrinks to 0 as $\tau_2\downarrow\frac{1}{5}$, and no certificate is given for $\tau_2<\frac{1}{5}$, in particular not for the anchor-free scheme $\tau_2=0$. The paper does not claim that the LMI is infeasible for $\tau_2<\frac{1}{5}$ (it stresses that "Lemma 11 works for general choices of ζ and τ_2") and reports no search over other certificates, so the absence of an anchor-free certificate in this family is consistent with, but does not prove, Propositions A3.18–A3.19. (The companion jump-system/LMI analysis of SAGA, Finito and SDCA without momentum is [41], whose model cannot represent SVRG; the LMI analysis of SVRG is in [42] itself.)

- *Driggs, Ehrhardt and Schönlieb (2022): anchor-free acceleration with mini-batches.* [21] analyze, for generic estimators, the anchor-free scheme (A3.22) when the strong convexity can be split off the components as a proximal term: their Algorithm 1 is $x_{k+1} = \tau z_k + (1-\tau) y_k$, $z_{k+1} = \operatorname{prox}_{\gamma g}(z_k - \gamma \tilde{\nabla}_{k+1})$, $y_{k+1} = \tau z_{k+1} + (1-\tau) y_k$, with the strong convexity assumed on the proximal term g ("If g is μ-strongly convex"). For ℓ_2-regularized objectives ($f_i = \ell_i + \frac{\lambda}{2}\|\cdot\|^2$ with ℓ_i convex, $\mu = \lambda$, $g = \frac{\lambda}{2}\|\cdot\|^2$), their Algorithm 1 with $\tau = \theta_1$, $\gamma = \eta/L$ and their SVRG estimator with $b = 1$ is exactly (A3.22), except that their reference point is reset to the query point (with probability $1/p$ in their notation) instead of to y_t (Remark A3.16). They explain the difficulty in the same terms as above: "In the analysis of Katyusha, negative momentum cancels part of the variance term, leaving telescoping terms that decrease at an accelerated rate. Without negative momentum, we must handle the variance term differently." For the SVRG estimator with mini-batch size b and full gradient with probability $1/p$, their Theorem 4 (Theorem 14 of arXiv:1910.09494v3) gives, under the condition $b \le 32p^2$ and for g μ-strongly convex, $\mathbb{E}\|z_T - x^*\|^2 \le \big(1 + \min\{\frac{\sqrt{b\mu}}{4p\sqrt{6L}}, \frac{b}{4p}\}\big)^{-T} K_2$, at an expected cost of "$n/p + b$" stochastic gradient evaluations per iteration; with $b = 1$ and $p = n$ (admissible since $1 \le 32n^2$; one full pass every n iterations on average, as in L-Katyusha) this is a contraction $1 - \Theta\big(\frac{1}{n\sqrt{\kappa}}\big)$ per iteration, i.e. $\mathcal{O}(n\sqrt{\kappa}\log\frac{1}{\varepsilon})$ gradient evaluations: worse than SVRG's $n + \kappa$ unless $\kappa \ge n^2$ and worse than $\sqrt{n\kappa}$ by the factor $\sqrt{n}$. Their Remark 4 is stated for the SAGA estimator and transferred to SVRG: "when using the SAGA gradient estimator on a strongly convex objective with $b = \mathcal{O}(n^{2/3})$ and $\gamma = \mathcal{O}(1/\sqrt{\mu L})$, Algorithm 1 finds a point satisfying $\mathbb{E}\|z_T - x^*\|^2 \le \epsilon$ in $\mathcal{O}(n^{2/3}\sqrt{\kappa}\log(1/\epsilon))$ iterations. This is compared to the complexity lower bound of $\mathcal{O}(\sqrt{n\kappa}\log(1/\epsilon))$ achieved by Katyusha [37]. SVRG has a similar complexity when $b = n^{2/3}$ and $p = n^{1/3}$" (again $b \le 32p^2$). So, as for Nitanda, acceleration without the anchor is proved only with mini-batches of size $b \asymp n^{2/3}$ and remains suboptimal by $n^{1/6}$.

- *Indirect acceleration.* Catalyst [54] accelerates SAGA/SVRG as inner solvers with complexity $\tilde{\mathcal{O}}(\sqrt{n\kappa}\log\frac{1}{\varepsilon})$ when $n \le \kappa$ (their Table 1), with an extra logarithmic factor; this is a different mechanism (proximal-point outer loop), not momentum on the estimator.

To our knowledge, no theorem shows that (A3.22) with $b = 1$ attains $\mathcal{O}((n + \sqrt{n\kappa})\log\frac{1}{\varepsilon})$, and no lower bound excludes it; the honest statement of (A2) is therefore: *the Lyapunov proof of L-Katyusha closes only because of the cancellation* (A3.19)*; without the anchor the best available certificates (Nitanda, Driggs et al.) give no improvement over* $\min\{n + \kappa, n\sqrt{\kappa}\}$ *for* $b = 1$.

Test plan for (A2) (partly executed; see the outcome at the end of Section A3.5). The plan was to compare, on the same instances as hypotheses (A4) and (B3) (the hard finite-sum instance with $\kappa \gg n$ of the A1 note, and ℓ_2-regularized logistic regression with $d = 40$ Gaussian inputs as in Fig. 5.4 of the book, $\lambda \in \{R^2/n, R^2/(10n), R^2/(100n)\}$ to vary κ at fixed n):

1. SAGA with $\gamma = 1/(4R^2)$ (Prop. 5.9) and L-SVRG with $\eta = 1/(6L)$, $p = 1/n$ (Theorem 5 of [47]);
2. L-Katyusha with (A3.3);
3. the naive scheme (A3.22) with the *same* θ_1, η, p as L-Katyusha (so that the only difference is the anchor), with the smaller steps $\eta \in \{\frac{1}{3\theta_1}, \frac{1}{3}, \frac{1}{3\kappa\theta_1}\}$, and in the two-sequence form of the book (Remark A3.16) with $\beta = \frac{1-\sqrt{\mu/L}}{1+\sqrt{\mu/L}}$ and step $1/L$;

4. the anchor-free scheme tuned as in Theorem 4 of [21] with $b = 1$ and $b = \lceil n^{2/3} \rceil$.

For each run we measure the linear rate per gradient evaluation, $\hat{\rho} := -$slope of $\log(F(y_t) - F_*)$ against the number of f_i' evaluations, and fit the exponent ξ in $\hat{\rho} \propto \kappa^{-\xi}$ over the three values of λ. Predictions: (P1) $\xi \approx 1$ for SAGA and L-SVRG, $\xi \approx \frac{1}{2}$ for L-Katyusha once $\kappa \geq 8n/3$ (regime where $\rho = 1/\sqrt{24n\kappa}$), $\xi \approx 0$ for all methods when $\kappa \ll n$ (the $n \log \frac{1}{\varepsilon}$ term dominates); (P2) the naive scheme with the Katyusha parameters ($\theta_1 = \sqrt{2n/(3\kappa)}$ small, mirror step $\frac{1}{3\theta_1 L}$ large) is non-monotone and may diverge for $\kappa \gg n$, because without the anchor $\|x_t - w_t\|$, hence the uncontrolled term $\frac{\nu}{\theta_1} D_F(w_t, x_t)$, is not kept small (this is the instability of Allen-Zhu's footnote 1); (P3) with a step small enough to be stable, $\xi \approx 1$ for the naive scheme (no $\sqrt{\kappa}$ scaling), and its rate is at best that of L-SVRG (*neither part is borne out*: see the outcome at the end of Section A3.5); (P4) for the naive scheme the ratio $v(t)$ of the uncontrolled term of (A3.24) to the certified decrease (Section A3.5) is frequently ≥ 1, i.e. the one-step inequality certifies no decrease at all, while for L-Katyusha the size $q(t)$ of the cancelled inner product relative to the certified one-step decrease (which is at least $\rho\Psi_t$) is of order one or larger; on the steps with $c(t) > 0$, where $q(t) = q_+(t)$ (Table 6), this means that the term removed by the anchor could not have been absorbed into the certified decrease, which is the mechanism behind (P2)–(P3) (on the steps with $c(t) \leq 0$ the removed term has the favourable sign). A positive result for the batch-$n^{2/3}$ variant of item 4 and a negative one for $b = 1$ would be consistent with [21, 67].

A3.5 What to instrument

Each inequality of the proof is a pair (left-hand side, right-hand side) whose ratio can be computed along a run (with η_* computed in closed form for the hard instance and by Newton's method for logistic regression; conditional expectations $\mathbb{E}_{i(t)}$ are computed exactly by enumerating the n values of $i(t)$, which costs one pass over the data except for $\mathbb{E}_{i(t)} F(y_{t+1})$, which costs n evaluations of F; this is affordable on the instances of (B3)). Tables 5 and 6 list them with the ratio to record; a ratio close to 1 means the inequality is tight on the trajectory. The last two rows of Table 6 concern the naive scheme. Three names of ratios are shared with the B3 note. Our r_{sc} follows the reverse convention (see its row); our r_{var} is the ratio of the whole variance bound (A3.9), whereas the r_{var} of the B3 note is the ratio of its step (I3), "variance $\leq$ second moment", the analogue of our r_{cen}; and our r_{coco} is the analogue, in the Bregman form and between x_t and w_t, of the ratio $r_{\mathrm{coco}} = L_{\mathrm{eff}}/R^2$ of the B3 note, which is in the symmetric form and between the iterate and η_*. Convention: a ratio is not recorded (left undefined; `NaN` in the code) at the steps where its denominator vanishes. At $t = 0$ the reference point is the initial point, $y_0 = z_0 = w_0 = \theta_0$, and the weights in (K1) sum to one, so $x_0 = w_0$: hence $D_F(w_0, x_0) = 0$, $\frac{1}{n}\sum_i \|f_i'(x_0) - f_i'(w_0)\|^2 = 0$ and $g_0 = F'(x_0)$ (exact estimator). The ratios with one of these quantities in the denominator, namely r_{cen}, r_{coco}, $r_{\mathrm{var}} = u$, c and $1 - c$, are then of the form $0/0$ and are undefined at $t = 0$ (the code records them only when $D_F(w_t, x_t) > 10^{-14}\,|F(x_t)|$, above the rounding error of $F(w_t) - F(x_t)$). If $\theta_0 \neq \eta_*$, every other ratio has a positive denominator at $t = 0$; in particular $q(0) = 0$ is defined (its numerator contains $x_0 - w_0 = 0$ and its denominator is a positive multiple of $\Psi_0 > 0$), and so are $r_{\mathrm{Y}}(0) = \bar{r}_{\mathrm{Y}}(0) = 0$ (as $g_0 = F'(x_0)$), $s_{\mathrm{cvx}}(0) = 0$ (as $y_0 = x_0$) and $v(0) = 0$ (the naive scheme also has $x_0 = w_0$). The ratios r_{sc} and μ_{eff} are undefined when $x_t = \eta_*$. For the Young row of Table 5, note that the step contains the noise: by (K3)–(K4), $y_{t+1} - x_t = -\gamma_y\big(g_t - \mu(x_t - z_t)\big)$ with $\gamma_y := \frac{\theta_1 \eta}{L(1+\eta\sigma)}$ (the drift coefficient $\frac{\theta_1\eta\sigma}{1+\eta\sigma}$ equals $\gamma_y L\sigma = \gamma_y \mu$), so that, with $\mathcal{V}_t := \mathbb{E}_{i(t)} \|g_t - F'(x_t)\|^2$, unbiasedness gives $\mathbb{E}_{i(t)}[-\langle g_t - F'(x_t), y_{t+1} - x_t\rangle] = \gamma_y \mathcal{V}_t$ and $\mathbb{E}_{i(t)} \|y_{t+1} - x_t\|^2 = \gamma_y^2\big(\mathcal{V}_t + \|F'(x_t) - \mu(x_t - z_t)\|^2\big)$; this gives the closed form of the averaged ratio $\bar{r}_{\mathrm{Y}}(t)$ in the table. For the cancellation row of Table 6: only when $c(t) > 0$ does the inner product removed by the anchor have the sign that a

bound without the anchor would have to absorb; if $c(t) \le 0$, then $D_F(w_t, x_t) \le F(w_t) - F(x_t)$, a difference of function values that needs no conversion of a squared distance. Hence $q_+(t) \ge 1$ means that the cancelled term could not have been absorbed into the certified decrease, whereas $q(t) \ge 1$ with $c(t) < 0$ only means that the two copies of the inner product in (A3.19) could not have been bounded separately.

Outcome of the experiments. (A3): L-Katyusha with (A3.3), 3 seeds, on the block-separable hard instance (H) of the report (Nesterov chains with the normalization of [50]; $n = 16$ blocks of size 64, $\kappa = 10^4$; the runs of (A4) use blocks of size 256) and on logistic regression ($n = 1000$, $d = 50$, covariance eigenvalues j^{-1}, $\kappa = 10^3$, 10^4), in this order (Figure A3 of the report). The statistics pool the instrumented states of the three seeds from $t = 0$ until Ψ_t first falls below 10^{-10} (rounding floor), i.e. over the first 16 to 60 passes. The contraction (A3.5) holds at every recorded state, componentwise (largest $(\mathbb{E}_{i(t)}\Psi_{t+1} - c_Z Z_t - c_Y Y_t - c_W W_t)/\Psi_t$: $-6.5 \cdot 10^{-4}$, $-2.3 \cdot 10^{-4}$, $-1.7 \cdot 10^{-4}$) and hence in the form $\mathbb{E}_{i(t)}\Psi_{t+1} \le \max\{c_Z, c_Y, c_W\}\Psi_t$ (largest $\mathbb{E}_{i(t)}\Psi_{t+1}/\Psi_t - \max\{c_Z, c_Y, c_W\}$: $-1.7 \cdot 10^{-3}$, $-4.0 \cdot 10^{-4}$, $-2.8 \cdot 10^{-4}$); the exact one-step contraction exponent averages 4.7, 2.8, 4.1 times the exponent $-\log\max\{c_Z, c_Y, c_W\}$ certified by (A3.5) (9.3, 3.7, 8.2 times the exponent $-\log(1-\rho)$ of (A3.6)). Geometric means: $r_{\text{cen}} = 0.94, 0.97, 0.99$; $r_{\text{coco}} = 0.13, 0.15, 0.15$; $r_{\text{var}} = 0.125, 0.146, 0.149$. $c(t) < 0$ on 77%, 77%, 64% of the steps; $q(t)$ has medians 1.42, 0.64, 1.40 and is ≥ 1 on 63%, 5%, 65% of them, but $c(t) < 0$ on 84%, 0%, 70% of these steps, on which the removed inner product has the sign that helps a bound without the anchor: the sign-aware ratio $q_+(t)$, obtained from the stored values of $q(t)$ and $\langle F'(x_t), x_t - w_t\rangle$, is ≥ 1 on only 9.9%, 5.2%, 19.3% of the recorded states, i.e. on 43%, 23%, 54% of the states with $c(t) > 0$ (where $q(t)$ has medians 0.84, 0.56, 1.09); cf. (P4). Other rows, $v(t)$ included, were not evaluated.

(A2): one seed, $\kappa = 10^4$ (Figure A2 of the report), on (H) with blocks of size 64 and on logistic regression with $d = 50$, to the target $\|\theta - \eta_*\|^2 \le 10^{-10}\|\theta_0 - \eta_*\|^2$; the gradient counts are those without storage of Section A3.6 (3 per iteration in expectation for L-SVRG, L-Katyusha and the anchor-free schemes, 1 for SAGA), read at the first record below the target. Without the anchor ($\theta_2 = 0$, same θ_1, η, p, i.e. the scheme $\mathrm{N}(\theta_1, \frac{1}{3\theta_1}, \frac{1}{n})$ of Definition A3.15), L-Katyusha diverges on (H) ($\|y_t - \eta_*\|^2 = 8.8 \cdot 10^{27}\|\theta_0 - \eta_*\|^2$ after $3.4 \cdot 10^3$ gradient evaluations), as (P2) predicts; in the runs of (A4) (Figure A4 of the report: blocks of size 256, (H) and block quadratics with random spectra, three seeds, relative accuracy 10^{-8}) it diverges for every $\kappa \ge 10^3$ and converges for $\kappa \le 3 \cdot 10^2$. On logistic regression it is stable and needs $4.90 \cdot 10^4$ gradient evaluations, against $1.06 \cdot 10^5$ for L-Katyusha, $8.78 \cdot 10^4$ for SAGA and $3.98 \cdot 10^5$ for L-SVRG. The two-sequence form $v_t = \theta_t + \beta(\theta_t - \theta_{t-1})$, $\theta_{t+1} = v_t - \gamma g_t$, with the estimator (K2) at v_t and the reference point reset to the query point v_t with probability $\frac{1}{n}$ (`nesterov_svrg` in `vrlab`; unlike Definition A3.15, it has no drift term (Remark A3.16), its reference point is reset to the query point instead of the gradient-step iterate, and (β, γ) are free), run with the setting of the grid $\beta \in \{0.5, 0.8, 0.9, 0.95, \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\}$, $\gamma R^2 \in \{\frac{1}{6}, \frac{1}{20}, \frac{1}{60}\}$ that is fastest to 10^{-6} ($\gamma R^2 = \frac{1}{6}$, the step of L-SVRG, on both instances), needs $4.66 \cdot 10^4$ gradient evaluations on logistic regression and $9.64 \cdot 10^4$ on (H), where L-Katyusha needs $3.20 \cdot 10^4$, SAGA $4.32 \cdot 10^5$ and L-SVRG $1.94 \cdot 10^6$. Prediction (P3) is therefore refuted in its second half: the stable anchor-free schemes need 8 to 20 times fewer gradient evaluations than L-SVRG at $\kappa = 10^4$, and in (A4) the two-sequence form, re-tuned at every κ over $\beta \in \{0.8, 0.95, \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\}$ and $\gamma R^2 \in \{\frac{1}{6}, \frac{1}{20}\}$ (on random spectra the grid is re-run and re-selected for every seed, the instance depending on the seed; on (H) the selected setting is common to the seeds), needs 7.8 to 48 times fewer gradient evaluations than L-SVRG and 1.7 to 11 times fewer than SAGA at every $\kappa \in [10^2, 10^5]$. Its first half, $\xi \approx 1$, is not supported either. In (A4) the slopes of log(gradient count) against $\log\kappa$ (least-squares fits of the per-κ geometric means over the seeds, over the $K = 6$ values $\kappa \ge 10n$, with 95%

Label	Inequality	Tightness ratio to measure (and expected behaviour)
(A3.7) centering	$\mathbb{E}_{i(t)} \left\|g_t - F'(x_t)\right\|^2 \leq \frac{1}{n}\sum_i \left\|f_i'(x_t) - f_i'(w_t)\right\|^2$	$r_{\mathrm{cen}}(t) = 1 - \dfrac{\left\|F'(x_t) - F'(w_t)\right\|^2}{\frac{1}{n}\sum_i \left\|f_i'(x_t) - f_i'(w_t)\right\|^2}$; close to 1 when the per-sample gradient differences are not aligned (expected for logistic regression).
(A3.8) co-coercivity	$\frac{1}{n}\sum_i \left\|f_i'(x_t) - f_i'(w_t)\right\|^2 \leq 2L\, D_F(w_t, x_t)$	$r_{\mathrm{coco}}(t) = \dfrac{\frac{1}{n}\sum_i \left\|f_i'(x_t) - f_i'(w_t)\right\|^2}{2L\, D_F(w_t, x_t)}$; for logistic regression expected $\approx L_{\mathrm{eff}}/L \ll 1$ where L is the worst-case constant $\max_i \left\|x_i\right\|^2/4 + \lambda$ (link with (B2)).
(A3.9) variance	$\mathbb{E}_{i(t)} \left\|g_t - F'(x_t)\right\|^2 \leq 2L\, D_F(w_t, x_t)$	$r_{\mathrm{var}}(t) = r_{\mathrm{cen}}(t)\, r_{\mathrm{coco}}(t)$ (not the r_{var} of the B3 note, the analogue of r_{cen}).
(A3.10) mirror step	slack $= \frac{\mu}{2}\left\|x_t - z_{t+1}\right\|^2$	$s_z(t) = \dfrac{\frac{\mu}{2}\left\|x_t - z_{t+1}\right\|^2}{\rho\, \Psi_t}$ (slack relative to the certified decrease; small means the dropped term is irrelevant).
(A3.12) smoothness	$F(y_{t+1}) - F(x_t) - \langle F'(x_t), y_{t+1} - x_t\rangle \leq \frac{L}{2}\left\|y_{t+1} - x_t\right\|^2$	$r_{\mathrm{sm}}(t) = \dfrac{D_F(y_{t+1}, x_t)}{\frac{L}{2}\left\|y_{t+1} - x_t\right\|^2} \in \left[\frac{\mu}{L}, \frac{L_F}{L}\right]$ ($L_F \leq L$ the smoothness constant of F): curvature of F along the step over L, an average of Rayleigh quotients of F'' on $[x_t, y_{t+1}]$ (cf. the μ_{eff} of (B2), Lemma B2.5); not the L_{eff} of (B2), built from per-sample gradients (co-coercivity row).
(A3.13) Young	$\langle g_t - F'(x_t), y_{t+1} - x_t\rangle \geq -\frac{\theta_2}{2L}\left\|g_t - F'(x_t)\right\|^2 - \frac{L}{2\theta_2}\left\|y_{t+1} - x_t\right\|^2$	$r_{\mathrm{Y}}(t) = \dfrac{-\langle g_t - F'(x_t), y_{t+1} - x_t\rangle}{\frac{\theta_2}{2L}\left\|g_t - F'(x_t)\right\|^2 + \frac{L}{2\theta_2}\left\|y_{t+1} - x_t\right\|^2} \in [-1, 1]$ per realization, and the ratio of conditional means $\bar{r}_{\mathrm{Y}}(t) = \dfrac{\gamma_y \mathcal{V}_t}{\frac{\theta_2}{2L}\mathcal{V}_t + \frac{L}{2\theta_2}\gamma_y^2\left(\mathcal{V}_t + \left\|F'(x_t) - \mu(x_t - z_t)\right\|^2\right)}$, with $\mathcal{V}_t$ and γ_y as in the text above. With (A3.3), $\bar{r}_{\mathrm{Y}}(t) \leq \frac{12(1+\eta\sigma)}{9(1+\eta\sigma)^2+4} \leq \frac{12}{13}$, with near-equality in the noise-dominated regime $\mathcal{V}_t \gg \left\|F'(x_t) - \mu(x_t - z_t)\right\|^2$: the Young step is then nearly tight, and it is loose only when the full gradient or the drift dominates the step.
(A3.16) strong convexity	$F_* - F(x_t) - \langle F'(x_t), \eta_* - x_t\rangle \geq \frac{\mu}{2}\left\|x_t - \eta_*\right\|^2$	$r_{\mathrm{sc}}(t) = \dfrac{2D_F(\eta_*, x_t)}{\mu\left\|x_t - \eta_*\right\|^2} \geq 1$ (the reverse of the convention $r_{\mathrm{sc}} = \mu/\mu_{\mathrm{eff}} \leq 1$ of the B3 note): the Bregman analogue of the μ_{eff}/μ of (B2). For comparability with (B2) also record the inner-product form $\mu_{\mathrm{eff}}(x_t)/\mu$ with $\mu_{\mathrm{eff}}(x_t) := \langle F'(x_t), x_t - \eta_*\rangle / \left\|x_t - \eta_*\right\|^2 \geq \mu$; the two differ, since $D_F(\eta_*, x_t) \leq \langle F'(x_t), x_t - \eta_*\rangle$ (because $F_* \leq F(x_t)$), so that $r_{\mathrm{sc}}(t) \leq 2\mu_{\mathrm{eff}}(x_t)/\mu$.
(A3.18) convexity	$\langle F'(x_t), x_t - y_t\rangle \geq F(x_t) - F(y_t)$	$s_{\mathrm{cvx}}(t) = \dfrac{D_F(y_t, x_t)}{F(y_t) - F_*}$ (slack relative to the Y-potential).

Table 5: Inequalities of the one-step lemmas L1–L3 and of the convexity steps of L5, with the ratio to record along a trajectory.

Label	Inequality	Tightness ratio to measure (and expected behaviour)
(A3.19) cancellation	identity	$c(t) = \dfrac{\langle F'(x_t), x_t - w_t\rangle}{D_F(w_t, x_t)} \in \mathbb{R}$ and $1 - c(t) = \dfrac{F(w_t) - F(x_t)}{D_F(w_t, x_t)}$: the *signed* shares of the Bregman gap removed by the anchor and left as a difference of function values. They sum to 1 by definition of D_F but are not fractions in $[0,1]$: $c(t) < 0$ when $\langle F'(x_t), x_t - w_t\rangle < 0$ (the anchor term then adds to the variance budget on that step), and $c(t) > 1$ when $F(w_t) < F(x_t)$ (the reference point is better than the query point and the function-value share is negative). Record $c(t)$ together with the sign of $F(w_t) - F(x_t)$, and the size of the cancelled term relative to the certified one-step decrease, $q(t) = \dfrac{\left\lvert \frac{\theta_2}{\theta_1} \langle F'(x_t), x_t - w_t\rangle \right\rvert}{(1 - \max\{c_Z, c_Y, c_W\})\,\Psi_t}$ (the denominator is the decrease certified by the next row; it is at least $\rho\,\Psi_t$), and its sign-aware part $q_+(t)$, the same ratio with $\left[\langle F'(x_t), x_t - w_t\rangle\right]_+ := \max\{\langle F'(x_t), x_t - w_t\rangle, 0\}$ in place of the absolute value (both are read as explained in the text of this section); and $u(t) = \dfrac{\frac{\theta_2}{2L\theta_1}\mathbb{E}_{i(t)} \left\lVert g_t - F'(x_t)\right\rVert^2}{\frac{\theta_2}{\theta_1} D_F(w_t, x_t)} = r_{\mathrm{var}}(t)$, the fraction of the cancelled budget actually used by the noise.
(A3.5) contraction	$\mathbb{E}_{i(t)}\Psi_{t+1} \le \max\{c_Z, c_Y, c_W\}\Psi_t$	$r_\Psi(t) = \dfrac{\mathbb{E}_{i(t)}\Psi_{t+1}}{\max\{c_Z, c_Y, c_W\}\,\Psi_t}$ and the three component ratios $\mathbb{E}_{i(t)}Z_{t+1}/Z_t$, $\mathbb{E}_{i(t)}Y_{t+1}/Y_t$, $\mathbb{E}_{i(t)}W_{t+1}/W_t$ against c_Z, c_Y, c_W (which of Z, Y, W is the bottleneck).
(A3.6) global rate	$\mathbb{E}\Psi_t \le (1-\rho)^t \Psi_0$	$\hat\rho_T(t) = -\frac{1}{T}\log\frac{\Psi_{t+T}}{\Psi_t}$ against ρ (expected $\hat\rho \gg \rho$ on logistic regression, cf. $r_{\mathrm{coco}}, r_{\mathrm{sc}}$).
(A3.24) naive, uncontrolled term	$\mathbb{E}_{i(t)}[Z_{t+1} + Y_{t+1}] \le \frac{Z_t}{1+\eta\sigma} + (1-\theta_1)Y_t + \frac{\nu}{\theta_1} D_F(w_t, x_t)$	$v(t) = \dfrac{\frac{\nu}{\theta_1} D_F(w_t, x_t)}{\frac{\eta\sigma}{1+\eta\sigma} Z_t + \theta_1 Y_t}$, the uncontrolled term over the certified decrease of $Z + Y$: $v(t) \ge 1$ exactly when the one-step inequality certifies no decrease of $Z_t + Y_t$ at step t.
(A3.25) naive, κ-cost	$\frac{\nu}{\theta_1} D_F \le 2\kappa\nu[\dots]$	ratio of the two sides; expected $\ll 1$ (the bound is loose by $\approx \kappa$), confirming that the obstruction is the proof, not the quantity.

Table 6: The cancellation, the contraction of the potential, the global rate, and the two inequalities of the naive scheme, with the ratio to record along a trajectory.

t-intervals with $K-2$ degrees of freedom) are, on (H) and on random spectra, 0.806 $[0.618, 0.995]$ and 0.814 $[0.649, 0.980]$ for the two-sequence form, whose profile is irregular (local slopes from 0.28 to 1.30), against 0.525 $[0.498, 0.551]$ and 0.500 $[0.496, 0.505]$ for L-Katyusha, 1.022 $[0.997, 1.047]$ and 1.003 $[1.002, 1.003]$ for SAGA, and 1.021 $[0.996, 1.047]$ and 1.002 $[1.001, 1.003]$ for L-SVRG; the last three confirm (P1) for $\kappa \geq 10n$. Propositions A3.18–A3.19 thus describe a limitation of the certificate (Remark A3.20(iv)), not the observed behaviour. Of the plan above, only items 1–2, the naive scheme of item 3 with $\eta = \frac{1}{3\theta_1}$ and the grid-tuned two-sequence form were run (at $\kappa = 10^4$ only), on (H) instead of the shared-regularizer instance of the A1 note and on logistic regression with $d = 50$ instead of $d = 40$ with three values of λ; the smaller steps $\eta \in \{\frac{1}{3}, \frac{1}{3\kappa\theta_1}\}$ of item 3, the two-sequence form with the book's β and the step $1/L$, and item 4 (the scheme of [21]) were not run; the number of gradient evaluations to the target replaced the slope $\hat{\rho}$, and the exponents were fitted only in (A4), on block quadratics with $\kappa \geq 10n$, so that the part $\kappa \ll n$ of (P1) was not tested.

A3.6 Complexity comparison

Table 7 collects the number of evaluations of a single f_i' needed to reach accuracy ε (on $\mathbb{E}[F - F_*]$ or $\mathbb{E}\,\|\cdot - \eta_*\|^2$, which differ by factors L or μ inside the logarithm), with the constants of the cited statements; $\kappa = L/\mu$ with L the per-sample smoothness constant, as everywhere in this note. Counting convention: the evaluation of f_i' at a stored reference point (w_t for L-SVRG and L-Katyusha, $\tilde{x}$ for Katyusha) counts as one gradient unless the n reference gradients are stored when the full gradient is computed ($\mathcal{O}(nd)$ memory, as for the table of SAGA), in which case it is a look-up; where the two conventions differ we give both counts, the first one without storage. The counts measured in the experiments of the report follow the first convention (the implementation stores the reference gradients but charges $f_{i(t)}'(w_t)$ as recomputed: n initially, then 2 per iteration and n per refresh, i.e. 3 per iteration in expectation for L-SVRG, L-Katyusha and the anchor-free schemes), so that they are about $\frac{3}{2}$ of the counts with storage; slopes in κ are unaffected. The leading n of the L-SVRG and L-Katyusha rows is the initial full gradient $F'(w_0)$.

Quotations used in Table 7. Gradient counts. [2, footnote 11]: "The claim 'SVRG or Katyusha computes 1.5 stochastic gradients' requires one to store $\nabla f_i(\tilde{x})$ in the memory for each $i \in [n]$, and this costs $O(dn)$ space in the most general setting. If one does not store $\nabla f_i(\tilde{x})$ in the memory, then each iteration of SVRG or Katyusha computes 2.5 stochastic gradients for the choice $m = 2n$." [47, Section 4]: "Algorithm 2 calls stochastic gradient oracle $\mathcal{O}(1 + pn)$ times per iteration in expectation." Lower bounds. [50, Corollary 3], in arXiv:1507.02000v3: "The number of gradient evaluations performed by any randomized incremental gradient methods for finding a solution $\bar{x} \in X$ of problem (1.1) such that $\mathbb{E}[\|\bar{x} - x^*\|_2^2] \leq \epsilon$ cannot be smaller than $\Omega\big((\sqrt{mC} + m)\log\frac{\|x^0 - x^*\|_2^2}{\epsilon}\big)$ if n is sufficiently large, where $C = L/\mu$ and $L = \sum_{i=1}^m L_i$" (their n is the dimension; in this sum convention their f_i is our f_i/n, while the journal version cited averages the components, $\frac{1}{m}\sum_i f_i + h + \mu\omega$, with $L = \frac{1}{m}\sum_i L_i$; in both versions L is our L when all L_i are equal, as on their hard instance), the worst-case instance being analyzed "for any arbitrary distribution used to choose f_i at each iteration". [87, Theorem 8], for m components that are γ-smooth and λ-strongly convex with $\gamma/\lambda > 161m$ and initial gap ϵ_0: any randomized algorithm $\mathcal{A}$ "must make $\Omega\big(m + \sqrt{m\frac{\gamma}{\lambda}}\log\big(\frac{\epsilon_0}{\epsilon}\sqrt{\frac{m\lambda}{\gamma}}\big)\big)$ queries to h_F".

In the regime $\kappa \gg n$ of hypothesis (A4), the worst-case rows read $n\kappa$ (GD, step $1/R^2$), $n\sqrt{\kappa}$ (AGD, step $1/R^2$), κ (SAGA, SVRG, L-SVRG) and $\sqrt{n\kappa}$ (Katyusha, L-Katyusha; optimal by the lower bounds), and Catalyst reaches $\sqrt{n\kappa}$ up to a factor $\log\kappa$ through a proximal-point outer loop. Among the methods of Table 7 that apply momentum directly to a variance-reduced estimator ($\mathcal{O}(1)$

Method	Gradient evaluations to accuracy ε	Source and constants
Gradient descent, $\gamma = 1/R^2$	$n\,\kappa \log \frac{1}{\varepsilon}$	Book, Section 5.4.4 ("running time $dn\frac{R^2}{\mu}\log\frac{1}{\varepsilon}$"); with $\gamma = 1/L_F$, $n\frac{L_F}{\mu}\log\frac{1}{\varepsilon}$ where $L_F \le R^2$.
Accelerated GD (5.16)–(5.17)	$n\sqrt{\kappa}\log\frac{1}{\varepsilon}$; with the step $1/L_F$, $n\sqrt{L_F/\mu}\log\frac{1}{\varepsilon}$, where $L_F \le R^2$	Book, Section 5.2.5: rate $(1-\sqrt{\mu/L})^t$; Proposition A3.17: $(1-\frac{1}{2}\sqrt{\mu/L})^t$.
SGD, $\gamma_t = 1/(\mu t)$	$\mathcal{O}(\kappa/\varepsilon)$ (no logarithm, sublinear)	Book, Section 5.4.4 ("the convergence rate has been shown to be $\mathcal{O}(\kappa/t)$ in Section 5.4.1") and the table of Section 5.5 (smooth case, rate $L/(t\mu)$); Prop. 5.8 is the Lipschitz case, $2B^2(1+\log t)/(\mu t)$.
SAGA, $\gamma = 1/(4R^2)$	$\left(3n + \frac{16}{3}\kappa\right)\log\frac{2n}{\varepsilon}$	Prop. 5.9 of the book: contraction $1-\min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}$, one gradient per iteration; [16].
SVRG (epoch length m)	$\mathcal{O}\big((n+\kappa)\log\frac{1}{\varepsilon}\big)$ with $m \asymp \kappa$	[43]; as quoted in [47, Section 1]: "the iteration complexity of SVRG is $\mathcal{O}((n+L/\mu)\log 1/\varepsilon)$".
L-SVRG, $\eta = 1/(6L)$, $p = 1/n$	$n + 3\lceil \max\{6\kappa, 2n\}\log\frac{\Phi_0}{\varepsilon}\rceil$ without storage, $n + 2\lceil \max\{6\kappa, 2n\}\log\frac{\Phi_0}{\varepsilon}\rceil$ with	[47, Theorem 5]: $\mathbb{E}\Phi^k \le \max\{1-\frac{\mu}{6L}, 1-\frac{1}{2n}\}^k\Phi^0$, $\Phi^k = \|x^k - x^*\|^2 + \frac{4\eta^2}{pn}\sum_i \|\nabla f_i(w^k) - \nabla f_i(x^*)\|^2$; $2 + pn = 3$ gradients per iteration without storage, $1 + pn = 2$ with.
Katyusha, $m = 2n$	$\mathcal{O}\big((n+\sqrt{n\kappa})\log\frac{F(\theta_0)-F_*}{\varepsilon}\big)$	[2, Theorem 2.1, Remark 2.2 and footnote 11]: 2.5 stochastic gradients per iteration without storage, 1.5 with (quoted below).
L-Katyusha, (A3.3)	$n + \left(12n + 15\sqrt{n\kappa}\right)\log\frac{3L(n+3)\|\theta_0-\eta_*\|^2}{8\varepsilon}$ without storage, $n + \left(8n + 10\sqrt{n\kappa}\right)\log\frac{3L(n+3)\|\theta_0-\eta_*\|^2}{8\varepsilon}$ with	Corollary A3.12; [47, Theorem 11]: $\rho = \min\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\}$; $2+pn = 3$ gradients per iteration without storage, $1 + pn = 2$ with (quoted below); the leading n is the initial $F'(w_0)$.
Catalyst on SAGA/SVRG	$\tilde{\mathcal{O}}\big(\sqrt{n\kappa}\log\frac{1}{\varepsilon}\big)$ for $n \le \kappa$ (extra $\log\kappa$ factor)	[54, Table 1].
Naive AGD+SVRG, $b = 1$	no bound better than $\min\{n+\kappa, n\sqrt{\kappa}\}\log\frac{1}{\varepsilon}$: Nitanda $\tilde{\mathcal{O}}(n + \frac{n-b}{n-1}\kappa + b\sqrt{\kappa})$; Driggs et al. $\mathcal{O}(n\sqrt{\kappa}\log\frac{1}{\varepsilon})$	[2, Remark 5.7] on [67]; [21, Theorem 4] with $b = 1$, $p = n$ (Section A3.4).
Lower bound, randomized incremental gradient methods (any sampling distribution)	$\Omega\big((n+\sqrt{n\kappa})\log\frac{\|\theta_0-\eta_*\|^2}{\varepsilon}\big)$, for a large enough dimension	[50, Corollary 3] (quoted below, with the conventions of the two versions; their m is our n).
Lower bound, any randomized algorithm, $\kappa > 161\,n$	$\Omega\big(n + \sqrt{n\kappa}\log\left(\frac{\varepsilon_0}{\varepsilon}\sqrt{\frac{n}{\kappa}}\right)\big)$, with no $n\log\frac{1}{\varepsilon}$ term	[87, Theorem 8] (quoted below); see hypothesis (A1) and [38] for the $n\log\frac{1}{\varepsilon}$ term and the role of the span assumption.

Table 7: Gradient complexity of the methods discussed in the book and in this note, in terms of n, $\kappa = L/\mu$ and ε.

gradients per iteration), the order $\sqrt{n\kappa}$ is certified only for Katyusha and L-Katyusha, whose proofs contain the cancellation (A3.19) (for Katyusha, its analogue in the epoch-based proof); without the anchor, the best certificates for $b = 1$ give $\min\{n + \kappa, n\sqrt{\kappa}\}$, i.e. the exponent $\frac{1}{2}$ of κ only together with the prefactor n [21], as for accelerated gradient descent, which computes n gradients per iteration. These rows are worst-case bounds over the instances with given (n, R^2, μ): with the step $1/L_F$ the full-gradient methods cost $n\kappa_F$ and $n\sqrt{\kappa_F}$ with $\kappa_F = L_F/\mu \leq \kappa$, and on the block-separable hard instance (H), where $L_F = R^2/n$, GD costs of order κ and AGD of order $\sqrt{n\kappa}$ gradient evaluations, the orders of SAGA and L-Katyusha (Section A1.4).

R Reproducibility

Code and data. The code, the small cached results, the summaries and the figures are available at

https://github.com/louisDGL/
Accelerating-Variance-Reduction---Optimal-Complexity-of-Finite-Sum-Optimization

The repository contains the Python package `vrlab` (problems, high-precision optimum, algorithms, exact instrumentation, bounds, gap attribution), a test suite (`tests/test_vrlab.py`), the experiment scripts in `experiments/`, the theory notes in `theory/`, the bibliography in `bibliography/`, this report in `report/`, the summaries (`results/*_summary.json` and `results/digest.md`) and their analysis (`results/findings.md`), the small cached runs (`A3.pkl`, `A4.pkl`, `A4b.pkl`, `B2.pkl`, `GammaDelta.pkl` and `validation_book.pkl` in `results/`) and the figures in `figures/`. The large caches `results/A2.pkl`, `results/B1.pkl` and `results/B3.pkl` are not versioned and are rebuilt by the scripts that produce them. Dependencies: numpy, scipy, matplotlib (see `requirements.txt`); Python 3.14 was used.

Commands. The commands of `README.md`, each run from the root of the repository, install the dependencies, run the tests and the experiments, rebuild the digest and compile the documents:

```
pip install -r requirements.txt
python tests/test_vrlab.py              # or: python -m pytest tests -q
python experiments/run_all.py --quick   # smoke test (a few minutes)
python experiments/run_all.py           # full experiments (many CPU hours)
python experiments/retry_timeouts.py A4 A4b   # drop the entries that hit the CPU budget
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 VRLAB_MAX_TIME=4000
python experiments/exp_A4_hard_instance.py
python experiments/exp_A4b_instances.py
python experiments/make_digest.py
cd theory && pdflatex main.tex && bibtex main && pdflatex main.tex && pdflatex main.tex
python report/build.py                  # make_restated.py, main.pdf, synthesis.pdf
```

`experiments/run_all.py` runs the scripts of all the experiments, those of hypotheses B3, B2, B1, A3, A2 and A4 (`exp_A4_hard_instance.py` and `exp_A4b_instances.py`), of Section 5.4 and of Section 5.2, together with `check_B2_fp_heuristic.py`, the sanity check of Remark B2.20 (gradient descent beyond quadratics: a heuristic) (it writes `results/B2_fp_check_summary.json`), and then `experiments/make_digest.py`; with `--quick` it runs a smoke test, whose figures get a `_quick` suffix, and skips the digest. Results are cached in `results/*.pkl` (delete a cache to recompute its experiment) and summarised in `results/*_summary.json`, which `experiments/make_digest.py` collects in `results/digest.md`; every number quoted in this report is read from these files or computed from them and from the cached runs. The statements restated in Section 4 are not typed in the report: `report/make_restated.py` copies them from the theory notes into `report/restated/` (one file per labelled result of the notes, and `report/restated/names.tex` with the type, appendix and name of each), so that the notes remain their only source. It must be run before `report/main.tex` and `report/synthesis.tex`, the standalone extract of Section 4, are compiled. `report/build.py` runs it, then compiles `main.tex` and `synthesis.tex` in this order (pdflatex, bibtex and pdflatex twice for each), since the cross-references of the extract are read from `main.aux`.

Tests. The test suite (`tests/test_vrlab.py`) checks the gradients and Hessians of logistic regression against finite differences and its per-sample smoothness constants; the Newton solver, on

a small problem ($\|F'(\eta_*)\| < 10^{-12}$) and on the problem $\kappa = 10$, $\alpha = 1$ of Section 5.3, where the corrected line search must reach $\|F'(\eta_*)\| \le 10^{-15}$ within ten iterations; the constants and the exact minimizers of the instances (H), (S) and (A), including the geometric minimizers of (S) and (A); that the exact conditional expectations used in the tightness analysis coincide with the average over the n possible indices of the realized next state, for SAGA in the Euclidean metric, in a dense and in a diagonal quadratic metric and on (S), and for the Lyapunov function of L-Katyusha (both outcomes of the coin included); that the tightness ratios of I1–I5 lie in $(0, 1]$ at a random state; that the attribution chains of orders A and B (Proposition B3.11) are monotone and connect the exact contraction to the certified rate; that, for a pair (γ, Δ) other than the textbook one, the a-posteriori factors of the trajectory-dependent bound (Proposition B3.19) dominate the exact one-step contractions; that the exact one-step contraction of the Lyapunov function of L-Katyusha stays below the largest coefficient of the one-step inequality of Theorem A3.4 (contraction and rate of L-Katyusha; Lemma 10 and Theorem 11 of [47]), at a random state and along a short run; that the mean of $\|\theta_t - \eta_*\|^2$ over six SAGA runs on a small problem stays below the bound of Proposition 5.9; that SAGA, SAGA in a diagonal and in a dense quadratic metric, L-SVRG, L-Katyusha, GD and AGD converge on small problems, and so do L-Katyusha with $\theta_2 = 0$ on two well-conditioned problems ($\kappa = 5$ and 20) and the naive scheme (Nesterov momentum with the SVRG estimator) on an easy problem, whose gradient count is also checked; that the error of SGD decreases; and the log–log slope fit with its interval. The divergence of L-Katyusha with $\theta_2 = 0$ on ill-conditioned instances is observed in the experiments only.

Experiments. Seeds: the logistic data sets are generated with seed 0 (for each α, one data set shared by all values of κ, which only set λ), the random block quadratics (R) with the sampling seed (one instance per seed), and the chain instances (H), (S) and (A) are deterministic; the algorithms are run with seeds $0, \dots, S-1$ (the number of seeds S is stated with each figure). Every "expectation" of a one-step quantity is computed exactly by enumerating the sampled index; curves labelled "mean over seeds" are empirical averages over the S runs. Trajectories are analysed only above the rounding floor (10^{-18} on V_t for SAGA and on $\|\theta_t - \eta_*\|^2$ in Section 5.5, a floor set by the precision of η_*; 10^{-10} for the Lyapunov function of L-Katyusha, whose W-term amplifies the rounding error of $F(w) - F(\eta_*)$ by the factor $\theta_2(1 + \theta_1)/(p\theta_1)$, from about 250 to about 2400 in our runs). Running times on one core range from tens of seconds per logistic configuration of Section 5.3 to several hours for each κ sweep.

CPU-time budget of the κ sweeps. In the sweeps of Section 5.9, every solver run is capped by `VRLAB_MAX_TIME` seconds of CPU time, measured with `time.process_time` (default 600; a suspended machine consumes no budget), and a run that reaches the cap does not enter the fits. The reported sweeps were run with single-threaded BLAS and a budget of 4000 s per run, raised to 8000 s for the last runs of the A4b sweep, among them SAGA at the nominal $\kappa = 10^6$ on (S); every run reached its target, except the runs of L-Katyusha without negative momentum, which diverge for $\kappa \ge 10^3$. The grid search of the naive scheme gives each grid run 45 s of CPU time; the grid runs have the same cost per iteration, so a run cut by this budget had already used more gradients than the selected setting needed in total, and the cap does not affect the selection.

Minimizers. For the quadratic instances, η_* is obtained by a direct solve. For logistic regression, the cached results were computed with a damped Newton method whose Armijo test, decided by the rounding error of F near the optimum, stalled at $\|F'(\eta_*)\| = 1.3 \cdot 10^{-12}$ on one problem ($\kappa = 10$, $\alpha = 1$; at most $9 \cdot 10^{-16}$ on the fourteen other problems of Section 5.3). By strong convexity the

squared error of η_* is at most $\|F'(\eta_*)\|^2/\mu^2$: $2.1\cdot10^{-21}$ on that problem, where the corrected solver moves η_* by a squared distance of $1.3\cdot10^{-21}$, and $7\cdot10^{-20}$ at most over the fifteen problems, attained at $\kappa = 10^5$, $\alpha = 0$, whose runs stay above 10^{-13}. On the problems whose runs reach the analysis floor, these errors change $\|\theta_t - \eta_*\|^2 = 10^{-18}$ by at most about 10%. The line search has since been corrected in `vrlab/solvers.py` (near the optimum, a step is accepted as soon as it decreases $\|F'\|$); the results reported here were computed before this correction.

Figures. Colours follow a fixed, colour-blind-safe categorical palette: a given algorithm or a given inequality keeps its colour across all figures; sequential magnitudes use a single-hue ramp.